



Intelligent Security API

Developer Guide

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

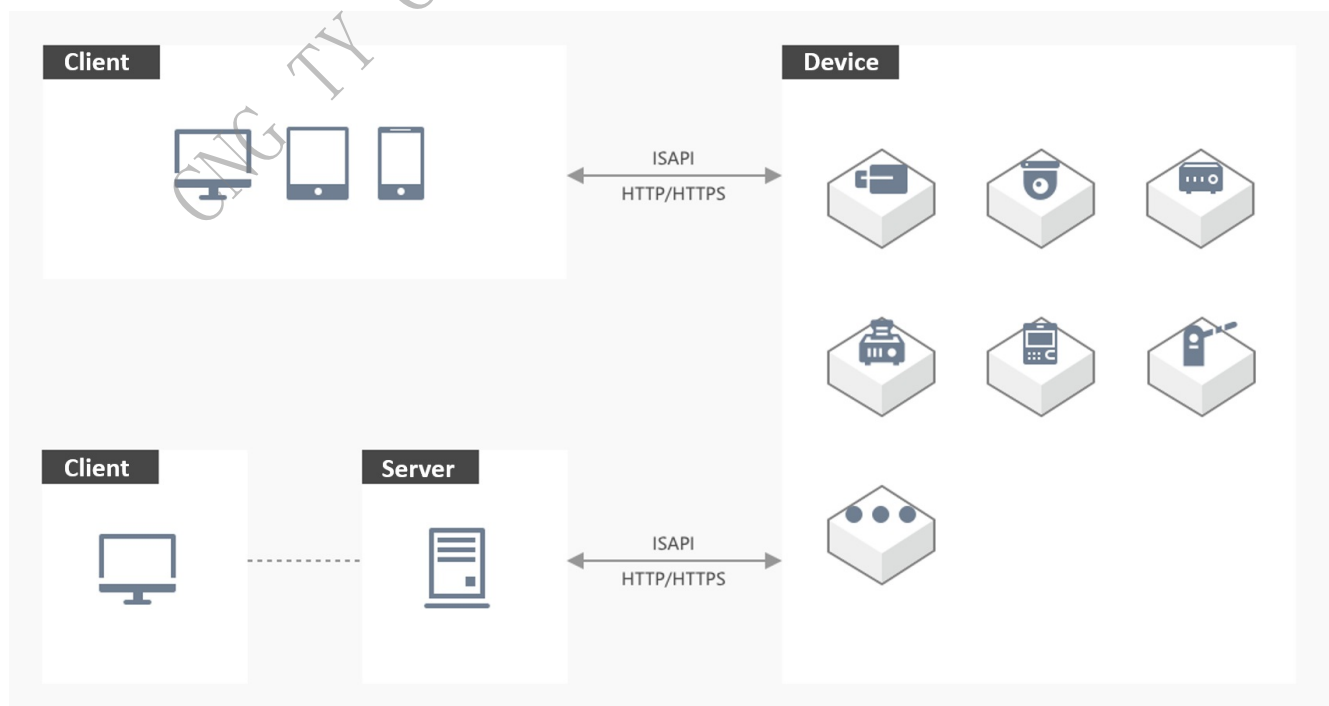
2.1 Introduction

Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on [HTTP \(Hypertext Transfer Protocol\)](#) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP. Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, [RTSP \(Real-Time Streaming Protocol\)](#) based on TCP/UDP for live view and video playback of devices, etc.

Application Layer	Device Discovery SADP	Signaling Interaction ISAPI (HTTP)	Media Stream RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- Controller
 - Videowall Controller

DS-C30S-02DPI/4K, DS-C30S-02HI/4K, DS-C30S-02HO/4K, DS-C30S-04DI, DS-C30S-04DO, DS-C30S-04HI, DS-C30S-04HO, DS-C30S-04VI, DS-C30S-DEC, DS-C30S-L104, DS-C30S-MCU, DS-C30S-PWR, DS-C30S-S11, DS-C30S-S23, DS-C30S-SW, DS-C60S-02DPI/4K , DS-C60S-02HI/4K, DS-C60S-02HO/4K, DS-C60S-04DI, DS-C60S-04DO, DS-C60S-04HI, DS-C60S-04HO, DS-C60S-16NO/2FO, DS-C60S-20NO, DS-C60S-DEC, DS-C60S-MCU, DS-C60S-PRE, DS-C60S-S6, DS-C66S-02DPI/4K, DS-C66S-02HI/4K, DS-C66S-02HO/4K, DS-C66S-04DI, DS-C66S-04DO, DS-C66S-04HI, DS-C66S-04HO, DS-C66S-04SDI/4K, DS-C66S-16NO/2FO, DS-C66S-20NO, DS-C66S-DEC, DS-C66S-MCU, DS-C66S-PRE, DS-C66S-PWR, DS-C66S-S12, DS-C66S-S6

2.3 Terms And Definitions

2.3.2 Device Operation Log

Logs generated during the operation of the device's firmware. These logs are recorded in log files or log systems in text format, and are primarily used by device developers and maintenance personnel to identify device issues. #

2.4 Symbols And Acronyms

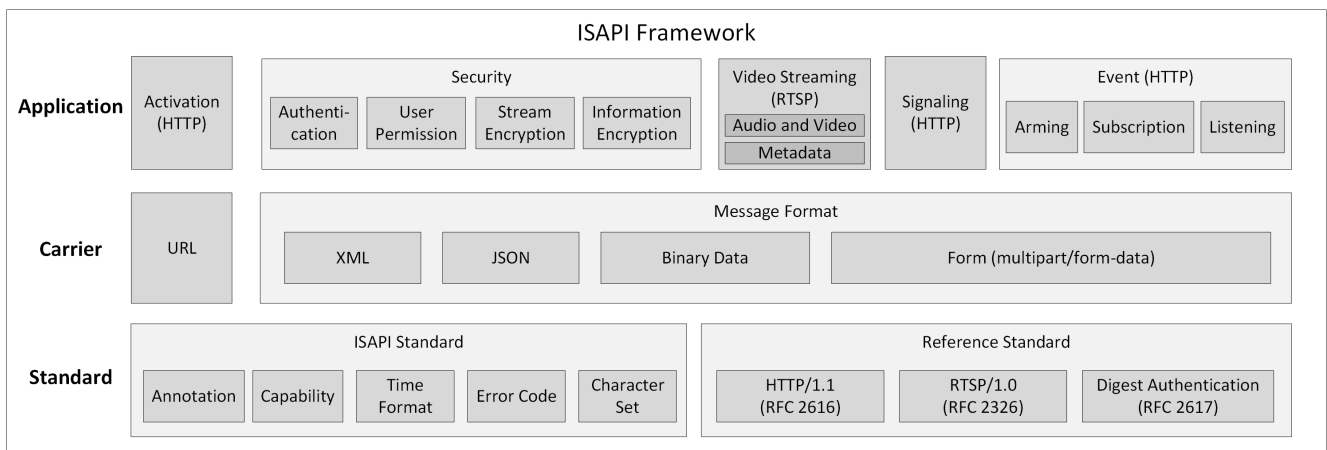
- admin: administrator
- HC: Connect Mobile Client
- HPP: Partner Pro
- NVR: Network Video Recorder
- IPC: IP Camera

2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview



Notes:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

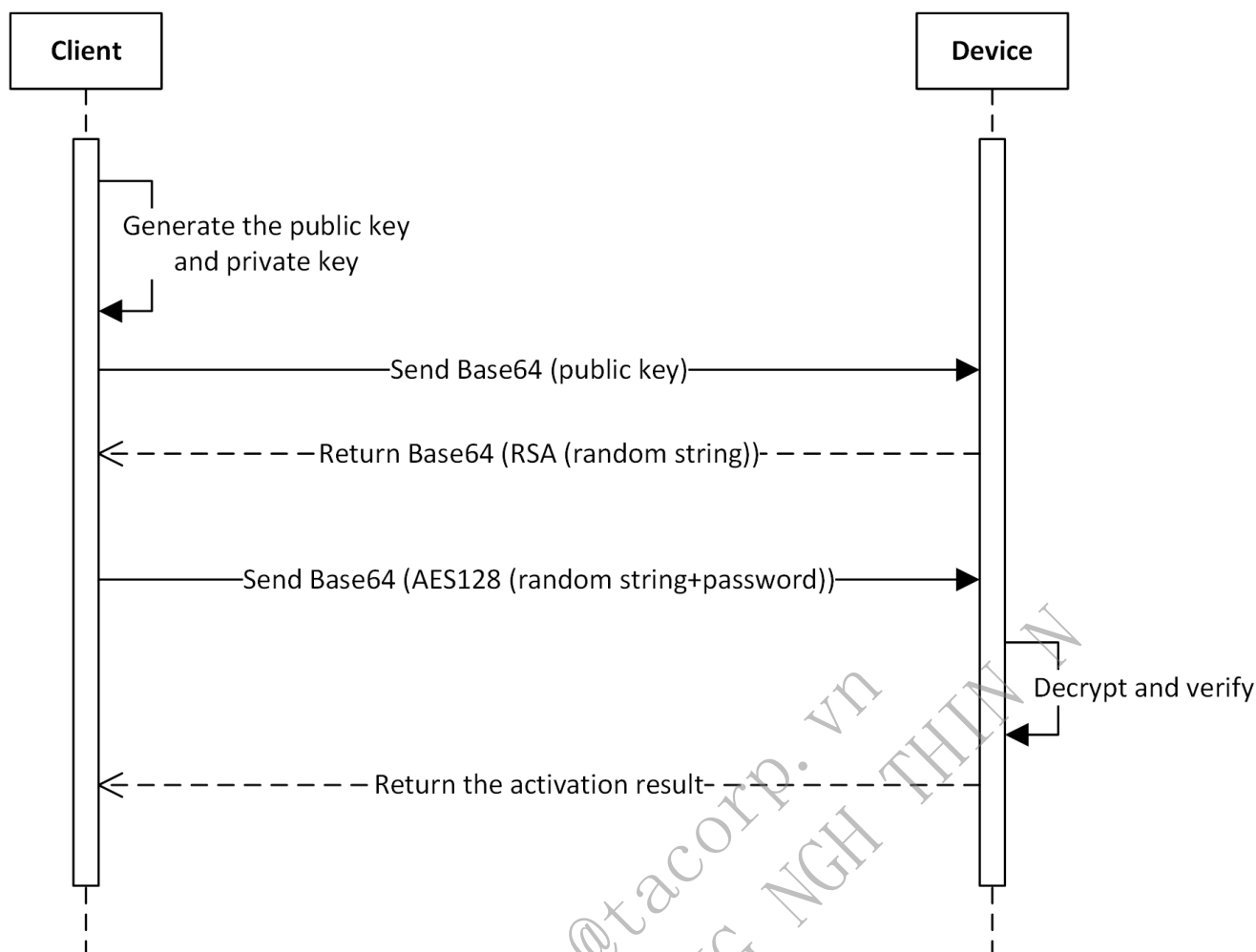
3.2 Activation

The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.



Firstly, two operations are defined:

- `bytesToHexstring`: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23 -> 7f0a17`
- `hexStringToBytes`: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17 -> 127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via `bytesToHexstring` and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via `hexStringToBytes`. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via `bytesToHexstring`, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via `hexStringToBytes` to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via `hexStringToBytes` to get a 16-byte AES key. Then the client uses the AES key to encrypt the "string consisting of the first 16 characters of the random string and the real password" by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via `bytesToHexstring`, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`).
Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via `hexStringToBytes` to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.

7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in [RFC 7616](#)) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in [Authentication](#) of Quick Start Guide.

3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

3.4 Video Streaming

3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in [RFC 7826](#)). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in [RFC 7616](#)) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)


```
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

```
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the Content-Type in the headers of the HTTP request is application/xml; charset="UTF-8".

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace http://www.isapi.org/ver20/XMLSchema and ISAPI version number 2.0 of XML messages are configured by default, see the example below.

```
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The Content-Type in the headers of the HTTP request is often application/json.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g.,

http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json . Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data`, `boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to [RFC 1867 \(Form-Based File Upload in HTML\)](#). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

```

HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
  "ipAddress": "172.6.64.7",
  "macAddress": "01:17:24:45:D9:F4",
  "channelID": 1,
  "dateTime": "2009-11-14T15:27+08:00",
  "eventType": "mixedTargetDetection",
  "eventDescription": "Mixed target detection",
  "deviceID": "123456789",
  "CaptureResult": [{
    "targetID": 1,
    "Human": {
      "Rect": {
        "height": 1.0,
        "width": 1.0,
        "x": 0.0,
        "y": 0.0
      },
      "contentID1": "humanImage", /*human body thumbnail*/
      "contentID2": "humanBackgroundImage", /*human body background picture*/
      "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
      "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
    }
  ]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--

```

4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```

<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#level 1,level2#level 2,level3#level 3]-->level1
    </level>
  </Node>
</NodeList>

```

```

{
  "name": "test",
  /*ro, req, string, name, range:[1,32]*/
  "type": "type1",
  /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
  "enabled": true,
  /*ro, opt, bool, enable or not, desc:xxxxxx*/
  "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
    "scene": 1,
    /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
    "ID": 1
  }
  /*req, int, No., range:[1,8], step:, unit:, unitType:*/
}

```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupporttxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```

{
  "isSupportMap": true,
  /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
  "isSupportAlgTrainResourceInfo": true,
  /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
  desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
  "isSupportAlgTrainAuthInfo": true,
  /*ro, opt, bool, whether it supports only getting the authorization information of the algorithm training platform,
  desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
  "isSupportAlgTrainNodeList": true,
  /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
  format=json*/
  "isSupportNAS": true
  /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}

```

The capability message example in XML format is shown below:

```

<isSupportNetworkStatus>
  <!--ro, opt, bool, whether it supports searching the network status, desc: reLated API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>

```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```

{
  "boolType": {
    /*req, object, example of the capability of type bool*/
    "@opt": [true, false]
    /*req, array, options, subType: bool*/
  },
  "integerType": {
    /*req, object, example of the capability of type integer*/
    "@min": 0,
    /*ro, req, int, the minimum value*/
    "@max": 100
    /*ro, req, int, the maximum value*/
  },
  "stringType": {
    /*req, object, example of the capability of type string*/
    "@min": 0,
    /*ro, req, int, the minimum string size*/
    "@max": 32
    /*ro, req, int, the maximum string size*/
  },
  "enumType": {
    /*req, object, capability example of type enum*/
    "@opt": ["enum1", "enum2", "enum3"]
    /*req, array, options, subType: string*/
  }
}

```

The example of XML format is shown below:

```

<boolType opt="true,false" def="true">
  <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
  <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
  <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
  <!--ro, opt, string, example of the capability of type enum-->white
</enumType>

```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts [ISO 8601 Standard Time Format](#), which is the same as [W3C Standard Date and Time Formats](#).

Format: YYYY-MM-DDThh:mm:ss.sTZD

YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
  "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the
device's local time and UTC*/
  "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert
the dateTime into the local time for use*/
}
```

4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to [RFC 2044](#)).

No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	\$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	'
12	Underscore	_	29	Vertical Bar	
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	\
17	Octothorpe	#			

The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in [RFC 2616](#). For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

```

HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
  "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
  "statusCode": 4,
  "statusString": "Invalid Operation",
  "subStatusCode": "notSupport",
  "errorCode": 1073741825,
  "errorMsg": "notSupport"
}

```

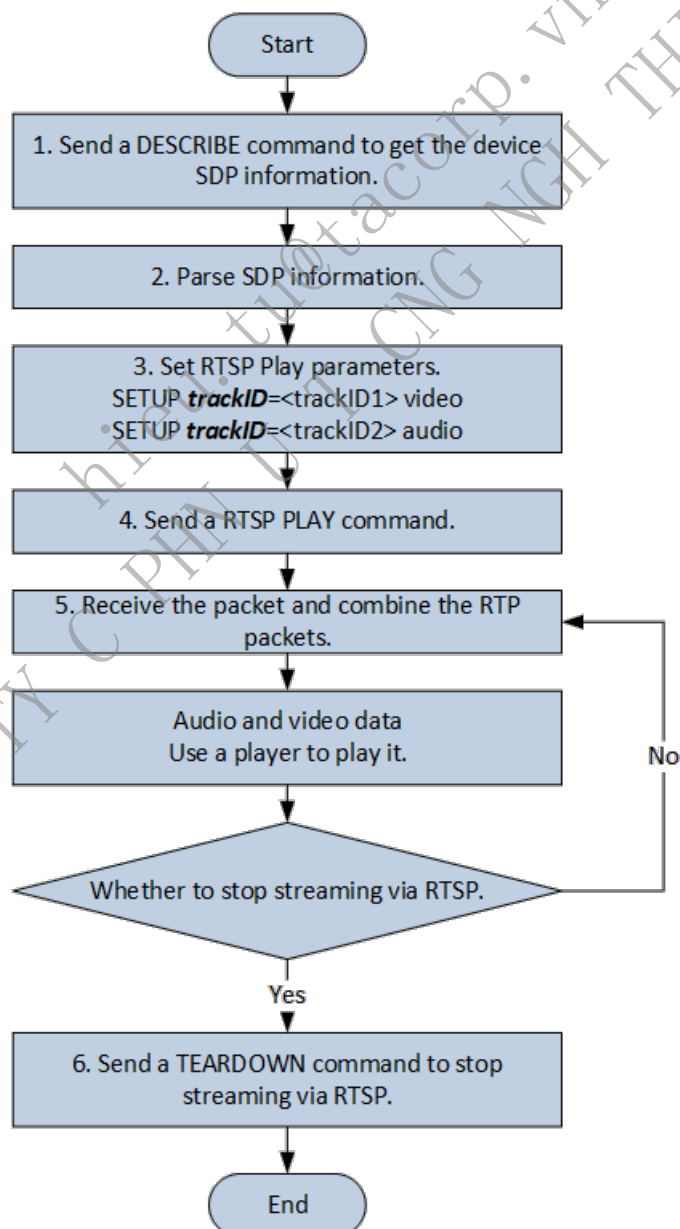
4.3 Real-Time Live View

4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in [RFC 7826](#)).

4.3.2 API Calling Flow



1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`. Digest authentication with devices is required before this step.
2. The client parses the media SDP information returned by the device.

3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2=value2...)`, such as `/Streaming/Channels/101?transportmode=unicast`.

4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.


```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. ....d1.w....C....".T....g....).i.....a....7.S...~J.....X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

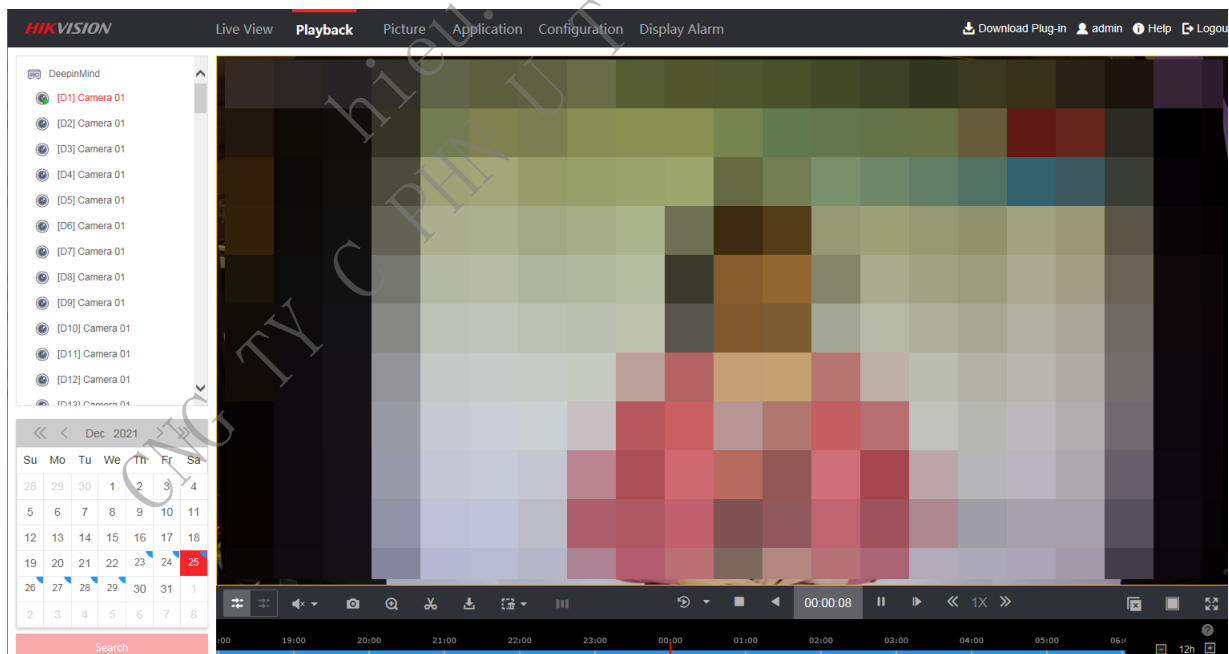
```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

4.4 Playback

4.4.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in [RFC 7826](#)).

Below is an example of playback on a web client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos are shown on the calendar and you can click a date to play back searched videos of the corresponding day.



4.4.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.

3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. Refer to streaming via RTSP in real-time streaming. See details in [RFC 7826](#) (Real Time Streaming Protocol).

Notes:

- Playback via RTSP needs digest authentication. The method is the same as the one of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in [RFC 7826](#) (Real Time Streaming Protocol).

4.4.3 Example

4.4.3.1 Search for Videos According to Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
  <year>2021</year>
  <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

```

HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <dayList>
    <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
    <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
    <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
    <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
    <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
    <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
    <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
    <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
    <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
    <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
    <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
    <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
    <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
    <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
    <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
    <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
    <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
  </dayList>
</trackDailyDistribution>

```

4.4.3.2 Search for Videos

Request from Client

```

POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
  <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
  <trackList>
    <trackID>101</trackID>
  </trackList>
  <timeSpanList>
    <timeSpan>
      <startTime>2021-08-16T00:00:00Z</startTime>
      <endTime>2021-08-18T23:59:59Z</endTime>
    </timeSpan>
  </timeSpanList>
  <maxResults>100</maxResults>
  <searchResultPostion>0</searchResultPostion>
  <metadataList>
    <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
  </metadataList>
</CMSearchDescription>

```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
  <responseStatus>true</responseStatus>
  <responseStatusStrg>OK</responseStatusStrg>
  <numOfMatches>1</numOfMatches>
  <matchList>
    <searchMatchItem>
      <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
      <trackID>101</trackID>
      <timeSpan>
        <startTime>2021-08-18T15:18:15Z</startTime>
        <endTime>2021-08-18T15:19:08Z</endTime>
      </timeSpan>
      <mediaSegmentDescriptor>
        <contentType>video</contentType>
        <codecType>H.264-BP</codecType>
        <playbackURI>rtsp://10.14.97.40/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788</playbackURI>
        <lockStatus>unlock</lockStatus>
        <name>00000004667000100</name>
      </mediaSegmentDescriptor>
      <metadataMatches>
        <metadataDescriptor>recordType.meta..com/timing</metadataDescriptor>
      </metadataMatches>
    </searchMatchItem>
  </matchList>
</CMSearchResult>
```

4.4.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

P commands two times, and the server responds to them.

```
acks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
ing Media v2016.11.28)
.048-61049
.048-61049;server_port=62000-62001,ssrc=d6d3ed5
limited, immutable,scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
```

acks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000004667000100&size=14007
realme="323852ae0234c718f2d4198b", nonce="55e585b9", uri="rtsp://10.14.97.40:554/Streaming/track
18T151908Z&name=0000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
ng Media v2016.11.28)
.048-61049
048-61049;server_port=62000-62001;ssrc=d6d3ed5
limited, Immutable, Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"

048-61049;server_port=62000-62001,ssrc=d6d3ed5

limited,Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"

acks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788

realtime="333852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/track/1/18T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"

ng Media v2016.11.28)

050-61051

050-61051;server_port=62002-62003;ssrc=d6d3ed6

limited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"

AY request, and the server starts to push streams for playback.

acks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=140078

realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/track

18T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"

ng Media v2016.11.28)

```
acks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", "response": "4d8b7ceb301c06b40a9ae411189a909f"
RealM="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
Long Media v2016.11.28)
```

```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

4.5 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

4.5.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

Notes

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

4.5.1.1 Arming without Subscription

1. Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
2. Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
3. Terminate the arming connection when no event message needs to be received.

Event Message Parsing:


```

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP
Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8" <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format according to Content-Type when parsing event messages-->
Content-Length: text_length

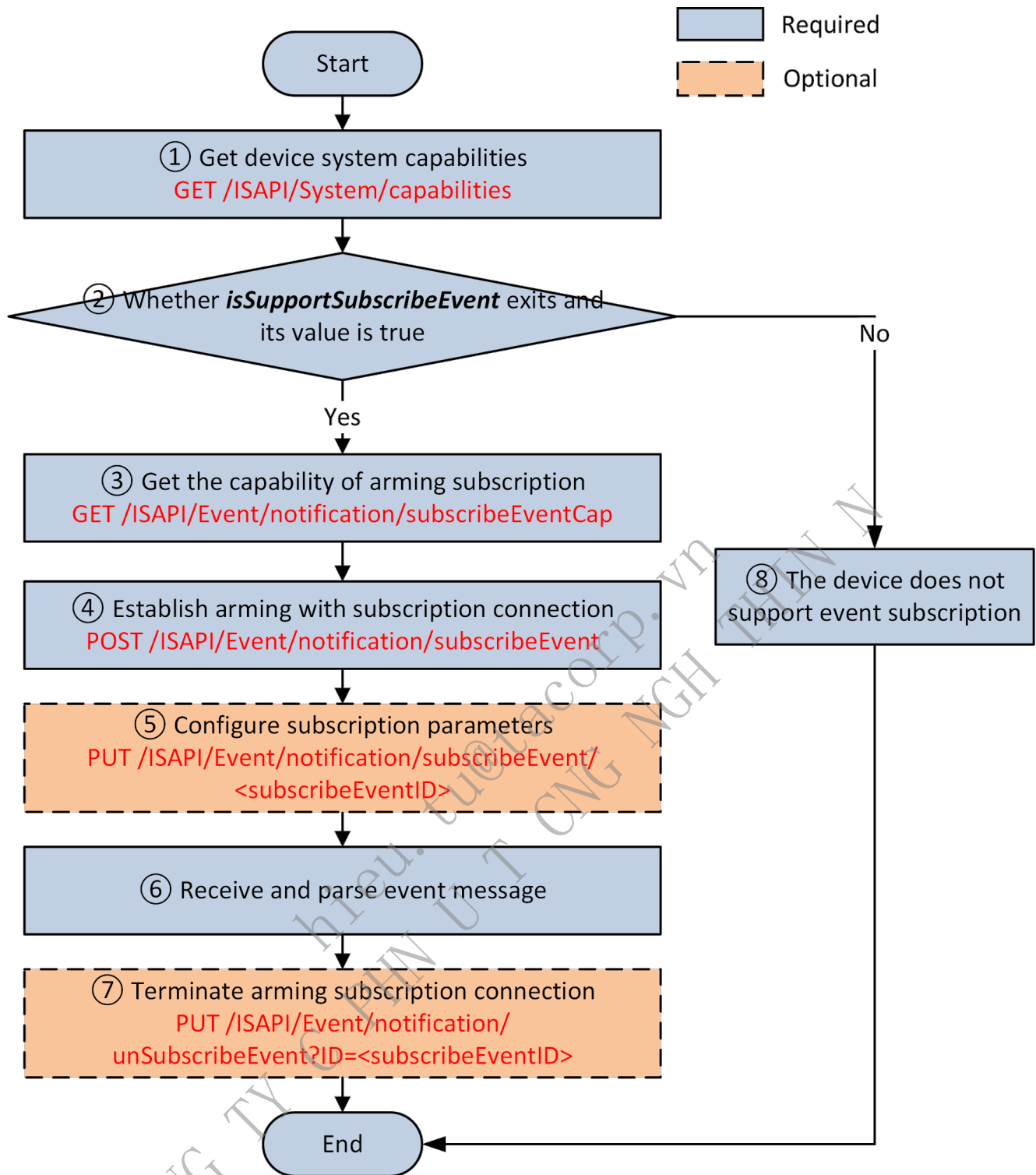
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

[Picture Data]
--<frontier>

```

Note: <data_gateway_ip> and <frontier> are variables, [Picture Data] indicates the raw data of a picture.

4.5.1.2 Arming with Subscription



1. Get device system capabilities: GET /ISAPI/System/capabilities.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: GET /ISAPI/Event/notification/subscribeEventCap.
4. Establish a connection of arming with subscription: POST /ISAPI/Event/notification/subscribeEvent. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>. Then, edit the parameters based on the existing subscription configurations: PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: <SubscribeEventResponse/>, <EventNotificationAlert/>, and picture data. <SubscribeEventResponse/> is the data of first form sent by the device after arming established, see the response parameters of URL (POST /ISAPI/Event/notification/subscribeEvent) for details; and <EventNotificationAlert/> is the event content or heartbeat, you can identify the event type via field eventType, e.g., for heartbeat, the value of eventType is heartBeat.

Event Messages Parsing:

Client Creates Arming with Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin", realm="IP
Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

Server Responds to Request and Push Event Message

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the Content-Type is multipart/form-data, boundary=AaB03x, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see [RFC 1867 \(Form-based File Upload in HTML\)](#). An example is shown below. Please note two hyphens -- before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark --AaB03x-- will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

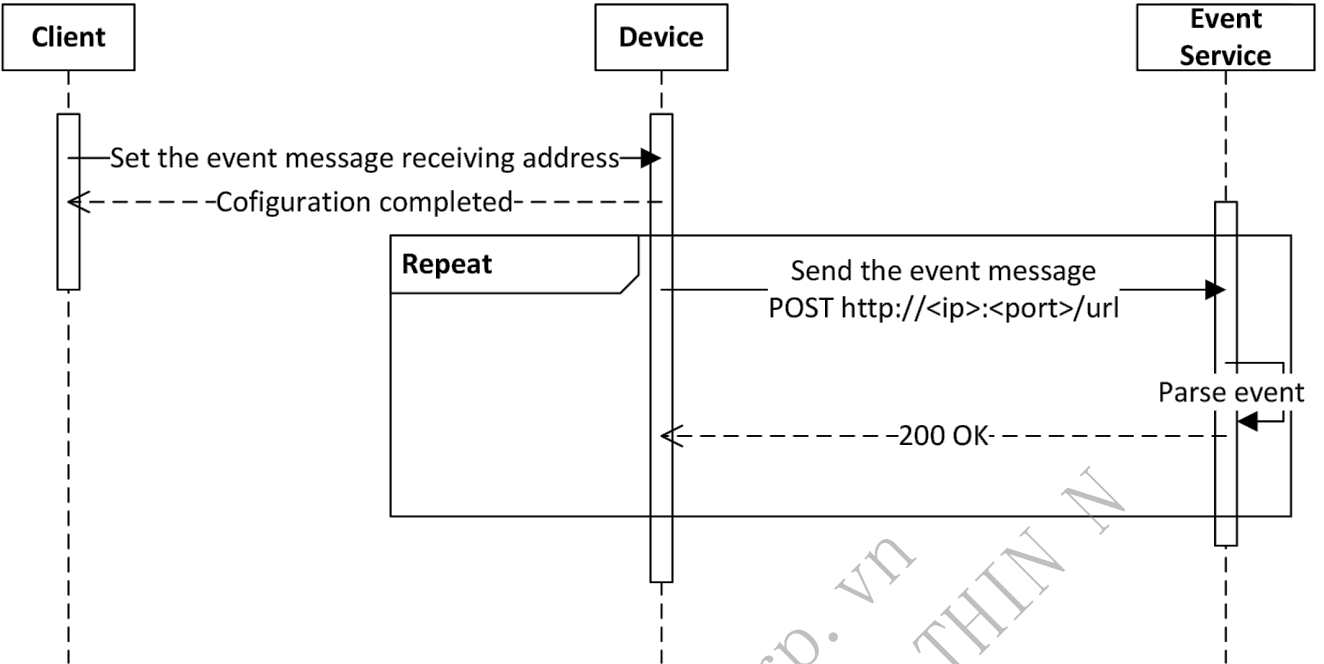
Picture Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. A form message which starts with --boundary and ends with --boundary--.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description. form-data is a piece of form data.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

4.5.2 Listening

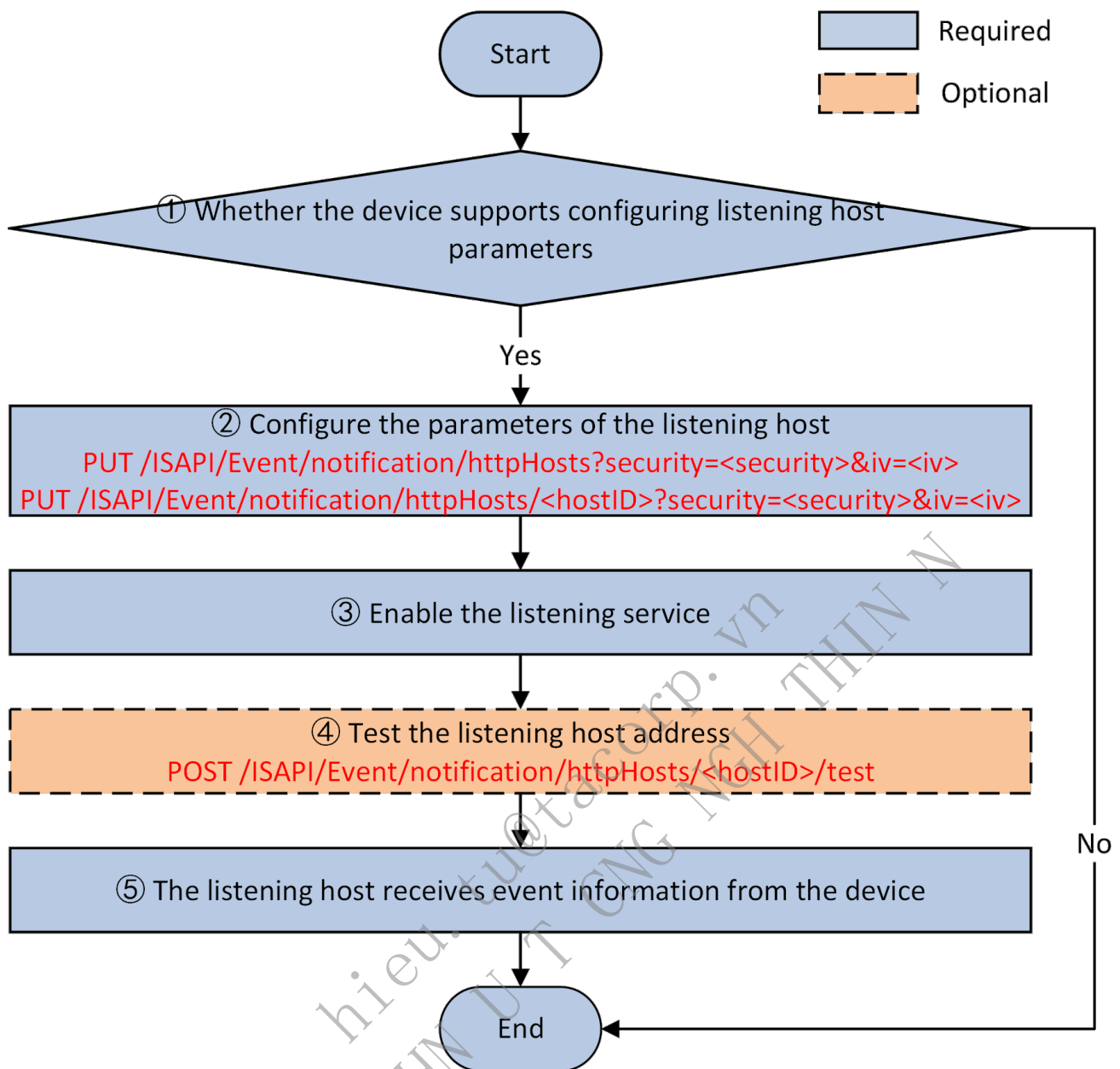
After a client enables the listening service, when an event occurs, the device will send the event information actively to the configured event receiving address. The event receiving address should be valid and configured on the device.



Notes:

- The client and event service can be the same program.
- In listening mode, no heartbeat information is generated on devices.

4.5.2.1 API Calling Flow



1. Check whether the device supports configuring listening host parameters.

Get the configuration capability of the listening host: `GET /ISAPI/Event/notification/httpHosts/capabilities`.
 If the node `<HttpHostNotificationCap>` is returned and its value is true, it indicates that the device supports configuring listening host parameters.

2. Configure parameters of the listening host.

Configure parameters of all listening hosts: `PUT /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

Get parameters of all listening hosts: `GET /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

Configure parameters of a listening host: `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

Get parameters of a listening host: `GET /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`.

3. Enable the listening service.

You need to enable the listening service of the listening host.

4. (Optional) Test the listening service.

The platform applies the command to the device to test whether the listening host is available for the device: `POST /ISAPI/Event/notification/httpHosts/<hostID>/test`.

5. The listening host receives event information from the device.

When an event occurs, the device creates an connection with the client and uploads alarm information actively. Meanwhile, the listening host receives data from the device. See details in **Event Messages**.

Note: You can also configure the listening parameters such as the time out via URL

`/ISAPI/Event/notification/httpHosts/<hostID>/uploadCtrl`.

4.5.2.2 Event Messages

When an event occurs or an alarm is triggered in listening mode, the event/alarm information can be uploaded with binary data (such as pictures) and without binary data.

1. Without Binary Data:

The `Content-Type` in headers of the HTTP request sent by the device is usually `application/xml` or `application/json` as follows:

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: data_gateway_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: application/xml; <!--Content Type, which is used for the upper Layer to distinguish different formats when parsing the message-->
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

<EventNotificationAlert/>
```

Response by the Listening Host

```
HTTP/1.1 200 OK
Date: YourDate
Connection: close
```

2. With Binary Data:

The format of the data sent by the device is HTTP form (multipart/form-data). The `Content-Type` in headers of the HTTP request is usually `multipart/form-data; boundary=<frontier>`, of which boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. See details in [RFC 1867 \(Form-based File Upload in HTML\)](#). An example is shown below. Please note two hyphens `--` before and after the boundary.

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, , related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: device_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml <!--some event messages are uploaded in JSON format, and the upper Layer needs to distinguish the message format according to Content-Type when parsing event messages-->

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture Data]
--<frontier>--
```

Response by the Listening Host

HTTP/1.1 200 OK
Date: YourDate
Connection: close

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=frontier	Content type, multipart/form-data refers to data in form format.
boundary	frontier	Delimiter of the form message. A form message which starts with <code>--boundary</code> and ends with <code>--boundary--</code> .
Content-Disposition	form-data; name="Picture_Name";	Content description. <code>form-data</code> is a piece of form data.
filename	"Picture_Name"	File name. The file refers to the form message.
Content-Length	10	Content length, starting from the next <code>\r\n</code> to the next <code>--boundary</code> .

4.5.2.3 Exception Handling

Error Codes

statusCode	statusString	subStatusCode	errorCode	Description
6	Invalid Content	eventNotSupport	0x60001024	

5 Device Management (General)

5.1 Calling Flow of Device Packet Capture

5.1.1 Function Introduction

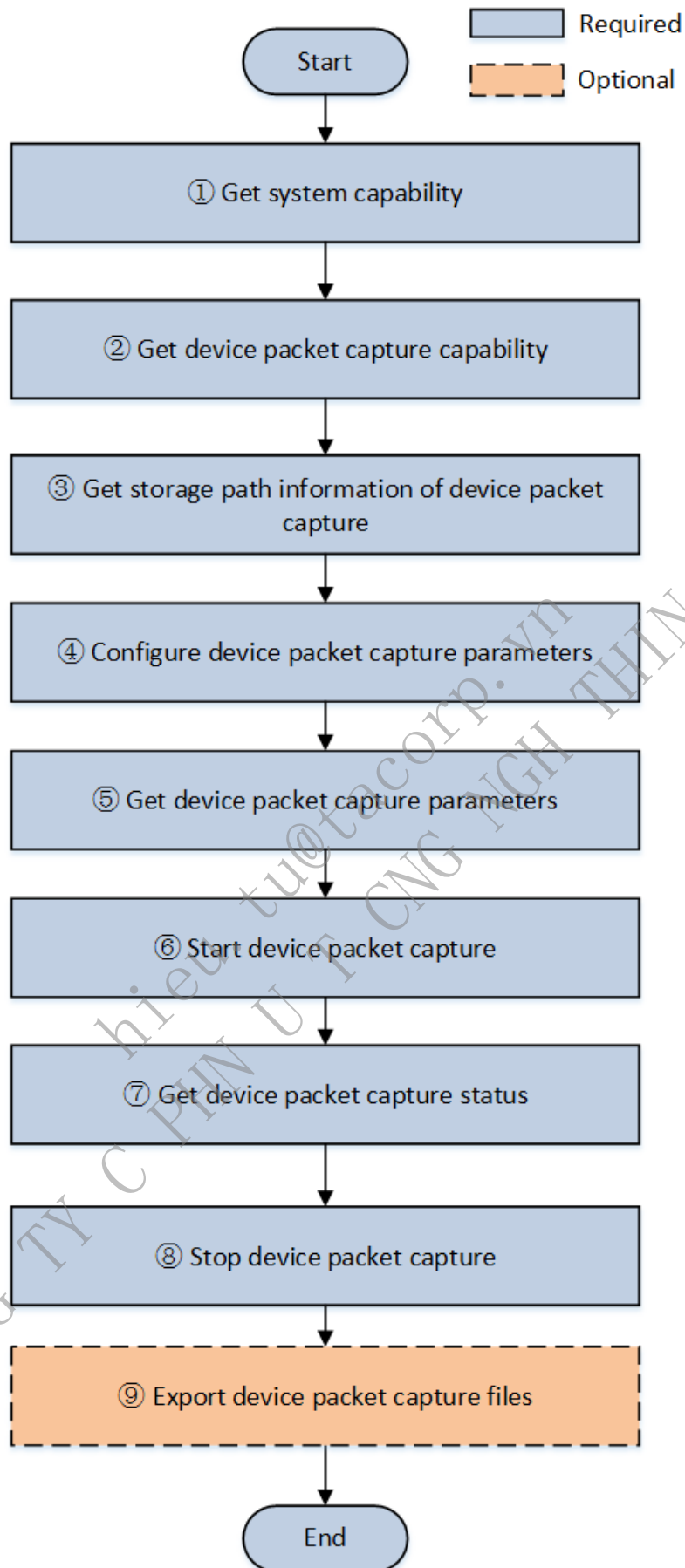
When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.1.2 API Calling Flow

5.1.2.1 Device Packet Capture



1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: GET /ISAPI/System/networkCapture/StoragePathInfo?format=json.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: PUT /ISAPI/System/networkCapture/captureParams?format=json.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: GET /ISAPI/System/networkCapture/captureParams?format=json.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: GET /ISAPI/System/networkCapture/manualStatus?format=json.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

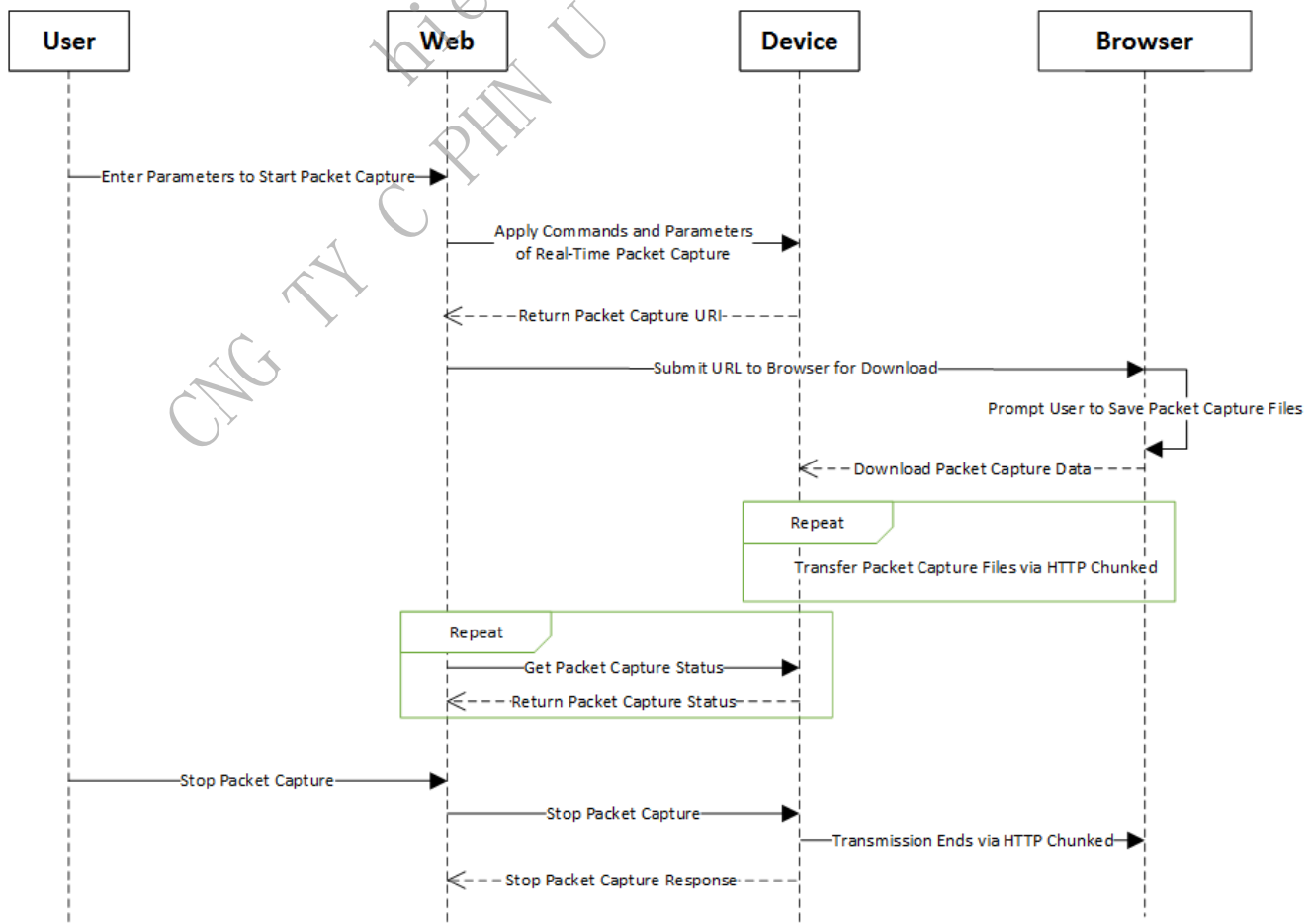
Stop device packet capture: PUT /ISAPI/System/networkCapture/manualStop?format=json.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

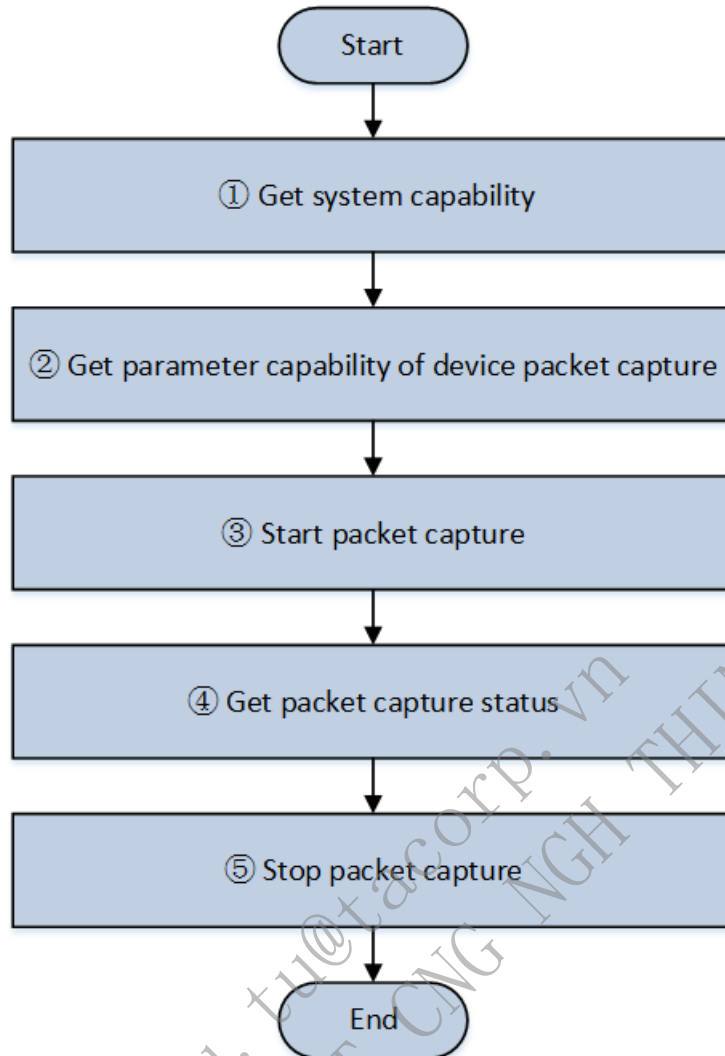
Export device packet capture files: GET /ISAPI/System/networkCapture/exportFile?format=json.

5.1.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



See the following figure for the calling flow:



1. Get device system capabilities: GET /ISAPI/System/capabilities.

<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture> indicates the device supports starting packet capture. <isSupportStopNetworkCapture>true</isSupportStopNetworkCapture> indicates the device supports stopping packet capture. <isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus> indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json. The realTimeEnabled field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URL, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>.

Note:

- If realTimeEnabled=true is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: GET /ISAPI/System/GetNetworkCaptureStatus?format=json.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: POST /ISAPI/System/StopNetworkCapture?format=json.

5.2 Device Hardware Asset Management

5.2.1 Introduction to the Function

Hardware assets: the host assets (CPU, memory, and HDD) and peripheral assets (hardware connected to the host including camera, sensor, and USB flash drive).

Typical application: financial industry, where the head office needs to regularly count the security device assets of each branch to collect the number, operation status, and other basic information of deployed host/storage/peripheral cameras.

Extended application: management of industry-related service applications for assets customized by the industry platform (e.g., information and software assets, person assets, and service assets) based on the hardware asset integration.

5.2.2 API Calling Flow

1. Get the search capability of the device hardware asset data via `GET /ISAPI/System/deviceInfo/capabilities`: when `isSupportSearchHardwareAssets` is true, it indicates the device supports searching for asset data.
2. Search for device hardware asset data via `GET /ISAPI/System/deviceInfo/ExportDeviceAssets?format=json` to get the information of hardware assets on device including host assets, connected sub-device assets, HDD assets, etc.
3. Export the device hardware asset information in binary data in Excel format.

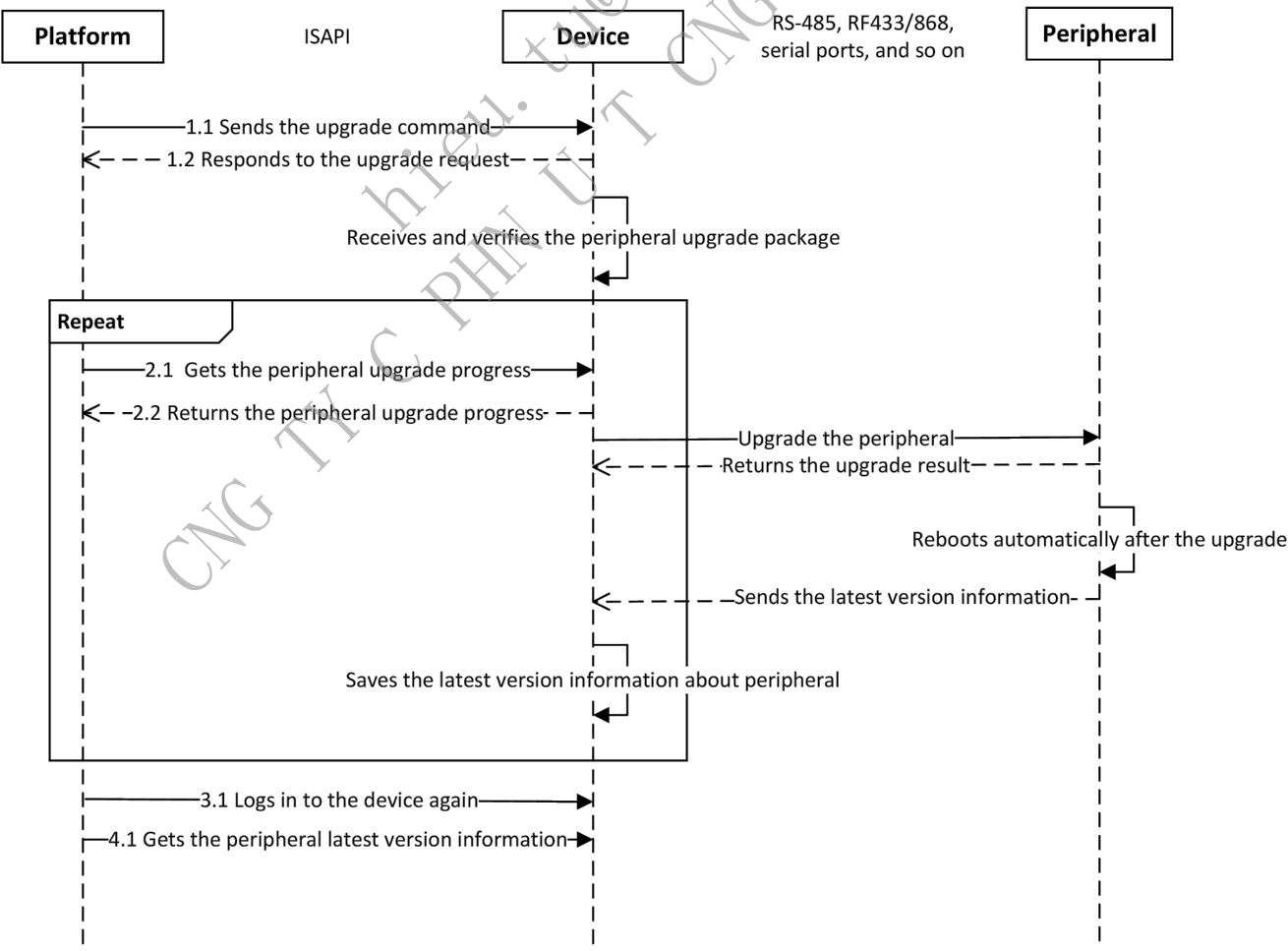
5.3 Device Peripherals Upgrade

5.3.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.3.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.

2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.4 Device Time Sync

5.4.1 Introduction to the Function

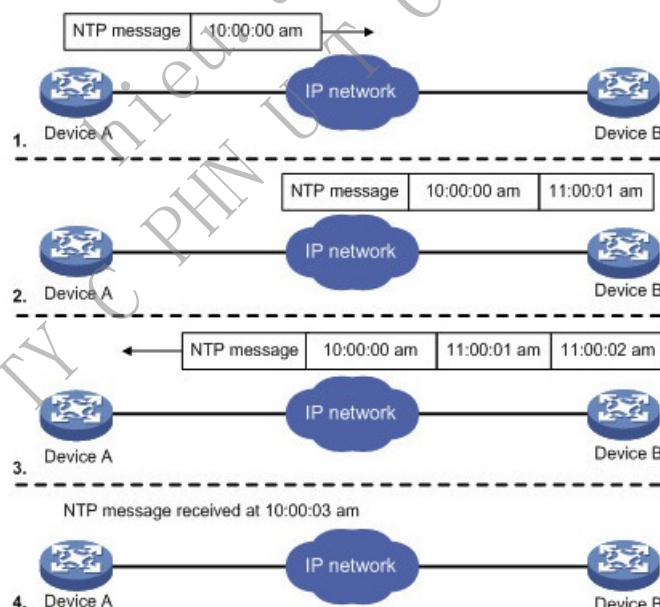
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.4.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B. Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: $\text{Delay} = (T4 - T1) - (T3 - T2) = 2 \text{ seconds}$.

Time difference between Device A and Device B: $\text{offset} = ((T2 - T1) + (T3 - T4)) / 2 = 1 \text{ h}$.

Device A can sync its own time with that of Device B according to calculation results.

5.4.2 API Calling Flow

5.4.2.1 Time Sync Configuration

1. Get the Capability of Device Time Synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows:

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`, `timeZone`.

Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note: For manual time synchronization (time offset including time zone offset): Set manual time synchronization: `localTime` refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); `timeZone` refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: `localTime` refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); `timeZone` refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.4.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

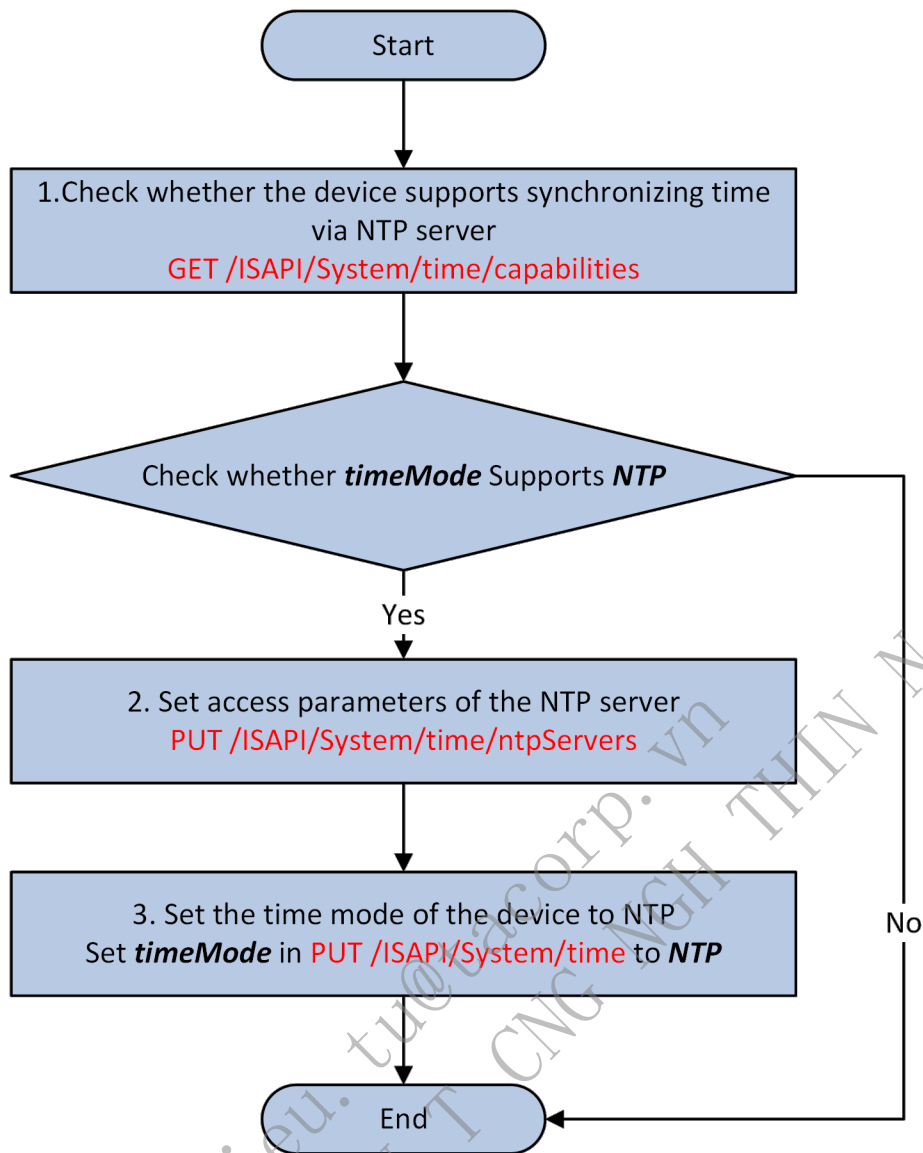
2. Configure time zone parameters

Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.4.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):



1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: GET /ISAPI/System/time/capabilities; and check whether timeMode supports NTP.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: GET /ISAPI/System/time/ntpServers/capabilities

Set access parameters of the NTP server: PUT /ISAPI/System/time/ntpServers

Get access parameters of the NTP server: GET /ISAPI/System/time/ntpServers

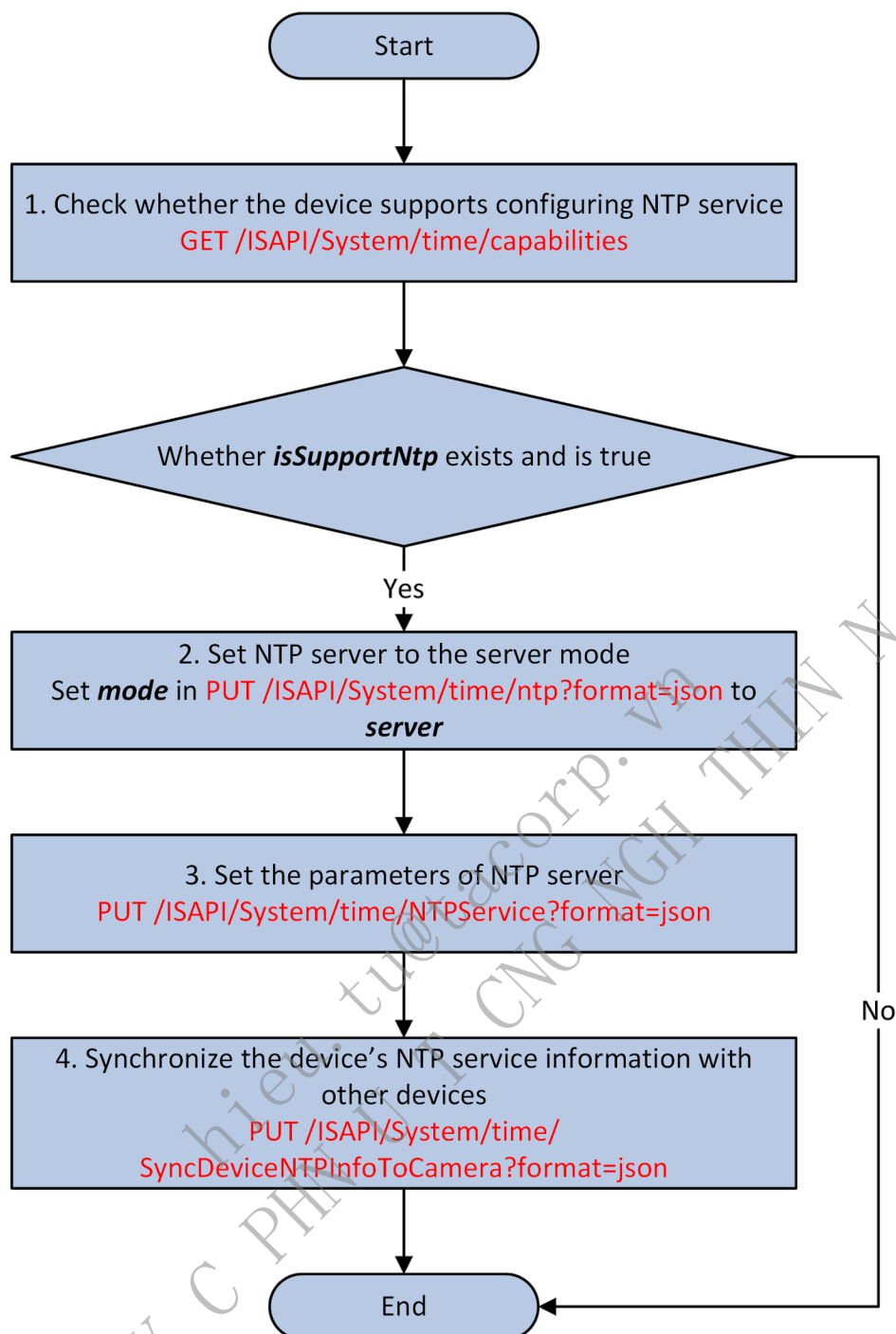
3. Set the time mode of the device to NTP

Supports setting the value of timeMode to NTP.

Get device time synchronization management parameters: GET /ISAPI/System/time Set device time synchronization management parameters: PUT /ISAPI/System/time

5.4.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):



1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: GET /ISAPI/System/time/capabilities; If isSupportNtp is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of mode to server.

Get the capability of server mode: GET /ISAPI/System/time/ntp/capabilities?format=json

Set NTP to server mode: PUT /ISAPI/System/time/ntp?format=json

Get parameters of NTP server mode: GET /ISAPI/System/time/ntp?format=json

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: GET /ISAPI/System/time/NTPService/capabilities?format=json

Set the NTP server parameters: PUT /ISAPI/System/time/NTPService?format=json

Get the parameters of the NTP server: GET /ISAPI/System/time/NTPService?format=json

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: GET
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json

Synchronize device's NTP service information with the camera: PUT
/ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json

Get the progress of synchronizing device's NTP service information with the camera: GET
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json

Search for the results of synchronizing device's NTP service information with the camera: POST
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json

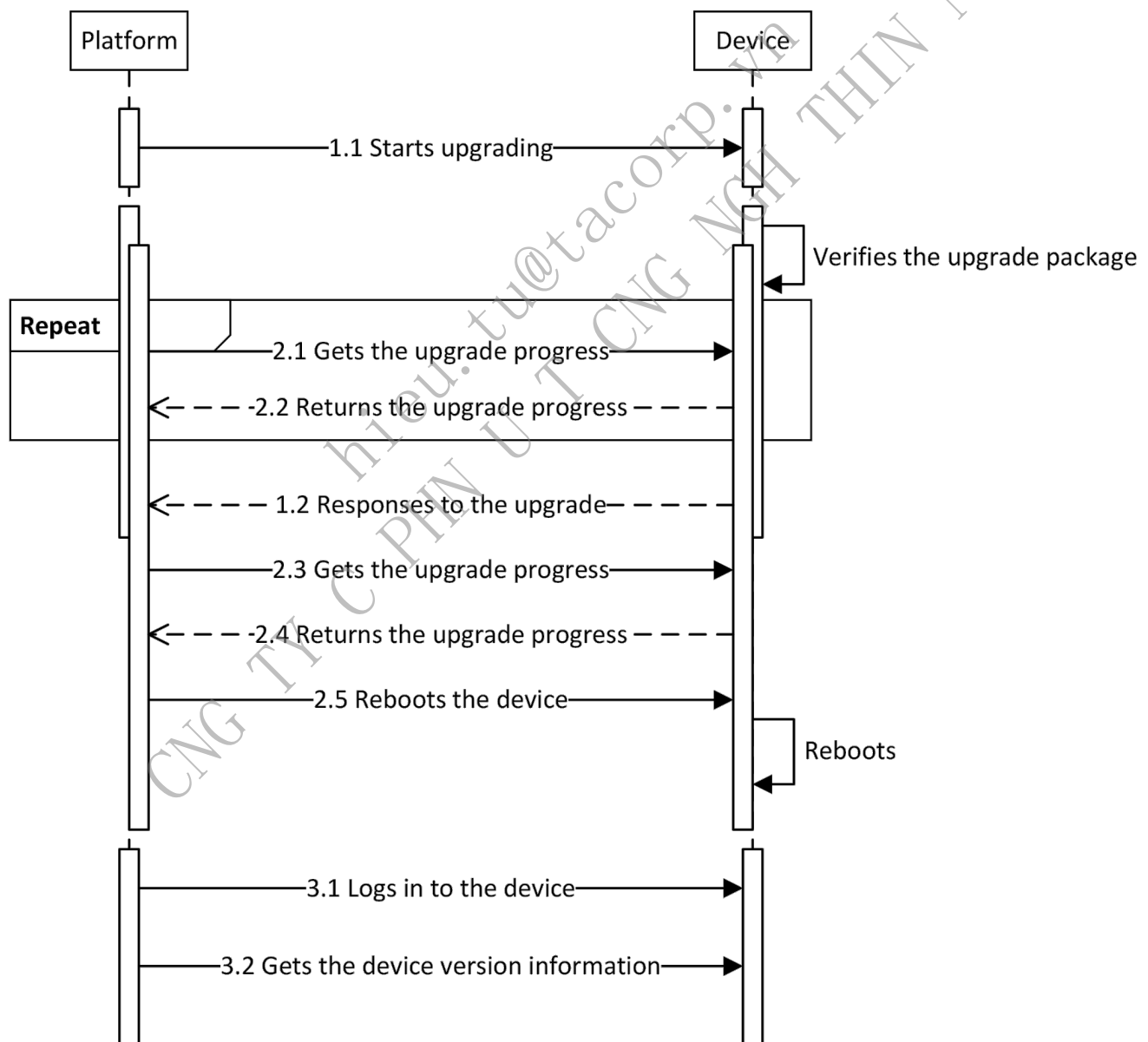
5.5 Device Upgrade

5.5.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.5.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



1. Upgrade devices.

Upgrade the device firmware: POST /ISAPI/System/updateFirmware.

2. Get the device upgrade progress.

Get the device upgrade progress: GET /ISAPI/System/upgradeStatus.

3. Reboot devices.

Reboot devices: PUT /ISAPI/System/reboot.

5.6 Mutually Exclusive Functions

5.6.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e., only one of them is allowed at one time).

5.6.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: GET /ISAPI/System/mutexFunction/capabilities?format=json. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exclusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).
2. Search for the functions that are mutually exclusive with a specified function: POST /ISAPI/System/mutexFunction?format=json. Based on the list of mutually exclusive functions returned by GET /ISAPI/System/mutexFunction/capabilities?format=json, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disable the mutually exclusive function.
3. Get the mutual exclusion information when device function exception occurs: GET /ISAPI/System/mutexFunctionErrorMsg. After getting the error code, you can call this API to get the current mutually exclusive functions.

5.7 Query Device Operation Log

5.7.1 Introduction to the Function

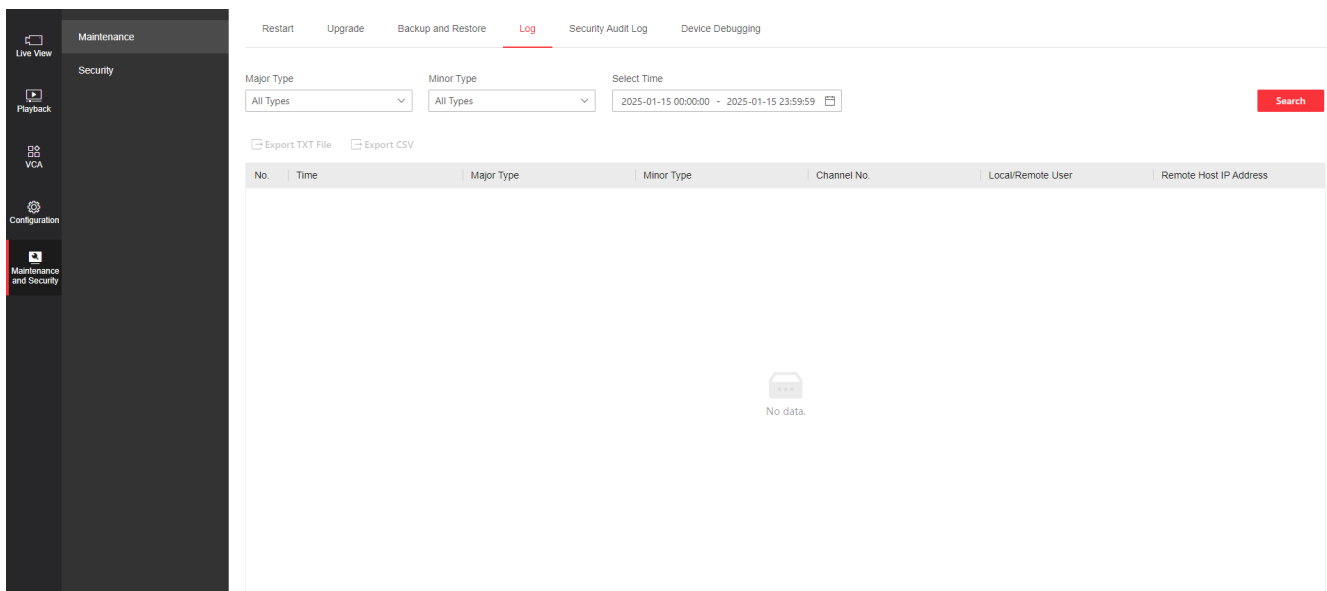
Device operation logs primarily include:

- Operation records for the device, such as startup, restart, PTZ control, etc.
- Events from the device, such as the start and end of motion detection, face capture, etc.
- Device status exceptions, such as network disconnection, network recovery, IP address conflict, etc.
- Device trigger information, such as starting recording, stopping recording, and periodically recording hard drive status.

Log query functions include log query and security audit log query, which overlap to some extent. The differences are as follows:

- Log query is mainly designed to query all operational status and operation records of the device.
- Security audit log query is mainly designed to query security-related status and operation records of the device, such as user login/logout records, device SSH service start/stop records, etc.

Operation logs can be queried and displayed item by item through the device web page, remote client, or platform. The device web page log query is shown in the following figure.



5.7.2 API Calling Flow

1. Call `POST /ISAPI/ContentMgmt/logSearch` to query logs except security audit logs.

Note 1:

- When you query logs of a specified type, first set the `metaId` to `log.std-cgi.com/Information`. If the total number returned is not 0, it indicates that the log query is standardized, and subsequent queries should follow the standard. If the total number returned is 0, use `log.std-cgi.com/Infomation` to query again. If there is a return value, subsequent queries should use `log.std-cgi.com/Infomation`.
- When you set the `metaId` to `all`, the response results should be parsed for both `log.std-cgi.com/Infomation` and `log.std-cgi.com/Information`.
- When you query logs of a specified type, the recommended format for the `searchID` field is the GUID (8-4-4-4-12) format, such as `812F04E0-4089-11A3-9A0C-0305E82C2906`.

2. Call `POST /ISAPI/ContentMgmt/security/logSearch` to query security audit logs.

**Note **:

- Standard definition: Query log `POST /ISAPI/ContentMgmt/logSearch: log.std-cgi.com/Information` indicates querying information logs.

Device Type	Implementation Differences
Encoding Devices (General Cameras/Storage Devices/Traffic Cameras/Thermal Cameras/Security Inspection)	Input and output log information are <code>log.std-cgi.com/Infomation</code>
Non-Encoding Devices (Access Control Devices/Transmission Devices/Display & Control Devices)	Input and output log information are <code>log.std-cgi.com/Information</code>

5.7.Export Device Operation Log

5.7.1 Introduction to the Function

Operation logs can be exported to a client over the network or to an external storage medium (USB drive, external hard drive, etc.) via a USB interface.

5.7.2 API Calling Flow

5.7.2.1 Export to Client

1. Call `POST /ISAPI/ContentMgmt/logSearch/dataPackage` to get the URL to download the exported device log.

2. The client downloads the log files via the URL returned by the device.

5.7.2.2 Export to USB Storage Medium

1. Call `GET /ISAPI/System/exportLogToUSB/capabilities?format=json` to determine if the device supports log output to a USB drive.
2. Call `PUT /ISAPI/System/exportLogToUSB/mode?format=json` to set the parameters of exporting logs to a USB drive.
3. Call `GET /ISAPI/System/exportLogToUSB/status?format=json` to get the status of log output to a USB drive.

5.7. Device Operation Log

5.7.1 Introduction to the Function

Device operation logs are used in the following two scenarios:

Scenario 1: View the detailed operation log files to locate the issue when a device fails.

Scenario 2: Before a device fails, basic operation status information is recorded in the logs. Maintenance personnel can proactively query the device operation status to promptly identify potential issues and faults in the device.

- The operation status logs generally include host information (network bandwidth, online users, video output port/USB port status), linked device information (online status, video recording schedule/status), hard disk information (capacity, status, runtime, temperature), etc.
- For example, to ensure the normal operation of bank equipment and facilities, and to promptly identify hidden dangers and faults in equipment operations, maintenance personnel of bank outlets are required to check the equipment operation status at regular intervals each day and manually record data for subsequent data traceability. This management approach consumes manpower and may result in data omissions. Therefore, you can configure the equipment to automatically record operational status by schedule for querying and exporting history data, thus saving labor and ensuring data integrity.

5.7.2 API Calling Flow

5.7.2.1 Upload to Specified FTP/HTTP(S) Server

1. Call `PUT /ISAPI/System/ diagnosedData/server?format=json` to allow devices to upload operation log files to a specified FTP/HTTP(S) Server.
2. Check the device logs on the FTP/HTTP(S) server.

Note:

- See the "Device Operation Status Diagnosis" module for FTP/HTTP(S) Server API details.
- We recommend that use this method to remotely obtain device operation logs for device operation and maintenance platforms.

5.7.2.2 Upload to Syslog Server

Syslog is a standard log transmission protocol widely used in system logs. It is defined in [RFC 5424](#) (The Syslog Protocol). After you configure the Syslog service address and enable the Syslog function on the device, the device will send its operation logs to the Syslog server using the Syslog protocol. The Syslog server can manage the operation logs of multiple devices.

1. Get the Syslog management capabilities to determine if the device supports the Syslog function and the parameter value range.
2. Call `PUT /ISAPI/System/logServer` to configure Syslog server settings such as the IP address, port No., and certificate.
3. When the device upload logs to the Syslog server using the Syslog protocol, the logs can be viewed on the Syslog server.

5.7.2.3 Query Device Operation Status Logs

1. Call `/ISAPI/ContentMgmt/RuningLogPlan?format=json` to configure the device operation log recording schedule.

Supports configuring up to 8 time points per day. The device will automatically store and back up the operation status data at the corresponding time points for subsequent historical log viewing or export. The operation status data includes host information (network bandwidth, online users, video output port/USB port status), external device information (online status, video device recording plan/status), disk information (capacity, status, runtime, temperature).

2. Query historical device operation logs: `/ISAPI/ContentMgmt/SearchRuningLogData?format=json`.
3. Export historical device operation logs: `/ISAPI/ContentMgmt/ExportRuningLogData?format=json`. Export based on the query conditions set in step 2.

5.8 Serial Port Accessed External Device Management

5.8.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

5.8.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information of devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port: `POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

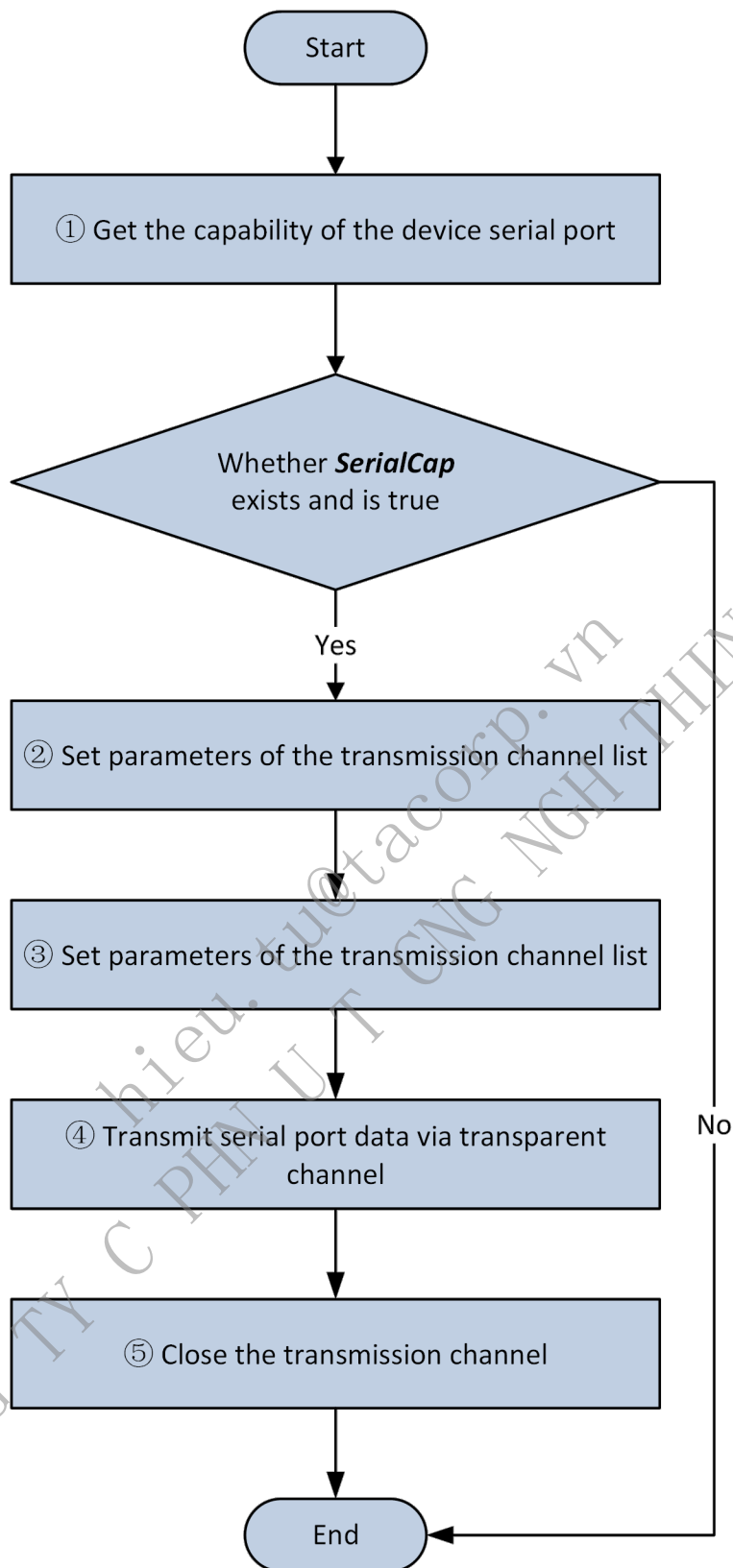
5.9 Serial Port Data Transparent Transmission

5.9.1 Introduction to the Function

RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of transparently transmitted serial data.

5.9.2 API Calling Flow



1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: PUT

/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: GET
/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData.
- Send data to device serial port through transmission channel: PUT
/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData.

5. Close the transmission channel: PUT

/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close.

5.10 Serial Port Parameter Configuration

5.10.1 Introduction to the Function

Serial port parameter configuration.

5.10.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: GET /ISAPI/System/Serial/capabilities.
- Get control parameters of all serial ports: GET /ISAPI/System/Serial/ports?permissionController=<indexID>.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>.
- Configure control parameters of single serial port: PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>.

5. Get the status of single serial port: GET /ISAPI/System/Serial/ports/<portID>/status.

5.11 Sub-device Batch Upgrade

5.11.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.11.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: /ISAPI/System/capabilities.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgrading sub-devices.

Sub-device batch upgrade: POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json.

Batch sub-device upgrade progress search: POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json.

Sub-device Batch Upgrade Task Management

Get device system capabilities /ISAPI/System/capabilities, If the node isSupportBulkUpgradeChildDevice exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node isSupportSearchBulkUpgradeChildDeviceProgress exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node isSupportSearchBulkUpgradeChildDeviceTask exists and is true, it indicates that batch upgrading devices can be searched,

If the node isSupportModifyBulkUpgradeChildDeviceTask exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node isSupportDeleteBulkUpgradeChildDeviceTask exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json.

Batch upgrade sub devices: POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json.****

Search for tasks of batch sub-device upgrade: POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json.

Edit tasks of batch sub-device upgrade: PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json.****

Delete tasks of batch sub-device upgrade: PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json.

Note: 1. The two methods for batch upgrading child devices are the same API: POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json. If the BulkUpgradeChildDeviceListCap.taskID is returned in GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json. Sub devices with the tag isUpgradable and value true indicate that the sub devices support upgrades.

5.12 User Management

5.12.1 Introduction to the Function

When the device is activated, you can log in to it via the admin account and corresponding password, and manage users as needed, including:

1. Change the password of the admin account. The user name cannot be edited.
2. Add, edit, and delete other users, including the user type, password, user name, and so on. A Non-admin user can log in to and operate the device after being created.

Remarks

1. Common user types:
 - o Administrator (admin): has the permission of accessing all resources supported by the device, and can operate all functions of the device. The admin account cannot be deleted.
 - o Operator (operator): has the permission to view. Their operation permissions are assigned by admin. Operator accounts are created by the administrator only.
 - o User (viewer): has the permission to view only. They have no operation permission. User accounts are created by the administrator only.
2. User types related to the installer:
 - o Local Administrator (localAdmin) : Activated by the device owner on the local software side (e.g., WEB) that accompanies the device. By default, a localAdmin has the permission to access the device and perform all functions supported by the device.
 - o Local Installer (localInstaller) : Activated by localAdmin. The identity can be given to the installer for device installation and debugging on site. By default, a localInstaller has the permission to access and perform all

functions supported by the device.

- Local Operator (`localOperator`) : Created by `localAdmin` or `cloudAdmin`. A password is required to be set by `localAdmin`, `cloudAdmin`, `localInstaller`, `installerAdmin`, or `installerEmployee` before the local operator can log in via the local keypad. A local operator's business functions are limited, and cannot configure device parameters. By default, only arming, disarming (including alarm clearing), and relay control operations are allowed. According to the actual business needs, there can be two types of local operator: one-time local operator (can only log in once and the identity will expire after logging out), and temporary local operator (can log in within the validity period), both created by `localAdmin` or `cloudAdmin`.
- Cloud Administrator (`cloudAdmin`) : Created by the device owner in the mobile software that accompanies the device (e.g. HC over the cloud) and then synchronized to the device. By default, a `CloudAdmin` has the permission to access and perform all functions supported by the device.
- Cloud Operator (`cloudOperator`) : Created by `cloudAdmin` in the mobile software that accompanies the device (e.g. HC over the cloud) and then synchronized to the device for ordinary mobile users. A cloud operator's business functions are limited, and cannot configure device parameters, access and view the device, arm/disarm (including alarm clearing), or operate relays.
- Installer Administrator (`installerAdmin`) : User on the installer business application, created on the business platform (e.g. HPC deployed in the cloud) and then synchronized to the device for remote device management by the business platform's admin. By default, an `installerAdmin` has the permission to access the device and perform all functions supported by the device. To facilitate installation management, when the business platform sets `installerAdmin` as the highest authority and synchronizes to the device, `installerAdmin` is allowed to modify user permissions of `localAdmin` and `cloudAdmin`.
- Installer Employee (`installerEmployee`) : User on the installer business application, created on the business platform (e.g. HPC deployed in the cloud) and then synchronized to the device for the employees on the business platform to remotely install and debug the device. By default, an `installerEmployee` has the permission to access the device and perform all functions supported by the device.

3. User password:

- To ensure the security of account information, it is recommended to create a password using eight to sixteen characters, including at least two kinds of the following categories: digits, lower case letters, upper case letters, and special characters, and the user name is not allowed in the password.
- Risky passwords include the following categories: less than 8 characters, containing only one type of characters, same as the user name or reversed user name. To protect user data privacy and improve security, it is recommended to use a strong password.
- The password strength rule is as follows:
 - a. Strong password: including at least three kinds of the categories (digits, lower case letters, upper case letters, and special characters).
 - b. Medium password: a combination of digits and special characters, lower case letters and special characters, upper case letters and special characters, or lower case letters and upper case letters.
 - c. Weak password: a combination of digits and lower case letters or digits and upper case letters.

5.12.2 API Calling Flow

- Get the user management capability of devices on the client: `GET /ISAPI/Security/users/<indexID>/capabilities`.
- Add device users on the client: `POST /ISAPI/Security/users?security=<security>&iv=<iv>`.

Remarks:

- Only `admin` can create other types of users, and creating users requires login password verification (`<loginPassword>`) of `admin`.
- If the user account is inactivated, it's required to log in to the account and change the user password (`PUT /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>`). The account is activated when the password is changed.
- When the account is inactivated, it's not allowed to perform any operations except changing the user

password. Otherwise, an error (0x0020000f) will be returned.

3. Edit the user information on the client: `PUT /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>`.

Remarks:

- It requires password verification of `admin` when `admin` changes the user password. The account turns inactivated when the user password is changed by `admin`. Once logging out, the user needs to change the password first before the next login.
- When `non-admin` users changed their passwords, the account remains activated.

4. Delete users on the client: `DELETE /ISAPI/Security/users?loginPassword=<loginPassword>&security=<security>&iv=<iv>`.

Remarks:

Only `admin` can delete users, and deleting users requires login password verification of `admin`.

5. Get the user information, including user name, activation status (`<userActivationStatus>`), and so on.

Get a single user information: `GET /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>`.

Get the information of all users: `GET /ISAPI/Security/users?security=<security>&iv=<iv>`.

Get the information of online users: `GET /ISAPI/Security/onlineUser`. Online users refer to users who have logged in to the device. The information such as user name, user type, and IP address can be obtained.

Remarks:

- If multiple attempts of `admin` login password verification failed in the process of adding, editing, or deleting users, the `admin` will be locked. The remaining attempts are defined by the field `retryTimes` in the response message.
- The new password cannot be the same as the last password. Otherwise, an error (0x400010E8) will be returned.

5.12.3 Exception Handling

Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Descript
4	Invalid Operation	theAccountIsNotActivated	0x0020000f	The account is not activated.	
4	Invalid Operation	loginPasswordError	0x4000000C	Incorrect login password.	
4	Invalid Operation	theAnswerToTheUserSecurityQuestionIsDuplicate	0x4000A0B6	The answer to the user security question is duplicate.	Please se different answer.
4	Invalid Operation	theAnswerToTheUserSecurityQuestionIsTooShort	0x4000A0B7	The answer to the user security question is too short.	Please se longer answer.
4	Invalid Operation	cannotSameAsOldPassword	0x400010E8	New password cannot be the same as the old one.	Please se different password
6	Invalid Content	administratorPasswordError	0x60000042	Incorrect administrator password.	Please en the corre password you forgot the password you can reset the password

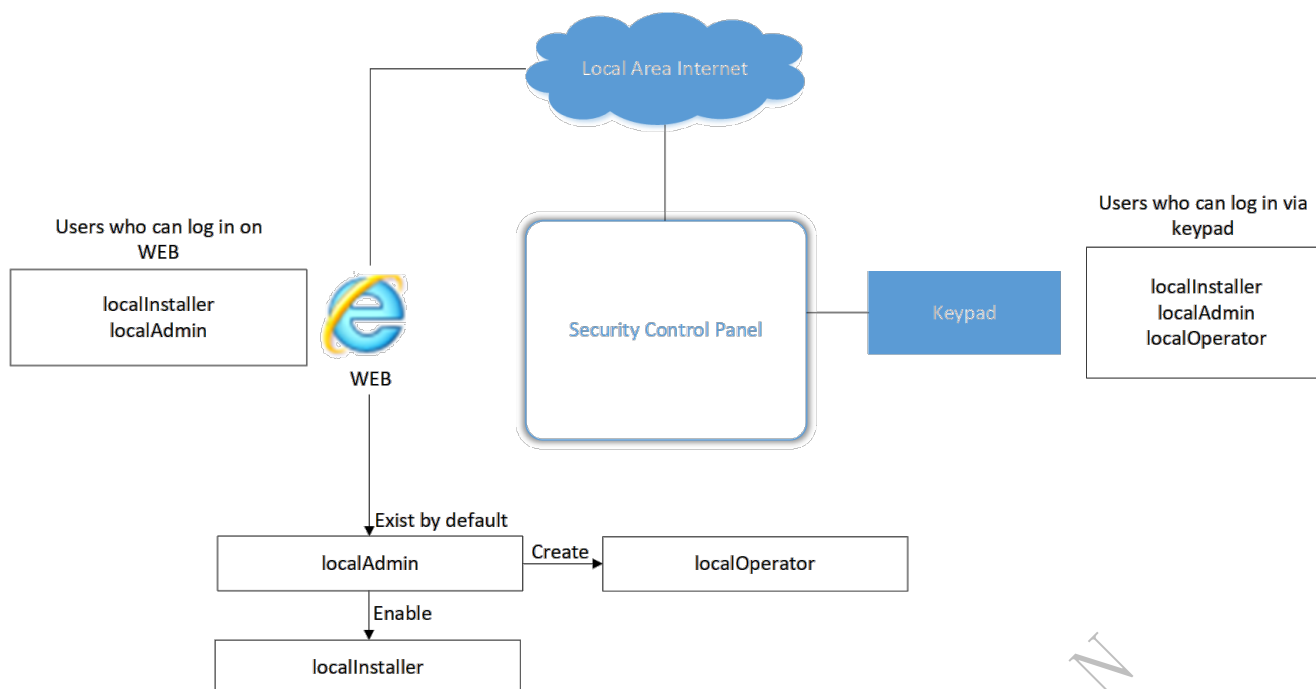
5.13 User Types Related to the Installer (supported by the security control panel)

For different application scenarios, e.g., local environment, cloud environment, and HPP installer environment, users related to the installer can be classified into seven types: `localAdmin`, `localInstaller`, `localOperator`, `cloudAdmin`, `installerAdmin`, `installEmployee`, and `cloudOperator`.

5.13.1 Create the User

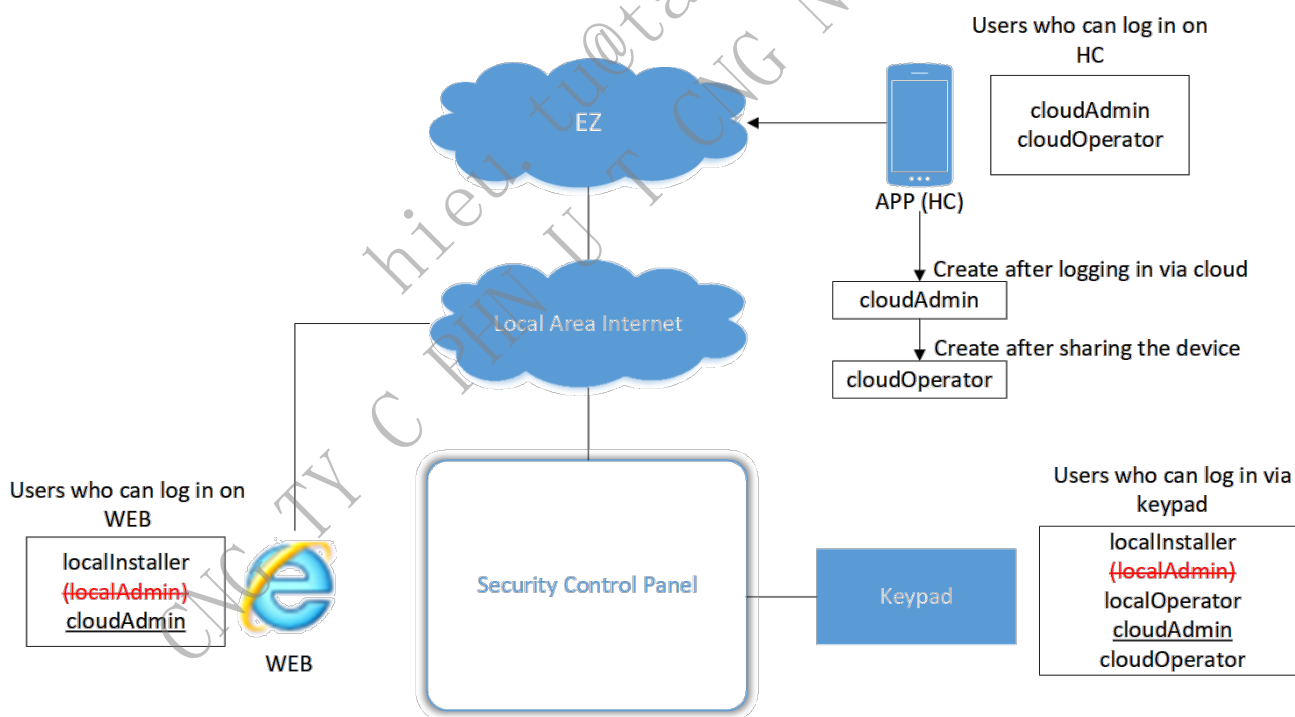
5.13.1.1 Local Environment

In a local environment (LAN), three types of users are involved: `localAdmin`, `localInstaller`, and `localOperator`. By default, two types of users exist on the device: `localAdmin` and `localInstaller`, but `localInstaller` is not enabled, which means that although a user ID is created, but the user cannot log in as `localInstaller` before the user type is enabled by `localAdmin`. `localOperator` is created by `localAdmin`, and can only log in via local keypad. Users who have been set with the keypad password can log in via keypad.



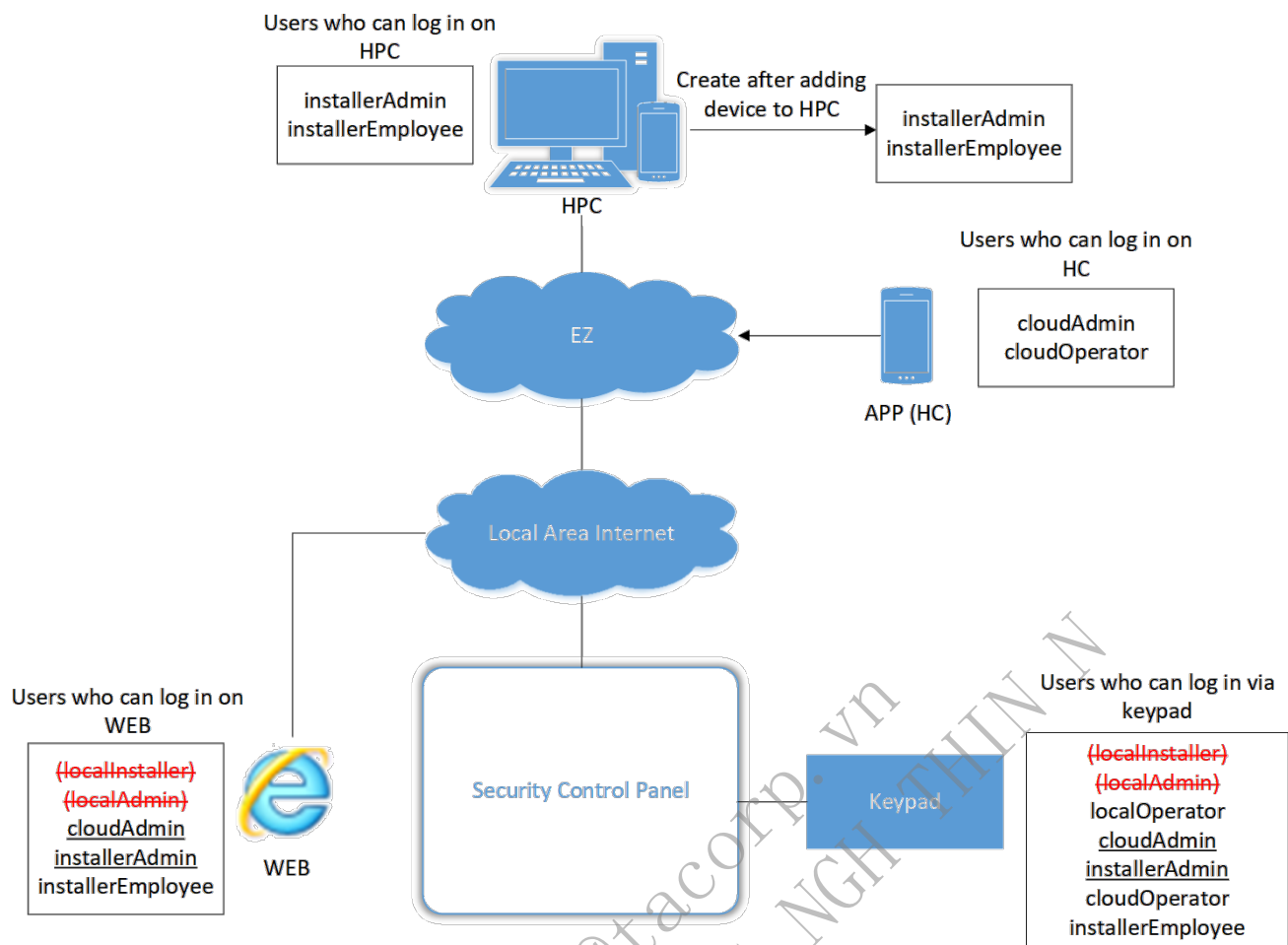
5.13.1.2 Cloud Environment

In a cloud environment, four types of users are involved: `cloudAdmin`, `cloudOperator`, `localInstaller`, and `localOperator`. After logging in to the device on HC application via cloud, a `cloudAdmin` can be created. Then, the `cloudAdmin` can share the device to create a `cloudOperator`. Note that after a `cloudAdmin` is created, the existing `localAdmin` will expire. Users who have been set with the keypad password can log via keypad.



5.13.1.3 HPP Installer Environment

In an HPP installer environment, five types of users are involved: `cloudAdmin`, `cloudOperator`, `localOperator`, `installerAdmin`, and `installerEmployee`. After a device is added to HPC, the user adding protocol will be applied, and `installerAdmin` and `installerEmployee` can be created (batch creating is supported). Note that after an `installerAdmin` is created, the existing `localInstaller` will expire. Users who have been set with the keypad password can log via keypad.



5.13.2 User Permissions

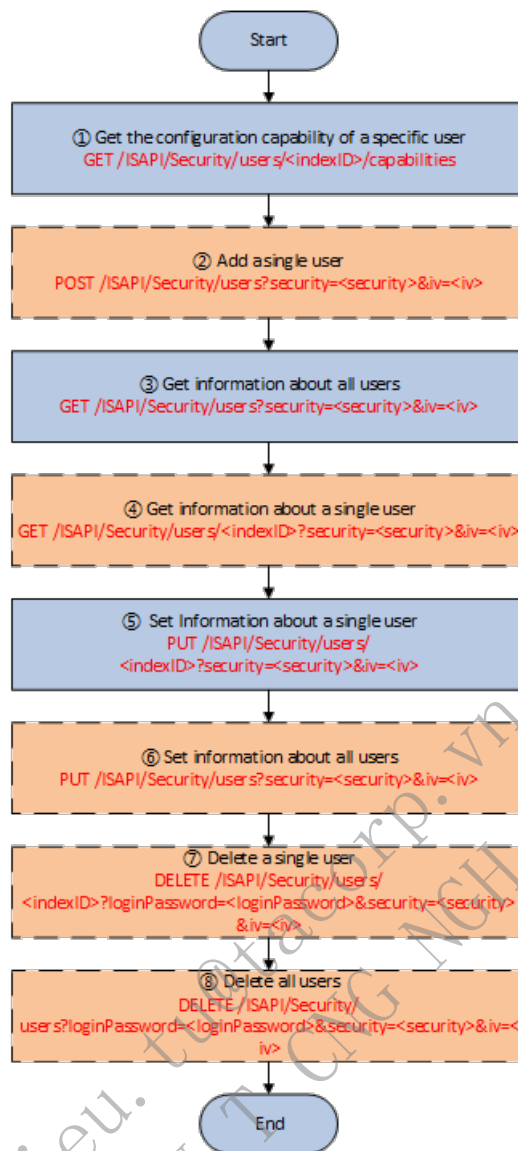
User Type / Permission	Local Admin	Local Installer	Local Operator	Cloud Admin	Cloud Operator	Installer Admin	Installer Employee
Arming	√	√	√	√	√	√	√
Disarming (Alarm Clearing)	√	√	√	√	√	√	√
Bypass	√	√	×	√	√	√	√
View logs and status	√	√	×	√	√	√	√
Configure parameters	√	√	×	√	×	√	√
Manage partitions	√	√	×	√	×	√	√
Operate relays	√	√	one-time local operator: × temporary local operator: √	√	√	√	√
Edit localAdmin's keypad password	√	√	×	×	×	√	√
Edit cloudAdmin's keypad password	×	√	×	√	×	√	√
Edit localInstaller's keypad password	×	√	×	×	×	×	×
Edit							

installerAdmin's keypad password	x	x	x	x	x	√	x
Edit installerEmployee's keypad password	x	x	x	x	x	√	√ (only the employee's own password)
Edit localOperator's keypad password	√	√	x	√	x	√	√
Edit cloudOperator's keypad password	x	√	x	√	√ (only the operator's own password)	√	√
Edit localAdmin's permission	x	x	x	x	x	x	x
Edit cloudAdmin's permission	x	x	x	x	x	x	x
Edit localInstaller's permission	√	x	x	√	x	x	x
Edit installerAdmin's permission	√	x	x	√	x	x	x
Edit installerEmployee's permission	√	x	x	√	x	x	x
Edit localOperator's permission	√	√	x	√	x	√	√
Edit cloudOperator's permission	x	√	x	√	x	√	√

Note: "Configure parameters" includes parameters of zones, sounders, keypads, card readers, keyfobs, cards, relays, repeaters, transmitters, network cameras, partitions, and so on.

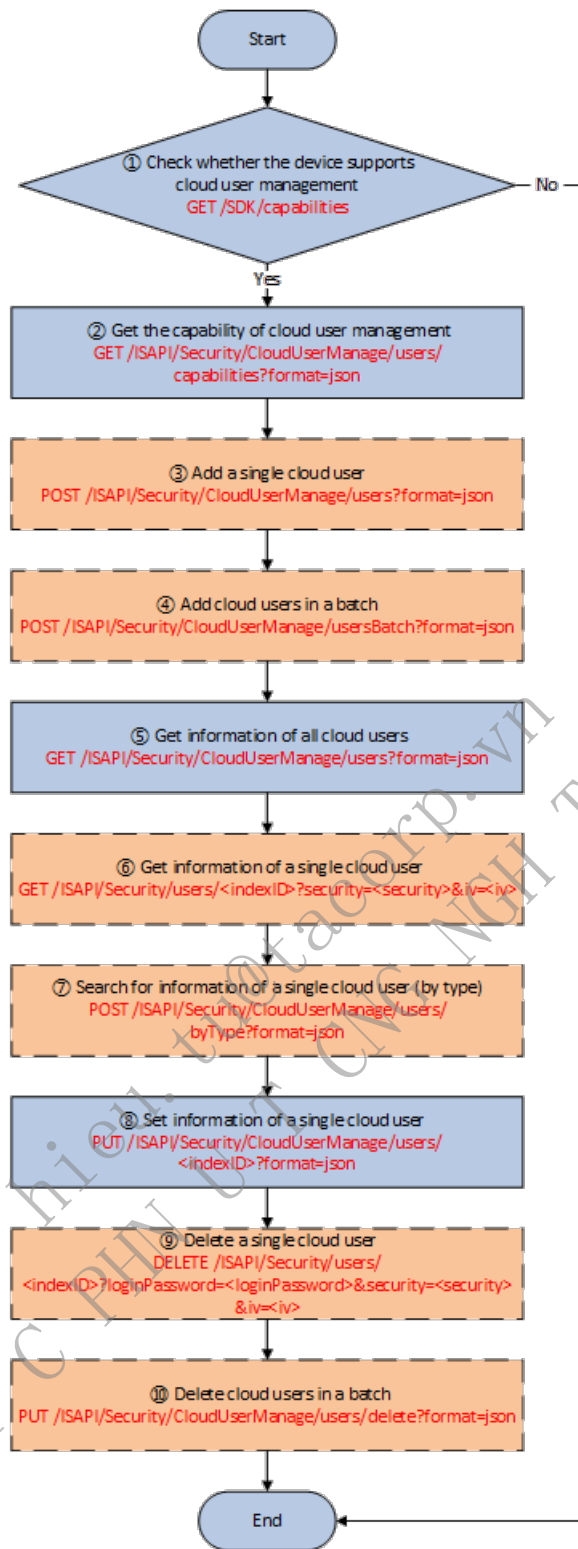
5.13.3 Manage User Information

5.13.3.1 Local User



1. Get the configuration capability of a specific user: GET /ISAPI/Security/users//capabilities
2. (Optional) Add a user: POST /ISAPI/Security/users?security=&iv=. The nodes userName, password, keypadPassword, and loginPassword in the message will be encrypted.
3. Get information of all users: GET /ISAPI/Security/users?security=&iv=. The nodes phoneNum, emailAddress, password, duressPassword, keypadPassword, and loginPassword in the message will be encrypted.
4. Get information of a single user: GET /ISAPI/Security/users/?security=&iv=. The index in the URL is the user ID.
5. Set permissions for a single user: PUT /ISAPI/Security/users/?security=&iv=. The index in the URL is the user ID.
6. (Optional) Set information of all users: PUT /ISAPI/Security/users?security=&iv=. This API can be called for batch configuring user information.
7. (Optional) Delete a single user: DELETE /ISAPI/Security/users/?loginPassword=&security=&iv=. Delete the user by indexID, and loginPassword is required for deleting the user. The loginPassword should be encrypted.
8. (Optional) Delete all users: DELETE /ISAPI/Security/users?loginPassword=&security=&iv=. loginPassword should be encrypted.

5.13.3.2 Cloud User



1. Check whether the device supports cloud user management: GET /SDK/capabilities. If the node value of isSupportCloudUserManage is true, cloud user management is supported in these URLs:

/ISAPI/Security/CloudUserManage/users/capabilities?format=json

/ISAPI/Security/CloudUserManage/users/?format=json

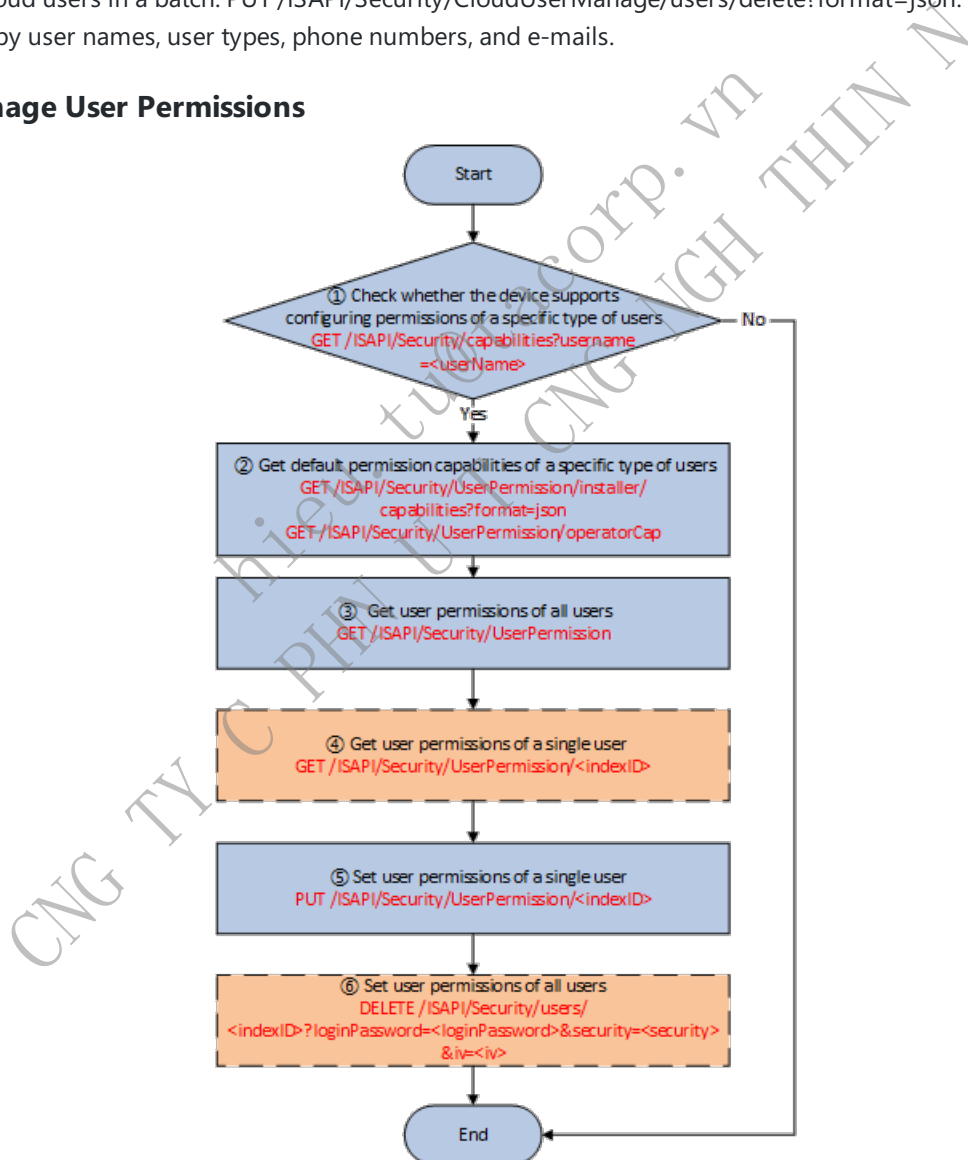
/ISAPI/Security/CloudUserManage/users?format=json

2. Get the capability of cloud user management: GET /ISAPI/Security/CloudUserManage/users/capabilities?format=json. If the node value of isSupportAddCloudUserList is true, it supports batch adding cloud users.
3. Add a single cloud user: POST /ISAPI/Security/CloudUserManage/users?format=json. The following cloud users need to create the user accounts on their own: coludAdmin, installerAdmin, installerEmployee, and cloudOperator, which is different from localOperator who needs to be created by admin users. Enter information such as user

name, password, e-mail, and phone number to create a user account and put into use.

4. (Optional) Add cloud users in a batch: POST /ISAPI/Security/CloudUserManage/usersBatch?format=json. For installerEmployee users that has been created on HPC, use this URL to apply them to the device and synchronize user information.
5. Get information of all cloud users: GET /ISAPI/Security/CloudUserManage/users?format=json
6. Get information of a single cloud user: GET /ISAPI/Security/CloudUserManage/users/?format=json
7. Search for information of a single cloud user (by type): /ISAPI/Security/CloudUserManage/users/byType?format=json. Support searching by e-mail, phone number, and user name.
8. Set information of a single cloud user: PUT /ISAPI/Security/CloudUserManage/users/?format=json
9. Delete a single cloud user: DELETE /ISAPI/Security/CloudUserManage/users/?format=json. Delete the user by specifying the user's indexID.
10. Delete cloud users in a batch: PUT /ISAPI/Security/CloudUserManage/users/delete?format=json. Support batch deleting by user names, user types, phone numbers, and e-mails.

5.13.4 Manage User Permissions



1. Check whether the device supports configuring permissions of a specific type of users: GET /ISAPI/Security/capabilities?username=. If the node value of isSupportInstallerCap is true, the installer users' permissions can be configured. There is no specific capability node for operator type users, and by default their permissions can be configured.
2. Get the default permission capabilities of a specific type of users.
 - For installer users: GET /ISAPI/Security/UserPermission/installer/capabilities?format=json

- For operator users: GET /ISAPI/Security/UserPermission/operatorCap
- 3. Get user permissions of all users: GET /ISAPI/Security/UserPermission
- 4. (Optional) Get user permissions of a single user: GET /ISAPI/Security/UserPermission/. The index in the URL is the user ID.
- 5. Set user permissions of a single user: PUT /ISAPI/Security/UserPermission/
- 6. (Optional) Set user permissions of all users: PUT /ISAPI/Security/UserPermission

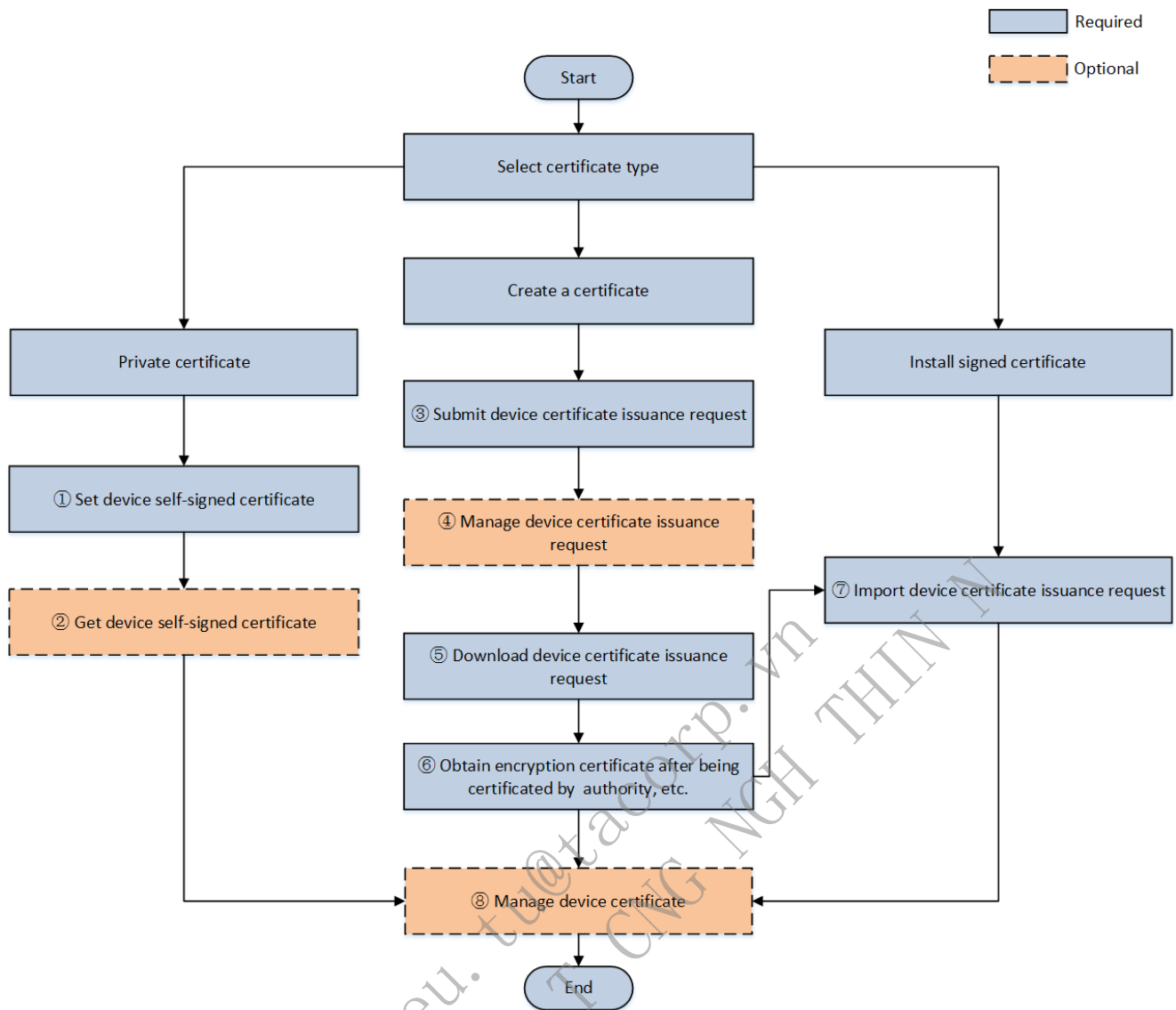
6 Information Security

6.1 . Certificate Management

6.1.1 Introduction to the Function

Certificate management is for device encryption, where certificates are used to verify the identities of the two communication sides. Certificates here are divided into client certificates and server certificates based on the direction of verification. The certificate contains the public key of the server, and the encryption protocol uses this public key to encrypt the communication content. The client uses the certificate to verify the server's identity and obtain the server's public key. The client and server use the encryption protocol to negotiate a symmetric encryption key for the following operations including encryption and decryption. Certificate management mainly involves creating, deleting, editing, and searching for certificates, as well as the request and import of various certificates.

6.1.2 API Calling Flow



Certificates are divided into three types: one is a private certificate, which is a certificate generated by the device that is not issued by an official certification authority, and it generates public and private keys; one is a custom certificate, which has been certified by another platform or authority. For this type of certificate, users need to create a certificate on the device, upload the generated certificate to the authority or platform for certification issuance, and then import the issued certificate into the device; one is to install a signed certificate, which has already been certified and can be installed directly.

Private Certificate

1. Set device self-signed certificate: `PUT /ISAPI/Security/serverCertificate/selfSignCert.`
2. Get device self-signed certificate: `GET /ISAPI/Security/serverCertificate/selfSignCert.`

Create Certificate

3. Submit a request for device certificate issuance: `POST /ISAPI/Security/serverCertificate/certSignReq?platformNo=<platformNo>&type=<type>&security=<security>&iv=<iv>.`

Note: `type` indicates certificate types.

4. Manage device certificate issuance request

Get device certificate issuance request: `GET /ISAPI/Security/serverCertificate/certSignReq?platformNo=<platformNo>&type=<type>&security=<security>&iv=<iv>.`

Generate device certificate issuance request: `PUT /ISAPI/Security/serverCertificate/certSignReq?platformNo=<platformNo>&type=<type>&security=<security>&iv=<iv>.`

Delete device certificate issuance request: DELETE /ISAPI/Security/serverCertificate/certSignReq?type=<type>.

5. Request for downloading device certificate issuance: GET /ISAPI/Security/serverCertificate/downloadCertSignReq?type=<type>&platformNo=<platformNo>&customID=<certificateID>.

6. Obtain encryption certificate after being certificated by the authority

Upload the downloaded device certificate assurance request to the corresponding authority for authentication to obtain an encryption certificate, the private key and other files.

Install a Signed Certificate

7. Import device certificate issuance

Import device certificate issuance (binary): PUT /ISAPI/Security/serverCertificate/certificate?type=<type>&platformNo=<platformNo>.

Import device certificate issuance: POST /ISAPI/Security/serverCertificate/certificate?customID=<certificateID>&security=<security>&iv=<iv>.

type indicates certificate types, allowing to import different certificate types.

8. Device certificate management

Manage certificates such as getting and deleting after importing them to the device.

Get certificate information of all devices: GET /ISAPI/Security/serverCertificate/certificate?type=<type>&platformNo=<platformNo>.

Get certificate information of a single device: GET /ISAPI/Security/serverCertificate/certificates/<certificateID>?format=json.

Delete certificate information of a single device: DELETE /ISAPI/Security/serverCertificate/certificates/<certificateID>?format=json.

Delete device certificate: DELETE /ISAPI/Security/serverCertificate/certificate?type=<type>&platformNo=<platformNo>.

Certificate Expiration

Get device system capabilities /ISAPI/System/capabilities, if node isSupportCertificateRevocation exists and is true, the device supports this feature.

Get the capability of configuring certificate expiration: GET /ISAPI/Security/deviceCertificate/certificateRevocation/capabilities?format=json.

Get certificate expiration configuration: GET /ISAPI/Security/deviceCertificate/certificateRevocation?format=json.

Configure certificate expiration parameters: PUT /ISAPI/Security/deviceCertificate/certificateRevocation?format=json.

Get linkage parameters of certificate expiration: GET /ISAPI/Event/triggers/certificateRevocation-<channelID>.

Set linkage parameters of certificate expiration: PUT /ISAPI/Event/triggers/certificateRevocation-<channelID>.

Get triggering method of certificate expiration linkage: GET /ISAPI/Event/triggers/certificateRevocation-<channelID>/notifications.

Set triggering method of certificate expiration linkage: PUT /ISAPI/Event/triggers/certificateRevocation-<channelID>/notifications.

Relate event: "eventType": "certificateRevocation".

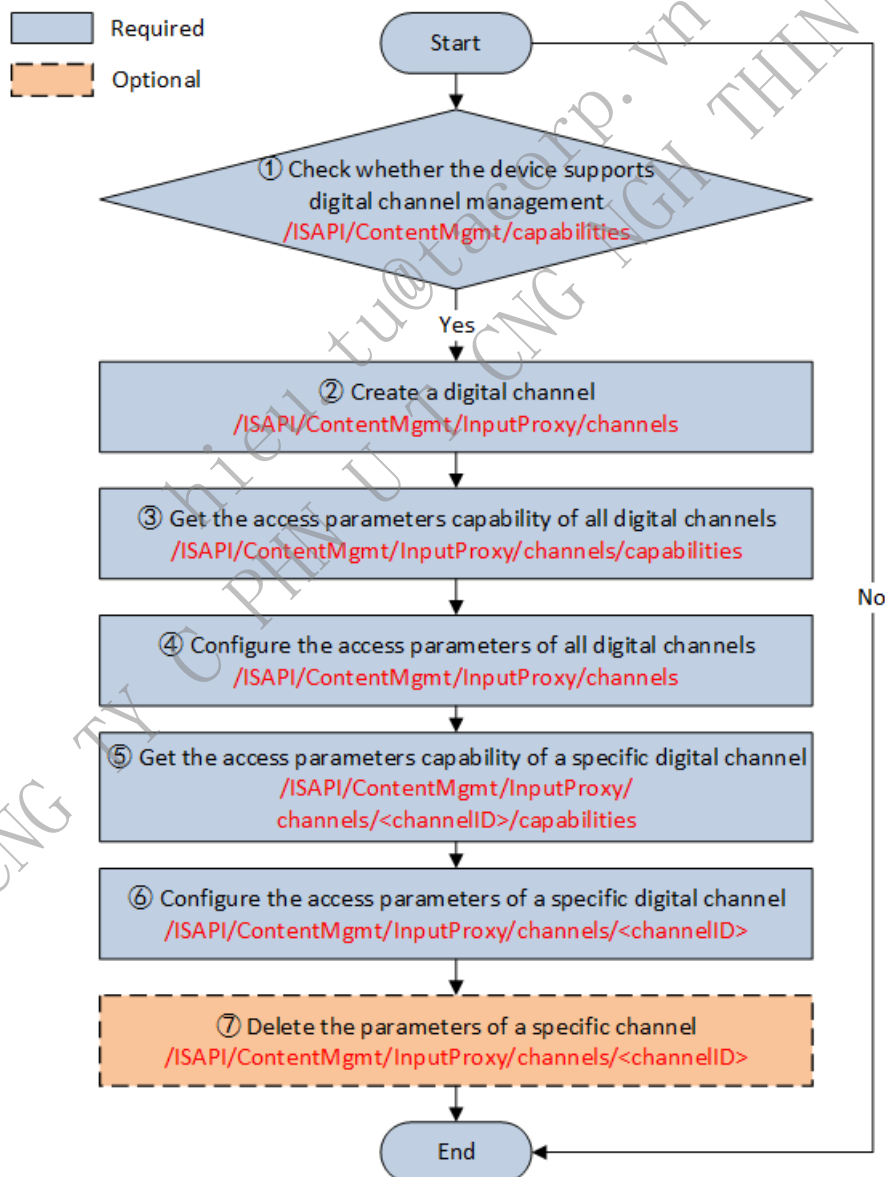
7 Video (General)

7.1 Digital Channel Access Management

7.1.1 Introduction to the Function

- Video channel includes analog channel and digital channel. The service involves channel access protocols, streaming protocols, address information, version No. of channel access firmware, and so on.
- Analog channel refers to the local video channels of the device.
- Digital channel refers to the video channels accessed by the device by IP address or POE port (e.g., NVR accessing the video channel of network camera).
- Analog channel No. and digital channel No. are unique. Devices that do not support digital channels will only return analog channel No. For example, analog channel No. ranges from 1 to 8 (determined by the number of analog channel ports of the device), digital channel No. ranges from 9 to 32 (within the maximum number of digital channels, determined by the number of added network cameras).

7.1.2 API Calling Flow



1. Get the capability node to check whether the device supports digital channel management: GET /ISAPI/ContentMgmt/capabilities. If the value of the node `iSptInputProxyChanCap` is true, it indicates that the device supports digital channel management.
2. Add digital channels: POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>.

3. Get the access parameters capability of all digital channels: GET
/ISAPI/ContentMgmt/InputProxy/channels/capabilities?security=<security>&iv=<iv>.
4. Configure the access parameters of all digital channels:

Get: GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>.

Configure: PUT /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>.
5. Get the capability of configuring the access parameters of a specific digital channel: GET
/ISAPI/ContentMgmt/InputProxy/channels/<channelID>/capabilities.
6. Configure the access parameters of a specific digital channel:

Get: GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>.

Configure: PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>.
7. Delete the parameters of a specific digital channel: DELETE
/ISAPI/ContentMgmt/InputProxy/channels/<channelID>.

7.1.3 Video Channel Application

1. Get the list of video channels, which is the sum of digital channels and analog channels. Digital channels and analog channels need to be obtained separately:

- 1) Call "Get all video input channel parameters" to get all analog channels via GET
/ISAPI/System/Video/inputs/channels and VideoInputChannelList parameter is returned. The id represents the analog channel No.
- 2) Call the "Get all digital channel access parameters" to get all digital channels via GET
/ISAPI/ContentMgmt/InputProxy/channels?security=&iv= and InputProxyChannelList parameter is returned. The id represents the digital channel No.

2. Video channel function configuration supports unified configuration of analog and digital channels. Both the analog channel interface starting with /ISAPI/System/Video/inputs/channels/<channelID>/ and the digital channel interface starting with /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/ are supported. **It is recommended to use the channel configuration interface starting with /ISAPI/System/Video/inputs/channels/<channelID>/, where <channelID> represents all channel No.s supported by the device, including analog channels and digital channels.**

For example, configure text overlay parameters for a channel: Both GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/overlays and GET/PUT
/ISAPI/ContentMgmt/InputProxy/channels/<channelID>/overlays are supported, and it is recommended to use
/ISAPI/System/Video/inputs/channels/<channelID>/overlays.

7.2 Sub-device's Transparent Transmission

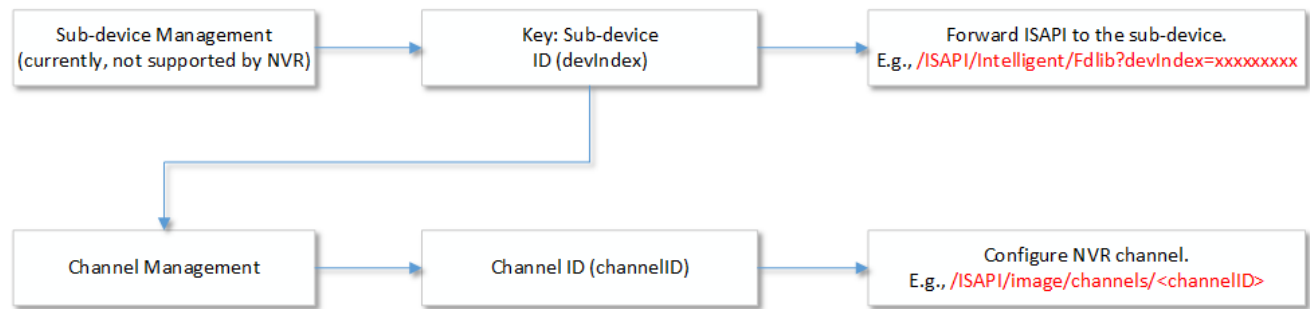
7.2.1 Introduction to the Function

In actual application scenarios, the NVR being used is generally not the latest version, if an IPC with latest functions is added to the system, the NVR will fail to recognize the IPC's new functions due to version incompatibility, so the platform (including device access platforms, client softwares, device's web plug-ins, etc, hereinafter referred to as the platform) will fail to operate new functions of the IPC.

Besides, in video channel resource management, the platform can only get information about digital channels (also known as IP channels) added to the NVR, but not the device that the channels belong to. For example, two video channels (channel 13 and channel 18) of a dual-lens people counting camera are added to the NVR. The platform can only get that digital channel 13 and 18 are added to the NVR, but not that the two channels belong to one camera. Such information loss may lead to repeated operations, which will further cause problems in actual practices. For example, users may fail to upgrade IPC via digital channel No. if not knowing the relationship between camera and digital channel.

To settle above problems, the sub-device's transmission function is introduced to the video channel resources management. In this way, users can directly access to the device of the digital channels via the devIndex (sub-device index) allocated to the device by NVR.

Notably, despite the introduction of the sub-device's transparent transmission function, the NVR does not function as an access gateway, and so it does not function with sub-device management on gateway. Refer to the Sub-device Management of Device Management for details.

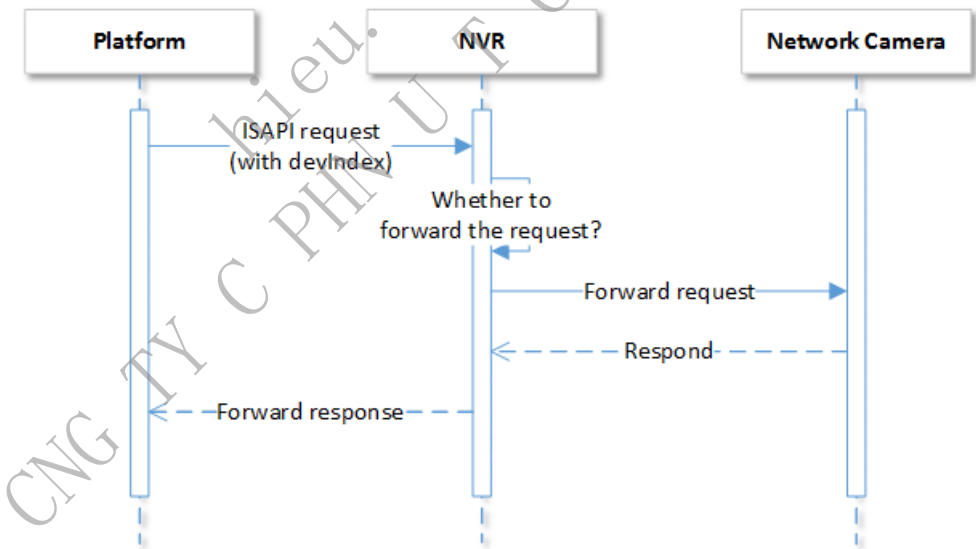


Steps are as follows:

1. First, the platform will check whether the NVR supports sub-device's transparent transmission. If it is supported, follow the next step. Second, add devIndex to the request URL.
2. For the ISAPI request sent to the NVR by the platform, if it contains devIndex, it will be forwarded to the specific sub-device; otherwise, the NVR will respond to the request.
3. When the request is forwarded to the sub-device by the NVR, delete the devIndex in the request URL so as to realize ISAPI interaction.

For example:

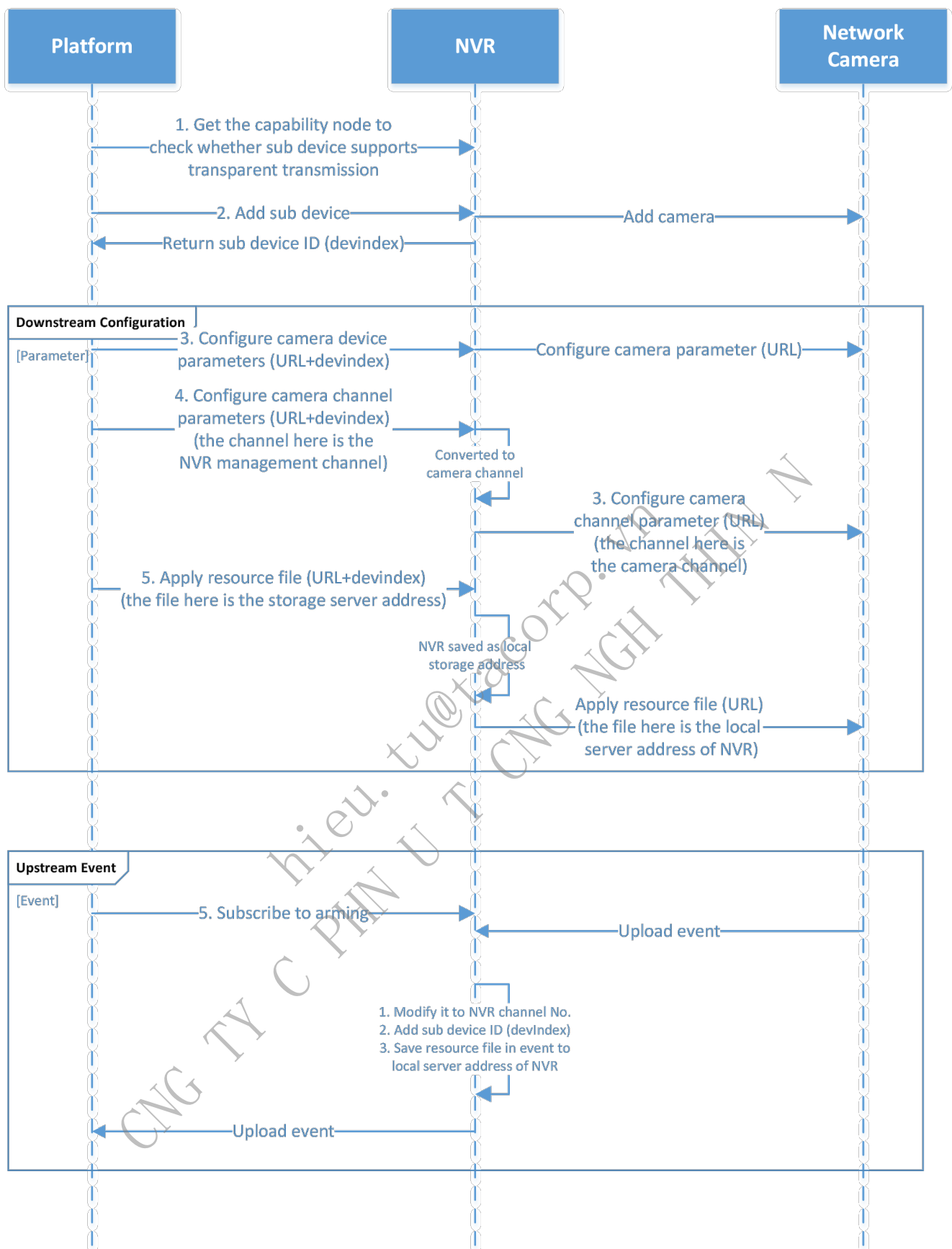
1. Send the request URL (GET /ISAPI/System/deviceInfo) to get the information of the platform's NVR device.
2. Send the request URL (GET /ISAPI/System/deviceInfo?devIndex=<devIndex>) to get the information of a specific sub-device of the NVR.



Note:

1. This function is applicable to transitory connections, while inapplicable to persistent connections (e.g., arming subscription, two-way audio).
2. It is required to delete the devIndex in the URL before NVR forwards the ISAPI request.
3. For the VCA function of NVR, the transmission solution of sub-devices is not adopted. You can switch the analysis unit by the following URI: /ISAPI/Smart/analysisUnitSwitch/channels/<channelID>/event/<EventType>.

7.2.2 API Calling Flow



1. Get the capability node to check whether the device supports transparent transmission. If the value of the node `isSupportDevIndex` is true, it indicates that the device supports transparent transmission.
2. Create digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`. The sub device ID `devindex` will be returned.
3. Get access parameters of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`. Each digital channel information contains one parameter `devIndex`, which is the same for digital channels of a same physical device.

4. When splicing ISAPI request, if the URL contains `devIndex=<devIndex>`, it represents a request for the sub-device. For example, you can send `12345678` (the request URL) to get the information of the sub-device `GET /ISAPI/System/deviceInfo?devIndex=12345678` of the target device. Note: 1) For the downstream configuration of a video channel, the platform will apply the channel managed by NVR, and before applying, the NVR will transform channel into the video channel. 2) When the platform applies the storage address of resource files, the camera cannot access the storage service to get resource files because of the network isolation between the camera and platform, so the storage address must be saved as a local storage address by the NVR before being applied.

8 Decoding and Video Wall

8.1 Video Wall Scenes

8.1.1 Function Introduction

For example: A security monitoring center needs to view two different monitoring feeds.

- Save Feed 1 as Scene 1
- Save Feed 2 as Scene 2

Switching between feeds is implemented via scene switching.

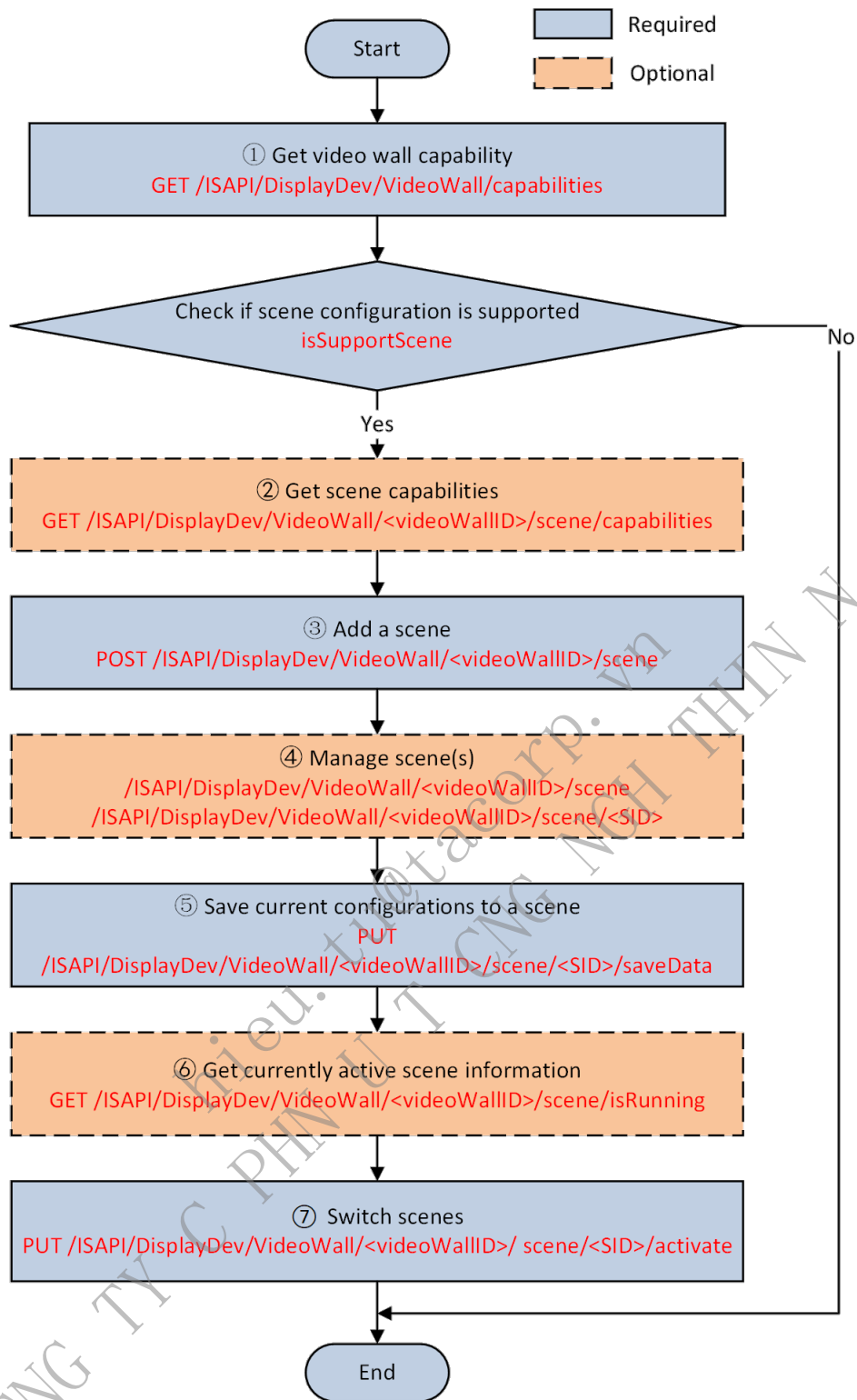


Feed 1



Feed 2

8.1.2 API Calling Flow



1. Get video wall capability.

URL: GET /ISAPI/DisplayDev/VideoWall/capabilities . If isSupportScene is returned with true value, the device supports scene configuration and switching.

2. (Optional) Get scene capabilities.

URL: GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/capabilities. The maximum scenes supported by device is defined by field (maxSceneNums) .

3. Add a scene.

URL: POST /ISAPI/DisplayDev/VideoWall/<ID>/scene. If succeeded, the new created scene ID is returned in ResponseStatus.

4. (Optional) Scene management.

- Get, set, or delete all scenes of a specified video wall: GET or PUT or DELETE /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene.
- Get, set, or delete a single scene: GET or PUT or DELETE /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>.

5. Save current configurations to a scene.

URL: PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>/saveData.

6. (Optional) Get the information about currently active scene on the video wall.

URL: GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/isRunning.

7. Switch scenes.

URL: PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>/activate.

8.2 Window Opening on Video Wall

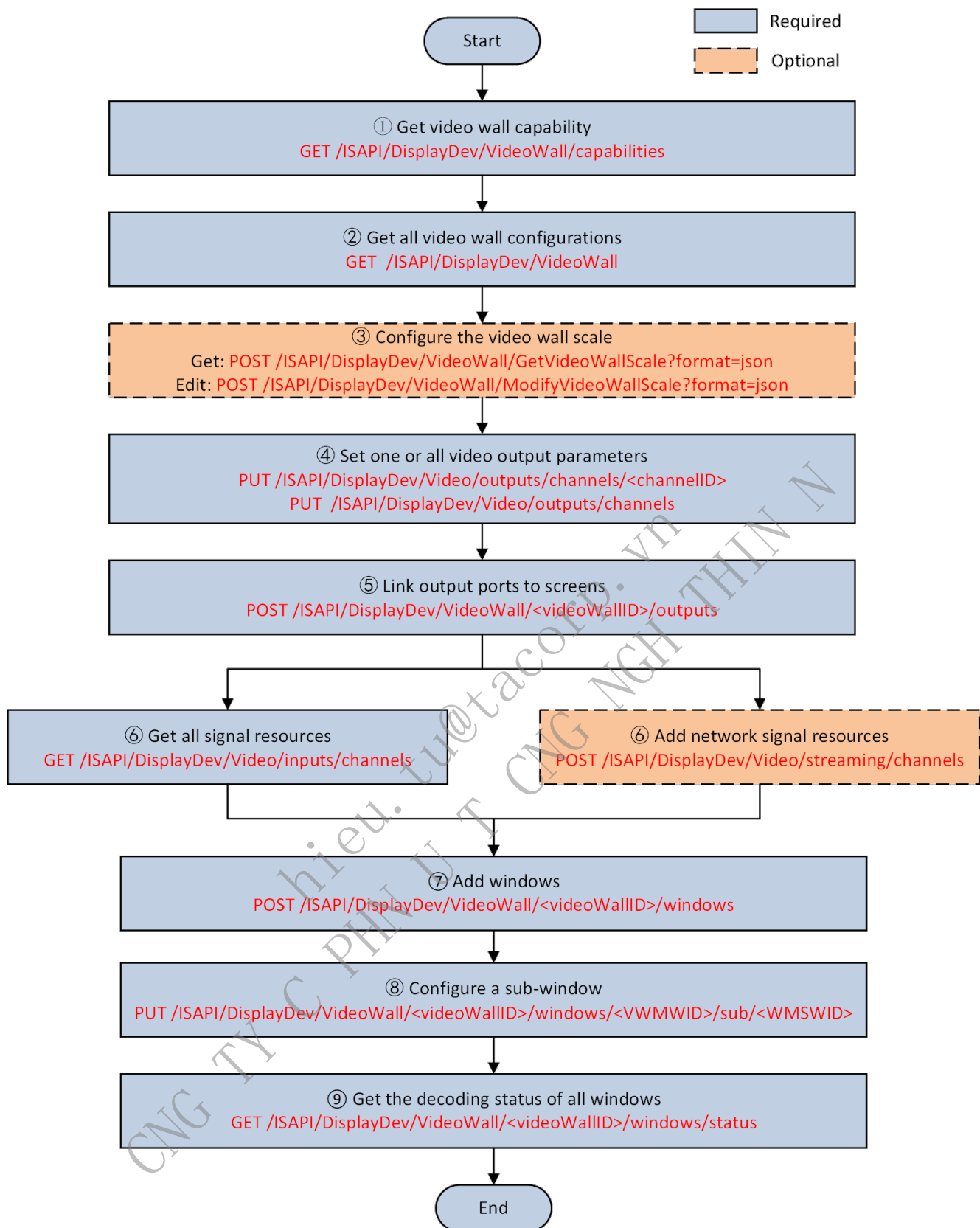
8.2.1 Function Introduction

Security monitoring centers display real-time videos from multiple cameras or local signal sources on video wall.

The image below shows the effect of displaying multiple signal sources via window layouts on a video wall.



8.2.2 API Calling Flow



1. Get video wall capabilities.

URL: `GET /ISAPI/DisplayDev/VideoWall/capabilities`. The supported maximum video walls is returned in field (`maxWallNums`), and maximum windows is (`maxWindowNums`).

2. Get all video wall configurations.

URL: `GET /ISAPI/DisplayDev/VideoWall`.

3. (Optional) Configure the video wall scale (such as the video wall row count, column count, row height, and column width).

- Get the video wall scale: `POST /ISAPI/DisplayDev/VideoWall/GetVideoWallScale?format=json`.
- Edit the video wall scale: `POST /ISAPI/DisplayDev/VideoWall/ModifyVideoWallScale?format=json`.

4. Set one or all video output parameters.

- Get or set all video output parameters: `GET` or `PUT /ISAPI/DisplayDev/Video/outputs/channels`.
- Get or set a video output parameters: `GET` or `PUT /ISAPI/DisplayDev/Video/outputs/channels/<channelID>`.

5. Link output ports to screens.

URL: `POST /ISAPI/DisplayDev/VideoWall/<videoWallID>/outputs`.

6. Manage signal sources, including local and network signal sources.

- Get all signal resources from devices: `POST /ISAPI/DisplayDev/Video/streaming/channels`.
- (Optional) Add network signal resources: `POST /ISAPI/DisplayDev/Video/streaming/channels`.

7. Add windows. Specify a signal resource, and open windows on a specified video wall.

URL: `POST /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows`.

8. Configure a sub-window. Link a signal resource to a sub-window to get, decode, and display the stream.

URL: `PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VMMWID>/sub/<WMSWID>`.

9. Get the decoding status of all windows.

URL: `GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/status`.

9 API Reference

9.1 Device Management (General)

9.1.1 Accessing via Protocol

9.1.1.1 Get security parameters of a single access protocol

Request URL

`GET /ISAPI/Security/adminAccesses/<indexID>`

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<AdminAccessProtocol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <enabled>
    <!--ro, opt, bool, "true,false"-->true
  </enabled>
  <protocol>
    <!--ro, req, enum, protocol type, subType:string, desc:protocol type-->HTTP
  </protocol>
  <redirectToHttps>
    <!--ro, opt, bool, whether to enable HTTPS redirection, desc:this node is valid when the protocol type is "HTTPS"-->true
  </redirectToHttps>
  <streamOverTls>
    <!--ro, opt, bool, whether to enable encryption of stream, desc:this node is valid when the protocol type is "SDK_OVER_TLS"-->true
  </streamOverTls>
  <portNo>
    <!--ro, req, int, Port No., range:[1,65535], desc:Port No.-->80
  </portNo>
  <TLS1_1Enable>
    <!--ro, opt, bool, dep:and,{$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,HTTPS},
    {$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,WebSocket},{$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,WebSockets},
    {$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,SDK_OVER_TLS}-->true
  </TLS1_1Enable>
  <TLS1_2Enable>
    <!--ro, opt, bool, dep:and,{$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,HTTPS},
    {$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,WebSocket},{$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,WebSockets},
    {$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,SDK_OVER_TLS}-->true
  </TLS1_2Enable>
  <TLS1_3Enable>
    <!--ro, opt, bool, dep:and,{$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,HTTPS},
    {$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,WebSocket},{$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,WebSockets},
    {$.AdminAccessProtocolList.AdminAccessProtocol.protocol,eq,SDK_OVER_TLS}-->true
  </TLS1_3Enable>
</AdminAccessProtocol>

```

9.1.1.2 Get device protocol capability

Request URL

GET /ISAPI/Security/adminAccesses/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AdminAccessProtocolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <AdminAccessProtocol>
    <!--ro, req, object-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <protocol opt="HTTP,HTTPS,SDK,RTSP,DEV_MANAGE,WebSocket,WebSocketS,SDK_OVER_TLS,MATRIX_GATEWAY,IOT,GB35114,104Proto,SRTP,ModbusTCP">
      <!--ro, req, string, HTTPS, attr:opt{req, string}-->GB35114
    </protocol>
    <redirectToHttps opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </redirectToHttps>
    <streamOverTls opt="true,false">
      <!--ro, opt, bool, this node is valid when the protocol type is "SDK_OVER_TLS", attr:opt{req, string}-->true
    </streamOverTls>
    <portNo min="1" max="65535" default="8443">
      <!--ro, req, int, range:[1,65535], attr:min{req, int},max{req, int},default{opt, int}-->1
    </portNo>
    <TLS1_1Enable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </TLS1_1Enable>
    <TLS1_2Enable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </TLS1_2Enable>
    <TLS1_3Enable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </TLS1_3Enable>
    <TLS1_1SecurityPrompt opt="Only when the video output mode is Panorama, the device supports the VCA functions except Monitoring">
      <!--ro, opt, string, attr:opt{req, string}-->Only when the video output mode is Panorama, the device supports the VCA functions except Monitoring
    </TLS1_1SecurityPrompt>
    <connectTime min="1" max="999" def="10">
      <!--ro, opt, int, range:[1,999], step:1, unit:min, attr:min{req, int},max{req, int},def{req, int}-->10
    </connectTime>
  </AdminAccessProtocol>
</AdminAccessProtocolList>
```

9.1.2 Detection Discovery

9.1.2.1 Get the device discovery mode

Request URL

GET /ISAPI/System/discoveryMode

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DiscoveryMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device discovery mode, attr:version{req, string, protocolVersion}-->
  <ONVIFDiscoveryMode>
    <!--ro, opt, enum, control mode of discovering devices via Open Network Video Interface Protocol, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </ONVIFDiscoveryMode>
  <ISDiscoveryMode>
    <!--ro, opt, enum, control mode of discovering devices via IS-Discovery of SADP, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </ISDiscoveryMode>
  <PcapMode>
    <!--ro, opt, enum, control mode of discovering devices via winpcap or libpcap or raw sockets, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </PcapMode>
  <SADP>
    <!--ro, opt, enum, control mode of discovering devices via SADP, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </SADP>
  <activeDiscoveryEnabled>
    <!--ro, opt, bool-->true
  </activeDiscoveryEnabled>
</DiscoveryMode>
```

9.1.2.2 Set the device discovery mode

Request URL

PUT /ISAPI/System/discoveryMode

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DiscoveryMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, device discovery mode, attr:version{req, string, protocolVersion}-->
  <ONVIFDiscoveryMode>
    <!--opt, enum, control mode of discovering devices via Open Network Video Interface Protocol, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </ONVIFDiscoveryMode>
  <ISDiscoveryMode>
    <!--opt, enum, control mode of discovering devices via IS-Discovery of SADP, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </ISDiscoveryMode>
  <PcapMode>
    <!--opt, enum, control mode of discovering devices via winpcap or libpcap or raw sockets, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </PcapMode>
  <SADP>
    <!--opt, enum, control mode of discovering devices via SADP, subType:string, desc:"discoverable", "nonDiscoverable"-->discoverable
  </SADP>
  <activeDiscoveryEnabled>
    <!--opt, bool-->true
  </activeDiscoveryEnabled>
</DiscoveryMode>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	pleaseOpenAtLeastOneSadpVersion	0x4000A759	--

9.1.2.3 Get the configuration capability of device discovery mode

Request URL

GET /ISAPI/System/discoveryMode/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DiscoveryModeCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of device discovery mode, attr:version{req, string, protocolVersion}-->
  <ONVIFDiscoveryMode opt="discoverable,nonDiscoverable">
    <!--ro, opt, string, "discoverable,nonDiscoverable, attr:opt{req, string}-->test
  </ONVIFDiscoveryMode>
  <ISDiscoveryMode opt="discoverable,nonDiscoverable">
    <!--ro, opt, string, "discoverable,nonDiscoverable, attr:opt{req, string}-->test
  </ISDiscoveryMode>
  <PcapMode opt="discoverable,nonDiscoverable">
    <!--ro, opt, string, "discoverable,nonDiscoverable, attr:opt{req, string}-->test
  </PcapMode>
  <SADP opt="discoverable,nonDiscoverable">
    <!--ro, opt, string, "discoverable,nonDiscoverable, attr:opt{req, string}-->test
  </SADP>
  <activeDiscoveryEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </activeDiscoveryEnabled>
</DiscoveryModeCap>
```

9.1.3 Device Basic Information Management

9.1.3.1 Get device information parameters

Request URL

GET /ISAPI/System/deviceInfo?serialNumber=<serialNumber>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	--
controllerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--ro, req, string, device name, range:[0,132]-->test
  </deviceName>
  <deviceID>
    <!--ro, req, string, device No., range:[1,128]-->test
  </deviceID>
  <deviceDescription>
    <!--ro, opt, string, device description, range:[1,128]-->test
  </deviceDescription>
  <deviceLocation>
    <!--ro, opt, string, device Location, range:[1,128]-->hangzhou
  </deviceLocation>
  <deviceStatus>
    <!--ro, opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
  </deviceStatus>
  <DetailAbnormalStatus>
    <!--ro, opt, object, error status details, desc:it is valid only when deviceStatus is "abnormal"-->
    <hardDiskFull>
      <!--ro, opt, bool, disk full-->true
    </hardDiskFull>
    <hardDiskError>
      <!--ro, opt, bool, disk error-->true
    </hardDiskError>
    <ethernetBroken>
      <!--ro, opt, bool, network disconnected-->true
    </ethernetBroken>
    <ipaddrConflict>
      <!--ro, opt, bool, IP address conflict-->true
    </ipaddrConflict>
    <illegalAccess>
      <!--ro, opt, bool, illegal access-->true
    </illegalAccess>
    <recordError>
      <!--ro, opt, bool, recording exception-->true
    </recordError>
    <raidLogicDiskError>
      <!--ro, opt, bool, virtual disk exception in the array-->true
    </raidLogicDiskError>
```



```

</spareWorkDeviceError>
<!--ro, opt, bool, hot spare active device exception-->true
</spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact>
  <!--ro, opt, string, manufacturer, range:[1,32]-->STD-CGI
</systemContact>
<model>
  <!--ro, req, string, device model, range:[1,64]-->IDS-9632NX-I8/X
</model>
<serialNumber>
  <!--ro, req, string, device serial No., range:[1,48]-->IDS-9632NX-I8/X1620181209CCRRRC77605411WCVU
</serialNumber>
<macAddress>
  <!--ro, req, string, MAC address, range:[1,64]-->44:47:cc:c8:d9:e4
</macAddress>
<firmwareVersion>
  <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
</firmwareVersion>
<firmwareReleasedDate>
  <!--ro, opt, string, release date of the device firmware version-->2019-11-01
</firmwareReleasedDate>
<encoderVersion>
  <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
</encoderVersion>
<encoderReleasedDate>
  <!--ro, opt, string, release date of the device encoder version-->2019-11-02
</encoderReleasedDate>
<bootVersion>
  <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
</bootVersion>
<bootReleasedDate>
  <!--ro, opt, string, release date of boot-->2019-11-03
</bootReleasedDate>
<panelVersion>
  <!--ro, opt, string, panel version, range:[1,64]-->V1.0
</panelVersion>
<hardwareVersion>
  <!--ro, opt, string, hardware version, range:[1,24]-->0x0
</hardwareVersion>
<decoderVersion>
  <!--ro, opt, string, device decoder version, range:[1,32]-->V1.0
</decoderVersion>
<decoderReleasedDate>
  <!--ro, opt, string, release date of the device decoder version-->2019-01-01
</decoderReleasedDate>
<softwareVersion>
  <!--ro, opt, string, software version, range:[1,32]-->V1.23
</softwareVersion>
<capacity>
  <!--ro, opt, int, device capacity, range:[1,10240], desc:unit: MB-->1
</capacity>
<upgradePackageMD5>
  <!--ro, opt, string-->2f003d3a733bf219b7afc538ff16943d
</upgradePackageMD5>
<usedCapacity>
  <!--ro, opt, int, used capacity of the device, range:[1,10240], desc:unit: MB-->1
</usedCapacity>
<subDeviceType>
  <!--ro, opt, enum, device sub type, subType:string, desc:"accessControlTerminal" (access control terminal; it is valid when the value of deviceType is
  "ACS"), "attendanceCheckDevice" (attendance check devices; it is valid when the value of deviceType is "ACS"), "multiChannelAccessController" (multi-channel
  access controller; it is valid when the value of deviceType is "ACS"), "personAndIdCardDevice" (person and ID card device; it is valid when the value of
  deviceType is "ACS"), "doorStation" (door station; it is valid when the value of deviceType is "VIS"), "indoor" (indoor station; it is valid when the value
  of deviceType is "VIS"), "mainStation" (main station; it is valid when the value of deviceType is "VIS"), "dropGate" (swing barrier; it is valid when the
  value of deviceType is "PersonnelChannel"), "wingGate" (flap barrier; it is valid when the value of deviceType is "PersonnelChannel"), "threeRollerGate"
  (tripod turnstile; it is valid when the value of deviceType is "PersonnelChannel")-->accessControlTerminal
</subDeviceType>
<telecontrolID>
  <!--ro, opt, int, remote control ID, range:[1,255]-->1
</telecontrolID>
<supportBeep>
  <!--ro, opt, bool, whether it supports buzzer-->true
</supportBeep>
<supportVideoLoss>
  <!--ro, opt, bool, whether it supports video loss-->true
</supportVideoLoss>
<firmwareVersionInfo>
  <!--ro, opt, string, firmware version information, range:[1,32]-->B-R-H5-0
</firmwareVersionInfo>
<actualFloorNum>
  <!--ro, opt, int, actual number of floors, range:[1,128]-->1
</actualFloorNum>
<localZoneNum>
  <!--ro, opt, int, number of local zones, range:[0,16]-->1
</localZoneNum>
<alarmOutNum>
  <!--ro, opt, int, number of alarm outputs, range:[0,16]-->1
</alarmOutNum>
<alarmInNum>
  <!--ro, opt, int, number of alarm inputs, range:[0,16]-->1
</alarmInNum>
<distanceResolution>
  <!--ro, opt, float, distance resolution, range:[0.000,0.999]-->0.000

```

```

</distanceResolution>
<angleResolution>
  <!--ro, opt, float, angle resolution, range:[0.000,0.999]-->0.000
</angleResolution>
<speedResolution>
  <!--ro, opt, float, speed resolution, range:[0.000,0.999]-->0.000
</speedResolution>
<detectDistance>
  <!--ro, opt, float, detection distance, range:[0.000,0.999]-->0.000
</detectDistance>
<relayNum>
  <!--ro, opt, int, number of relays (local), range:[0,16]-->1
</relayNum>
<electroLockNum>
  <!--ro, opt, int, number of Local eElectronic Locks, range:[0,16]-->1
</electroLockNum>
<sirenNum>
  <!--ro, opt, int, number of sirens, range:[0,16]-->1
</sirenNum>
<alarmLamp>
  <!--ro, opt, int, number of alarm lights, range:[0,16]-->1
</alarmLamp>
<RS485Num>
  <!--ro, opt, int, number of Local RS-485, range:[0,16]-->1
</RS485Num>
<radarVersion>
  <!--ro, opt, string, radar version, range:[1,32]-->test
</radarVersion>
<cameraModuleVersion>
  <!--ro, opt, string, camera module version, range:[1,32]-->test
</cameraModuleVersion>
<mainversion>
  <!--ro, opt, int, main version No., range:[1,255]-->1
</mainversion>
<subversion>
  <!--ro, opt, int, sub version No., range:[1,255]-->1
</subversion>
<upgradeversion>
  <!--ro, opt, int, updated version No., range:[1,255]-->1
</upgradeversion>
<customizeversion>
  <!--ro, opt, int, customized version No., range:[1,255]-->1
</customizeversion>
<companyName>
  <!--ro, opt, string, manufacturer name, range:[1,32]-->test
</companyName>
<copyright>
  <!--ro, opt, string, copyright Information, range:[1,32]-->test
</copyright>
<systemName>
  <!--ro, opt, enum, storage system name, subType:string, desc:"storageManagement" (storage management system), "distributedStorageManagement"
(distributed storage management system)-->storageManagement
</systemName>
<systemStatus>
  <!--ro, opt, enum, system status, subType:string, desc:"configured", "unConfigured" (not configured)-->configured
</systemStatus>
<isLeaderDevice>
  <!--ro, opt, bool, whether it is the corresponding device of the resource IP address-->true
</isLeaderDevice>
<clusterVersion>
  <!--ro, opt, string, system cluster version, range:[1,32], desc:this node is valid when the value of isLeaderDevice is true-->test
</clusterVersion>
<centralStorageVersion>
  <!--ro, opt, string, center storage version, range:[1,16]-->test
</centralStorageVersion>
<powerOnMode>
  <!--ro, opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
</powerOnMode>
<customizedInfo>
  <!--ro, opt, string, customized project No., range:[1,32], desc:no value will be returned if it is baseline device. Custom project No. will be returned
if is custom device-->test
</customizedInfo>
<verificationCode>
  <!--ro, opt, string, device verification code-->test
</verificationCode>
<supportUrl>
  <!--ro, opt, string, service portal-->test
</supportUrl>
<subSerialNumber>
  <!--ro, opt, string, sub serial No.-->test
</subSerialNumber>
<languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese">
  <!--ro, opt, enum, language type, subType:string, attr:opt{req, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french",
"russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch",
"hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic",
"traditionalChinese", "lithuanian", "anglicism", "estonian"-->chinese
</languageType>
<DockStation>
  <!--ro, opt, object, dock station information configuration-->
  <Platform>
    <!--ro, opt, object, information configuration of the dock station accessing the platform-->
    <ip>
      <!--ro, opt, string, IP address, range:[1,32]-->test
    </ip>
  </Platform>
</DockStation>

```

```

</ip>
<port>
  <!--ro, opt, int, communication port, range:[1,65535]-->1
</port>
<userName>
  <!--ro, req, string, user name, range:[1,32]-->test
</userName>
</Platform>
<centralStorageBackupEnabled>
  <!--ro, opt, bool, whether to enable center storage backup-->true
</centralStorageBackupEnabled>
</DockStation>
<webVersion>
  <!--ro, opt, string, Web version No., range:[1,32]-->test
</webVersion>
<deviceRFProgramVersion>
  <!--ro, opt, string, device RF program version, range:[1,32]-->test
</deviceRFProgramVersion>
<securityModuleSerialNo>
  <!--ro, opt, string, security module serial No., range:[1,32]-->test
</securityModuleSerialNo>
<securityModuleVersion>
  <!--ro, opt, string, security module version, range:[1,32]-->test
</securityModuleVersion>
<securityChipVersion>
  <!--ro, opt, string, security chip version, range:[1,32]-->test
</securityChipVersion>
<securityModuleKeyVersion>
  <!--ro, opt, string, security module key version, range:[1,32]-->test
</securityModuleKeyVersion>
<UIDLampRecognition>
  <!--ro, opt, object, recognize the device information by UID-->
  <enabled>
    <!--ro, opt, bool, whether to enable the function-->true
  </enabled>
</UIDLampRecognition>
<confDeviceIdPrefix>
  <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
</confDeviceIdPrefix>
<OEMCode>
  <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
</OEMCode>
<simpleAlgorithmVersion>
  <!--ro, opt, string, single algorithm version, desc:for fire control analyzer, it only supports one recognition algorithm if this node is returned to
the device information-->test
</simpleAlgorithmVersion>
<bootTime>
  <!--ro, opt, datetime, system boot time, desc:ISO8601 time (TD-format, Local time and time difference)-->1970-01-01T08:00:00+08:00
</bootTime>
<intelligentAnalysisEngineModel>
  <!--ro, opt, string, engine model, range:[1,32]-->test
</intelligentAnalysisEngineModel>
<marketType>
  <!--ro, opt, enum, market type, subType:int, desc:0 (invalid), 1 (distribution type), 2 (industry type)-->0
</marketType>
<algorithmVersion>
  <!--ro, opt, string, HCNN algorithm version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
</algorithmVersion>
<firmware>
  <!--ro, opt, string, firmware version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
</firmware>
<engineList>
  <!--ro, opt, object, List of device computing power, desc:it returns engine No. by number of GPU chips. For example, there are 3 chips whose numbers are
1, 5, and 11, then the returned values are 1, 5, 11-->
  <engine>
    <!--ro, opt, int, returned engine No.-->1
  </engine>
</engineList>
<platform>
  <!--ro, opt, enum, platform type, subType:int, desc:1 (TX1), 2 (P4), 3 (3559), 4 (3519), 5 (3516)-->1
</platform>
<platformName>
  <!--ro, opt, string, the platform where the algorithm runs, desc:"TX1/K81" (NVR), "KT" (data center), "H5" (front-end device), "H7" (front-end device),
"K82" (NVR), "G3" (front-end device)-->test
</platformName>
<touchScreenVersionInfo>
  <!--ro, opt, string, touch screen module version-->test
</touchScreenVersionInfo>
<protocolFileURL>
  <!--ro, opt, string, protocol notice URI, range:[1,32]-->test
</protocolFileURL>
<recycleRecordEnabled>
  <!--ro, opt, bool, whether to enable overwritten recording-->false
</recycleRecordEnabled>
<decordChannelNums>
  <!--ro, opt, int, number of decoding channels-->0
</decordChannelNums>
<VGANums>
  <!--ro, opt, int, number of VGA ports-->0
</VGANums>
<USBNums>
  <!--ro, opt, int, number of USB ports-->0
</USBNums>
<auxoutNums>

```

```

    <!--ro, opt, int, number of auxiliary ports-->0
</auxoutNums>
<expansionBoardVersion>
    <!--ro, opt, string, extension board version information, range:[1,32]-->test
</expansionBoardVersion>
<initWizzardDisplay>
    <!--ro, opt, bool, whether it displays initialization wizard, desc:it is only used for displaying configuration of applications integrated to the wizard
such as Local GUI. By default, the value is true; it is false after the wizard is configured on the Local GUI-->true
</initWizzardDisplay>
<beaconID>
    <!--ro, opt, string, device RF program version No., range:[0,32], desc:by default, it is the current value-->test
</beaconID>
<isResetDeviceLanguage>
    <!--ro, opt, bool, whether to reset the language, desc:only Admin and Installer have the permission to switch the device language. If the value is true,
the Hik-Connect Mobile Client and the device's web page display device information in the target language; if the value is false, the language switching
takes no effect on the Hik-Connect Mobile Client and the device's web page-->false
</isResetDeviceLanguage>
<displyNum>
    <!--ro, opt, int, number of windows-->0
</displyNum>
<bspVersion>
    <!--ro, opt, string, BSP software version-->test
</bspVersion>
<dspVersion>
    <!--ro, opt, string, DSP software version-->test
</dspVersion>
<localUIVersion>
    <!--ro, opt, string, Local UI version-->test
</localUIVersion>
<OPCASubType>
    <!--ro, opt, enum, sub type of optical physical and chemical analyzer, subType:string, desc:"pump", "autoSampler" (auto sampler),"detector",
"columnOven" (column oven), "SOC" (gas chromatography)-->detector
</OPCASubType>
<wiegandOutNum>
    <!--ro, opt, int, number of Wiegand outputs, range:[0,1]-->1
</wiegandOutNum>
<ChipVersionInfoList>
    <!--ro, opt, array, chip version information list, subType:object, range:[1,6]-->
<ChipVersionInfo>
    <!--ro, opt, object, chip version information-->
    <ID>
        <!--ro, req, int, chip No., range:[1,10], desc:the No.s of chips of different types can be duplicate. Users can distinguish them by chip type and
chip No. together-->1
    </ID>
    <firmwareVersion>
        <!--ro, req, string, chip firmware version, range:[0,64]-->1.0.0
    </firmwareVersion>
    <algorithmVersion>
        <!--ro, opt, string, algorithm version, desc:according to the chip name to determine the specific algorithm type, such as the chip name is
temperature control system, then it is the temperature control system algorithm-->1.0.0
    </algorithmVersion>
    <chipName>
        <!--ro, opt, string, chip name, range:[0,32], desc:N/A-->test
    </chipName>
</ChipVersionInfo>
</ChipVersionInfoList>
<personBagLinkAlgoEngineVersion>
    <!--ro, opt, string, engine version of the person & package Linkage module, range:[0,64], desc:this node is for analyzer (security inspection)-->1.0.0
</personBagLinkAlgoEngineVersion>
<BIOSVersion>
    <!--ro, opt, string, BIOS version, range:[0,16]-->V1.3.4
</BIOSVersion>
<contactInformation>
    <!--ro, opt, string, contact information, range:[0,64]-->test
</contactInformation>
<temperatureModuleVersionInfo>
    <!--ro, opt, string, temperature measurement component version-->test
</temperatureModuleVersionInfo>
<pedestrianWarningModuleVersion>
    <!--ro, opt, object, pedestrian warning module information, desc:this node is returned only for pedestrian warning devices-->
    <pedestrianWarningMCUVersion>
        <!--ro, opt, string, MCU program version, range:[0,32]-->V1.3.4
    </pedestrianWarningMCUVersion>
    <pedestrianWarningRadarVersion>
        <!--ro, opt, string, the program version of pedestrian warning radar, range:[0,32]-->V1.3.4
    </pedestrianWarningRadarVersion>
    <pedestrianRangingModuleVersion>
        <!--ro, opt, string, the program version of the pedestrian distance measurement module, range:[0,32]-->V1.3.4
    </pedestrianRangingModuleVersion>
</PedestrianWarningModuleVersion>
<encryptionModel>
    <!--ro, opt, string, device encryption model, range:[0,16], desc:it cannot be modified and is a hexadecimal digit currently-->test
</encryptionModel>
<UMBVersion>
    <!--ro, opt, string, UMB version, range:[1,32]-->V1.3.0
</UMBVersion>
<audioBoard>
    <!--ro, opt, object, audio board information-->
    <audioBoardModel>
        <!--ro, opt, string, audio board model, range:[0,32], desc:currently only C8 and R8 are supported-->C8
    </audioBoardModel>
    <audioBoardVersion>
        <!--ro, opt, string, audio board version, range:[0,32]-->V1.0.0 build 211210
    </audioBoardVersion>

```

```

</audioBoardVersion>
</audioBoard>
<materialScanAlgorithmVersion>
  <!--ro, opt, string, material scanning algorithm version, range:[0,64], desc:material scanning algorithm version-->test
</materialScanAlgorithmVersion>
<regionVersion>
  <!--ro, opt, enum, N/A, subType:string, desc:N/A-->basic
</regionVersion>
<productionDate>
  <!--ro, opt, date, date of manufacture-->2022-04-02
</productionDate>
<wifiModuleMACAddress>
  <!--ro, opt, string, Mac address of Wi-Fi module, range:[1,48]-->44:47:cc:c8:d9:e4
</wifiModuleMACAddress>
<displayInterfaceSize>
  <!--ro, opt, float, display interface size, range:[0.0,16.0], unit:in, desc:diagonal length of display interface for video intercom devices-->0
</displayInterfaceSize>
<releaseRegion>
  <!--ro, opt, enum, N/A, subType:string, desc:N/A-->China
</releaseRegion>
<shortSerialNumber>
  <!--ro, opt, string, short serial No., range:[1,9]-->test
</shortSerialNumber>
<audioVersion>
  <!--ro, opt, string, audio version, range:[1,32]-->test
</audioVersion>
<communicationFrequency>
  <!--ro, opt, enum, device wireless communication frequency, subType:string, desc:"433MHz", "868MHz"-->433MHz
</communicationFrequency>
<OSCoreVersionInfo>
  <!--ro, opt, object, OS/core component version information-->
  <OSCoreVersion>
    <!--ro, req, string, OS/core component version No., range:[1,32]-->1.0.0
  </OSCoreVersion>
  <minisysVersion>
    <!--ro, opt, string, minisys version No., range:[1,32]-->1.0.0
  </minisysVersion>
  <networkServiceVersion>
    <!--ro, opt, string, network service version No., range:[1,32]-->1.0.0
  </networkServiceVersion>
  <upgradeServiceVersion>
    <!--ro, opt, string, upgrade service version No., range:[1,32]-->1.0.0
  </upgradeServiceVersion>
</OSCoreVersionInfo>
<xTransVersion>
  <!--ro, opt, string, version No. of xtrans chip communication component, range:[1,32], desc:this node is for analyzer-->1.0.0
</xTransVersion>
<deviceMaintenanceInfoQRCode>
  <!--ro, opt, string, Get the QR code of the device maintenance information., range:[0,64], desc:scan to get more device information, including
  maintenance and documentation materials-->test
</deviceMaintenanceInfoQRCode>
<batteryFirmwareVersion>
  <!--ro, opt, string, battery firmware version No., range:[1,32]-->1.0.0
</batteryFirmwareVersion>
<radarVideoMapInfoList>
  <!--ro, opt, array, information list of radar-assisted video coordinate conversion matrix, subType:object, range:[1,32], desc:multiple cameras are added
  as sub-devices by terminal device(the transmission solution of sub-devices is adopted, refer to Sub-device Transparent Transmission in the Video Channel
  Resource Management domain). Each camera has its own radar-assisted video coordinate conversion matrix table, while the terminal device has multiple ones
  corresponding to each camera. When getting the terminal device information parameters via GET /ISAPI/System/deviceInfo, an information list of multiple
  matrix tables of the terminal device is returned. When getting the camera information parameters via GET /ISAPI/System/deviceInfo?devIndex=<devIndex>, the
  matrix table information of the specified camera is returned-->
  <radarVideoMapInfo>
    <!--ro, opt, object, radar-assisted video coordinate conversion matrix information-->
    <devIndex>
      <!--ro, opt, string, sub-device ID, range:[0,64], desc:used for corresponding sub device-->123456789
    </devIndex>
    <radarVideoMapVersion>
      <!--ro, opt, string, radar-assisted video coordinate conversion matrix version, range:[1,32], desc:video coordinate and map GPS conversion matrix
      version-->1.0.0
    </radarVideoMapVersion>
  </radarVideoMapInfo>
</radarVideoMapInfoList>
<screenVersionInfo>
  <!--ro, opt, object, device screen version information, desc:the device (such as wall-mounted dock station) is a dual-system device, that is, an
  embedded device and an Android system screen-->
  <screenAndroidOSVersion>
    <!--ro, opt, string, Android system version number of device screen, range:[1,64], desc:system program based on Android kernel,-->1.2.0 build20230101
  </screenAndroidOSVersion>
  <screenAndroidAPPVersion>
    <!--ro, opt, string, Android application version number of device screen, range:[1,64]-->1.0.0 build20230101
  </screenAndroidAPPVersion>
</screenVersionInfo>
<isSupportBattery>
  <!--ro, opt, bool, whether the device supports battery installing, desc:if the node is not returned or the returned value is false, it indicates no--
  >true
</isSupportBattery>
<networkConfigSettingEnabled>
  <!--ro, opt, bool, whether to enter the network configuration interface after login, desc:application scenario: the default device IP address is
  192.168.0.100, which is different from the network segment or IP address of the field environment. This is because it is a static IP address, and the
  default IP address cannot be accessed on the local area network. The network cable is directly connected to the device. The technical support can open the
  device web page to check if the IP address is the default IP address 192.168.0.100. If it is default IP address, the network configuration interface will be
  auto displayed for IP address configuration. When it is true, it indicates that you will enter the configuration page after login. When it is false or does
  not exist, the login follows the previous logic-->true
</networkConfigSettingEnabled>

```

```

<ptzVersion>
  <!--ro, opt, string, PTZ version number, range:[1,32]-->test
</ptzVersion>
<movementSoftVersion>
  <!--ro, opt, string, module software version number, range:[1,32]-->test
</movementSoftVersion>
<movementHardVersion>
  <!--ro, opt, string, module hardware version number, range:[1,32]-->test
</movementHardVersion>
<loraModuleList>
  <!--ro, opt, array, Lora module information list, subType:object-->
  <loraModule>
    <!--ro, opt, object, Lora module information-->
    <id>
      <!--ro, opt, int, sequence number, desc:currently only one-->1
    </id>
    <firmwareVersion>
      <!--ro, opt, string, firmware version, range:[1,64]-->test
    </firmwareVersion>
  </loraModule>
</loraModuleList>
<deviceMode>
  <!--ro, opt, enum, device mode, subType:string, desc:"factoryMode", "maintenanceMode"-->factoryMode
</deviceMode>
<maintenanceDuration>
  <!--ro, opt, int, maintenance mode duration, range:[1,30], unit:d, dep:and,{$.DeviceInfo.deviceMode,eq,maintenance}-->1
</maintenanceDuration>
<pirAlgorithmVersion>
  <!--ro, opt, string, PIR algorithm version No., range:[1,32]-->V1.0
</pirAlgorithmVersion>
<customSerialNumber>
  <!--ro, opt, string, range:[1,9]-->B0000021
</customSerialNumber>
<deviceCabinID>
  <!--ro, opt, int, range:[0,128]-->1
</deviceCabinID>
<detectorVersion>
  <!--ro, opt, string, range:[1,32]-->test
</detectorVersion>
<internalModel>
  <!--ro, opt, string, range:[1,64]-->DS-AT1000S-2U4U
</internalModel>
<batteryFirmwareCodeVersion>
  <!--ro, opt, string, range:[1,32]-->12345
</batteryFirmwareCodeVersion>
<batteryFirmwareSerialNumber>
  <!--ro, opt, string, range:[1,32]-->DS-2FGHASKGHKWHBGKAHSG
</batteryFirmwareSerialNumber>
<mainModulePictureUrl>
  <!--ro, opt, string-->test
</mainModulePictureUrl>
<mainModuleDialUpPictureUrl>
  <!--ro, opt, string-->test
</mainModuleDialUpPictureUrl>
<controlInfo>
  <!--ro, opt, object-->
  <controlType>
    <!--ro, req, enum, subType:string-->single
  </controlType>
  <singleControlInfo>
    <!--ro, opt, object, dep:or,{$.DeviceInfo.controlInfo.controlType,eq,single}-->
    <serialMatchState>
      <!--ro, opt, enum, subType:string-->match
    </serialMatchState>
  </singleControlInfo>
  <doubleControlInfo>
    <!--ro, opt, object, dep:or,{$.DeviceInfo.controlInfo.controlType,eq,double}-->
  </doubleControlInfo>
  <controlInfoList>
    <!--ro, req, array, subType:object, range:[2,2]-->
    <controlInfo>
      <!--ro, opt, object-->
      <childDevID>
        <!--ro, req, string, range:[9,32]-->12345
      </childDevID>
      <role>
        <!--ro, opt, enum, subType:string-->main
      </role>
    </controlInfo>
  </controlInfoList>
  <supportModuleList>
    <!--ro, opt, array, subType:object-->
    <supportModule>
      <!--ro, opt, enum, subType:string-->interfaces
    </supportModule>
  </supportModuleList>
</doubleControlInfo>
</controlInfo>
<opticalFiberNumber>
  <!--ro, opt, int, range:[1,10]-->10
</opticalFiberNumber>
<devMaterialNumber>
  <!--ro, opt, string-->336800009
</devMaterialNumber>

```

```

<middlePlateMCUVersion>
  <!--ro, opt, string-->test
</middlePlateMCUVersion>
<panelMCUVersion>
  <!--ro, opt, string-->test
</panelMCUVersion>
<middlePlateCPLDVersion>
  <!--ro, opt, string-->test
</middlePlateCPLDVersion>
<backboardCPLDVersion>
  <!--ro, opt, string-->test
</backboardCPLDVersion>
<DFDVersion>
  <!--ro, opt, string-->test
</DFDVersion>
<CPUModel>
  <!--ro, opt, enum, subType:string-->K51
</CPUModel>
<deviceAssembleType>
  <!--ro, opt, enum, subType:string-->endDevice
</deviceAssembleType>
<deviceCategory>
  <!--ro, opt, enum, subType:string-->thermometryDevice
</deviceCategory>
<storageType>
  <!--ro, opt, enum, subType:string-->TF
</storageType>
<role>
  <!--ro, opt, enum, subType:string-->managementNode
</role>
<SRRCModel>
  <!--ro, opt, string-->ACS-A-U-07-002-A
</SRRCModel>
</DeviceInfo>

```

9.1.3.2 Set device information parameters

Request URL

PUT /ISAPI/System/deviceInfo?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--req, string, device name, range:[1,132]-->test
  </deviceName>
  <deviceId>
    <!--opt, string, device No., range:[1,128]-->test
  </deviceId>
  <deviceDescription>
    <!--opt, string, device description, range:[1,128]-->test
  </deviceDescription>
  <deviceLocation>
    <!--opt, string, device location, range:[1,128]-->hangzhou
  </deviceLocation>
  <deviceStatus>
    <!--ro, opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
  </deviceStatus>
  <DetailAbnormalStatus>
    <!--ro, opt, object, error status details, desc:it is valid only when deviceStatus is "abnormal"-->
    <hardDiskFull>
      <!--ro, opt, bool, disk full-->true
    </hardDiskFull>
    <hardDiskError>
      <!--ro, opt, bool, disk error-->true
    </hardDiskError>
    <ethernetBroken>
      <!--ro, opt, bool, network disconnected-->true
    </ethernetBroken>
    <ipaddrConflict>
      <!--ro, opt, bool, IP address conflict-->true
    </ipaddrConflict>
    <illegalAccess>
      <!--ro, opt, bool, illegal access-->true
    </illegalAccess>
    <recordError>
      <!--ro, opt, bool, recording exception-->true
    </recordError>

```

```
<raidLogicDiskError>
  <!--ro, opt, bool, virtual disk exception in the array-->true
</raidLogicDiskError>
<spareWorkDeviceError>
  <!--ro, opt, bool, hot spare active device exception-->true
</spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact>
  <!--ro, opt, string, manufacturer, range:[1,32]-->STD-CGI
</systemContact>
<model>
  <!--ro, req, string, device model, range:[1,64]-->iDS-9632NX-I8/X
</model>
<serialNumber>
  <!--ro, req, string, device serial No., range:[1,48]-->iDS-9632NX-I8/X1620181209CCRR77605411WCWU
</serialNumber>
<macAddress>
  <!--ro, req, string, MAC Address, range:[1,64]-->44:47:cc:c8:d9:e4
</macAddress>
<firmwareVersion>
  <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
</firmwareVersion>
<firmwareReleasedDate>
  <!--ro, opt, string, release date of the device firmware version-->2019-11-01
</firmwareReleasedDate>
<encoderVersion>
  <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
</encoderVersion>
<encoderReleasedDate>
  <!--ro, opt, string, release date of the device encoder version-->2019-11-02
</encoderReleasedDate>
<bootVersion>
  <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
</bootVersion>
<bootReleasedDate>
  <!--ro, opt, string, release date of boot-->2019-11-03
</bootReleasedDate>
<panelVersion>
  <!--ro, opt, string, panel version, range:[1,32]-->V1.0
</panelVersion>
<hardwareVersion>
  <!--ro, opt, string, hardware version, range:[1,16]-->0x0
</hardwareVersion>
<decoderVersion>
  <!--ro, opt, string, device decoder version, range:[1,32]-->V1.0
</decoderVersion>
<decoderReleasedDate>
  <!--ro, opt, string, release date of the device decoder version-->2019-01-01
</decoderReleasedDate>
<softwareVersion>
  <!--ro, opt, string, software version, range:[1,32]-->V1.23
</softwareVersion>
<capacity>
  <!--ro, opt, int, device capacity, range:[1,10240], desc:unit: MB-->1
</capacity>
<usedCapacity>
  <!--ro, opt, int, used capacity of the device, range:[1,10240], desc:unit: MB-->1
</usedCapacity>
<telecontrolID>
  <!--opt, int, remote control ID, range:[1,255]-->1
</telecontrolID>
<supportBeep>
  <!--ro, opt, bool, whether to support buzzer-->true
</supportBeep>
<supportVideoLoss>
  <!--ro, opt, bool, whether the device supports video loss-->true
</supportVideoLoss>
<firmwareVersionInfo>
  <!--ro, opt, string, firmware version information, range:[1,32]-->test
</firmwareVersionInfo>
<actualFloorNum>
  <!--ro, opt, int, actual number of floors, range:[1,128]-->1
</actualFloorNum>
<localZoneNum>
  <!--ro, opt, int, number of zones (Local), range:[0,16]-->1
</localZoneNum>
<alarmOutNum>
  <!--ro, opt, int, number of alarm outputs, range:[0,16]-->1
</alarmOutNum>
<distanceResolution>
  <!--ro, opt, float, distance resolution, range:[0.000,0.999]-->0.000
</distanceResolution>
<angleResolution>
  <!--ro, opt, float, angle resolution, range:[0.000,0.999]-->0.000
</angleResolution>
<speedResolution>
  <!--ro, opt, float, speed resolution, range:[0.000,0.999]-->0.000
</speedResolution>
<detectDistance>
  <!--ro, opt, float, detection distance, range:[0.000,0.999]-->0.000
</detectDistance>
<relayNum>
  <!--ro, opt, int, number of relays (Local), range:[0,16]-->1
</relayNum>
```



```

<electroLockNum>
  <!--ro, opt, int, number of locks (local), range:[0,16]-->1
</electroLockNum>
<sirenNum>
  <!--ro, opt, int, number of sirens, range:[0,16]-->1
</sirenNum>
<alarmLamp>
  <!--ro, opt, int, number of alarm lamps, range:[0,16]-->1
</alarmLamp>
<RS485Num>
  <!--ro, opt, int, number of Local 485, range:[0,16]-->1
</RS485Num>
<radarVersion>
  <!--ro, opt, string, radar version, range:[1,32]-->test
</radarVersion>
<cameraModuleVersion>
  <!--ro, opt, string, camera module version, range:[1,32]-->test
</cameraModuleVersion>
<mainversion>
  <!--ro, opt, int, main version No., range:[1,255]-->1
</mainversion>
<subversion>
  <!--ro, opt, int, sub version No., range:[1,255]-->1
</subversion>
<upgradeversion>
  <!--ro, opt, int, update version No., range:[1,255]-->1
</upgradeversion>
<customizeversion>
  <!--ro, opt, int, custom version No., range:[1,255]-->1
</customizeversion>
<companyName>
  <!--ro, opt, string, manufacturer name, range:[1,32]-->test
</companyName>
<copyright>
  <!--ro, opt, string, version information, range:[1,32]-->test
</copyright>
<systemName>
  <!--ro, opt, enum, storage system name, subType:string, desc:"storageManagement" (storage management system), "distributedStorageManagement"
(distributed storage management system)-->storageManagement
</systemName>
<systemStatus>
  <!--ro, opt, enum, system status, subType:string, desc:"configured", "unConfigured" (not configured)-->configured
</systemStatus>
<isLeaderDevice>
  <!--ro, opt, bool, whether it is the corresponding device of the resource IP address-->true
</isLeaderDevice>
<clusterVersion>
  <!--ro, opt, string, cluster version, range:[1,32], desc:it is valid when isLeaderDevice returns true-->test
</clusterVersion>
<centralStorageVersion>
  <!--ro, opt, string, center storage version, range:[1,16]-->test
</centralStorageVersion>
<powerOnMode>
  <!--ro, opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
</powerOnMode>
<customizedInfo>
  <!--ro, opt, string, custom project No., range:[1,32], desc:no value will be returned if it is baseline device. Custom project No. will be returned if
is custom device-->test
</customizedInfo>
<verificationCode>
  <!--ro, opt, string, device verification code-->test
</verificationCode>
<supportUrl>
  <!--ro, opt, string, service portal-->test
</supportUrl>
<subSerialNumber>
  <!--ro, opt, string, sub serial No.-->test
</subSerialNumber>
<languageType> opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese">
  <!--ro, opt, enum, language type, subType:string, attr:opt{req, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french",
"russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch",
"hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic",
"traditionalChinese"-->chinese
</languageType>
<DockStation>
  <!--ro, opt, object, dock station configuration-->
  <Platform>
    <!--ro, opt, object, information configuration of the dock station accessing the platform-->
    <ip>
      <!--ro, opt, string, IP address, range:[1,32]-->test
    </ip>
    <port>
      <!--ro, opt, int, communication port, range:[1,65535]-->1
    </port>
    <userName>
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--ro, req, string, password, range:[1,16]-->test
    </password>
  </Platform>
  <centralStorageBackupEnabled>
    <!--ro, opt, bool, whether to enable central storage backup-->true
  </centralStorageBackupEnabled>

```

```

    </centralStorageBackupEnabled>
</DockStation>
<webVersion>
    <!--ro, opt, string, Web version No., range:[1,32]-->test
</webVersion>
<deviceRFProgramVersion>
    <!--ro, opt, string, device RF program version, range:[1,32]-->test
</deviceRFProgramVersion>
<securityModuleSerialNo>
    <!--ro, opt, string, security module serial No., range:[1,32]-->test
</securityModuleSerialNo>
<securityModuleVersion>
    <!--ro, opt, string, security module version, range:[1,32]-->test
</securityModuleVersion>
<securityChipVersion>
    <!--ro, opt, string, security chip version, range:[1,32]-->test
</securityChipVersion>
<securityModuleKeyVersion>
    <!--ro, opt, string, security module key version, range:[1,32]-->test
</securityModuleKeyVersion>
<UIDLampRecognition>
    <!--ro, opt, object, recognize the device information by UID-->
    <enabled>
    <!--ro, opt, bool, whether to enable the function-->true
    </enabled>
</UIDLampRecognition>
<confDeviceIdPrefix>
    <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
</confDeviceIdPrefix>
<OEMCode>
    <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
</OEMCode>
<simpleAlgorithmVersion>
    <!--ro, opt, string, single algorithm version, desc:for fire control analyzer, it only supports one recognition algorithm if this node is returned to
the device information-->test
</simpleAlgorithmVersion>
<bootTime>
    <!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
</bootTime>
<intelligentAnalysisEngineModel>
    <!--ro, opt, string, engine model, range:[1,32]-->test
</intelligentAnalysisEngineModel>
<marketType>
    <!--ro, opt, enum, market type, subType:int, desc:0 (invalid), 1 (distribution type), 2 (industry type)-->0
</marketType>
<algorithmVersion>
    <!--ro, opt, string, HCNW algorithm version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
</algorithmVersion>
<firmware>
    <!--ro, opt, string, firmware version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
</firmware>
<engineList>
    <!--ro, opt, object, List of device computing power, desc:it returns engine No. by the number of GPU chips. For example, there are 3 chips whose numbers
are 1, 5, and 11, then the returned values are 1, 5, 11-->
    <engine>
    <!--ro, opt, int, returned engine No.-->1
    </engine>
</engineList>
<platform>
    <!--ro, opt, enum, platform type, subType:int, desc:1 (TX1), 2 (P4), 3 (3559), 4 (3519), 5 (3516)-->1
</platform>
<platformName>
    <!--ro, opt, string, the platform where the algorithm runs, desc:TX1/K81: NVR, KT: data center, H5: front-end device, H7: front-end device, K82: NVR, G3:
front-end device-->test
</platformName>
<touchScreenVersionInfo>
    <!--ro, opt, string, touch screen module version-->test
</touchScreenVersionInfo>
<protocolFileURL>
    <!--ro, opt, string, protocol notice URI, range:[1,32]-->test
</protocolFileURL>
<recycleRecordEnabled>
    <!--ro, opt, bool, whether to enable overwritten recording-->false
</recycleRecordEnabled>
<decordChannelNums>
    <!--ro, opt, int, number of decoding channels-->0
</decordChannelNums>
<VGANums>
    <!--ro, opt, int, number of VGA ports-->0
</VGANums>
<USBNums>
    <!--ro, opt, int, number of USB ports-->0
</USBNums>
<auxoutNums>
    <!--ro, opt, int, number of auxiliary ports-->0
</auxoutNums>
<expansionBoardVersion>
    <!--ro, opt, string, extension board version information, range:[1,32]-->test
</expansionBoardVersion>
<initWizzardDisplay>
    <!--ro, opt, bool, whether it displays initialization wizard, desc:it is only used for displaying configuration of applications integrated to the
wizard, such as Local GUI and Hik-Connect app by default, the value is true. it is false after the Local GUI configures the wizard-->true
</initWizzardDisplay>
<hasConnTH>

```

```

<!--ro, opt, string, device RF program version No., range:[0,32], desc:by default, it is the current value-->test
</beaconID>
<isResetDeviceLanguage>
  <!--ro, opt, bool, whether it supports resetting the device Language, desc:only Admin and Installer have the permission to switch the device Language.
  If the value is true, the Hik-Connect Mobile Client and the device's web page display device information in the target Language; if the value is false, the
  Language switching takes no effect on the Hik-Connect Mobile Client and the device's web page-->false
</isResetDeviceLanguage>
<dispayNum>
  <!--ro, opt, int, number of windows-->0
</dispayNum>
<bspVersion>
  <!--ro, opt, string, BSP software version-->test
</bspVersion>
<dspVersion>
  <!--ro, opt, string, DSP software version-->test
</dspVersion>
<localUIVersion>
  <!--ro, opt, string, Local UI version-->test
</localUIVersion>
<detectorType>
  <!--ro, opt, enum, detector type, subType:string, desc:"PDA" (photo-diode array)-->PDA
</detectorType>
<wiegandOutNum>
  <!--ro, opt, int, number of Wiegand outputs, range:[0,1]-->1
</wiegandOutNum>
<ChipVersionInfoList>
  <!--ro, opt, array, subType:object, range:[1,6]-->
  <ChipVersionInfo>
    <!--ro, opt, object-->
    <ID>
      <!--ro, req, int, range:[0,1]-->1
    </ID>
    <firmwareVersion>
      <!--ro, req, string, range:[0,64]-->1.0.0
    </firmwareVersion>
  </ChipVersionInfo>
</ChipVersionInfoList>
<personBagLinkAlgoEngineVersion>
  <!--ro, opt, string, engine version of the person & package Linkage module, range:[0,64], desc:this node is for analyzer (security inspection)-->1.0.0
</personBagLinkAlgoEngineVersion>
<BIOSVersion>
  <!--ro, opt, string, BIOS version, range:[0,16]-->test
</BIOSVersion>
<contactInformation>
  <!--ro, opt, string, contact information, range:[0,64]-->test
</contactInformation>
<materialScanAlgorithmVersion>
  <!--opt, string, material scan algorithm version, range:[0,64], desc:algorithm version of imaging radar panel for material scan-->test
</materialScanAlgorithmVersion>
<audioVersion>
  <!--ro, opt, string, range:[1,32]-->test
</audioVersion>
<loginPassword>
  <!--opt, string, range:[1,16]-->test
</loginPassword>
<deviceMode>
  <!--opt, enum, subType:string-->factoryMode
</deviceMode>
<maintenanceDuration>
  <!--opt, int, range:[1,30], unit:d, dep:and,{$.DeviceInfo.deviceMode,eq,maintenance}-->1
</maintenanceDuration>
<deviceCabinID>
  <!--opt, int, range:[0,128]-->1
</deviceCabinID>
</DeviceInfo>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	theCurrentControllerCannotBeConfiguredAsAHost	0x4000A827	--

9.1.3.3 Get firmware code list

Request URL

GET /ISAPI/System/firmwareCodeV2?startIndex=<startIndex>&maxNumber=<maxNumber>

Query Parameter

Parameter Name	Parameter Type	Description
startIndex	string	To control the message size, the quantity of returned firmware IDs returned by device depends on the client request. For example, if there are 100 firmware IDs of a device, and up to 32 IDs are allowed in every request, then the device returns only 32 IDs according to the start index (startIndex). Examples: GET /ISAPI/System/firmwareCodeV2?startIndex=1&maxNumber=32
maxNumber	string	To control the message size, the quantity of returned firmware IDs returned by device depends on the client request. For example, if there are 100 firmware IDs of a device, and up to 32 IDs are allowed in every request, then the device returns only 32 IDs according to the start index (startIndex). Examples: GET /ISAPI/System/firmwareCodeV2?startIndex=1&maxNumber=32

Request Message

None

Response Message

`screen)-->cardReader`

`No., desc:it depends on the value of moduleType-->1`

gement

ability

```
<?xml version="1.0" encoding="UTF-8"?>

<EventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device event capability, attr:version{req, string, protocolVersion}-->
  <isSupportHDFull>
    <!--ro, opt, bool, whether the device supports full disk event-->true
  </isSupportHDFull>
  <isSupportHDError>
    <!--ro, opt, bool, whether the device supports disk error event-->true
  </isSupportHDError>
  <isSupportNicBroken>
    <!--ro, opt, bool, whether the device supports network disconnected event-->true
  </isSupportNicBroken>
  <isSupportIpConflict>
    <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
  </isSupportIpConflict>
  <isSupportIILAccess>
    <!--ro, opt, bool, whether the device supports illegal access event-->true
  </isSupportIILAccess>
  <isSupportViException>
    <!--ro, opt, bool, whether the device supports video signal exception event-->true
  </isSupportViException>
  <isSupportViMismatch>
    <!--ro, opt, bool, whether the device supports video standard mismatch event-->true
  </isSupportViMismatch>
  <isSupportRecordException>
    <!--ro, opt, bool, whether the device supports video error event-->true
  </isSupportRecordException>
  <isSupportRaidException>
    <!--ro, opt, bool, whether the device supports disk RAID error event-->true
  </isSupportRaidException>
  <isSupportViResMismatch>
    <!--ro, opt, bool, whether the device supports disk RAID mismatch event-->true
  </isSupportViResMismatch>
  <isSupportPOCException>
```

```
<!--ro, opt, bool, whether the device supports PoC error event-->true
</isSupportPOCException>
<isSupportSmartDetection>
  <!--ro, opt, bool, whether the device supports smart event detection event-->true
</isSupportSmartDetection>
<isSupportSpareException>
  <!--ro, opt, bool, whether the device supports back exception event-->true
</isSupportSpareException>
<isSupportPoePowerException>
  <!--ro, opt, bool, whether the device supports POE power exception event-->true
</isSupportPoePowerException>
<isSupportTriggerFocus>
  <!--ro, opt, bool, whether the device supports defocus detection event-->true
</isSupportTriggerFocus>
<isSupportMotionDetection>
  <!--ro, opt, bool, whether the device supports motion detection event-->true
</isSupportMotionDetection>
<isSupportVideoLoss>
  <!--ro, opt, bool, whether the device supports video Loss event-->true
</isSupportVideoLoss>
<isSupportTamperDetection>
  <!--ro, opt, bool, whether the device supports video tempering event-->true
</isSupportTamperDetection>
<isSupportFireDetection>
  <!--ro, opt, bool, whether the device supports fire detection event-->true
</isSupportFireDetection>
<isSupportRegionEntrance>
  <!--ro, opt, bool, whether the device supports region entrance event-->true
</isSupportRegionEntrance>
<isSupportRegionExiting>
  <!--ro, opt, bool, whether the device supports region exiting event-->true
</isSupportRegionExiting>
<isSupportLoitering>
  <!--ro, opt, bool, whether the device supports loitering detection event-->true
</isSupportLoitering>
<isSupportGroup>
  <!--ro, opt, bool, whether the device supports people gathering event-->true
</isSupportGroup>
<isSupportRapidMove>
  <!--ro, opt, bool, whether the device supports fast moving event-->true
</isSupportRapidMove>
<isSupportParking>
  <!--ro, opt, bool, whether the device supports parking detection event-->true
</isSupportParking>
<isSupportUnattendedBaggage>
  <!--ro, opt, bool, whether the device supports unattended baggage event-->true
</isSupportUnattendedBaggage>
<isSupportAttendedBaggage>
  <!--ro, opt, bool, whether the device supports object removal event-->true
</isSupportAttendedBaggage>
<isSupportHumanAttribute>
  <!--ro, opt, bool, whether the device supports human body feature detection event-->true
</isSupportHumanAttribute>
<isSupportFaceContrast>
  <!--ro, opt, bool, whether the device supports face comparison event-->true
</isSupportFaceContrast>
<isSupportFaceLib>
  <!--ro, opt, bool, whether the front-end devices support linkage by face picture library-->true
</isSupportFaceLib>
<isSupportBlackListFaceContrast>
  <!--ro, opt, bool, whether the device supports face comparison in the blockList-->true
</isSupportBlackListFaceContrast>
<isSupportWhitelListFaceContrast>
  <!--ro, opt, bool, whether the device supports face comparison in the allowList-->true
</isSupportWhitelListFaceContrast>
<isSupportIllegalParking>
  <!--ro, opt, bool, whether the device supports illegal parking detection-->true
</isSupportIllegalParking>
<isSupportPedestrian>
  <!--ro, opt, bool, whether the device supports pedestrian detection-->true
</isSupportPedestrian>
<isSupportTrafficAccident>
  <!--ro, opt, bool, whether the device supports traffic incident detection-->true
</isSupportTrafficAccident>
<isSupportConstruction>
  <!--ro, opt, bool, whether the device supports construction detection-->true
</isSupportConstruction>
<isSupportRoadBlock>
  <!--ro, opt, bool, whether the device supports roadblock detection-->true
</isSupportRoadBlock>
<isSupportAbandonedObject>
  <!--ro, opt, bool, whether the device supports fallen object detection-->true
</isSupportAbandonedObject>
<isSupportParallelParking>
  <!--ro, opt, bool, whether the device supports parallel parking detection-->true
</isSupportParallelParking>
<isSupportParkingState>
  <!--ro, opt, bool, whether the device supports parking space status detection-->true
</isSupportParkingState>
<isSupportTrafficJam>
  <!--ro, opt, bool, whether the device supports traffic jam detection (old version)-->true
</isSupportTrafficJam>
<isSupportCongestion>
  <!--ro, opt, bool, whether the device supports traffic jam detection-->true
</isSupportCongestion>
```

```
<!--ro, opt, bool, whether the device supports trigger gun detection-->true
</isSupportCongestion>
<isSupportFogDetection>
  <!--ro, opt, bool, whether the device supports fog detection-->true
</isSupportFogDetection>
<isSupportTurnRound>
  <!--ro, opt, bool, whether the device supports U-turning detection-->true
</isSupportTurnRound>
<isSupportCrossLane>
  <!--ro, opt, bool, whether the device supports driving on the Lane Line detection-->true
</isSupportCrossLane>
<isSupportLaneChange>
  <!--ro, opt, bool, whether the device supports illegal Lane change detection-->true
</isSupportLaneChange>
<isSupportVehicleExist>
  <!--ro, opt, bool, whether the device supports motor vehicle on non-motor vehicle Lane detection-->true
</isSupportVehicleExist>
<isSupportStudentsStoodUp>
  <!--ro, opt, bool, whether the device supports student standing up event-->true
</isSupportStudentsStoodUp>
<isSupportFramesPeopleCounting>
  <!--ro, opt, bool, whether the device supports single-frame people counting event-->true
</isSupportFramesPeopleCounting>
<isSupportPersonQueueDetection>
  <!--ro, opt, bool, whether the device supports queue detection event-->true
</isSupportPersonQueueDetection>
<isSupportHumanRecognition>
  <!--ro, opt, bool, whether the device supports human object detection event-->true
</isSupportHumanRecognition>
<isSupportFaceSnap>
  <!--ro, opt, bool, whether the device supports face capture event-->true
</isSupportFaceSnap>
<isSupportPersonDensityDetection>
  <!--ro, opt, bool, whether the device supports people density detection event-->true
</isSupportPersonDensityDetection>
<HFPD>
  <!--ro, opt, bool, whether the device supports frequently appeared person detection event-->true
</HFPD>
<isSupportVehicleStatistics>
  <!--ro, opt, bool, whether it supports traffic flow data statistics event-->true
</isSupportVehicleStatistics>
<isSupportFaceThermometry>
  <!--ro, opt, bool, whether the device supports face temperature screening event-->true
</isSupportFaceThermometry>
<isSupportSafetyHelmetDetection>
  <!--ro, opt, bool, whether the device supports hard hat detection event-->true
</isSupportSafetyHelmetDetection>
<isSupportPowerSupplyCabinet>
  <!--ro, opt, bool, whether the device supports power cabinet event-->true
</isSupportPowerSupplyCabinet>
<isSupportSensorDetection>
  <!--ro, opt, bool, whether the device supports sensor detection event-->true
</isSupportSensorDetection>
<isSupportSignAbnormal>
  <!--ro, opt, bool, whether the device supports signal exception event-->true
</isSupportSignAbnormal>
<isSupportTeacherBehaviorDetect>
  <!--ro, opt, bool, whether the device supports teacher abnormal behavior detection event-->true
</isSupportTeacherBehaviorDetect>
<isSupportCityManagement>
  <!--ro, opt, bool, whether the device supports intelligent city management detection event-->true
</isSupportCityManagement>
<isSupportMixedTargetDetection>
  <!--ro, opt, bool, whether the device supports multi-target-type detection event-->true
</isSupportMixedTargetDetection>
<isSupportFaceSnapModeling>
  <!--ro, opt, bool, whether the device supports face capture modeling event-->true
</isSupportFaceSnapModeling>
<isSupportIntersectionAnalysis>
  <!--ro, opt, bool, whether the device supports intersection analysis event-->true
</isSupportIntersectionAnalysis>
<isSupportTriggerCapCheck>
  <!--ro, opt, bool, whether the device supports triggering capture Line detection event-->true
</isSupportTriggerCapCheck>
<isSupportLuma>
  <!--ro, opt, bool, whether the device supports VQD brightness exception event-->true
</isSupportLuma>
<isSupportChroma>
  <!--ro, opt, bool, whether the device supports VQD image color cast event-->true
</isSupportChroma>
<isSupportSnow>
  <!--ro, opt, bool, whether the device supports VQD snow noise event-->true
</isSupportSnow>
<isSupportStreak>
  <!--ro, opt, bool, whether the device supports VQD stripe noise event-->true
</isSupportStreak>
<isSupportFreeze>
  <!--ro, opt, bool, whether the device supports VQD image freeze event-->true
</isSupportFreeze>
<isSupportSigLose>
  <!--ro, opt, bool, whether the device supports VQD signal Loss event-->true
</isSupportSigLose>
<isSupportClarity>
  <!--ro, opt, bool, whether the device supports VQD resolution exception event-->true
```

```

</isSupportClarity>
<isSupportJitter>
  <!--ro, opt, bool, whether the device supports VQD image flutter event-->true
</isSupportJitter>
<isSupportBlock>
  <!--ro, opt, bool, whether the device supports VQD video tampering event-->true
</isSupportBlock>
<isSupportFlowers>
  <!--ro, opt, bool, whether the device supports VQD blurred image alarm event-->true
</isSupportFlowers>
<isSupportNoise>
  <!--ro, opt, bool, whether the device supports VQD blurred screen event-->true
</isSupportNoise>
<isSupportGhost>
  <!--ro, opt, bool, whether the device supports VQD image noise event-->true
</isSupportGhost>
<isSupportPurple>
  <!--ro, opt, bool, whether the device supports VQD purple fringing event-->true
</isSupportPurple>
<isSupportICR>
  <!--ro, opt, bool, whether the device supports VQD ICP event-->true
</isSupportICR>
<isSupportProtectiveFilm>
  <!--ro, opt, bool, whether the device supports VQD protective film unremoved event-->true
</isSupportProtectiveFilm>
<isSupportVoltageInstable>
  <!--ro, opt, bool, whether the device supports the voltage instable exception detection event-->true
</isSupportVoltageInstable>
<isSupportCertificateRevocation>
  <!--ro, opt, bool, whether the device supports the certificate expiry alarm event-->true
</isSupportCertificateRevocation>
<isSupportSysStorFull>
  <!--ro, opt, bool, whether the device supports the storage full alarm of the cluster system event-->true
</isSupportSysStorFull>
<isSupportVersionAbnormal>
  <!--ro, opt, bool, whether the device supports cluster version exception event-->true
</isSupportVersionAbnormal>
<isSupportNodeOffline>
  <!--ro, opt, bool, whether the device supports offline exception event-->true
</isSupportNodeOffline>
<isSupportRecordCycleAbnormal>
  <!--ro, opt, bool, whether the device supports insufficient recording period event-->true
</isSupportRecordCycleAbnormal>
<isSupportIpcTransferAbnormal>
  <!--ro, opt, bool, whether the device supports IPC migration failure event-->true
</isSupportIpcTransferAbnormal>
<isSupportDrivingBehaviorDetect>
  <!--ro, opt, bool, whether the device supports driving behavior detection event-->true
</isSupportDrivingBehaviorDetect>
<isSupportWasteGasDetection>
  <!--ro, opt, bool, whether the device supports waste gas monitoring detection event-->true
</isSupportWasteGasDetection>
<isSupportGreyScaleAlarm>
  <!--ro, opt, bool, whether the device supports gray scale alarm event-->true
</isSupportGreyScaleAlarm>
<isSupportPTLocking>
  <!--ro, opt, bool, whether the device supports the PT Locking event-->true
</isSupportPTLocking>
<isSupportNoMaskDetection>
  <!--ro, opt, bool, whether the device supports not wearing mask detection event-->true
</isSupportNoMaskDetection>
<isSupportTMPA>
  <!--ro, opt, bool, whether the device supports the warning linkage of face temperature screening event-->true
</isSupportTMPA>
<RuleScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by rule-->
  <isSupportCityManagement>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by rule for the city management event-->true
  </isSupportCityManagement>
</RuleScheduleCap>
<RegionScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by area-->
  <isSupportPersonDensityDetection>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the people density detection,
desc:related URLs: /ISAPI/Event/schedules/personDensityDetection/personDensityDetection-<channelID>/regions/<ID>,
/ISAPI/Event/schedules/personDensityDetection/personDensityDetection-<channelID>/regions-->true
  </isSupportPersonDensityDetection>
  <isSupportPersonQueueDetection>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the queue detection (area popularity analytics),
desc:related URLs: /ISAPI/Event/schedules/personQueueDetection/personQueueDetection-<channelID>/regions/<ID>,
/ISAPI/Event/schedules/personQueueDetection/personQueueDetection-<channelID>/regions-->--&true
  </isSupportPersonQueueDetection>
  <isSupportObjectsThrownDetection>
    <!--ro, opt, bool, whether the detection of objects falling from heights supports arming by region time,
desc:related URLs: /ISAPI/Event/schedules/objectsThrownDetection/objectsThrownDetection-<channelID>/regions/<regionID>,
/ISAPI/Event/schedules/objectsThrownDetection/objectsThrownDetection-<channelID>/regions-->true
  </isSupportObjectsThrownDetection>
</RegionScheduleCap>
<isSupportAbnormalReboot>
  <!--ro, opt, bool, whether the device supports abnormal reboot event-->true
</isSupportAbnormalReboot>
<isSupportChannelOccupy>
  <!--ro, opt, bool, whether the device supports fire escape occupied by vehicle event: /ISAPI/Event/triggers/channelOccupy-->true
</isSupportChannelOccupy>

```



```

</isSupportChannelOccupancy>
<isSupportIndoorPasswayBlock>
  <!--ro, opt, bool, whether the device supports indoor fire escape occupied event: /ISAPI/Event/triggers/indoorPasswayBlock-->true
</isSupportIndoorPasswayBlock>
<isSupportFireSmartFireDetect>
  <!--ro, opt, bool, whether the device supports fire detection event: /ISAPI/Event/triggers/fireSmartFireDetect-->true
</isSupportFireSmartFireDetect>
<isSupportOffDuty>
  <!--ro, opt, bool, whether the device supports absence on duty event in the fire control room: /ISAPI/Event/triggers/offDuty-->true
</isSupportOffDuty>
<isSupportNoCertificate>
  <!--ro, opt, bool, whether the device supports insufficient person with certificate in the fire control room event:
/ISAPI/Event/triggers/fireSmartFireDetect-->true
</isSupportNoCertificate>
<isSupportSmokeAlarm>
  <!--ro, opt, bool, whether the device supports smoke alarm event: /ISAPI/Event/triggers/smokeAlarm-->true
</isSupportSmokeAlarm>
<isSupportBatteryCarDisobey>
  <!--ro, opt, bool, whether the device supports electric moped parking violation event: /ISAPI/Event/triggers/batteryCarDisobey-->true
</isSupportBatteryCarDisobey>
<isSupportNoFireExtinguisherRecog>
  <!--ro, opt, bool, whether the device supports fire extinguisher missing event: /ISAPI/Event/triggers/noFireExtinguisherRecog-->true
</isSupportNoFireExtinguisherRecog>
<isSupportSmokingDetectAlarm>
  <!--ro, opt, bool, whether the device supports smoke alarm event: /ISAPI/Event/triggers/smokeDetectAlarm-->true
</isSupportSmokingDetectAlarm>
<isSupportThermalCalibrationFileException>
  <!--ro, opt, bool, whether the device supports the temperature calibration file exception event-->true
</isSupportThermalCalibrationFileException>
<isSupportTemperatureIntervalMeasurement>
  <!--ro, opt, bool, whether the device supports the temperature range detection event-->true
</isSupportTemperatureIntervalMeasurement>
<isSupportObjectsThrownDetection>
  <!--ro, opt, bool, whether the device supports the objects falling from heights detection event, desc:related URL:
/ISAPI/Intelligent/channels/<channelID>/objectsThrownDetection?format=json, and related event: objectsThrownDetection. This not will not be returned if not
supported.-->true
</isSupportObjectsThrownDetection>
<isSupportThermalVehicleDetection>
  <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
</isSupportThermalVehicleDetection>
<isSupportSlagTruckWashingDetection>
  <!--ro, opt, bool, whether the device supports dump truck cleaning detection event-->true
</isSupportSlagTruckWashingDetection>
<isSupportLensTemperatureException>
  <!--ro, opt, bool, whether the device supports uploading lens abnormal temperature event-->true
</isSupportLensTemperatureException>
<isSupportAlertQuality>
  <!--ro, opt, bool, whether the device supports the event of getting the device's aiming quality information,
desc:/ISAPI/Event/notification/alertQuality?format=json-->true
</isSupportAlertQuality>
<isSupportPTZFXFinish>
  <!--ro, opt, bool, whether the device supports the event of railway robot and PTZ going to the preset, desc:PTZFXFinish-->true
</isSupportPTZFXFinish>
<isSupportVideoAccessAbnormal>
  <!--ro, opt, bool, whether the device supports video access exception event, desc:/ISAPI/Event/triggers/VideoAccessAbnormal-->true
</isSupportVideoAccessAbnormal>
<isSupportSubSystemAbnormal>
  <!--ro, opt, bool, whether the device supports partition exception event, desc:/ISAPI/Event/triggers/SubSystemAbnormal-->true
</isSupportSubSystemAbnormal>
<isSupportPTEventCfg>
  <!--ro, opt, bool, whether the device supports transparent transmission event configuration event, desc:/ISAPI/Event/PTEventCfg/capabilities?
format=json-->true
</isSupportPTEventCfg>
<isSupportFanFailure>
  <!--ro, opt, bool, whether the device supports fan exception linkage configuration event, desc:/ISAPI/Event/triggers/fanFailure-->true
</isSupportFanFailure>
<isSupportNormalTemperatureAlarm>
  <!--ro, opt, bool, whether the device supports normal temperature alarm-->true
</isSupportNormalTemperatureAlarm>
<isSupportLowTemperatureAlarm>
  <!--ro, opt, bool, whether the device supports low temperature alarm-->true
</isSupportLowTemperatureAlarm>
<isSupportLowCoolerLifeEvent>
  <!--ro, opt, bool, whether the device supports short life exception event of refrigerating machine-->true
</isSupportLowCoolerLifeEvent>
<isSupportSSDTemperatureAbnormal>
  <!--ro, opt, bool, whether the device supports SSD temperature exception detection event, desc:SSDTemperatureAbnormal-->true
</isSupportSSDTemperatureAbnormal>
<isSupportSSDLifeTimeAbnormal>
  <!--ro, opt, bool, whether the device supports SSD life detection event, desc:SSDLifeTimeAbnormal-->true
</isSupportSSDLifeTimeAbnormal>
<isSupportObstacleAvoidanceAlarm>
  <!--ro, opt, bool, whether the device supports obstacle avoidance alarm, desc:SSDLifeTimeAbnormal-->true
</isSupportObstacleAvoidanceAlarm>
<isSupportStaticEventParams>
  <!--ro, opt, bool, whether the device supports capability of static event configuration on channel, desc:/ISAPI/Event/staticEventParams/capabilities?
format=json-->true
</isSupportStaticEventParams>
<isSupportPTZRotationAbnormal>
  <!--ro, opt, bool-->true
</isSupportPTZRotationAbnormal>
</EventCap>

```

9.1.5 Exception Detection

9.1.5.1 Set the linkage parameters of illegal login

Request URL

PUT /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

hieu.tu@tacorp.vn
CNG TY C PHN U T CNG NGH THIN N

```

<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->illaccess
  </id>
  <eventType>
    <!--req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
        <ptzAction>
        <cellphoneNumber>
          <!--opt, string, range:[0,11]-->test
        </cellphoneNumber>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>

```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

9.1.5.2 Get the linkage parameters of illegal login

Request URL

GET /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->illaccess
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <wLSensorID>
    <!--ro, opt, string-->1
  </wLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>

```

9.1.6 Input and Output Management

9.1.6.1 Get the alarm input and output information

Request URL

GET /ISAPI/System/IO

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IOPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, see details in XML_IOOutputPortList, attr:version{req, string, protocolVersion}-->
  <IOInputPortList>
    <!--ro, opt, object-->
    <IOInputPort>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <triggering>
        <!--ro, req, enum, condition for triggering alarm output, subType:string, desc:"high"-high Level, "Low"-Low Level. Both the high Level and the Low Level can trigger continuously-->high
      </triggering>
      <name>
        <!--ro, opt, string-->name
      </name>
      <IOUseType>
        <!--ro, opt, enum, subType:string-->disable
      </IOUseType>
      <inputType>
        <!--ro, opt, enum, input manner, subType:string-->switch
      </inputType>
      <CombinationAlarm>
        <!--ro, opt, object-->
        <channel>
          <!--ro, req, int, channel No.-->1
        </channel>
        <EventTypeList>
          <!--ro, req, array, event type List, subType:object-->
          <eventType>
            <!--ro, opt, enum, event type, subType:string-->VMD
          </eventType>
        </EventTypeList>
      </CombinationAlarm>
    </IOInputPort>
  </IOInputPortList>
  <IOOutputPortList>
    <!--ro, opt, object-->
    <IOOutputPort>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <IODescriptor>
        <!--ro, opt, object-->
        <userName>
          <!--ro, req, string, user name-->name
        </userName>
        <addressingFormatType>
          <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
        </addressingFormatType>
        <hostName>
          <!--ro, opt, string, host name-->hostName
        </hostName>
        <ipAddress>
          <!--ro, opt, string, IPv4 address-->172.6.64.7
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, IPv6 address-->1080:0:0:0:8:800:200C:417A
        </ipv6Address>
        <managePortNo>
          <!--ro, req, int-->1
        </managePortNo>
        <innerIOPortID>
          <!--ro, req, int-->1
        </innerIOPortID>
      </IODescriptor>
      <PowerOnState>
        <!--ro, req, object-->
        <defaultState>
          <!--ro, req, enum, subType:string, desc:"high"-high Level, "Low"-Low Level. Both the high Level and the Low Level can trigger continuously-->high
        </defaultState>
        <outputState>
          <!--ro, req, enum, subType:string-->high
        </outputState>
      </PowerOnState>
    </IOOutputPort>
  </IOOutputPortList>
</IOPortList>
```

```

    <pulseDuration>
      <!--ro, opt, int-->1000
    </pulseDuration>
  </PowerOnState>
  <name>
    <!--ro, opt, string-->test
  </name>
  <IOUseType>
    <!--ro, opt, enum, subType:string-->disable
  </IOUseType>
  <normalStatus>
    <!--ro, opt, enum, subType:string-->open
  </normalStatus>
  <enabled>
    <!--ro, opt, bool, whether to enable the function-->true
  </enabled>
  <IOType>
    <!--ro, opt, enum, subType:string-->local
  </IOType>
</IOOutputPort>
</IOOutputPortList>
</IOPortList>

```

9.1.6.2 Get device input and output capability

Request URL

GET /ISAPI/System/IO/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<IOCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device input and output capability, attr:version(req, string, protocolVersion)-->
  <IOInputPortNums>
    <!--ro, opt, int, number of device local inputs-->0
  </IOInputPortNums>
  <IOOutputPortNums>
    <!--ro, opt, int, number of device local outputs-->0
  </IOOutputPortNums>
  <isSupportStrobeLamp>
    <!--ro, opt, bool, whether the device supports configuring the output of the gate's light, desc:whether the device supports configuring the output of
the gate's light-->true
  </isSupportStrobeLamp>
  <SoftIOInputPortNums>
    <!--ro, opt, int, number of device soft inputs, desc:number of device soft inputs-->0
  </SoftIOInputPortNums>
  <isSupportIOOutputAdvanceParameter>
    <!--ro, opt, bool, whether supports configuration of alarm input advanced parameters, desc:related URI:
/ISAPI/System/IO/outputs/advanceParameter/capabilities?format=json-->true
  </isSupportIOOutputAdvanceParameter>
  <isSupportSetAllIOOutput>
    <!--ro, opt, bool, whether the device supports configuring all outputs, desc:whether the device supports configuring all outputs-->true
  </isSupportSetAllIOOutput>
  <isSupportCombinationAlarm>
    <!--ro, opt, bool, whether supports combined alarm, desc:see details in related URI: /ISAPI/System/IO/inputs/capabilities/#CombinationAlarmCap-->true
  </isSupportCombinationAlarm>
  <enabledIOOutputPortNums>
    <!--ro, opt, int, number of enabled alarm output ports-->0
  </enabledIOOutputPortNums>
  <isSupportAlarmKeyParam>
    <!--ro, opt, bool, whether supports configuring the physical keys of one-push alarm, desc:related URI:
/ISAPI/System/IO/inputs/<IOInputID>/AlarmKeyParam/capabilities?format=json-->true
  </isSupportAlarmKeyParam>
</IOCap>

```

9.1.6.3 Get alarm input and output status

Request URL

GET /ISAPI/System/IO/status

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IOPortStatusList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <IOPortStatus>
    <!--ro, opt, object-->
    <ioPortID>
      <!--ro, req, int, I/O port No.-->0
    </ioPortID>
    <ioPortType>
      <!--ro, req, enum, I/O type, subType:string, desc:input, output-->input
    </ioPortType>
    <ioState>
      <!--ro, req, enum, I/O status, subType:string, desc:active, inactive-->active
    </ioState>
  </IOPortStatus>
</IOPortStatusList>
```

9.1.7 Log Management

9.1.7.1 Search for log information

Request URL

POST /ISAPI/ContentMgmt/logSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, Log search, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, range:[1,32], desc:UUID/GUID for record search. It is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device for faster search next time.-->test
  </searchID>
  <trackIDList>
    <!--opt, array, channel list, subType:object-->
    <trackID>
      <!--req, int, channel No., desc:channel No.-->1
    </trackID>
  </trackIDList>
  <metaId>
    <!--req, enum, Log type, subType:string, desc:{Log.std-cgi.com/<majorType>/<minorType>}, <majorType>-major type of Log, <minorType>-minor type of Log, see remarks for detailed Log type definitions; "Log.std-cgi.com/Alarm/methaneConcentrationException", "Log.std-cgi.com/Alarm/methaneLightIntensityException", "Log.std-cgi.com/Alarm/fishingShipDetection"-->log.std-cgi.com/Alarm/methaneConcentrationException
  </metaId>
  <timeSpanList>
    <!--opt, array, List of time periods, subType:object, desc:List of time periods-->
    <timeSpan>
      <!--opt, object, time period, desc:time period-->
      <startTime>
        <!--req, datetime, start time-->1970-01-01T00:00:00Z
      </startTime>
      <endTime>
        <!--req, datetime, end time-->1970-01-01T01:00:00Z
      </endTime>
    </timeSpan>
  </timeSpanList>
  <searchResultPostion>
    <!--opt, int, the start position of the search result, desc:the start position of the search result in the result list-->100
  </searchResultPostion>
  <maxResults>
    <!--opt, int, the maximum number of records supported in this search, range:[0,100]-->0
  </maxResults>
  <onlySmart>
    <!--opt, bool, whether to search for Logs with smart information, desc:the value of this node is false by default-->>false
  </onlySmart>
  <logLevel>
    <!--opt, enum, subType:string-->emergency
  </logLevel>
  <alarmLevel>
    <!--opt, enum, subType:string-->high
  </alarmLevel>
</CMSearchDescription>
```

Response Message

<?xml version="1.0" encoding="UTF-8"?>

<CMSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">

<!--ro, req, object, log search result, attr:version{opt, string, protocolVersion}-->

<searchID>

<!--ro, req, string, search ID, range:[1,32], desc:UUID/GUID for record search. It is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device for faster search next time-->test

</searchID>

<responseStatus>

<!--ro, req, bool, search status-->true

</responseStatus>

<responseStatusStrg>

<!--ro, req, enum, search status string, subType:string, desc:"OK" (search completed), "NO MATCHES" (no matched data), "MORE" (more data waiting to be searched)-->OK

</responseStatusStrg>

<totalMatches>

<!--ro, opt, int, total number of matched results-->12

</totalMatches>

<numOfMatches>

<!--ro, opt, int, number of results returned this time, range:[0,100]-->24

</numOfMatches>

<matchList>

<!--ro, opt, array, List of matched data records, subType:object-->

<searchMatchItem>

<!--ro, opt, object, single matched record-->

<logDescriptor>

<!--ro, opt, object, log description-->

<metaId>

<!--ro, req, string, Log type for search, desc:Log.std-cgi.com/<majorType>/<minorType>, "<majorType>" (major type of Log), "<minorType>" (minor type of Log), see remarks for detailed Log type definitions-->log.std-cgi.com/Alarm/24HZoneAlarm

</metaId>

<startDateTime>

<!--ro, req, datetime, Log time-->1970-01-01T00:30:00Z

</startDateTime>

<localID>

<!--ro, opt, string, channel No.-->1

</localID>

<paraType>

<!--ro, opt, string, parameter type-->test

</paraType>

<userName>

<!--ro, opt, string, user name-->Admin

</userName>

<infoContent>

<!--ro, opt, string, Log information content-->test

</infoContent>

<logInfo>

<!--ro, opt, object, Log information-->

<OpenDoorRecord>

<!--ro, opt, object, video intercom unlocking record-->

<type>

<!--ro, req, enum, unlocking type, subType:string, desc:"password" (unlocking by password), "hijack" (unlocking by duress), "card" (unlocking by card), "resident" (unlocking by resident), "center" (unlocking by management center)-->password

</type>

</OpenDoorRecord>

<VisAlarmRecord>

<!--ro, opt, object, video intercom alarm record-->

<type>

<!--ro, req, enum, alarm type, subType:string, desc:"zone" (zone alarm), "dismantle" (tampering alarm), "hijack" (duress alarm), "passwordErr" (password error alarm), "doorNotOpen" (unlocking-door-failed alarm), "doorNotClose" (locking-door-failed alarm), "SOS" (SOS alarm), "callReq" (device call request alarm), "smartLockHijackFingerPrint" (smart Lock duress alarm (fingerprint)), "smartLockHijackPassword" (smart Lock duress alarm (password)), "smartLockBreaking" (forced-open door alarm), "smartLockBelocked" (Lock-up door alarm), "smartLockLowBattery" (smart-Lock-battery-Low alarm)-->zone

</type>

</VisAlarmRecord>

</logInfo>

<ipAddress>

<!--ro, opt, string, device IP address, range:[1,32]-->10.20.30.10

</ipAddress>

<object>

<!--ro, opt, enum, operation object, subType:string, desc:"network", "keypad", "remoteCtrl" (remote control), "card"-->network

</object>

<params>

<!--ro, opt, string, Log parameters, desc:parameters (zone No., etc.)-->test

</params>

<seq>

<!--ro, opt, string, device serial No., range:[1,32]-->test

</seq>

<additionInformation>

<!--ro, opt, string, additional information, range:[1,128], desc:additional information of Log-->test

</additionInformation>

<panelUser>

<!--ro, opt, string, user name of the operation panel, range:[0,16]-->admin

</panelUser>

<diskNumber>

<!--ro, opt, int, HDD No., range:[0,10000]-->0

</diskNumber>

<alarmInPort>

<!--ro, opt, int, alarm input port, range:[0,1000]-->0

</alarmInPort>

<alarmOutPort>

<!--ro, opt, int, alarm output port, range:[0,10000]-->0

</alarmOutPort>

```

<remoteHostIPAddress>
  <!--ro, opt, string, remote host IP address, range:[1,32]-->10.12.25.23
</remoteHostIPAddress>
<logLevel>
  <!--ro, opt, enum, subType:string-->emergency
</logLevel>
<alarmLevel>
  <!--ro, opt, enum, subType:string-->high
</alarmLevel>
<moduleName>
  <!--ro, opt, string, range:[1,32]-->test
</moduleName>
<logDescriptor>
</searchMatchItem>
</matchList>
</CMSearchResult>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	searchLogFailed	0x4000A063	--

9.1.8 Serial Port Management

9.1.8.1 Get the serial port capability of the device

Request URL

GET /ISAPI/System/Serial/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<SerialCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device, attr:version(req, string, protocolVersion)-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->0
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->0
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->0
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger,transparent">
    <!--ro, opt, string, range of RS-485 serial port No. supported by the device, attr:opt(req, string)-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, int, range of RS-232 serial port numbers supported by the device, attr:opt(req, string)-->1
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, int, range of RS-485 serial port numbers supported by the device, attr:opt(req, string)-->1
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>

```

9.1.8.2 Get the capability of a specific serial port

Request URL

GET /ISAPI/System/Serial/ports/<portID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <encryptKey min="0" max="10">
    <!--ro, opt, string, key, attr:min{req, int},max{req, int}, desc:key-->test
  </encryptKey>
  <serialNumber opt="1,2,3,4,5">
    <!--ro, opt, int, serial No., attr:opt{req, string}-->0
  </serialNumber>
  <serialPortType opt="RS485,RS422,RS232">
    <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
  </serialPortType>
  <serialAddress min="0" max="10">
    <!--ro, opt, int, serial port address, attr:min{req, int},max{req, int}, desc:serial port address-->1
  </serialAddress>
  <duplexMode opt="half,full">
    <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
  </duplexMode>
  <direction opt="monodirectional,bidirectional">
    <!--ro, opt, string, attr:opt{req, string}-->monodirectional
  </direction>
  <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
    <!--ro, opt, int, attr:opt{req, string}-->1200
  </baudRate>
  <dataBits opt="6,7,8">
    <!--ro, opt, int, attr:opt{req, string}-->6
  </dataBits>
  <parityType opt="none,even,odd,mark,space">
    <!--ro, opt, string, attr:opt{req, string}-->none
  </parityType>
  <stopBits opt="1,1.5,2">
    <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
  </stopBits>
  <workMode opt="console,transparent,audiomix,screenCtrl,ptzCtrl,keyboard,matrix,audioMixers">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->console
  </workMode>
  <flowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </flowCtrl>
  <deviceName min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </deviceName>
  <deviceProtocol min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </deviceProtocol>
  <mode opt="readerMode,clientMode,externMode,stairsControl,accessControlHost,disabled,custom,cardReceiver,QRCodeReader">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->readerMode
  </mode>
  <modeChangeReboot>
    <!--ro, opt, bool-->true
  </modeChangeReboot>
  <outputDataType opt="cardNo,employeeNo,auto">
    <!--ro, opt, string, output data type, attr:opt{req, string}, desc:data type output from the door station to the elevator controller: fFloorNumber (fFloor
    No.,default),cardNo (card No.)-->cardNo
  </outputDataType>
  <copyToAll opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </copyToAll>
  <isVariable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isVariable>
  <PTZInfo>
    <!--ro, opt, object-->
    <bindChannelNo min="0" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--ro, req, object-->
      <PTZProtocol>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protocolName>
```

```

    </protoDesc>
    <!--ro, req, string, protocol description-->test
  </protoDesc>
</PTZProtocol>
</PTZProtocolList>
</PTZInfo>
<ScreenCtrlInfo>
  <!--ro, opt, object, ScreenCtrlInfo, desc:this field is valid when the value of workMode is "screenCtrl"-->
  <ScreenCtrlProtoList>
    <!--ro, req, object, ScreenCtrlProtoList-->
    <ScreenCtrlProto>
      <!--ro, opt, object, ScreenCtrlProto-->
      <id>
        <!--ro, req, int, protocol index-->0
      </id>
      <protoDesc>
        <!--ro, req, string, protocol description-->test
      </protoDesc>
    </ScreenCtrlProto>
  </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
  <!--ro, opt, object, MatrixCtrlInfo-->
  <MatrixCtrlProtoList>
    <!--ro, req, object, MatrixCtrlProtoList-->
    <MatrixCtrlProto>
      <!--ro, opt, object, MatrixCtrlProto-->
      <id>
        <!--ro, req, int, protocol index-->0
      </id>
      <protoDesc>
        <!--ro, req, string, protocol description-->test
      </protoDesc>
    </MatrixCtrlProto>
  </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyboardCtrlInfo>
  <!--ro, opt, object, KeyboardCtrlInfo-->
  <KeyboardCtrlProtoList>
    <!--ro, req, object, KeyboardCtrlProtoList-->
    <KeyboardCtrlProto>
      <!--ro, opt, object, KeyboardCtrlProto-->
      <id>
        <!--ro, req, int, protocol index-->0
      </id>
      <protoDesc>
        <!--ro, req, string, protocol description-->test
      </protoDesc>
    </KeyboardCtrlProto>
  </KeyboardCtrlProtoList>
</KeyboardCtrlInfo>
<ControlAddress>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <address min="0" max="255">
    <!--ro, req, int, IP address, attr:min{req, int},max{req, int}-->0
  </address>
</ControlAddress>
<rs485WorkMode
opt="Led,sensor,CaptureTrigger,transparent,GPS,ModbusRTU,supplementLight,audibleVisualAlarm,columnSpeaker,commModeUsing433MHz,signalLampControl">
  <!--ro, opt, string, working mode of RS-485 serial port, dep:and,{$.SerialPortList.SerialPort.serialPortType,eq,RS485}, attr:opt{req, string},
  desc:working mode of RS-485 serial port, which is used for LED display or triggering transmission of captured pictures. This node is valid only when
  <serialPortType> is "RS485"-->sensor
</rs485WorkMode>
<SensorCtrlList>
  <!--ro, opt, object-->
  <Sensor>
    <!--ro, opt, object-->
    <id min="1" max="3">
      <!--ro, req, int, index, attr:min{req, int},max{req, int}-->1
    </id>
    <name min="1" max="32">
      <!--ro, req, string, attr:min{req, int},max{req, int}-->visibilitySensor
    </name>
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}, desc:whether to enable the serial port: "true,false"-->false
    </enabled>
  </Sensor>
</SensorCtrlList>
<stairsControl>
  <!--ro, opt, object, elevator control parameters-->
  <outputDataType opt="floorNumber,cardNo">
    <!--ro, opt, enum, data type output from the door station to the elevator controller, subType:string, attr:opt{req, string}, desc:floorNumber (floor
  No.,default),cardNo (card No.)-->floorNumber
  </outputDataType>
</stairsControl>
<comMode opt="active,passive">
  <!--ro, opt, enum, communication mode, subType:string, attr:opt{req, string}, desc:communication mode-->active
</comMode>
<ModbusRTU>
  <!--ro, opt, object-->
  <ModbusRTUBaudRate opt="1200,2400,4800,9600,19200" def="9600">

```

```

    <!--ro, opt, int, attr:opt{req, string},def{req, string}-->9600
  </ModbusRTUBaudRate>
  <ModbusRTUDataBits opt="5,6,7,8" def="8">
    <!--ro, opt, int, attr:opt{req, string},def{req, string}-->6
  </ModbusRTUDataBits>
  <ModbusRTUParityType opt="none,even,odd" def="none">
    <!--ro, opt, string, attr:opt{req, string},def{req, string}-->none
  </ModbusRTUParityType>
  <ModbusRTUStopBits opt="1,2" def="1">
    <!--ro, opt, string, stop bit, attr:opt{req, string},def{req, string}-->1
  </ModbusRTUStopBits>
  <ModbusRTUFlowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </ModbusRTUFlowCtrl>
  <decoderAddress min="0" max="255" def="0">
    <!--ro, opt, int, decoder IP address, attr:min{req, int},max{req, int},def{req, int}-->1
  </decoderAddress>
  <online opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </online>
</ModbusRTU>
<accessName min="1" max="32">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</accessName>
<serialPortUse opt="acquAndCtrl,swipe,strongElectric">
  <!--ro, opt, string, attr:opt{req, string}-->acquisition
</serialPortUse>
<acquisitionProtocol opt="modbus,custom">
  <!--ro, opt, string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}, attr:opt{req, string}-->modbus
</acquisitionProtocol>
<serialProtocol opt="FECbus,GB26875">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB26875
</serialProtocol>
<SupplementLightList size="5">
  <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}, attr:size{req, int}-->
  <SupplementLight>
    <!--ro, opt, object-->
    <SupplementLightID min="1" max="5">
      <!--ro, req, int, range:[1,5], attr:min{req, int},max{req, int}-->1
    </SupplementLightID>
  </SupplementLight>
</SupplementLightList>
<permissionController opt="0,1">
  <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
</permissionController>
<deviceModel min="0" max="32">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</deviceModel>
<deviceSoftwareVersion min="0" max="32">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</deviceSoftwareVersion>
<audibleVisualAlarmType opt="CIRCLE,AFE80,AFE82,XY">
  <!--ro, opt, enum, CIRCLE; AFE80; AFE82; XY, subType:string, attr:opt{req, string}, desc:CIRCLE; AFE80; AFE82; XY-->CIRCLE
</audibleVisualAlarmType>
<cardReaderType opt="K1108AM">
  <!--ro, opt, string, attr:opt{req, string}-->test
</cardReaderType>
<strongElectricType opt="19M04-RO/T,19M08-RO/T">
  <!--ro, opt, string, attr:opt{req, string}-->test
</strongElectricType>
<signalLampControl>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
  <protocolType opt="GB920,HisenSe,LESER,JERE">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB920
  </protocolType>
</signalLampControl>
</SerialPort>

```

9.1.8.3 Get the configuration parameters of a specified transparent transmission channel

Request URL

GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TransparentChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, transparent transmission channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the channel-->true
  </enabled>
  <serialPortID>
    <!--ro, req, string, serial port No.-->1
  </serialPortID>
</TransparentChannel>
```

9.1.8.4 Set the parameters of a specific serial port supported by the device

Request URL

PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TransparentChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, serial port ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the serial port-->true
  </enabled>
  <serialPortID>
    <!--req, string-->1
  </serialPortID>
</TransparentChannel>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

9.1.8.5 Set the parameters of a specific serial port supported by the device

Request URL

PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	--
childDevID	string	--
deviceIndex	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, serial port ID-->0
  </id>
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <encryptKey>
    <!--opt, string-->test
  </encryptKey>
  <serialNumber>
    <!--opt, int, serial No.-->0
  </serialNumber>
  <serialPortType>
    <!--opt, enum, serial port type: "RS485","RS422","RS232", subType:string, desc:serial port type: "RS485","RS422","RS232"-->RS485
  </serialPortType>
  <serialAddress>
    <!--opt, int-->1
  </serialAddress>
  <duplexMode>
    <!--opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <direction>
    <!--opt, enum, "monodirectional,bidirectional", subType:string, desc:"monodirectional, bidirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--opt, int-->6
  </dataBits>
  <parityType>
    <!--opt, enum, parity type, subType:string, desc:"none, even, odd, mark, space"-->none
  </parityType>
  <stopBits>
    <!--opt, string, stop bit: "1,1.5,2"-->1
  </stopBits>
  <workMode>
    <!--opt, enum, work mode, subType:string, desc:working mode: "console","transparent","audiomixer","stairsControl"-elevator control,"cardReader"-card reader,"disabled","custom". This node is required only when <serialPortType> is set to "RS232"-->console
  </workMode>
  <flowCtrl>
    <!--opt, enum, "none,software,hardware", subType:string, desc:"none, software, hardware"-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--opt, enum, working mode of RS-485 serial port, which is used for LED display or triggering transmission of captured pictures: "Led, CaptureTrigger". This node is valid only when <serialPortType> is "RS485", subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
  <copyToAll>
    <!--opt, bool-->true
  </copyToAll>
  <deviceName>
    <!--ro, opt, string-->test
  </deviceName>
  <deviceProtocol>
    <!--ro, opt, int-->1
  </deviceProtocol>
  <mode>
    <!--opt, enum, work mode, subType:string, desc:deq,working mode: "readerMode,clientMode,externMode,accessControlHost,disabled",this node is valid only when <serialPortType> is "RS485"-->readerMode
  </mode>
  <outputDataType>
    <!--opt, enum, output data type, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}, desc:"cardNo,employeeNo", this node is valid when <mode>is "accessControlHost"-->cardNo
  </outputDataType>
  <isVariable>
    <!--opt, bool-->true
  </isVariable>
  <PTZInfo>
    <!--opt, object, PTZ information-->
  <bindChannelNo>
    <!--req, int-->0
  </bindChannelNo>
</SerialPort>
```

```

</bindChannelNo>
<PTZProcollList>
  <!--req, object-->
  <PTZProtocol>
    <!--opt, object-->
    <id>
      <!--req, int, serial port ID-->0
    </id>
    <protoDesc>
      <!--req, string-->test
    </protoDesc>
  </PTZProtocol>
</PTZProcollList>
</PTZInfo>
<ScreenCtrlInfo>
  <!--opt, object-->
  <ScreenCtrlProtoList>
    <!--req, object-->
    <ScreenCtrlProto>
      <!--opt, object-->
      <id>
        <!--req, int, serial port ID-->0
      </id>
      <protoDesc>
        <!--req, string-->test
      </protoDesc>
    </ScreenCtrlProto>
  </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
  <!--opt, object-->
  <MatrixCtrlProtoList>
    <!--req, object-->
    <MatrixCtrlProto>
      <!--opt, object-->
      <id>
        <!--req, int, serial port ID-->0
      </id>
      <protoDesc>
        <!--req, string-->test
      </protoDesc>
    </MatrixCtrlProto>
  </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyBoardCtrlInfo>
  <!--opt, object-->
  <KeyBoardCtrlProtoList>
    <!--req, object-->
    <KeyBoardCtrlProto>
      <!--opt, object-->
      <id>
        <!--req, int, serial port ID-->0
      </id>
      <protoDesc>
        <!--req, string-->test
      </protoDesc>
    </KeyBoardCtrlProto>
  </KeyBoardCtrlProtoList>
</KeyBoardCtrlInfo>
<ControlAddress>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <address>
    <!--req, int, address, range:[0,255]-->1
  </address>
</ControlAddress>
<SensorCtrlList>
  <!--opt, object-->
  <Sensor>
    <!--opt, object-->
    <id>
      <!--req, int, ID, range:[1,3]-->1
    </id>
    <name>
      <!--req, string, sensor name, range:[1,32]-->visibilitySensor
    </name>
    <enabled>
      <!--req, bool, whether to enable the function, desc:whether to enable the serial port: "true,false"-->>false
    </enabled>
  </Sensor>
</SensorCtrlList>
<stairsControl>
  <!--opt, object, elevator control parameters, this node is valid when the value of the node <mode> is stairsControl, dep:and,
  {$.SerialPort.mode:eq,stairsControl}-->
  <outputDataType>
    <!--opt, enum, output data type, subType:string, desc:"cardNo,employeeNo", this node is valid when <mode>is "accessControlHost"-->floorNumber
  </outputDataType>
</stairsControl>
<comMode>
  <!--opt, enum, subType:string-->active
</comMode>

```



```

<ModbusRTU>
  <!--opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <ModbusRTUBaudRate>
    <!--opt, enum, subType:string-->9600
  </ModbusRTUBaudRate>
  <ModbusRTUDataBits>
    <!--opt, enum, subType:int-->6
  </ModbusRTUDataBits>
  <ModbusRTUParityType>
    <!--opt, enum, subType:string-->none
  </ModbusRTUParityType>
  <ModbusRTUStopBits>
    <!--opt, enum, subType:string-->1
  </ModbusRTUStopBits>
  <ModbusRTUFlowCtrl>
    <!--opt, enum, subType:string-->none
  </ModbusRTUFlowCtrl>
  <decoderAddress>
    <!--opt, int, range:[0,255]-->0
  </decoderAddress>
</ModbusRTU>
<accessName>
  <!--opt, string, range:[1,32]-->test
</accessName>
<serialPortUse>
  <!--opt, enum, subType:string-->acquisition
</serialPortUse>
<acquisitionProtocol>
  <!--opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}-->modbus
</acquisitionProtocol>
<serialProtocol>
  <!--opt, enum, subType:string-->GB26875
</serialProtocol>
<SupplementLightList>
  <!--opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
  <SupplementLight>
    <!--opt, object-->
    <SupplementLightID>
      <!--req, int, range:[1,5]-->1
    </SupplementLightID>
  </SupplementLight>
</SupplementLightList>
<cardReaderType>
  <!--opt, enum, subType:string-->K1108AM
</cardReaderType>
</SerialPort>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	unableToSwitchTheSerialPortUsageWhenHotSparesIsEnabled	0x4000A613	--
4	Invalid Operation	unableToControlTheStandbyDeviceWhenHotSparesIsEnabled	0x4000A614	--
4	Invalid Operation	hotStandbyIsInProgressOperationsAreUnavailable	0x4000A76B	--

9.1.8.6 Get the parameters of a specific port supported by the device

Request URL

GET /ISAPI/System/Serial/ports/<portID>?permissionController= <indexID> &childDevID= <childDevID> &deviceIndex= <deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	0 (main permission controller), 1 (sub permission controller) When this parameter does not exist, it indicates getting the serial port information of the main permission controller (i.e., the device serial port information).
childDevID	string	--
deviceIndex	string	For C60S devices connected to multiple LED sending cards, the URL needs to include deviceIndex= to distinguish which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, serial port ID-->0
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the serial port-->true
  </enabled>
  <encryptKey>
    <!--ro, opt, string-->test
  </encryptKey>
  <serialNumber>
    <!--ro, opt, int, serial No.-->0
  </serialNumber>
  <serialPortType>
    <!--ro, opt, enum, serial port type, subType:string, desc:"RS485", "RS422", "RS232"-->RS485
  </serialPortType>
  <serialAddress>
    <!--ro, opt, int-->1
  </serialAddress>
  <duplexMode>
    <!--ro, opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <direction>
    <!--ro, opt, enum, "monodirectional,bidirectional", subType:string, desc:"monodirectional", "bidirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--ro, opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--ro, opt, int-->6
  </dataBits>
  <parityType>
    <!--ro, opt, enum, parity type, subType:string, desc:parity type: "none,even,odd,mark,space"-->none
  </parityType>
  <stopBits>
    <!--ro, opt, enum, stop bit, subType:string, desc:stop bit-->1
  </stopBits>
  <workMode>
    <!--ro, opt, enum, working mode, subType:string, desc:"console", "transparent", "audiomixer", "screenCtrl", "ptzCtrl", "keyboard", "matrix", "audioMixers"-->console
  </workMode>
  <flowCtrl>
    <!--ro, opt, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--ro, opt, enum, working mode of RS-485 serial port, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial port, which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
  <copyToAll>
    <!--ro, opt, bool-->true
  </copyToAll>
  <deviceName>
```

```

    <!--ro, opt, string-->test
  </deviceName>
  <deviceProtocol>
    <!--ro, opt, int-->1
  </deviceProtocol>
  <mode>
    <!--ro, opt, enum, working mode, subType:string, desc:deg,working mode: "readerMode,clientMode,externMode,accessControlHost,disabled",this node is valid
only when <serialPortType> is "RS485"-->readerMode
  </mode>
  <outputDataType>
    <!--ro, opt, enum, output data type, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}, desc:output data type: "cardNo,employeeNo",this
node is valid when <mode>is "accessControlHost"-->cardNo
  </outputDataType>
  <isVariable>
    <!--ro, opt, bool-->true
  </isVariable>
  <PTZInfo>
    <!--ro, opt, object, PTZ information-->
    <bindChannelNo>
      <!--ro, req, int-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--ro, req, object-->
      <PTZProtocol>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </PTZProtocol>
    </PTZProtocolList>
  </PTZInfo>
  <ScreenCtrlInfo>
    <!--ro, opt, object-->
    <ScreenCtrlProtoList>
      <!--ro, req, object-->
      <ScreenCtrlProto>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </ScreenCtrlProto>
    </ScreenCtrlProtoList>
  </ScreenCtrlInfo>
  <MatrixCtrlInfo>
    <!--ro, opt, object-->
    <MatrixCtrlProtoList>
      <!--ro, req, object-->
      <MatrixCtrlProto>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </MatrixCtrlProto>
    </MatrixCtrlProtoList>
  </MatrixCtrlInfo>
  <KeyboardCtrlInfo>
    <!--ro, opt, object-->
    <KeyboardCtrlProtoList>
      <!--ro, req, object-->
      <KeyboardCtrlProto>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </KeyboardCtrlProto>
    </KeyboardCtrlProtoList>
  </KeyboardCtrlInfo>
  <ControlAddress>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the serial port-->true
    </enabled>
    <address>
      <!--ro, req, int, address, range:[0,255]-->0
    </address>
  </ControlAddress>
  <SensorCtrlList>
    <!--ro, opt, object-->
    <Sensor>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, serial port ID, range:[1,10]-->1
      </id>

```

```

</id>
<name>
  <!--ro, req, enum, sensor name, subType:string, desc:sensor name-->visibilitySensor
</name>
<enabled>
  <!--ro, req, bool, whether to enable the function, desc:whether to enable the serial port: "true,false"-->>false
</enabled>
<senorOnlineStatus>
  <!--ro, opt, enum, subType:string-->online
</senorOnlineStatus>
</Sensor>
</SensorCtrlList>
<stairsControl>
  <!--ro, opt, object, elevator control parameters, dep:and,{$.SerialPort.mode,eq,stairsControl}-->
  <outputDataType>
    <!--ro, opt, enum, output data type, subType:string, desc:output data type: "cardNo,employeeNo",this node is valid when <mode>is "accessControlHost"-->
  </outputDataType>
</stairsControl>
<comMode>
  <!--ro, opt, enum, subType:string-->active
</comMode>
<ModbusRTU>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <ModbusRTUBaudRate>
    <!--ro, opt, enum, subType:string-->9600
  </ModbusRTUBaudRate>
  <ModbusRTUDataBits>
    <!--ro, opt, enum, subType:int-->6
  </ModbusRTUDataBits>
  <ModbusRTUParityType>
    <!--ro, opt, enum, subType:string-->none
  </ModbusRTUParityType>
  <ModbusRTUStopBits>
    <!--ro, opt, enum, subType:string-->1
  </ModbusRTUStopBits>
  <ModbusRTUFlowCtrl>
    <!--ro, opt, enum, subType:string-->none
  </ModbusRTUFlowCtrl>
  <decoderAddress>
    <!--ro, opt, int, range:[0,255]-->0
  </decoderAddress>
  <online>
    <!--ro, opt, bool-->true
  </online>
</ModbusRTU>
<accessName>
  <!--ro, opt, string, range:[1,32]-->test
</accessName>
<serialPortUse>
  <!--ro, opt, enum, subType:string-->acquAndCtrl
</serialPortUse>
<serialProtocol>
  <!--ro, opt, enum, subType:string-->GB26875
</serialProtocol>
<SupplementLightList>
  <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
  <SupplementLight>
    <!--ro, opt, object-->
    <SupplementLightID>
      <!--ro, req, int, range:[1,5]-->1
    </SupplementLightID>
  </SupplementLight>
</SupplementLightList>
<cardReaderType>
  <!--ro, opt, enum, subType:string-->K1108AM
</cardReaderType>
<strongElectricType>
  <!--ro, opt, enum, subType:string-->19M04-RO/T
</strongElectricType>
<signalLampControl>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
  <protocolType>
    <!--ro, opt, enum, subType:string-->GB920
  </protocolType>
</signalLampControl>
<passingMode>
  <!--ro, opt, enum, subType:string, dep:or,{$.SerialPort.mode,eq,fingerprintReader}-->entrance
</passingMode>
<modbusProtocol>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <checkBitOrder>
    <!--ro, opt, enum, subType:string-->bigEndian
  </checkBitOrder>
  <deviceMode>
    <!--ro, opt, enum, subType:string-->subordinate
  </deviceMode>
</modbusProtocol>
<serialPortName>
  <!--ro, opt, string, range:[1,32]-->test
</serialPortName>
<accessProtocolType>
  <!--ro, opt, enum, subType:string-->PELCO

```

```

</accessProtocolType>
<weightSenseInfo>
  <!--ro, opt, object-->
  <countTimes>
    <!--ro, opt, enum, subType:float-->0.01
  </countTimes>
  <countUnit>
    <!--ro, opt, enum, subType:string-->kg
  </countUnit>
</weightSenseInfo>
<PDUCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PDUCtrl}-->DS-65A2T0008
</PDUCtrlType>
<paperLessOfficeType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,paperLess}-->DS-65PL0015-T
</paperLessOfficeType>
<videoWallControllerType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,spellCtrl}-->matrixProtocol
</videoWallControllerType>
<audioCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,audioCtrl}-->DS-QAPDM16081C
</audioCtrlType>
<interactiveTabletType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,interactiveTablet}-->interactiveTabletProtocol
</interactiveTabletType>
<humanSensorType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,humanSensor}-->HM-TD1017-2/QW-HS121
</humanSensorType>
<transparentOverWebsocket>
  <!--ro, opt, bool-->true
</transparentOverWebsocket>
<PTZCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PTZCtrl}-->VISCA
</PTZCtrlType>
<airConditionerCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,airConditionerCtrl}-->JGL
</airConditionerCtrlType>
<acquisitionLinkageType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisitionLinkage}-->general
</acquisitionLinkageType>
<externalInterfaceBoard>
  <!--ro, opt, object, dep:or,{$.SerialPort.mode,eq,externalInterfaceBoard}-->
  <serialPortList>
    <!--ro, opt, array, subType:object-->
    <serialPort>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <serialPortType>
        <!--ro, opt, enum, subType:string-->RS485
      </serialPortType>
      <RS485serialProtocol>
        <!--ro, opt, enum, subType:string-->private485
      </RS485serialProtocol>
      <private485Mode>
        <!--ro, opt, enum, subType:string-->cardReceiver
      </private485Mode>
      <RS232Mode>
        <!--ro, opt, enum, subType:string-->readerMode
      </RS232Mode>
    </serialPort>
  </serialPortList>
</externalInterfaceBoard>
</SerialPort>

```

9.1.8.7 Get the list of serial ports supported by the device

Request URL

GET /ISAPI/System/Serial/ports?permissionController= <indexID> &deviceIndex= <deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	0 (main permission controller), 1 (sub permission controller) If this parameter does not exist, it indicates getting the serial port information of main permission controller, that is, the serial port information of the device.
deviceIndex	string	For C60S device connecting multiple LED sending cards, the URL needs to include the parameter deviceIndex= to specify which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SerialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <encryptKey>
      <!--ro, opt, string-->test
    </encryptKey>
    <serialNumber>
      <!--ro, opt, int-->0
    </serialNumber>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <serialAddress>
      <!--ro, opt, int-->1
    </serialAddress>
    <duplexMode>
      <!--ro, opt, enum, subType:string-->half
    </duplexMode>
    <direction>
      <!--ro, opt, enum, subType:string-->monodirectional
    </direction>
    <baudRate>
      <!--ro, opt, enum, subType:int-->2400
    </baudRate>
    <dataBits>
      <!--ro, opt, int-->6
    </dataBits>
    <parityType>
      <!--ro, opt, enum, subType:string-->none
    </parityType>
    <stopBits>
      <!--ro, opt, enum, subType:string-->1
    </stopBits>
    <workMode>
      <!--ro, opt, enum, subType:string-->console
    </workMode>
    <flowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </flowCtrl>
    <rs485WorkMode>
      <!--ro, opt, enum, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}-->sensor
    </rs485WorkMode>
    <copyToAll>
      <!--ro, opt, bool-->true
    </copyToAll>
    <deviceName>
      <!--ro, opt, string-->test
    </deviceName>
    <deviceProtocol>
      <!--ro, opt, int-->1
    </deviceProtocol>
    <mode>
      <!--ro, opt, enum, subType:string-->readerMode
    </mode>
    <outputDataType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}-->cardNo
    </outputDataType>
    <isVariable>
      <!--ro, opt, bool-->true
    </isVariable>
  </SerialPort>
</SerialPortList>
```

```

<!--ro, opt, bool-->true
</isVariable>
<PTZInfo>
  <!--ro, opt, object-->
  <bindChannelNo>
    <!--ro, req, int-->0
  </bindChannelNo>
  <PTZProtocolList>
    <!--ro, req, object-->
    <PTZProtocol>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </PTZProtocol>
  </PTZProtocolList>
</PTZInfo>
<ScreenCtrlInfo>
  <!--ro, opt, object-->
  <ScreenCtrlProtoList>
    <!--ro, req, object-->
    <ScreenCtrlProto>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </ScreenCtrlProto>
  </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
  <!--ro, opt, object-->
  <MatrixCtrlProtoList>
    <!--ro, req, object-->
    <MatrixCtrlProto>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </MatrixCtrlProto>
  </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyboardCtrlInfo>
  <!--ro, opt, object-->
  <KeyboardCtrlProtoList>
    <!--ro, req, object-->
    <KeyboardCtrlProto>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </KeyboardCtrlProto>
  </KeyboardCtrlProtoList>
</KeyboardCtrlInfo>
<ControlAddress>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <address>
    <!--ro, req, int, range:[0,255]-->1
  </address>
</ControlAddress>
<SensorCtrlList>
  <!--ro, opt, object-->
  <Sensor>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, range:[1,10]-->1
    </id>
    <name>
      <!--ro, req, enum, subType:string-->visibilitySensor
    </name>
    <enabled>
      <!--ro, req, bool-->false
    </enabled>
    <senorOnlineStatus>
      <!--ro, opt, enum, subType:string-->online
    </senorOnlineStatus>
  </Sensor>
</SensorCtrlList>
<stairsControl>
  <!--ro, opt, object, dep:and,{$.SerialPort.mode,eq,stairsControl}-->

```

```

<outputDataType>
  <!--ro, opt, enum, subType:string-->floorNumber
</outputDataType>
</stairsControl>
<comMode>
  <!--ro, opt, enum, subType:string-->active
</comMode>
<ModbusRTU>
  <!--ro, opt, object, dep:and,{$.SerialPortList.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <ModbusRTUBaudRate>
    <!--ro, opt, enum, subType:string-->9600
  </ModbusRTUBaudRate>
  <ModbusRTUDataBits>
    <!--ro, opt, enum, subType:int-->6
  </ModbusRTUDataBits>
  <ModbusRTUParityType>
    <!--ro, opt, enum, subType:string-->none
  </ModbusRTUParityType>
  <ModbusRTUStopBits>
    <!--ro, opt, enum, subType:string-->1
  </ModbusRTUStopBits>
  <ModbusRTUFlowCtrl>
    <!--ro, opt, enum, subType:string-->none
  </ModbusRTUFlowCtrl>
  <decoderAddress>
    <!--ro, opt, int, range:[0,255]-->0
  </decoderAddress>
  <online>
    <!--ro, opt, bool-->true
  </online>
</ModbusRTU>
<accessName>
  <!--ro, opt, string, range:[1,32]-->test
</accessName>
<serialPortUse>
  <!--ro, opt, enum, subType:string-->acqAndCtrl
</serialPortUse>
<serialProtocol>
  <!--ro, opt, enum, subType:string-->GB26875
</serialProtocol>
<accessProtocolType>
  <!--ro, opt, enum, subType:string-->PELCO
</accessProtocolType>
<SupplementLightList>
  <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
  <SupplementLight>
    <!--ro, opt, object-->
    <SupplementLightID>
      <!--ro, req, int, range:[1,5]-->1
    </SupplementLightID>
  </SupplementLight>
</SupplementLightList>
<cardReaderType>
  <!--ro, opt, enum, subType:string-->K1108AW
</cardReaderType>
<strongElectricType>
  <!--ro, opt, enum, subType:string-->19M04-R0/T
</strongElectricType>
<passingMode>
  <!--ro, opt, enum, subType:string, dep:or,{$.SerialPortList.SerialPort.mode,eq,fingerprintReader}-->entrance
</passingMode>
<modbusProtocol>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <checkBitOrder>
    <!--ro, opt, enum, subType:string-->bigEndian
  </checkBitOrder>
  <deviceMode>
    <!--ro, opt, enum, subType:string-->subordinate
  </deviceMode>
</modbusProtocol>
<serialPortName>
  <!--ro, opt, string, range:[1,32]-->test
</serialPortName>
<PDUCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PDUCtrl}-->DS-65A2T0008
</PDUCtrlType>
<paperLessOfficeType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,paperLess}-->DS-65PL0015-T
</paperLessOfficeType>
<videoWallControllerIType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,spellIngCtrl}-->matrixProtocol
</videoWallControllerIType>
<audioCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,audioCtrl}-->DS-QAPDM16081C
</audioCtrlType>
<interactiveTabletType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,interactiveTablet}-->interactiveTabletProtocol
</interactiveTabletType>
<humanSensorType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,humanSensor}-->HM-TD1017-2/QW-HS121
</humanSensorType>
<transparentOverWebsocket>
  <!--ro, opt, bool-->true
</transparentOverWebsocket>

```



```

</transparentUverwebsocket>
<PTZCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PTZCtrl}-->VISCA
</PTZCtrlType>
<airConditionerCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,airConditionerCtrl}-->GL
</airConditionerCtrlType>
<acquisitionLinkageType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisitionLinkage}-->general
</acquisitionLinkageType>
<externalInterfaceBoard>
  <!--ro, opt, object, dep:or,{$.SerialPort.mode,eq,externalInterfaceBoard}-->
<serialPortList>
  <!--ro, opt, array, subType:object-->
  <serialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <RS485serialProtocol>
      <!--ro, opt, enum, subType:string-->private485
    </RS485serialProtocol>
    <private485Mode>
      <!--ro, opt, enum, subType:string-->cardReceiver
    </private485Mode>
    <RS232Mode>
      <!--ro, opt, enum, subType:string-->readerMode
    </RS232Mode>
    </serialPort>
  </serialPortList>
</externalInterfaceBoard>
</SerialPort>
</SerialPortList>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	dataTransmissionFunctionOfTheSerialportEnabled	0x4000A4A2	--

9.1.9 System Maintenance

9.1.9.1 Get the storage capability of the device

Request URL

GET /ISAPI/ContentMgmt/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
  <isSupportZeroChan>
    <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
  </isSupportZeroChan>
  <inputProxyNums>
    <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->0
  </inputProxyNums>
  <eSATANums>
    <!--ro, opt, int, supported number of eSATA interfaces-->0
  </eSATANums>
  <miniSASNums>
    <!--ro, opt, int, supported number of miniSAS interfaces-->0
  </miniSASNums>
  <nasNums>
    <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->0
  </nasNums>
  <ipSanNums>
    <!--ro, opt, int, supported number of IPSANS-->0
  </ipSanNums>
  <isSupportRaid>
    <!--ro, opt, bool, whether the device supports RAID-->true
  </isSupportRaid>
  <isSupportExtHdCfg>
    <!--ro, opt, bool, whether the device supports advanced HDD management-->true
  </isSupportExtHdCfg>

```

```

<isSupportTransCode>
  <!--ro, opt, bool, whether the device supports Live view transcoding-->true
</isSupportTransCode>
<isSupportIpcImport>
  <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/IPCConfig-->true
</isSupportIpcImport>
<NasMountType>
  <!--ro, opt, bool, NAS type-->true
<isNFSSupportAuthentication>
  <!--ro, opt, bool, whether the device supports NFS authentication-->true
</isNFSSupportAuthentication>
<isCIFSsupportAuthentication>
  <!--ro, opt, bool, whether the device supports CIFS authentication-->true
</isCIFSsupportAuthentication>
</NasMountType>
<isSupportIpcStreamType>
  <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
</isSupportIpcStreamType>
<isSupportIOInputProxy>
  <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
</isSupportIOInputProxy>
<isSupportIOOutputProxy>
  <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
</isSupportIOOutputProxy>
<isSupportPTZrs485Proxy>
  <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
</isSupportPTZrs485Proxy>
<isSupportSrcIDSearch>
  <!--ro, opt, bool, whether the device supports stream ID search-->true
</isSupportSrcIDSearch>
<isSupportReversePlayback>
  <!--ro, opt, bool, whether the device supports reverse playback-->true
</isSupportReversePlayback>
<isSupportSMARTTest>
  <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
</isSupportSMARTTest>
<isSupportDownloadByTime>
  <!--ro, opt, bool, whether the device supports downloading by time, desc:related URI: /ISAPI/ContentMgmt/download/capabilities-->true
</isSupportDownloadByTime>
<pictureSearchType>

```

opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual,manualSnapshot,playSnapshot,safetyHelmet, personQueueCounting,personQueueTime,violentMotion,LeavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,roadblock, abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,getUp,toiletTarry, playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePhone,fatigueDriving, notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringWarning, PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad, illegalHeap,illegalParkingofNon-MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel, ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,temperatureIntervalMeasurement,AIOP,tail,abnormalFace,normalFace,multiFace,callFace,sunglasses Face,humanEnter,operateOverTime,stickUp,installScanner,objectPlacedOrTaken,fakeCard,securityChecking,pothole,crack,roadMaint,imageCapture,methaneConcentrationException,methaneLightIntensityException,bannderDetection,variableDataEvent,shipChannelAbnormal,radarVideoDetection,smokeAndFireDetection,smokeDetection,sensorAlarm,alarmResult,AIOP_Video,AIOP_Polling_Snap,swipePatrol,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMotionAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapture,UVPhotonDetection,DevPowerStatusReport,HFPD,audioAnalysis,IndoorPasswayRecovery,BatteryCarRecovery,detectorTemperatureRecovery,RadarFallenObjectsOnTracks,VegetationSpectrumDetection,personDensityDetection,ChannelClearEvent,channelOccupancy,ChannelPassingEvent,FallenObjectsOnTracksEvent,audioexception,CarWindow,shipsDetection,InsulatorAbnormalAlarm,PackingSpaceRecognition,smokeAlarm,SmokeAlarmRecovery,fireSmartFireDetect,FirePointAlarmRecovery,offDuty,backDuty,sleepDuty,batteryCarDisobey,SmokingDetectionRecoverEvent,noFireExtinguisherRecog,FireExtinguisherRecoveryEvent,TelephoningBehaviorAlarmEvent,TelephoningBehaviorRecoveryEvent,GasTankMonitoringEvent,TemperatureMeasurementPreviewRecovery,TMPA,TMA, TemperatureMeasurementPreviewAlarmRecovery, TemperatureMeasurementAlarmRecovery,DetectorShelterAlarmRecovery, TemperatureUp, TemperatureDown, TemperatureEvent, TemperatureUpRecover, TemperatureDownRecover, TemperatureEventRecover, ElectricFenceAlarmEvent,cityManagement">

```

  <!--ro, opt, enum, picture search type, subType:string, attr:opt(req, string), desc:"AllEvent", "CMR", "MOTION", "ALARM", "EDR", "ALARMANDMOTION",
"pir:PIR", "wlsensor", "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection", "regionEntrance", "regionExiting",
"loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual", "manualSnapshot", "playSnapshot",
"safetyHelmet", "personQueueCounting", "personQueueTime", "violentMotion", "LeavePosition", "advReachHeight", "peopleNumChange", "mixedTargetDetection",
"illegalParking", "pedestrian", "construction", "roadblock", "evidence", "crossLane", "vehicleExist", "Lanechange", "wrongdirection", "congestion",
"turnround", "parallelParking", "abandonedObject", "trafficAccident", "fogDetection", "smokeDetection", "personDensity", "failDown", "accessController",
"videoInterCom", "GJD", "Luminite", "OPTEX", "securityControlPanel", "getUp", "toiletTarry", "playCellphone", "vehicleMonitor", "running", "retention",
"spacingChange", "yardTarry", "keyPersonGetUp", "sitQuietly", "standup", "plateRecognition", "smoking", "onThePhone", "fatigueDriving",
"notLookingStraightAhead", "LeaveThePost", "seatbelt", "sunglasses", "identityAbnormal", "ForwardCollisionWarning", "LaneDepartureWarning",
"BlindSpotDetection", "HeadwayMonitoringWarning", "PedestrianCollisionWarning", "SuddenSpeedUp", "SuddenSpeedDown", "SuddenLeftTurn", "SuddenRightTurn",
"Turnover", "NotYieldToPedestrian;allPic", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin", "smokeDetectAlarm", "temperatureIntervalMeasurement",
"AIOP", "tail", "abnormalFace", "normalFace;callFace", "sunglassesFace", "humanEnter", "operateOverTime", "stickup", "installScanner",
"objectPlacedOrTaken", "fakeCard", "securityChecking", "pothole", "crack", "roadMaint", "imageCapture", "ReID", "thermalVehicleDetection",
"methaneConcentrationException", "methaneLightIntensityException", "bannderDetection", "waterQualityDetection", "waterLevelDetection", "parallelParking",
"methaneConcentrationException", "variableDataEvent", "VOCsGasDetection", "shipChannelAbnormal", "radarVideoDetection", "AIDfogDetection:AID",
"smokeAndFireDetection", "smokeDetection", "sensorAlarm", "GasTankDetection", "alarmResult", "AIOP_Video", "AIOP_Polling_Snap", "swipePatrol",
"BatteryCarInfoEvent", "indoorPasswayBlock", "detectorTemperatureAlarm", "fireNoRegulation", "detectorMotionAlarm", "detectorShelterAlarm",
"detectorRunningStatus", "faceCapture", "UVPhotonDetection", "DevPowerStatusReport", "HFPD"-->AllEvent
</pictureSearchType>
<recordSearchType>

```

opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual,manualSnapshot,playSnapshot,safetyHelmet, failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofNon-MotorVehicle, illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin ,temperatureIntervalMeasurement,AIOP,securityChecking,scheduledRecordUpload,variableDataEvent,smokeAndFireDetection,smokeDetection,AIOP_Video,AIOP_Polling_Snap,AudioAnalysis,IndoorPasswayRecovery,BatteryCarRecovery,detectorTemperatureRecovery,RadarFallenObjectsOnTracks,VegetationSpectrumDetection,personDensityDetection,ChannelClearEvent,channelOccupancy,ChannelPassingEvent,FallenObjectsOnTracksEvent,audioexception,CarWindow,shipsDetection,InsulatorAbnormalAlarm,PackingSpaceRecognition,smokeAlarm,SmokeAlarmRecovery,fireSmartFireDetect,FirePointAlarmRecovery,offDuty,backDuty,sleepDuty,batteryCarDisobey,SmokingDetectionRecoverEvent,noFireExtinguisherRecog,FireExtinguisherRecoveryEvent,TelephoningBehaviorAlarmEvent,TelephoningBehaviorRecoveryEvent,GasTankMonitoringEvent,TemperatureMeasurementPreviewRecovery,TMPA,TMA, TemperatureMeasurementPreviewAlarmRecovery, TemperatureMeasurementAlarmRecovery,DetectorShelterAlarmRecovery, TemperatureUp, TemperatureDown, TemperatureEvent, TemperatureUpRecover, TemperatureDownRecover, TemperatureEventRecover, ElectricFenceAlarmEvent,cityManagement">

```
nap, batteryLowEvent, indoorPasswayBlock, detectorTemperatureAlarm, fireNoRegulation, detectorMotionAlarm, detectorShelterAlarm, detectorKunngStatus, faceCapture, UVPhotonDetection, DevPowerStatusReport, ChannelClearEvent, channelOccupy, ChannelPassingEvent, personQueueCounting, personQueueTime, leavePosition, localButtonRecord, callRecord">
```

```
<!--ro, opt, enum, video search type, subType:string, attr:opt(req, string), desc:"AllEvent" (all events), "CMR" (capture by schedule), "MOTION" (motion detection), "ALARM", "EDR", "ALARMANDMOTION", "pir", "wlsensor", "callhelp", "facedetection", "FieldDetection", "scenexchangedetection", "LineDetection", "regionEntrance", "regionExiting", "loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual", "manualSnapshot", "playSnapshot", "safetyHelmet", "personDensityDetection", "smokeDetection", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin", "temperatureIntervalMeasurement", "AIOP", "securityChecking", "scheduledRecordUpload", "thermalVehicleDetection", "waterQualityDetection", "waterLevelDetection", "methaneConcentrationException", "variableDataEvent"-->AllEvent
```

```
</recordSearchType>
```

```
<isSupportActivateIpc>
```

```
<!--ro, opt, bool, whether the device supports activating the network camera-->true
```

```
</isSupportActivateIpc>
```

```
<isSupportCheckIpcSecurity>
```

```
<!--ro, opt, object, whether the device supports IPC security verification-->
```

```
<isSupportCheckPassword>
```

```
<!--ro, opt, bool, whether the device supports password detection-->true
```

```
</isSupportCheckPassword>
```

```
<isSupportCheckIpcSecurity>
```

```
<isSupportMainAndSubRecord>
```

```
<!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
```

```
</isSupportMainAndSubRecord>
```

```
<isSupportSyncIPCPassword>
```

```
<!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
```

```
</isSupportSyncIPCPassword>
```

```
<isSupportTransferIPC>
```

```
<!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
```

```
</isSupportTransferIPC>
```

```
<isSupportSmartSearch>
```

```
<!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
```

```
</isSupportSmartSearch>
```

```
<isSupportPOS>
```

```
<!--ro, opt, bool, whether the device supports POS-->true
```

```
</isSupportPOS>
```

```
<isSupportLinkNum>
```

```
<!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/LinkNum-->true
```

```
</isSupportLinkNum>
```

```
<isSupportPassBackBasicCfg>
```

```
<!--ro, opt, bool, whether the device supports configuring basic parameters of the CVR copy-back function, desc:related URI:
```

```
/ISAPI/ContentMgmt/record/passback/basicCfg/capabilities-->true
```

```
</isSupportPassBackBasicCfg>
```

```
<PassBackTaskConfig>
```

```
<!--ro, opt, object, capability of managing tasks of the CVR copy-back function-->
```

```
<isSupportAddTask>
```

```
<!--ro, opt, bool, whether the device supports adding a task, desc:related URI: /ISAPI/ContentMgmt/record/passback/addTask/capabilities-->true
```

```
</isSupportAddTask>
```

```
<isSupportSearchTask>
```

```
<!--ro, opt, bool, whether the device supports searching tasks, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchTask/capabilities-->true
```

```
</isSupportSearchTask>
```

```
<isSupportControlTask>
```

```
<!--ro, opt, bool, whether the device supports controlling the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/controlTask/capabilities-->
```

```
true
```

```
</isSupportControlTask>
```

```
<isSupportDeleteTask>
```

```
<!--ro, opt, bool, whether the device supports deleting the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/deleteTask/capabilities-->true
```

```
</isSupportDeleteTask>
```

```
</PassBackTaskConfig>
```

```
<PassBackPlanConfig>
```

```
<!--ro, opt, object, capability of configuring the CVR copy-back schedule-->
```

```
<isSupportAddPlan>
```

```
<!--ro, opt, bool, whether the device supports adding a schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/addPlan/capabilities-->true
```

```
</isSupportAddPlan>
```

```
<isSupportSearchPlan>
```

```
<!--ro, opt, bool, whether the device supports searching schedules, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchPlan/capabilities-->
```

```
true
```

```
</isSupportSearchPlan>
```

```
<isSupportDeletePlan>
```

```
<!--ro, opt, bool, whether the device supports deleting the schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/deletePlan/capabilities-->
```

```
true
```

```
</isSupportDeletePlan>
```

```
</PassBackPlanConfig>
```

```
<IRAIDCap>
```

```
<!--ro, opt, object, network array capability-->
```

```
<isSupportIRAIDRapidConfiguration>
```

```
<!--ro, opt, bool, whether the device supports quick configuration of iRAID, desc:related URI:
```

```
/ISAPI/ContentMgmt/Storage/IRAID/RapidConfiguration/capabilities-->true
```

```
</isSupportIRAIDRapidConfiguration>
```

```
</IRAIDCap>
```

```
<isSupportStorageExtraInfo>
```

```
<!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
```

```
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
```

```
</isSupportStorageExtraInfo>
```

```
<isSupportRecordStatus>
```

```
<!--ro, opt, bool, whether the device supports getting the recording status, desc:related URI: /ISAPI/ContentMgmt/channels/<ID>/record/capabilities-->
```

```
true
```

```
</isSupportRecordStatus>
```

```
<isSupportRacmChannelsCap opt="true,false">
```

```
<!--ro, opt, bool, whether the device supports the Racm capability by channel, attr:opt(opt, string), desc:related URI:
```

```
/ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
```

```
</isSupportRacmChannelsCap>
```

```
<LockCap>
```

```
<!--ro, opt, object, capability of locking and unlocking video, desc:/ISAPI/ContentMgmt/record/control/Locks/capabilities-->
```

```
<LockByName>
```

```
<!--ro, opt, object, lock/unlock video by file name-->
```

```

<TrackId>
  <!--ro, req, string, channel stream ID, desc:channel No. * 100 + stream type (-1: main stream, -2: sub-stream)-->101
</TrackId>
<recordName min="1" max="100">
  <!--ro, req, string, video file name, range:[1,100], attr:min{req, int},max{req, int}-->test
</recordName>
<command opt="lock,unlock">
  <!--ro, req, enum, operation command, subType:string, attr:opt{req, string}, desc:"Lock", "unlock"-->lock
</command>
<duration min="1" max="10">
  <!--ro, opt, int, Lock time, unit:s, dep:and,{$.LockByName.command,eq,lock}, attr:min{req, int},max{req, int}, desc:0: Lock permanently-->1
</duration>
<recorderCode min="1" max="100">
  <!--ro, opt, string, video recorder No., attr:min{req, int},max{req, int}-->test
</recorderCode>
<policeCode min="1" max="100">
  <!--ro, opt, string, police No., attr:min{req, int},max{req, int}-->test
</policeCode>
</LockByName>
<LockByTime>
  <!--ro, opt, object, Lock/unlock video by time-->
  <TrackId>
    <!--ro, req, string, channel stream ID, desc:channel No. * 100 + stream type (-1: main stream, -2: sub-stream)-->101
  </TrackId>
  <recordType opt="ALL,MANUAL,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,COMMAND,SMART">
    <!--ro, req, enum, recording type, subType:string, attr:opt{req, string}, desc:"ALL" (all videos), "MANUAL" (manual recording), "CMR" (capture by
    schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR"(capture alarm pictures or motion detection pictures),
    "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "Command" (command-based recording), "SMART" (smart recording)-->ALL
  </recordType>
  <startDateTime>
    <!--ro, req, datetime, video segment start time-->1970-01-01T00:00:00+08:00
  </startDateTime>
  <endDateTime>
    <!--ro, req, datetime, video segment end time-->1970-01-01T00:00:00+08:00
  </endDateTime>
  <command opt="lock,unlock">
    <!--ro, req, enum, operation command, subType:string, attr:opt{req, string}, desc:"Lock", "unlock"-->lock
  </command>
  <duration min="1" max="10">
    <!--ro, opt, int, Lock time, unit:s, dep:and,{$.LockByName.command,eq,lock}, attr:min{req, int},max{req, int}, desc:0: Lock permanently-->1
  </duration>
</LockByTime>
</LockCap>
<isSupportFormatAll>
  <!--ro, opt, bool, whether the device supports formatting all HDDs, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/format-->true
</isSupportFormatAll>
<isSupportExtendCabinetCfg>
  <!--ro, opt, bool, whether it supports enclosure configuration, desc:related URI: /ISAPI/ContentMgmt/Storage/ExtendCabinet/capabilities-->true
</isSupportExtendCabinetCfg>
<diskGroupNums>
  <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->0
</diskGroupNums>
<ExternalDevice>
  <!--ro, opt, object, external device storage-->
  <USB>
    <!--ro, opt, object, USB information-->
    <debugLogOutput>
      <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
    </debugLogOutput>
    <isSupportCapturePackage>
      <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
      /ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
    </isSupportCapturePackage>
    <isSupportStreamStorage>
      <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
      /ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
    </isSupportStreamStorage>
  </USB>
</ExternalDevice>
<customProtocolNums>
  <!--ro, opt, int, number of custom protocols for accessing network cameras, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols-->0
</customProtocolNums>
<isSupportIPCTiming>
  <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:relate to enableTiming on URI:
  /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
</isSupportIPCTiming>
<isSupportLogConfig>
  <!--ro, opt, bool, whether the device supports log configuration, desc:related URI: /ISAPI/ContentMgmt/LogConfig/capabilities-->true
</isSupportLogConfig>
<SecurityLog>
  <!--ro, opt, object, security log-->
  <isSupportSecurityLog>
    <!--ro, opt, bool, whether the device supports the security log-->true
  </isSupportSecurityLog>
  <isSupportLogServer>
    <!--ro, opt, bool, whether it supports security log server configuration-->true
  </isSupportLogServer>
  <isSupportLogServerTest>
    <!--ro, opt, bool, whether it supports security log server test-->true
  </isSupportLogServerTest>
  <SecurityLogTypeList>
    <!--ro, req, array, List of security log types, subType:object-->
  </SecurityLogType>

```

```

    <!--ro, opt, object, List of Log types-->
    <primaryType>
    <!--ro, req, enum, main Log type, subType:string, desc:"Event", "Operation", "Other" (other types), "All" (all types)-->Event
    </primaryType>
    <secondaryType opt="test">
    <!--ro, req, enum, sub type, subType:string, attr:opt{req, string}, desc:see other types on the appendix. "all" (all types)-->all
    </secondaryType>
    </SecurityLogType>
    </SecurityLogTypeList>
  </SecurityLog>
  <isSupportGetBackupDevice>
    <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json-->true
  </isSupportGetBackupDevice>
  <isSupportAutoBackup>
    <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/autos/capabilities?format=json-->true
  </isSupportAutoBackup>
  <certificateValidationEnabled>
    <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <certificateValidationEnabled> node supports-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
  </defaultAdminPortEnabled>
  <isSptInputProxyChanCap>
    <!--ro, opt, bool, whether the device supports getting the capability set of the digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/capabilities-->true
  </isSptInputProxyChanCap>
  <isSupportLogDataPackage>
    <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/LogSearch/dataPackage-->true
  </isSupportLogDataPackage>
  <logSearchTimeSpanNums>
    <!--ro, opt, int, number of time periods for log search, desc:related URI: /ISAPI/ContentMgmt/LogSearch; corresponding node: <timeSpanList> in <CMSearchDescription>-->0
  </logSearchTimeSpanNums>
  <CloudStorageServerCap>
    <!--ro, opt, object, capability of the cloud storage service-->
    <isSupportCloudStorageParameter>
    <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI: /ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
    </isSupportCloudStorageParameter>
    <isSupportCloudStoragePool>
    <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI: /ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
    </isSupportCloudStoragePool>
    <supportChannelsNum>
    <!--ro, opt, int, number of channels supported by the cloud storage-->0
    </supportChannelsNum>
    <protocolType opt="videoStreamingOS">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->videoStreamingOS
    </protocolType>
  </CloudStorageServerCap>
  <isSupportWebPrivatePlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback (via private protocol) according to UTC-->true
  </isSupportWebPrivatePlaybackByUTC>
  <isSupportFindCommonFileByUTC>
    <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
  </isSupportFindCommonFileByUTC>
  <isSupportFindEventFileByUTC>
    <!--ro, opt, bool, whether it supports searching for event related videos (via private protocol) according to UTC-->true
  </isSupportFindEventFileByUTC>
  <isSupportSmartSearchRecordByUTC>
    <!--ro, opt, bool, whether it supports VCA search by UTC time-->true
  </isSupportSmartSearchRecordByUTC>
  <isSupportMRDSearchByTimezone>
    <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
  </isSupportMRDSearchByTimezone>
  <isSupportSearchRecordLabelByUTC>
    <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
  </isSupportSearchRecordLabelByUTC>
  <isSupportSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
  </isSupportSearchPictureByUTC>
  <isSupportSmartSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
  </isSupportSmartSearchPictureByUTC>
  <isSupportFindLogByUTC>
    <!--ro, opt, bool, whether it supports searching log by UTC time-->true
  </isSupportFindLogByUTC>
  <isSupportUploadRecordByUTC>
    <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
  </isSupportUploadRecordByUTC>
  <isSupportPlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
  </isSupportPlaybackByUTC>
  <isSupportFaceDetectionAlarmByTimezone>
    <!--ro, opt, bool, whether it supports reporting face detection by UTC time, desc:the time should be before isSupportCountingSearchByUTC and after isSupportPlaybackByUTC-->true
  </isSupportFaceDetectionAlarmByTimezone>
  <isSupportSourceModify>
    <!--ro, opt, bool, whether it supports configuring network camera parameters on the LAN, desc:related URI: /ISAPI/ContentMgmt/InputProxy/sourceModify/capabilities-->true
  </isSupportSourceModify>

```

```

<!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
</isSupportCountingSearchByUTC>
<isSupportCluster>
  <!--ro, opt, bool, whether it supports cluster function, desc:related URI: /ISAPI/ContentMgmt/Cluster-->true
</isSupportCluster>
<supportAIDTFSType opt="illegalParking,wrongdirection,crosslane,vehicleexist,lanchange,turnround,evidence">
  <!--ro, opt, enum, supported AIDTFS types, subType:string, attr:opt{req, string}, desc:"illegalParking", "wrongdirection" (wrong-way driving),
  "crosslane"(driving on the Lane Line), "vehicleexist" (vehicle on non-motor vehicle Lane), "lanchange" (illegal Lane change detection), "turnround"
  (illegal U-turning detection), "evidence" (face picture collection)-->illegalParking
</supportAIDTFSType>
<isSupportMisinfoStateSearch>
  <!--ro, opt, bool, whether it supports searching by false alarm status-->true
</isSupportMisinfoStateSearch>
<isSupportOneKeyOperationStorageData>
  <!--ro, opt, bool, whether it supports one-touch configuration for storage data, desc:related URI: /ISAPI/ContentMgmt/Storage/oneKey/capabilities?
  format=json-->true
</isSupportOneKeyOperationStorageData>
<eventRecordSearch>
  <!--ro, opt, object, whether the device supports searching videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
  <eventType
  opt="all,batch,alarmResult,faceSnap,vehicleMatchResult,hawkResult,VMD,alarmIn,inquestAlarm,behavior,posAlarm,accessControllerEvent,IOTd,trialAlarm,rapidMove
  ,defocus,scenechangedetection,PIR,fireSmartFireDetect,shipsDetection,TMA,TDA,framesPeopleCounting,personQueueTime,peopleDetection,safetyHelmetDetection,mixe
  dTargetDetection,TMPA,smokeDetection,smokeAndFireDetection,abnormalReboot,audioexception,ANPR,objectsThrownDetection,audioAnalysis,HFPD,shelteralarm,cityMan
  agement,hideStart,hideStop,sleepOnduty,PerimeterTargetDetEvent">
  <!--ro, opt, enum, event type, subType:string, attr:opt{req, string}, desc:"vehicleMatchResult" (vehicle comparison), "hawkResult" (video analysis
  task), "faceContrast" (face picture comparison), "faceSnap" (face capture)-->alarmResult
  </eventType>
  <recordChannelListSize>
    <!--ro, opt, int, maximum Length of the value of recordChannelList (recoding channel List)-->16
  </recordChannelListSize>
  <triggerChannelsSize>
    <!--ro, opt, int, maximum Length of the value of triggerChannels (List of video channel numbers which trigger recording)-->16
  </triggerChannelsSize>
  <alarmResultAuxEventType opt="all,alarmResultFailure,alarmResultSuccess,whitelist">
    <!--ro, opt, enum, sub-type of face picture comparison alarm, subType:string, dep:and,{$.RoomCap.eventRecordSearch.eventType,eg,alarmResult},
    attr:opt{req, string}, desc:"alarmResultFailure" (face picture comparison failed), "alarmResultSuccess" (face picture comparison succeeded), "whitelist"
    (allowlist alarm, i.e., strangers alarm)-->alarmResultFailure
  </alarmResultAuxEventType>
  <inquestParam>
    <!--ro, opt, object, interrogation information, dep:and,{$.EventSearchDescription.eventType,eg,inquestAlarm}-->
    <inquestEventType opt="any,inquestStart,inquestStop,inquestTag,inquestSegment">
      <!--ro, opt, enum, interrogation information, subType:string, attr:opt{req, string}, desc:"any", "inquestStart" (interrogation start), "inquestStop"
      (interrogation stop), "inquestTag" (marked information), "inquestSegment" (interrogation segment status)-->any
    </inquestEventType>
    <roomIndex min="1" max="65535">
      <!--ro, opt, int, interrogation room No., attr:min{req, int},max{req, int}, desc:it starts from 1-->1
    </roomIndex>
  </inquestParam>
  <behavior>
    <!--ro, opt, object, abnormal event detection, dep:and,{$.EventSearchDescription.eventType,eg,behavior}-->
    <behaviorEventType
    opt="any,linedetection,regionEntrance,regionExiting,fielddetection,loitering,parking,violentMotion,group,unattendedBaggage,attendedBaggage,audioAbnormal,lea
    vePosition,peopleNumChange,retention,failDown,running,spacingChange,playCellphone,objectPlacedOrTaken,tailing">
      <!--ro, req, enum, abnormal event detection event type, subType:string, attr:opt{req, string}, desc:"any", "Linedetection" (Line crossing
      detection), "regionEntrance", "regionExiting", "fielddetection" (intrusion detection), "Loitering" (Loitering detection), "parking" (parking detection),
      "violentMotion", "group" (people gathering), "unattendedBaggage", "attendedBaggage" (object removal), "audioAbnormal" (sudden change of sound intensity
      detection)-->any
    </behaviorEventType>
    <falseAlarmType opt="all,falseAlarmFilterate" def="all">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->all
    </falseAlarmType>
  </behavior>
  <posAlarm>
    <!--ro, opt, object, pos event, dep:and,{$.EventSearchDescription.eventType,eg,posAlarm}, desc:it is recorded in the device, not in the internet-->
    <caseSensitive opt="true,false">
      <!--ro, req, bool, whether it is case-sensitive, attr:opt{req, string}-->false
    </caseSensitive>
    <combinateMode opt="and,or">
      <!--ro, req, enum, keyword combination, subType:string, attr:opt{req, string}, desc:"and", "or"-->and
    </combinateMode>
    <keyword size="3">
      <!--ro, opt, string, keyword, attr:size{req, int}-->test
    </keyword>
  </posAlarm>
  <trialParam>
    <!--ro, opt, object, trial event, dep:and,{$.EventSearchDescription.eventType,eg,trialAlarm}-->
    <trialEventType opt="any,trialStart,trialStop,trialTag,trialSegment,trialJudge">
      <!--ro, opt, enum, trial information, subType:string, attr:opt{req, string}, desc:"any", "trialStart" (trial start), "trialStop" (trial stop),
      "trialTag" (marked information), "trialSegment" (trial segment status), "trialJudge" (trial case information)-->any
    </trialEventType>
    <caseNo min="0" max="56">
      <!--ro, req, string, case No., range:[0,56], attr:min{req, int},max{req, int}-->test
    </caseNo>
    <caseName>
      <!--ro, opt, string, case name, range:[0,100]-->test
    </caseName>
    <litigant1 min="0" max="32">
      <!--ro, opt, string, Litigant 1, range:[0,32], attr:min{req, int},max{req, int}-->test
    </litigant1>
    <litigant2 min="0" max="32">
      <!--ro, opt, string, Litigant 2, range:[0,32], attr:min{req, int},max{req, int}-->test
    </litigant2>

```



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<chiefJudge min="0" max="32">
  <!--ro, opt, string, judge, range:[0,32], attr:min{req, int},max{req, int}-->test
</chiefJudge>
<caseType opt="0,1,2">
  <!--ro, opt, enum, case type, subType:int, attr:opt{req, string}, desc:0 (all), 1 (criminal case), 2 (civil case)-->0
</caseType>
</trialParam>
<accessControllerEvent>
  <!--ro, opt, object, access control event, dep:and,{$.EventSearchDescription.eventType,eq,accessControllerEvent}-->
  <majorEventType>
    <!--ro, opt, int, alarm main type, desc:see details in macro definition-->0
  </majorEventType>
  <subEventType>
    <!--ro, opt, int, alarm sub type, desc:see details in macro definition-->0
  </subEventType>
  <cardNo min="0" max="32">
    <!--ro, opt, string, card No., range:[0,32], attr:min{req, int},max{req, int}-->test
  </cardNo>
  <name min="0" max="32">
    <!--ro, opt, string, name, range:[0,32], attr:min{req, int},max{req, int}-->test
  </name>
  <MACAddr min="0" max="12">
    <!--ro, opt, string, MAC address, range:[0,12], attr:min{req, int},max{req, int}-->test
  </MACAddr>
</accessControllerEvent>
<IOTD>
  <!--ro, opt, object, upload events related with IoT devices, dep:and,{$.EventSearchDescription.eventType,eq,IOTD}-->
  <DeviceEventTypeList>
    <!--ro, req, array, sub event type list, subType:object-->
    <DeviceEventType>
      <!--ro, opt, object, sub event type-->
      <subEventType>
        <!--ro, opt, string, sub event type, desc:see details in the node IoTEventType of Data Dictionary-->test
      </subEventType>
    </DeviceEventType>
  </DeviceEventTypeList>
  <IoTChannelID size="256">
    <!--ro, opt, string, channel No., attr:size{req, int}, desc:search all channels if the node does not exists-->test
  </IoTChannelID>
  <caseSensitive opt="true,false">
    <!--ro, req, bool, whether it is case-sensitive, attr:opt{req, string}-->false
  </caseSensitive>
  <combineMode opt="and,or">
    <!--ro, req, enum, keyword combination, subType:string, attr:opt{req, string}, desc:"and", "or"-->and
  </combineMode>
  <keyword size="3">
    <!--ro, opt, string, keyword, attr:size{req, int}-->test
  </keyword>
  <zono min="0" max="32">
    <!--ro, opt, int, zone No., attr:min{req, int},max{req, int}, desc:it is valid only when deviceType==2&&subEventType=="zone"-->1
  </zono>
</IOTD>
<AudioAnalysis>
  <!--ro, opt, object, audio analysis event, dep:and,{$.EventSearchDescription.eventType,eq,audioAnalysis}-->
  <targetType>
    opt="all,robbery,help,remittance,comment,fraud,cardNo,confirm,certificateID,certificateNumber,password,breakevenInterest,invest,noRisk,riskLow,riskSmall,transferMoney,sureEarnNoLose,onlyEarnNoLose,financing,noLose,borrowMoney,highReturn,highInterestRates,highYield"
    <!--ro, req, enum, audio analysis type, subType:string, attr:opt{req, string}, desc:"all", "robbery", "help", "remittance", "comment", "fraud", "cardNo" (card No.), "confirm", "certificateID", "certificateNumber", "password", "breakevenInterest" (break-even), "invest", "noRisk", "riskLow" (Low risk), "riskSmall" (small risk), "transfer", "transferMoney", "sureEarnNoLose" (earn without Losing), "onlyEarnNoLose" (earn without Losing), "financing", "noLose" (not Lose), "borrowMoney", "highReturn", "highInterestRates" (high interest rate), "highYield"-->all
  </targetType>
</AudioAnalysis>
<objectsThrownDetection>
  <!--ro, opt, object, objects thrown from heights detection, dep:and,{$.EventSearchDescription.eventType,eq,objectsThrownDetection}-->
  <alarmLevel opt="1,2,3">
    <!--ro, req, enum, alarm level, subType:string, attr:opt{req, string}, desc:"1" (Low), "2" (medium), "3" (high). It is categorized into three levels according to the target track and the actual event credibility. The higher the level, the greater the credibility of the alarm being triggered by an actual object thrown from building event. This node should be configured in the rule to filter alarm event uploading.-->1
  </alarmLevel>
  <floorID min="-100" max="100">
    <!--ro, opt, int, floor range, attr:min{req, int},max{req, int}-->1
  </floorID>
</objectsThrownDetection>
<HFPDSearch>
  <!--ro, opt, object, dep:and,{$.EventSearchDescription.eventType,eq,HFPD}-->
  <conditionType opt="greaterThan,equal,lessThan,notEqual,greaterThanOrEqual,lessThanOrEqual">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->greaterThan
  </conditionType>
  <occurrences min="0" max="100" def="1">
    <!--ro, opt, int, attr:min{req, int},max{req, int},def{req, int}-->1
  </occurrences>
</HFPDSearch>
<cityManagement>
  <!--ro, opt, object, dep:and,{$.RacmCap.eventRecordSearch.eventType,eq,cityManagement}-->
  <subEventType opt="unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad">
    <!--ro, opt, string, attr:opt{req, string}-->unregisteredStreetVendor
  </subEventType>
</cityManagement>
<ANPRSearch>
  <!--ro, opt, object-->
  <licensePlate min="1" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->A283KY77
  </licensePlate>

```

```

</deviceType>
<province opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34">
  <!--ro, opt, string, attr:opt{req, string}-->0
</province>
<country opt="0,1,2,3,4,5,6,7,8,9,10,11,12,169,172,277,255">
  <!--ro, opt, int, attr:opt{req, string}-->0
</country>
</ANPRSearch>
<PerimeterTargetDetEvent>
  <!--ro, opt, object-->
  <analysisStatusList opt="analyzed,unanalyzed,analysisException">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </analysisStatusList>
  <taskID min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </taskID>
</PerimeterTargetDetEvent>
</eventRecordSearch>
<isSupportManualRecord>
  <!--ro, opt, bool, whether the device supports manual recording, desc:related URI: /ISAPI/ContentMgmt/record/control/manual/capabilities?format=json-->true
</isSupportManualRecord>
<isSupportRemark>
  <!--ro, opt, bool, whether the device supports file remarks-->true
</isSupportRemark>
<FileUpload>
  <!--ro, opt, object, automatically upload file-->
  <enabled>
    <!--ro, req, bool, whether the device supports uploading the file-->true
  </enabled>
</FileUpload>
<PoliceInfoUpload>
  <!--ro, opt, object, automatically upload police officer information-->
  <enabled>
    <!--ro, req, bool, whether the device supports uploading the police officer information-->true
  </enabled>
</PoliceInfoUpload>
<PoliceInfo>
  <!--ro, opt, object, police officer information-->
  <policeID>
    <!--ro, opt, bool, whether the device supports police officer ID-->true
  </policeID>
  <policeCode>
    <!--ro, opt, bool, whether the device supports police officer number-->true
  </policeCode>
  <policeName>
    <!--ro, opt, bool, whether the device supports police officer name-->true
  </policeName>
  <password>
    <!--ro, opt, bool, whether the device supports police officer password-->true
  </password>
  <deviceID>
    <!--ro, opt, bool, whether the device supports device ID-->true
  </deviceID>
  <isSupportPoliceFacePicture>
    <!--ro, opt, bool, whether the device supports managing the face picture of police officer,
desc:related URL:
/ISAPI/ContentMgmt/police/<policeID>?format=json,
/ISAPI/ContentMgmt/police?format=json,
/ISAPI/ContentMgmt/police/deleteFacePicture?format=json-->true
  </isSupportPoliceFacePicture>
</PoliceInfo>
<isSupportAssignChannelID>
  <!--ro, opt, bool, whether the device supports specifying the access channel No. when adding the network camera, desc:related to channelID on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportAssignChannelID>
<isSupportAssignStreamID>
  <!--ro, opt, bool, whether the device supports specifying stream ID of the channel when adding the network camera, desc:related to streamID on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportAssignStreamID>
<PictureSearchSubTypeList>
  <!--ro, opt, object, list of picture searching sub type-->
  <PictureSearchSubType>
    <!--ro, opt, object, the sub type of picture search-->
    <mainType>
      <!--ro, req, enum, main event type of the searched picture, subType:string, desc:"vehicleDetection", "HVTVehicleDetection" (mixed-traffic
detection)-->vehicleDetection
    </mainType>
    <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
      <!--ro, req, enum, sub type of each main event type, subType:string, attr:opt{req, string}, desc:"all", "motorVehicle", "nonMotorVehicle",
"pedestrian"-->all
    </subType>
  </PictureSearchSubType>
</PictureSearchSubTypeList>
<isSupportPlaybackStreamIDByUTC>
  <!--ro, opt, bool, whether it supports playback by stream ID according to UTC time-->true
</isSupportPlaybackStreamIDByUTC>
<isSupportTimeLockStreamIDByUTC>
  <!--ro, opt, bool, whether it supports locking and unlocking the video segment by steam ID according to UTC time-->true
</isSupportTimeLockStreamIDByUTC>
<isSupportTimeSearch>
  <!--ro, opt, bool, whether it supports searching for start and end time of recording by channel, desc:/ISAPI/ContentMgmt/time/search?format=json-->true
</isSupportTimeSearch>
<behaviorEventPicSearch opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retention,sleepOnDuty">

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    <!--ro, opt, string, abnormal event detection events supported by picture search, attr:opt{req, string}, desc:"allBehaviorEvent" represents all events
    which are Listed in the node "behaviorEventPicSearch"-->test
    </behaviorEventPicSearch>
    <perimeterEventPicSearch opt="allPerimeterEvent,linedetection,fielddetection,regionEntrance,regionExiting,loitering">
    <!--ro, opt, string, perimeter protection events supported by picture search, attr:opt{req, string}, desc:"allPerimeterEvent" represents all events
    which are Listed in the node "perimeterEventPicSearch"-->test
    </perimeterEventPicSearch>
    <isSupportSourceStatus>
    <!--ro, opt, bool, whether it supports getting the status of access source-->true
    </isSupportSourceStatus>
    <isSupportSSDSMARTTest>
    <!--ro, opt, bool, whether the device supports S.M.A.R.T. detection on SSD, desc:related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start-->true
    </isSupportSSDSMARTTest>
    <MobileStorage>
    <!--ro, opt, object, storage capability of the mobile HDD-->
    <isSupportGetFileTime>
    <!--ro, opt, bool, whether it supports getting the file time of the mobile HDD, desc:it supports mobile HDD currently.
    /ISAPI/ContentMgmt/mobileStorage/fileTime/capabilities?format=json-->true
    </isSupportGetFileTime>
    </MobileStorage>
    <isSupportGetChannelList>
    <!--ro, opt, bool, whether it supports getting channels of the HDD, desc:it supports HDD currently.
    /ISAPI/ContentMgmt/Storage/hdd/channelList/capabilities?format=json-->true
    </isSupportGetChannelList>
    <isSupportCascade>
    <!--ro, opt, bool, whether it supports configuring parameters for NVR cascading topology-->true
    </isSupportCascade>
    <isSupportFileManagement>
    <!--ro, opt, bool, whether it supports configuring file management, desc:related URI: /ISAPI/ContentMgmt/fileManagement/capabilities?format=json-->true
    </isSupportFileManagement>
    <isSupportPictureInfo>
    <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
    </isSupportPictureInfo>
    <isSupportChanStatusPage>
    <!--ro, opt, string, whether it supports searching the digital channel by page, desc:related URI:
    /ISAPI/ContentMgmt/InputProxy/channels/statusPage/capabilities-->test
    </isSupportChanStatusPage>
    <isSupportRecordSearchByTargetType>
    <!--ro, opt, bool, whether it supports searching for event recordings by target types (pedestrians and vehicles)-->true
    </isSupportRecordSearchByTargetType>
    <storIpcStartNo>
    <!--ro, opt, int, the first IP channel No. of the recording, desc:it is supported by Hikmed Imaging Technology-->0
    </storIpcStartNo>
    <maxRecordIpcNum>
    <!--ro, opt, int, number of the IP channels which support recording, desc:it is supported by Hikmed Imaging Technology-->0
    </maxRecordIpcNum>
    <maxConferenceIpcNum>
    <!--ro, opt, int, number of the meeting IP channels, desc:it is supported by Hikmed Imaging Technology-->0
    </maxConferenceIpcNum>
    <decIpcStartNo>
    <!--ro, opt, int, start No. of screen mirroring IP channels, desc:it is supported by Hikmed Imaging Technology-->0
    </decIpcStartNo>
    <maxDecodeIpcNum>
    <!--ro, opt, int, number of screen mirroring IP channels, desc:it is supported by Hikmed Imaging Technology-->0
    </maxDecodeIpcNum>
    <pipIpcStartNo>
    <!--ro, opt, int, start No. of the integrated IP channels, desc:it is supported by Hikmed Imaging Technology-->0
    </pipIpcStartNo>
    <maxPipIpcNum>
    <!--ro, opt, int, number of the integrated IP channels, desc:it is supported by Hikmed Imaging Technology-->0
    </maxPipIpcNum>
    <isSupportLogSearch>
    <!--ro, opt, bool, whether it supports Log search-->true
    </isSupportLogSearch>
    <isSupportRecordSearch>
    <!--ro, opt, bool, whether it supports video search-->true
    </isSupportRecordSearch>
    <isSupportRecordDownload>
    <!--ro, opt, bool, whether it supports video downloading-->true
    </isSupportRecordDownload>
    <isSupportPnp opt="true,false">
    <!--ro, opt, bool, whether it supports configuring plug-and-play for the network camera, attr:opt{req, string}-->true
    </isSupportPnp>
    <esataUseage>
    <!--ro, opt, enum, storage strategy, subType:string, desc:"backup", "record"-->backup
    </esataUseage>
    <subType opt="all,leave,on,sleep">
    <!--ro, opt, enum, absence sub type, subType:string, attr:opt{req, string}, desc:"all", "Leave" (absence), "on" (on duty), "sleep" (sleeping on duty)--
    >all
    </subType>
    <isSupportManualBackup>
    <!--ro, opt, bool, whether it supports manual backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/manual/capabilities?format=json-->true
    </isSupportManualBackup>
    <isSupportLinkToIPCWeb>
    <!--ro, opt, bool, whether it supports going to the network camera web page (the digital channel management page), desc:N/A-->true
    </isSupportLinkToIPCWeb>
    <isSupportIPCConfigCSV>
    <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
    /ISAPI/ContentMgmt/InputProxy/ipcConfigCSV?format=json-->true
    </isSupportIPCConfigCSV>
    <isSupportExperimentExam>
    <!--ro, opt, bool, whether it supports the related function of the experiment exam, desc:related URI: /ISAPI/ContentMgmt/experimentExam/capabilities?
    format=json-->true
    </isSupportExperimentExam>

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</isSupportExperimentExam>
<isSupportPOSRecordSearch>
  <!--ro, opt, bool, whether it supports POS video research, desc:related URI: /ISAPI/ContentMgmt/POSRecord/search/capabilities-->true
</isSupportPOSRecordSearch>
<isSupportSpare>
  <!--ro, opt, bool, whether the device supports hot spare configuration, desc:related capability set: /ISAPI/ContentMgmt/spare/capabilities?format=json-->true
</isSupportSpare>
<isSupportTracksDates>
  <!--ro, opt, bool, whether it supports configuring recording schedule by calendar, desc:related URI: /ISAPI/ContentMgmt/record/tracks/dates/capabilities?format=json-->true
</isSupportTracksDates>
<isSupportRecordAudio>
  <!--ro, opt, bool, whether the device supports audio recording, desc:only the audio will be recorded and the video will not be recorded. Related URI: /ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json-->true
</isSupportRecordAudio>
<isSupportSearchAudio>
  <!--ro, opt, bool, whether the device supports searching the audio, desc:related URI: /ISAPI/ContentMgmt/recordAudio/search/capabilities?format=json-->true
</isSupportSearchAudio>
<isSupportAudioDailyDistribution>
  <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI: /ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
</isSupportAudioDailyDistribution>
<isSupportDownloadAudio>
  <!--ro, opt, bool, whether it supports downloading audio files by file name, desc:related URI: /ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
</isSupportDownloadAudio>
<isSupportDeleteAudio>
  <!--ro, opt, bool, whether the device supports deleting the recorded audio file, desc:Related URI: /ISAPI/ContentMgmt/recordAudio/deleteFile?format=json-->true
</isSupportDeleteAudio>
<isSupportEventRecordCfg>
  <!--ro, opt, bool, whether it supports configuring parameters of event recordings, desc:related URI: /ISAPI/ContentMgmt/eventRecordCfg?format=json-->true
</isSupportEventRecordCfg>
<isSupportSearchRemoteAlbum>
  <!--ro, opt, bool, whether the device supports searching the remote album information, desc:related URI: /ISAPI/ContentMgmt/searchRemoteAlbum?format=json-->true
</isSupportSearchRemoteAlbum>
<isNotSupportTracks>
  <!--ro, opt, bool, whether it does not support recording schedule, desc:true will be returned if it is not supported-->true
</isNotSupportTracks>
<isSupportTFCardConfig>
  <!--ro, opt, bool, whether the device supports function switch of the extended TF card: /ISAPI/ContentMgmt/TFCardConfig/capabilities?format=json-->true
</isSupportTFCardConfig>
<isSupportDockStationFileStorageDuration>
  <!--ro, opt, bool, whether the device supports configuring the retention period of files, desc:related URI: /ISAPI/ContentMgmt/dockStationFile/DockStationFileStorageDuration?format=json-->true
</isSupportDockStationFileStorageDuration>
<isSupportDockStationFileAcquisitionMode>
  <!--ro, opt, bool, whether the device supports configuring the file collection mode, desc:related URI: /ISAPI/ContentMgmt/dockStationFile/DockStationFileAcquisitionMode?format=json-->true
</isSupportDockStationFileAcquisitionMode>
<isSupportSSDAbnormalCfg>
  <!--ro, opt, bool, whether the device supports SSD exception detection parameters, desc:/ISAPI/ContentMgmt/Storage/ssd/SSDAbnormalCfg/capabilities?format=json-->true
</isSupportSSDAbnormalCfg>
<isSupportAudioQuota>
  <!--ro, opt, bool, whether the device supports audio recording quota, desc:GET/PUT /ISAPI/ContentMgmt/channels/<audioID>/AudioQuota?format=json-->true
</isSupportAudioQuota>
<targetRegionEventPicSearch opt="linedetection,fielddetection,regionEntrance,regionExiting,WMD">
  <!--ro, opt, string, event type supported to return target region in the picture when searching for pictures, attr:opt(req, string)-->test
</targetRegionEventPicSearch>
<isSupportAudioChanCap>
  <!--ro, opt, bool, whether the device supports managing audio channel resources, desc:related URL: /ISAPI/ContentMgmt/audioChannel/capabilities?format=json-->true
</isSupportAudioChanCap>
<isSupportImportEnforcementWorkFile>
  <!--ro, opt, bool, whether the device supports importing the Law enforcement working file, desc:POST /ISAPI/ContentMgmt/ImportEnforcementWorkFile?format=json-->true
</isSupportImportEnforcementWorkFile>
<isSupportEnforcementProjectInfo>
  <!--ro, opt, bool, whether the device supports getting the project information of the body camera for Law enforcement, desc:GET/PUT /ISAPI/ContentMgmt/EnforcementProjectInfo?format=json-->true
</isSupportEnforcementProjectInfo>
<isSupportTriggerVoiceMonitoringHistoryEventRetrans>
  <!--ro, opt, bool, whether the device supports reloading audio files of the history event of triggering human voice monitoring, desc:POST /ISAPI/ContentMgmt/channels/<audioID>/TriggerVoiceMonitoringHistoryEventRetrans?format=json-->true
</isSupportTriggerVoiceMonitoringHistoryEventRetrans>
<isSupportDockStationAudioFileToText>
  <!--ro, opt, bool, whether the device supports converting dock station's audio to text, desc:POST /ISAPI/ContentMgmt/DockStationAudioFileToText?format=json-->true
</isSupportDockStationAudioFileToText>
<isSupportDockStationAudioFileAnalysisKeyword>
  <!--ro, opt, bool, whether the device supports keyword analysis of dock station's audio file, desc:GET/PUT /ISAPI/ContentMgmt/DockStationAudioFileAnalysisKeyword?format=json-->true
</isSupportDockStationAudioFileAnalysisKeyword>
<isSupportStartingUpRecordParams>
  <!--ro, opt, bool-->true
</isSupportStartingUpRecordParams>
<isSupportSendLawEnforcementFileUploadInfo>
  <!--ro, opt, bool-->true
</isSupportSendLawEnforcementFileUploadInfo>

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<isSupportQuotaParams>
  <!--ro, opt, bool-->true
</isSupportQuotaParams>
<isSupportSearchAudioRecordPlan>
  <!--ro, opt, bool-->true
</isSupportSearchAudioRecordPlan>
<isSupportAddQuotaParams>
  <!--ro, opt, bool-->true
</isSupportAddQuotaParams>
<isSupportModifyQuotaParams>
  <!--ro, opt, bool-->true
</isSupportModifyQuotaParams>
<isSupportGetQuotaParams>
  <!--ro, opt, bool-->true
</isSupportGetQuotaParams>
<isSupportDeleteQuotaParams>
  <!--ro, opt, bool-->true
</isSupportDeleteQuotaParams>
<isSupportAutoRecordSynthesisParams>
  <!--ro, opt, bool-->true
</isSupportAutoRecordSynthesisParams>
<isSupportGetSSDSystemConfigRecoverProgress>
  <!--ro, opt, bool-->true
</isSupportGetSSDSystemConfigRecoverProgress>
<isSupportSSDSystemConfigRecover>
  <!--ro, opt, bool-->true
</isSupportSSDSystemConfigRecover>
<isSupportSSDSystemConfigState>
  <!--ro, opt, bool-->true
</isSupportSSDSystemConfigState>
<isSupportPicSearchByCoordinate>
  <!--ro, opt, bool-->true
</isSupportPicSearchByCoordinate>
<AcuSearchCap>
  <!--ro, opt, object-->
  <targetType opt="face,human,vehicle">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->face
  </targetType>
  <dataType opt="url,binary,modelData">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->url
  </dataType>
</AcuSearchCap>
<isSupportRecordAsyncParams>
  <!--ro, opt, bool-->true
</isSupportRecordAsyncParams>
<isSupportRedirectStorageUrl>
  <!--ro, opt, bool-->true
</isSupportRedirectStorageUrl>
<isSupportUploadFileToCloud>
  <!--ro, opt, bool-->true
</isSupportUploadFileToCloud>
<isSupportAddRecordAgeParams>
  <!--ro, opt, bool-->true
</isSupportAddRecordAgeParams>
<isSupportDeleteRecordAgeParams>
  <!--ro, opt, bool-->true
</isSupportDeleteRecordAgeParams>
<isSupportSearchRecordAgeParams>
  <!--ro, opt, bool-->true
</isSupportSearchRecordAgeParams>
<isSupportModifyRecordAgeParams>
  <!--ro, opt, bool-->true
</isSupportModifyRecordAgeParams>
<isSupportEstimateAgeRecordCycle>
  <!--ro, opt, bool-->true
</isSupportEstimateAgeRecordCycle>
<isSupportSendDiskPowerOnCtrl>
  <!--ro, opt, bool-->true
</isSupportSendDiskPowerOnCtrl>
<isSupportSearchAgeCondChannel>
  <!--ro, opt, bool-->true
</isSupportSearchAgeCondChannel>
<isSupportDiskMaterielInfoDetect>
  <!--ro, opt, bool-->true
</isSupportDiskMaterielInfoDetect>
<isNotSupportHDDBadSectorsTestStart>
  <!--ro, opt, bool-->false
</isNotSupportHDDBadSectorsTestStart>
<isSupportSearchAcuCollectPics>
  <!--ro, opt, bool-->true
</isSupportSearchAcuCollectPics>
<isSupportModifyAcuCollectPics>
  <!--ro, opt, bool-->true
</isSupportModifyAcuCollectPics>
<isSupportSearchRAIDStoragePerformance>
  <!--ro, opt, bool-->true
</isSupportSearchRAIDStoragePerformance>
<isSupportSearchDiskConfigurationSpecificationList>
  <!--ro, opt, bool-->true
</isSupportSearchDiskConfigurationSpecificationList>
<isSupportOneKeyRebuildArray>
  <!--ro, opt, bool-->true
</isSupportOneKeyRebuildArray>
```

```

<isSupportRecordStorageCalculate>
  <!--ro, opt, bool-->true
</isSupportRecordStorageCalculate>
<isSupportStartStorageEnvDetection>
  <!--ro, opt, bool-->true
</isSupportStartStorageEnvDetection>
<isSupportSearchStorageEnvDetectionProcess>
  <!--ro, opt, bool-->true
</isSupportSearchStorageEnvDetectionProcess>
<isSupportSearchAddRaidNeedDiskNum>
  <!--ro, opt, bool-->true
</isSupportSearchAddRaidNeedDiskNum>
<isSupportInitQuotaParams>
  <!--ro, opt, bool-->true
</isSupportInitQuotaParams>
</RacmCap>

```

9.1.9.2 Get the system security capability

Request URL

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
userName	string	user name
deviceIndex	string	For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, system security capability, attr:version{req, string, protocolVersion}-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->0
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->0
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->0
  </userBondMacNums>
  <isSupCertificate>
    <!--ro, opt, bool, whether the device supports authentication-->true
  </isSupCertificate>
  <issupIllegalLoginLock>
    <!--ro, opt, bool, whether the device supports Locking Login-->true
  </issupIllegalLoginLock>
  <isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
  </isSupportOnlineUser>
  <isSupportAnonymous>
    <!--ro, opt, bool, whether the device supports anonymous Login-->true
  </isSupportAnonymous>
  <isSupportStreamEncryption>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamEncryption>
  <securityVersion opt="1,2,3,4">
    <!--ro, opt, string, encryption capability set, attr:opt{req, string}, desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is described in each protocol-->1
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends on the node securityVersion, the range is between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the Login password when editing (editing/adding/deleting) user parameters, dep:or,{$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the Login password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>

```

```

</isSupportUserCheck>
<isSupportGUIDFileDataExport>
  <!--ro, opt, bool, whether the device supports exporting the device's GUID file, desc:related URI: /ISAPI/Security/GUIDFileData-->true
</isSupportGUIDFileDataExport>
<isSupportSecurityQuestionConfig>
  <!--ro, opt, bool, whether the device supports answering security questions, desc:related URI: /ISAPI/Security/questionConfiguration-->true
</isSupportSecurityQuestionConfig>
<supportSecurityNode opt="wirelessServer,wirelessDial">
  <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt(req, string),
desc:in the earlier version of the solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI),
some URIs containing sensitive information are left out now these URIs need to be encrypted for transmission, this capability node is added to be compatible
with old devices _x000D_
    if wirelessServer in this node is returned, it indicates that the following sensitive information should be encrypted _x000D_
    /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities, /ISAPI/System/Network/interfaces/<ID>/wirelessServer, if wirelessDial in this
node is returned, it indicates that the following sensitive information should be encrypted _x000D_
    /ISAPI/System/Network/WirelessDial/Interfaces,_x000D_
    /ISAPI/System/Network/WirelessDial/Interfaces/<ID>-->wirelessServer
</supportSecurityNode>
<isSupportGetOnlineUserListSC>
  <!--ro, opt, bool, whether the device supports searching the online user list, desc:related URI: /ISAPI/Security/onlineUser-->true
</isSupportGetOnlineUserListSC>
<SecurityLimits>
  <!--ro, opt, object, capability of configuring security limit parameters-->
  <LoginPasswordLenLimit min="1" max="16">
    <!--ro, opt, string, length limit of the user's login password, attr:min(req, int),max(req, int)-->1
  </LoginPasswordLenLimit>
  <SecurityAnswerLenLimit min="1" max="128">
    <!--ro, opt, string, length limit of the security question's answer, attr:min(req, int),max(req, int)-->1
  </SecurityAnswerLenLimit>
</SecurityLimits>
<RSAKeyLength opt="512,1024,2048" def="2048">
  <!--ro, opt, enum, HTTPS certificate length, subType:int, attr:opt(req, string),def(req, string), desc:512, 1024, 2048-->2048
</RSAKeyLength>
<isSupportONVIFUserManagement>
  <!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol-->true
</isSupportONVIFUserManagement>
<WebCertificateCap>
  <!--ro, opt, object, HTTP authentication capability, desc:if this node is not returned, it indicates that device supports basic and digest
authentication-->
  <CertificateType opt="basic,digest,digest/basic">
    <!--ro, req, enum, certificate type, subType:string, attr:opt(req, string), desc:"basic" (basic authentication), "digest" (digest authentication),
"digest/basic" (digest/basic authentication)-->basic
  </CertificateType>
  <SecurityAlgorithm>
    <!--ro, opt, object, security algorithm, dep:or,{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest},
{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256">
      <!--ro, opt, enum, algorithm type, subType:string, attr:opt(req, string), desc:"MD5", "SHA256", "MD5/SHA256"-->MD5
    </algorithmType>
  </SecurityAlgorithm>
</WebCertificateCap>
<isSupportConfigFileImport>
  <!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)-->true
</isSupportConfigFileImport>
<isSupportConfigFileExport>
  <!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)-->true
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
  <!--ro, opt, string, length limit of the configuration file's verification key, attr:min(req, int),max(req, int)-->1
</cfgFileSecretKeyLenLimit>
<supportIPActivatePassword>
  <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
</supportIPActivatePassword>
<isIrreversible>
  <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user
information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
</isIrreversible>
<salt>
  <!--ro, opt, string, the specific salt used by the user to log in-->test
</salt>
<isSupportOnvifInfo>
  <!--ro, opt, bool, whether the device supports getting the Open Network Video Interface Protocol, desc:true (support), this node is not returned (not
support)-->true
</isSupportOnvifInfo>
<isSupportPictureURLCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt(req, string), desc:if this node is not
returned, it indicates the digest authentication for picture URL by default-->true
</isSupportPictureURLCertificate>
<isSupportGB35114Certificate>
  <!--ro, opt, bool, whether supports GB35114 protocol and CA/SIP authentication certificate-->true
</isSupportGB35114Certificate>
<isSupportAuthenticationMode opt="uKey,private">
  <!--ro, opt, enum, verification mode, subType:string, attr:opt(req, string), desc:"uKey", "private"-->private
</isSupportAuthenticationMode>
<isSupportUserWithIssuerDN>
  <!--ro, opt, bool-->true
</isSupportUserWithIssuerDN>
<isSupportUnloggedUserPermissionConfig opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt(req, string)-->true
</isSupportUnloggedUserPermissionConfig>
<isSupportUserNamePasswordCheckUpgrade>
  <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password, desc:if this node is not returned, it indicates
that device does not support this function-->true
</isSupportUserNamePasswordCheckUpgrade>

```

```

<isSupportDeviceCertificatesManagement>
  <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not
support this function-->true
</isSupportDeviceCertificatesManagement>
<isSupportSIPCertificatesManagement>
  <!--ro, opt, bool, whether the device supports SIP certificate management, desc:this node is returned if the device supports this function and not
returned if device does not support this function, related URI: /ISAPI/Security/SIPCertificate, additional node: <SIPCertificatesManagementCap-->true
</isSupportSIPCertificatesManagement>
<isSupportDeviceSelfSignCertExport>
  <!--ro, opt, bool, whether the device supports exporting self-signed certificate-->true
</isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail>
  <!--ro, opt, bool, whether the device supports configuring the security email, desc:if this node is not returned, it indicates that device does not
support this function-->true
</isSupportSecurityEmail>
<isSupportRTSPCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports RTSP certificate management, attr:opt{req, string}-->true
</isSupportRTSPCertificate>
<maxIllegalLoginTimes min="3" max="20" def="5">
  <!--ro, opt, int, maximum failed illegal login attempts, range:[3,20], attr:min{req, int},max{req, int},def{req, int}, desc:this node is valid when the
value of the node issupportIllegalLoginLock is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->3
</maxIllegalLoginTimes>
<SecurityAdvanced>
  <!--ro, opt, object, security advanced parameters-->
  <securityEnhanced>
    <!--ro, opt, bool, whether the device supports security reinforcement-->true
  </securityEnhanced>
  <noOperationEnabled>
    <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
  </noOperationEnabled>
  <noOperationTime min="1" max="60" def="15">
    <!--ro, req, int, control timeout when there is no operation, range:[1,60], unit:min, attr:min{req, int},max{req, int},def{req, int}-->15
  </noOperationTime>
  <isSupportDigestStatus>
    <!--ro, opt, bool, digest status-->true
  </isSupportDigestStatus>
  <passwordLimitLevel opt="midding,high">
    <!--ro, opt, enum, password strength limit level, subType:string, attr:opt{req, string},
desc:the configured password is required by the following limit _x0000_
"midding" (middle level), "high" (high level), middle level password contains 8 to 16 characters, including at least 2 types of the following characters:
digits, lower case letters, upper case letters, and special characters; high level password contains 8 to 16 characters, including at least 3 types of the
following characters: digits, lower case letters, upper case letters, and special characters-->midding
  </passwordLimitLevel>
</SecurityAdvanced>
<LoginLinkNum>
  <!--ro, opt, object, number of controls logging in at the same time-->
  <maxLinkNum min="1" max="128" def="50">
    <!--ro, req, int, maximum number of controls logging in at the same time, range:[1,128], attr:min{req, int},max{req, int},def{req, int}-->50
  </maxLinkNum>
</LoginLinkNum>
<isSupportCCClientCertificate>
  <!--ro, opt, bool, whether the device supports the CC (Client Certificate) function-->true
</isSupportCCClientCertificate>
<passwordValidity min="0" max="365">
  <!--ro, opt, string, range of the password validity period, range:[0,365], attr:min{req, int},max{req, int}, desc:if this node is returned, it indicates
that the device supports configuring the password validity period, and only the administrator can edit the configuration-->1
</passwordValidity>
<isSupportResetChannelPassword>
  <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
</isSupportResetChannelPassword>
<DoubleVerificationCap>
  <!--ro, opt, object, secondary authentication capability-->
  <isSupportUsersConfig>
    <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
  </isSupportUsersConfig>
  <isSupportUsersPermissionConfig>
    <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
  </isSupportUsersPermissionConfig>
  <isSupportFactoryReset>
    <!--ro, opt, bool, whether the device supports restoring secondary authentication to defaults-->true
  </isSupportFactoryReset>
</DoubleVerificationCap>
<isSupportUserSearch>
  <!--ro, opt, bool, whether the device supports searching users-->true
</isSupportUserSearch>
<keypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length, attr:min{req, int},max{req, int}, desc:When the password length of different user types is different, this
node only indicates the keypad password length of the administrator; otherwise, the keypad password length of all user types is the same and defined by the
value of this node-->1
</keypadPassword>
<installerKeypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length of the installer, attr:min{req, int},max{req, int}-->1
</installerKeypadPassword>
<operatorKeypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length of the operator, attr:min{req, int},max{req, int}-->1
</operatorKeypadPassword>
<userOperateType opt="1,2,3">
  <!--ro, opt, enum, user type, subType:int, attr:opt{req, string}, desc:1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
</userOperateType>
<isSptUserEnabled>
  <!--ro, opt, bool, whether the device supports configuring parameters for enabling the user-->true
</isSptUserEnabled>

```



```

<!--ro, opt, bool, whether the device supports getting the admin permission capability-->true
</isSptAdminCap>
<isSupportBackLogCfg>
  <!--ro, opt, bool, whether the device supports configuring basic parameters of the background log-->true
</isSupportBackLogCfg>
<isSupportBackLogExport>
  <!--ro, opt, bool, whether the device supports exporting the background log-->true
</isSupportBackLogExport>
<isSupportModifyInnerPassword opt="true,false">
  <!--ro, opt, bool, whether the device supports editing its built-in passwords,, attr:opt{req, string}, desc:such as the database password, system
account password-->true
</isSupportModifyInnerPassword>
<isSupportEncryptCertificate>
  <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
</isSupportEncryptCertificate>
<maxIllegalLoginLockTime min="1" max="120" def="30.00">
  <!--ro, opt, int, lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
</maxIllegalLoginLockTime>
<cloudSensitiveInfoEncryptType>
  <!--ro, opt, string, encryption type of the cloud sensitive information, desc:the current device returns the value "2.1", which indicates that the
method of generating the key for encrypting the sensitive information is SHA256 (calc_sha256 (user, salt, SHA256 (user+salt+SHA256 (user+salt+password))) +
AaBbCcDd1234!@#%)-->2.1
</cloudSensitiveInfoEncryptType>
<isSupportCertificateCustomID>
  <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
</isSupportCertificateCustomID>
<isSupportResetLANPassword>
  <!--ro, opt, bool, whether the device supports resetting the device's password in the LAN, desc:related URI:
/ISAPI/Security/resetPassword/certification?format=json-->true
</isSupportResetLANPassword>
<isSupportPicHttpsCertificatesManagement>
  <!--ro, opt, bool, whether the device supports managing the HTTPS certificate for pictures, desc:true (support), this node is not returned (not
support)-->true
</isSupportPicHttpsCertificatesManagement>
<isSupportAKSKImport>
  <!--ro, opt, bool, whether the device supports importing AKSK, desc:true (support), this node is not returned (not support)-->true
</isSupportAKSKImport>
<isSupportAKSKExport>
  <!--ro, opt, bool, whether the device supports exporting AKSK, desc:true (support), this node is not returned (not support)-->true
</isSupportAKSKExport>
<isSupportAKSKReset>
  <!--ro, opt, bool, whether the device supports resetting AKSK, desc:true (support), this node is not returned (not support)-->true
</isSupportAKSKReset>
<ExportCertificateFile>
  <!--ro, opt, object, whether the device supports exporting the certificate, desc:related URI: POST
/ISAPI/Security/serverCertificate/exportCertificateFile-->
  <type opt="GB35114,GB35114Service">
    <!--ro, req, enum, types of certificates that can be exported, subType:string, attr:opt{req, string}, desc:"GB35114" (support exporting GB35114
certificate for the client), "GB35114Service" (support exporting GB35114 certificate for the server)-->GB35114
  </type>
</ExportCertificateFile>
<isSupportGB35114FDWSFCertificate>
  <!--ro, opt, bool, whether the device supports FDWSF certificate management, desc:true (support), this node is not returned (not support), related URIs:
/ISAPI/Security/GB35114/FDWSFCertificate/capabilities?format=json, POST /ISAPI/Security/GB35114/FDWSFCertificate?format=json, GET
/ISAPI/Security/GB35114/FDWSFCertificate?format=json, and DELETE /ISAPI/Security/GB35114/FDWSFCertificate?format=json-->true
</isSupportGB35114FDWSFCertificate>
<isSupportMediaObjectEncryption>
  <!--ro, opt, bool, whether supports setting encryption parameters of the private target in media data for the specified video channel, desc:if this
function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI:
/ISAPI/Security/MediaObjectEncryption/capabilities?format=json-->true
</isSupportMediaObjectEncryption>
<isSupportDisclaimer>
  <!--ro, opt, bool, whether support configuring the product disclaimer, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/disclaimer/capabilities?format=json-->true
</isSupportDisclaimer>
<isSupportSoftwareLicense>
  <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
</isSupportSoftwareLicense>
<isSupportInstallerCap>
  <!--ro, opt, bool, whether the device supports getting the installer permission, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/UserPermission/installer/capabilities?format=json-->true
</isSupportInstallerCap>
<isSupportResetPasswordFile>
  <!--ro, opt, bool, whether the device supports exporting and importing the secret key file of device in the LAN, desc:Related URI:
/ISAPI/Security/resetPassword/keyFile/capabilities?format=json-->true
</isSupportResetPasswordFile>
<isNotSupportLogSeverCertificate>
  <!--ro, opt, bool, whether the device support importing Log server certificate, desc:related URI: /ISAPI/Security/serverCertificate/certificate-->true
</isNotSupportLogSeverCertificate>
<isSupportLocalLoginCfg>
  <!--ro, opt, bool, whether it supports the capability of device Local Login parameters, desc:GET/PUT /ISAPI/Security/LocalLoginCfg?format=json-->true
</isSupportLocalLoginCfg>
<isSupportGetProtocolFirewallRulesFileImportStatus>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
</isSupportGetProtocolFirewallRulesFileImportStatus>
<isSupportImportProtocolFirewallRulesFile>
  <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
</isSupportImportProtocolFirewallRulesFile>
<isSupportGetProtocolFirewallEngineUpgradeStatus>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
</isSupportGetProtocolFirewallEngineUpgradeStatus>
<isSupportUpgradeProtocolFirewallEngine>

```

```

    <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
</isSupportUpgradeProtocolFirewallEngine>
<isSupportProtocolFirewallVersionInfo>
    <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
</isSupportProtocolFirewallVersionInfo>
<isSupportProtocolFirewallParams>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
</isSupportProtocolFirewallParams>
<isSupportResetPasswordBySecurityPhoneNo>
    <!--ro, opt, bool, whether it supports resetting device password with phone number verification,
desc:1. attendance check devices support applying a preset phone number during device activation. The user can reset the device password with the preset
phone number, which is similar with password reset via email. The phone number can be changed on the website or local UI of the device _x0000_
2. if the node is not returned or the returned value is false, it indicates "no"-->true
    <isSupportResetPasswordBySecurityPhoneNo>
    <isSupportGetIntrusionPreventionEngineUpgradeStatus>
    <!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET
/ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
    <isSupportGetIntrusionPreventionEngineUpgradeStatus>
    <isSupportIntrusionPreventionEngineVersionInfo>
    <!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET
/ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
    <isSupportIntrusionPreventionEngineVersionInfo>
    <isSupportIntrusionPreventionParams>
    <!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json--
>true
    <isSupportIntrusionPreventionParams>
    <isSupportUpgradeIntrusionPreventionEngine>
    <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT /ISAPI/Security/UpgradeIntrusionPreventionEngine?
format=json-->true
    <isSupportUpgradeIntrusionPreventionEngine>
    <notSupportImportCertificateType opt="CA,CClntCertificate,GB35114,GB35114Service">
    <!--ro, opt, string, file type that cannot be imported, attr:opt{req, string}, desc:if this node does not exist, all supported certificate types can be
imported-->test
    </notSupportImportCertificateType>
    <isSupportPasswordProtection>
    <!--ro, opt, bool, whether it supports configuring SDK protocol authentication mode parameters, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    <isSupportPasswordProtection>
    <isSupportGMT0028SecretStandardParam>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/GMT0028SecretStandard/GMT0028SecretStandardParam?format=json-->true
    <isSupportGMT0028SecretStandardParam>
    <isSupportCipherCardRunningPolicy>
    <!--ro, opt, bool-->true
    </isSupportCipherCardRunningPolicy>
    <isSupportGetSecurityScanStatus>
    <!--ro, opt, bool-->true
    </isSupportGetSecurityScanStatus>
    <isSupportStartSecurityScan>
    <!--ro, opt, bool-->true
    </isSupportStartSecurityScan>
    <isSupportStopSecurityScan>
    <!--ro, opt, bool-->true
    </isSupportStopSecurityScan>
    <isSupportModifySecurityItems>
    <!--ro, opt, bool-->true
    </isSupportModifySecurityItems>
</SecurityCap>

```

9.1.9.3 Get device system capability

Request URL

GET /ISAPI/System/capabilities?type=<all>

Query Parameter

Parameter Name	Parameter Type	Description
all	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
    <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
    <isSupportPreview>
    <!--ro, opt, bool, whether the device supports live view-->true
    </isSupportPreview>
    <isSupportSystemMaintain>
    <!--ro, opt, bool, whether the device supports system maintenance, desc:for traffic devices, this node is required and must be true-->true
    </isSupportSystemMaintain>
    <isSupportReboot>
    <!--ro, opt, bool, whether the device supports rebooting, desc:for traffic devices, this node is required and must be true-->true

```



```

<!--ro, opt, bool, whether the device supports restoring default traffic settings, this node is required and must be true-->true
</isSupportReboot>
<isSupportFactoryReset>
  <!--ro, opt, bool, whether the device supports restoring to default settings, desc:for traffic devices, this node is required and must be true-->true
</isSupportFactoryReset>
<isSupportConfigurationData>
  <!--ro, opt, bool, whether the device supports importing and exporting the configuration files safely, desc:for traffic devices, this node is required
and must be true-->true
</isSupportConfigurationData>
<isSupportUpdateFirmware>
  <!--ro, opt, bool, whether the device supports upgrading, desc:for traffic devices, this node is required and must be true-->true
</isSupportUpdateFirmware>
<isSupportDeviceInfo>
  <!--ro, opt, bool, whether the device supports getting the device information, desc:for traffic devices, this node is required and must be true-->true
</isSupportDeviceInfo>
<isSupportTime>
  <!--ro, opt, bool, whether the device supports time configuration, desc:for traffic device, this node is required and must be true. For other devices,
the isSupportTimeCap is used to configure time-->true
</isSupportTime>
<SysCap>
  <!--ro, opt, object, system capability-->
  <isSupportDst>
    <!--ro, opt, bool, whether the device supports DST (DayLight Saving Time)-->true
  </isSupportDst>
  <NetworkCap>
    <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
    <isSupportWireless>
      <!--ro, req, bool, whether the device supports wireless network-->true
    </isSupportWireless>
    <isSupportWired>
      <!--ro, opt, bool, whether it supports wired network-->true
    </isSupportWired>
    <isSupportWAN>
      <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
    </isSupportWAN>
    <isSupportPPPoE>
      <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
    </isSupportPPPoE>
    <isSupportBond>
      <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
    </isSupportBond>
    <isSupport802_1x>
      <!--ro, req, bool, whether the device supports 802.1x protocol-->true
    </isSupport802_1x>
    <isSupportNtp>
      <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
    </isSupportNtp>
    <isSupportFtp>
      <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
    </isSupportFtp>
    <isSupportUpnp>
      <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
    </isSupportUpnp>
    <isSupportPNP>
      <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
    </isSupportPNP>
    <isSupportDdns>
      <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
    </isSupportDdns>
    <isSupportHttps>
      <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
    </isSupportHttps>
    <isSupport28181>
      <!--ro, opt, bool, N/A-->true
    </isSupport28181>
    <SnmpCap>
      <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
      <isSupport>
        <!--ro, req, bool, whether the device supports SNMP-->true
      </isSupport>
    </SnmpCap>
    <isSupportExtNetCfg>
      <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
    </isSupportExtNetCfg>
    <isSupportIPFilter>
      <!--ro, opt, bool, whether the device supports IP filtering-->true
    </isSupportIPFilter>
    <isSupportSSH>
      <!--ro, opt, bool, whether the device supports SSH-->true
    </isSupportSSH>
    <isSupportNetPreviewStrategy>
      <!--ro, opt, bool, whether the device supports network preview strategy-->true
    </isSupportNetPreviewStrategy>
    <isSupportEZVIZ>
      <!--ro, opt, bool, whether the device supports EZ protocol-->true
    </isSupportEZVIZ>
    <isSupportEhome>
      <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
    </isSupportEhome>
    <isSupportWirelessDial>
      <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
    </isSupportWirelessDial>
    <isSupportWirelessServer>
      <!--ro, opt, bool, whether the device supports wireless server-->true
    </isSupportWirelessServer>

```

```

</isSupportWirelessServer>
<isSupportWPS>
  <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
</isSupportWPS>
<isWirelessMutexWithWirelessServer>
  <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
</isWirelessMutexWithWirelessServer>
<isSupportMACFilter>
  <!--ro, opt, bool, whether the device supports physical address filtering-->true
</isSupportMACFilter>
<GB28181Cap>
  <!--ro, opt, object, N/A-->
  <isSupportGB28181Service>
    <!--ro, opt, bool, N/A-->true
  </isSupportGB28181Service>
</GB28181Cap>
<WPS>
  <!--ro, opt, object, Wi-Fi Protected Setup-->
  <NetworkInterfaceList size="2">
    <!--ro, opt, array, List of network nodes, subType:object, attr:size{opt, string}-->
    <NetworkInterface>
      <!--ro, opt, object, network node-->
      <id>
        <!--ro, req, string, NIC index-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable or not-->true
      </enabled>
      <isSupportAutoConnect>
        <!--ro, opt, bool, whether the device supports automatic connection-->true
      </isSupportAutoConnect>
      <isSupportDevicePinCode>
        <!--ro, opt, bool, whether the device supports PIN code-->true
      </isSupportDevicePinCode>
      <isSupportDevicePinCodeUpdate>
        <!--ro, opt, bool, whether the device supports updating the PIN code-->true
      </isSupportDevicePinCodeUpdate>
      <ApPinCode>
        <!--ro, opt, object, PIN code of the access point-->
        <ssid min="1" max="1">
          <!--ro, opt, string, SSID (Service Set Identifier), attr:min{opt, string},max{opt, string}-->test
        </ssid>
        <pinCode min="1" max="1">
          <!--ro, opt, string, PIN code, attr:min{opt, string},max{opt, string}-->test
        </pinCode>
      </ApPinCode>
    </NetworkInterface>
  </NetworkInterfaceList>
</WPS>
<isSupportRFIDData>
  <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
</isSupportRFIDData>
<isSupportwifiProbeSSID>
  <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
</isSupportwifiProbeSSID>
<isSupportWifiProbe>
  <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
</isSupportWifiProbe>
<verificationCode min="1" max="1">
  <!--ro, opt, string, verification code, attr:min{opt, string},max{opt, string}-->test
</verificationCode>
<WPSCap>
  <!--ro, opt, object, capability of configuring Wi-Fi Protected Setup-->
  <isSupport>
    <!--ro, req, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupport>
  <isSupportAutoConnect>
    <!--ro, req, bool, whether the device supports automatic connection-->true
  </isSupportAutoConnect>
</WPSCap>
<NetworkMode>
  <!--ro, opt, object, network mode configuration-->
  <workMode>
    <!--ro, opt, string, network mode-->true
  </workMode>
</NetworkMode>
<VerificationCodeModification>
  <!--ro, opt, object, device verification code configuration-->
  <verificationCodeType opt="normal,empty">
    <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
  </verificationCodeType>
  <isSupportDeclarationURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
  </isSupportDeclarationURL>
  <isSupportPrivacyPolicyURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
  </isSupportPrivacyPolicyURL>
  <verificationCodeModify opt="true,false">
    <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{opt, string}-->true
  </verificationCodeModify>
  <Hyperlinks>
    <!--ro, opt, object, hyperlink configuration-->
    <declarationURL>
      <!--ro, opt, string, hyperlink to the Terms of Service, attr:opt{opt, string}-->
    </declarationURL>
  </Hyperlinks>

```

```

<!--ro, opt, string, hyperlink to the terms of service-->test
</declarationURL>
<privacyPolicyURL>
<!--ro, opt, string, hyperlink to the Privacy Policy-->test
</privacyPolicyURL>
</Hyperlinks>
<isSupportVerificationCodeCheck>
<!--ro, opt, bool, whether the device supports verifying the verification code-->true
</isSupportVerificationCodeCheck>
<isSupportOldVerificationCode>
<!--ro, opt, bool, whether the device supports the EZ password configuration of old version (the password of old version consists of six
characters)-->true
</isSupportOldVerificationCode>
<maintenanceVerificationCodeType opt="normal,empty">
<!--ro, opt, string, maintenance verification code type, attr:opt{opt, string}, desc:used to identify the verificationCode type in
ISAPI/System/Network/EZVIZ?security=<security>&iv=<iv>&moduleType=<moduleType>, valid only when the moduleType in the URL is maintenanceModule-->empty
</maintenanceVerificationCodeType>
<maintenanceVerificationCodeModify opt="true,false">
<!--ro, opt, bool, whether the device maintenance verification code has been modified, attr:opt{opt, string}, desc:used to identify the
verificationCode type in ISAPI/System/Network/EZVIZ?security=<security>&iv=<iv>&moduleType=<moduleType>, valid only when the moduleType in the URL is
maintenanceModule-->true
</maintenanceVerificationCodeModify>
</VerificationCodeModification>
<isSupportIntegrate>
<!--ro, opt, bool, whether the device supports access protocol configuration-->true
</isSupportIntegrate>
<isSupportPlatformAccess>
<!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome and GB28181 allLowList configuration))-->true
</isSupportPlatformAccess>
<isSupportIntelligentBoost>
<!--ro, opt, bool, whether the device supports intelligent acceleration-->true
</isSupportIntelligentBoost>
<isSupportResourceStatistics>
<!--ro, opt, bool, whether the device supports getting the network resource information-->true
</isSupportResourceStatistics>
<isSupportBandwidthLimit>
<!--ro, opt, bool, whether the device supports bandwidth limit-->true
</isSupportBandwidthLimit>
<isSupportPOEPortsDisableAdaptiveServer>
<!--ro, opt, bool, whether the device supports the POE protocol-->true
</isSupportPOEPortsDisableAdaptiveServer>
<isSupportPOEConfiguration>
<!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
</isSupportPOEConfiguration>
<isSupportGetLinkSocketIP>
<!--ro, opt, bool, whether the device supports getting the SocketIP of the current Link-->true
</isSupportGetLinkSocketIP>
<isSupportWebSocket>
<!--ro, opt, bool, whether the device supports WebSocket-->true
</isSupportWebSocket>
<isSupportWebSockets>
<!--ro, opt, bool, whether the device supports WebSockets-->true
</isSupportWebSockets>
<Adaption>
<!--ro, opt, object, network self-adaption configuration-->
<streamType opt="0,1,2,3,4,5,7,8,9,10">
<!--ro, opt, string, stream type that supports network self-adaption during Live view, attr:opt{opt, string}, desc:0 (main stream), 1 (sub-
stream), 2 (third stream), 3 (fourth stream/virtual stream), 4 (fifth stream), 5 (sixth stream), 6 (seventh stream), 7 (eighth stream), 8 (ninth stream), 9 (tenth stream),
and so forth. The index value here is the same as that in the Live view structure NET_DVR_PREVIEWINFO of Device Network SDK. Note that 6 is
an invalid value, and from the seventh stream (index value: 7), the index value is the same as that for Live view via RTSP-->test
</streamType>
<isSupportPlayback>
<!--ro, opt, bool, whether the device supports playback-->true
</isSupportPlayback>
</Adaption>
<isSupportVideoImgDB>
<!--ro, opt, bool, whether the device supports image and video library configuration-->true
</isSupportVideoImgDB>
<isSupportVideoImgLibService>
<!--ro, opt, bool, whether the device supports image and video library service-->true
</isSupportVideoImgLibService>
<isSupportWirelessWifi>
<!--ro, opt, bool, whether the device supports Wi-Fi status-->true
</isSupportWirelessWifi>
<isSupportPrivatePlatomo>
<!--ro, opt, bool, whether the device supports accessing third party platform-->true
</isSupportPrivatePlatomo>
<isSupportGantryETCServer>
<!--ro, opt, bool-->true
</isSupportGantryETCServer>
<isSupportEventDataOverWebSocket opt="true,false">
<!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
</isSupportEventDataOverWebSocket>
<isSupportDynamicHostName>
<!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
</isSupportDynamicHostName>
<isSupportTCPAcceleration>
<!--ro, opt, bool, whether the device supports TCP acceleration-->true
</isSupportTCPAcceleration>
<isSupportFlowControl>
<!--ro, opt, bool, whether the device supports traffic control-->true
</isSupportFlowControl>
<isSupportEZVIZTiming>

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    <!--ro, opt, bool, whether the device supports synchronizing time with EZ platform, desc:related URI: enabledTiming in the message of
/ISAPI/System/Network/EZVIZ-->true
    </isSupportEZVIZTiming>
    <isSupportTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json-->true
    </isSupportTrafficMonitor>
    <isSupportDMSAuthInfo>
    <!--ro, opt, bool, whether the device supports DMS authentication information management-->true
    </isSupportDMSAuthInfo>
    <isSupportWiredandWirelessTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC), desc:related URI:
/ISAPI/System/Network/Interfaces/<ID>/trafficMonitor?format=json-->true
    </isSupportWiredandWirelessTrafficMonitor>
    <isSupport4G>
    <!--ro, opt, bool, whether the device supports 4G module-->true
    </isSupport4G>
    <isSupport4GConfig>
    <!--ro, opt, bool, whether the device supports 4G configuration-->true
    </isSupport4GConfig>
    <isSupportGRIDServer>
    <!--ro, opt, bool, N/A-->true
    </isSupportGRIDServer>
    <isSupportGRIDInfo>
    <!--ro, opt, bool, N/A-->true
    </isSupportGRIDInfo>
    <isSupportSipTransmit>
    <!--ro, opt, bool, whether the device supports configuring SIP transmission parameters-->true
    </isSupportSipTransmit>
    <isSupportSip>
    <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
    </isSupportSip>
    <isSupportWifiCascade>
    <!--ro, opt, bool, whether the device supports enabling automatic Wi-Fi cascading, desc:related URI: /ISAPI/System/Network/wifiCascade?format=json--
>true
    </isSupportWifiCascade>
    <isSupportRelativeInfo>
    <!--ro, opt, bool, whether the device supports getting the information about the connected devices, desc:related URI:
/ISAPI/System/Network/wifiCascade/relativeInfo?format=json-->true
    </isSupportRelativeInfo>
    <isSupportEmailEncrypt>
    <!--ro, opt, bool, whether the device supports email encryption configuration-->true
    </isSupportEmailEncrypt>
    <isSupportPortFilter>
    <!--ro, opt, bool, whether the device supports port filtering-->true
    </isSupportPortFilter>
    <isSupportPingDeny>
    <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
    </isSupportPingDeny>
    <isSupportDOSDeny>
    <!--ro, opt, bool, whether the device supports configuring parameters to prevent the DoS attack-->true
    </isSupportDOSDeny>
    <isSupportNetworkStatus>
    <!--ro, opt, bool, whether the device supports searching for the network status, desc:related URI: /ISAPI/System/Network/status?format=json-->true
    </isSupportNetworkStatus>
    <isSupportEZVIZQRCode>
    <!--ro, opt, bool, whether the device supports getting the EZ QR code, desc:related URI: /ISAPI/System/Network/EZVIZ/QRCode?format=json-->true
    </isSupportEZVIZQRCode>
    <isSupportEZVIZUnbind>
    <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
    </isSupportEZVIZUnbind>
    <isSupportPingTest>
    <!--ro, opt, bool, whether the device supports ping test, desc:Related URI: /ISAPI/System/Network/pingtest/capabilities-->true
    </isSupportPingTest>
    <isSupportMOTCloudPlatform>
    <!--ro, opt, bool, whether the device supports configuring parameters for accessing the cloud platform service, desc:related URI:
/ISAPI/System/Network/Traffic/MOTCloudPlatform/capabilities?format=json-->true
    </isSupportMOTCloudPlatform>
    <isSupportSpectrumScan>
    <!--ro, opt, bool, whether the device supports frequency spectrum scanning, desc:related URI:
/ISAPI/System/Network/interface/<netCardID>/spectrumScan/capabilities?format=json-->true
    </isSupportSpectrumScan>
    <isSupportFRP>
    <!--ro, opt, bool, whether the device supports FRP (Fast Reverse Proxy), desc:related URI: /ISAPI/Syetem/Network/FRP?format=json-->true
    </isSupportFRP>
    <isSupport5GModule>
    <!--ro, opt, bool, whether the device supports 5G module configuration, desc:related URI: /ISAPI/System/Network/5GModule/capabilities?format=json--
>true
    </isSupport5GModule>
    <isSupportWirelessDialRoute>
    <!--ro, opt, bool, whether the device supports configuring parameters of route forwarding rules of wireless dial-up, desc:related URI:
/ISAPI/System/Network/WirelessDial/route/capabilities?format=json-->true
    </isSupportWirelessDialRoute>
    <isSupportNPlusOne>
    <!--ro, opt, bool, whether the device supports N+1 protocol-->true
    </isSupportNPlusOne>
    <supportEnvironmentalStandards opt="HJ212">
    <!--ro, opt, string, environmental monitoring standard, attr:opt{req, string}, desc:N/A-->true
    </supportEnvironmentalStandards>
    <isSupportIntranetSegment>
    <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/intranetSegment/capabilities?format=json-->true

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</isSupportIntranetSegment>
<isSupportDHCPServer>
    <!--ro, opt, bool, whether the device supports DHCP service configuration, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI: /ISAPI/System/Network/DHCP/Server/capabilities?
format=json-->true
</isSupportDHCPServer>
<isSupportClusterIntercom>
    <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is true or not be returned.-->true
</isSupportClusterIntercom>
<isSupportHydrologicMonitoring>
    <!--ro, opt, bool, whether the device supports hydrologic monitoring data communication, desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/HydrologicMonitoring/capabilities?format=json-->true
</isSupportHydrologicMonitoring>
<isWirelessDialMutexWithWlanAndWifi>
    <!--ro, opt, bool, whether the wireless dial-up is mutually exclusive with the WLAN hotspot and Wi-Fi, desc:before a function is enabled, another
one must be disabled first-->true
</isWirelessDialMutexWithWlanAndWifi>
<isSupportSpeechTranscription>
    <!--ro, opt, bool, whether the device supports accessing the speech-to-text transcription platform, desc:if this function is supported, this node
will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/SpeechTranscriptionPlatform/capabilities?format=json-->true
</isSupportSpeechTranscription>
<isSupportAsynPingTest>
    <!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and packet
loss rate of the network using the ping command for knowing about the network condition. Related URI: /ISAPI/System/Network/asynPingTest/capabilities?
format=json-->true
</isSupportAsynPingTest>
<isSupportBasebandCapture>
    <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and
the base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
</isSupportBasebandCapture>
<isSupportCellularNetworkSpeedTest>
    <!--ro, opt, bool, whether the device supports testing the connection speed between the device and the cellular network, desc:you can test the
connection speed of the network where the current device is. Related URI: /ISAPI/System/Network/cellularNetwork/speedTest/capabilities?format=json-->true
</isSupportCellularNetworkSpeedTest>
<isSupportCellularNetworkCellInfo>
    <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the
cellular network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&security=<security>&iv=<iv>-->true
</isSupportCellularNetworkCellInfo>
<isSupportMultiBandFrequencyScanning>
    <!--ro, opt, bool, whether the device supports scanning all frequency bands of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/multiBandFrequencyScanning/capabilities?format=json-->true
</isSupportMultiBandFrequencyScanning>
<isSupportCellularNetworkLockBand>
    <!--ro, opt, bool, whether the device accessing the locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/lockBand/capabilities?format=json-->true
</isSupportCellularNetworkLockBand>
<isSupportRadioFrequencyFluctuation>
    <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
</isSupportRadioFrequencyFluctuation>
<isSupportBasebandBlock>
    <!--ro, opt, bool, whether the device supports the unauthorized list of the base station, desc:related URI:
/ISAPI/System/Network/cellularNetwork/basebandBlock/capabilities?format=json-->true
</isSupportBasebandBlock>
<isSupportRFIDTagCheckTest>
    <!--ro, opt, bool, whether the device supports the RFID tag disk storage performance test, desc:related URI: GET
/ISAPI/System/Network/RFIDData/GetTagCheckTestInfo?format=json-->true
</isSupportRFIDTagCheckTest>
<isSupportQoSdSCPtoLoaclPriorityMapping>
    <!--ro, opt, bool, whether the device supports DSCP and local mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/dSCPtoLoaclPriorityMapping/capabilities?format=json-->true
</isSupportQoSdSCPtoLoaclPriorityMapping>
<isSupportQoS802.1pTo802.1pPriorityMapping>
    <!--ro, opt, bool, whether the device supports 802.1p mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/802.1pTo802.1pPriorityMapping/capabilities?format=json-->true
</isSupportQoS802.1pTo802.1pPriorityMapping>
<isSupportLinkTestStart>
    <!--ro, opt, bool, whether the network link test is enabled, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
</isSupportLinkTestStart>
<isSupportLinkTestResult>
    <!--ro, opt, bool, whether the device supports getting the result of network link test, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
</isSupportLinkTestResult>
<isSupportNetworkModuleUpgrade>
    <!--ro, opt, bool, whether it supports upgrading network module, desc:including importing upgrade package
(/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
(/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrade
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json)-->true
</isSupportNetworkModuleUpgrade>
<isSupportNetworkModuleVersionInfo>
    <!--ro, opt, bool, whether it supports getting the version information of network module, desc:GET
/ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
</isSupportNetworkModuleVersionInfo>
<isSupportGreenShieldPlatform>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/GreenShieldPlatform/GreenShieldPlatform?format=json-->true
</isSupportGreenShieldPlatform>
<isSupportFusionUniversalApplicationPlatCfg>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/FusionUniversalApplicationPlatform/FusionUniversalApplicationPlatCfg?format=json-->true
</isSupportFusionUniversalApplicationPlatCfg>
<isSupportModbusAccess>

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<!--ro, opt, bool, whether the device supports Modbus accessing parameters, desc:GET/PUT /ISAPI/System/Network/ModbusAccess?format=json-->true
</isSupportModbusAccess>
<isSupportModbusTestSubordinateStatus>
<!--ro, opt, bool, whether it supports Modbus testing specified client status, desc:POST /ISAPI/System/Network/ModbusTestSubordinateStatus?
format=json-->true
</isSupportModbusTestSubordinateStatus>
<isSupportGetModbusAndRegisterMap>
<!--ro, opt, bool, whether it supports getting the mapping relationship between Modbus uploaded content and register quantity, desc:GET
/ISAPI/System/Network/GetModbusAndRegisterMap?format=json-->true
</isSupportGetModbusAndRegisterMap>
<isSupportTSOverWebSocket>
<!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
</isSupportTSOverWebSocket>
<isSupportRequestOutlookAuth>
<!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
</isSupportRequestOutlookAuth>
<isSupportOutlookEmailList>
<!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
</isSupportOutlookEmailList>
<isSupportDSTPServerCfg>
<!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET or PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
</isSupportDSTPServerCfg>
<isSupportDSTPProtocolsCap>
<!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
</isSupportDSTPProtocolsCap>
<isSupportSearchOutlookAuthResult>
<!--ro, opt, bool-->true
</isSupportSearchOutlookAuthResult>
<RTDROverWebSocket>
<!--ro, opt, object-->
<isSupportGasSpectrumRealTimeData>
<!--ro, opt, bool-->true
</isSupportGasSpectrumRealTimeData>
<isSupportDangerousGasSensorRealTimeData>
<!--ro, opt, bool-->true
</isSupportDangerousGasSensorRealTimeData>
<isSupportComposeWindConditionRealTimeData>
<!--ro, opt, bool-->true
</isSupportComposeWindConditionRealTimeData>
<isSupportMeteorologySensorRealTimeData>
<!--ro, opt, bool-->true
</isSupportMeteorologySensorRealTimeData>
<RTDROverWebSocket>
<isSupportServerCfg>
<!--ro, opt, bool-->true
</isSupportServerCfg>
<isSupportDelimiterCfg>
<!--ro, opt, bool-->true
</isSupportDelimiterCfg>
<isSupportStartCellNetSelFdiagnosis>
<!--ro, opt, bool-->true
</isSupportStartCellNetSelFdiagnosis>
<isSupportCellNetSelFdiagnosisStatus>
<!--ro, opt, bool-->true
</isSupportCellNetSelFdiagnosisStatus>
</NetworkCap>
<IOCap>
<!--ro, opt, object, I/O capability, desc:related URI: /ISAPI/System/IO/capabilities-->
<IOInputPortNums>
<!--ro, opt, int, number of local input ports of the device-->true
</IOInputPortNums>
<IOOutputPortNums>
<!--ro, opt, int, number of local output ports of the device-->true
</IOOutputPortNums>
<isSupportStrobeLamp>
<!--ro, opt, bool, whether the device supports configuring the output of the gate's light, desc:related URI:
/ISAPI/System/IO/outputs/strobeLampConf-->true
</isSupportStrobeLamp>
<SoftIOInputPortNums>
<!--ro, opt, int, number of soft input ports of the device, desc:this node depends on whether the URI /ISAPI/System/IO/softInputs/capabilities?
format=json is supported-->true
</SoftIOInputPortNums>
<isSupportIOOutputAdvanceParameter>
<!--ro, opt, bool, whether the device supports configuring advanced parameters of the output port, desc:this node determines whether the URI
/ISAPI/System/IO/outputs/advanceParameter/capabilities?format=json is supported-->true
</isSupportIOOutputAdvanceParameter>
<isSupportSetAllIOOutput>
<!--ro, opt, bool, whether the device supports configuring all output ports, desc:this node determines whether the PUT method of the URI
/ISAPI/System/IO/outputs is supported-->true
</isSupportSetAllIOOutput>
<isSupportCombinationAlarm>
<!--ro, opt, bool, whether the device supports the combined alarm, desc:CombinationAlarmCap in /ISAPI/System/IO/inputs/capabilities-->true
</isSupportCombinationAlarm>
<enabledIOOutputPortNums>
<!--ro, opt, int, number of enabled alarm outputs-->0
</enabledIOOutputPortNums>
<isSupportAlarmKeyParam>
<!--ro, opt, bool, whether supports configuring the physical keys of one-push alarm, desc:related URI:
/ISAPI/System/IO/inputs/<IOInputID>/AlarmKeyParam/capabilities?format=json-->true
</isSupportAlarmKeyParam>

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<isSupportWebDisplayAlarmStatus>
  <!--ro, opt, bool, whether the web supports displaying the alarm output status, desc:used for web interface display, true indicates that the web
  supports displaying the alarm output status; otherwise, it does not support-->true
</isSupportWebDisplayAlarmStatus>
</IOCap>
<SerialCap>
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
  </rs232PortNums>
  <rs485WorkMode opt="Led,CaptureTrigger">
    <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:Led (external LED screen, which is used for LED
    display), CaptureTrigger (external device, which is used to trigger the transmission of the captured data)-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, string, range of RS-232 serial port numbers supported by the device, attr:opt{opt, string}-->test
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, string, range of RS-485 serial port numbers supported by the device, attr:opt{opt, string}-->test
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the Linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
<VideoCap>
  <!--ro, opt, object, video encoding capability, desc:related URI: /ISAPI/System/Video/capabilities-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->true
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->true
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of Local menus that can be displayed-->true
  </menuNums>
  <externalChannelNum>
    <!--ro, opt, bool, number of extended analog channels-->true
  </externalChannelNum>
  <isSupportHeatmap>
    <!--ro, opt, bool, whether the device supports heat map-->true
  </isSupportHeatmap>
  <isSupportCounting>
    <!--ro, opt, bool, whether the device supports people counting-->true
  </isSupportCounting>
  <countingType>
    <!--ro, opt, string, statistics type, desc:human (people counting); object (counting)-->test
  </countingType>
  <isSupportPicture>
    <!--ro, opt, bool, whether the device supports OSD picture overlay-->true
  </isSupportPicture>
  <isSupportCapturePicOverlays>
    <!--ro, opt, bool, whether the device supports text overlay on a single picture-->true
  </isSupportCapturePicOverlays>
  <isSupportMergePicOverlays>
    <!--ro, opt, bool, whether the device supports text overlay on composite picture-->true
  </isSupportMergePicOverlays>
  <isSupportPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view-->true
  </isSupportPreviewSwitch>
  <isSupportMultiChannelCounting>
    <!--ro, opt, bool, whether the device supports multichannel statistics-->true
  </isSupportMultiChannelCounting>
  <isSupportCountingCollection>
    <!--ro, opt, bool, whether the device supports people counting ANR-->true
  </isSupportCountingCollection>
  <isSupportHeatmapCollection>
    <!--ro, opt, bool, whether the device supports heat map data ANR-->true
  </isSupportHeatmapCollection>
  <channelFlexible opt="name,online,linknum">
    <!--ro, opt, string, this node is located after the node <isSupportHeatmapCollection>., attr:opt{opt, string}-->true
  </channelFlexible>
  <isSupportOutputsResource>
    <!--ro, opt, bool-->true
  </isSupportOutputsResource>
  <isSupportMixedChannel>
    <!--ro, opt, bool, this node is located after the node <channelFlexible>.-->true
  </isSupportMixedChannel>
  <isSupportMixedChannelStatus>
    <!--ro, opt, bool, this node is located after the node <isSupportMixedChannel>.-->true
  </isSupportMixedChannelStatus>

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</isSupportMixedChannelStatus>
<isSupportRecodStatus>
  <!--ro, opt, bool, whether the device supports searching for the recording status-->true
</isSupportRecodStatus>
<isSupportPrivacyMask>
  <!--ro, opt, bool, whether the device supports privacy mask-->true
</isSupportPrivacyMask>
<isSupportBinocularPreviewSwitch>
  <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
</isSupportBinocularPreviewSwitch>
<isSupportCalibCheck>
  <!--ro, opt, bool, whether the device supports calibration verification-->true
</isSupportCalibCheck>
<isSupportPIP>
  <!--ro, opt, bool, whether the device supports PIP-->true
</isSupportPIP>
<isSupportFocusVideoMode>
  <!--ro, opt, bool, whether the device supports the focus mode of the video-->true
</isSupportFocusVideoMode>
<isSupportExternalChannel>
  <!--ro, opt, bool, whether the device supports extended analog channels-->true
</isSupportExternalChannel>
<isSupportCompressedTranscode>
  <!--ro, opt, bool, whether the device supports configuring the encoding compression of transcoding stream, desc:related URL:
/ISAPI/Streaming/channels/<channelID>/TranscodeCfg/capabilities?format=json-->true
</isSupportCompressedTranscode>
<isSupportCompressedStream>
  <!--ro, opt, bool, whether the device supports transcoding compressed stream, desc:it is valid when isSupportCompressedTranscode is true. The device
supports getting real-time stream and real-time compressed stream. The streaming channel stream type can be obtained via id when the serviceType is
transcode (URL: /ISAPI/ContentMgmt/StreamingProxy/channels)-->true
</isSupportCompressedStream>
<isSupportLocalStreamCfg>
  <!--ro, opt, bool, whether the device supports configuring the streaming parameters of Local device, desc:related URL:
/ISAPI/System/Video/inputs/channels/<channelID>/LocalStreamCfg/capabilities?format=json-->true
</isSupportLocalStreamCfg>
<isSupportMenuStatus>
  <!--ro, opt, bool, whether it supports getting the menu status, desc:corresponding URI: /ISAPI/System/Video/Menu/<indexID>/status?format=json, the
range of indexID comes from VideoCap.menuNums-->true
</isSupportMenuStatus>
<isSupportVideoOutputMode>
  <!--ro, opt, bool, whether the device supports video output pattern parameters-->true
</isSupportVideoOutputMode>
<pictureChannelList>
  <!--ro, opt, array, subType:object-->
  <channelID>
    <!--ro, opt, int-->0
  </channelID>
</pictureChannelList>
<transcodeStreamType opt="previewTranscode,recordTranscode">
  <!--ro, opt, string, attr:opt{req, string}-->true
</transcodeStreamType>
<isSupportImportPanoramaIOOverlayPicture>
  <!--ro, opt, bool, POST /ISAPI/System/Video/channels/<channelID>/ImportPanoramaIOOverlayPicture?format=json, desc:POST
/ISAPI/System/Video/channels/<channelID>/ImportPanoramaIOOverlayPicture?format=json-->true
</isSupportImportPanoramaIOOverlayPicture>
<isSupportPanoramaIOOverlayPictureParams>
  <!--ro, opt, bool, GET or PUT /ISAPI/System/Video/channels/<channelID>/PanoramaIOOverlayPictureParams?format=json, desc:GET or PUT
/ISAPI/System/Video/channels/<channelID>/PanoramaIOOverlayPictureParams?format=json-->true
</isSupportPanoramaIOOverlayPictureParams>
</VideoCap>
<AudioCap>
  <!--ro, opt, object, audio encoding capability, desc:related URL: /ISAPI/System/Audio/capabilities-->
  <audioInputNums>
    <!--ro, req, int, number of audio inputs-->1
  </audioInputNums>
  <audioOutputNums>
    <!--ro, req, int, number of audio outputs-->1
  </audioOutputNums>
  <mixAudioInSet>
    <!--ro, opt, bool, N/A-->true
  </mixAudioInSet>
  <mixAudioOutSet>
    <!--ro, opt, bool, N/A-->true
  </mixAudioOutSet>
  <isSupportAudioMixing>
    <!--ro, opt, bool, whether the device supports audio mixing-->true
  </isSupportAudioMixing>
  <isSupportRemoveFrequencyPoint>
    <!--ro, opt, bool, whether the device supports clearing the howling frequency point-->true
  </isSupportRemoveFrequencyPoint>
  <isSupportAudioInConfig>
    <!--ro, opt, bool, whether the device supports configuring parameters of all audio inputs, desc:related URI:
/ISAPI/System/Audio/AudioIn/capabilities-->true
  </isSupportAudioInConfig>
  <muteDuringPanTilt opt="MicIn">
    <!--ro, opt, string, whether the device supports muting during motion, attr:opt{opt, string}, desc:muteDuringPanTilt in
/ISAPI/System/TwoWayAudio/channels/ID-->true
  </muteDuringPanTilt>
  <isSupportAudioOutConfig>
    <!--ro, opt, bool, whether the device supports configuring parameters of all audio outputs, desc:related URI:
/ISAPI/System/Audio/AudioOut/capabilities?format=json-->true
  </isSupportAudioOutConfig>
  <isSupportSAIP>

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    <!--ro, opt, bool, whether the device supports connecting to the independent audio track channel-->true
</isSupportSAIP>
<SAIPChannelsNum>
    <!--ro, opt, int, number of devices that supports connecting to the independent audio track channel, dep:and,{$.AudioCap.isSupportSAIP,eq,true}-->1
</SAIPChannelsNum>
<isSupportDataStorage>
    <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URL: /ISAPI/System/Audio/dataStorage?format=json-->true
</isSupportDataStorage>
<isSupportDataRelease>
    <!--ro, opt, bool, whether the device supports releasing the audio file, desc:related URL: /ISAPI/System/Audio/dataRelease?format=json-->true
</isSupportDataRelease>
<isSupportAudioPlan>
    <!--ro, opt, bool, whether the device supports the audio scheme plan, desc:related URL: /ISAPI/System/Audio/plan?format=json-->true
</isSupportAudioPlan>
<isSupportAudioTask>
    <!--ro, opt, bool, whether the device supports the audio scheme task, desc:related URL: /ISAPI/System/Audio/plan/<PlanID>/task?format=json-->true
</isSupportAudioTask>
<isSupportaudioCollection>
    <!--ro, opt, bool, whether the device supports collecting audio files, desc:related URL: /ISAPI/System/Audio/audioCollection/capabilities?
format=json; maintenance function, it is used for collecting audio to analyze audio questions-->true
</isSupportaudioCollection>
<isSupportWirelessPickUp>
    <!--ro, opt, bool, whether the device supports connecting wireless pickup, desc:related URL: /ISAPI/System/Audio/wirelessPickUp/capabilities?
format=json-->true
</isSupportWirelessPickUp>
<audioOutputBroadcastNums opt="1,2,3,4">
    <!--ro, opt, int, number of audio broadcast outputs, attr:opt{opt, string}, desc:at present, this node is only supported by broadcast audio devices
when opt="1,2,3,4" is returned and , the output depends on the actual node; when not returned, the output starts from 1 by default according to business
type, the audio devices classify the output ports into broadcast output port and audio output port considering the start No. of every output starts from 1,
it should return the specific number of supported broadcast outputs if this node is returned, it indicates that the device supports the audio output volume
configuration; otherwise, the device does not support. Corresponding URI: /ISAPI/System/Audio/AudioOut/broadcast/capabilities?format=json-->0
</audioOutputBroadcastNums>
<isSupportAudioOutMixing>
    <!--ro, opt, bool, whether the device supports outputting audio mix, desc:related URL: /ISAPI/System/Audio/AudioOut/mixing/capabilities?format=json-
->true
</isSupportAudioOutMixing>
<isSupportRTSPAUDIO>
    <!--ro, opt, bool, whether the device supports listening to the audio transmitted via RTSP in real time, desc:RTSP URI:
/ISAPI/System/Audio/AudioIn/channels/<audioID>-->true
</isSupportRTSPAUDIO>
<isSupportRTSPAUDIOPlaybackByFile>
    <!--ro, opt, bool, whether the device supports playing back audio transmitted via RTSP according to files, desc:the RTSP streaming URL is
playbackURI returned by /ISAPI/ContentMgmt/recordAudio/search?format=json-->true
</isSupportRTSPAUDIOPlaybackByFile>
<isSupportDeviceAudioCapture>
    <!--ro, opt, bool, whether the device supports audio collection, desc:GET/PUT /ISAPI/System/Audio/DeviceAudioCapture?format=json-->true
</isSupportDeviceAudioCapture>
</AudioCap>
<isSupportHoliday>
    <!--ro, opt, bool, this node is located after the node <AudioCap>-->true
</isSupportHoliday>
<RebootConfigurationCap>
    <!--ro, opt, object, capability of configuring parameters for rebooting the device, desc:related URI: /ISAPI/System/reboot/capabilities. This node is
located before the node <isSupportSubscribeEvent> and after the node <isSupportHoliday>-->
<Genetec>
    <!--ro, opt, bool, whether to enable the Genetec protocol-->true
</Genetec>
<ONVIF>
    <!--ro, opt, bool, whether to enable the ONVIF protocol-->true
</ONVIF>
<RTSP>
    <!--ro, opt, bool, whether to enable the RTSP protocol-->true
</RTSP>
<HTTP>
    <!--ro, opt, bool, whether to enable the HTTP protocol-->true
</HTTP>
<SADP>
    <!--ro, opt, object, SADP (Search Active Devices Protocol, including the Link Layer and multicast discovery)-->
<ISDiscoveryMode>
    <!--ro, opt, bool, mode of the SADP multicast discovery protocol-->true
</ISDiscoveryMode>
<PcapMode>
    <!--ro, opt, bool, mode of the SADP Link Layer discovery protocol-->true
</PcapMode>
</SADP>
<IPCAddStatus>
    <!--ro, opt, bool, IP address status-->true
</IPCAddStatus>
</RebootConfigurationCap>
<isSupportExternalDevice>
    <!--ro, opt, bool, whether the device supports connecting to peripherals: true (support), desc:related URI: /ISAPI/System/externalDevice/capabilities-
->true
</isSupportExternalDevice>
<isSupportSoundCfg>
    <!--ro, opt, bool, whether the device supports the sound configuration, desc:related URI: /ISAPI/System/SoundCfg/capabilities-->true
</isSupportSoundCfg>
<isSupportSubscribeEvent>
    <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
</isSupportSubscribeEvent>
<isSupportSubscribeIOTInfo>
    <!--ro, opt, bool, whether the device supports subscribing to the IoT information-->true
</isSupportSubscribeIOTInfo>
<isSupportDiagnosedData>
    <!--ro, opt, bool, whether the device supports exporting the device diagnostic data, desc:true (support), this node is not returned (not support)
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<!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support).
Related URI: /ISAPI/System/diagnosedData-->true
</isSupportDiagnosedData>
<isSupportChangedUpload>
  <!--ro, opt, bool, whether the device supports uploading changes-->true
</isSupportChangedUpload>
<isSupportGettingWorkingStatus>
  <!--ro, opt, bool, whether the device supports getting the device status-->true
</isSupportGettingWorkingStatus>
<isSupportGettingChannelInfoByCondition>
  <!--ro, opt, bool, whether the device supports getting the channel information by condition-->true
</isSupportGettingChannelInfoByCondition>
<isSupportDiagnosedDataParameter>
  <!--ro, opt, bool, whether the device supports enabling or disabling exporting the diagnosis information-->true
</isSupportDiagnosedDataParameter>
<isSupportSimpleDevStatus>
  <!--ro, opt, bool, whether the device supports getting the device's simple status, desc:related URI: /ISAPI/System/workingstatus?format=json-->true
</isSupportSimpleDevStatus>
<isSupportFlexible>
  <!--ro, opt, bool, whether the device supports the flexible combination configuration. Related URI: /ISAPI/System/Video/inputs/channels?format=json--
>true
</isSupportFlexible>
<isSupportPTZChannels>
  <!--ro, opt, bool, whether the device supports returning the PTZ channel (it is distinguished from the video channel)-->true
</isSupportPTZChannels>
<isSupportMetadata>
  <!--ro, opt, bool, whether the device supports metadata, desc:related URI: /ISAPI/Streaming/channels/<ID>/metadata/capabilities-->true
</isSupportMetadata>
<isSupportTimeCap>
  <!--ro, opt, bool, whether the device supports the time configuration, desc:related URI: /ISAPI/System/time/capabilities-->true
</isSupportTimeCap>
<isSupportThermalStreamData>
  <!--ro, opt, bool, whether the device supports uploading thermal imaging stream data in real time-->true
</isSupportThermalStreamData>
<isSupportShutdown>
  <!--ro, opt, bool, whether the device supports the shutdown configuration, desc:related URI: /ISAPI/System/shutdown?format=json-->true
</isSupportShutdown>
<isSupportConsumptionMode>
  <!--ro, opt, bool, whether the device supports switching the consumption mode, desc:related URI: /ISAPI/System/consumptionMode/capabilities?
format=json-->true
</isSupportConsumptionMode>
<isSupportManualPowerConsumption>
  <!--ro, opt, bool, whether the device supports controlling the device consumption manually-->true
</isSupportManualPowerConsumption>
<isSupportPostUpdateFirmware>
  <!--ro, opt, bool, whether the device supports upgrading firmware by POST method-->true
</isSupportPostUpdateFirmware>
<isSupportPostConfigData>
  <!--ro, opt, bool, whether the device supports importing and exporting the configuration file by POST method-->true
</isSupportPostConfigData>
<isSupportUserLock>
  <!--ro, opt, bool, whether the device supports locking the user, desc:related URI: /ISAPI/System/userLock/config/capabilities?format=json-->true
</isSupportUserLock>
<isSupportModuleLock>
  <!--ro, opt, bool, whether the device supports locking the module, desc:related URI: /ISAPI/System/moduleLock-->true
</isSupportModuleLock>
<supportSmartOverlapChannels opt="1">
  <!--ro, opt, bool, whether the device supports configuring the smart event stream, attr:opt[opt, string], desc:if this function is supported, this
node and the corresponding channel ID will be returned; otherwise, this node will not be returned-->true
</supportSmartOverlapChannels>
<isSupportNetworkCapture>
  <!--ro, opt, bool, whether the device supports capturing network packets, desc:this node is used to determine whether the device supports capturing
communication packets transmitted between the device and the external. Related URI: /ISAPI/System/networkCapture/capabilities?format=json-->true
</isSupportNetworkCapture>
<isSupportOperationRecord>
  <!--ro, opt, bool, whether the device supports the device operation records, desc:the charging voltage and monthly flow can be recorded in the device
in real-time low-consumption mode and full-consumption mode. Related URI: /ISAPI/System/operationRecord/capabilities?format=json-->true
</isSupportOperationRecord>
<isSupportSearchModuleLock>
  <!--ro, opt, bool, whether the device support search module locking parameters, desc:/ISAPI/System/moduleLock/searchconfig?format=json-->true
</isSupportSearchModuleLock>
<devSupportUpdateVersion>
  <!--ro, opt, string, supported version of Local device upgrade, desc:the device returns the upgrade version to guide tool users (e.g. service
technician) to select an upgrade package for Local upgrade-->1
</devSupportUpdateVersion>
<notSupportCatalogUpgrade>
  <!--ro, opt, bool-->true
</notSupportCatalogUpgrade>
</SysCap>
<voicetalkNums>
  <!--ro, opt, int, number of two-way audio channels-->2
</voicetalkNums>
<isSupportSnapshot>
  <!--ro, opt, bool, whether the device supports capturing pictures-->true
</isSupportSnapshot>
<SecurityCap>
  <!--ro, opt, object, encryption capability set-->
<supportUserNums>
  <!--ro, opt, int, supported maximum number of users-->1
</supportUserNums>
<userBondIpNums>
  <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->1
</userBondIpNums>
<userBondMacNums>

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    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->1
</userBondMacNums>
<isSupCertificate>
    <!--ro, opt, bool, whether the device supports authentication-->true
</isSupCertificate>
<issupIllegalLoginLock>
    <!--ro, opt, bool, whether the device supports Locking Login-->true
</issupIllegalLoginLock>
<isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
</isSupportOnlineUser>
<isSupportAnonymous>
    <!--ro, opt, bool, whether the device supports anonymous Login-->true
</isSupportAnonymous>
<isSupportStreamEncryption>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
</isSupportStreamEncryption>
<securityVersion opt="1,2,3,4,7">
    <!--ro, opt, string, encryption capability, attr:opt{opt, string}, desc:the encryption capability of each version consists of two parts: encryption
algorithm and the range of encrypted nodes. Currently 1 refers to AES128 encryption and 2 refers to AES256 encryption. The range of encrypted nodes is
described in each protocol-->test
</securityVersion>
<keyIterateNum>
    <!--ro, opt, int, iteration times, desc:the value is usually between 100 and 1000-->100
</keyIterateNum>
<isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the Login password when editing (editing/adding/deleting) user parameters, desc:this node
depends on the node <securityVersion>, which means that it is only valid for the versions that support encrypting the sensitive information-->true
</isSupportUserCheck>
<isSupportGUIDFileDataExport>
    <!--ro, opt, bool, whether the device supports exporting the device's GUID file-->true
</isSupportGUIDFileDataExport>
<isSupportSecurityQuestionConfig>
    <!--ro, opt, bool, whether the device supports answering security questions-->true
</isSupportSecurityQuestionConfig>
<supportSecurityNode opt="wirelessServer,wirelessDial">
    <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{opt, string}, desc:in the earlier version of the
solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI), some URIs containing sensitive
information are left out. Now these URIs need to be encrypted for transmission. This capability node is added to be compatible with old devices-->test
</supportSecurityNode>
<isSupportGetOnlineUserListSC>
    <!--ro, opt, bool, whether the device supports searching the online user list-->true
</isSupportGetOnlineUserListSC>
<SecurityLimits>
    <!--ro, opt, object, capability of configuring security limit parameters-->
    <LoginPasswordLenLimit min="1" max="16">
        <!--ro, opt, string, length limit of the user's login password, attr:min{opt, string},max{opt, string}-->test
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
        <!--ro, opt, string, length limit of the security question's answer, attr:min{opt, string},max{opt, string}-->test
    </SecurityAnswerLenLimit>
</SecurityLimits>
<RSAKeyLength opt="512,1024,2048,3072" def="3072">
    <!--ro, opt, enum, HTTPS certificate length, subType:string, attr:opt{opt, string},def{opt, string}, desc:512, 1024, 2048-->3072
</RSAKeyLength>
<isSupportONVIFUserManagement>
    <!--ro, opt, bool, whether the device supports ONVIF user management-->true
</isSupportONVIFUserManagement>
<WebCertificateCap>
    <!--ro, opt, object, HTTP authentication capability-->
    <CertificateType opt="basic,digest,digest/basic">
        <!--ro, req, string, certificate type: basic (authentication), attr:opt{opt, string}-->test
    </CertificateType>
    <SecurityAlgorithm>
        <!--ro, opt, object, encryption algorithm, dep:or,{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest},
{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
        <algorithmType opt="MD5,SHA256,MD5/SHA256">
            <!--ro, opt, string, algorithm type, attr:opt{req, string}-->test
        </algorithmType>
    </SecurityAlgorithm>
</WebCertificateCap>
<isSupportConfigFileImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file-->true
</isSupportConfigFileImport>
<isSupportConfigFileExport>
    <!--ro, opt, bool, whether the device supports exporting the configuration file-->true
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
    <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{opt, string},max{opt, string}-->0
</cfgFileSecretKeyLenLimit>
<supportIPCActivatePassword>
    <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
</supportIPCActivatePassword>
<isIrreversible>
    <!--ro, opt, bool, whether the device supports irreversible password storage, desc:if this function is not supported, the plaintext password of the
user information will be stored in the device; otherwise, the password will be hashed for storage in the device-->true
</isIrreversible>
<salt>
    <!--ro, opt, string, the specific salt used by the user to log in-->test
</salt>
<isSupportOnvifInfo>
    <!--ro, opt, bool, whether the device supports getting the ONVIF protocol information-->true
</isSupportOnvifInfo>

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<isSupportUserNamePasswordCheckUpgrade>
  <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password-->true
</isSupportUserNamePasswordCheckUpgrade>
<isSupportDeviceCertificatesManagement>
  <!--ro, opt, bool, whether the device supports certificate management-->true
</isSupportDeviceCertificatesManagement>
<isSupportSIPCertificatesManagement>
  <!--ro, opt, bool, whether the device supports SIP certificate management-->true
</isSupportSIPCertificatesManagement>
<isSupportDeviceSelfSignCertExport>
  <!--ro, opt, bool, whether the device supports exporting the self-signed certificate-->true
</isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail>
  <!--ro, opt, bool, whether the device supports configuring the security email-->true
</isSupportSecurityEmail>
<isSupportRTSPCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports RTSP authentication, attr:opt{opt, string}-->true
</isSupportRTSPCertificate>
<isSupportPictureURLCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{opt, string}-->true
</isSupportPictureURLCertificate>
<isSupportGB35114Certificate>
  <!--ro, opt, bool, N/A-->true
</isSupportGB35114Certificate>
<isSupportAuthenticationMode opt="uKey,private">
  <!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:"uKey", "private"-->private
</isSupportAuthenticationMode>
<isSupportUnloggedUserPermissionConfig opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{opt, string}-->true
</isSupportUnloggedUserPermissionConfig>
<maxIllegalLoginTimes min="3" max="20" def="5">
  <!--ro, opt, string, maximum failed illegal login attempts, attr:min{opt, string},max{opt, string},def{opt, string}, desc:this node is valid when the
value of the node <issupportIllegalLoginLock> is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->1
</maxIllegalLoginTimes>
<SecurityAdvanced>
  <!--ro, opt, object, advanced security configuration-->
  <securityEnhanced>
    <!--ro, opt, bool, whether the device supports security reinforcement-->true
  </securityEnhanced>
  <noOperationEnabled>
    <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
  </noOperationEnabled>
  <noOperationTime min="1" max="60" def="15">
    <!--ro, req, int, control timeout when there is no operation, unit:分#, attr:min{opt, string},max{opt, string},def{opt, string}-->1
  </noOperationTime>
  <isSupportDigestStatus>
    <!--ro, opt, bool, digest status-->true
  </isSupportDigestStatus>
</SecurityAdvanced>
<LoginLinkNum>
  <!--ro, opt, object, number of users logging in at the same time-->
  <maxLinkNum min="1" max="128" def="50">
    <!--ro, req, int, number of controls logging in at the same time, attr:min{opt, string},max{opt, string},def{opt, string}-->64
  </maxLinkNum>
</LoginLinkNum>
<isSupportCCClientCertificate>
  <!--ro, opt, bool, whether the device supports the CC (Client Certificate) function-->true
</isSupportCCClientCertificate>
<passwordValidity min="0" max="365">
  <!--ro, opt, string, range of the password validity period, attr:min{opt, string},max{opt, string}, desc:unit: day. If this node is returned, it
indicates that the device supports configuring the password validity period, and only the administrator can edit the configuration-->1
</passwordValidity>
<isSupportResetChannelPassword>
  <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
</isSupportResetChannelPassword>
<DoubleVerificationCap>
  <!--ro, opt, object, secondary authentication capability-->
  <isSupportUsersConfig>
    <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
  </isSupportUsersConfig>
  <isSupportUsersPermissionConfig>
    <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
  </isSupportUsersPermissionConfig>
  <isSupportFactoryReset>
    <!--ro, opt, bool, whether the device supports restoring secondary authentication to defaults-->true
  </isSupportFactoryReset>
  <isSupportQuestionConfiguration>
    <!--ro, opt, bool, whether it supports configuring a single security question for secondary authentication, desc:PUT
/ISAPI/Security/questionConfiguration/<indexID>-->true
  </isSupportQuestionConfiguration>
</DoubleVerificationCap>
<isSupportUserSearch>
  <!--ro, opt, bool, whether the device supports searching users-->true
</isSupportUserSearch>
<keypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length, attr:min{req, int},max{req, int}, desc:when the password length of different user types is different, this
node only indicates the keypad password length of the administrator; otherwise, the keypad password length of all user types is the same and defined by the
value of this node-->1
</keypadPassword>
<installerKeypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length of the installer, attr:min{req, int},max{req, int}-->1
</installerKeypadPassword>
<operatorKeypadPassword min="1" max="10">

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    <!--ro, opt, int, keypad password length of the operator, attr:min{req, int},max{req, int}-->1
</operatorKeypadPassword>
<userOperateType opt="1,2,3">
    <!--ro, opt, enum, user type: 1 (network user), subType:int, attr:opt{req, string}, desc:1 (network user), 2 (keypad user), 3 (network user and keypad
user)-->1
</userOperateType>
<isSptUserEnabled>
    <!--ro, opt, bool, whether the device supports configuring parameters for enabling the user-->true
</isSptUserEnabled>
<isSptAdminCap>
    <!--ro, opt, bool, whether the device supports getting the admin permission-->true
</isSptAdminCap>
<isSupportBackLogCfg>
    <!--ro, opt, bool, whether the device supports configuring basic parameters of the background Log-->true
</isSupportBackLogCfg>
<isSupportBackLogExport>
    <!--ro, opt, bool, whether the device supports exporting the background Log-->true
</isSupportBackLogExport>
<isSupportModifyInnerPassword opt="true,false">
    <!--ro, opt, bool, whether the device supports editing the built-in password, attr:opt{req, string}, desc:the device's built-in passwords include the
database password, system account password and so on-->true
</isSupportModifyInnerPassword>
<isSupportEncryptCertificate>
    <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
</isSupportEncryptCertificate>
<maxIllegalLoginLockTime min="1" max="120" def="30">
    <!--ro, opt, int, Lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
</maxIllegalLoginLockTime>
<cloudSensitiveInfoEncryptType>
    <!--ro, opt, string, encryption type of the cloud sensitive information, desc:the current device returns the value "2.1", which means that the method
of generating the key for encrypting the sensitive information is SHA256(calC_sha256(user, salt, SHA256(user+salt+SHA256(user+salt+password))))+
AaBbCcDd1234!@#$.-->2.1
</cloudSensitiveInfoEncryptType>
<isSupportCertificateCustomID>
    <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
</isSupportCertificateCustomID>
<isSupportResetLANPassword>
    <!--ro, opt, bool, whether the device supports resetting the device's password in the LAN, desc:related URI:
/ISAPI/Security/resetPassword/certification?format=json-->true
</isSupportResetLANPassword>
<isSupportPicHttpsCertificatesManagement>
    <!--ro, opt, bool, whether the device supports managing the HTTPS certificate for pictures, desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will not be returned-->true
</isSupportPicHttpsCertificatesManagement>
<isSupportAKSKImport>
    <!--ro, opt, bool, whether the device supports importing AKSK, desc:if this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned-->true
</isSupportAKSKImport>
<isSupportAKSKExport>
    <!--ro, opt, bool, whether the device supports exporting AKSK, desc:If this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned-->true
</isSupportAKSKExport>
<isSupportAKSKReset>
    <!--ro, opt, bool, whether the device supports resetting AKSK, desc:If this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned.-->true
</isSupportAKSKReset>
<ExportCertificateFile>
    <!--ro, opt, object, whether the device supports exporting the certificate, desc:related URI: POST
/ISAPI/Security/serverCertificate/exportCertificateFile-->
</ExportCertificateFile>
<isSupportGB351114FDWSFCertificate>
    <!--ro, opt, bool, whether the device supports FDWSF certificate management, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URIs: /ISAPI/Security/GB351114/FDWSFCertificate/capabilities?format=json, POST
/ISAPI/Security/GB351114/FDWSFCertificate?format=json, GET /ISAPI/Security/GB351114/FDWSFCertificate?format=json, and DELETE
/ISAPI/Security/GB351114/FDWSFCertificate?format=json.-->true
</isSupportGB351114FDWSFCertificate>
<isSupportMediaObjectEncryption>
    <!--ro, opt, bool, whether the device supports setting encryption parameters of the private target in media data for the specified video channel,
desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI:
/ISAPI/Security/MediaObjectEncryption/capabilities?format=json-->true
</isSupportMediaObjectEncryption>
<isSupportDisclaimer>
    <!--ro, opt, bool, whether support configuring the product disclaimer, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/disclaimer/capabilities?format=json-->true
</isSupportDisclaimer>
<isSupportSoftwareLicense>
    <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and
its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
</isSupportSoftwareLicense>
<isSupportInstallerCap>
    <!--ro, opt, bool, whether the device supports getting the installer permission, desc:if this function is supported, this node will be returned and
its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/UserPermission/installer/capabilities?format=json-->true
</isSupportInstallerCap>
<isSupportResetPasswordFile>
    <!--ro, opt, bool, whether the device supports exporting and importing the secret key file of device in the LAN, desc:Related URI:
/ISAPI/Security/resetPassword/keyFile/capabilities?format=json-->true
</isSupportResetPasswordFile>
<isNotSupportLogSeverCertificate>
    <!--ro, opt, bool, whether the device support importing log server certificate, desc:related URI: /ISAPI/Security/serverCertificate/certificate-->true
</isNotSupportLogSeverCertificate>
<isSupportLocalLoginCfg>
    <!--ro, opt, bool, whether it supports the capability of device Local Login parameters, desc:GET/PUT /ISAPI/Security/LocalLoginCfg?format=json-->true
</isSupportLocalLoginCfg>

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<isSupportGetProtocolFirewallRulesFileImportStatus>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
</isSupportGetProtocolFirewallRulesFileImportStatus>
<isSupportImportProtocolFirewallRulesFile>
  <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
</isSupportImportProtocolFirewallRulesFile>
<isSupportGetProtocolFirewallEngineUpgradeStatus>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
</isSupportGetProtocolFirewallEngineUpgradeStatus>
<isSupportUpgradeProtocolFirewallEngine>
  <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
</isSupportUpgradeProtocolFirewallEngine>
<isSupportProtocolFirewallVersionInfo>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
</isSupportProtocolFirewallVersionInfo>
<isSupportProtocolFirewallParams>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
</isSupportProtocolFirewallParams>
<isSupportResetPasswordBySecurityPhoneNo>
  <!--ro, opt, bool, whether it supports resetting device password with phone number verification,
desc:1. attendance check devices support applying a preset phone number during device activation. The user can reset the device password with the preset
phone number, which is similar with password reset via email. The phone number can be changed on the website or Local UI of the device
2. if the node is not returned or the returned value is false, it indicates "no"-->true
  <isSupportResetPasswordBySecurityPhoneNo>
  <isSupportGetIntrusionPreventionEngineUpgradeStatus>
    <!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET
/ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
    </isSupportGetIntrusionPreventionEngineUpgradeStatus>
  <isSupportIntrusionPreventionEngineVersionInfo>
    <!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET
/ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
    </isSupportIntrusionPreventionEngineVersionInfo>
  <isSupportIntrusionPreventionParams>
    <!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json--
>true
    </isSupportIntrusionPreventionParams>
  <isSupportUpgradeIntrusionPreventionEngine>
    <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT
/ISAPI/Security/UpgradeIntrusionPreventionEngine?format=json-->true
    </isSupportUpgradeIntrusionPreventionEngine>
  <notSupportImportCertificateType opt="CA,CClntCertificate,GB35114,GB35114Service,https">
    <!--ro, opt, string, file type that cannot be imported, attr:opt(req, string), desc:if this node does not exist, all supported certificate types can
be imported-->test
  </notSupportImportCertificateType>
  <isSupportPasswordProtection>
    <!--ro, opt, bool, whether it supports parameter configuration of SDK protocol authentication mode, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    </isSupportPasswordProtection>
  <SecurityCap>
  <EventCap>
    <!--ro, opt, object, event capability-->
    <isSupportHDFull>
      <!--ro, opt, bool, whether the device supports HDD full alarm-->true
    </isSupportHDFull>
    <isSupportHDError>
      <!--ro, opt, bool, whether the device supports HDD error alarm-->true
    </isSupportHDError>
    <isSupportNicBroken>
      <!--ro, opt, bool, whether the device supports the network disconnected event-->true
    </isSupportNicBroken>
    <isSupportIpConflict>
      <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
    </isSupportIpConflict>
    <isSupportIllaAccess>
      <!--ro, opt, bool, whether the device supports the illegal login event-->true
    </isSupportIllaAccess>
    <isSupportViException>
      <!--ro, opt, bool, whether the device supports video input exception (only for analog channels)-->true
    </isSupportViException>
    <isSupportViMismatch>
      <!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
    </isSupportViMismatch>
    <isSupportRecordException>
      <!--ro, opt, bool, whether the device supports the recording exception event-->true
    </isSupportRecordException>
    <isSupportRaidException>
      <!--ro, opt, bool, whether the device supports the array exception event-->true
    </isSupportRaidException>
    <isSupportViResMismatch>
      <!--ro, opt, bool, whether the device supports the event of video resolution mismatch-->true
    </isSupportViResMismatch>
    <isSupportPOCException>
      <!--ro, opt, bool, whether the device supports the event of powering over coaxial exception-->true
    </isSupportPOCException>
    <isSupportSmartDetection>
      <!--ro, opt, bool, whether the device supports the Smart event-->true
    </isSupportSmartDetection>
    <isSupportSpareException>
      <!--ro, opt, bool, whether the device supports hot spare exception-->true
    </isSupportSpareException>
    <isSupportPoePowerException>
      <!--ro, opt, bool, whether the device supports the POE exception event-->true
    </isSupportPoePowerException>
    <isSupportTriggerFocus>

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<!--ro, opt, bool, whether the device supports defocus exception-->true
</isSupportTriggerFocus>
<isSupportMotionDetection>
  <!--ro, opt, bool, whether the device supports motion detection-->true
</isSupportMotionDetection>
<isSupportVideoLoss>
  <!--ro, opt, bool, whether the device supports video Loss events-->true
</isSupportVideoLoss>
<isSupportTamperDetection>
  <!--ro, opt, bool, whether the device supports tampering detection-->true
</isSupportTamperDetection>
<isSupportFireDetection>
  <!--ro, opt, bool, whether the device supports fire detection events-->true
</isSupportFireDetection>
<isSupportRegionEntrance>
  <!--ro, opt, bool, whether the device supports region entrance events-->true
</isSupportRegionEntrance>
<isSupportRegionExiting>
  <!--ro, opt, bool, whether the device supports region exiting events-->true
</isSupportRegionExiting>
<isSupportLoitering>
  <!--ro, opt, bool, whether the device supports loitering events-->true
</isSupportLoitering>
<isSupportGroup>
  <!--ro, opt, bool, whether the device supports people gathering events-->true
</isSupportGroup>
<isSupportRapidMove>
  <!--ro, opt, bool, whether the device supports fast moving events-->true
</isSupportRapidMove>
<isSupportParking>
  <!--ro, opt, bool, whether the device supports parking detection events-->true
</isSupportParking>
<isSupportUnattendedBaggage>
  <!--ro, opt, bool, whether the device supports unattended baggage events-->true
</isSupportUnattendedBaggage>
<isSupportAttendedBaggage>
  <!--ro, opt, bool, whether the device supports object removal events-->true
</isSupportAttendedBaggage>
<isSupportHumanAttribute>
  <!--ro, opt, bool, whether the device supports human body features-->true
</isSupportHumanAttribute>
<isSupportHumanRecognition>
  <!--ro, opt, bool, whether the device supports target capture events-->true
</isSupportHumanRecognition>
<isSupportFaceContrast>
  <!--ro, opt, bool, whether the device supports face picture comparison events-->true
</isSupportFaceContrast>
<isSupportFaceLib>
  <!--ro, opt, bool, Front-end devices support Linkage by face picture Library., desc:related URI: /ISAPI/Event/schedules/faceLib/<ID>/<FDID>-->true
</isSupportFaceLib>
<isSupportBlackListFaceContrast>
  <!--ro, opt, bool, whether the device supports events of blacklist face picture comparison-->true
</isSupportBlackListFaceContrast>
<isSupportWhiteListFaceContrast>
  <!--ro, opt, bool, whether the device supports events of allowlist face picture comparison-->true
</isSupportWhiteListFaceContrast>
<isSupportIllegalParking>
  <!--ro, opt, bool, whether the device supports illegal parking detection-->true
</isSupportIllegalParking>
<isSupportPedestrian>
  <!--ro, opt, bool, whether the device supports pedestrian detection-->true
</isSupportPedestrian>
<isSupportTrafficAccident>
  <!--ro, opt, bool, whether the device supports traffic incident detection-->true
</isSupportTrafficAccident>
<isSupportConstruction>
  <!--ro, opt, bool, whether the device supports construction detection-->true
</isSupportConstruction>
<isSupportRoadBlock>
  <!--ro, opt, bool, whether the device supports roadblock detection-->true
</isSupportRoadBlock>
<isSupportAbandonedObject>
  <!--ro, opt, bool, whether the device supports fallen object detection-->true
</isSupportAbandonedObject>
<isSupportParallelParking>
  <!--ro, opt, bool, whether the device supports parallel parking detection-->true
</isSupportParallelParking>
<isSupportParkingState>
  <!--ro, opt, bool, whether the device supports parking space status detection-->true
</isSupportParkingState>
<isSupportTrafficJam>
  <!--ro, opt, bool, whether the device supports traffic jam detection (old version)-->true
</isSupportTrafficJam>
<isSupportCongestion>
  <!--ro, opt, bool, whether the device supports traffic jam detection-->true
</isSupportCongestion>
<isSupportStudentsStoodUp>
  <!--ro, opt, bool, whether the device supports the student standing up detection event-->true
</isSupportStudentsStoodUp>
<isSupportFramesPeopleCounting>
  <!--ro, opt, bool, whether the device supports the regional people counting event-->true
</isSupportFramesPeopleCounting>
<isSupportFaceSnap>
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    <!--ro, opt, bool, whether the device supports face capture events-->true
</isSupportFaceSnap>
<isSupportPersonDensityDetection>
    <!--ro, opt, bool, whether the device supports people density detection-->true
</isSupportPersonDensityDetection>
<HFPD>
    <!--ro, opt, bool, whether the device supports frequently appeared person detection-->true
</HFPD>
<isSupportPersonQueueDetection>
    <!--ro, opt, bool, whether the device supports queue detection events-->true
</isSupportPersonQueueDetection>
<isSupportVehicleStatistics>
    <!--ro, opt, bool, whether the device supports vehicle statistics events-->true
</isSupportVehicleStatistics>
<isSupportFaceThermometry>
    <!--ro, opt, bool, whether the device supports face temperature screening-->true
</isSupportFaceThermometry>
<isSupportSafetyHelmetDetection>
    <!--ro, opt, bool, whether the device supports hard hat detection-->true
</isSupportSafetyHelmetDetection>
<isSupportPowerSupplyCabinet>
    <!--ro, opt, bool, whether the device supports cabinet powering detection-->true
</isSupportPowerSupplyCabinet>
<isSupportSensorDetection>
    <!--ro, opt, bool, whether the device supports sensor detection-->true
</isSupportSensorDetection>
<isSupportSignAbnormal>
    <!--ro, opt, bool, whether the device supports the physical sign exception detection-->true
</isSupportSignAbnormal>
<isSupportTeacherBehaviorDetect>
    <!--ro, opt, bool, whether the device supports teacher's behavior detection-->true
</isSupportTeacherBehaviorDetect>
<isSupportCityManagement>
    <!--ro, opt, bool, whether the device supports the city management function-->true
</isSupportCityManagement>
<isSupportMixedTargetDetection>
    <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
</isSupportMixedTargetDetection>
<isSupportFaceSnapModeling>
    <!--ro, opt, bool, whether the device supports face modeling events-->true
</isSupportFaceSnapModeling>
<isSupportIntersectionAnalysis>
    <!--ro, opt, bool, whether the device supports the intersection analysis-->true
</isSupportIntersectionAnalysis>
<isSupportTriggerCapCheck>
    <!--ro, opt, bool, whether the device supports verifying the capability of triggering linkage actions, desc:this node indicates that the capability
message returned by calling the URI GET /ISAPI/Event/triggersCap has been verified, and the event linkage actions can be displayed exactly according to the
capability-->true
</isSupportTriggerCapCheck>
<isSupportLuma>
    <!--ro, opt, bool, VQD brightness exception-->true
</isSupportLuma>
<isSupportChroma>
    <!--ro, opt, bool, VQD image color cast-->true
</isSupportChroma>
<isSupportSnow>
    <!--ro, opt, bool, VQD snow noise-->true
</isSupportSnow>
<isSupportStreak>
    <!--ro, opt, bool, VQD stripe noise-->true
</isSupportStreak>
<isSupportFreeze>
    <!--ro, opt, bool, VQD image freeze-->true
</isSupportFreeze>
<isSupportSigLose>
    <!--ro, opt, bool, VQD signal loss-->true
</isSupportSigLose>
<isSupportClarity>
    <!--ro, opt, bool, VQD resolution exception-->true
</isSupportClarity>
<isSupportJitter>
    <!--ro, opt, bool, VQD image flutter-->true
</isSupportJitter>
<isSupportBlock>
    <!--ro, opt, bool, VQD video tampering-->true
</isSupportBlock>
<isSupportFlowers>
    <!--ro, opt, bool, VQD blurred image alarm-->true
</isSupportFlowers>
<isSupportNoise>
    <!--ro, opt, bool, VQD blurred screen-->true
</isSupportNoise>
<isSupportGhost>
    <!--ro, opt, bool, VQD image noise-->true
</isSupportGhost>
<isSupportPurple>
    <!--ro, opt, bool, VQD purple fringing-->true
</isSupportPurple>
<isSupportICR>
    <!--ro, opt, bool, VQD ICP-->true
</isSupportICR>
<isSupportProtectiveFilm>
    <!--ro, opt, bool, VQD protective film unremoved-->true
</isSupportProtectiveFilm>

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</isSupportVoltageInstable>
<isSupportVoltageInstable>
  <!--ro, opt, bool, whether the device supports the voltage instable exception detection-->true
</isSupportVoltageInstable>
<isSupportCertificateRevocation>
  <!--ro, opt, bool, whether the device supports the certificate expiry alarm-->true
</isSupportCertificateRevocation>
<isSupportSysStorFull>
  <!--ro, opt, bool, whether the device supports the storage full alarm of the cluster system-->true
</isSupportSysStorFull>
<isSupportVersionAbnormal>
  <!--ro, opt, bool, whether the device supports cluster version exception-->true
</isSupportVersionAbnormal>
<isSupportNodeOffline>
  <!--ro, opt, bool, whether the device supports device offline detection-->true
</isSupportNodeOffline>
<isSupportRecordCycleAbnormal>
  <!--ro, opt, bool, whether the device supports the event of insufficient recording period-->true
</isSupportRecordCycleAbnormal>
<isSupportIpcTransferAbnormal>
  <!--ro, opt, bool, whether the device supports the event of network camera migration failure-->true
</isSupportIpcTransferAbnormal>
<isSupportDrivingBehaviorDetect>
  <!--ro, opt, bool, whether the device supports driving behavior detection-->true
</isSupportDrivingBehaviorDetect>
<isSupportWasteGasDetection>
  <!--ro, opt, bool, whether the device supports waste gas monitoring-->true
</isSupportWasteGasDetection>
<isSupportGreyScaleAlarm>
  <!--ro, opt, bool, whether the device supports the gray scale alarm-->true
</isSupportGreyScaleAlarm>
<isSupportPTLocking>
  <!--ro, opt, bool, whether the device supports the PTZ Locking exception-->true
</isSupportPTLocking>
<isSupportNoMaskDetection>
  <!--ro, opt, bool, whether the device supports not wearing mask detection-->true
</isSupportNoMaskDetection>
<isSupportTMPA>
  <!--ro, opt, bool, whether the device supports the warning linkage of face temperature screening-->true
</isSupportTMPA>
<RuleScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by rule-->
<isSupportCityManagement>
  <!--ro, opt, bool, whether the device supports configuring the arming schedule by rule for the city management event-->true
</isSupportCityManagement>
</RuleScheduleCap>
<RegionScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by area-->
<isSupportPersonDensityDetection>
  <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the people density detection-->true
</isSupportPersonDensityDetection>
<isSupportPersonQueueDetection>
  <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the queue detection (area popularity analytics)-->true
</isSupportPersonQueueDetection>
</RegionScheduleCap>
<isSupportThermalCalibrationFileException>
  <!--ro, opt, bool, whether the device supports the temperature calibration file exception-->true
</isSupportThermalCalibrationFileException>
<isSupportTemperatureIntervalMeasurement>
  <!--ro, opt, bool, whether the device supports the temperature range detection-->true
</isSupportTemperatureIntervalMeasurement>
<isSupportThermalVehicleDetection>
  <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
</isSupportThermalVehicleDetection>
<isSupportWaterQualityDetection>
  <!--ro, opt, bool, whether the device supports water quality detection-->true
</isSupportWaterQualityDetection>
<isSupportWaterLevelDetection>
  <!--ro, opt, bool, whether the device supports water level detection-->true
</isSupportWaterLevelDetection>
<isSupportDeviceTemperatureAbnormal>
  <!--ro, opt, bool, whether the device supports temperature exception detection-->true
</isSupportDeviceTemperatureAbnormal>
<isSupportObjectAccumulationDetection>
  <!--ro, opt, bool, whether the device supports object accumulation detection-->true
</isSupportObjectAccumulationDetection>
<isSupportAppAbnormal>
  <!--ro, opt, bool, whether the device supports the App exception alarm or event-->true
</isSupportAppAbnormal>
<isSupportPTEventCfg>
  <!--ro, opt, bool, whether the device supports transparent transmission event configuration event, desc:/ISAPI/Event/PTEventCfg/capabilities?
format=json-->true
</isSupportPTEventCfg>
<isSupportLensTemperatureException>
  <!--ro, opt, bool, whether the device supports uploading lens abnormal temperature event-->true
</isSupportLensTemperatureException>
<isSupportPTZXFYFinish>
  <!--ro, opt, bool, the event of railway robot and PTZ going to the preset, desc:PTZXFYFinish-->true
</isSupportPTZXFYFinish>
<isSupportNormalTemperatureAlarm>
  <!--ro, opt, bool, whether the device supports normal temperature alarm-->true
</isSupportNormalTemperatureAlarm>
<isSupportLowTemperatureAlarm>
  <!--ro, opt, bool, whether the device supports low temperature alarm-->true

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</isSupportLowTemperatureAlarm>
<isSupportLowCoolerLifeEvent>
  <!--ro, opt, bool, whether it supports short life exception event of refrigerating machine-->true
</isSupportLowCoolerLifeEvent>
<isSupportObstacleAvoidanceAlarm>
  <!--ro, opt, bool, whether it supports obstacle avoidance alarm-->true
</isSupportObstacleAvoidanceAlarm>
<isSupportPTZRotationAbnormal>
  <!--ro, opt, bool, whether it supports PTZ rotation exception event-->true
</isSupportPTZRotationAbnormal>
<isSupportBlackListVehContrast>
  <!--ro, opt, bool, whether the vehicle blacklist event is supported-->true
</isSupportBlackListVehContrast>
<isSupportWhitelListVehContrast>
  <!--ro, opt, bool, whether the vehicle allowlist event is supported-->true
</isSupportWhitelListVehContrast>
</EventCap>
<ITCCap>
  <!--ro, opt, object, related node and URIs: the node <isSupportVehicleDetection> in /ISAPI/ITC/capability and /ISAPI/ITC/capabilities-->
  <isSupportITC>
    <!--ro, opt, bool, whether the device supports ITC-->true
  </isSupportITC>
  <isSupportIllegalDictionary>
    <!--ro, opt, bool, whether the device supports the violation dictionary-->true
  </isSupportIllegalDictionary>
  <isSupportVehicleDetection>
    <!--ro, opt, bool, whether the device supports vehicle detection-->true
  </isSupportVehicleDetection>
  <isSupportHVTVehicleDetection>
    <!--ro, opt, bool, whether the device supports mixed-traffic vehicle detection-->true
  </isSupportHVTVehicleDetection>
  <isSupportLicencePlateAuditData>
    <!--ro, opt, bool, whether the device supports importing or exporting the license plate blacklist and allowlist data-->true
  </isSupportLicencePlateAuditData>
  <isSupportSearchLPListAudit>
    <!--ro, opt, bool, whether the device supports searching for the license plate blacklist and allowlist-->true
  </isSupportSearchLPListAudit>
  <isSupportVehicleDetectionManualTrigger opt="false,true">
    <!--ro, opt, bool, whether the device supports manually triggering vehicle detection, attr:opt{opt, string}-->true
  </isSupportVehicleDetectionManualTrigger>
  <isSupportSyncSignalOutput>
    <!--ro, opt, bool, whether the device supports configuring output parameters-->true
  </isSupportSyncSignalOutput>
</ITCCap>
<ImageCap>
  <!--ro, opt, object, capability of configuring image parameters-->
  <isSupportRegionalExposure>
    <!--ro, opt, bool, whether the device supports regional exposure-->true
  </isSupportRegionalExposure>
  <isSupportRegionalFocus>
    <!--ro, opt, bool, whether the device supports regional focus-->true
  </isSupportRegionalFocus>
</ImageCap>
<RacmCap>
  <!--ro, opt, object, UI before picture search-->
  <isSupportZeroChan>
    <!--ro, opt, bool, whether the device supports channel=zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
  </isSupportZeroChan>
  <inputProxyNums>
    <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->1
  </inputProxyNums>
  <eSATANums>
    <!--ro, opt, int, supported number of eSATA interfaces-->1
  </eSATANums>
  <miniSASNums>
    <!--ro, opt, int, supported number of miniSAS interfaces-->1
  </miniSASNums>
  <nasNums>
    <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->1
  </nasNums>
  <ipSanNums>
    <!--ro, opt, int, supported number of IPSANS-->1
  </ipSanNums>
  <isSupportRaid>
    <!--ro, opt, bool, whether the device supports RAID-->true
  </isSupportRaid>
  <isSupportExtHdCfg>
    <!--ro, opt, bool, whether the device supports advanced HDD management-->true
  </isSupportExtHdCfg>
  <isSupportTransCode>
    <!--ro, opt, bool, whether the device supports live view transcoding-->true
  </isSupportTransCode>
  <isSupportIpcImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
  </isSupportIpcImport>
  <NasMountType>
    <!--ro, opt, object, NAS mounting type-->
  <isNFSSupportAuthentication>
    <!--ro, opt, bool, whether the device supports CIFS authentication-->true
  </isNFSSupportAuthentication>
  <isCIFSsupportAuthentication>
    <!--ro, opt, bool, whether the device supports CIFS authentication-->true
  </isCIFSsupportAuthentication>

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</isCIFSupportAuthentication>
</NasMountType>
<isSupportIpcStreamType>
  <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
</isSupportIpcStreamType>
<isSupportIOInputProxy>
  <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
</isSupportIOInputProxy>
<isSupportIOOutputProxy>
  <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
</isSupportIOOutputProxy>
<isSupportPTZRs485Proxy>
  <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
</isSupportPTZRs485Proxy>
<isSupportSrcIDSearch>
  <!--ro, opt, bool, whether the device supports stream ID search-->true
</isSupportSrcIDSearch>
<isSupportReversePlayback>
  <!--ro, opt, bool, whether the device supports reverse playback-->true
</isSupportReversePlayback>
<isSupportSMARTTest>
  <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
</isSupportSMARTTest>
<isSupportDownloadByTime>
  <!--ro, opt, bool, whether the device supports downloading by time, desc:related URI: /ISAPI/ContentMgmt/download/capabilities-->true
</isSupportDownloadByTime>
<pictureSearchType>
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEx,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,
illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,alarmResult">
  <!--ro, opt, string, picture search type, attr:opt{opt, string}-->test
</pictureSearchType>
<recordSearchType>
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
">
  <!--ro, opt, string, video search type, attr:opt{opt, string}, desc:"AllEvent" (all events), "CMR" (capture by schedule), "MOTION" (capture motion
detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures
and motion detection pictures), "Command", "pir" (pir alarm), "wlsensor" (wireless alarm), "callhelp" (panic alarm), "facedetection" (face detection),
"FieldDetection" (intrusion detection), "scenechangedetection" (scene change detection), "LineDetection" (line crossing detection)-->test
</recordSearchType>
<isSupportActivateIpc>
  <!--ro, opt, bool, whether the device supports activating the network camera-->true
</isSupportActivateIpc>
<isSupportCheckIpcSecurity>
  <!--ro, opt, object, device password detection-->
<isSupportCheckPassword>
  <!--ro, opt, bool, whether the device supports password detection-->true
</isSupportCheckPassword>
</isSupportCheckIpcSecurity>
<isSupportMainAndSubRecord>
  <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
</isSupportMainAndSubRecord>
<isSupportSyncIPCPassword>
  <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
</isSupportSyncIPCPassword>
<isSupportTransferIPC>
  <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
</isSupportTransferIPC>
<isSupportSmartSearch>
  <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
</isSupportSmartSearch>
<isSupportPOS>
  <!--ro, opt, bool, whether the device supports POS-->true
</isSupportPOS>
<isSupportLinkNum>
  <!--ro, opt, bool, whether the device supports getting the number of Links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/LinkNum--
>true
</isSupportLinkNum>
<isSupportPassBackBasicCfg>
  <!--ro, opt, bool, whether the device supports configuring basic parameters of the CVR copy-back function, desc:related URI:
/ISAPI/ContentMgmt/record/passback/basicCfg/capabilities-->true
</isSupportPassBackBasicCfg>
<PassBackTaskConfig>
  <!--ro, opt, object, capability of managing tasks of the CVR copy-back function-->
<isSupportAddTask>
  <!--ro, opt, bool, whether the device supports adding a task, desc:related URI: /ISAPI/ContentMgmt/record/passback/addTask/capabilities-->true
</isSupportAddTask>

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<isSupportSearchTask>
  <!--ro, opt, bool, whether the device supports searching tasks, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchTask/capabilities-->true
</isSupportSearchTask>
<isSupportControlTask>
  <!--ro, opt, bool, whether the device supports controlling the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/controlTask/capabilities--
>true
  </isSupportControlTask>
<isSupportDeleteTask>
  <!--ro, opt, bool, whether the device supports deleting the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/deleteTask/capabilities--
>true
  </isSupportDeleteTask>
</PassBackTaskConfig>
<PassBackPlanConfig>
  <!--ro, opt, object, capability of configuring the DVR copy-back schedule-->
  <isSupportAddPlan>
    <!--ro, opt, bool, whether the device supports adding a schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/addPlan/capabilities-->true
  </isSupportAddPlan>
  <isSupportSearchPlan>
    <!--ro, opt, bool, whether the device supports searching schedules, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchPlan/capabilities--
>true
  </isSupportSearchPlan>
  <isSupportDeletePlan>
    <!--ro, opt, bool, whether the device supports deleting the schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/deletePlan/capabilities--
>true
  </isSupportDeletePlan>
</PassBackPlanConfig>
<IRAIDCap>
  <!--ro, opt, object, network array capability-->
  <isSupportIRAIDRapidConfiguration>
    <!--ro, opt, bool, whether the device supports the rapid IRAID configuration, desc:related URI:
/ISAPI/ContentMgmt/Storage/IRAID/RapidConfiguration/capabilities-->true
  </isSupportIRAIDRapidConfiguration>
</IRAIDCap>
<isSupportStorageExtraInfo>
  <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
</isSupportStorageExtraInfo>
<isSupportRecordStatus>
  <!--ro, opt, bool, whether the device supports getting the recording status, desc:related URI: /ISAPI/ContentMgmt/channels/<ID>/record/capabilities--
>true
  </isSupportRecordStatus>
  <isSupportRacmChannelsCap opt="true,false">
    <!--ro, opt, bool, whether the device supports the Racm capability by channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
  </isSupportRacmChannelsCap>
  <LockCap>
    <!--ro, opt, object, video locking and unLocking, desc:related URI: /ISAPI/ContentMgmt/record/control/Locks/capabilities-->
  </LockCap>
  <isSupportFormatAll>
    <!--ro, opt, bool, whether the device supports formatting all HDDs, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/format-->true
  </isSupportFormatAll>
  <isSupportExtendCabinetCfg>
    <!--ro, opt, bool, whether the device supports configuring parameters of the extended cabinet. Related URI:
/ISAPI/ContentMgmt/Storage/ExtendCabinet/capabilities-->true
  </isSupportExtendCabinetCfg>
  <diskGroupNums>
    <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->1
  </diskGroupNums>
  <ExternalDevice>
    <!--ro, opt, object, external device storage-->
    <USB>
      <!--ro, opt, object, USB information-->
      <debugLogOutput>
        <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
      </debugLogOutput>
      <isSupportCapturePackage>
        <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
      </isSupportCapturePackage>
      <isSupportStreamStorage>
        <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
      </isSupportStreamStorage>
    </USB>
  </ExternalDevice>
  <customProtocolNums>
    <!--ro, opt, int, number of custom protocols connected via IPC, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols, supported number of
custom protocols-->1
  </customProtocolNums>
  <isSupportIPTiming>
    <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <enableTiming> node supports-->true
  </isSupportIPTiming>
  <isSupportLogConfig>
    <!--ro, opt, bool, whether the device supports Log configuration, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
  </isSupportLogConfig>
  <SecurityLog>
    <!--ro, opt, object, device's capability of security Log-->
    <isSupportSecurityLog>
      <!--ro, opt, bool, whether the device supports the security Log-->true
    </isSupportSecurityLog>
    <isSupportLogServer>
      <!--ro, opt, bool, whether it supports security Log server configuration-->true
    </isSupportLogServer>
  </SecurityLog>

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</isSupportLogServer>
<isSupportLogServerTest>
  <!--ro, opt, bool, whether it supports security log server test-->true
</isSupportLogServerTest>
<SecurityLogTypeList>
  <!--ro, req, array, List of security log types, subType:object-->
  <SecurityLogType>
    <!--ro, opt, object, security log type-->
    <primaryType>
      <!--ro, req, string, main log type, desc:"Event", "Operation", "Other", "ALL"-->test
    </primaryType>
    <secondaryType opt="all">
      <!--ro, req, string, minor log type, attr:opt{opt, string}, desc:for other types, refer to the appendix-->test
    </secondaryType>
  </SecurityLogType>
</SecurityLogTypeList>
<isSupportLogServerList>
  <!--ro, opt, bool, whether the device supports batch configuring security log server-->true
</isSupportLogServerList>
<SecurityLog>
<isSupportGetBackupDevice>
  <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json-->true
>true
</isSupportGetBackupDevice>
<isSupportAutoBackup>
  <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/auto/capabilities?format=json-->true
>true
</isSupportAutoBackup>
<certificateValidationEnabled>
  <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <certificateValidationEnabled> node supports-->true
</certificateValidationEnabled>
<defaultAdminPortEnabled>
  <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
</defaultAdminPortEnabled>
<isSupportLogDataPackage>
  <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/LogSearch/dataPackage-->true
</isSupportLogDataPackage>
<LogSearchTimeSpanNums>
  <!--ro, opt, int, number of time periods for log search, desc:related URI: /ISAPI/ContentMgmt/LogSearch; corresponding node: timeSpanList in CMSearchDescription-->1
</LogSearchTimeSpanNums>
<CloudStorageServerCap>
  <!--ro, opt, object, capability of configuring the cloud storage server-->
  <isSupportCloudStorageParameter>
    <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI: /ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
  </isSupportCloudStorageParameter>
  <isSupportCloudStoragePool>
    <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI: /ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
  </isSupportCloudStoragePool>
  <supportChannelsNum>
    <!--ro, opt, int, number of channels supported by the cloud storage, desc:if this node is not returned, all channels of the device support; if this node is returned, several channels of the device supports cloud storage (at present, it will not return the specific number of the channels supporting cloud storage)-->1
  </supportChannelsNum>
</CloudStorageServerCap>
<isSupportWebPrivatePlaybackByUTC>
  <!--ro, opt, bool, whether it supports playback (via private protocol) by UTC-->true
</isSupportWebPrivatePlaybackByUTC>
<isSupportFindCommonFileByUTC>
  <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
</isSupportFindCommonFileByUTC>
<isSupportFindEventFileByUTC>
  <!--ro, opt, bool, whether it supports searching for event files (private protocols) by UTC time-->true
</isSupportFindEventFileByUTC>
<isSupportSmartSearchRecordByUTC>
  <!--ro, opt, bool, whether it supports smart video search by UTC time-->true
</isSupportSmartSearchRecordByUTC>
<isSupportMRDSearchByTimeZone>
  <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
</isSupportMRDSearchByTimeZone>
<isSupportSearchRecordLabelByUTC>
  <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
</isSupportSearchRecordLabelByUTC>
<isSupportSearchPictureByUTC>
  <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
</isSupportSearchPictureByUTC>
<isSupportSmartSearchPictureByUTC>
  <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
</isSupportSmartSearchPictureByUTC>
<isSupportFindLogByUTC>
  <!--ro, opt, bool, whether it supports searching log by UTC time-->true
</isSupportFindLogByUTC>
<isSupportUploadRecordByUTC>
  <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
</isSupportUploadRecordByUTC>
<isSupportPlaybackByUTC>
  <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
</isSupportPlaybackByUTC>
<isSupportFaceDetectionAlarmByTimeZone>
  <!--ro, opt, bool, no file is found-->true

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    </isSupportFaceDetectionAlarmByTimeZone>
    <isSupportSourceModify>
        <!--ro, opt, bool, whether it supports configuring the parameters of the LAN IPC, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/sourceModify/capabilities-->true
    </isSupportSourceModify>
    <isSupportCountingSearchByUTC>
        <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
    </isSupportCountingSearchByUTC>
    <isSupportCluster>
        <!--ro, opt, bool, whether the device supports the cluster function, desc:related URI: /ISAPI/ContentMgmt/Cluster-->true
    </isSupportCluster>
    <supportAIDTFSType opt="illegalParking,wrongdirection,crosslane,vehicleexist,lanechange,turnround,evidence">
        <!--ro, opt, string, the event picture search type supported by AID and TFS, attr:opt{opt, string}-->test
    </supportAIDTFSType>
    <isSupportMisinfoStateSearch>
        <!--ro, opt, bool, whether it supports searching by false alarm status-->true
    </isSupportMisinfoStateSearch>
    <isSupportOneKeyOperationStorageData>
        <!--ro, opt, bool, whether it supports one-touch configuration for storage data, desc:related URI: /ISAPI/ContentMgmt/Storage/oneKey/capabilities?
format=json-->true
    </isSupportOneKeyOperationStorageData>
    <eventRecordSearch>
        <!--ro, opt, object, whether the device supports searching for videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
    <eventType opt="faceContrast,faceSnap,vehicleMatchResult,hawkResult">
        <!--ro, opt, string, event type, attr:opt{opt, string}-->test
    </eventType>
    <auxEventType opt="faceContrastFailure,faceContrastSuccess">
        <!--ro, opt, string, search sub type of the event, attr:opt{opt, string}-->test
    </auxEventType>
    </eventRecordSearch>
    <isSupportManualRecord>
        <!--ro, opt, bool, whether the device supports manual recording, desc:related URI: /ISAPI/ContentMgmt/record/control/manual/capabilities?format=json--
>true
    </isSupportManualRecord>
    <isSupportRemark>
        <!--ro, opt, bool, whether it supports file remarks-->true
    </isSupportRemark>
    <FileUpload>
        <!--ro, opt, object, automatically upload file-->
    <enabled>
        <!--ro, req, bool, whether it supports uploading the file-->true
    </enabled>
    </FileUpload>
    <PoliceInfoUpload>
        <!--ro, opt, object, automatically upload police officer information-->
    <enabled>
        <!--ro, req, bool, whether the device supports uploading the police officer information-->true
    </enabled>
    </PoliceInfoUpload>
    <PoliceInfo>
        <!--ro, opt, object, police officer information-->
    <policeID>
        <!--ro, opt, bool, whether the device supports police officer ID-->true
    </policeID>
    <policeCode>
        <!--ro, opt, bool, whether the device supports police officer number-->true
    </policeCode>
    <policeName>
        <!--ro, opt, bool, whether the device supports police officer name-->true
    </policeName>
    <password>
        <!--ro, opt, bool, whether the device supports police officer password-->true
    </password>
    <deviceID>
        <!--ro, opt, bool, whether the device supports device ID-->true
    </deviceID>
    </PoliceInfo>
    <isSupportAssignChannelID>
        <!--ro, opt, bool, whether the device supports specifying the access channel No. when adding the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
    </isSupportAssignChannelID>
    <isSupportAssignStreamID>
        <!--ro, opt, bool, whether it supports specifying stream ID of the channel when adding the network camera in channel mode, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
    </isSupportAssignStreamID>
    <PictureSearchSubTypeList>
        <!--ro, opt, array, list of picture search sub type, subType:object-->
    <PictureSearchSubType>
        <!--ro, opt, object, the sub type of picture search-->
    <mainType>
        <!--ro, req, string, main event type of the searched picture-->test
    </mainType>
    <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
        <!--ro, req, string, sub type of each main event type, attr:opt{opt, string}-->test
    </subType>
    </PictureSearchSubType>
    </PictureSearchSubTypeList>
    <isSupportPlaybackStreamIDByUTC>
        <!--ro, opt, bool, whether it supports playback by stream ID according to UTC time-->true
    </isSupportPlaybackStreamIDByUTC>
    <isSupportTimeLockStreamIDByUTC>
        <!--ro, opt, bool, whether it supports locking and unlocking the video segment by steam ID according to UTC time-->true
    </isSupportTimeLockStreamIDByUTC>

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<isSupportTimeSearch>
  <!--ro, opt, bool, whether it supports searching for recording start and end time by channel, desc:related URI: /ISAPI/ContentMgmt/time/search?
format=json-->true
</isSupportTimeSearch>
<behaviorEventPicSearch opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retention,sleepOnduty">
  <!--ro, opt, string, abnormal event detection supported by picture search, attr:opt{opt, string}-->test
</behaviorEventPicSearch>
<perimeterEventPicSearch opt="allPerimeterEvent,linedetection,fielddetection,regionEntrance,regionExiting,loitering">
  <!--ro, opt, string, perimeter protection events supported by picture search, attr:opt{opt, string}-->test
</perimeterEventPicSearch>
<isSupportSourceStatus>
  <!--ro, opt, bool, whether it supports getting the status of access source-->true
</isSupportSourceStatus>
<isSupportSSDSMARTTest>
  <!--ro, opt, bool, whether the device supports S.M.A.R.T. detection on SSD, desc:related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start--
>true
</isSupportSSDSMARTTest>
<MobileStorage>
  <!--ro, opt, object, storage capability of the mobile HDD-->
  <isSupportGetFileTime>
    <!--ro, opt, bool, whether it supports getting the file time of the mobile HDD, desc:it supports mobile HDD currently. Related URI:
/ISAPI/ContentMgmt/mobileStorage/fileTime/capabilities?format=json-->true
  </isSupportGetFileTime>
</MobileStorage>
<isSupportGetChannelList>
  <!--ro, opt, bool, whether it supports getting channels of the HDD, desc:it supports mobile HDD currently. Related URI:
/ISAPI/ContentMgmt/Storage/hdd/channelList/capabilities?format=json-->true
</isSupportGetChannelList>
<isSupportCascade>
  <!--ro, opt, bool, whether it supports configuring parameters for NVR cascading topology-->true
</isSupportCascade>
<isSupportPictureInfo>
  <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
</isSupportPictureInfo>
<isSupportCloudStorageModel>
  <!--ro, opt, bool, whether the device supports configuring the cloud storage mode,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>/capabilities?format=json-->true
</isSupportCloudStorageModel>
<isSupportCloudStorageModelList>
  <!--ro, opt, bool, whether the device supports configuring cloud storage modes in a batch,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>?format=json-->true
</isSupportCloudStorageModelList>
<isSupportRecordAudio>
  <!--ro, opt, bool, whether the device supports recording the audio, desc:only the audio will be recorded and the video will not be recorded. Related
URI: /ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json-->true
</isSupportRecordAudio>
<isSupportSearchAudio>
  <!--ro, opt, bool, whether the device supports searching for the audio, desc:related URI: /ISAPI/ContentMgmt/recordAudio/search/capabilities?
format=json-->true
</isSupportSearchAudio>
<isSupportAudioDailyDistribution>
  <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
</isSupportAudioDailyDistribution>
<isSupportDownloadAudio>
  <!--ro, opt, bool, whether the device supports downloading the audio by file name, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
</isSupportDownloadAudio>
<isSupportDeleteAudio>
  <!--ro, opt, bool, whether the device supports deleting the recorded audio file, desc:related URI: /ISAPI/ContentMgmt/recordAudio/deleteFile?
format=json-->true
</isSupportDeleteAudio>
<isSupportSearchRemoteAlbum>
  <!--ro, opt, bool, whether the device supports remotely searching for album information, desc:related URI: /ISAPI/ContentMgmt/searchRemoteAlbum?
format=json-->true
</isSupportSearchRemoteAlbum>
<isSupportStreamParam>
  <!--ro, opt, bool, whether the device supports configuring the access stream parameters of all digital channels, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/streamParam/capabilities?format=json-->true
</isSupportStreamParam>
<isSupportFusionChan>
  <!--ro, opt, bool, whether the device supports fusion channel, desc:if this node does not exist, it indicates that the device does not support the
function; interaction IP channel: it is a concept from the education sharing system; the education sharing system will allocate IP channels to images of the
connected sub classrooms or meeting rooms. fusion channel: a channel generated when adding the NVR locally or remotely; it supports live view, playback,
storage, and recording schedule/linkage configuration; the channel will take effect by enabling the fusion channel-->true
</isSupportFusionChan>
<isSupportRecordMultipleDownload>
  <!--ro, opt, bool, whether the device supports downloading videos in multiples, desc:if this node does not exist, it indicates that the device does
not support the function by default-->true
</isSupportRecordMultipleDownload>
</RacmCap>
<isSupportMI3704>
  <!--ro, opt, bool-->true
</isSupportMI3704>
<isSupportLogServer>
  <!--ro, opt, bool-->true
</isSupportLogServer>
<isSupportLogServerTest>
  <!--ro, opt, bool-->true
</isSupportLogServerTest>
<PTZCtrlCap>
  <!--ro, opt, object, device's PTZ capability-->
  <isSupportPatrols>
    <!--ro, opt, bool, whether the device supports patrol status
  </isSupportPatrols>

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<!--ro, opt, bool, whether the device supports patrol-->true
</isSupportPatrols>
<isSupportCombinedPath>
  <!--ro, opt, bool, whether the device supports the PTZ combined path-->true
</isSupportCombinedPath>
<isSupportRailwayRobot>
  <!--ro, opt, bool, whether the device supports railway robot, desc:/ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json-->true
</isSupportRailwayRobot>
</PTZCtrlCap>
<SmartCap>
  <!--ro, opt, object, compact and smart capability, desc:/ISAPI/Smart/capabilities-->
  <isSupportROI>
    <!--ro, opt, bool, whether the device supports ROI encoding configuration-->true
  </isSupportROI>
  <isSupportFaceDetect>
    <!--ro, opt, bool, whether the device supports face detection-->true
  </isSupportFaceDetect>
  <isSupportIntelliTrace>
    <!--ro, opt, bool, whether the device supports smart tracking-->true
  </isSupportIntelliTrace>
  <isSupportFieldDetection>
    <!--ro, opt, bool, whether the device supports intrusion detection-->true
  </isSupportFieldDetection>
  <isSupportDefocusDetection>
    <!--ro, opt, bool, whether the device supports defocus detection-->true
  </isSupportDefocusDetection>
  <isSupportAudioDetection>
    <!--ro, opt, bool, whether the device supports audio detection-->true
  </isSupportAudioDetection>
  <isSupportSceneChangeDetection>
    <!--ro, opt, bool, whether the device supports scene change detection-->true
  </isSupportSceneChangeDetection>
  <isSupportLineDetection>
    <!--ro, opt, bool, whether the device supports line crossing detection-->true
  </isSupportLineDetection>
  <isSupportRegionEntrance>
    <!--ro, opt, bool, whether the device supports region entrance detection-->true
  </isSupportRegionEntrance>
  <isSupportRegionExiting>
    <!--ro, opt, bool, whether the device supports region exiting detection-->true
  </isSupportRegionExiting>
  <isSupportLoitering>
    <!--ro, opt, bool, whether the device supports loitering detection-->true
  </isSupportLoitering>
  <isSupportGroup>
    <!--ro, opt, bool, whether the device supports people gathering detection-->true
  </isSupportGroup>
  <isSupportRapidMove>
    <!--ro, opt, bool, whether the device supports fast moving detection-->true
  </isSupportRapidMove>
  <isSupportParking>
    <!--ro, opt, bool, whether the device supports parking detection-->true
  </isSupportParking>
  <isSupportUnattendedBaggage>
    <!--ro, opt, bool, whether the device supports unattended baggage detection-->true
  </isSupportUnattendedBaggage>
  <isSupportAttendedBaggage>
    <!--ro, opt, bool, whether the device supports object removal detection-->true
  </isSupportAttendedBaggage>
  <isSupportPeopleDetection>
    <!--ro, opt, bool, whether the device supports human body detection-->true
  </isSupportPeopleDetection>
  <isSupportSmartCalibration>
    <!--ro, opt, bool, whether the device supports calibration filter size-->true
  </isSupportSmartCalibration>
  <isSupportStorageDetection>
    <!--ro, opt, bool, whether the device supports health monitoring storage-->true
  </isSupportStorageDetection>
  <isSupportShipsDetection>
    <!--ro, opt, bool, whether the device supports ship detection-->true
  </isSupportShipsDetection>
  <isSupportShield>
    <!--ro, opt, bool, whether the device supports configuring the compact and smart shielded regions-->true
  </isSupportShield>
  <isSupportAlgVersion>
    <!--ro, opt, bool, whether the device supports algorithm version-->true
  </isSupportAlgVersion>
  <isSupportVideoOverlap>
    <!--ro, opt, bool, whether the device supports configuring information overlay for video stream-->true
  </isSupportVideoOverlap>
  <isSupportParkingState>
    <!--ro, opt, bool, whether the device supports parking status-->true
  </isSupportParkingState>
  <isSupportChannelResource>
    <!--ro, opt, bool, whether the device supports channel resources-->true
  </isSupportChannelResource>
  <isSupportAnalysisUnitSwitch opt="true,false">
    <!--ro, opt, bool, whether the device supports switching analysis unit, attr:opt{opt, string}-->true
  </isSupportAnalysisUnitSwitch>
  <isSupportHFPD>
    <!--ro, opt, bool, whether the device supports frequently appeared person detection, desc:related URI: /ISAPI/SDT/HFPD?format=json-->true
  </isSupportHFPD>
  <isSupportImageROI>
    <!--ro, opt, bool, whether the device supports smartJpeg (image ROI)-->true
```



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</isSupportImageROI>
<isSupportLFPD>
  <!--ro, opt, bool, whether the device supports rarely appeared person detection, desc:related URI: /ISAPI/SDT/LFPD?format=json-->true
</isSupportLFPD>
<isSupportSmartOverlapParams>
  <!--ro, opt, bool, whether it supports smart overlay parameters, desc:related URI: /ISAPI/Smart/channels/<channelID>/SmartOverlapParams?format=json-->true
</isSupportSmartOverlapParams>
</SmartCap>
<isSupportEhome>
  <!--ro, opt, bool, whether the device supports ISUP functions-->true
</isSupportEhome>
<isSupportStreamingEncrypt>
  <!--ro, opt, bool, whether the device supports stream encryption-->true
</isSupportStreamingEncrypt>
<TestCap>
  <!--ro, opt, object, test capability-->
  <isSupportFTPTest>
    <!--ro, opt, bool, whether it supports FTP test-->true
  </isSupportFTPTest>
  <isSupportPingTest>
    <!--ro, opt, bool, whether it supports PING test-->true
  </isSupportPingTest>
  <isSupportNTPTest>
    <!--ro, opt, bool, whether it supports NTP test-->true
  </isSupportNTPTest>
  <isSupportNASTest>
    <!--ro, opt, bool, whether it supports NAS test-->true
  </isSupportNASTest>
  <isSupportEmailTest>
    <!--ro, opt, bool, whether the device supports email test-->true
  </isSupportEmailTest>
</TestCap>
<ImageMiscCap>
  <!--ro, opt, object, DPC capability-->
  <isSupportDPC>
    <!--ro, opt, bool, whether the device supports DPC-->true
  </isSupportDPC>
</ImageMiscCap>
<PanoramaCap>
  <!--ro, opt, object, capability of panoramic channels, desc:the object cannot be extended. You can extend it in /ISAPI/Panorama/capabilities-->
  <isSupportGeneratePanorama>
    <!--ro, opt, bool, whether the device supports generating panoramic images-->true
  </isSupportGeneratePanorama>
  <isSupportPanoramaPosition3D>
    <!--ro, opt, bool, whether the device supports panoramic 3D position-->true
  </isSupportPanoramaPosition3D>
  <isSupportPreset>
    <!--ro, opt, bool, whether the device supports configuring panoramic preset-->true
  </isSupportPreset>
  <isSupportGetPanoramicSplicingStatus>
    <!--ro, opt, bool-->true
  </isSupportGetPanoramicSplicingStatus>
  <isSupportStopPanoramicSplicing>
    <!--ro, opt, bool-->true
  </isSupportStopPanoramicSplicing>
  <isSupportSplicingChannelMotionControl>
    <!--ro, opt, bool-->true
  </isSupportSplicingChannelMotionControl>
  <isSupportGetPanoramaSplicingPictures>
    <!--ro, opt, bool-->true
  </isSupportGetPanoramaSplicingPictures>
  <isSupportImportPanoramaSplicingParamFile>
    <!--ro, opt, bool-->true
  </isSupportImportPanoramaSplicingParamFile>
  <isSupportExportPanoramaSplicingParamFile>
    <!--ro, opt, bool-->true
  </isSupportExportPanoramaSplicingParamFile>
  <isSupportImportPanoramicSplicingPicture>
    <!--ro, opt, bool-->true
  </isSupportImportPanoramicSplicingPicture>
  <isSupportExportPanoramicSplicingPicture>
    <!--ro, opt, bool-->true
  </isSupportExportPanoramicSplicingPicture>
</PanoramaCap>
<ThermalCap>
  <!--ro, opt, object, thermal capability, desc:related URI: /ISAPI/Thermal/capabilities-->
  <isSupportFireDetection>
    <!--ro, opt, bool, whether the device supports fire detection, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
  </isSupportFireDetection>
  <isFireFocusZoomSupport>
    <!--ro, opt, bool, whether the device supports visible light lens zooming, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireFocusZoom-->true
  </isFireFocusZoomSupport>
  <isSupportThermometry>
    <!--ro, opt, bool, whether the device supports thermography, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
  </isSupportThermometry>
  <isSupportRealtimeThermometry>
    <!--ro, opt, bool, whether the device supports uploading real-time temperature measurement data-->true
  </isSupportRealtimeThermometry>
  <isSupportThermIntell>
    <!--ro, opt, bool, whether the device supports configuring smart thermal mutual exclusion, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermIntell/capabilities-->true
  </isSupportThermIntell>

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<isSupportNVR>
  <!--ro, opt, bool, it is used for client to check-->true
</isSupportNVR>
<isSupportThermometrySchedule>
  <!--ro, opt, bool, whether the device supports the arming schedule of temperature measurement, desc:related URI: /ISAPI/Event/schedules/thermometry-->true
>true
</isSupportThermometrySchedule>
<isSupportTemperatureSchedule>
  <!--ro, opt, bool, whether the device supports the arming schedule of temperature difference, desc:related URI: /ISAPI/Event/schedules/temperature-->true
>true
</isSupportTemperatureSchedule>
<isSupportFireDetectionSchedule>
  <!--ro, opt, bool, whether the device supports the arming schedule of fire detection, desc:related URI: /ISAPI/Event/schedules/fireDetections-->true
</isSupportFireDetectionSchedule>
<isSupportPower>
  <!--ro, opt, bool, whether the device supports power on/off capability, desc:related URI: /ISAPI/Thermal/Power/capabilities-->true
</isSupportPower>
<isSupportRealtimeTempHumi>
  <!--ro, opt, bool, whether the device supports real-time detection of temperature and humidity, desc:related URI:
/ISAPI/Thermal/TempHumi/capabilities-->true
</isSupportRealtimeTempHumi>
<ManualThermCap>
  <!--ro, opt, object, the capability of manual temperature measurement-->
<manualThermRuleNum>
  <!--ro, opt, int, if this node is not returned, it indicates manual temperature measurement is not supported, desc:related URI:
/ISAPI/Thermal/manualThermometry/capabilities-->true
</manualThermRuleNum>
</ManualThermCap>
<isSupportManualThermBasic>
  <!--ro, opt, bool, whether the device supports basic configuration of manual temperature measurement, desc:related URI:
/ISAPI/Thermal/manualThermBasic/capabilities-->true
</isSupportManualThermBasic>
<isSupportFireShieldMask>
  <!--ro, opt, bool, whether the device supports configuring the parameters of fire shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/fireShieldMask/capabilities-->true
</isSupportFireShieldMask>
<isSupportSmokeShieldMask>
  <!--ro, opt, bool, whether the device supports configuring the parameters of smoke shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/smokeShieldMask/capabilities-->true
</isSupportSmokeShieldMask>
<isSupportThermometryMode>
  <!--ro, opt, bool, whether the device supports configuring temperature measurement mode, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities-->true
</isSupportThermometryMode>
<isSupportThermalPip>
  <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/pip/capabilities-->true
</isSupportThermalPip>
<isSupportThermalIntelRuleDisplay>
  <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/IntelRuleDisplay/capabilities-->true
>true
</isSupportThermalIntelRuleDisplay>
<AlgVersionInfo>
  <!--ro, opt, object, whether the device supports getting the version information of thermal algorithms library, desc:related URI:
/ISAPI/Thermal/AlgVersion-->
<thermometryAlgName min="1" max="128">
  <!--ro, opt, string, version information of temperature measurement algorithms library, attr:min{opt, string},max{opt, string}-->test
</thermometryAlgName>
<shipsAlgName min="1" max="128">
  <!--ro, opt, string, version name of ship detection algorithms library, attr:min{opt, string},max{opt, string}-->test
</shipsAlgName>
<fireAlgName min="1" max="128">
  <!--ro, opt, string, algorithm library version name of fire detection, attr:min{opt, int},max{opt, int}-->test
</fireAlgName>
<smokeAlgName min="1" max="128">
  <!--ro, opt, string, algorithm library version name of smoke detection, attr:min{opt, int},max{opt, int}-->test
</smokeAlgName>
<wasteGasAlgName min="1" max="128">
  <!--ro, opt, string, algorithm library version name of waste gas detection, attr:min{opt, string},max{opt, string}-->test
</wasteGasAlgName>
<greyScaleAlarmAlgName min="1" max="128">
  <!--ro, opt, string, algorithm library version name of greyscale alarm, attr:min{opt, string},max{opt, string}-->test
</greyScaleAlarmAlgName>
</AlgVersionInfo>
<isSupportFaceThermometry>
  <!--ro, opt, bool, whether supports temperature screening configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities-->true
>true
</isSupportFaceThermometry>
<isSupportThermalBlackBody>
  <!--ro, opt, bool, whether the device supports configuring black body parameters of thermal imaging, desc:related URI:
/ISAPI/Thermal/channels/<ID>/blackBody/capabilities-->true
</isSupportThermalBlackBody>
<isSupportThermalStreamParam>
  <!--ro, opt, bool, whether the device supports configuring stream parameters of thermal imaging, desc:related URI:
/ISAPI/Thermal/channels/<ID>/streamParam/capabilities-->true
</isSupportThermalStreamParam>
<isSupportBodyTemperatureCompensation>
  <!--ro, opt, bool, whether the device supports temperature compensation configuration, desc:related URI:
/ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities-->true
</isSupportBodyTemperatureCompensation>
<isSupportClickToThermometry>
  <!--ro, opt, bool, whether the device supports click-to-thermometry, desc:related URI: ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization-->true
>true
</isSupportClickToThermometry>
<isSupportThermometryHistorySearch>

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    <!--ro, opt, bool, whether the device supports searching for historical temperature data, desc:related URI:
/ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities-->true
    </isSupportThermometryHistorySearch>
    </isSupportThermometryShieldMask>
    <!--ro, opt, bool, whether the device supports configuring shield parameters of thermography region, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometryShieldMask/capabilities-->true
    </isSupportThermometryShieldMask>
    </isSupportTemperatureCorrection>
    <!--ro, opt, bool, whether device supports temperature measurement correction, desc:related URI:
/ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities-->true
    </isSupportTemperatureCorrection>
    </isSupportBurningPrevention>
    <!--ro, opt, bool, whether device supports burning prevention, desc:related URI: /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities-->true
    </isSupportBurningPrevention>
    </isSupportTemperatureCollection>
    <!--ro, opt, bool, whether device supports temperature ANR, desc:related URI: ISAPI/Thermal/temperature/collection/capabilities?format=json-->true
    </isSupportTemperatureCollection>
    </isSupportJpegPicWithAppendData>
    <!--ro, opt, bool, whether the device supports capturing pictures, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?
format=json-->true
    </isSupportJpegPicWithAppendData>
    </isSupportRealTimeThermometryForHTTP>
    <!--ro, opt, bool, whether device supports real-time temperature measurement, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometry/realTimeThermometry/rules?format=json-->true
    </isSupportRealTimeThermometryForHTTP>
    </isSupportThermometryOfflineCapture>
    <!--ro, opt, bool, whether the device supports capturing pictures offLine, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/OfflineCapture?
format=json-->true
    </isSupportThermometryOfflineCapture>
    </isSupportShipsDetectionWithScene>
    <!--ro, opt, bool, whether the device supports ship detection by scene, desc:related URI: /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?
format=json-->true
    </isSupportShipsDetectionWithScene>
    </isSupportWasteGasDetection>
    <!--ro, opt, bool, whether the device supports waste gas monitoring-->true
    </isSupportWasteGasDetection>
    </isSupportGreyScaleAlarm>
    <!--ro, opt, bool, whether the device supports the gray scale alarm-->true
    </isSupportGreyScaleAlarm>
    </isSupportFaceSnapThermometry>
    <!--ro, opt, bool, whether the device supports temperature screening when capturing face pictures-->true
    </isSupportFaceSnapThermometry>
    </isSupportThermalTemperatureCorrect>
    <!--ro, opt, bool, whether device supports temperature calibration, desc:related URI: /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json--
>true
    </isSupportThermalTemperatureCorrect>
    </isSupportThermalVehicleDetection>
    <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
    </isSupportThermalVehicleDetection>
    </isSupportSensorParam>
    <!--ro, opt, bool, whether the device supports configuring the sensor temperature measurement parameters, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/thermometry/sensorParam/capabilities?format=json-->true
    </isSupportSensorParam>
    </isSupportTemperatureIntervalMeasurement>
    <!--ro, opt, bool, whether the device supports temperature range detection-->true
    </isSupportTemperatureIntervalMeasurement>
    </isSupportIndustryThermometryTemperatureRise>
    <!--ro, opt, bool, whether the device supports setting temperature rise compensation in thermal industrial temperature measurement, desc:related URI:
/ISAPI/Thermal/channels/<ID>/industryThermometry/temperatureRise?format=json-->true
    </isSupportIndustryThermometryTemperatureRise>
    </isSupportThermalDataLinkInfo>
    <!--ro, opt, bool, whether the device supports getting the temperature data Link information, desc:related URI: /ISAPI/Thermal/thermalDataLinkInfo?
format=json; /ISAPI/Thermal/thermalDataLinkStatus?format=json-->true
    </isSupportThermalDataLinkInfo>
    </isSupportTemperatureCompensationCalib>
    <!--ro, opt, bool, whether the device supports temperature compensation, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/temperatureCompensationCalibration?format=json-->true
    </isSupportTemperatureCompensationCalib>
    </isSupportWebDisplayPixelToPixelParam>
    <!--ro, opt, bool, whether the web needs to display full-screen temperature parameters, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/thermometry/pixelToPixelParam/capabilities-->true
    </isSupportWebDisplayPixelToPixelParam>
    </isSupportThermalInspection>
    <!--ro, opt, bool, whether the device supports thermal patrol, desc:related URI: /ISAPI/Thermal/inspection/capabilities?format=json-->true
    </isSupportThermalInspection>
    </isSupportBlackBodyTemperature>
    <!--ro, opt, bool, whether the device supports searching for black body center point temperature, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/blackBodyTemperature?format=json-->true
    </isSupportBlackBodyTemperature>
    </isSupportExportThermalCalibrationFile>
    <!--ro, opt, bool, whether the device supports exporting the temperature calibration file, desc:POST /ISAPI/Thermal/ExportThermalCalibrationFile?
format=json-->true
    </isSupportExportThermalCalibrationFile>
    </isSupportSearchSmokeAndFireDetection>
    <!--ro, opt, bool, whether the device supports searching for history records of smoke and fire detection, desc:POST
/ISAPI/Thermal/fireDetection/SearchSmokeAndFireDetection?format=json-->true
    </isSupportSearchSmokeAndFireDetection>
    </isSupportColumnOven>
    <!--ro, opt, bool, whether the device supports column oven parameters, desc:GET/PUT /ISAPI/OPCA/GC/ColumnOven?format=json-->true
    </isSupportColumnOven>
    </isSupportImportThermalCalibrationFile>
    <!--ro, opt, bool, whether the device supports importing the temperature calibration file, desc:POST /ISAPI/Thermal/ImportThermalCalibrationFile?
format=json-->true

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</isSupportImportThermalCalibrationFile>
<isSupportThermalCalibrationFileCreatParam>
  <!--ro, opt, bool, whether the device supports the parameters of generating temperature calibration file, desc:GET
  /ISAPI/Thermal/ThermalCalibrationFileCreatParam?format=json-->true
</isSupportThermalCalibrationFileCreatParam>
<isSupportNormalRuleParams>
  <!--ro, opt, bool, whether the device supports the parameters of smoke and fire detection normal rules, desc:GET/PUT
  /ISAPI/Thermal/fireDetection/channels/<channelID>/NormalRuleParams?format=json-->true
</isSupportNormalRuleParams>
<isSupportEnvironmentTemperatureMonitorScheduleParams>
  <!--ro, opt, bool, whether the device supports the parameters of environment temperature monitor arming schedule, desc:GET/PUT
  /ISAPI/Thermal/EnvironmentTemperatureMonitorScheduleParams?format=json-->true
</isSupportEnvironmentTemperatureMonitorScheduleParams>
<isSupportEnvironmentTemperatureMonitorParam>
  <!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring, desc:GET/PUT
  /ISAPI/Thermal/EnvironmentTemperatureMonitorParam?format=json-->true
</isSupportEnvironmentTemperatureMonitorParam>
<isSupportEnvironmentTemperatureMonitorTriggerParams>
  <!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring linkage action, desc:GET/PUT
  /ISAPI/Thermal/EnvironmentTemperatureMonitorTriggerParams?format=json-->true
</isSupportEnvironmentTemperatureMonitorTriggerParams>
<isSupportSearchAsynThermalInspectionTaskTestPoint>
  <!--ro, opt, bool, whether the device supports searching for information of specified asynchronous thermal inspection point, desc:POST
  /ISAPI/Thermal/SearchAsynThermalInspectionTaskTestPoint?format=json-->true
</isSupportSearchAsynThermalInspectionTaskTestPoint>
<isSupportSearchAsynThermalInspectionTask>
  <!--ro, opt, bool, whether the device supports searching for specified asynchronous thermal inspection task, desc:POST
  /ISAPI/Thermal/SearchAsynThermalInspectionTask?format=json-->true
</isSupportSearchAsynThermalInspectionTask>
<isSupportAddAsynThermalInspectionTaskTestPoint>
  <!--ro, opt, bool, whether the device supports adding information of specified asynchronous thermal inspection point, desc:POST
  /ISAPI/Thermal/AddAsynThermalInspectionTaskTestPoint?format=json-->true
</isSupportAddAsynThermalInspectionTaskTestPoint>
<isSupportAddAsynThermalInspectionTask>
  <!--ro, opt, bool, whether the device supports adding asynchronous thermal inspection task, desc:POST /ISAPI/Thermal/AddAsynThermalInspectionTask?
  format=json-->true
</isSupportAddAsynThermalInspectionTask>
<isSupportImportAsynThermalInspectionTaskBaseMap>
  <!--ro, opt, bool, whether the device supports importing asynchronous thermal inspection task base map, desc:POST
  /ISAPI/Thermal/ImportAsynThermalInspectionTaskBaseMap?format=json-->true
</isSupportImportAsynThermalInspectionTaskBaseMap>
<isSupportShutterFreeze>
  <!--ro, opt, bool, whether the device supports freezing the shutter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ShutterFreeze?format=json-->true
</isSupportShutterFreeze>
</ThermalCap>
<WLAlarmCap>
  <!--ro, opt, object, wireless alarm capability-->
  <isSupportTeleControl>
    <!--ro, opt, bool, whether the device supports remote control-->true
  </isSupportTeleControl>
  <isSupportPIR>
    <!--ro, opt, bool, whether the device supports PIR configuration-->true
  </isSupportPIR>
  <isSupportWLSensors>
    <!--ro, opt, bool, N/A-->true
  </isSupportWLSensors>
  <isSupportCallHelp>
    <!--ro, opt, bool, whether the device supports calling for help-->true
  </isSupportCallHelp>
  <WLSensorsNum>
    <!--ro, opt, int, N/A-->1
  </WLSensorsNum>
</WLAlarmCap>
<isSupportGIS>
  <!--ro, opt, bool, whether the device supports GIS, desc:related URI: /ISAPI/GIS/channels-->true
</isSupportGIS>
<isSupportCompass>
  <!--ro, opt, bool, whether the device supports compass configuration, desc:related URI: /ISAPI/Compass/channels/<ID>/capabilities-->true
</isSupportCompass>
<isSupportRoadInfoOverlays>
  <!--ro, opt, bool, whether the device supports overlying the Lane information-->true
</isSupportRoadInfoOverlays>
<isSupportFaceCaptureStatistics>
  <!--ro, opt, bool, whether the device supports face capture statistics, desc:related URI: /ISAPI/Intelligent/channels/<ID>/faceCaptureStatistics/search-
  -->true
</isSupportFaceCaptureStatistics>
<isSupportElectronicsEnlarge>
  <!--ro, opt, bool, whether the device supports electronics enlarge-->true
</isSupportElectronicsEnlarge>
<isSupportRemoveStorage>
  <!--ro, opt, bool, whether the device supports remote storage-->true
</isSupportRemoveStorage>
<isSupportCloud>
  <!--ro, opt, bool, whether the device supports cloud storage-->true
</isSupportCloud>
<isSupportRecordHost>
  <!--ro, opt, bool, whether the device supports configuring the education sharing server, desc:related URI: /ISAPI/RecordHost/capabilities-->true
</isSupportRecordHost>
<isSupportViewshedOverlay>
  <!--ro, opt, bool, whether the device supports overlying viewshed camera target, desc:related URI:
  /ISAPI/System/Video/inputs/channels/<ID>/viewshedOverLayer-->true
</isSupportViewshedOverlay>
<isSupportGPSLabelTracking>
  <!--ro, opt, bool, whether the device supports GPS tag tracking, desc:related URI: /ISAPI/GTS/channel/<ID>/GPSLabelTracking?format=json-->true

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</isSupportGPSTracking>
<isSupportEagleEye>
  <!--ro, opt, bool, whether the device supports the PanoVu series camera-->true
</isSupportEagleEye>
<isSupportPanorama>
  <!--ro, opt, bool, whether the device supports panoramic mode-->true
</isSupportPanorama>
<isSupportFirmwareVersionInfo>
  <!--ro, opt, bool, whether the device supports displaying the firmware version information-->true
</isSupportFirmwareVersionInfo>
<isSupportExternalWirelessServer>
  <!--ro, opt, bool, whether the device supports external wireless servers: true (support)-->true
</isSupportExternalWirelessServer>
<isSupportSetupCalibration>
  <!--ro, opt, bool, whether the device supports configuring calibration-->true
</isSupportSetupCalibration>
<isSupportGetmutexFuncErrMsg>
  <!--ro, opt, bool, whether the device supports the mutual exclusion information-->true
</isSupportGetmutexFuncErrMsg>
<isSupportTokenAuthenticate>
  <!--ro, opt, bool, whether the device supports token authentication-->true
</isSupportTokenAuthenticate>
<isSupportStreamDualVCA>
  <!--ro, opt, bool, whether the device supports overlying intelligent private data on the stream, desc:related URI: in the
  /ISAPI/Intelligent/channels/<ID>/intelliResource; corresponding node: <isOverLapIntelli-->true
</isSupportStreamDualVCA>
<isSupportLaserSpotManual>
  <!--ro, opt, bool, whether the device supports configuring Laser Light spot-->true
</isSupportLaserSpotManual>
<isSupportRTMP>
  <!--ro, opt, bool, whether the device supports RTMP configuration, desc:related URI: /ISAPI/Streaming/channels/<ID>/RTMPCfg/capabilities-->true
</isSupportRTMP>
<isSupportTraffic>
  <!--ro, opt, bool, traffic service capability-->true
</isSupportTraffic>
<isSupportLaserSpotAdjustment>
  <!--ro, opt, bool, adjust the size of Laser spot, desc:related URI: /ISAPI/Image/channels/<ID>/LaserLight/LaserSpotAdjustment/capabilities-->true
</isSupportLaserSpotAdjustment>
<VideoIntercomCap>
  <!--ro, opt, object, video intercom capability, desc:related URI: /ISAPI/VideoIntercom/capabilities-->
  <isSupportSceneManage>
    <!--ro, opt, bool, scene management (the configuration port needs to be switched according to various scenes)-->true
  </isSupportSceneManage>
  <isSupportDeviceId>
    <!--ro, opt, bool, whether the device supports device ID-->true
  </isSupportDeviceId>
  <isSupportOperationTime>
    <!--ro, opt, bool, whether the device supports the operation time-->true
  </isSupportOperationTime>
  <isSupportCallerInfo>
    <!--ro, opt, bool, whether the device supports the information management of the calling persons-->true
  </isSupportCallerInfo>
  <isSupportCallSignal>
    <!--ro, opt, bool, whether the device supports the call signaling interaction, desc:related URI: /ISAPI/VideoIntercom/callSignal?format=json-->true
  </isSupportCallSignal>
  <isSupportCallStatus>
    <!--ro, opt, bool, whether the device supports getting call status, desc:related URI: /ISAPI/VideoIntercom/callStatus?format=json-->true
  </isSupportCallStatus>
  <isSupportCallElevator>
    <!--ro, opt, bool, whether the device supports calling for elevator-->true
  </isSupportCallElevator>
  <isSupportRelatedDeviceAddress>
    <!--ro, opt, bool, whether the device supports linking device address-->true
  </isSupportRelatedDeviceAddress>
  <isSupportCardSectorCheck>
    <!--ro, opt, bool, sector encryption verification-->true
  </isSupportCardSectorCheck>
  <isSupportRemoteOpenDoor>
    <!--ro, opt, bool, whether it supports remote unlocking-->true
  </isSupportRemoteOpenDoor>
  <isSupportKeyCfg>
    <!--ro, opt, bool, configuration for pressing button to call for resident-->true
  </isSupportKeyCfg>
  <isSupportWorkModeCfg>
    <!--ro, opt, bool, whether the device supports configuring the working mode, desc:related URI: URL/ISAPI/VideoIntercom/workMode/capabilities-->true
  </isSupportWorkModeCfg>
  <isSupportSystemSwitchCfg>
    <!--ro, opt, bool, configuring the switch of video intercom system-->true
  </isSupportSystemSwitchCfg>
  <isSupportPrivilegePasswordStatus>
    <!--ro, opt, bool, getting the status of the device's permission password-->true
  </isSupportPrivilegePasswordStatus>
  <isSupportSmartHomeRoomsCfg>
    <!--ro, opt, bool, whether the device supports configuring room information-->true
  </isSupportSmartHomeRoomsCfg>
  <isSupportSmartHomeDevices>
    <!--ro, opt, bool, whether the device supports intelligent devices management-->true
  </isSupportSmartHomeDevices>
  <isSupportScenesCfg>
    <!--ro, opt, bool, whether the device supports scene configuration (adding, deleting, editing and searching for)-->true
  </isSupportScenesCfg>
  <isSupportNameModify>
    <!--ro, opt, bool, whether the device supports editing the zone name-->true
  </isSupportNameModify>
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</isSupportNameModify>
<isSupportElevatorControlCfg>
  <!--ro, opt, bool, whether the device supports elevator control-->true
</isSupportElevatorControlCfg>
<isSupportPrivilegePassword>
  <!--ro, opt, bool, whether the device supports configuring passwords for different scenes-->true
</isSupportPrivilegePassword>
<isSupportSubModules>
  <!--ro, opt, bool, whether the device supports sub-module management-->true
</isSupportSubModules>
<isSupportIssueCards>
  <!--ro, opt, bool, whether the device supports configuring the card issuing function of the master card-->true
</isSupportIssueCards>
<isSupportSendCardCfg>
  <!--ro, opt, bool, whether the door station supports configuring card issuing function-->true
</isSupportSendCardCfg>
<isSupportRingBackTone>
  <!--ro, opt, bool, whether the device supports importing the ring back tone-->true
</isSupportRingBackTone>
<isSupportCallPriority>
  <!--ro, opt, bool, whether the device supports calling permission-->true
</isSupportCallPriority>
<isSupportPhoneCfg>
  <!--ro, opt, bool, whether the device supports configuring resident phones-->true
</isSupportPhoneCfg>
<isSupportWatchPoint>
  <!--ro, opt, bool, whether the device supports watch point, desc:/ISAPI/VideoIntercom/WatchPoint-->true
</isSupportWatchPoint>
<isSupportVoiceConfiguration>
  <!--ro, opt, bool, whether the device supports the configuration of enabling the voice recognition and prompt-->true
</isSupportVoiceConfiguration>
<isSupportSubModulesCfg>
  <!--ro, opt, bool, whether the device supports configuring editing the sub module parameters-->true
</isSupportSubModulesCfg>
<isSupportDisplayRegion>
  <!--ro, opt, bool, whether the device supports configuring the area to be displayed-->true
</isSupportDisplayRegion>
<isSupportUpdateData>
  <!--ro, opt, bool, whether the device supports importing upgrade files-->true
</isSupportUpdateData>
<isSupportUpdateStrategy>
  <!--ro, opt, bool, whether the device supports upgrade configuration-->true
</isSupportUpdateStrategy>
<isSupportMetaInfo>
  <!--ro, opt, bool, whether the device supports getting the information of upgrade files-->true
</isSupportMetaInfo>
<isSupportStreamAbility>
  <!--ro, opt, bool, whether the device supports IPC management-->true
</isSupportStreamAbility>
<platform opt="R0,R1,S29,N10,N8,N4">
  <!--ro, opt, enum, platform type, subType:string, attr:opt{reg, string}, desc:"R0", "R1"; it will be returned if the device supports-->R0
</platform>
<isSupportSecurityMode>
  <!--ro, opt, bool, whether the device supports switching the version of security protocols-->true
</isSupportSecurityMode>
<isSupportDeviceCommunication>
  <!--ro, opt, bool, whether the device supports configuring the parameters of device group-net communication-->true
</isSupportDeviceCommunication>
<isSupportAutoSync>
  <!--ro, opt, bool, whether the device supports automatically synchronous parameters configuration-->true
</isSupportAutoSync>
<isSupportDifferentKeystrokes>
  <!--ro, opt, bool, whether the device supports distinguishing long press and short press-->true
</isSupportDifferentKeystrokes>
<isSupportKeyCfgWeekPlan>
  <!--ro, opt, bool, the week schedule configuration for pressing to call-->true
</isSupportKeyCfgWeekPlan>
<isSupportKeyCfgHolidayPlan>
  <!--ro, opt, bool, the holiday schedule configuration for pressing to call-->true
</isSupportKeyCfgHolidayPlan>
<isSupportKeyCfgHolidayGroup>
  <!--ro, opt, bool, the holiday group configuration for pressing to call-->true
</isSupportKeyCfgHolidayGroup>
<isSupportKeyCfgPlanTemplate>
  <!--ro, opt, bool, the schedule template configuration for pressing to call-->true
</isSupportKeyCfgPlanTemplate>
<isSupportCallCfg>
  <!--ro, opt, bool, device call configuration-->true
</isSupportCallCfg>
<isSupportAccessParams>
  <!--ro, opt, bool, the application access parameters-->true
</isSupportAccessParams>
<isSupportTransmitRequestEventConfirm>
  <!--ro, opt, bool, response to the transparent transmission request-->true
</isSupportTransmitRequestEventConfirm>
<isSupportIncomingCallLinkage>
  <!--ro, opt, bool, whether the device supports configuring incoming calls linkage-->true
</isSupportIncomingCallLinkage>
<isSupportAdbControl>
  <!--ro, opt, bool, whether the device supports remotely enabling and disabling ADB-->true
</isSupportAdbControl>
<isSupportProtocolFileManager>
  <!--ro, opt, bool, whether the device supports the management of the protocol notice files, desc:/ISAPI/VideoIntercom/protocolFileManager/capabilities?
  format=json, status=

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format=json-->true
</isSupportProtocolFileManage>
<isSupportStreamSwitch>
  <!--ro, opt, bool, whether the device supports stream switch, desc:/ISAPI/VideoIntercom/streamSwitch/capabilities?format=json-->true
</isSupportStreamSwitch>
<isSupportKeywordAudioRecognition>
  <!--ro, opt, bool, whether the device supports keyword voice recognition, desc:/ISAPI/VideoIntercom/keywordAudioRecognition/capabilities?format=json--
>true
</isSupportKeywordAudioRecognition>
<isSupportOneWaySpeaking>
  <!--ro, opt, bool, whether the device supports speaking loud, desc:true (support), this node is not returned (not support)-->true
</isSupportOneWaySpeaking>
<isSupportPositionInfo>
  <!--ro, opt, bool, whether the device supports configuring the device location information, desc:/ISAPI/VideoIntercom/positionInfo?format=json-->true
</isSupportPositionInfo>
<isSupportElevatorControlDisplay>
  <!--ro, opt, bool, whether the device supports configuring displaying elevator control, desc:/ISAPI/VideoIntercom/elevatorControlDisplay?format=json--
>true
</isSupportElevatorControlDisplay>
<isSupportRingManage>
  <!--ro, opt, bool, whether the device supports ring tone management, desc:related URI: /ISAPI/VideoIntercom/ringTone?format=json;
/ISAPI/VideoIntercom/ringTone/<ringID>?format=json-->true
</isSupportRingManage>
<isSupportPasswordAuthentication>
  <!--ro, opt, bool, whether the device supports verifying the video intercom password, desc:related URI:
/ISAPI/VideoIntercom/passwordAuthentication/capabilities-->true
</isSupportPasswordAuthentication>
<isSupportAlarmControlByPhone>
  <!--ro, opt, bool, whether the device supports phone arming control, desc:related URI: /ISAPI/SecurityCP/AlarmControlByPhone/capabilities-->true
</isSupportAlarmControlByPhone>
<isSupportZoneCfgByScene>
  <!--ro, opt, bool, whether the device supports configuring the zones parameter by scene, desc:related URI:
/ISAPI/SecurityCP/BasicParam/sceneZoneCfg/capabilities-->true
</isSupportZoneCfgByScene>
<isSupportAppKeyConfiguration>
  <!--ro, opt, bool, whether the device supports configuring APPKey, desc:related URI: /ISAPI/VideoIntercom/AppKeyConfiguration/capabilities-->true
</isSupportAppKeyConfiguration>
<isSupportIntercomProtocolChange>
  <!--ro, opt, bool, whether the device supports configuring the switch of device VoIP protocols, desc:related URI:
/ISAPI/VideoIntercom/IntercomProtocolType/capabilities-->true
</isSupportIntercomProtocolChange>
<isSupportGetAnnouncementMessage>
  <!--ro, opt, bool, whether the device supports getting announcement message, desc:related URI: /ISAPI/VideoIntercom/AnnouncementMessage/capabilities--
>true
</isSupportGetAnnouncementMessage>
<isSupportGetDeployInfo>
  <!--ro, opt, bool, whether the device supports getting arming information, desc:related URI: /ISAPI/AccessControl/DeployInfo/capabilities-->true
</isSupportGetDeployInfo>
<detectorType opt="panicButton,magneticContact,smokeDetector,activeInfraredDetector,passiveInfraredDetector,combustibleGasDetector,doorbell">
  <!--ro, opt, string, detector types supported by the device, attr:opt(req, string), desc:"panicButton" (panic button), "magneticContact" (magnetic
contact), "smokeDetector" (smoke detector), "activeInfraredDetector" (active infrared detector), "passiveInfraredDetector" (passive infrared detector),
"combustibleGasDetector" (combustible gas detector)-->test
</detectorType>
<isSupportBroadcastRealtime>
  <!--ro, opt, bool, whether the device supports real-time audio broadcast-->true
</isSupportBroadcastRealtime>
<isSupportOrganizationNodes>
  <!--ro, opt, bool, whether the device supports organization architecture management-->true
</isSupportOrganizationNodes>
<isSupportAudioPlayStatus>
  <!--ro, opt, bool, whether the device supports uploading the audio playing status-->true
</isSupportAudioPlayStatus>
<isSupportDeviceIdAnalysisRuleCfg>
  <!--ro, opt, bool, whether the device supports configuring the analysis rule of the device No. (related URI:
/ISAPI/VideoIntercom/AnalysisRuleCfg/capabilities?format=json)-->true
</isSupportDeviceIdAnalysisRuleCfg>
<isSupportOperationTimeByChannel>
  <!--ro, opt, bool, whether the device supports configuring the operation time by channel-->true
</isSupportOperationTimeByChannel>
<isSupportCallAuthorization>
  <!--ro, opt, bool, whether the device supports configuring permission for incoming calls, desc:related URI:
/ISAPI/VideoIntercom/CallAuthorization/capabilities?format=json-->true
</isSupportCallAuthorization>
<isSupportRecordControl>
  <!--ro, opt, bool, whether the device supports audio/video control, desc:related URI: /ISAPI/VideoIntercom/recordControl/capabilities?format=json--
>true
</isSupportRecordControl>
<isSupportTrusteeship>
  <!--ro, opt, bool, whether the device supports configuring trusteeship, desc:related URI: /ISAPI/VideoIntercom/trusteeship/capabilities?format=json--
>true
</isSupportTrusteeship>
<isSupportAnalogIndoorStation>
  <!--ro, opt, bool, whether the device supports the management of the analog indoor station, desc:related URI:
/ISAPI/VideoIntercom/analogIndoorStation/capabilities?format=json-->true
</isSupportAnalogIndoorStation>
<isSupportImportPartitions>
  <!--ro, opt, bool, whether the device supports importing the terminal List, desc:(partition list for the paging microphone devices distributed by the
platform) (Related URI: /ISAPI/VideoIntercom/ImportTerminals/capabilities?format=json)-->true
</isSupportImportPartitions>
<isSupportBroadcastPlan>
  <!--ro, opt, bool, whether the device supports full application of schedule broadcast, desc:related URI:
ISAPI/VideoIntercom/broadcast/plan/capabilities?format=json-->true
</isSupportBroadcastPlan>
<isSupportBroadcastPlanSearch>

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<!--ro, opt, bool, whether the device supports searching for full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/planSearch?
format=json-->true
</isSupportBroadcastPlanSearch>
<isSupportBroadcastPlanPause>
<!--ro, opt, bool, whether the device supports pausing full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/pause?
format=json-->true
</isSupportBroadcastPlanPause>
<isSupportBroadcastPlanResume>
<!--ro, opt, bool, whether the device supports restoring full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/resume?
format=json-->true
</isSupportBroadcastPlanResume>
<isSupportBroadcastPlanTest>
<!--ro, opt, bool, whether the device supports the test run of full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/test?
format=json-->true
</isSupportBroadcastPlanTest>
<isSupportBroadcastPlanEx>
<!--ro, opt, bool, whether the device supports incremental application of schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planEx/capabilities?format=json-->true
</isSupportBroadcastPlanEx>
<isSupportBroadcastPlanExSearch>
<!--ro, opt, bool, whether the device supports searching for incremental schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planExSearch?format=json-->true
</isSupportBroadcastPlanExSearch>
<isSupportBroadcastPlanModify>
<!--ro, opt, bool, whether the device supports editing a single schedule broadcast, desc:it is valid only during incremental application-->true
</isSupportBroadcastPlanModify>
<isSupportBroadcastPlanDelete>
<!--ro, opt, bool, whether the device supports deleting a single schedule broadcast, desc:it is valid only during incremental application-->true
</isSupportBroadcastPlanDelete>
<isSupportAudioInPlayCfg>
<!--ro, opt, bool, whether the device supports the configuration of playing the audio input sources of the broadcast, desc:URI for getting the
capability: /ISAPI/VideoIntercom/broadcast/AudioInPlayCfg/capabilities?format=json-->true
</isSupportAudioInPlayCfg>
<isSupportSearchWardBedInfo>
<!--ro, opt, bool, whether the device supports searching for ward bed information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/searchWardBedInfo/capabilities?format=json. It is used for the main station to search for whether the bed connected to the sub machine
in the ward is already occupied-->true
</isSupportSearchWardBedInfo>
<isSupportPatient>
<!--ro, opt, bool, whether the device supports applying patient information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/patient/capabilities?format=json-->true
</isSupportPatient>
<isSupportIntroduction>
<!--ro, opt, bool, whether the device supports applying the medical introduction, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/introduction/capabilities?format=json-->true
</isSupportIntroduction>
<isSupportConstantVoltage>
<!--ro, opt, bool, whether the device supports constant voltage broadcast, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/ConstantVoltage/capabilities?format=json-->true
</isSupportConstantVoltage>
<isSupportRecordMode>
<!--ro, opt, bool, whether it supports configuring recording mode, desc:URI for getting the capability: /ISAPI/VideoIntercom/RecordMode/capabilities?
format=json-->true
</isSupportRecordMode>
<isSupportCallForwarding>
<!--ro, opt, bool, whether the device supports active call transfer, desc:URI for getting the capability:
/ISAPI/VideoIntercom/callForwarding/capabilities?format=json-->true
</isSupportCallForwarding>
<isSupportMediaLibrary>
<!--ro, opt, bool, whether the device supports applying media library information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/MediaLibrary/capabilities?format=json-->true
</isSupportMediaLibrary>
<isSupportBroadcastControl>
<!--ro, opt, bool, whether the device supports configuring broadcast control parameters, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/control/capabilities?format=json-->true
</isSupportBroadcastControl>
<isSupportGetImportPartitions>
<!--ro, opt, bool, whether the device supports getting partition list, desc:POST /ISAPI/VideoIntercom/broadcast/GetImportPartitions?format=json-->true
</isSupportGetImportPartitions>
<isSupportSearchMediaLibrary>
<!--ro, opt, bool, whether the device supports searching media library, desc:POST /ISAPI/VideoIntercom/broadcast/SearchMediaLibrary?format=json-->true
</isSupportSearchMediaLibrary>
<isSupportImportPartitionsBatch>
<!--ro, opt, bool, whether the device supports batch importing of partitions, desc:PUT /ISAPI/VideoIntercom/broadcast/ImportPartitionsBatch?
format=json-->true
</isSupportImportPartitionsBatch>
<isSupportMediaLibraryBatch>
<!--ro, opt, bool, whether the device supports batch applying of media library information, desc:PUT /ISAPI/VideoIntercom/broadcast/MediaLibraryBatch?
format=json-->true
</isSupportMediaLibraryBatch>
<isSupportPlanScheme>
<!--ro, opt, bool, whether the device supports broadcast schedules, desc:GET /ISAPI/VideoIntercom/broadcast/PlanScheme/capabilities?format=json-->true
</isSupportPlanScheme>
<isSupportRealTimeInsertion>
<!--ro, opt, bool, whether the device supports real-time cut-in, desc:POST /ISAPI/VideoIntercom/broadcast/RealTimeInsertion?format=json-->true
</isSupportRealTimeInsertion>
</VideoIntercomCap>
<SecurityCPCapabilities>
<!--ro, opt, object, [TBC], desc:related URI: /ISAPI/SecurityCP/capabilities?format=json-->true
</SecurityCPCapabilities>
<isSupportSafetyCabin>
<!--ro, opt, bool, safety cabinet controller, desc:related URI: /ISAPI/SafetyCabin/capabilities-->true
</isSupportSafetyCabin>

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<isSupportAcUpdate>
  <!--ro, opt, bool, whether the device supports peripheral module upgrade (true, support), desc:related URI: /ISAPI/System/AcsUpdate/capabilities-->true
</isSupportAcUpdate>
<isSupportSecurityCP>
  <!--ro, opt, bool, N/A-->true
</isSupportSecurityCP>
<isSupportPEA>
  <!--ro, opt, bool, the host capability of one-push alarm-->true
</isSupportPEA>
<isSupportCurrentLock>
  <!--ro, opt, bool, whether the device supports the configuration of Locking current-->true
</isSupportCurrentLock>
<isSupportGuardAgainstTheft>
  <!--ro, opt, bool, whether the device supports configuring device guard against theft, desc:related URI: /ISAPI/System/guardAgainstTheft/capabilities-->true
</isSupportGuardAgainstTheft>
<isSupportPicInfoOverlap>
  <!--ro, opt, bool, whether the device supports overlaying pictures and texts-->true
</isSupportPicInfoOverlap>
<isSupportPlay>
  <!--ro, opt, bool, shielded live view-->true
</isSupportPlay>
<isSupportPlayback>
  <!--ro, opt, bool, shielded playback-->true
</isSupportPlayback>
<UHFRFIDReader>
  <!--ro, opt, object, capability supported by the RFID reader-->
  <isSupportBasicInformation>
    <!--ro, opt, bool, whether the device supports the basic parameters of the reader-->true
  </isSupportBasicInformation>
  <isSupportHardDiskStorageTest>
    <!--ro, opt, bool, whether the device supports disk storage performance test of the reader-->true
  </isSupportHardDiskStorageTest>
  <isSupportCarReaderConfig>
    <!--ro, opt, bool, whether the device supports configuring the parameters of the electronic license plate reader, desc:related URI: /ISAPI/Traffic/UHFRFIDReader/carReaderConfig/capabilities?format=json-->true
  </isSupportCarReaderConfig>
</UHFRFIDReader>
<isSupportMasterSlaveConfig>
  <!--ro, opt, bool, whether the device supports configuring main and sub device, desc:related URI: /ISAPI/System/masterSlaveConfig?format=json-->true
</isSupportMasterSlaveConfig>
<isSupportIntelligentStructureAnalysis>
  <!--ro, opt, bool, whether the device supports intelligent structuralization analysis, desc:related URI: /ISAPI/Intelligent/structureAnalysis/capabilities-->true
</isSupportIntelligentStructureAnalysis>
<isSupportIntelligentAnalysisEngines>
  <!--ro, opt, bool, whether the device supports configuring the intelligent engine, desc:related URI: /ISAPI/Intelligent/analysisEngines/capabilities-->true
</isSupportIntelligentAnalysisEngines>
<PreviewDisplayNum>
  <!--ro, opt, int, number of the browser live view window-->1
</PreviewDisplayNum>
<isSupportBoard opt="true,false">
  <!--ro, opt, bool, whether the device supports related sub-board protocols, attr:opt{opt, string}-->true
</isSupportBoard>
<ResourceSwitch>
  <!--ro, opt, object, configuration of resource conversion-->
  <workMode opt="4KPreview,educationRecord">
    <!--ro, req, string, the working mode: education record mode, attr:opt{opt, string}-->test
  </workMode>
</ResourceSwitch>
<isSupportCustomStream>
  <!--ro, opt, bool, whether the device supports custom stream, desc:related URI: /ISAPI/Streaming/channels/<ID>/customStream/capabilities-->true
</isSupportCustomStream>
<isSupportTriggerCapCheck>
  <!--ro, opt, bool, it indicates that the capability of the URL (/ISAPI/Event/triggersCap) is already verified-->true
</isSupportTriggerCapCheck>
<isSupportChannelEventCap>
  <!--ro, opt, bool, whether the device supports getting the event capability by channel, desc:related URI: /ISAPI/Event/channels/<ID>/capabilities-->true
</isSupportChannelEventCap>
<SensorCap>
  <!--ro, opt, object, capability of configuring sensor parameters, desc:related URI: /ISAPI/System/Sensor/capabilities-->
  <SensorNums>
    <!--ro, opt, int, the maximum number of sensors that can be added-->1
  </SensorNums>
  <isSupportSensorPortConfiguration>
    <!--ro, opt, bool, supports configuring the detection parameters-->true
  </isSupportSensorPortConfiguration>
  <isSupportSensorPortControl>
    <!--ro, opt, bool, supports detection control-->true
  </isSupportSensorPortControl>
  <isSupportAcquireSensorStatus>
    <!--ro, opt, bool, supports getting the real-time status of the sensor-->true
  </isSupportAcquireSensorStatus>
  <isSupportSearchSensorData>
    <!--ro, opt, bool, supports searching the sensor data-->true
  </isSupportSearchSensorData>
</SensorCap>
<isSupportEncryption>
  <!--ro, opt, bool, stream encryption capability-->true
</isSupportEncryption>
<isSupportTrial>
  <!--ro, opt, bool, whether the device supports configuring inquest device-->true
</isSupportTrial>

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<isSupportSignInstrument>
  <!--ro, opt, bool, the capability of physical sign monitor-->true
</isSupportSignInstrument>
<isSupportLocate>
  <!--ro, opt, bool, whether the device supports location configuration-->true
</isSupportLocate>
<isSupportVibration>
  <!--ro, opt, bool, whether the device supports configuring vibration parameters-->true
</isSupportVibration>
<isSupportMixedTargetDetection>
  <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
</isSupportMixedTargetDetection>
<SDCardCtrlCap>
  <!--ro, opt, object, the remote control capability of the memory card-->
  <isSupportIOCtrl>
    <!--ro, opt, bool, whether the device supports the capability of control and command upgrade-->true
  </isSupportIOCtrl>
  <isSupportReadCardStatus>
    <!--ro, opt, bool, whether the device supports the capability of card recognition operation-->true
  </isSupportReadCardStatus>
  <isSupportStartUpdate>
    <!--ro, opt, bool, whether the device supports the capability of upgrade start control-->true
  </isSupportStartUpdate>
  <isSupportEndUpdate>
    <!--ro, opt, bool, whether the device supports the capability of upgrade end control-->true
  </isSupportEndUpdate>
  <isSupportPowerCtrl>
    <!--ro, opt, bool, whether the device supports the capability of power on and off control-->true
  </isSupportPowerCtrl>
</SDCardCtrlCap>
<isSupportMultiChannelCalibration>
  <!--ro, opt, bool, whether the device supports multi-channel calibration-->true
</isSupportMultiChannelCalibration>
<isSupportElevatorFloorCalibration>
  <!--ro, opt, bool, whether the device supports configuring floor display-->true
</isSupportElevatorFloorCalibration>
<isSupportPictureCaptureComparison>
  <!--ro, opt, bool, whether the device supports 1V1 comparison of the pictures in the library and from camera-->true
</isSupportPictureCaptureComparison>
<isSupportGPSCalibration>
  <!--ro, opt, bool, whether the device supports the GPS calibration function-->true
</isSupportGPSCalibration>
<isSupportCityManagement>
  <!--ro, opt, bool, whether the device supports city management detection event-->true
</isSupportCityManagement>
<isSupportBVCorrect>
  <!--ro, opt, bool, whether the device supports configuring the parameters of camera correction-->true
</isSupportBVCorrect>
<isSupportSteamingCodingModeSwitch>
  <!--ro, opt, bool, encoding output mode (true-support), desc:related URI: /ISAPI/System/streamingCoding/mode/capabilities?format=json-->true
</isSupportSteamingCodingModeSwitch>
<supportImageChannel opt="1,2,3,4">
  <!--ro, opt, string, whether the device supports configuring the image channel, attr:opt(req, string),
  desc:the configuration of actual image channel supported by the device (URI for picture parameters: /ISAPI/Image/channels/<channelID>), if this node is not
  returned, the image channel is the same with the encoding channel (/ISAPI/Streaming/channels) by default
  this node is mainly for the situation when the image channel is inconsistent with the actual stream channel. For example, some device, which altogether
  has four sensors and they are all spliced into one encoding channel for output. Every sensor supports configuring image parameters; by default, the image
  channel 1 is only valid for the images of sensor1-->test
</supportImageChannel>
<isSupportPanoramaCalibration>
  <!--ro, opt, bool, whether the device supports configuring the calibration of the panoramic camera-->true
</isSupportPanoramaCalibration>
<isSupportPanoramaMode>
  <!--ro, opt, bool, whether the device supports the mode switch of the panoramic camera-->true
</isSupportPanoramaMode>
<supportSnapshotChannel opt="1">
  <!--ro, opt, string, the capability supported by the capture schedule channel, attr:opt(opt, string)-->test
</supportSnapshotChannel>
<isSupportPanoramaCorrections>
  <!--ro, opt, bool, whether the device supports configuring the correction algorithm parameters-->true
</isSupportPanoramaCorrections>
<isSupportActiveMulticast>
  <!--ro, opt, bool, whether the device supports multicast configuration (/ISAPI/Streaming/channels/<ID>/capabilities)-->true
</isSupportActiveMulticast>
<isSupportChannelEventListCap>
  <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
</isSupportChannelEventListCap>
<VCAResourceChannelsCap>
  <!--ro, opt, array, subType:object-->
  <ChannelsList>
    <!--ro, opt, object-->
    <channelsID>
      <!--ro, req, int-->1
    </channelsID>
  </ChannelsList>
</VCAResourceChannelsCap>
<isSupportTimerVCAResource>
  <!--ro, opt, bool, whether the device supports scheduled switch of intelligent resources-->true
</isSupportTimerVCAResource>
<MixedTargetDetection>
  <!--ro, opt, object, the capability of recognizing specified target among mixed targets supported by the device-->
  <isSupportFaceRecognition>
    <!--ro, opt, bool, whether the device supports facial recognition-->true

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</isSupportFaceRecognition>
<isSupportHumanRecognition>
  <!--ro, opt, bool, whether the device supports human body recognition-->true
</isSupportHumanRecognition>
<isSupportVehicleRecognition>
  <!--ro, opt, bool, whether the device supports vehicle recognition-->true
</isSupportVehicleRecognition>
</MixedTargetDetection>
<isSupportDiscoveryMode>
  <!--ro, opt, bool, whether the device supports discovery mode, desc:related URI: /ISAPI/System/discoveryMode/capabilities-->true
</isSupportDiscoveryMode>
<isSupportDumpData>
  <!--ro, opt, bool, whether the device supports exporting the Dump file-->true
</isSupportDumpData>
<isSupportCameraAccessConfiguration>
  <!--ro, opt, bool, whether the device supports the camera access configuration, desc:related URI: /ISAPI/System/cameraAccess/configuration/capabilities?
format=json-->true
</isSupportCameraAccessConfiguration>
<isSupportIntelligentSearch>
  <!--ro, opt, bool, whether the device supports intelligent search, desc:related URI: /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json--
>true
</isSupportIntelligentSearch>
<isSupportAutoMaintenance>
  <!--ro, opt, bool, whether the device supports automatic maintenance, desc:related URI: /ISAPI/System/autoMaintenance/capabilities?format=json-->true
</isSupportAutoMaintenance>
<isSupportRtspOverHTTPS>
  <!--ro, opt, bool, supports tspOverHTTPS-->true
</isSupportRtspOverHTTPS>
<isSupportIntersectionAnalysis>
  <!--ro, opt, bool, whether the device supports the intersection analysis, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/intersectionAnalysis/capabilities?format=json-->true
</isSupportIntersectionAnalysis>
<isSupportAudioInteract>
  <!--ro, opt, bool, whether the device supports audio interaction, desc:related URI: /ISAPI/Streaming/interact/capabilities?format=json-->true
</isSupportAudioInteract>
<isSupportChannelFullEventListCap>
  <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
</isSupportChannelFullEventListCap>
<isSupportAUXInfoCap>
  <!--ro, opt, bool, it indicates that whether the device supports getting the capability of getting the attributes of device's all channels-->true
</isSupportAUXInfoCap>
<supportPanoramaChannel opt="2">
  <!--ro, opt, string, the panoramic channels supported by the device, attr:opt{opt, string}-->test
</supportPanoramaChannel>
<regionClip>
  <!--ro, opt, object, the channels and streams that supports area cropping-->
  <channels opt="101">
    <!--ro, opt, string, the channels and streams that supports area cropping, attr:opt{opt, string}, desc:101: channel No.*100+stream type-->test
  </channels>
</regionClip>
<AIDEventSupport
opt="abandonedObject,pedestrian,congestion,roadBlock,construction,trafficAccident,fogDetection,wrongDirection,illegalParking,SSharpDriving,lowSpeed,dragRaci
ng,obstacle,vehNoYieldPedest,illegalMannedVeh,illegalMannedNonMotorVeh,umbrellaTentInstall,nonMotorVehOnVehLane,wearingNoHelmet,pedestRedLightRunning,pedest
OnNonMotorVehLane,pedestOnVehLane,conflagration,smoke,VehicleSpeedDrop,lowVisibility,animalsOccupy,targetSpeedDetection,emergencyLane,gasser,changeLaneConti
nuously">
  <!--ro, opt, string, supported AID event, attr:opt{opt, string}-->test
</AIDEventSupport>
<TFSEventSupport opt="illegalParking
,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,notSlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,chan
geLaneContinuously,SSharpDriving,largeVehicleOccupyLine,jamCrossLine,obstacle,blackSmokeVehicle,turnRightStop,occupyDedicatedLane,notDriveInDedicatedLane,TFS
_nonZipperMerge,overSpeed,emergencyLane,gasser">
  <!--ro, opt, string, supported TFS, attr:opt{opt, string}-->test
</TFSEventSupport>
<isVehicleStatisticsSupport>
  <!--ro, opt, bool, whether the device supports configuring the parameters for traffic data statistics-->true
</isVehicleStatisticsSupport>
<isSupportElevatorFloorName>
  <!--ro, opt, bool, whether the device supports configuring floor custom name-->true
</isSupportElevatorFloorName>
<isSupportVehicleFaceRecognition>
  <!--ro, opt, bool, whether the device supports checkpoint vehicle recognition, desc:background: the traffic devices support "facial recognition"
(driver), however, the corresponding node is not added to the device to notify the integration of this capability-->true
</isSupportVehicleFaceRecognition>
<VQDEventSupport opt="luma,chroma,snow,streak,freeze,siglose,clarity,jitter,block,flowers,noise,ghost,purple,ICR,protectiveFilm,blur,contrast">
  <!--ro, opt, string, VQD event List, attr:opt{opt, string}, desc:"Luma" (brightness exception); "Chroma" (image color cast); "Snow" (snow noise);
"Streak" (strip noise); "Freeze" (frozen screen); "Siglose" (signal loss); "Clarity" (resolution exception); "Jitter" (image flutter); "Block"
(video tampering); "Flowers" (blurred screen); "Noise" (image noise); "Ghost" (ghost image); "Purple" (purple fringing); "ICR" (ICP); "protectiveFilm"
(protective film unremoved)-->test
</VQDEventSupport>
<IOTCap>
  <!--ro, opt, object, supports searching for the device status List by page, desc:not returned (not support, get the device status List via GET)-->
  <supportChannelNum opt="5000">
    <!--ro, opt, int, the number of supported IOT device channels, attr:opt{opt, string}-->1
  </supportChannelNum>
  <startChannelNo>
    <!--ro, opt, int, the channel No. of start IOT device-->1
  </startChannelNo>
  <isSupportLinkageChannelsSearch>
    <!--ro, opt, bool, true (support), desc:related URI: (POST /ISAPI/System/IOT/LinkageChannels?format=json)-->true
  </isSupportLinkageChannelsSearch>
  <isSupportIOTChannelCap>
    <!--ro, opt, bool, whether the device supports the channel configuration capability, desc:it returns true for support, related URI: (GET
/ISAPI/System/IOT/channels/<IoTChannelID>/capabilities?format=json)-->true
  </isSupportIOTChannelCap>

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<!--ro, opt, object, task search capability-->
<satusSearchTaskCap>
  <!--ro, opt, object, task search capability-->
  <maxResults>
    <!--ro, opt, int, the maximum number of returned results-->1
  </maxResults>
  <filterType opt="all,online,offline,updateFailed,notPartitioned">
    <!--ro, opt, enum, filtering type, subType:string, attr:opt{req, string}, desc:all, online, offline, updateFailed (upgrading failed)-->all
  </filterType>
  <deviceStatus opt="calling,noCall,medicationChangePrompt,alarming,offline,free,callWaiting">
    <!--ro, opt, string, the value range for the device status search condition, range:[0,32], attr:opt{req, string}, desc:"calling" (calling), "noCall"
(non-calling status), "medicationChangePrompt", "alarming", "offline", "free"; if multiple conditions are specified, their relationship is "and" by default-->calling
  </deviceStatus>
  <logicRelationship opt="and,or">
    <!--ro, opt, string, the value range for the logic relationship of device status search condition, range:[0,16], attr:opt{req, string}, desc:and-
the relationship between them is "and" (by default), or- the relationship between them is "or"; it is valid when deviceStatus exists-->and
  </logicRelationship>
  <alarmType opt="SOS,toilet">
    <!--ro, opt, string, value range of alarm type, range:[0,16], attr:opt{req, string}, desc:SOS-SOS call, toilet-toilet alarm. It is valid when the
deviceStatus contains Alarming-->test
  </alarmType>
  <callWaitingType opt="normal,cisit">
    <!--ro, opt, string, alarm type, attr:opt{req, string}, desc:normal-normal call waiting, visit-visit call waiting. It is valid when the deviceStatus
contains callWaiting-->test
  </callWaitingType>
</satusSearchTaskCap>
<channelConfigList>
  <!--ro, opt, array, capability list returned according to unitType, subType:object-->
  <channelConfig>
    <!--ro, opt, object, device type capability-->
    <unitType>
      <!--ro, opt, enum, unit type, subType:string, desc:"indoor" (indoor station), "villa" (villa door station), "confirm" (doorphone), "outdoor" (door
station), "fence" (outer door station), "doorbell" (doorbell), "manage" (main station), "acs" (access control devices), "wardStation", "bedheadExtension",
"bedsideExtension", "terminal", "netAudio" (network sound box), "analogIndoor" (analog indoor station), "decoder" (decoder), "networkFireAlarmMatrix",
"amplifier" (network amplifier), "pagingMicrophone" (SIP phone)-->outdoor
    </unitType>
    <floorNumber min="-10" max="999">
      <!--ro, opt, int, floor No., attr:min{req, int},max{req, int}-->0
    </floorNumber>
    <roomNumber min="0" max="999">
      <!--ro, opt, int, room No., attr:min{req, int},max{req, int}-->0
    </roomNumber>
    <isSupportSlaveDevice>
      <!--ro, opt, bool, whether the device supports sub-devices-->true
    </isSupportSlaveDevice>
    <isSupportZeroFloorNumber>
      <!--ro, opt, bool, whether the zero floor is supported, desc:it is used together with node floorNumber-->true
    </isSupportZeroFloorNumber>
    <periodNumber min="1" max="999">
      <!--ro, opt, int, community number, attr:min{req, int},max{req, int}-->0
    </periodNumber>
    <buildingNumber min="1" max="999">
      <!--ro, opt, int, building number, attr:min{req, int},max{req, int}-->0
    </buildingNumber>
    <unitNumber min="0" max="999">
      <!--ro, opt, int, unit number, attr:min{req, int},max{req, int}-->0
    </unitNumber>
    <deviceIndex min="1" max="32">
      <!--ro, opt, string, device No. Length range, attr:min{req, int},max{req, int}-->test
    </deviceIndex>
    <industryType opt="buildings,prison,medicalTreatment,broadcasting,general">
      <!--ro, opt, string, industry (scene) type, range:[0,32], attr:opt{req, string}, desc:"buildings" (building), "prison", "medicalTreatment"
(medical), "broadcasting", "general"-->test
    </industryType>
    <alarmEnabled opt="true,false">
      <!--ro, opt, bool, whether to enable the arming function of IoT devices, attr:opt{req, string}, desc:this field is used to check if arming is
supported when adding devices. (Related field: alarmEnabled; related URI: /ISAPI/System/IOT/channels?format=json)-->true
    </alarmEnabled>
    <priority min="1" max="15">
      <!--ro, opt, int, calling priority, attr:min{req, int},max{req, int}-->0
    </priority>
    <transportType opt="unicast,multicast">
      <!--ro, opt, string, broadcasting method, attr:opt{req, string}, desc:unicast, multicast-->test
    </transportType>
    <audioVolume min="0" max="15">
      <!--ro, opt, int, volume, attr:min{req, int},max{req, int}-->0
    </audioVolume>
    <callForwardingEnabled>
      <!--ro, opt, bool, whether to enable call transfer, desc:true (enable), false (disable). The default value is false-->true
    </callForwardingEnabled>
    <callForwardingType opt="unconditional,busy,noReply,noReachable,calling">
      <!--ro, opt, string, call transfer type, range:[0,32], attr:opt{req, string}, desc:"unconditional" (call transfer under any condition), "busy"
(call transfer when busy), "noReply" (call transfer when there is no reply), "noReachable" (call transfer when it is unreachable), "calling" (call transfer
when calling)-->test
    </callForwardingType>
    <isSupportCallTerminalNumber>
      <!--ro, opt, bool, whether the device supports configuring the target master station No. for call forwarding-->true
    </isSupportCallTerminalNumber>
    <customCommunicationName min="0" max="128">
      <!--ro, opt, string, custom contact name, range:[0,128], attr:min{req, int},max{req, int}, desc:when this node is configured, the custom contact
name will be displayed during device calling prior to the terminal number-->test
    </customCommunicationName>
    <isSupportMoreFeatures>

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    <!--ro, opt, bool, whether the device supports more functions, desc:it is not supported by default if this node is not returned-->true
  </isSupportMoreFeatures>
  <isNotSupportRegPassword>
    <!--ro, opt, bool, whether the password registration configuration is not supported, desc:it is supported if this node is not returned or false is
returned-->true
    </isNotSupportRegPassword>
    <mappingRoomNumber min="0" max="5">
      <!--ro, opt, int, mapping room No., attr:min{req, int},max{req, int}, desc:the mapping room No. is unique and can be called without exposing the
actual room No., for privacy requirement for overseas users-->0
    </mappingRoomNumber>
    <deviceID min="0" max="64">
      <!--ro, opt, string, device ID, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
    </deviceID>
    <ISUPKey min="1" max="32">
      <!--ro, opt, string, ISUP Key, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
    </ISUPKey>
    <dialUpAddress min="1" max="1024">
      <!--ro, opt, string, dial address, attr:min{req, int},max{req, int}, desc:this node is required when <unitType> is "analogIndoor"-->test
    </dialUpAddress>
  </channelConfig>
</channelConfigList>
<isSupportConfigTemplate>
  <!--ro, opt, bool, whether the device supports exporting the device list template, desc:related URL: GET /ISAPI/System/IOT/channelConfig/template?
format=json-->true
  </isSupportConfigTemplate>
  <isSupportConfigReport>
    <!--ro, opt, bool, whether the device supports getting the importing details, desc:related URL: GET /ISAPI/System/IOT/channelConfig/report?
format=json-->true
    </isSupportConfigReport>
    <unitType
opt="indoor,villa,confirm,outdoor,fence,doorbell,manage,acs,interactive,wardStation,bedheadExtension,bedsideExtension,analogIndoor,decoder,networkFireAlarmM
atrix,amplifier,pagingMicrophone,PDA,netSwitch,entrywayStation,alarmManage,alarmTerminal">
      <!--ro, opt, string, device type, range:[0,32], attr:opt{req, string}, desc:terminal, netAudio (network sound box)-->test
    </unitType>
    <serialNumber min="1" max="32">
      <!--ro, opt, string, the value range of device serial No. Length, range:[0,32], attr:min{req, int},max{req, int}-->test
    </serialNumber>
    <isSupportAsyncStatus>
      <!--ro, opt, bool, whether the device supports asynchronously getting the status of non-video channels, desc:URI for device update: PUT
/ISAPI/System/IOT/channels/updateStatusTask?format=json; URI for getting the update progress: GET /ISAPI/System/IOT/channels/updateStatusProgress?
format=json-->true
      </isSupportAsyncStatus>
      <isSupportAddIOTChannelList>
        <!--ro, opt, bool, whether the device supports batch adding channels of IoT devices, desc:URI for batch adding channels of IoT devices: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
        </isSupportAddIOTChannelList>
        <isSupportSearchMoreIOTSource>
          <!--ro, opt, bool, whether the device supports searching for the IoT devices which can be added (by multiple conditions), desc:related URI: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
          </isSupportSearchMoreIOTSource>
          <channelConfigReportMode>
            <!--ro, opt, enum, mode of importing the list of IoT devices, subType:string, desc:different devices (main station and door station) process the
imported device list in a different way, which should be returned to the users for prompt. URL for importing the device list:
/ISAPI/System/IOT/channelConfig?format=json; cover; add-->cover
            </channelConfigReportMode>
            <isSupportEncryptImportIOTChannelFile>
              <!--ro, opt, bool, whether it supports encrypting the imported non-video channel files, desc:corresponding URI: PUT /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
              </isSupportEncryptImportIOTChannelFile>
              <isSupportEncryptExportIOTChannelFile>
                <!--ro, opt, bool, whether it supports encrypting exported non-video channel files, desc:corresponding URI: GET /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
                </isSupportEncryptExportIOTChannelFile>
                <isSupportExportAbnormalReport>
                  <!--ro, opt, bool, whether it supports exporting error reports for non-video channel devices, desc:corresponding URI: GET
/ISAPI/System/IOT/ChannelConfig/ExportAbnormalReport?format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned--
>true
                  </isSupportExportAbnormalReport>
                  <isSupportSyncDeviceInfoTask>
                    <!--ro, opt, bool-->true
                  </isSupportSyncDeviceInfoTask>
                </IOTCap>
                <isSupportLCDScreen>
                  <!--ro, opt, bool, whether the device supports LCD display screen-->true
                </isSupportLCDScreen>
                <isSupportBluetooth>
                  <!--ro, opt, bool, whether the device supports bluetooth, desc:related URI: /ISAPI/System/Bluetooth/capabilities-->true
                </isSupportBluetooth>
                <isSupportNetScreen>
                  <!--ro, opt, bool, whether the device supports network screen (Nova screen), desc:related URI: /ISAPI/System/netScreens/capabilities?format=json-->true
                </isSupportNetScreen>
                <isSupportLocation>
                  <!--ro, opt, bool, whether the device supports getting the device location information, desc:related URI: /ISAPI/System/Location?format=json-->true
                </isSupportLocation>
                <isSupportDebugInfoExport>
                  <!--ro, opt, bool, whether the device supports exporting the debugging information, desc:related URI: /ISAPI/System/debugInfoExport/capabilities?
format=json-->true
                  </isSupportDebugInfoExport>
                  <isSupportIoTGateway>
                    <!--ro, opt, bool, whether the device supports IoT gateway functions, desc:related URI: /ISAPI/System/IoTGateway/capabilities?format=json-->true
                  </isSupportIoTGateway>
                  <isSupportImageCap>
                    <!--ro, opt, bool, whether the device supports configuring image parameters-->true
                  </isSupportImageCap>

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</isSupportImageCap>
<isSupportLogToUSB opt="true,false">
  <!--ro, opt, bool, whether the device supports exporting logs to USB, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/exportLogToUSB/capabilities?format=json-->true
</isSupportLogToUSB>
<isSupportWiresharkToUSB opt="true,false">
  <!--ro, opt, bool, whether the device supports exporting packet capture information to USB, attr:opt{opt, string}, desc:related URI:
ISAPI/System/wiresharkToUSB/capabilities?format=json-->true
</isSupportWiresharkToUSB>
<CBurningStreamEncryptionChannels opt="1,2">
  <!--ro, opt, string, return the capability of device recording stream encryption channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/Streaming/channels/<ID>/capabilities-->true
</CBurningStreamEncryptionChannels>
<isSupportOCR>
  <!--ro, opt, bool, whether the device supports OCR detection, desc:related URI: /ISAPI/Intelligent/channels/<ID>/OCR/capabilities?format=json-->true
</isSupportOCR>
<isSupportInstallationAngleCalibration>
  <!--ro, opt, bool, whether the device supports the capability of installation angle calibration, desc:related URI:
/ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities-->true
</isSupportInstallationAngleCalibration>
<isSupportZeroBiasCalibration>
  <!--ro, opt, bool, whether the device supports the zero angle calibration capability node, desc:related URI:
/ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json-->true
</isSupportZeroBiasCalibration>
<isSupportCalibrationFile>
  <!--ro, opt, bool, whether the device supports importing the calibration file, desc:related URI: /ISAPI/System/configurationData-->true
</isSupportCalibrationFile>
<isSupportDisplayTrajectory>
  <!--ro, opt, bool, whether the device supports target track overlay (playing library)-->true
</isSupportDisplayTrajectory>
<maximumSuperPositionTime opt="5,10,20,30">
  <!--ro, opt, string, the maximum overlay time, attr:opt{opt, string}, desc:unit: second; it is only returned when the target track overlay is supported-
-->true
</maximumSuperPositionTime>
<isSupportUnitConfig>
  <!--ro, opt, bool, whether the device supports the unit configuration, desc:related URI: /ISAPI/System/unitConfig/capabilities?format=json-->true
</isSupportUnitConfig>
<isSupportShipsDetectionWithScene>
  <!--ro, opt, bool, whether the device supports ship detection by scene-->true
</isSupportShipsDetectionWithScene>
<isSupportPictureServer>
  <!--ro, opt, bool, whether the device supports configuring the picture storage server, desc:related URI: /ISAPI/System/PictureServer/capabilities?
format=json-->true
</isSupportPictureServer>
<isSupportEmmcInfo>
  <!--ro, opt, bool, the capability of supporting the EMMC configuration, desc:related URI: /ISAPI/System/emmcInfo/capabilities?format=json-->true
</isSupportEmmcInfo>
<isSupportVehicleIllegalType opt="1,2...29">
  <!--ro, opt, string, the vehicle violation types supported by the device, attr:opt{opt, string}, desc:the content value is the same as the index value,
refer to NET ITS_PLATE_RESULT.NET_DVR_VEHICLE_INFO.byIllegalType the capture camera is connected to the new platform, and the device supports "traffic
violation alarm"; however, the corresponding node is not added to the device to notify the integration of this capability; that is, the traffic violation
alarm capability lacks details. If later the platform needs to develop capability with details, the current device will be incompatible with the capability;
and detailed capability is hidden in the configuration of all trigger modes, which will influence overall access-->test
</isSupportVehicleIllegalType>
<isSupportRFIDChannels>
  <!--ro, opt, bool, whether the device supports getting the RFID channel, desc:related URI: /ISAPI/System/RFIDChannels?format=json-->true
</isSupportRFIDChannels>
<isSupportRfidCollection>
  <!--ro, opt, bool, whether the device supports the event of uploading RFID tag information, desc:related URI:
/ISAPI/System/Network/RFIDData/<channelID>/capabilities, eventType: rfidCollection-->true
</isSupportRfidCollection>
<DevInfoSearch opt="tiltAngle">
  <!--ro, opt, string, whether the device supports searching for the device information, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/workingstatus/devInfoSearch?format=json-->test
</DevInfoSearch>
<isSupportAIOpenPlatform>
  <!--ro, opt, bool, whether the device supports AI open platform device capability, desc:related URI: /ISAPI/Intelligent/AIOpenPlatform/capabilities?
format=json-->true
</isSupportAIOpenPlatform>
<isSupportPictureDownloadError>
  <!--ro, opt, bool, whether the device supports uploading the exception that downloading the picture failed, desc:eventType: pictureDownloadError-->true
</isSupportPictureDownloadError>
<isSupportStudentBehavior>
  <!--ro, opt, bool, whether the device supports uploading the data of students in-class behavior, desc:eventType: studentBehavior-->true
</isSupportStudentBehavior>
<guideEventSupport opt="linkageCapture">
  <!--ro, opt, string, the list of web quick configuration guide, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/guideConfig/guideEvent/capabilities?format=json-->test
</guideEventSupport>
<supportRemoteCtrl opt="up,down,left,right,enter,menu,num,power,esc,edit,F1,.prev,rec,play,stop,notSupport">
  <!--ro, opt, string, whether the device supports remote control, attr:opt{opt, string}, desc:related URI: /ISAPI/System/remoteCtrl/XXX-->test
</supportRemoteCtrl>
<isSupportSSDFileSystemUpgrade>
  <!--ro, opt, bool, whether the device supports updating the SSD file system, desc:related URI: /ISAPI/System/SSDFileSystem/upgrade?format=json-->true
</isSupportSSDFileSystemUpgrade>
<isSupportSSDFileSystemFormat>
  <!--ro, opt, bool, whether the device supports formatting SSD, desc:related URI: /ISAPI/System/SSDFileSystem/format?format=json-->true
</isSupportSSDFileSystemFormat>
<FaceCounting>
  <!--ro, opt, object, whether the device supports the customer traffic (face)-->
<ChannelsList>
  <!--ro, opt, array, channel list, subType:object-->
<channelsID>
  <!--ro, req, int, No. of channels supported by the device-->1

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    </channelsID>
  </ChannelsList>
</FaceCounting>
<isSupportCluster>
  <!--ro, opt, bool, related URI:, desc:/ISAPI/ContentMgmt/Cluster/capabilities-->true
</isSupportCluster>
<DeviceConfigurationSupport opt="time,network,accessProtocol,recordSchedule,channel,customProtocol">
  <!--ro, opt, string, content supported by device parameters configuration, attr:opt{opt, string}, desc:related URI: /ISAPI/System/deviceConfiguration-->
>test
  </DeviceConfigurationSupport>
  <behaviorEventSupport opt="getUp,toiletTarry,playCellphone">
    <!--ro, opt, string, abnormal event detection types, attr:opt{opt, string}-->test
  </behaviorEventSupport>
  <isSupportBVCorrectPassthrough>
    <!--ro, opt, bool, whether the device supports transparent transmission of camera correction data, desc:related URI:
/ISAPI/System/BVCorrect/Passthrough/channels/<ID>/capabilities?format=json-->true
  </isSupportBVCorrectPassthrough>
  <isSupportChangedCallStatus>
    <!--ro, opt, bool-->true
  </isSupportChangedCallStatus>
  <VehiclePositionControl>
    <!--ro, opt, object, whether the device supports vehicle position arming-->
    <license min="0" max="16">
      <!--ro, req, string, license plate number, attr:min{opt, string},max{opt, string}-->test
    </license>
    <intervalTime min="0" max="65535" def="1">
      <!--ro, req, int, uploading interval, units:s, attr:min{opt, string},max{opt, string},def{opt, string}-->1
    </intervalTime>
  </VehiclePositionControl>
  <supportGISEvent opt="AID,TPS,ANPR,mixedTargetDetection">
    <!--ro, opt, string, the capability of supporting getting the GIS information, attr:opt{opt, string}-->test
  </supportGISEvent>
  <isSupportGPSPTZCalibration>
    <!--ro, opt, bool, whether the device supports GPS PTZ calibration, desc:related URI: /ISAPI/System/GPSPTZCalibration/channels/<ID>/points?
format=json-->true
  </isSupportGPSPTZCalibration>
  <isSupportRadar>
    <!--ro, opt, bool, whether the device supports radar functions, desc:related URI: /ISAPI/Radar/capabilities?format=json-->true
  </isSupportRadar>
  <isSupportISUPHttpPassthrough>
    <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
  </isSupportISUPHttpPassthrough>
  <isSupportMaterialListQuery>
    <!--ro, opt, bool, whether the device supports uploading the material list, desc:event: materialListQuery-->true
  </isSupportMaterialListQuery>
  <isSupportDebugLogAsync>
    <!--ro, opt, bool, whether the device supports exporting the debugging log asynchronously-->true
  </isSupportDebugLogAsync>
  <isSupportAlgorithmsInfo>
    <!--ro, opt, bool, whether the device supports getting the algorithm version of the device-->true
  </isSupportAlgorithmsInfo>
  <isSupportSecurityChecking>
    <!--ro, opt, bool, whether the device supports configuring security checking, desc:true (support); otherwise, not support-->true
  </isSupportSecurityChecking>
  <isSupportSecurityCheckException>
    <!--ro, opt, bool, whether the device supports the security checking exception event, desc:eventType: securityCheckException-->true
  </isSupportSecurityCheckException>
  <isSupportPictureManage>
    <!--ro, opt, bool, whether the device supports picture management function, desc:related URI: /ISAPI/System/isSupportPictureManage/capabilities?
format=json-->true
  </isSupportPictureManage>
  <isSupportAutoSwitch>
    <!--ro, opt, bool, whether the device supports automatically powering on and powering off, desc:related URI: /ISAPI/System/autoSwitch/capabilities?
format=json-->true
  </isSupportAutoSwitch>
  <SHMCap>
    <!--ro, opt, object, SHM alarm-->
    <isSupportHighHDDTemperature>
      <!--ro, opt, bool, whether the device supports HDD high temperature detection, desc:eventType: highHDDTemperature-->true
    </isSupportHighHDDTemperature>
    <isSupportLowHDDTemperature>
      <!--ro, opt, bool, whether the device supports HDD low temperature detection, desc:eventType: LowHDDTemperature-->true
    </isSupportLowHDDTemperature>
    <isSupportHDImpact>
      <!--ro, opt, bool, whether the device supports HDD impact detection, desc:eventType: hdImpact-->true
    </isSupportHDImpact>
    <isSupportHDBadBlock>
      <!--ro, opt, bool, whether the device supports HDD bad sector detection, desc:eventType: hdBadBlock-->true
    </isSupportHDBadBlock>
    <isSupportSevereHDFailure>
      <!--ro, opt, bool, whether the device supports HDD major fault detection, desc:eventType: severeHDFailure-->true
    </isSupportSevereHDFailure>
  </SHMCap>
  <isSupportLinkStatusEvent>
    <!--ro, opt, bool, whether the device supports uploading the information about the network connection status of the device, desc:eventType: LinkEvent-->
>true
  </isSupportLinkStatusEvent>
  <isSupportVideoDealStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the video data processing status, desc:eventType: videoDealStatus-->true
  </isSupportVideoDealStatus>
  <isSupportAudioDealStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the audio data processing status, desc:eventType: audioDealStatus-->true
  </isSupportAudioDealStatus>

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<isSupportNetPackLossStatus>
  <!--ro, opt, bool, whether the device supports uploading the information about the network lost package status, desc:eventType: netPacketLoss-->true
</isSupportNetPackLossStatus>
<isSupportVideoConferenceStatus>
  <!--ro, opt, bool, whether the device supports uploading the information about the video conference status, desc:eventType: videoConference-->true
</isSupportVideoConferenceStatus>
<isSupportDataPreAlarm>
  <!--ro, opt, bool, whether the device supports network traffic pre-alarm event, desc:eventType: dataPreAlarm-->true
</isSupportDataPreAlarm>
<isSupportClearCache>
  <!--ro, opt, bool, whether the device supports clearing the cache, desc:related URI: /ISAPI/System/clearCache?format=json-->true
</isSupportClearCache>
<isSupportIntelligentMode>
  <!--ro, opt, bool, whether the device supports switching the intelligent mode, desc:related URI: /ISAPI/System/IntelligentMode?format=json-->true
</isSupportIntelligentMode>
<isSupportFaceTemperatureMeasurementEvent>
  <!--ro, opt, bool, whether the device supports the skin-surface temperature measurement event, desc:eventType: FaceTemperatureMeasurementEvent-->true
</isSupportFaceTemperatureMeasurementEvent>
<isSupportQRCodeEvent>
  <!--ro, opt, bool, whether the device supports the QR code event, desc:eventType: QRCodeEvent-->true
</isSupportQRCodeEvent>
<isSupportVisitorEvent>
  <!--ro, opt, bool, whether the device supports the visitor event, desc:eventType: VisitorEvent-->true
</isSupportVisitorEvent>
<isSupportEmployeeAuthEvent>
  <!--ro, opt, bool, whether the device supports the employee authorization event, desc:eventType: EmployeeAuthEvent-->true
</isSupportEmployeeAuthEvent>
<isSupportVisitorInfoSearchEvent>
  <!--ro, opt, bool, whether the device supports the visitor information search event, desc:eventType: VisitorInfoSearchEvent-->true
</isSupportVisitorInfoSearchEvent>
<isSupportConsumptionEvent>
  <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: ConsumptionEvent-->true
</isSupportConsumptionEvent>
<isSupportTransactionRecordEvent>
  <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: TransactionRecordEvent-->true
</isSupportTransactionRecordEvent>
<isSupportFaceContrastResultDisplay>
  <!--ro, opt, bool, whether the device supports displaying the face picture comparison result, desc:related URI:
/ISAPI/Intelligent/faceContrast/resultDisplay/capabilities?format=json-->true
</isSupportFaceContrastResultDisplay>
<isSupportPatchUpgrade>
  <!--ro, opt, bool, whether it supports incremental upgrade, desc:related URI: /ISAPI/Custom/OpenPlatform/patchUpgrade/capabilities?format=json-->true
</isSupportPatchUpgrade>
<isSupportSnapshotAsync>
  <!--ro, opt, bool, whether the device supports capturing pictures asynchronously (the picture URL will be returned), desc:related URI:
/ISAPI/Streaming/channels/<ID>/picture/async?format=json-->true
</isSupportSnapshotAsync>
<isSupportLogExportByUSB>
  <!--ro, opt, bool, exporting logs to PC from USB flash, desc:related URI: /ISAPI/System/exportUSBLog?format=json-->true
</isSupportLogExportByUSB>
<isSupportShellCmd>
  <!--ro, opt, bool, whether the device supports shell command control, desc:related URI: /ISAPI/System/shellCmd?format=json-->true
</isSupportShellCmd>
<faceMonitor opt="dailyFaceMonitor,dailyFaceTracking,VCAFaceMonitor,VCAFaceTracking>manualFaceMonitor,advanceConfiguration,">
  <!--ro, opt, bool, whether the device supports face arming, attr:opt(opt, string), desc:related URI: /ISAPI/Intelligent/channels/<ID>/dailyFaceMonitor?
format=json-->true
</faceMonitor>
<isSupportRelatedVCAResource>
  <!--ro, opt, bool, whether the device supports the system-level intelligent resources configuration protocol, desc:related URI:
/ISAPI/System/RelatedVCAResource/capabilities?format=json; this protocol conflicts with the previous intelligent resources switch protocol-->true
</isSupportRelatedVCAResource>
<isSupportVehicleMonitor>
  <!--ro, opt, bool, whether the device supports intelligent arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/capabilities?format=json-->true
</isSupportVehicleMonitor>
<isSupportManualVehicleMonitor>
  <!--ro, opt, bool, whether the device supports manual arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/manual/capabilities?format=json-->true
</isSupportManualVehicleMonitor>
<isSupportDailyVehicleMonitor>
  <!--ro, opt, bool, whether the device supports daily arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleDetect/capabilities-->true
</isSupportDailyVehicleMonitor>
<isSupportInputProxyDeviceInfo>
  <!--ro, opt, bool, whether the device supports getting the information of the devices added to the proxy device channel, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities-->true
</isSupportInputProxyDeviceInfo>
<isSupportAccessControlCap>
  <!--ro, opt, bool, access control capability, desc:related URI: /ISAPI/AccessControl/capabilities-->true
</isSupportAccessControlCap>
<isSupportDebugLog>
  <!--ro, opt, bool, whether the device supports exporting the debugging log, desc:related URI: /ISAPI/System/debugLog?format=json-->true
</isSupportDebugLog>
<isSupportFaceAppear>
  <!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: faceAppear-->true
</isSupportFaceAppear>
<isSupportFaceDisappear>
  <!--ro, opt, bool, whether the device supports face disappearing detection event, desc:eventType: faceDisappear-->true
</isSupportFaceDisappear>
<isSupportTeacherBehavior>
  <!--ro, opt, bool, whether the device supports the abnormal event detection of teachers, desc:eventType: teacherBehavior-->true
</isSupportTeacherBehavior>
<isSupportHeadRiseRate>
  <!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: headRiseRate-->true

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</isSupportHeadRiseRate>
<isSupportIPCUpgrade>
  <!--ro, opt, bool, whether the platform supports upgrading network cameras via NVR and other devices, desc:related URI: /ISAPI/System/IPCUpgrade/<ID>?
format=json-->true
</isSupportIPCUpgrade>
<isSupportFaceLibRebuildEvent>
  <!--ro, opt, bool, whether the device supports uploading the modeling rebuilding of the face picture library event, desc:eventType: faceLibRebuild--
>true
</isSupportFaceLibRebuildEvent>
<isSupportCalibrationStream>
  <!--ro, opt, bool, supported capability of calibration stream, desc:related URI: /ISAPI/Streaming/calibStream/capabilities?format=json-->true
</isSupportCalibrationStream>
<isSupportChannelOccupy>
  <!--ro, opt, bool, whether the device supports outdoor fire engine access occupied by vehicles, desc:eventType: channelOccupy-->true
</isSupportChannelOccupy>
<isSupportOffDuty>
  <!--ro, opt, bool, whether to support fire control room person absence detection, desc:eventType: offDuty-->true
</isSupportOffDuty>
<isSupportNoCertificate>
  <!--ro, opt, bool, whether to support fire control room person without certificate detection, desc:eventType: noCertificate-->true
</isSupportNoCertificate>
<isSupportSmokeAlarm>
  <!--ro, opt, bool, whether the device supports smoke alarm, desc:eventType: smokeAlarm-->true
</isSupportSmokeAlarm>
<isSupportBatteryCarDisobey>
  <!--ro, opt, bool, whether the device supports illegal parking detection of electric moped, desc:eventType: batteryCarDisobey-->true
</isSupportBatteryCarDisobey>
<isSupportNoFireExtinguisherRecog>
  <!--ro, opt, bool, whether the device supports fire extinguisher loss detection, desc:eventType: noFireExtinguisherRecog-->true
</isSupportNoFireExtinguisherRecog>
<isSupportIndoorPasswayBlock>
  <!--ro, opt, bool, whether the device supports indoor access blocking detection, desc:eventType: indoorPasswayBlock-->true
</isSupportIndoorPasswayBlock>
<isSupportFireSmartFireDetect>
  <!--ro, opt, bool, whether the device supports fire detection, desc:eventType: fireSmartFireDetect-->true
</isSupportFireSmartFireDetect>
<isSupportDetectorRunningStatus>
  <!--ro, opt, bool, whether the device supports getting the detector running status, desc:eventType: detectorRunningStatus-->true
</isSupportDetectorRunningStatus>
<isSupportDetectorOperationStatus>
  <!--ro, opt, bool, whether the device supports getting the detector operation status, desc:eventType: detectorOperationStatus-->true
</isSupportDetectorOperationStatus>
<isSupportDetectorTemperatureAlarm opt="highTemperature, riseTemperature, Flame">
  <!--ro, opt, bool, whether the device supports the detector temperature alarm, attr:opt{req, string}-->true
</isSupportDetectorTemperatureAlarm>
<isSupportDetectorShelterAlarm>
  <!--ro, opt, bool, whether the device supports detector sheltering alarm, desc:eventType: detectorShelterAlarm-->true
</isSupportDetectorShelterAlarm>
<isSupportDetectorMotionAlarm>
  <!--ro, opt, bool, whether the device supports the detector moving alarm, desc:eventType: detectorMotionAlarm-->true
</isSupportDetectorMotionAlarm>
<isSupportDetectorTamperAlarm>
  <!--ro, opt, bool, whether the device supports detector tampering alarm, desc:eventType: detectorTamperAlarm-->true
</isSupportDetectorTamperAlarm>
<isSupportDetectorEmergencyAlarm>
  <!--ro, opt, bool, whether the device supports the detector emergency alarm, desc:eventType: detectorEmergencyAlarm-->true
</isSupportDetectorEmergencyAlarm>
<isSupportDetectorSmokeAlarm>
  <!--ro, opt, bool, whether the device supports the detector smoke alarm, desc:eventType: detectorSmokeAlarm-->true
</isSupportDetectorSmokeAlarm>
<isSupportDetectorCombustibleGasAlarm>
  <!--ro, opt, bool, whether the device supports the detector combustible gas alarm, desc:eventType: detectorCombustibleGasAlarm-->true
</isSupportDetectorCombustibleGasAlarm>
<isSupportFireControlData>
  <!--ro, opt, bool, whether the device supports uploading real-time fire control data, desc:eventType: FireControlData-->true
</isSupportFireControlData>
<isSupportFireNoRegulation>
  <!--ro, opt, bool, whether to support people leaving event uploading, desc:eventType: fireNoRegulation-->true
</isSupportFireNoRegulation>
<isSupportSmokeFireRecognize>
  <!--ro, opt, bool, whether the device supports smoke and fire recognition event uploading, desc:eventType: smokeFireRecognize-->true
</isSupportSmokeFireRecognize>
<isSupportClientProxyWEB>
  <!--ro, opt, bool, whether to support WEB page jumping, desc:whether to support WEB page jumping-->true
</isSupportClientProxyWEB>
<WEBLocation>
  <!--ro, opt, string, WEB page placement Location, range:[1,32], desc:Local-Local server, remote-remote server; (if this node is not returned, the WEB
page will be placed in the local server by default)-->local
</WEBLocation>
<isSupportWEBLocalPlug>
  <!--ro, opt, bool, whether to support local download of web plug-in, desc:true=yes (by default), false=no; when this node is not returned, it is
supported by default-->true
</isSupportWEBLocalPlug>
<deviceId>
  <!--ro, opt, string, device ID, range:[1,32], desc:if WEBLocation is "remote", the unique device ID should be returned to identify the unique
corresponding web package. This device ID is managed by the Web configuration package. For a specified device of different versions, the same deviceId will
be returned-->R0101webs_4.0_isapi
</deviceId>
<isSupportTimeZone>
  <!--ro, opt, bool, whether the device supports configuring the DST (DayLight Saving Time), desc:related URI: /ISAPI/System/time/timeZone-->true
</isSupportTimeZone>
<isSptHealthDetection>
  <!--ro, opt, bool, whether the device supports health monitoring, desc:related URI: /ISAPI/System/healthDetection-->true

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</isSptHealthDetection>
<isSptDiagnosis>
  <!--ro, opt, bool, whether the device supports diagnosis, desc:related URI: /ISAPI/System/diagnosis-->true
</isSptDiagnosis>
<isSptSerialLogCfg>
  <!--ro, opt, bool, whether the device supports configuring parameters of serial port log redirection, desc:related URI: /ISAPI/System/serialLogCfg?
format=json-->true
</isSptSerialLogCfg>
<isSptFileExport>
  <!--ro, opt, bool, whether the device supports exporting the device file, desc:related URI: /ISAPI/System/fileExport?format=json-->true
</isSptFileExport>
<isSptCertificationStandard>
  <!--ro, opt, bool, whether the device supports configuring host verification standard, desc:related URI: /ISAPI/SecurityCP/certificationStandard-->true
</isSptCertificationStandard>
<isSptKeypadLock>
  <!--ro, opt, bool, whether the device supports locking the keypad, desc:related URI: /ISAPI/System/keypadLock/config-->true
</isSptKeypadLock>
<isSupportVibrationDetection>
  <!--ro, opt, bool, whether the device supports the vibration detection, desc:related URI: /ISAPI/System/channels/<ID>/vibrationDetection?format=json--
>true
</isSupportVibrationDetection>
<isSupportSmokingDetectAlarm>
  <!--ro, opt, bool, whether the device supports the smoking alarm, desc:eventType: smokeDetectAlarm-->true
</isSupportSmokingDetectAlarm>
<isSupportRadarChannels>
  <!--ro, opt, bool, whether the device supports getting the radar channel, desc:related URI: /ISAPI/System/RadarChannels?format=json-->true
</isSupportRadarChannels>
<radarIPDForm>
  <!--ro, opt, enum, radar-camera status, subType:string, desc:"single" (single radar); "double_diagonal" (double radars arranged diagonally);
"double_vertical" (double radars arranged vertically)-->single
</radarIPDForm>
<isSupportRadarFieldDetection>
  <!--ro, opt, bool, whether the device supports radar's area intrusion detection, desc:related URI:
/ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection/capabilities?format=json-->true
</isSupportRadarFieldDetection>
<isSupportRadarLineDetection>
  <!--ro, opt, bool, whether the device supports radar line crossing detection, desc:related URI:
/ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection/capabilities?format=json-->true
</isSupportRadarLineDetection>
<mixedTargetDetectionWebNoDisplay>
  <!--ro, opt, bool, multiple targets recognition is not displayed-->true
</mixedTargetDetectionWebNoDisplay>
<isSupportRemovePlaybackRecord>
  <!--ro, opt, bool, whether to support removing playback (the device does not support playback), desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will not be returned. This node will be returned when the value of isSupportRemoveStorage is true;
otherwise, this node will not be returned-->true
</isSupportRemovePlaybackRecord>
<isSupportAudioUpload>
  <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URI: /ISAPI/System/Audio/audioUpload?format=json-->true
</isSupportAudioUpload>
<VCAResourceList>
  <!--ro, opt, array, supported smart resources list, subType:object-->
<VCAResource>
  <!--ro, opt, object, smart resources-->
<VCAResourceType>
  <!--ro, opt, string, smart resources type-->test
</VCAResourceType>
<ChannelVCAResourceList>
  <!--ro, opt, array, supported smart resources list, subType:object-->
<ChannelVCAResource>
  <!--ro, opt, object, smart channel resources-->
<channel>
  <!--ro, opt, int, channel No.-->1
</channel>
<channelVCAResourceType>
  <!--ro, opt, string, smart channel resources type-->test
</channelVCAResourceType>
</ChannelVCAResource>
</ChannelVCAResourceList>
</VCAResource>
</VCAResourceList>
<isSupportSlotInsertion>
  <!--ro, opt, bool, whether to support slot plugging in and out, desc:eventType: slotInsertion-->true
</isSupportSlotInsertion>
<isSupportNetworkLoop>
  <!--ro, opt, bool, whether to support network loop alarm, desc:eventType: networkLoop-->true
</isSupportNetworkLoop>
<isSupportApOnOffline>
  <!--ro, opt, bool, whether to support AP online and offline alarm event uploading, desc:eventType: apOnOffline-->true
</isSupportApOnOffline>
<isSupportAbnormalWANPort>
  <!--ro, opt, bool, whether to support WAN abnormal alarm event uploading, desc:eventType: abnormalWANPort-->true
</isSupportAbnormalWANPort>
<isSupportAPException>
  <!--ro, opt, bool, whether to support AP exception alarm information uploading, desc:eventType: APException-->true
</isSupportAPException>
<isSupportACException>
  <!--ro, opt, bool, whether to support AC exception alarm information uploading, desc:eventType: ACException-->true
</isSupportACException>
<isSupportCartridgeCapacity>
  <!--ro, opt, bool, whether the device supports the configuration of notifying cartridge capacity, desc:related URL:
/ISAPI/System/printer/cartridgeCapacity/capabilities?format=json-->true
</isSupportCartridgeCapacity>
<isSupportTapeBoxCapacity>

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<!--ro, opt, bool, whether the device supports notifying CD capacity, desc:related URL: /ISAPI/System/printer/CDCakeBoxCapacity/capabilities?
format=json-->true
</isSupportCDCakeBoxCapacity>
<isSupportSleep>
<!--ro, opt, bool, whether the device supports configuring sleep, desc:related URL: /ISAPI/System/sleep/capabilities?format=json-->true
</isSupportSleep>
<isSupportContainerDetection>
<!--ro, opt, bool, whether the device supports container detection, desc:/ISAPI/Intelligent/containerDetection/capabilities?format=json (for devices
that does not return this node, you can check according to the node returned by the following protocol:
/ISAPI/Traffic/ContentMgmt/InputProxy/channels/<ID>/ocrScene/capabilities)-->true
</isSupportContainerDetection>
<isSupportRadarVideoDetection>
<!--ro, opt, bool, related URI: /ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
</isSupportRadarVideoDetection>
<isSupportRadarVideoDetectionRealtime>
<!--ro, opt, bool, whether the device supports radar video detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
</isSupportRadarVideoDetectionRealtime>
<isSupportBatteryStatusUpload>
<!--ro, opt, bool, whether the device supports uploading the battery status, desc:eventType: batteryStatus-->true
</isSupportBatteryStatusUpload>
<isSupportPowerConsumptionStatusUpload>
<!--ro, opt, bool, whether the device supports uploading device consumption status-->true
</isSupportPowerConsumptionStatusUpload>
<isSupportOnlineUpgradeTask>
<!--ro, opt, bool, whether device online upgrading is supported-->true
</isSupportOnlineUpgradeTask>
<isSupportPersonArmingTrack>
<!--ro, opt, bool, whether the device supports person arming configuration, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/personArmingTrack/capabilities?format=json-->true
</isSupportPersonArmingTrack>
<isSupportManualPersonArmingTrack>
<!--ro, opt, bool, whether the device supports manual person arming, desc:related URI: /ISAPI/Intelligent/channels/<ID>/manualPersonArmingTrack?
format=json-->true
</isSupportManualPersonArmingTrack>
<isSupportGPSVerification>
<!--ro, opt, bool, whether the device supports GPS verification points, desc:related URI: /ISAPI/System/GPSVerification/channels/<ID>/points?
format=json-->true
</isSupportGPSVerification>
<isSupportHBDLib>
<!--ro, opt, bool, whether the device supports the human body library, desc:related URI: /ISAPI/Intelligent/HBDLib/capabilities?format=json-->true
</isSupportHBDLib>
<isSupportDisinfectionInfo>
<!--ro, opt, bool, whether the device supports uploading disinfection information, desc:eventType: disinfectionInfo-->true
</isSupportDisinfectionInfo>
<isSupportTempQueryDelete>
<!--ro, opt, bool, whether the device supports deleting temperature query, desc:eventType: tempQueryDelete-->true
</isSupportTempQueryDelete>
<isSupportMobilePayment>
<!--ro, opt, bool, whether the device supports mobile payment, desc:/ISAPI/System/mobilePayment?format=json-->true
</isSupportMobilePayment>
<isSupportFileSearch>
<!--ro, opt, bool, whether the device supports searching files, desc:related URI: /ISAPI/System/fileSearch?format=json-->true
</isSupportFileSearch>
<isSupportFileExportAsync>
<!--ro, opt, bool, whether the device supports exporting files asynchronously, desc:related URI: /ISAPI/System/fileExport/async?format=json-->true
</isSupportFileExportAsync>
<isSupportSSDFileSystemCapacity>
<!--ro, opt, bool, whether the device supports getting the capacity distribution of the SSD system, desc:related URI:
/ISAPI/System/SSDFileSystem/capacity?format=json-->true
</isSupportSSDFileSystemCapacity>
<isSupport4GSafetyHelmetInfo>
<!--ro, opt, bool, whether the device supports the hard hat event, desc:eventType: 4GSafetyHelmetInfo-->true
</isSupport4GSafetyHelmetInfo>
<isSupportICCID>
<!--ro, opt, bool, whether the device supports getting the ICCID information, desc:related URI: /ISAPI/System/deviceInfo/ICCID?format=json-->true
</isSupportICCID>
<isSupportFireEscapeDetection>
<!--ro, opt, bool, whether the device supports fire escape detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/fireEscapeDetection/capabilities?format=json-->true
</isSupportFireEscapeDetection>
<isSupportTakingElevatorDetection>
<!--ro, opt, bool, whether the device supports taking elevator detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/takingElevatorDetection/capabilities?format=json-->true
</isSupportTakingElevatorDetection>
<isSupportVehicleMatchResult>
<!--ro, opt, bool, whether the device supports license plate comparison, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/LicensePlateContrast/capabilities?format=json-->true
</isSupportVehicleMatchResult>
<isSupportDataAware>
<!--ro, opt, bool, the data aware capability, desc:related URI: /ISAPI/System/dataAware/capabilities?format=json-->true
</isSupportDataAware>
<isSupportAbnormalReboot>
<!--ro, opt, bool, whether the device supports abnormal reboot event, desc:eventType: abnormalReboot-->true
</isSupportAbnormalReboot>
<isSupportVehicleThermometry>
<!--ro, opt, bool, whether the device supports vehicle temperature measurement detection, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleDetect/thermometry/capabilities?format=json-->true
</isSupportVehicleThermometry>
<AssociationScenesList>
<!--ro, opt, array, the List of scene adaptive linkage, subType:object, desc:it indicates the device supports the scene adaptive linkage between
channels. The current device has two channels: main channel (box camera) and sub channel (radar camera), however, the device only opens a main channel to
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users for scene configuration; the scene configuration of the sub channel is not open for the users; the scene of the sub channel is automatically connected to the main channel scene; currently, the front-end returns this capability-->

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<AssociationScenes>
  <!--ro, req, object, the channel List of scene adaptive Linkage between channels-->
  <channelID>
    <!--ro, req, int, supported scene channel ID, desc:supported scene channel ID-->1
  </channelID>
  <isSupportScene>
    <!--ro, opt, bool, whether the device supports the scene configuration, desc:whether the current channel is the main channel and supports scene
    configuration , true (main channel and support), this node is not returned (sub channel and not support)-->true
  </isSupportScene>
</AssociationScenes>
</AssociationScenesList>
<supportRelationVideoEvent opt="PIR,cityManagement,roadMaint">
  <!--ro, opt, string, supports uploading the video of Linked triggering event, attr:opt(req, string), desc:"PIR" (PIR event), "cityManagement" (city
  management event), "roadMaint" (road maintenance event)-->test
</supportRelationVideoEvent>
<isSupportVCAResourceList>
  <!--ro, opt, bool, whether the device supports configuring intelligent resources in a batch, desc:related URI:
  /ISAPI/System/Video/inputs/channels/VCAResourceList?format=json-->true
</isSupportVCAResourceList>
<isSupportConvergenceCloud>
  <!--ro, opt, bool, whether the device supports the URI for EZ interconnection channels, desc:URI for capability of all channels:
  /ISAPI/Event/convergenceCloud/channels/capabilities?format=json; URI for capability of specified channel:
  /ISAPI/Event/convergenceCloud/channels/<ID>/capabilities?format=json-->true
</isSupportConvergenceCloud>
<isSupportInterconnection>
  <!--ro, opt, bool, whether the device supports the URI for EZ interconnection devices, desc:related URI: /ISAPI/Event/convergenceCloud/capabilities?
  format=json-->true
</isSupportInterconnection>
<isSupportHPVehicleStatus>
  <!--ro, opt, bool, whether the device supports the platform vehicle status event, desc:eventType: HPVehicleStatus-->true
</isSupportHPVehicleStatus>
<isSupportHPWorkingStatus>
  <!--ro, opt, bool, whether the device supports the scheduled platform working status event, desc:eventType: HPWorkingStatus-->true
</isSupportHPWorkingStatus>
<isSupportHPPeopleDetection>
  <!--ro, opt, bool, whether the device supports the platform people detection event, desc:eventType: HPPeopleDetection-->true
</isSupportHPPeopleDetection>
<isSupportHPVehicleDoorDetection>
  <!--ro, opt, bool, whether the device supports the platform door detection event, desc:eventType: HPVehicleDoorDetection-->true
</isSupportHPVehicleDoorDetection>
<isSupportRoadMaint>
  <!--ro, opt, bool, whether the device supports road maintenance, desc:related URI: /ISAPI/Intelligent/channels/<ID>/roadMaint?format=json-->true
</isSupportRoadMaint>
<isSupportDockStationInfo>
  <!--ro, opt, bool, whether the device supports the event of uploading the dock station information to the media, desc:eventType: dockStationInfo; this
  node is not returned if not supported-->true
</isSupportDockStationInfo>
<isSupportBodyCameraManualForensics>
  <!--ro, opt, bool, whether the device supports the event of uploading the manual evidence information of the body camera, desc:eventType:
  bodyCameraManualForensics; this node is not returned if not supported-->true
</isSupportBodyCameraManualForensics>
<isSupportBodyCameraLogin>
  <!--ro, opt, bool, whether the device supports the event of police officer logging in to the body camera, desc:eventType: bodyCameraLogin; this node is
  not returned if not supported-->true
</isSupportBodyCameraLogin>
<isSupportBodyCameraRunStatus>
  <!--ro, opt, bool, whether the device supports the status of body camera, desc:eventType: bodyCameraRunStatus; this node is not returned if not
  supported-->true
</isSupportBodyCameraRunStatus>
<isSupportBodyCameraSignIn>
  <!--ro, opt, bool, whether the device supports the event of police officer signing in the body camera, desc:eventType: bodyCameraSignIn; this node is
  not returned if not supported-->true
</isSupportBodyCameraSignIn>
<isSupportAlertDistribution>
  <!--ro, opt, bool, whether the device supports the alert distribution, desc:eventType: alertDistribution; this node is not returned if not supported--
  >true
</isSupportAlertDistribution>
<isSupportImageCapture>
  <!--ro, opt, bool, whether the device supports the event of scheduled capture, desc:related URI: /ISAPI/System/Video/inputs/channels/<ID>/imageCapture?
  format=json-->true
</isSupportImageCapture>
<isSupportScheduledRecordUpload>
  <!--ro, opt, bool, whether the device supports the event of channel scheduled recording, desc:related URI:
  /ISAPI/System/Video/inputs/channels/<ID>/scheduledRecordUpload?format=json-->true
</isSupportScheduledRecordUpload>
<isSupportTargetPatternCollect>
  <!--ro, opt, bool, whether the device supports the target track collection event, desc:eventType: targetPatternCollect-->true
</isSupportTargetPatternCollect>
<isSupportAudioAnalysis>
  <!--ro, opt, bool, whether the device supports the audio analysis event, desc:eventType: audioAnalysis; this node is not returned if not supported--
  >true
</isSupportAudioAnalysis>
<isSupportDebuggingDataExport>
  <!--ro, opt, bool, whether the device supports importing the parameters (in-out parameters, rule configuration parameters), desc:related URI:
  /ISAPI/System/debuggingData/parameters?format=json; this node is not returned if not supported-->true
</isSupportDebuggingDataExport>
<isSupportInternalParaAdjust>
  <!--ro, opt, bool, whether the device supports trinocular device; whether the device supports editing in-parameters, desc:related URI:
  /ISAPI/System/internalParameters/adjustment?format=json; this node is not returned if not supported-->true
</isSupportInternalParaAdjust>
<isSupportManualImageCapture>
  <!--ro, opt, bool, whether the device supports manual capture of the video channel, desc:related URI:
  /ISAPI/System/Video/inputs/channels/<ID>/manualImageCapture?format=json; this node is not returned if not supported-->true
</isSupportManualImageCapture>
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<!--ro, opt, bool, whether the device supports manual capture of the video channel, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/manualImageCapture?format=json; this node is not returned if not supported-->true
</isSupportManualImageCapture>
<characteristicCode min="1" max="128">
  <!--ro, opt, string, device attribute code, range:[1,128], attr:min{opt, string},max{opt, string}, desc:related URI:
/ISAPI/System/deviceInfo/characteristicCode?format=json-->test
</characteristicCode>
<isSupportSIMCardStatus>
  <!--ro, opt, bool, whether the device supports getting the information of SIM card status, desc:related URI: /ISAPI/System/SIMCardStatus?format=json--
>true
</isSupportSIMCardStatus>
<OpenPlatformCap>
  <!--ro, opt, object, detailed HEOP capability supported by the device, desc:related URI: /ISAPI/Custom/OpenPlatform/capabilities-->
  <isSupportOpenPlatform>
    <!--ro, req, bool, whether the device supports HEOP (Embedded Open Platform)-->true
  </isSupportOpenPlatform>
  <maxAPPNum>
    <!--ro, opt, int, the maximum number of third-party application packages allowed to be installed-->0
  </maxAPPNum>
</OpenPlatformCap>
<isSupportLogException>
  <!--ro, opt, bool, whether the device supports the log exception alarm, desc:eventType: LogException-->true
</isSupportLogException>
<isSupportMEF>
  <!--ro, opt, bool, whether the device supports Mobile electronic fence - mobile electronic fence data detection, desc:related URI:
/ISAPI/System/Network/MEF/<ID>?format=json-->true
</isSupportMEF>
<manualEvent
opt="fieldDetection,linedetection,ANPR,faceCapture,mixedTargetDetection,studentsStoodUp,headRiseRate,teacherBehavior,ElevatorRunningDetectionEvent">
  <!--ro, opt, string, whether the device supports URI for manual triggering event, attr:opt{req, string}, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/manualEvent/<eventType>?format=json-->test
</manualEvent>
<isSupportAppCfg>
  <!--ro, opt, bool, whether the device supports application management, desc:related URI: /ISAPI/System/app/capabilities?format=json-->true
</isSupportAppCfg>
<isSupportCleanup>
  <!--ro, opt, bool, whether the device supports one-click clearing, desc:related URI: /ISAPI/System/cleanup?format=json-->true
</isSupportCleanup>
<isSupportEndUserCheckMsg opt="sleep,reboot,factoryReset,cleanup,parentPickUpQuery,schoolAttendanceQuery,screenShare,screenMirror">
  <!--ro, opt, bool, whether the device supports uploading the confirmation information of end users, attr:opt{req, string}, desc:eventType:
endUserCheckMsg-->true
</isSupportEndUserCheckMsg>
<isSupportScreenConfig>
  <!--ro, opt, bool, whether the device supports configuring screen parameters, desc:related URI: /ISAPI/System/screenConfig/capabilities?format=json--
>true
</isSupportScreenConfig>
<isSupportTemperatureMonitor>
  <!--ro, opt, bool, whether the device supports temperature monitoring configuration, desc:related URI: /ISAPI/System/temperatureMonitor/capabilities?
format=json-->true
</isSupportTemperatureMonitor>
<isSupportObjectServer>
  <!--ro, opt, bool, whether the device supports the object storage service, desc:related URI: /ISAPI/System/objectServer/capabilities?format=json-->true
</isSupportObjectServer>
<isSupportCdStatus>
  <!--ro, opt, bool, whether the device supports notifying disk status, desc:related URI for enabling the function: the bEnableCdStatus node of
/ISAPI/Trial/roompara/capabilities; corresponding event: CdStatus-->true
</isSupportCdStatus>
<isSupportCoalGangueDetection>
  <!--ro, opt, bool, whether the device supports coal gangue detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/coalGangueDetection;
this node is not returned if not supported-->true
</isSupportCoalGangueDetection>
<isSupportVacantSeatRate>
  <!--ro, opt, bool, whether the device supports seat vacancy detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/vacantSeatRate?
format=json; corresponding event: VacantSeatRate; this node is not returned if not supported-->true
</isSupportVacantSeatRate>
<isSupportObjectsThrownDetection>
  <!--ro, opt, bool, whether the device supports objects thrown detection, desc:related URI:
/ISAPI/Intelligent/channels/<channelID>/objectsThrownDetection?format=json; corresponding event: objectsThrownDetection; this node is not returned if not
supported-->true
</isSupportObjectsThrownDetection>
<isSupportBVCorrectDataSearch>
  <!--ro, opt, bool, whether the device supports searching for the parameters of camera correction, desc:related URI: /ISAPI/System/BVCorrectData/search?
format=json; this node is not returned if not supported-->true
</isSupportBVCorrectDataSearch>
<isSupportSlagTruckWashingDetection>
  <!--ro, opt, bool, whether the device supports dump truck cleaning detection event, desc:eventType: slagTruckWashingDetection-->true
</isSupportSlagTruckWashingDetection>
<isSupportWOL>
  <!--ro, opt, bool, whether the device supports configuring the LAN wakeup parameters, desc:it can be realized via the WOL (Wake On-Lan) function.
Related URL: /ISAPI/System/WOL/capabilities?format=json-->true
</isSupportWOL>
<isSupportPersonQueueTimingStatistics>
  <!--ro, opt, bool, whether the device supports counting the number of persons in each queue, desc:eventType: isSupportPersonQueueTimingStatistics-->true
</isSupportPersonQueueTimingStatistics>
<isSupportLensParamFile>
  <!--ro, opt, bool, whether the device supports importing and exporting camera parameters files-->true
</isSupportLensParamFile>
<isSupportVehiclePassingInParkingLot>
  <!--ro, opt, bool, whether the device supports the detection of vehicle passing in parking lot entrance and exit, desc:eventType:
vehiclePassingInParkingLot-->true
</isSupportVehiclePassingInParkingLot>
<isSupportPowerOutput>
  <!--ro, opt, bool, whether power supply output configuration is supported, desc:related URL: /ISAPI/System/powerOutput/capabilities?format=json-->true
</isSupportPowerOutput>

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<isSupportIPCUpgradeByFTP>
  <!--ro, opt, bool, it indicates that the device supports upgrading the proxy device channel, such as IPC upgrade via NVR, desc:related URI:
  /ISAPI/System/IPCUpgrade/<channelID>/FTP?format=json&security=<security>&iv=<iv>-->true
</isSupportIPCUpgradeByFTP>
<isSupportMethaneConcentrationException>
  <!--ro, opt, bool, whether the device supports methane concentration exception detection, desc:eventType: methaneConcentrationException-->true
</isSupportMethaneConcentrationException>
<isSupportMethaneLightIntensityException>
  <!--ro, opt, bool, whether the device supports methane light intensity exception detection, desc:eventType: methaneLightIntensityException-->true
</isSupportMethaneLightIntensityException>
<isSupportLocalButtonPermission>
  <!--ro, opt, bool, whether the device supports controlling local button permission, desc:related URI: /ISAPI/System/LocalButtonPermission/capabilities?
  format=json-->true
</isSupportLocalButtonPermission>
<isSupportLocalButtonRecord>
  <!--ro, opt, bool, whether the device supports button triggering record, desc:eventType: LocalButtonRecord-->true
</isSupportLocalButtonRecord>
<isSupportLocalButtonSnapshot>
  <!--ro, opt, bool, whether the device supports button triggering snapshot, desc:eventType: LocalButtonSnapshot-->true
</isSupportLocalButtonSnapshot>
<isSupportVoiceTalkEvent>
  <!--ro, opt, bool, whether the device supports voice talk, desc:eventType: voiceTalkEvent-->true
</isSupportVoiceTalkEvent>
<isSupportPowerSupply>
  <!--ro, opt, bool, whether the device supports external power supply, desc:related URI: /ISAPI/System/powerSupply/capabilities?format=json-->true
</isSupportPowerSupply>
<isSupportBarrierGateStatus>
  <!--ro, opt, bool, whether the device supports barrier gate status, desc:eventType: barrierGateStatus-->true
</isSupportBarrierGateStatus>
<DeployCap>
  <!--ro, opt, object, the capability of arming non-subscription events-->
  <deployID opt="1,2">
    <!--ro, opt, enum, arming type, subType:int, attr:opt{req, string}, desc:0 (client arming); 1 (real-time arming)-->0
  </deployID>
</DeployCap>
<isSupportSignalControllerStatus>
  <!--ro, opt, bool, whether the device supports uploading the signal controller status, desc:eventType: signalControllerStatus-->true
</isSupportSignalControllerStatus>
<isSupportSignalControllerFault>
  <!--ro, opt, bool, whether the device supports uploading the signal controller fault, desc:eventType: signalControllerFault-->true
</isSupportSignalControllerFault>
<isSupportNTPConfig>
  <!--ro, opt, bool, whether the device supports configuring NTP parameters, desc:/ISAPI/System/time/ntp/capabilities?format=json-->true
</isSupportNTPConfig>
<isSupportConsumptionQuery>
  <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption, desc:eventType: ConsumptionQuery-->true
</isSupportConsumptionQuery>
<StreamingCodingRegionPositionCap>
  <!--ro, opt, object, whether the device supports getting all stream splice area positions information capability,
  desc:/ISAPI/System/streamingCoding/channels/<channelID>/regionPosition-->
  <ChannelsList>
    <!--ro, opt, array, channel list, subType:object-->
    <channelsID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelsID>
  </ChannelsList>
</StreamingCodingRegionPositionCap>
<isSupportLowPowerPlan>
  <!--ro, opt, bool, whether the device supports configuring low consumption mode schedule, desc:related URI:
  /ISAPI/System/consumptionMode/LowPowerPlan/capabilities?format=json-->true
</isSupportLowPowerPlan>
<isSupportPlateQuaAlarm>
  <!--ro, opt, bool, whether the device supports plate quality detection, desc:eventType: pLateQuaAlarm-->true
</isSupportPlateQuaAlarm>
<isSupportWiegand>
  <!--ro, opt, bool, whether the device supports the Wiegand protocol, desc:related URI: /ISAPI/System/Wiegand/<wiegandID>/capabilities?format=json-->true
</isSupportWiegand>
<zoomFocusWebDisplay
opt="ROI,roadTrafficDetection,SMD,mixedTargetDetection,faceCapture,personCounting,multidimensionalPeopleCounting,AIOP,heelPDC,packingSpaceRecognition,tunnel
Detection,cityManagement">
  <!--ro, opt, string, whether the web supports displaying the zoomfocus page, attr:opt{req, string},
  desc:ROI page path: Configuration-Vedio/Audio-ROI
  roadTrafficDetection:Configuration-Road Traffic-Detection Configuration; SMD; mixedTargetDetection; faceCapture-->ROI
</zoomFocusWebDisplay>
<isSupportInternalParamsAdjustment>
  <!--ro, opt, bool, whether the device supports in-parameters correction by region, desc:true (support), false or this node is not returned (not
  support). Related URI: /ISAPI/System/internalParameters/regionAdjustment/capabilities?format=json-->true
</isSupportInternalParamsAdjustment>
<isSupportDiagnosticReport>
  <!--ro, opt, bool, whether the device supports diagnostic report function, desc:true (support), false or this node is not returned (not support).
  Related URI: /ISAPI/System/diagnosticReport/capabilities?format=json-->true
</isSupportDiagnosticReport>
<isSupportWaterQualitySearch>
  <!--ro, opt, bool, whether the device supports the function of searching for water quality, desc:true (support), false or this node is not returned (not
  support). Related URI: /ISAPI/Event/waterQualityDetection/search?format=json-->true
</isSupportWaterQualitySearch>
<isSupportExhaustDetection>
  <!--ro, opt, bool, whether the device supports configuring gas detection parameters, desc:true (support), false or this node is not returned (not
  support). Related URI: /ISAPI/Event/exhaustDetection/channels/<channelID>/capabilities?format=json-->true
</isSupportExhaustDetection>
<isSupportTPMSecretKey>
  <!--ro, opt, bool, whether the device supports enabling stream encryption for all TPM keys Lists, desc:true (support), false or this node is not
  returned (not support). Related URI: /ISAPI/Streaming/encryption/TPMSecretKeyList?format=json&security=<security>&iv=<iv>-->true

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</isSupportTPMSecretKey>
<isSupportWeatherStation>
  <!--ro, opt, bool, whether the device supports uploading weather station information, desc:eventType: weatherStation; this node is not returned if not supported-->true
</isSupportWeatherStation>
<isSupportRoadVisibilityAnalysis>
  <!--ro, opt, bool, whether the device supports road visibility detection analysis, desc:eventType: weatherStation; this node is not returned if not supported-->true
</isSupportRoadVisibilityAnalysis>
<isSupportRoadWeatherAnalysis>
  <!--ro, opt, bool, whether the device supports road weather detection analysis, desc:eventType: roadWeatherAnalysis; this node is not returned if not supported-->true
</isSupportRoadWeatherAnalysis>
<isSupportRoadVisibilityAbnormal>
  <!--ro, opt, bool, whether the device supports uploading road visibility detection exception, desc:eventType: roadVisibilityAbnormal; this node is not returned if not supported-->true
</isSupportRoadVisibilityAbnormal>
<isSupportRoadWeatherAbnormal>
  <!--ro, opt, bool, whether the device supports uploading road weather detection exception, desc:eventType: roadWeatherAbnormal; this node is not returned if not supported-->true
</isSupportRoadWeatherAbnormal>
<isSupportChannelsTimeSync>
  <!--ro, opt, bool, whether the device supports batch device time synchronization of the IPC/IOT channel, desc:related URL: /ISAPI/System/time/channelsTimeSync/capabilities?format=json-->true
</isSupportChannelsTimeSync>
<isSupport433Sensor>
  <!--ro, opt, bool, whether the device supports configuring 433 sensor transmission, desc:related URLs: /ISAPI/System/Sensor/433sensor/433wireLessTransfer/capabilities?format=json /ISAPI/System/Sensor/433sensor/capabilities?format=json /ISAPI/System/Sensor/433sensor/sensorDetection/capabilities?format=json /ISAPI/System/Sensor/433sensor/sensorPicOverlay/capabilities?format=json /ISAPI/System/Sensor/433sensor/rfParamTemplate/capabilities?format=json-->true
</isSupport433Sensor>
<isSupportSignalSource>
  <!--ro, opt, bool, whether the device supports configuring signal source parameters, desc:related URL: /ISAPI/System/signalSource/capabilities?format=json-->true
</isSupportSignalSource>
<isSupportUSBConfig>
  <!--ro, opt, bool, whether the device supports configuring USB parameters, desc:related URL: /ISAPI/System/USBConfig/capabilities?format=json-->true
</isSupportUSBConfig>
<isSupportStandbyStrategy>
  <!--ro, opt, bool, whether the device supports configuring standby strategy parameters, desc:related URL: /ISAPI/System/standbyStrategy/capabilities?format=json-->true
</isSupportStandbyStrategy>
<isSupportWakeupStrategy>
  <!--ro, opt, bool, whether the device supports configuring wakeup strategy parameters, desc:related URL: /ISAPI/System/wakeupStrategy/capabilities?format=json-->true
</isSupportWakeupStrategy>
<isSupportSeachMutexFunction>
  <!--ro, opt, bool, whether the device supports searching for disabling the mutual exclusion function, desc:related URL: GET /ISAPI/System/mutexFunction/capabilities?format=json; function description: whether the device supports searching for mutual exclusion function-->true
</isSupportSeachMutexFunction>
<isSupportMultiDeviceID>
  <!--ro, opt, bool, whether the device supports multiple Long numbers configuration, desc:related URL: /ISAPI/VideoIntercom/deviceId/batchCfg/capabilities?format=json-->true
</isSupportMultiDeviceID>
<isNotSupportMultiDeviceIDCustomName>
  <!--ro, opt, bool, whether the device supports configuring custom name for multiple Long numbers, desc:if this function is not supported, the user cannot configure custom name for the device via the Long number configuration API; true (not support), false or this node is not returned (support)-->true
</isNotSupportMultiDeviceIDCustomName>
<isSupportMixedSecurityChecking>
  <!--ro, opt, bool, whether the device supports uploading the event of mixed security checking, desc:eventType: mixedSecurityChecking-->true
</isSupportMixedSecurityChecking>
<isSupportVisiblePackageEvent>
  <!--ro, opt, bool, whether the device supports visible package event, desc:eventType: visiblePackageEvent-->true
</isSupportVisiblePackageEvent>
<isSupportBarCodeEvent>
  <!--ro, opt, bool, whether the device supports the barcode event-->true
</isSupportBarCodeEvent>
<isSupportGaugeReading>
  <!--ro, opt, bool, whether the device supports the gauge reading function, desc:related URL: GET /ISAPI/Event/channels/<channelID>/gaugeReading/capabilities?format=json; this node is not returned if not supported-->true
</isSupportGaugeReading>
<isSupportUncertificateCompareEvent>
  <!--ro, opt, bool, whether the device supports the UncertificateCompareEvent event, desc:eventType: UncertificateCompareEvent; this node is not returned if not supported-->true
</isSupportUncertificateCompareEvent>
<isSupportAudioDiacritical>
  <!--ro, opt, bool, whether the device supports audio voice changing function, desc:related URL: /ISAPI/System/Video/inputs/channels/<channelID>/audioDiacritical/capabilities?format=json-->true
</isSupportAudioDiacritical>
<isSupportMEFReportSearch>
  <!--ro, opt, bool, whether the device supports searching for electronic fence data statistics report, desc:URL for getting the capability: /ISAPI/System/Network/MEF/reportSearch/capabilities?format=json; this node is not returned if not supported-->true
</isSupportMEFReportSearch>
<isSupportBroadcastTriggerEvent>
  <!--ro, opt, bool, whether the device supports broadcast triggering event, desc:after the event is uploaded to the platform via the paging microphone, the platform will broadcast and check according to the uploaded terminal No. List; eventType: broadcastTriggerEvent. URL for checking: /ISAPI/VideoIntercom/broadcastTriggerConfirm?format=json-->true
</isSupportBroadcastTriggerEvent>
<isSupportFishingShipDetection>
  <!--ro, opt, bool, whether the device supports fishing ship detection event, desc:eventType: fishingShipDetection; related URL: /ISAPI/Thermal/channels/<channelID>/fishingShipDetection/<SID>?format=json-->true
</isSupportFishingShipDetection>

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<isSupportHPVehicleLoadingRate>
  <!--ro, opt, bool, whether the device supports the platform vehicle loading rate event, desc:eventType: vehicleLoadingRate, this node is not returned if
not supported. Related URL: /ISAPI/Intelligent/channels/<channelID>/HPDetection/capabilities?format=json-->true
</isSupportHPVehicleLoadingRate>
<isSupportVideoBarrierGateDetection>
  <!--ro, opt, bool, whether the device supports port and gate video detection, desc:eventType: videoBarrierGateDetection, this node is not returned if
not supported. Related URL: /ISAPI/Traffic/ContentMgmt/videoBarrierGate/capabilities?format=json-->true
</isSupportVideoBarrierGateDetection>
<isSupportLogicalResource>
  <!--ro, opt, bool, whether the device supports camera organization and management, desc:related URL: /ISAPI/System/LogicalResource/capabilities?
format=json, this node is not returned if not supported-->true
</isSupportLogicalResource>
<isSupportSlagTruckCoverPlate>
  <!--ro, opt, bool, whether the device supports detecting whether back cover the slag truck is covered, desc:eventType: slagTruckCoverPlateDetection,
this node is not returned if not supported. Related URL: /ISAPI/Event/channels/<channelID>/slagTruckCoverPlate/capabilities?format=json-->true
</isSupportSlagTruckCoverPlate>
<isSupportTrainDetection>
  <!--ro, opt, bool, whether the device supports train detection, desc:If train detection is not supported, this node will not be returned. Corresponding
eventType: trainDetection-->true
</isSupportTrainDetection>
<isNotSupportVideoOverlays>
  <!--ro, opt, bool, whether the device does not support configuring OSD overlay on the device web configuration page, desc:this node is only valid for
checking whether the OSD overlay configuration of the image menu on the person and vehicle verification station is supported or not (this configuration is
supported by default in the protocol and it does not have node control capability). If true is returned, it indicates that the OSD overlay is not supported;
if false or no node is returned, it indicates the OSD overlay function is supported-->true
</isNotSupportVideoOverlays>
<isSupportScheduledCaptureTask>
  <!--ro, opt, bool, whether the device supports scheduled capturing tasks, desc:related URI: /ISAPI/Sytem/Video/scheduledCapture/capabilities-->true
</isSupportScheduledCaptureTask>
<isSupportGuideCfg>
  <!--ro, opt, bool, whether the device supports the wizard configuration, desc:this node is only valid for the indoor station to check whether to enable
wizard configuration, which is provided for the users to perform basic function configuration after activating and logging in to the device (including
Internet parameters configuration, dialog configuration and batch adding analog indoor stations). If true is returned, it indicates wizard configuration is
supported; if false or no node is returned, it indicates the function is not supported-->true
</isSupportGuideCfg>
<isSupportCameraGroupPeopleCounting>
  <!--ro, opt, bool, whether the device supports passenger flow statistics of multiple cameras, desc:/ISAPI/Intelligent/counting/cameraGroup/capabilities?
format=json-->true
</isSupportCameraGroupPeopleCounting>
<isSupportAngleDeviationDetection>
  <!--ro, opt, bool, whether the device supports tilting detection, desc:related URL:
/ISAPI/Event/channels/<channelID>/angleDeviationDetection/capabilities?format=json; this node is not returned if not supported-->true
</isSupportAngleDeviationDetection>
<isSupportCityCode>
  <!--ro, opt, bool, whether the device supports getting city code, desc:related URI: /ISAPI/System/cityCode?format=json-->true
</isSupportCityCode>
<isSupportAlcoholDetectionResult>
  <!--ro, opt, bool, whether the device supports uploading alcohol test, desc:eventType: alcoholDetectionResult-->true
</isSupportAlcoholDetectionResult>
<isSupportAlcoholDetectionRodBind>
  <!--ro, opt, bool, whether the device supports the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBind-->true
</isSupportAlcoholDetectionRodBind>
<isSupportAlcoholDetectionRodBindQuery>
  <!--ro, opt, bool, whether the device supports the query for the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBindQuery--
-->true
</isSupportAlcoholDetectionRodBindQuery>
<isSupportAlcoholDetectionRodUnbind>
  <!--ro, opt, bool, whether the device supports the event of unbinding alcohol test stick, desc:eventType: alcoholDetectionRodUnbind-->true
</isSupportAlcoholDetectionRodUnbind>
<isSupportAbnormalDriving>
  <!--ro, opt, bool, whether the device supports abnormal driving behaviors detection, desc:eventType: abnormalDriving-->true
</isSupportAbnormalDriving>
<isSupportEmanationsEvent>
  <!--ro, opt, bool, whether the device supports the event of emanation test, desc:eventType: emanationsEvent; the security inspection system is equipped
with an emanation tester, and when the emanation exceeds the threshold, an alarm event will be uploaded. This event is only triggered by the emanation
tester, and its configuration is not required-->true
</isSupportEmanationsEvent>
<isSupportClientUpdatePackage>
  <!--ro, opt, bool, whether the device supports the client upgrade package management, desc:related URL: /ISAPI/System/clientUpdatePackage/capabilities?
format=json-->true
</isSupportClientUpdatePackage>
<isSupportDeviceStatusUploadStrategy>
  <!--ro, opt, bool, whether the device supports configuring the parameters of device status uploading strategy, desc:related URL:
/ISAPI/System/status/uploadStrategy/capabilities?format=json-->true
</isSupportDeviceStatusUploadStrategy>
<isSupportDialIndicatorReading>
  <!--ro, opt, bool, whether the devices supports dial indicator reading, desc:related URL:
/ISAPI/Intelligent/channels/<channelID>/dialIndicatorReading/capabilities?format=json; the function refers to reading single pointer dial and uploading the
result (currently, it can only recognize dials that do not accumulate the readings such as thermometer and barometer)-->true
</isSupportDialIndicatorReading>
<isNotSupportAudioVideoCompress>
  <!--ro, opt, bool, whether the device does not support audio/video encoding, desc:this node is only valid for the person and vehicle verification
station; if true is returned, the video/audio page will not be displayed on the device web page. If false or no node is returned, the video/audio page will
be displayed on the device web page by default-->true
</isNotSupportAudioVideoCompress>
<isSupportUPLExceptionEvent>
  <!--ro, opt, bool, whether the device supports the event of Liquid chromatograph system exception, desc:eventType: UPLException-->true
</isSupportUPLExceptionEvent>
<isSupportUrgentTurn>
  <!--ro, opt, bool, whether the device supports the sharp turn event, desc:eventType: urgentTurn-->true
</isSupportUrgentTurn>
<isSupportUrgentBrake>
  <!--ro, opt, bool, whether the device supports the sudden brake event, desc:eventType: urgentBrake-->true

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</isSupportUrgentBrake>
<isSupportUrgentAcceleration>
  <!--ro, opt, bool, whether the device supports the rapid acceleration event, desc:eventType: urgentAcceleration-->true
</isSupportUrgentAcceleration>
<isSupportVehiclePanicButton>
  <!--ro, opt, bool, whether the device supports the alarm/event of mobile emergency button, desc:eventType: vehiclePanicButton-->true
</isSupportVehiclePanicButton>
<isSupportOverSpeed>
  <!--ro, opt, bool, whether the device supports the vehicle speeding alarm/event, desc:eventType: overSpeed-->true
</isSupportOverSpeed>
<isSupportAbnormalAcceleration>
  <!--ro, opt, bool, whether the device supports the event of abnormal vehicle custom acceleration alarm, desc:eventType: abnormalAcceleration-->true
</isSupportAbnormalAcceleration>
<isSupportRollOver>
  <!--ro, opt, bool, whether the device supports the vehicle turnover alarm/event, desc:eventType: rollover-->true
</isSupportRollOver>
<isSupportCollision>
  <!--ro, opt, bool, whether the device supports the vehicle collision alarm/event, desc:eventType: collision-->true
</isSupportCollision>
<isSupportVehicleOBUInfo>
  <!--ro, opt, bool, whether the device supports uploading the vehicle OBU information, desc:eventType: vehicleOBUInfo-->true
</isSupportVehicleOBUInfo>
<isSupportPowerUpPicture>
  <!--ro, opt, bool, whether the device supports importing and exporting pictures displayed during startup,
desc:/ISAPI/System/powerUpPicture/capabilities?format=json-->true
</isSupportPowerUpPicture>
<isSupportPowerUpPictureCfg>
  <!--ro, opt, bool, whether the device supports configuring parameters of pictures displayed during startup,
desc:/ISAPI/System/powerUpPictureCfg/capabilities?format=json-->true
</isSupportPowerUpPictureCfg>
<isSupportAlgoPackageScheduling>
  <!--ro, opt, bool, whether the device supports algorithm package scheduling, desc:Related URIs: /ISAPI/System/AlgoPackageScheduling/capabilities?
format=json and /ISAPI/System/AlgoPackageScheduling/task/capabilities?format=json-->true
</isSupportAlgoPackageScheduling>
<isSupportVoiceTalkRequestingAlarm>
  <!--ro, opt, bool, whether the device supports the two-way audio request alarm, desc:eventType: voiceTalkRequestingAlarm-->true
</isSupportVoiceTalkRequestingAlarm>
<isSupportSetupStructure>
  <!--ro, opt, bool, whether the device supports configuring construction structure parameters, desc:related URI: /ISAPI/System/setupStructure?
format=json-->true
</isSupportSetupStructure>
<isSupportPassNumInfoEvent>
  <!--ro, opt, bool, whether the device supports uploading the number of people passed, desc:eventType: passNumInfoEvent-->true
</isSupportPassNumInfoEvent>
<isShowClientRemote>
  <!--ro, opt, bool, whether the device supports displaying remote configuration on the client,
desc:1. If this node is not returned or it is returned and its value is true, it indicates that the device supports this function.
2. If this node is returned and its value is false, it indicates that the device does not support this function.
iVMS-4200 client software supports remote configuration by default. However, there are still some devices that do not support remote configuration via iVMS-
4200 and they can only be configured remotely via the web browser. The capability determined by this node is used to control whether to display the remote
configuration on iVMS-4200 client software-->true
</isShowClientRemote>
<isSupportUserManualQRCode>
  <!--ro, opt, bool, whether the device supports getting the QR code of the user manual, desc:related URI: /ISAPI/System/userManualQRCode/capabilities?
format=json-->true
</isSupportUserManualQRCode>
<isSupportUnclosedFrontPanel>
  <!--ro, opt, bool, whether the device supports uploading the event of the front panel not closed, desc:related URI for getting the capability:
/ISAPI/Trial/roompara/capabilities; corresponding node: <unclosedFrontPanelEnabled>. Related event: unclosedFrontPanel-->true
</isSupportUnclosedFrontPanel>
<isSupportVehicleWeight>
  <!--ro, opt, bool, whether the device supports configuring parameters for vehicle weight and axle detection,
desc:/ISAPI/Traffic/channels/<channelID>/vehicleWeight?format=json-->true
</isSupportVehicleWeight>
<isSupportWeightInterface>
  <!--ro, opt, bool, whether it supports integrated overLoad control parameters, desc:/ISAPI/Traffic/ContentMgmt/WeightInterface?format=json-->true
</isSupportWeightInterface>
<isSupportSecurityPanicEvent>
  <!--ro, opt, bool, whether the device supports emergency events of security inspection, desc:eventType: securityPanicEvent-->true
</isSupportSecurityPanicEvent>
<isSupportTrafficLightMalfunction>
  <!--ro, opt, bool, whether the device supports the event of detecting faults of traffic lights, desc:eventType: trafficLightMalfunction-->true
</isSupportTrafficLightMalfunction>
<isSupportTOFShelterAlarm>
  <!--ro, opt, bool, whether the device supports TOF tampering alarm, desc:/ISAPI/Event/channels/<channelID>/TOFShelterAlarm/capabilities?format=json--
>true
</isSupportTOFShelterAlarm>
<isSupportSetMealQuery>
  <!--ro, opt, bool, whether the device supports uploading the event of searching set meals, desc:eventType: SetMealQuery-->true
</isSupportSetMealQuery>
<isSupportMultiScreenControlAlarm>
  <!--ro, opt, bool, whether the device supports uploading the multi-screen controller alarm, desc:eventType: multiScreenControlAlarm-->true
</isSupportMultiScreenControlAlarm>
<isSupportKeyFrameLoss>
  <!--ro, opt, bool, whether the device supports uploading the event of key frame loss, desc:related URI: /ISAPI/Event/keyFrameLoss/capabilities?
format=json; eventType: keyFrameLoss.-->true
</isSupportKeyFrameLoss>
<isSupportWorkStatusCfg>
  <!--ro, opt, bool, whether the device supports working status configuration, desc:/ISAPI/System/WorkStatusCfg?format=json-->true
</isSupportWorkStatusCfg>
<isSupportHealthInfoSyncQuery>
  <!--ro, opt, bool, whether the device supports the event of searching for health information, desc:eventType: HealthInfoSyncQuery-->true
</isSupportHealthInfoSyncQuery>
<isSupportVisitorPictureSearchEvent>

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<!--ro, opt, bool, whether the device supports the event of searching for visitor profiles, desc:eventType: VisitorPictureSearchEvent-->true
</isSupportVisitorPictureSearchEvent>
<isSupportEZVIZParameterTest>
  <!--ro, opt, bool, whether the device supports testing EZ parameters, desc:related URI: /ISAPI/System/Network/EZVIZ/parameterTest?format=json-->true
</isSupportEZVIZParameterTest>
<isSupportUnattendedPackageEvent>
  <!--ro, opt, bool, whether the device supports the unattended package event, desc:eventType: unattendedPackageEvent-->true
</isSupportUnattendedPackageEvent>
<isSupportPackageFragmentEvent>
  <!--ro, opt, bool, whether the device supports the event of uploading package fragment data, desc:eventType: packageFragmentEvent-->true
</isSupportPackageFragmentEvent>
<isSupportPackageBarcodeMismatchEvent>
  <!--ro, opt, bool, whether the device supports the event of package and barcode mismatching, desc:eventType: packageBarcodeMismatchEvent-->true
</isSupportPackageBarcodeMismatchEvent>
<isSupportCheckingModeSwitchEvent>
  <!--ro, opt, bool, whether the device supports the event of switching the image checking mode, desc:eventType: checkingModeSwitchEvent-->true
</isSupportCheckingModeSwitchEvent>
<FaceSnapPrivacyMask>
  <!--ro, opt, object, whether the device supports privacy mask on faces, desc:based on face picture capture, all faces will be masked to protect the
privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json-->
  <ChannelsList>
    <!--ro, opt, array, channel List, subType:object-->
    <channelsID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelsID>
  </ChannelsList>
</FaceSnapPrivacyMask>
<FaceContrastPrivacyMask>
  <!--ro, opt, object, whether the device supports privacy mask on faces of key persons, desc:based on face picture comparison, the faces of key persons
will be masked to protect the privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceContrast/privacyMask/capabilities?format=json-->
  <ChannelsList>
    <!--ro, opt, array, channel List, subType:object-->
    <channelsID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelsID>
  </ChannelsList>
</FaceContrastPrivacyMask>
<isSupportHumanBodyComparison>
  <!--ro, opt, bool, whether the device supports human body comparison events, desc:related URI:
/ISAPI/Event/channels/<channelID>/humanBodyComparison/capabilities?format=json; eventType: humanBodyComparison-->true
</isSupportHumanBodyComparison>
<isSupportMedicalInfoSearchEvent>
  <!--ro, opt, bool, whether the device supports the event of searching for medical information, desc:medicalInfoSearchEvent-->true
</isSupportMedicalInfoSearchEvent>
<isSupportConsumptionStatusQuery>
  <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption status, desc:eventType: ConsumptionStatusQuery-->true
</isSupportConsumptionStatusQuery>
<isSupportShipChannelAbnormal>
  <!--ro, opt, bool, whether the device supports uploading ship channel exception detection event, desc:eventType: shipChannelAbnormal-->true
</isSupportShipChannelAbnormal>
<isSupportDepositRetrieveEvent>
  <!--ro, opt, bool, whether the device supports the event of object access in the intelligent cabinet-->true
</isSupportDepositRetrieveEvent>
<isSupportDepositTimeoutEvent>
  <!--ro, opt, bool, whether the device supports the event of overtime object deposit in the intelligent cabinet-->true
</isSupportDepositTimeoutEvent>
<isSupportCabinetStatusEvent>
  <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's changed status-->true
</isSupportCabinetStatusEvent>
<isSupportCabinetAuthorityExpired>
  <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's permission expiration-->true
</isSupportCabinetAuthorityExpired>
<isSupportT1Test>
  <!--ro, opt, bool, whether the device supports the configuration of controlling T1 tool-->true
</isSupportT1Test>
<isSupportDrivingDirectionAtIntersection>
  <!--ro, opt, bool, whether the device supports intersection driving direction detection, desc:eventType: drivingDirectionAtIntersection; this node is
not returned if not supported-->true
</isSupportDrivingDirectionAtIntersection>
<isSupportPlaceSceneExecuteStatus>
  <!--ro, opt, bool, whether the device supports the event of scene execution status, desc:eventType: placeSceneExecuteStatus; this node is not returned
if not supported-->true
</isSupportPlaceSceneExecuteStatus>
<isSupportSceneRecordingCtrl>
  <!--ro, opt, bool, whether the device supports the event of controlling scene recording, desc:eventType: sceneRecordingCtrl; this node is not returned
if not supported-->true
</isSupportSceneRecordingCtrl>
<isSupportStreamRecordStatus>
  <!--ro, opt, bool, whether the device supports getting the stream ID record status, desc:/ISAPI/System/Video/inputs/streams/recordStatus-->true
</isSupportStreamRecordStatus>
<isSupportManualSnapPicture>
  <!--ro, opt, bool, whether the device supports manual picture capturing, desc:related URI:
/ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json-->true
</isSupportManualSnapPicture>
<isSupportVerificationResultEvent>
  <!--ro, opt, bool, whether the device supports the event of human and vehicle verification result, desc:eventType: verificationResultEvent-->true
</isSupportVerificationResultEvent>
<OTAP>
  <!--ro, opt, object, the capability of Open Things Access Protocol, desc:OTAP capability-->
  <isSupportOTAPModel>
    <!--ro, opt, bool, whether the device supports getting the virtual No. and device No. of the OTAP, desc:GET /ISAPI/System/Network/OTAP/Model?
format=json-->true
  </isSupportOTAPModel>

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</OTAP>
<isSupportForeignObjectInRoadDetection>
  <!--ro, opt, bool, whether the device supports the detection of foreign object in road, desc:eventType: foreignObjectInRoadDetection-->true
</isSupportForeignObjectInRoadDetection>
<isSupportUPLCMethodOperatingStatus>
  <!--ro, opt, bool, whether the device supports the event of UPLC method operating status, desc:eventType: UPLCMethodOperatingStatus-->true
</isSupportUPLCMethodOperatingStatus>
<isSupportMaterialScanTaskStatus>
  <!--ro, opt, bool, whether the device supports material scanning task status, desc:eventType: materialScanTaskStatus-->true
</isSupportMaterialScanTaskStatus>
<isSupportSearchInputProxy>
  <!--ro, opt, bool, whether the device supports searching for accessible IPC of device in the LAN, desc:related URI: GET
/ISAPI/ContentMgmt/InputProxy/search?security=<security>&iv=<iv>; true (support), this node is not returned (call the specific API to confirm whether it
supports or not)-->true
</isSupportSearchInputProxy>
<isSupportPostureDetection>
  <!--ro, opt, bool, whether the device supports people posture detection, desc:eventType: MeetingLock-->true
</isSupportPostureDetection>
<isSupportFans>
  <!--ro, opt, bool, whether the device supports configuring fan, desc:/ISAPI/System/Fans?format=json-->true
</isSupportFans>
<pickUpCalibration>
  <!--ro, opt, object, whether the device supports wireless pickup calibration, desc:related URI:
/ISAPI/System/PickUpCalibration/channels/<channelID>/capabilities?format=json-->
  <ChannelsList>
    <!--ro, opt, array, channel List, subType:object-->
    <channelID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelID>
  </ChannelsList>
</pickUpCalibration>
<isSupportSerialLogUpload>
  <!--ro, opt, bool, whether the device supports uploading serial port log, desc:true (support), false (not support)-->true
</isSupportSerialLogUpload>
<isSupportLiquidDetectionEvent>
  <!--ro, opt, bool, whether the device supports liquid detection, desc:eventType: LiquidDetectionEvent-->true
</isSupportLiquidDetectionEvent>
<isSupportETCRSU>
  <!--ro, opt, bool, whether the device supports ETCRSU configuration, desc:related URI: /ISAPI/Traffic/ETCRSU Level protocol-->true
</isSupportETCRSU>
<isSupportVideoCodeConvertTool>
  <!--ro, opt, bool, whether the device supports getting the video encoding conversion tool, desc:related URI: /ISAPI/System/videoCodeConvertTool?
format=json protocol-->true
</isSupportVideoCodeConvertTool>
<supportQuickConfigType opt="icr,dayNightGate,syncSignalOutput">
  <!--ro, opt, string, the list of supported functions of the quick configuration pop-up on the web, attr:opt{opt, string},
desc:the web quick configuration does not distinguish the configuration order, and it displays all supported function lists on the pop-up on the tab page
icr refers to ICR switch configuration (related URL: /ISAPI/Image/channels/<channelID>/); icr
dayNightGate refers to day and night mode switching (related URL: /ISAPI/Image/channels/<channelID>/dayNightGate);
IOOutputs refers to synchronous signal output (related URL: /ISAPI/System/syncSignalOutput)-->syncSignalOutput
</supportQuickConfigType>
<isSupportVisualLocation>
  <!--ro, opt, bool, whether the device supports configuring visual location service, desc:related URI:
/ISAPI/System/LocationMgr/visualLocation/capabilities?format=json-->true
</isSupportVisualLocation>
<isSupportVideoAccessAbnormal>
  <!--ro, opt, bool, whether the device supports abnormal video access, desc:eventType: VideoAccessAbnormal; this node is not returned if not supported-->true
</isSupportVideoAccessAbnormal>
<isSupportSubSystemAbnormal>
  <!--ro, opt, bool, whether the device supports abnormal subsystem, desc:eventType: SubSystemAbnormal; this node is not returned if not supported-->true
</isSupportSubSystemAbnormal>
<isSupportSearchFileList>
  <!--ro, opt, bool, whether the device supports searching for list of device files, desc:related URI: /ISAPI/System/fileExport/SearchFileList?
format=json-->true
</isSupportSearchFileList>
<isSupportDistanceDetectionEvent>
  <!--ro, opt, bool, whether the device supports distance detection, desc:eventType: distanceDetectionEvent; currently it is only used for devices test
tools-->true
</isSupportDistanceDetectionEvent>
<isSupportMeetingQueryEvent>
  <!--ro, opt, bool, whether the device supports searching for meetings, desc:eventType: MeetingElementCancel-->true
</isSupportMeetingQueryEvent>
<isSupportMeetingLockEvent>
  <!--ro, opt, bool, whether the device supports locking meetings, desc:eventType: MeetingLock-->true
</isSupportMeetingLockEvent>
<isSupportMeetingAddEvent>
  <!--ro, opt, bool, whether the device supports reserving meetings, desc:eventType: MeetingAdd-->true
</isSupportMeetingAddEvent>
<isSupportMeetingModifyEvent>
  <!--ro, opt, bool, whether the device supports modifying meetings, desc:eventType: MeetingModify-->true
</isSupportMeetingModifyEvent>
<isSupportMeetingCancelEvent>
  <!--ro, opt, bool, whether the device supports canceling meetings, desc:eventType: ScreenBindIPC-->true
</isSupportMeetingCancelEvent>
<isSupportScreenBindIPCEvent>
  <!--ro, opt, bool, whether the device supports the screen binding IPC, desc:eventType: ScreenBindIPC-->true
</isSupportScreenBindIPCEvent>
<isSupportSensorDetection>
  <!--ro, opt, bool, whether the device supports uploading sensor data, desc:eventType: sensorDetection-->true
</isSupportSensorDetection>
<isSupportClearLogCache>
  <!--ro, opt, bool, whether the device supports clearing the log cache, desc:related URI: /ISAPI/ContentMgmt/ClearLogCache?format=json-->true
</isSupportClearLogCache>

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</isSupportClearLogAlarm>
<isSupportRoadOccupancyDetection>
  <!--ro, opt, bool, whether the device supports the entrance and exit block detection, desc:related URI:
  /ISAPI/ITC/Entrance/channels/<channelID>/roadOccupancyDetection/capabilities?format=json-->true
</isSupportRoadOccupancyDetection>
<isSupportMeetingRoomSignInEvent>
  <!--ro, opt, bool, whether the device supports the MeetingRoomSignIn event, desc:eventType: MeetingRoomSignIn-->true
</isSupportMeetingRoomSignInEvent>
<isSupportMeetingRoomSignOutEvent>
  <!--ro, opt, bool, whether the device supports the MeetingRoomSignOut event, desc:eventType: MeetingRoomSignOut-->true
</isSupportMeetingRoomSignOutEvent>
<isSupportTimeTypeSTD>
  <!--ro, opt, bool, whether the device supports the time type of ISO8601, desc:if "true" is returned, it indicates that all the devices containing
  ISO8601 parameters support adding the timeType=STD, such as /ISAPI/ContentMgmt/search?timeType=STD; the time will be implemented according to the standard--
  >true
</isSupportTimeTypeSTD>
<isSupportScanner>
  <!--ro, opt, bool, whether the device supports scanning events and configuration, desc:if true is returned, it indicates the device supports the
  /ISAPI/System/scanner/capabilities?format=json and scannerInfo event-->true
</isSupportScanner>
<isSupportRegionTargetNumberCounting>
  <!--ro, opt, bool, whether the device supports the region target counting, desc:related URI:
  /ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/Capabilities?format=json-->true
</isSupportRegionTargetNumberCounting>
<depthMapMaintainMode>
  <!--ro, opt, object, the maintenance mode of depth image, desc:it will influence the smart functions when using depth images for tests, such as vertical
  customer traffic, object accumulation detection and so on enabling the maintenance mode will change the audio and video parameters of the channels related
  URI: /ISAPI/Intelligent/channels/<channelID>/DepthMapMaintainModeCfg/capabilities?format=json-->
  <ChannelsList>
    <!--ro, opt, array, channel list, subType:object-->
    <channelID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelID>
  </ChannelsList>
</depthMapMaintainMode>
<isSupportExportPipCalibrationFile>
  <!--ro, opt, bool, whether it supports exporting visual and thermal image fusion calibration file, desc:related URI:
  /ISAPI/System/ExportPipCalibrationFile?format=json-->true
</isSupportExportPipCalibrationFile>
<isSupportCourseUploadInfo>
  <!--ro, opt, bool, whether it supports notifying the course uploading completion, desc:eventType: courseUploadInfo-->true
</isSupportCourseUploadInfo>
<isSupportPresetSwitchingInformation>
  <!--ro, opt, bool, whether the device supports uploading the preset switching information, desc:eventType: presetSwitchingInformation-->true
</isSupportPresetSwitchingInformation>
<isSupportCabinetChargingQuery>
  <!--ro, opt, bool, whether it supports the request of searching for the charging of intelligent cabinet, desc:eventType: cabinetChargingQuery-->true
</isSupportCabinetChargingQuery>
<isSupportSearchVoltageAndPowerConsumptionReport>
  <!--ro, opt, bool, whether the device supports searching for voltage and consumption report, desc:related URI:
  /ISAPI/System/SearchVoltageAndPowerConsumptionReport/capabilities?format=json-->true
</isSupportSearchVoltageAndPowerConsumptionReport>
<isSupportDeviceDrop>
  <!--ro, opt, bool, whether the device supports the deviceDrop event, desc:corresponding event: deviceDrop-->true
</isSupportDeviceDrop>
<isSupportDeviceNotMoved>
  <!--ro, opt, bool, whether the device supports the deviceNotMoved event, desc:corresponding event: deviceNotMoved-->true
</isSupportDeviceNotMoved>
<isSupportWalkWrongWayOnEscalator>
  <!--ro, opt, bool, whether the device supports the walkWrongWayOnEscalator event, desc:corresponding event: walkWrongWayOnEscalator-->true
</isSupportWalkWrongWayOnEscalator>
<isSupportFallOnEscalator>
  <!--ro, opt, bool, whether the device supports the fallOnEscalator event, desc:corresponding event: fallOnEscalator-->true
</isSupportFallOnEscalator>
<isSupportCarryLargeLuggageOnEscalator>
  <!--ro, opt, bool, whether the device supports the carryLargeLuggageOnEscalator event, desc:corresponding event: carryLargeLuggageOnEscalator-->true
</isSupportCarryLargeLuggageOnEscalator>
<isSupportPushStrollerOnEscalator>
  <!--ro, opt, bool, whether the device supports the pushStrollerOnEscalator event, desc:corresponding event: pushStrollerOnEscalator-->true
</isSupportPushStrollerOnEscalator>
<isSupportTeachingSpeechDetection>
  <!--ro, opt, bool, whether it supports teachingSpeechDetection event, desc:corresponding event: teachingSpeechDetection-->true
</isSupportTeachingSpeechDetection>
<updatefirmwareType opt="warningScreenFirmware,warningScreenFont,warningScreenVoice,ledReceiverCard">
  <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:corresponding to <type> in /ISAPI/System/updateFirmware?
  type=<type>; "warningScreenFirmware", "warningScreenFont", "warningScreenVoice"; note that <isSupportUpdatefirmware> is supported when this mode is
  supported-->warningScreenFirmware
</updatefirmwareType>
<isSupportPaymentQRCodeQuery>
  <!--ro, opt, bool, whether the device supports the paymentQRCodeQuery event, desc:corresponding event: paymentQRCodeQuery-->true
</isSupportPaymentQRCodeQuery>
<isSupportTopUpEvent>
  <!--ro, opt, bool, whether the device supports the topUpEvent event, desc:corresponding event: topUpEvent-->true
</isSupportTopUpEvent>
<isSupportTopUpRequest>
  <!--ro, opt, bool, whether the device supports the topUpRequest event, desc:corresponding event: topUpRequest-->true
</isSupportTopUpRequest>
<isSupportTopUpStatusQuery>
  <!--ro, opt, bool, whether the device supports the topUpStatusQuery event, desc:corresponding event: topUpStatusQuery-->true
</isSupportTopUpStatusQuery>
<isSupportLogOffRefundRequest>
  <!--ro, opt, bool, whether the device supports the LogOffRefundRequest event, desc:corresponding event: LogOffRefundRequest-->true
</isSupportLogOffRefundRequest>
<isSupportLogOffRefundStatusQuery>

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    <!--ro, opt, bool, whether the device supports searching for the refund status of Logged-off persons, desc:corresponding event: LogOffRefundStatusQuery-->true
  </isSupportLogOffRefundStatusQuery>
  <isSupportAuthenticationTimeoutEvent>
    <!--ro, opt, bool, whether the device supports the LogOffRefundStatusQuery event, desc:corresponding event: LogOffRefundStatusQuery-->true
  </isSupportAuthenticationTimeoutEvent>
  <isSupportCertificateCapture>
    <!--ro, opt, bool, whether the device supports the certificateCapture events, desc:corresponding event: certificateCapture-->true
  </isSupportCertificateCapture>
  <isSupportDeviceBatchActivation>
    <!--ro, opt, bool, whether the device supports batch activation, desc:POST /ISAPI/System/activate/DeviceBatchActivation?format=json-->true
  </isSupportDeviceBatchActivation>
  <isSupportPlaceEnvironmentMonitoringRule>
    <!--ro, opt, bool, whether the device supports place environment monitoring rule parameters,
desc:/ISAPI/Event/channels/<channelID>/PlaceEnvironmentMonitoringRule?format=json-->true
  </isSupportPlaceEnvironmentMonitoringRule>
  <isSupportEngineResource>
    <!--ro, opt, bool, whether the device supports engine resources, desc:/ISAPI/System/EngineResource?format=json-->true
  </isSupportEngineResource>
  <isSupportPTModulePowerCfg>
    <!--ro, opt, bool, whether the device supports modifying the power supply parameters of PTZ base, desc:PUT /ISAPI/System/ModifyPTModulePowerCfg?
format=json POST /ISAPI/System/GetPTModulePowerCfg?format=json-->true
  </isSupportPTModulePowerCfg>
  <isSupportGetBoxDoorDetectionCfg>
    <!--ro, opt, bool, whether the device supports getting chassis door status detection configuration, desc:POST /ISAPI/System/GetBoxDoorDetectionCfg?
format=json-->true
  </isSupportGetBoxDoorDetectionCfg>
  <isSupportModifyBoxDoorDetectionCfg>
    <!--ro, opt, bool, whether the device supports modifying chassis door status configuration, desc:PUT /ISAPI/System/ModifyBoxDoorDetectionCfg?
format=json-->true
  </isSupportModifyBoxDoorDetectionCfg>
  <isSupportSmartBoxStatus>
    <!--ro, opt, bool, whether the device supports intelligent chassis status, desc:/ISAPI/System/SmartBoxStatus?format=json-->true
  </isSupportSmartBoxStatus>
  <isSupportBatteryStatusCtrl>
    <!--ro, opt, bool, whether the device supports battery status event upload control parameters, desc:/ISAPI/System/BatteryStatusCtrl?format=json-->true
  </isSupportBatteryStatusCtrl>
  <isSupportCooler>
    <!--ro, opt, bool, whether the device supports cooling management, desc:/ISAPI/System/Cooler/-->true
  </isSupportCooler>
  <isSupportBoxDoorDetectionCfg>
    <!--ro, opt, bool, whether the device supports configuring chassis door status detection, desc:/ISAPI/System/BoxDoorDetectionCfg?format=json-->true
  </isSupportBoxDoorDetectionCfg>
  <isSupportBoxDoorStatus>
    <!--ro, opt, bool, whether the device supports chassis door status, desc:/ISAPI/System/BoxDoorStatus?format=json-->true
  </isSupportBoxDoorStatus>
  <isSupportLightningProtectionStatus>
    <!--ro, opt, bool, whether the device supports lightning protection status, desc:/ISAPI/System/LightningProtectionStatus?format=json-->true
  </isSupportLightningProtectionStatus>
  <isSupportDeviceVisualLocalizationDetectionEvent>
    <!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationDetectionEvent event, desc:corresponding event:
DeviceVisualLocalizationDetectionEvent-->true
  </isSupportDeviceVisualLocalizationDetectionEvent>
  <isSupportDeviceVisualLocalizationAnalysisEvent>
    <!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationAnalysisEvent event, desc:corresponding event:
DeviceVisualLocalizationAnalysisEvent-->true
  </isSupportDeviceVisualLocalizationAnalysisEvent>
  <isSupportGetPowerPortWorkParamList>
    <!--ro, opt, bool, whether the device supports getting the working parameter List of power supply interface, desc:POST
/ISAPI/System/GetPowerPortWorkParamList?format=json-->true
  </isSupportGetPowerPortWorkParamList>
  <isSupportUsersLoginNotification>
    <!--ro, opt, bool, whether the device supports parameters of notifications on user Login, desc:/ISAPI/System/UsersLoginNotification?format=json-->true
  </isSupportUsersLoginNotification>
  <isSupportLiquidLevelDetectorInfo>
    <!--ro, opt, bool, whether the device supports uploading information of Liquid Level detector, desc:eventType: LiquidLevelDetectorInfo-->true
  </isSupportLiquidLevelDetectorInfo>
  <isSupportWaterPressureDetectorInfo>
    <!--ro, opt, bool, whether the device supports uploading information of water pressure detector, desc:eventType: WaterPressureDetectorInfo-->true
  </isSupportWaterPressureDetectorInfo>
  <isSupportEncryptionFileUpgradeMgr>
    <!--ro, opt, bool, whether the device supports advanced management of encrypted files, desc:GET /ISAPI/System/EncryptionFileUpgradeMgr/capabilities?
format=json-->true
  </isSupportEncryptionFileUpgradeMgr>
  <isSupportInformedConsentTextParam>
    <!--ro, opt, bool, whether the device supports text parameters of consent confirmation document,
desc:/ISAPI/System/InformedConsentMgr/InformedConsentTextParam?format=json-->true
  </isSupportInformedConsentTextParam>
  <isSupportSendConsolePassword>
    <!--ro, opt, bool, whether the device supports applying control panel password, desc:POST /ISAPI/System/SendConsolePassword?format=json-->true
  </isSupportSendConsolePassword>
  <isSupportAutoLaserRanging>
    <!--ro, opt, bool, whether to support auto Laser ranging parameter configuration, desc:/ISAPI/System/AutoLaserRanging?format=json-->true
  </isSupportAutoLaserRanging>
  <isSupportDelayedReboot>
    <!--ro, opt, bool, whether the device supports delayed reboot after power disconnection, desc:POST /ISAPI/System/DelayedReboot?format=json-->true
  </isSupportDelayedReboot>
  <isSupportFalseAlarmMaterialLibMgr>
    <!--ro, opt, bool, whether the device supports false alarm library management, desc:GET /ISAPI/Intelligent/falseAlarmMaterialLibMgr/capabilities?
format=json-->true
  </isSupportFalseAlarmMaterialLibMgr>
  <isSupportGetServerRandom>
    <!--ro, opt, bool, whether the device supports getting random number of Chinese cryptographic server, desc:POST
/ISAPI/System/GetServerRandom?format=json-->true
  </isSupportGetServerRandom>

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/ISAPI/Security/ProtocolCertificate/GetServerRandom?format=json-->true
</isSupportGetServerRandom>
<isSupportCustomPreviewSplitCfg>
  <!--ro, opt, bool, whether the device supports custom parameters of divided window in live view, desc:GET/PUT /ISAPI/System/Video/CustomPreviewSplitCfg?
format=json-->true
</isSupportCustomPreviewSplitCfg>
<isSupportSendTelnetTestCmd>
  <!--ro, opt, bool, whether the device supports applying telnet testing command, desc:POST /ISAPI/System/Network/NetworkAdvancedCfg/SendTelnetTestCmd?
format=json-->true
</isSupportSendTelnetTestCmd>
<isSupportTelnetTestCmdStatus>
  <!--ro, opt, bool, whether the device supports the execution status of telnet testing command, desc:GET
/ISAPI/System/Network/NetworkAdvancedCfg/TelnetTestCmdStatus?format=json-->true
</isSupportTelnetTestCmdStatus>
<isSupportImportUSBKeyAuthInfo>
  <!--ro, opt, bool, whether the device supports Chinese cryptographic USBKey authentication information, desc:POST
/ISAPI/Security/ProtocolCertificate/ImportUSBKeyAuthInfo?format=json-->true
</isSupportImportUSBKeyAuthInfo>
<isSupportSendSupervisionRecordCollection>
  <!--ro, opt, bool, whether the device supports sending completing status of supervision recording replenishment, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SendSupervisionRecordCollection?format=json-->true
</isSupportSendSupervisionRecordCollection>
<isSupportSearchPersonnelOperationSupervisionTask>
  <!--ro, opt, bool, whether it supports searching for operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPersonnelOperationSupervisionTask?format=json-->true
</isSupportSearchPersonnelOperationSupervisionTask>
<isSupportDeletePersonnelOperationSupervisionTask>
  <!--ro, opt, bool, whether it supports deleting operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/DeletePersonnelOperationSupervisionTask?format=json-->true
</isSupportDeletePersonnelOperationSupervisionTask>
<isSupportAddPersonnelOperationSupervisionTask>
  <!--ro, opt, bool, whether it supports adding operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/AddPersonnelOperationSupervisionTask?format=json-->true
</isSupportAddPersonnelOperationSupervisionTask>
<isSupportUserLoginStatusChangeEvent>
  <!--ro, opt, bool, whether the device supports the UserLoginStatusChangeEvent event, desc:eventType: UserLoginStatusChangeEvent-->true
</isSupportUserLoginStatusChangeEvent>
<isSupportObstacleAvoidance>
  <!--ro, opt, bool, whether it supports obstacle avoidance management, desc:GET/PUT /ISAPI/System/ObstacleAvoidance/-->true
</isSupportObstacleAvoidance>
<isSupportManualSnapFishEyeballPanoramaView>
  <!--ro, opt, bool, whether the device supports manually capturing expanded panorama images by wall-mounted fisheye, desc:GET
/ISAPI/Streaming/channels/<channelID>/ManualSnapFishEyeballPanoramaView?format=json-->true
</isSupportManualSnapFishEyeballPanoramaView>
<isSupportPreviewSplitSwitchCfg>
  <!--ro, opt, bool, whether the device supports custom parameters of divided window switch, desc:GET/PUT /ISAPI/System/Video/PreviewSplitSwitchCfg?
format=json-->true
</isSupportPreviewSplitSwitchCfg>
<isSupportCountingTargetParam>
  <!--ro, opt, bool, whether the device supports target parameters of people counting, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/counting/CountingTargetParam?format=json-->true
</isSupportCountingTargetParam>
<isSupportDebugLogModuleType opt="playService,communicationService,attendanceService,faceService,LCD">
  <!--ro, opt, bool, whether the device supports exporting debug log by module, attr:opt{req, string}-->true
</isSupportDebugLogModuleType>
<isSupportLoginPasswordCfg>
  <!--ro, opt, bool, whether the device supports login password parameters, desc:GET/PUT /ISAPI/Security/users/LoginPasswordCfg?format=json-->true
</isSupportLoginPasswordCfg>
<isSupportUserDataFile>
  <!--ro, opt, bool, whether the device supports importing and exporting the user data file, desc:POST /ISAPI/Security/users/UserDataFile/capabilities?
format=json-->true
</isSupportUserDataFile>
<isSupportNtripAccessCfg>
  <!--ro, opt, bool, whether the device supports NTRIP protocol access parameters, desc:GET/PUT /ISAPI/System/Network/Ntrip/NtripAccessCfg?format=json--
>true
</isSupportNtripAccessCfg>
<isSupportNetworkWatchdogParams>
  <!--ro, opt, bool, whether the device supports parameters of watchdog for network performance, desc:GET/PUT /ISAPI/System/Network/WatchdogParams?
format=json-->true
</isSupportNetworkWatchdogParams>
<isSupportAudioRecordPlan>
  <!--ro, opt, bool, whether it supports recording schedule, desc:GET/PUT /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json-->true
</isSupportAudioRecordPlan>
<isSupportStartVoiceMonitor>
  <!--ro, opt, bool, whether it supports enabling audio listening, desc:POST
/ISAPI/System/TwoWayAudio/channels/channel/<twoWayAudioChannelID>/StartVoiceMonitor?format=json-->true
</isSupportStartVoiceMonitor>
<isSupportWebPTZCfg>
  <!--ro, opt, bool, whether it supports displaying the web page for PTZ configuration, desc:this node is to determine whether to display the web page for
PTZ configuration and is irrelevant to whether PTZ is supported; default process logic: when this node does not exist, whether to display the web page for
PTZ configuration is subject to the device type-->true
</isSupportWebPTZCfg>
<isSupportGaugeReadingEvent>
  <!--ro, opt, bool, whether the device supports the GaugeReadingEvent event, desc:eventType: GaugeReadingEvent-->true
</isSupportGaugeReadingEvent>
<isSupportVNCServerCfg>
  <!--ro, opt, bool, whether it supports VNC service configuration, desc:GET/PUT /ISAPI/System/VNCServerCfg?format=json-->true
</isSupportVNCServerCfg>
<isSupportResources>
  <!--ro, opt, bool, whether it supports resources, desc:GET /ISAPI/System/DynamicResource/Resources?format=json, Resources-->true
</isSupportResources>
<isSupportDeviceGroupMgr>
  <!--ro, opt, bool, whether it supports device application resources group management, desc:GET /ISAPI/System/DeviceGroupMgr/capabilities?format=json--
>true

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</isSupportDeviceGroupMgr>
<isSupportAsyncResetSubDevicePassword>
  <!--ro, opt, bool, whether the device supports resetting the password of sub-device asynchronously,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePassword?format=json-->true
</isSupportAsyncResetSubDevicePassword>
<isSupportAsyncResetSubDevicePasswordStatusList>
  <!--ro, opt, bool, whether the device supports the status List of asynchronously resetting sub-devices' passwords,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePasswordStatusList?format=json-->true
</isSupportAsyncResetSubDevicePasswordStatusList>
<isSupportEDRAccessList>
  <!--ro, opt, bool, whether the device supports accessing address parameters of EDR service, desc:GET/PUT /ISAPI/System/Network/EDRAccessList?
format=json-->true
</isSupportEDRAccessList>
<isSupportBatteryCarInfoEvent>
  <!--ro, opt, bool, whether to support the BatteryCarInfoEvent event, desc:eventType: BatteryCarInfoEvent-->true
</isSupportBatteryCarInfoEvent>
<isSupportChildmanage>
  <!--ro, opt, bool, whether the device supports gateway sub-device topology management, desc:GET /ISAPI/IoTGateway/Childmanage/capabilities?format=json--
>true
</isSupportChildmanage>
<isSupportSearchResources>
  <!--ro, opt, bool, whether the device supports searching the resource List, desc:POST /ISAPI/System/DynamicResource/SearchResources?format=json-->true
</isSupportSearchResources>
<isSupportResetPasswordPhoneNo>
  <!--ro, opt, bool, whether the device supports the phone No. for resetting password, desc:GET/PUT /ISAPI/Security/resetPassword/ResetPasswordPhoneNo?
format=json-->true
</isSupportResetPasswordPhoneNo>
<isSupportGetPhoneNoQRCode>
  <!--ro, opt, bool, whether the device supports getting the QR code to be scanned by phone for resetting password, desc:GET
/ISAPI/Security/resetPassword/GetPhoneNoQRCode?format=json-->true
</isSupportGetPhoneNoQRCode>
<isSupportResetPasswordByPhoneNoSecurityCode>
  <!--ro, opt, bool, whether it supports resetting device password with phone number verification, desc:PUT
/ISAPI/Security/resetPassword/ResetPasswordByPhoneNoSecurityCode?format=json-->true
</isSupportResetPasswordByPhoneNoSecurityCode>
<isSupportGaugeReadingOverlapParams>
  <!--ro, opt, bool, whether the device supports water Level gauge detection information overlay parameters, desc:GET/PUT
/ISAPI/Event/gaugeReading/channels/<channelID>/GaugeReadingOverlapParams?format=json-->true
</isSupportGaugeReadingOverlapParams>
<isSupportGaugeReadingRainfallEvent>
  <!--ro, opt, bool, whether the device supports the GaugeReadingRainfallEvent event, desc:eventType: GaugeReadingRainfallEvent-->true
</isSupportGaugeReadingRainfallEvent>
<isSupportSearchTrafficLightMalfunctionResult>
  <!--ro, opt, bool, whether the device supports searching for traffic light fault detection result, desc:POST
/ISAPI/Event/TrafficLightMalfunctionDetection/SearchTrafficLightMalfunctionResult?format=json-->true
</isSupportSearchTrafficLightMalfunctionResult>
<isSupportCheckPhoneNoSecurityCode>
  <!--ro, opt, bool, whether the device supports phone security code verification, desc:PUT /ISAPI/Security/resetPassword/CheckPhoneNoSecurityCode?
format=json-->true
</isSupportCheckPhoneNoSecurityCode>
<isSupportUserValidExpired>
  <!--ro, opt, bool, whether the device supports the UserValidExpired event, desc:eventType: UserValidExpired-->true
</isSupportUserValidExpired>
<isSupportSearchWaterLevelData>
  <!--ro, opt, bool, whether the device supports searching for water Level detection data, desc:POST
/ISAPI/Event/waterQualityDetection/SearchWaterLevelData?format=json-->true
</isSupportSearchWaterLevelData>
<isSupportSearchRecommendPerspective>
  <!--ro, opt, bool, whether the device supports searching for recommended FoV, desc:POST
/ISAPI/Streaming/channels/channels/<channelID>/SearchRecommendPerspective?format=json-->true
</isSupportSearchRecommendPerspective>
<isSupportGetLegendaryPerspectiveInfo>
  <!--ro, opt, bool, whether the device supports getting the information of camera FoV, desc:GET
/ISAPI/Streaming/channels/channels/<channelID>/GetLegendaryPerspectiveInfo?format=json-->true
</isSupportGetLegendaryPerspectiveInfo>
<configurationDataType opt="OSD,MODE,IMAGE,IOOUT,OTHERS">
  <!--ro, opt, string, "OSD"-import video and image OSD settings, "MODE"-import application mode, "IMAGE"-import image parameters, "IOOUT"-import
supplement Light Parameters, "OTHERS", attr:opt{req, string}-->test
</configurationDataType>
<isSupportRestoreVideoAdjustmentParam>
  <!--ro, opt, bool, whether the device supports restoring video image setting parameters, desc:GET/PUT
/ISAPI/System/Video/channels/<channelID>/RestoreVideoAdjustmentParam?format=json-->true
</isSupportRestoreVideoAdjustmentParam>
<isSupportWeatherStationInfo>
  <!--ro, opt, bool, whether the device supports weather station information, desc:GET/PUT /ISAPI/Traffic/channels/<channelID>/WeatherStationInfo?
format=json-->true
</isSupportWeatherStationInfo>
<isSupportRadarPerimeterAlgCap>
  <!--ro, opt, bool, whether the device supports the capability of radar perimeter algorithm parameters, desc:GET
/ISAPI/Smart/RadarPerimeterAlg/capabilities?format=json-->true
</isSupportRadarPerimeterAlgCap>
<isSupportBodyCameraOneKeyAlarmEvent>
  <!--ro, opt, bool, whether the device supports the BodyCameraOneKeyAlarmEvent event, desc:eventType: BodyCameraOneKeyAlarmEvent; this node is not
returned if not supported-->true
</isSupportBodyCameraOneKeyAlarmEvent>
<isSupportGetOnlineTrafficLight>
  <!--ro, opt, bool, whether the device supports getting online traffic Light information, desc:GET
/ISAPI/Event/TrafficLightMalfunctionDetection/GetOnlineTrafficLight?format=json-->true
</isSupportGetOnlineTrafficLight>
<isSupportTrafficLightMalfunctionDefaultParams>
  <!--ro, opt, bool, whether the device supports default parameters of traffic Light fault detection rules, desc:GET/PUT
/ISAPI/Event/TrafficLightMalfunctionDetection/TrafficLightMalfunctionDefaultParams?format=json-->true
</isSupportTrafficLightMalfunctionDefaultParams>

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<isSupportFallenObjectsOnTracksDetection>
  <!--ro, opt, bool, whether the device supports detection of fallen objects on tracks, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FallenObjectsOnTracksDetection/capabilities?format=json-->true
</isSupportFallenObjectsOnTracksDetection>
<isSupportMonitorAuthorization>
  <!--ro, opt, bool, whether the device supports device service monitoring authorization information, desc:GET
/ISAPI/Security/DeviceServiceMonitor/MonitorAuthorization?format=json-->true
</isSupportMonitorAuthorization>
<isSupportBodyCameraTakeEvent>
  <!--ro, opt, bool, whether the device supports the BodyCameraTakeEvent event, desc:eventType: BodyCameraTakeEvent; this node is not returned if not
supported-->true
</isSupportBodyCameraTakeEvent>
<isSupportCollaborationModeCfg>
  <!--ro, opt, bool, whether the device supports configuring multi-Lens coordination mode, desc:GET/PUT /ISAPI/Intelligent/CollaborationModeCfg?
format=json-->true
</isSupportCollaborationModeCfg>
<isSupportAudioChannelVCAResource>
  <!--ro, opt, bool, whether the device supports switching intelligent resources running on audio channels, desc:GET/PUT
/ISAPI/System/VCAResource/channels/<audioID>/AudioChannelVCAResource?format=json-->true
</isSupportAudioChannelVCAResource>
<isSupportGasTankMonitoringEvent>
  <!--ro, opt, bool, whether the device supports the GasTankMonitoringEvent event, desc:eventType: GasTankMonitoringEvent-->true
</isSupportGasTankMonitoringEvent>
<isSupportChannelClearEvent>
  <!--ro, opt, bool, whether the device supports the ChannelClearEvent event, desc:eventType: ChannelClearEvent-->true
</isSupportChannelClearEvent>
<isSupportBackDuty>
  <!--ro, opt, bool, whether the device supports the backDuty event, desc:eventType: backDuty-->true
</isSupportBackDuty>
<isSupportSleepDuty>
  <!--ro, opt, bool, whether the device supports the sleepDuty event, desc:eventType: sleepDuty-->true
</isSupportSleepDuty>
<isSupportSearchChildDevResourcesStatus>
  <!--ro, opt, bool, whether the device supports searching for sub-device resource status, desc:POST
/ISAPI/System/DynamicResource/SearchChildDevResourcesStatus?format=json-->true
</isSupportSearchChildDevResourcesStatus>
<isSupportDeleteResources>
  <!--ro, opt, bool, whether the device supports deleting resources, desc:PUT /ISAPI/System/DynamicResource/DeleteResources?format=json-->true
</isSupportDeleteResources>
<isSupportMenuFunctionList>
  <!--ro, opt, bool, whether the device supports the menu List, desc:GET /ISAPI/Thermal/MenuFunctionList?format=json-->true
</isSupportMenuFunctionList>
<isSupportKeyFunctionList>
  <!--ro, opt, bool, whether the device supports the button List, desc:GET /ISAPI/Thermal/KeyFunctionList?format=json-->true
</isSupportKeyFunctionList>
<isSupportKeyValueTypeList>
  <!--ro, opt, bool, whether the device supports button key List, desc:GET /ISAPI/Thermal/KeyValueTypeList?format=json-->true
</isSupportKeyValueTypeList>
<isSupportGetPseudoColorBackgroundPic>
  <!--ro, opt, bool, whether the device supports 3-channel palette background picture, desc:GET
/ISAPI/Streaming/channels/<channelID>/GetPseudoColorBackgroundPic?format=json-->true
</isSupportGetPseudoColorBackgroundPic>
<isSupportElevatorRunningDetection>
  <!--ro, opt, bool, whether the device supports elevator running status detection, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ElevatorRunningDetection/capabilities?format=json-->true
</isSupportElevatorRunningDetection>
<isSupportSearchPersonQueueRealTimeResult>
  <!--ro, opt, bool, whether the device supports searching for history records of person waiting detection event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueRealTimeResult?format=json-->true
</isSupportSearchPersonQueueRealTimeResult>
<isSupportSearchLeavePositionEventResult>
  <!--ro, opt, bool, whether the device supports searching for history records of absence detection event, desc:POST
/ISAPI/Intelligent/BehaviorAnalysis/SearchLeavePositionEventResult?format=json-->true
</isSupportSearchLeavePositionEventResult>
<isSupportImportElectronicBadgeBackgroundImage>
  <!--ro, opt, bool, whether the device supports importing the background picture of nameplate, desc:POST
/ISAPI/Traffic/ImportElectronicBadgeBackgroundImage?format=json-->true
</isSupportImportElectronicBadgeBackgroundImage>
<isSupportVoiceMonitoringEvent>
  <!--ro, opt, bool, whether the device supports uploading the VoiceMonitoringEvent event, desc:eventType: VoiceMonitoringEvent-->true
</isSupportVoiceMonitoringEvent>
<isSupportPseudoColorBackgroundParam>
  <!--ro, opt, bool, whether the device supports 3-channel palette background parameters, desc:GET/PUT
/ISAPI/Streaming/channels/<channelID>/PseudoColorBackgroundParam?format=json-->true
</isSupportPseudoColorBackgroundParam>
<isSupportGotoPerimeterShieldMaskScene>
  <!--ro, opt, bool, whether the device supports jumping to shielded scenes of specified radar perimeter, desc:POST
/ISAPI/Smart/RadarPerimeterAlg/GotoPerimeterShieldMaskScene?format=json-->true
</isSupportGotoPerimeterShieldMaskScene>
<isSupportSearchPersonQueueTimingStatistics>
  <!--ro, opt, bool, whether the device supports the SearchPersonQueueTimingStatistics event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueTimingStatistics?format=json-->true
</isSupportSearchPersonQueueTimingStatistics>
<isSupportLocalUserChange>
  <!--ro, opt, bool, whether the device supports the LocalUserChange event, desc:eventType: LocalUserChange-->true
</isSupportLocalUserChange>
<isSupportSendClientTask>
  <!--ro, opt, bool, whether the device supports applying tasks configured on the client, desc:POST /ISAPI/System/RailwayPatrol/SendClientTask?
format=json-->true
</isSupportSendClientTask>
<isSupportGetCurrentClientTaskStatus>
  <!--ro, opt, bool, whether the device supports getting the current execution progress of the client task, desc:GET
/ISAPI/System/RailwayPatrol/GetCurrentClientTaskStatus?format=json-->true
</isSupportGetCurrentClientTaskStatus>

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</isSupportTestEDRServerConnect>
  <!--ro, opt, bool, whether the device supports testing the connection status of the EDR platform, desc:POST /ISAPI/System/Network/TestEDRServerConnect?
format=json-->true
</isSupportTestEDRServerConnect>
<isSupportRealTimeStreamInfo>
  <!--ro, opt, bool, whether the device supports getting the real-time stream information, desc:GET /ISAPI/Streaming/RealTimeStreamInfo?format=json-->true
</isSupportRealTimeStreamInfo>
<isSupportIrcutFilterStatusChange>
  <!--ro, opt, bool, whether the device supports the IrcutFilterStatusChange event, desc:eventType: IrcutFilterStatusChange-->true
</isSupportIrcutFilterStatusChange>
<isSupportSendRoadMaintPictureAnalysisTask>
  <!--ro, opt, bool, whether the device supports applying tasks of road maintenance picture analysis, desc:POST
/ISAPI/Intelligent/SendRoadMaintPictureAnalysisTask?format=json-->true
</isSupportSendRoadMaintPictureAnalysisTask>
<isSupportFDRemoveDuplicateParams>
  <!--ro, opt, bool, whether the device supports deduplication parameters of custom face picture library, desc:GET/PUT
/ISAPI/System/Video/inputs/counting/FDRemoveDuplicateParams?format=json-->true
</isSupportFDRemoveDuplicateParams>
<isSupportRadarMixedTargetSensitivityParams>
  <!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via radar, desc:GET/PUT
/ISAPI/Smart/RadarMixedTargetAlg/RadarMixedTargetSensitivityParams?format=json-->true
</isSupportRadarMixedTargetSensitivityParams>
<isSupportMixedTargetRadarSensitivityParams>
  <!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarSensitivityParams?format=json-->true
</isSupportMixedTargetRadarSensitivityParams>
<isSupportMixedTargetRadarDebugParams>
  <!--ro, opt, bool, whether the device supports algorithm debugging parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarDebugParams?format=json-->true
</isSupportMixedTargetRadarDebugParams>
<isSupportHttpPreview>
  <!--ro, opt, bool, whether the device supports HTTP Live view,
desc:related URI: GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview;
there are 2 steps as follows:
1. check if this node is returned by device and if the returned value is true. If yes, the HTTP Live view function is supported by device.
2. if this node is not returned by device, you should go through the stream ID in the API (GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview) of all
channels. If notSupport is returned for all channels, HTTP Live view is not supported by device; if at least the success code is returned for at least one
channel, HTTP Live view is supported by device.
Note: whether the stream ID (trackStreamID) is supported is determined by the response parameters StreamingChannelList, StreamingChannel, and id from GET
/ISAPI/Streaming/channels-->true
</isSupportHttpPreview>
<isSupportGetRecordThumbnail>
  <!--ro, opt, bool, whether the device supports getting the video file thumbnail, desc:POST /ISAPI/ContentMgmt/search/GetRecordThumbnail?format=json--
>true
</isSupportGetRecordThumbnail>
<isSupportProxyAIOP>
  <!--ro, opt, bool, whether the device supports proxy access of AIOP (AIOP devices connected with network cameras via NVR transparent transmission),
desc:when this node is supported, full AIOP functions (device AIOP capability, model package management, task rule management, detection data, etc.) can be
accessed via proxy transmission of ISAPI sub-device-->true
</isSupportProxyAIOP>
<isSupportStaticObjectPositionOffsetComparison>
  <!--ro, opt, bool, whether the device supports position deviation comparison of static objects, desc:GET
/ISAPI/Intelligent/channels/<channelID>/StaticObjectPositionOffsetComparison/capabilities?format=json-->true
</isSupportStaticObjectPositionOffsetComparison>
<isSupportClearFDRemoveDuplicateFDLib>
  <!--ro, opt, bool, whether the device supports manually clearing the custom face picture library on device, desc:POST
/ISAPI/System/Video/inputs/counting/ClearFDRemoveDuplicateFDLib?format=json-->true
</isSupportClearFDRemoveDuplicateFDLib>
<isSupportDevPowerStatusReportCtrl>
  <!--ro, opt, bool, whether the device supports control parameters of device battery status upload, desc:GET/PUT /ISAPI/System/DevPowerStatusReportCtrl?
format=json-->true
</isSupportDevPowerStatusReportCtrl>
<isSupportIndicatorLight>
  <!--ro, opt, bool, whether the device supports indicator light configuration, desc:GET /ISAPI/System/IndicatorLight/capabilities?format=json-->true
</isSupportIndicatorLight>
<isSupportEnvironmentTemperatureMonitorEvent>
  <!--ro, opt, bool, whether the device supports the EnvironmentTemperatureMonitorEvent event, desc:eventType: EnvironmentTemperatureMonitorEvent-->true
</isSupportEnvironmentTemperatureMonitorEvent>
<isSupportPhysicalTraining>
  <!--ro, opt, bool, whether the device supports physical training, desc:GET /ISAPI/Intelligent/channels/<channelID>/PhysicalTraining/capabilities?
format=json-->true
</isSupportPhysicalTraining>
<isSupportPullUp>
  <!--ro, opt, bool, whether the device supports the PullUp event, desc:eventType: PullUp-->true
</isSupportPullUp>
<isSupportSitUpWithHandsBehindHead>
  <!--ro, opt, bool, whether the device supports the SitUpWithHandsBehindHead event, desc:eventType: SitUpWithHandsBehindHead-->true
</isSupportSitUpWithHandsBehindHead>
<isSupportSitUpWithArmsOnChest>
  <!--ro, opt, bool, whether the device supports the SitUpWithArmsOnChest event, desc:eventType: SitUpWithArmsOnChest-->true
</isSupportSitUpWithArmsOnChest>
<isSupportPushUp>
  <!--ro, opt, bool, whether the device supports the PushUp event, desc:eventType: PushUp-->true
</isSupportPushUp>
<isSupportPullUpHold>
  <!--ro, opt, bool, whether the device supports the PullUpHold event, desc:eventType: PullUpHold-->true
</isSupportPullUpHold>
<isSupportVerificationPlatformParams>
  <!--ro, opt, bool, whether the device supports accessing parameters for the verification and recording platform, desc:GET/PUT
/ISAPI/System/Network/VerificationPlatformAccess/VerificationPlatformParams?format=json-->true
</isSupportVerificationPlatformParams>
<isSupportVehicleMaintenanceDetection>
  <!--ro, opt, bool, whether the device supports vehicle maintenance detection, desc:GET /ISAPI/Intelligent/VehicleMaintenanceDetection/capabilities?

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format=json; detects the status of vehicles in the maintenance shop and triggers the alarm when the vehicle is lifted (when detecting chassis) or Lowered,
or the hood is opened-->true
</isSupportVehicleMaintenanceDetection>
<isSupportPhysicalTrainingStatus>
  <!--ro, opt, bool, whether the device supports the PhysicalTrainingStatus event, desc:eventType: PhysicalTrainingStatus-->true
</isSupportPhysicalTrainingStatus>
<isSupportDeviceAbnormalDebug>
  <!--ro, opt, bool, whether the device supports device operation exception debugging, desc:GET /ISAPI/System/DeviceAbnormalDebug/capabilities?
format=json; during device operation exception, the device operation debugging data (debugging Log, dump file, etc.) will be collected by technical support-
-->true
</isSupportDeviceAbnormalDebug>
<isSupportSetWaterLevelResetOfWaterTank>
  <!--ro, opt, bool, whether the device supports manually filling up the water tank and uploading the filled tank information for calculating remaining
water via Lens wiper usage, desc:PUT /ISAPI/System/Hardware/SetWaterLevelResetOfWaterTank?format=json-->true
</isSupportSetWaterLevelResetOfWaterTank>
<isSupportGetRemoteAlbumPicture>
  <!--ro, opt, bool, whether the device supports getting pictures from the remote album, desc:POST /ISAPI/ContentMgmt/GetRemoteAlbumPicture?format=json--
-->true
</isSupportGetRemoteAlbumPicture>
<isSupportUltravioletImagingFusionParams>
  <!--ro, opt, bool, whether the device supports UV imaging fusion parameters, desc:GET/PUT
/ISAPI/System/Video/channels/<channelID>/UltravioletImagingFusionParams?format=json-->true
</isSupportUltravioletImagingFusionParams>
<isSupportAudioAnalysisCfg>
  <!--ro, opt, bool, whether it supports audio analysis parameters, desc:GET/PUT
/ISAPI/Intelligent/SensitiveInfoAnalysis/AudioIn/<AudioInID>/AudioAnalysisCfg?format=json-->true
</isSupportAudioAnalysisCfg>
<isSupportRuningLogPlan>
  <!--ro, opt, bool, whether the device supports device operation Log schedule, desc:GET/PUT /ISAPI/ContentMgmt/RuningLogPlan?format=json-->true
</isSupportRuningLogPlan>
<isSupportSearchRuningLogData>
  <!--ro, opt, bool, whether the device supports searching for device operation Log data, desc:POST /ISAPI/ContentMgmt/SearchRuningLogData?format=json--
-->true
</isSupportSearchRuningLogData>
<isSupportExportRuningLogData>
  <!--ro, opt, bool, whether the device supports exporting device operation Log data, desc:POST /ISAPI/ContentMgmt/ExportRuningLogData?format=json-->true
</isSupportExportRuningLogData>
<isSupportCertificateCaptureEvent>
  <!--ro, opt, bool, whether the device supports the CertificateCaptureEvent event, desc:corresponding event: CertificateCaptureEvent-->true
</isSupportCertificateCaptureEvent>
<isSupportUpgradeKeywordModel>
  <!--ro, opt, bool, whether the device supports upgrading the sensitive information keyword model, desc:PUT
/ISAPI/Intelligent/SensitiveInfoAnalysis/UpgradeKeywordModel?format=json-->true
</isSupportUpgradeKeywordModel>
<isSupportGetKeywordModelStatus>
  <!--ro, opt, bool, whether the device supports getting algorithm running status of the sensitive information keyword model, desc:GET
/ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelStatus?format=json-->true
</isSupportGetKeywordModelStatus>
<isSupportCloseLensPrivacyCover>
  <!--ro, opt, bool, whether the device supports manually closing the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/CloseLensPrivacyCover?
format=json-->true
</isSupportCloseLensPrivacyCover>
<isSupportOpenLensPrivacyCover>
  <!--ro, opt, bool, whether the device supports manually opening the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/OpenLensPrivacyCover?
format=json-->true
</isSupportOpenLensPrivacyCover>
<isSupportLensPrivacyCoverModeParams>
  <!--ro, opt, bool, whether the device supports working mode parameters of the lens privacy cap, desc:GET/PUT
/ISAPI/System/channels/<channelID>/LensPrivacyCoverModeParams?format=json-->true
</isSupportLensPrivacyCoverModeParams>
<isSupportOTAPService>
  <!--ro, opt, bool, whether the device supports OTAP service configuration, desc:/ISAPI/System/Network/OTAPService/capabilities?format=json-->true
</isSupportOTAPService>
<isSupportGetPlaybackStreamInfo>
  <!--ro, opt, bool, whether the device supports getting video playback stream information, desc:GET /ISAPI/Streaming/GetPlaybackStreamInfo?format=json--
-->true
</isSupportGetPlaybackStreamInfo>
<isSupportNBioTPlatformAccessStatus>
  <!--ro, opt, bool, whether the device supports the NB-IoT platform access status, desc:GET /ISAPI/System/Network/NBIOtPlatformAccessStatus?format=json-
-->true
</isSupportNBioTPlatformAccessStatus>
<isSupportScannedListEvent>
  <!--ro, opt, bool, whether the device supports Bluetooth scanning List event, desc:eventType: ScannedListEvent-->true
</isSupportScannedListEvent>
<isSupportDownloadAudioRecordByTime>
  <!--ro, opt, bool, whether the device supports downloading audio files by time, desc:POST /ISAPI/System/RecordAudio/DownloadAudioRecordByTime?
format=json-->true
</isSupportDownloadAudioRecordByTime>
<isSupportBackupConfigurationFile>
  <!--ro, opt, bool, whether the device supports local backup of the configuration file, desc:PUT /ISAPI/System/BackupConfigurationFile?format=json-->true
</isSupportBackupConfigurationFile>
<isSupportRestoreConfigurationFile>
  <!--ro, opt, bool, whether the device supports restoring the device local configuration file, desc:PUT /ISAPI/System/RestoreConfigurationFile?
format=json-->true
</isSupportRestoreConfigurationFile>
<isSupportGetCfgFileBackupInfo>
  <!--ro, opt, bool, whether the device supports getting the local configuration file backup information, desc:GET /ISAPI/System/GetCfgFileBackupInfo?
format=json-->true
</isSupportGetCfgFileBackupInfo>
<isSupportSearchVideoCodeMealParams>
  <!--ro, opt, bool, whether the device supports searching for video encoding plan parameters, desc:POST
/ISAPI/Streaming/channels/<channelID>/SearchVideoCodeMealParams?format=json-->true
</isSupportSearchVideoCodeMealParams>
<isSupportHdmiCodeMealParams>
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</isSupportVideoCodeMealParams>
  <!--ro, opt, bool, whether the device supports video encoding plan parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/VideoCodeMealParams?
format=json-->true
</isSupportVideoCodeMealParams>
<isSupportFlameCombustionStatusRuleParams>
  <!--ro, opt, bool, whether the device supports flame burning status detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusRuleParams?format=json-->true
</isSupportFlameCombustionStatusRuleParams>
<isSupportFlameCombustionStatusOverlapParams>
  <!--ro, opt, bool, whether the device supports flame burning status detection information overlay parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusOverlapParams?format=json-->true
</isSupportFlameCombustionStatusOverlapParams>
<isSupportAcousticLeakDetection>
  <!--ro, opt, bool, whether the device supports acoustic leak detection capability, desc:related URL: /ISAPI/System/AcousticLeakDetection/capabilities?
format=json-->true
</isSupportAcousticLeakDetection>
<isSupportFlameCombustionStatusSceneList>
  <!--ro, opt, bool, whether the device supports flame burning status detection scene list, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusSceneList?format=json-->true
</isSupportFlameCombustionStatusSceneList>
<isSupportGasLeakageDetection>
  <!--ro, opt, bool, whether the device supports gas leak event, desc:eventType: GasLeakageDetection-->true
</isSupportGasLeakageDetection>
<isSupportDischargeDetection>
  <!--ro, opt, bool, whether the device supports discharging event, desc:eventType: DischargeDetection-->true
</isSupportDischargeDetection>
<isSupportSearchBaseMapIDList>
  <!--ro, opt, bool, whether the device supports searching for base map ID list, desc:POST /ISAPI/ContentMgmt/SearchBaseMapIDList?format=json-->true
</isSupportSearchBaseMapIDList>
<isSupportRestoreDefaultParam>
  <!--ro, opt, bool, whether the device supports restoring default parameters, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?format=json-->true
</isSupportRestoreDefaultParam>
<isSupportSearchEventTargetInfo>
  <!--ro, opt, bool, whether the device supports searching for event detection target information in FoV, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/SearchEventTargetInfo?format=json-->true
</isSupportSearchEventTargetInfo>
<isSupportElevatorRunningDetectionEvent>
  <!--ro, opt, bool, whether the device supports elevator running event, desc:eventType: ElevatorRunningDetectionEvent-->true
</isSupportElevatorRunningDetectionEvent>
<isSupportElevatorOperationStatistics>
  <!--ro, opt, bool, whether the device supports elevator running statistics event, desc:eventType: ElevatorOperationStatistics-->true
</isSupportElevatorOperationStatistics>
<isSupportStandingLongJump>
  <!--ro, opt, bool, whether the device supports standing long jump event, desc:eventType: StandingLongJump-->true
</isSupportStandingLongJump>
<isSupportJumpRope>
  <!--ro, opt, bool, whether the device supports jumping rope event, desc:eventType: JumpRope-->true
</isSupportJumpRope>
<isSupportUserAvailablePwQuery>
  <!--ro, opt, bool, whether the device supports person query through password, desc:eventType: UserAvailablePwQuery-->true
</isSupportUserAvailablePwQuery>
<isSupportAddHistoryEventSubscribeCfg>
  <!--ro, opt, bool, whether the device supports adding history event subscription, desc:POST /ISAPI/Event/notification/AddHistoryEventSubscribeCfg?
format=json-->true
</isSupportAddHistoryEventSubscribeCfg>
<isSupportHistoryEventSubscribeUploadCtrl>
  <!--ro, opt, bool, whether the device supports history event subscription uploading control, desc:PUT
/ISAPI/Event/notification/HistoryEventSubscribeUploadCtrl?format=json-->true
</isSupportHistoryEventSubscribeUploadCtrl>
<isSupportHLSSStreamCfg>
  <!--ro, opt, bool, whether the device supports HLS streaming address parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/HLSSStreamCfg?
format=json-->true
</isSupportHLSSStreamCfg>
<isSupportGetKeywordModelVersionInfo>
  <!--ro, opt, bool, whether the device supports getting sensitive information keyword model version information, desc:GET
/ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelVersionInfo?format=json-->true
</isSupportGetKeywordModelVersionInfo>
<isSupportManualTriggerRemoteAlbumSnapshot>
  <!--ro, opt, bool, whether the device supports manually triggering remote album capture, desc:PUT /ISAPI/ContentMgmt/ManualTriggerRemoteAlbumSnapshot?
format=json-->true
</isSupportManualTriggerRemoteAlbumSnapshot>
<isSupportRadarVideoAssistedFocusParams>
  <!--ro, opt, bool, whether the device supports radar-assisted focusing parameters, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusParams?format=json-->true
</isSupportRadarVideoAssistedFocusParams>
<isSupportCollectionResultUpload>
  <!--ro, opt, bool, whether the device supports the uploading collected result event, desc:eventType: CollectionResultUpload-->true
</isSupportCollectionResultUpload>
<isSupportDeveloperMode>
  <!--ro, opt, bool, whether the device supports developer mode, desc:GET/PUT /ISAPI/System/DeveloperMode?format=json-->true
</isSupportDeveloperMode>
<isSupportGetDetailsFunction>
  <!--ro, opt, bool, whether the device supports getting detailed functions supported in T1 test, desc:GET /ISAPI/System/T1Test/DetailsFunction?
format=json-->true
</isSupportGetDetailsFunction>
<isSupportSearchDeviceIANAList>
  <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
</isSupportSearchDeviceIANAList>
<isSupportModifySpecificROIStreamJointParams>
  <!--ro, opt, bool, whether the device supports setting ROI stream jointing parameters, desc:PUT
/ISAPI/Streaming/channels/<channelID>/ModifySpecificROIStreamJointParams?format=json-->true
</isSupportModifySpecificROIStreamJointParams>
<isSupportDeviceTamperParams>
  <!--ro, opt, bool, whether the device supports tampering parameters, desc:GET/PUT /ISAPI/System/DeviceTamperParams?format=json-->true
</isSupportDeviceTamperParams>
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</isSupportDeviceTamperParams>
<isSupportDeviceTamperEvent>
  <!--ro, opt, bool, whether the device supports tampering event, desc:eventType: DeviceTamperEvent-->true
</isSupportDeviceTamperEvent>
<isSupportRoadVariableLaneCtrlParams>
  <!--ro, opt, bool, whether the device supports dynamic management parameters of changeable Lane, desc:GET/PUT
/ISAPI/Traffic/RoadVariableLaneCtrl/channels/<channelID>/RoadVariableLaneCtrlParams?format=json-->true
</isSupportRoadVariableLaneCtrlParams>
<isSupportImportModelFile>
  <!--ro, opt, bool, whether the device supports importing the water quality detection model file, desc:POST
/ISAPI/Event/WaterQualityDetection/ImportModelFile?format=json-->true
</isSupportImportModelFile>
<isSupportSearchImportStatus>
  <!--ro, opt, bool, whether the device supports getting water quality detection model file importing progress, desc:GET
/ISAPI/Event/WaterQualityDetection/SearchImportStatus?format=json-->true
</isSupportSearchImportStatus>
<isSupportSafetyWorkwearDetectionOverlapParams>
  <!--ro, opt, bool, whether the device supports parameters of wearing safety work clothes detection information overlay, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionOverlapParams?format=json-->true
</isSupportSafetyWorkwearDetectionOverlapParams>
<isSupportSafetyWorkwearDetectionRuleParams>
  <!--ro, opt, bool, whether the device supports detection rule parameters of wearing safety work clothes, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
</isSupportSafetyWorkwearDetectionRuleParams>
<isSupportUpdateFirmwareMode>
  <!--ro, opt, bool, whether the device supports the upgrade mode, desc:GET/PUT /ISAPI/System/UpdateFirmwareMode?format=json-->true
</isSupportUpdateFirmwareMode>
<isSupportIndoorPasswayRecovery>
  <!--ro, opt, bool, whether the device supports indoor passway block alarm restoring event, desc:eventType: IndoorPasswayRecovery#-->true
</isSupportIndoorPasswayRecovery>
<isSupportBatteryCarRecovery>
  <!--ro, opt, bool, whether the device supports electric moped parking violation alarm restoring event, desc:eventType: BatteryCarRecovery-->true
</isSupportBatteryCarRecovery>
<isSupportDetectorTemperatureRecovery>
  <!--ro, opt, bool, whether the device supports detector temperature alarm restoring event, desc:eventType: DetectorTemperatureRecovery-->true
</isSupportDetectorTemperatureRecovery>
<isSupportGetRFIDReaderWorkStatus>
  <!--ro, opt, bool, whether the device supports getting RFID read/write working status, desc:GET /ISAPI/Traffic/UHF RFIDReader/GetRFIDReaderWorkStatus?
format=json-->true
</isSupportGetRFIDReaderWorkStatus>
<isSupportRFIDReaderWorkStatusCtrl>
  <!--ro, opt, bool, whether the device supports RFID read/write working status control, desc:PUT /ISAPI/Traffic/UHF RFIDReader/RFIDReaderWorkStatusCtrl?
format=json-->true
</isSupportRFIDReaderWorkStatusCtrl>
<isSupportFlameCombustionStatusAlgorithmsVersion>
  <!--ro, opt, bool, whether the device supports getting flame burning status detection algorithm version, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusAlgorithmsVersion?format=json-->true
</isSupportFlameCombustionStatusAlgorithmsVersion>
<isSupportTestResultUpload>
  <!--ro, opt, bool, whether the device supports test result uploading event, desc:eventType: TestResultUpload-->true
</isSupportTestResultUpload>
<isSupportAutoTestParams>
  <!--ro, opt, bool, whether the device supports automation testing parameters, desc:GET/PUT /ISAPI/System/AutoTest/AutoTestParams?format=json-->true
</isSupportAutoTestParams>
<isSupportExportATInstructRecord>
  <!--ro, opt, bool, whether the device supports exporting AT command records, desc:POST /ISAPI/System/ExportATInstructRecord?format=json-->true
</isSupportExportATInstructRecord>
<isSupportOfflineVideoFileManagement>
  <!--ro, opt, bool, whether the device supports offline video file management, desc:GET /ISAPI/ContentMgmt/OfflineVideoFileManagement/capabilities?
format=json-->true
</isSupportOfflineVideoFileManagement>
<isSupportChannelOccupySceneList>
  <!--ro, opt, bool, whether the device supports getting the scene list of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupySceneList?format=json-->true
</isSupportChannelOccupySceneList>
<isSupportChannelOccupyAlgorithmsVersion>
  <!--ro, opt, bool, whether the device supports getting the algorithm version of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyAlgorithmsVersion?format=json-->true
</isSupportChannelOccupyAlgorithmsVersion>
<isSupportChannelOccupyOverlapParams>
  <!--ro, opt, bool, whether the device supports overlay parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyOverlapParams?format=json-->true
</isSupportChannelOccupyOverlapParams>
<isSupportChannelOccupyRuleParams>
  <!--ro, opt, bool, whether the device supports rule parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyRuleParams?format=json-->true
</isSupportChannelOccupyRuleParams>
<isSupportGotoChannelOccupyScene>
  <!--ro, opt, bool, whether the device supports jumping to the specified fire escape occupation scene, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/GotoChannelOccupyScene?format=json-->true
</isSupportGotoChannelOccupyScene>
<isSupportChannelPassingEvent>
  <!--ro, opt, bool, whether the device supports vehicles passing outdoor fire escape, desc:eventType: channelPassingEvent-->true
</isSupportChannelPassingEvent>
<isSupportGunsHandoverRuleParams>
  <!--ro, opt, bool, whether the device supports object handover rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/GunsHandoverRuleParams?format=json-->true
</isSupportGunsHandoverRuleParams>
<isSupportGunsHandover>
  <!--ro, opt, bool, whether the device supports object handover event, desc:eventType: GunsHandover-->true
</isSupportGunsHandover>
<isSupportBulkUpgradeChildDevice>
  <!--ro, opt, bool, whether the device supports batch upgrading sub-devices, desc:POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json-->true
</isSupportBulkUpgradeChildDevice>

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</isSupportBulkUpgradeChildDevice>
<isSupportSearchBulkUpgradeChildDeviceProgress>
  <!--ro, opt, bool, whether the device supports checking the progress of batch upgrading sub-devices, desc:PUT
/ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json-->true
</isSupportSearchBulkUpgradeChildDeviceProgress>
<isSupportDash>
  <!--ro, opt, bool, whether the device supports the dash event, desc:eventType:Dash-->true
</isSupportDash>
<isSupportStopRadarVideoAssistedFocusCalibrating>
  <!--ro, opt, bool, whether the device supports stopping the calibration of radar-assisted focusing, desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoAssistedFocusCalibrating?format=json-->true
</isSupportStopRadarVideoAssistedFocusCalibrating>
<isSupportRadarVideoAssistedFocusCalibrating>
  <!--ro, opt, bool, whether the device supports the calibration of radar-assisted focusing, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibrating?format=json-->true
</isSupportRadarVideoAssistedFocusCalibrating>
<isSupportRadarVideoAssistedFocusCalibratingStatus>
  <!--ro, opt, bool, whether the device supports displaying the calibration status of radar-assisted focusing, desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibratingStatus?format=json-->true
</isSupportRadarVideoAssistedFocusCalibratingStatus>
<isSupportETCMatchResultUpload>
  <!--ro, opt, bool, whether the device supports uploading ETC vehicle comparison results, desc:eventType:ETCMatchResultUpload-->true
</isSupportETCMatchResultUpload>
<isSupportPartsLifeRemind>
  <!--ro, opt, bool, whether the device supports prompting the Life span of device component, desc:corresponding event: PartsLifeRemind-->true
</isSupportPartsLifeRemind>
<isSupportAuthorizeMicAcquisitionParam>
  <!--ro, opt, bool, whether the device supports audio authorization information, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/AuthorizeMicAcquisitionParam?format=json-->true
</isSupportAuthorizeMicAcquisitionParam>
<isSupportTOFDetectEvent>
  <!--ro, opt, bool, whether the device supports TOF detection event, desc:eventType:TOFDetectEvent-->true
</isSupportTOFDetectEvent>
<isSupportAudioAnalysisResultOSDOverlay>
  <!--ro, opt, bool, whether the device supports OSD parameters of audio analysis results, desc:GET/PUT
/ISAPI/Intelligent/SensitiveInfoAnalysis/AudioAnalysisResultOSDOverlay?format=json-->true
</isSupportAudioAnalysisResultOSDOverlay>
<isSupportChildPasswordStatus>
  <!--ro, opt, bool, whether the device supports getting the activation password status of sub-devices, desc:GET
/ISAPI/System/activate/ChildPasswordStatus?format=json-->true
</isSupportChildPasswordStatus>
<isSupportMutexFunctionClose>
  <!--ro, opt, bool, whether the device supports quickly disabling exclusive functions, desc:POST /ISAPI/System/mutexFunctionClose?format=json-->true
</isSupportMutexFunctionClose>
<isSupportFirePointAlarmRecovery>
  <!--ro, opt, bool, whether the device supports fire detection alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
</isSupportFirePointAlarmRecovery>
<isSupportSmokeAlarmRecovery>
  <!--ro, opt, bool, whether the device supports smoke alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
</isSupportSmokeAlarmRecovery>
<isSupportTemperatureMeasurementPreviewAlarmRecovery>
  <!--ro, opt, bool, whether the device supports temperature measurement pre-alarm restoring event, desc:eventType:TemperatureMeasurementPreviewRecovery--
>true
</isSupportTemperatureMeasurementPreviewAlarmRecovery>
<isSupportTemperatureMeasurementAlarmRecovery>
  <!--ro, opt, bool, whether the device supports temperature measurement alarm restoring event, desc:eventType:TemperatureMeasurementRecovery-->true
</isSupportTemperatureMeasurementAlarmRecovery>
<isSupportUVPhotonDetection>
  <!--ro, opt, bool, whether the device supports UV photon detection event, desc:eventType:UVPhotonDetection-->true
</isSupportUVPhotonDetection>
<isSupportUVPhotonTimeDetection>
  <!--ro, opt, bool, whether the device supports UV photon scheduled-detection event, desc:eventType:UVPhotonTimeDetection-->true
</isSupportUVPhotonTimeDetection>
<isSupportPatientExpensesInfoQuery>
  <!--ro, opt, bool, whether the device supports the patient cost search event, desc:eventType:PatientExpensesInfoQuery-->true
</isSupportPatientExpensesInfoQuery>
<isSupportReinforcementCallEvent>
  <!--ro, opt, bool, whether the device supports reinforcement call event, desc:eventType:ReinforcementCallEvent-->true
</isSupportReinforcementCallEvent>
<isSupportChargeMode>
  <!--ro, opt, bool, whether the device supports charging mode, desc:GET/PUT /ISAPI/System/ChargeMode?format=json-->true
</isSupportChargeMode>
<isSupportCombinationRuleParams>
  <!--ro, opt, bool, whether the device supports combination rule parameters of perimeter detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterCombinationDetection/CombinationRuleParams?format=json-->true
</isSupportCombinationRuleParams>
<isSupportSwitchHolidayPlanCfg>
  <!--ro, opt, bool, whether the device supports the regular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchHolidayPlanCfg?format=json--
>true
</isSupportSwitchHolidayPlanCfg>
<isSupportSwitchIrregularHolidayPlanCfg>
  <!--ro, opt, bool, whether the device supports irregular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchIrregularHolidayPlanCfg?
format=json-->true
</isSupportSwitchIrregularHolidayPlanCfg>
<isSupportSearchPresetConfigInfo>
  <!--ro, opt, bool, whether the device supports searching for the configuration of specified presets, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPresetConfigInfo?format=json-->true
</isSupportSearchPresetConfigInfo>
<isSupportSwitchPlanCfg>
  <!--ro, opt, bool, whether the device supports startup/shutdown schedule, desc:GET/PUT /ISAPI/System/SwitchPlanCfg?format=json-->true
</isSupportSwitchPlanCfg>
<isSupportVegetationSpectrumDetection>
  <!--ro, opt, bool, whether the device supports vegetation spectrum detection, desc:eventType:VegetationSpectrumDetection-->true
</isSupportVegetationSpectrumDetection>

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<isSupportCaptureResultAttributeSearch>
  <!--ro, opt, bool, whether the device supports searching for attributes of human body capture, desc:POST
  /ISAPI/Intelligent/CaptureResultAttributeSearch?format=json-->true
</isSupportCaptureResultAttributeSearch>
<isSupportHealthEducationInfoReadStatus>
  <!--ro, opt, bool, whether the device supports the event of uploading the reading status of health education information,
  desc:eventType:HealthEducationInfoReadStatus-->true
</isSupportHealthEducationInfoReadStatus>
<isSupportNurseLocationEvent>
  <!--ro, opt, bool, whether the device supports nurse locating event, desc:eventType:NurseLocationEvent-->true
</isSupportNurseLocationEvent>
<isSupportAddChildDevResource>
  <!--ro, opt, bool, whether the device supports adding gateway resource information, desc:POST /ISAPI/System/DynamicResource/AddChildDevResource?
  format=json-->true
</isSupportAddChildDevResource>
<isSupportModifyChildDevResource>
  <!--ro, opt, bool, whether the device supports editing gateway resource information, desc:POST /ISAPI/System/DynamicResource/ModifyChildDevResource?
  format=json-->true
</isSupportModifyChildDevResource>
<isSupportDiagnosisDataCollectionParams>
  <!--ro, opt, bool, whether the device supports diagnosis data statistics parameters, desc:GET/PUT /ISAPI/System/DiagnosisDataCollectionParams?
  format=json-->true
</isSupportDiagnosisDataCollectionParams>
<isSupportServicePlatformAccountAuth>
  <!--ro, opt, bool, whether the device supports platform account authentication, desc:GET /ISAPI/Security/ServicePlatformAccountAuth/capabilities?
  format=json-->true
</isSupportServicePlatformAccountAuth>
<isSupportSyncPatientInfoEvent>
  <!--ro, opt, bool, whether the device supports synchronizing patient information, desc:eventType:SyncPatientInfoEvent-->true
</isSupportSyncPatientInfoEvent>
<isSupportAuthenticationNetworkCfg>
  <!--ro, opt, bool, whether the device supports the authentication parameters of networking method, desc:GET/PUT
  /ISAPI/Security/ProtocolCertificate/AuthenticationNetworkCfg?format=json-->true
</isSupportAuthenticationNetworkCfg>
<isSupportPersonnelChannelLogParams>
  <!--ro, opt, bool, whether the device supports turnstile Log parameters, desc:GET/PUT /ISAPI/ContentMgmt/PersonnelChannelLogParams?format=json-->true
</isSupportPersonnelChannelLogParams>
<isSupportElectricSaveModeParams>
  <!--ro, opt, bool, whether the device supports power-saving mode parameters, desc:GET/PUT /ISAPI/System/ElectricSaveModeParams?format=json-->true
</isSupportElectricSaveModeParams>
<isSupportChannelOccupyPOSParams>
  <!--ro, opt, bool, whether the device supports overLaying POS information on the outdoor access blocking detection data, desc:GET/PUT
  /ISAPI/Intelligent/channels/<channelID>/ChannelOccupyPOSParams?format=jsonGET or PUT
  /ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
</isSupportChannelOccupyPOSParams>
<isSupportChildPowerTimePlanParams>
  <!--ro, opt, bool, whether the device supports power-off schedule parameters of sub-devices, desc:GET/PUT /ISAPI/System/ChildPowerTimePlanParams?
  format=json-->true
</isSupportChildPowerTimePlanParams>
<isSupportPowerIndicatorLightCtrlParams>
  <!--ro, opt, bool, whether the device supports parameters of device battery indicator control, desc:GET/PUT /ISAPI/System/PowerIndicatorLightCtrlParams?
  format=json-->true
</isSupportPowerIndicatorLightCtrlParams>
<isSupportRadarVideoMapDetectionParams>
  <!--ro, opt, bool, whether the device supports parameters of radar-assisted map detection, desc:GET
  /ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoMapDetectionParams?format=json-->true
</isSupportRadarVideoMapDetectionParams>
<isSupportRadarPOSParams>
  <!--ro, opt, bool, whether the device supports overLaying POS information on the radar detection data, desc:GET/PUT
  /ISAPI/System/Video/inputs/channels/<channelID>/RadarPOSParams?format=json-->true
</isSupportRadarPOSParams>
<isSupportVideoIntercomModeEvent>
  <!--ro, opt, bool, whether the device supports the two-way audio mode event, desc:eventType:VideoIntercomModeEvent-->true
</isSupportVideoIntercomModeEvent>
<isSupportSoftwareAndHardwareCtrlParams>
  <!--ro, opt, bool, whether the device supports parameters of hardware and software service control, desc:GET/PUT
  /ISAPI/System/FunctionSwitchCtrl/SoftwareAndHardwareCtrlParams?format=json-->true
</isSupportSoftwareAndHardwareCtrlParams>
<isSupportUserLoginAbnormal>
  <!--ro, opt, bool, whether the device supports user login exception information, desc:GET/PUT /ISAPI/Security/users/UserLoginAbnormal?format=json-->true
</isSupportUserLoginAbnormal>
<isSupportIndustryThermometryAdvancedParam>
  <!--ro, opt, bool, whether the device supports the advanced parameters of industrial temperature measurement, desc:GET/PUT
  /ISAPI/Thermal/channels/<channelID>/IndustryThermometryAdvancedParam?format=json-->true
</isSupportIndustryThermometryAdvancedParam>
<isSupportGotoVideoRecheckScene>
  <!--ro, opt, bool, whether the device supports jumping to the specified scene of water level recheck detection, desc:POST
  /ISAPI/Event/channels/<channelID>/GotoVideoRecheckScene?format=json-->true
</isSupportGotoVideoRecheckScene>
<isSupportVideoRecheckRuleParams>
  <!--ro, opt, bool, whether the device supports rule parameters of water level recheck scene, desc:PUT
  /ISAPI/Event/channels/<channelID>/VideoRecheckRuleParams?format=json-->true
</isSupportVideoRecheckRuleParams>
<isSupportVideoRecheckSceneList>
  <!--ro, opt, bool, whether the device supports getting the list of water level recheck detection scenes, desc:GET
  /ISAPI/Event/channels/<channelID>/VideoRecheckSceneList?format=json-->true
</isSupportVideoRecheckSceneList>
<isSupportPeopleSedentary>
  <!--ro, opt, bool, whether the device supports sedentary detection, desc:eventType:PeopleSedentary-->true
</isSupportPeopleSedentary>
<isSupportPeopleFallingDownDetection>
  <!--ro, opt, bool, whether the device supports person fall detection, desc:eventType: PeopleFallingDownDetection-->true
</isSupportPeopleFallingDownDetection>

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<isSupportBridgeDeflectionTargetCalibration>
  <!--ro, opt, bool, whether the device supports target calibration of bridge deflection, desc:PUT
  /ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetCalibration?format=json-->true
</isSupportBridgeDeflectionTargetCalibration>
<isSupportGetBridgeDeflectionTargetCalibrationStatus>
  <!--ro, opt, bool, whether the device supports getting the target calibration status of bridge deflection, desc:GET
  /ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetCalibrationStatus?format=json-->true
</isSupportGetBridgeDeflectionTargetCalibrationStatus>
<isSupportBridgeDeflectionMonitoringRuleParams>
  <!--ro, opt, bool, whether the device supports the rule parameters of bridge deflection monitoring, desc:GET/PUT
  /ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionMonitoringRuleParams?format=json-->true
</isSupportBridgeDeflectionMonitoringRuleParams>
<isSupportHardwareReboot>
  <!--ro, opt, bool, whether the device supports hardware rebooting, desc:PUT /ISAPI/System/HardwareReboot?format=json-->true
</isSupportHardwareReboot>
<isSupportSearchSceneCalibrationStatus>
  <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
  /ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibrationStatus?format=json-->true
</isSupportSearchSceneCalibrationStatus>
<isSupportSearchSceneCalibration>
  <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
  /ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibration?format=json-->true
</isSupportSearchSceneCalibration>
<isSupportSoundSourceLocationRuleParams>
  <!--ro, opt, bool, whether the device supports rule parameters of sound source locating, desc:GET/PUT
  /ISAPI/System/SoundSourceLocation/AudioIn/<AudioInID>/SoundSourceLocationRuleParams?format=json-->true
</isSupportSoundSourceLocationRuleParams>
<isSupportPersonnelChannelLogDebug>
  <!--ro, opt, bool, whether the device supports debugging commands of turnstile logs, desc:GET /ISAPI/ContentMgmt/PersonnelChannelLogDebug/capabilities?
  format=json-->true
</isSupportPersonnelChannelLogDebug>
<isSupportCheckDeviceExistNUCFile>
  <!--ro, opt, bool, whether the device supports checking the locally stored the NUC file, desc:GET
  /ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/CheckDeviceExistNUCFile?format=json-->true
</isSupportCheckDeviceExistNUCFile>
<isSupportExportSmartDataData>
  <!--ro, opt, bool, whether the device supports exporting smart data, desc:GET /ISAPI/System/VideoQualityMeasurementCollection/ExportSmartDataData?
  format=json-->true
</isSupportExportSmartDataData>
<isSupportGetCollectImageCalculateResult>
  <!--ro, opt, bool, whether the device supports getting the calculation result of image collection, desc:GET
  /ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetCollectImageCalculateResult?format=json-->true
</isSupportGetCollectImageCalculateResult>
<isSupportAIOPAlarmTrans>
  <!--ro, opt, bool, whether the device supports uploading AIOP-related events via ISAPI message transmitted by COMM_VCA_ALARM,
  desc:AIOP-related events include the following binary alarm types defined in HCNetsDK:
  AIOP-accessed video detection alarm information (binary alarm: COMM_UPLOAD_AIOP_VIDEO, ISAPI alarm: AIOP_Video), AIOP-accessed picture detection alarm
  information (binary alarm: COMM_UPLOAD_AIOP_PICTURE, ISAPI alarm: AIOP_Picture), AIOP-accessed capture polling detection alarm information (binary alarm:
  COMM_UPLOAD_AIOP_POLLING_SNAP, ISAPI alarm: AIOP_Polling_Snap), AIOP-accessed video polling detection alarm information (binary alarm:
  COMM_UPLOAD_AIOP_POLLING_VIDEO, ISAPI alarm: AIOP_Polling_Video)) -->true
</isSupportAIOPAlarmTrans>
<isSupportLongDistanceRace>
  <!--ro, opt, bool, whether the device supports long-distance race event, desc:eventType:LongDistanceRace-->true
</isSupportLongDistanceRace>
<isSupportIndicatorLightAbnormal>
  <!--ro, opt, bool, whether the device supports abnormal alarm indicator event, desc:eventType:IndicatorLightAbnormal-->true
</isSupportIndicatorLightAbnormal>
<isSupportVideoLostPictureStatus>
  <!--ro, opt, bool, whether the device supports the status of custom pictures displayed when video loss occurred, desc:GET
  /ISAPI/System/Video/videoLosses/VideoLostPictureStatus?format=json-->true
</isSupportVideoLostPictureStatus>
<DualVCA>
  <!--ro, opt, object, intelligent information overlay-->
  <isSupportdualVCAList>
    <!--ro, opt, bool, whether it supports setting the parameters for overlaying intelligent information of channels in a batch,
    desc:/ISAPI/Streaming/channels/dualVCA?format=json-->true
  </isSupportdualVCAList>
  <enabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable the function, attr:opt{req, string},def{req, bool}, desc:"ruleRegion" (overlay only rule area information)-->true
  </enabled>
  <channelID min="1" max="256">
    <!--ro, opt, int, channel No., attr:min{req, int},max{req, int}-->1
  </channelID>
  <overlapRangeType opt="ruleRegion,fullScreen" def="ruleRegion">
    <!--ro, opt, string, type of overlay area, attr:opt{req, string},def{req, string}-->ruleRegion
  </overlapRangeType>
</DualVCA>
<isSupportGetScopeCalibrationStatus>
  <!--ro, opt, bool, whether the device supports getting the scope calibration status, desc:GET
  /ISAPI/Intelligent/channels/<channelID>/GetScopeCalibrationStatus?format=json-->true
</isSupportGetScopeCalibrationStatus>
<isSupportSearchShootTableUserGroupParams>
  <!--ro, opt, bool, whether the device supports searching for the parameters of specified zeroing profiles, desc:POST
  /ISAPI/Intelligent/channels/<channelID>/SearchShootTableUserGroupParams?format=json-->true
</isSupportSearchShootTableUserGroupParams>
<isSupportModifyCurrentShootTableParams>
  <!--ro, opt, bool, whether the device supports editing the information of current valid zeroing profiles, desc:PUT
  /ISAPI/Intelligent/channels/<channelID>/ModifyCurrentShootTableParams?format=json-->true
</isSupportModifyCurrentShootTableParams>
<isSupportCalibrateScopeTargetPointPosition>
  <!--ro, opt, bool, whether the device supports calibrating the absolute position of the sight cross and drop point, desc:PUT
  /ISAPI/Intelligent/channels/<channelID>/CalibrateScopeTargetPointPosition?format=json-->true
</isSupportCalibrateScopeTargetPointPosition>
<isSupportGetCurrentShootTableParams>

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<!--ro, opt, bool, whether the device supports getting the information of the current valid zeroing profiles, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetCurrentShootTableParams?format=json-->true
</isSupportGetCurrentShootTableParams>
<isSupportVideoGunsightRuleParam>
<!--ro, opt, bool, whether the device supports rule parameters of video scope, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/VideoGunsightRuleParam?format=json-->true
</isSupportVideoGunsightRuleParam>
<isSupportFinishCollectImageData>
<!--ro, opt, bool, whether it supports completing the collection of image data, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/FinishCollectImageData?format=json-->true
</isSupportFinishCollectImageData>
<isSupportStartCollectImageData>
<!--ro, opt, bool, whether it supports starting collecting image data, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/StartCollectImageData?format=json-->true
</isSupportStartCollectImageData>
<isSupportSearchBridgeDeflectionMonitoringEvent>
<!--ro, opt, bool, whether it supports searching for the event result of bridge deflection detection, desc:POST
/ISAPI/Intelligent/SearchBridgeDeflectionMonitoringEvent?format=json-->true
</isSupportSearchBridgeDeflectionMonitoringEvent>
<isSupportGetAllPIPCofigParams>
<!--ro, opt, bool, whether it supports getting all picture-in-picture configuration parameters, desc:GET /ISAPI/System/Video/GetAllPIPCofigParams?
format=json-->true
</isSupportGetAllPIPCofigParams>
<isSupportManualDetectionCarriageClearStatus>
<!--ro, opt, bool, whether it supports manual detection of the clearing status of the carriage, desc:POST
/ISAPI/System/channels/<channelID>/ManualDetectionCarriageClearStatus?format=json-->true
</isSupportManualDetectionCarriageClearStatus>
<isSupportGetCarriageClearCalibrationStatus>
<!--ro, opt, bool, whether it supports getting the reference calibration carriage clearing status, desc:GET
/ISAPI/System/channels/<channelID>/GetCarriageClearCalibrationStatus?format=json-->true
</isSupportGetCarriageClearCalibrationStatus>
<isSupportModifyCarriageClearCalibration>
<!--ro, opt, bool, whether it supports setting the reference calibration of carriage clearing, desc:POST
/ISAPI/System/channels/<channelID>/ModifyCarriageClearCalibration?format=json-->true
</isSupportModifyCarriageClearCalibration>
<isSupportCarriageClearRuleParams>
<!--ro, opt, bool, whether it supports setting rule parameters of carriage clearing, desc:GET/PUT
/ISAPI/System/channels/<channelID>/CarriageClearRuleParams?format=json-->true
</isSupportCarriageClearRuleParams>
<isSupportRadarFallenObjectsOnTracks>
<!--ro, opt, bool, whether it supports radar detection of fallen objects on tracks, desc:eventType:RadarFallenObjectsOnTracks-->true
</isSupportRadarFallenObjectsOnTracks>
<isSupportTemperatureRiseCalibrationParams>
<!--ro, opt, bool, whether it supports temperature rise calibration parameters, desc:GET/PUT /ISAPI/Thermal/TemperatureRiseCalibrationParams?
format=json-->true
</isSupportTemperatureRiseCalibrationParams>
<isSupportStartTemperatureRiseCalibration>
<!--ro, opt, bool, whether it supports starting temperature rise calibration, desc:PUT /ISAPI/Thermal/StartTemperatureRiseCalibration?format=json-->true
</isSupportStartTemperatureRiseCalibration>
<isSupportGetTemperatureRiseCalibrationStatus>
<!--ro, opt, bool, whether it supports getting temperature rise calibration status, desc:GET /ISAPI/Thermal/GetTemperatureRiseCalibrationStatus?
format=json-->true
</isSupportGetTemperatureRiseCalibrationStatus>
<isSupportExportTemperatureRiseCalibrationFile>
<!--ro, opt, bool, whether it supports exporting temperature rise calibration file, desc:GET /ISAPI/Thermal/ExportTemperatureRiseCalibrationFile?
format=json-->true
</isSupportExportTemperatureRiseCalibrationFile>
<isSupportGetTemperatureRiseCorrectionVersion>
<!--ro, opt, bool, whether it supports getting temperature rise calibration version information, desc:GET
/ISAPI/Thermal/GetTemperatureRiseCorrectionVersion?format=json-->true
</isSupportGetTemperatureRiseCorrectionVersion>
<isSupportSearchImportDeviceDataTaskProgress>
<!--ro, opt, bool, whether it supports searching for the progress of device data importing task, desc:POST
/ISAPI/System/SearchImportDeviceDataTaskProgress?format=json-->true
</isSupportSearchImportDeviceDataTaskProgress>
<isSupportSearchExportDeviceDataTaskProgress>
<!--ro, opt, bool, whether it supports searching for the progress of device data exporting task, desc:POST
/ISAPI/System/SearchExportDeviceDataTaskProgress?format=json-->true
</isSupportSearchExportDeviceDataTaskProgress>
<isSupportAddImportDeviceDataTask>
<!--ro, opt, bool, whether it supports creating device data importing task, desc:POST /ISAPI/System/AddImportDeviceDataTask?format=json-->true
</isSupportAddImportDeviceDataTask>
<isSupportAddExportDeviceDataTask>
<!--ro, opt, bool, whether it supports creating device data exporting task, desc:POST /ISAPI/System/AddExportDeviceDataTask?format=json-->true
</isSupportAddExportDeviceDataTask>
<isSupportSearchRecordStorage>
<!--ro, opt, bool, whether it supports searching for record storage HDD information, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStorage?
format=json-->true
</isSupportSearchRecordStorage>
<isSupportDetectionPathRuleParams>
<!--ro, opt, bool, whether it supports hazardous gas detection patrol path rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/DetectionPathRuleParams?format=json-->true
</isSupportDetectionPathRuleParams>
<isSupportSearchObjectDistanceFocusInfoList>
<!--ro, opt, bool, whether it supports searching for the list of object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/SearchObjectDistanceFocusInfoList?format=json-->true
</isSupportSearchObjectDistanceFocusInfoList>
<isSupportMoveObjectDistanceFocusLocation>
<!--ro, opt, bool, whether it supports moving to the point where the focus of the object distance is calibrated, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/MoveObjectDistanceFocusLocation?format=json-->true
</isSupportMoveObjectDistanceFocusLocation>
<isSupportAddObjectDistanceFocusInfo>
<!--ro, opt, bool, whether it supports adding object distance focus calibration information, desc:POST

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/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/AddObjectDistanceFocusInfo?format=json-->true
</isSupportAddObjectDistanceFocusInfo>
<isSupportModifyObjectDistanceFocusInfo>
  <!--ro, opt, bool, whether it supports editing object distance focus calibration information, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ModifyObjectDistanceFocusInfo?format=json-->true
</isSupportModifyObjectDistanceFocusInfo>
<isSupportObjectDistanceCalibrateAutoFocusParams>
  <!--ro, opt, bool, whether it supports auto-focus parameters of object distance calibration, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
</isSupportObjectDistanceCalibrateAutoFocusParams>
<isSupportExpresswayNetworkTollAccessParams>
  <!--ro, opt, bool, whether it supports protocol access parameters of network parking fee charging on highway, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
</isSupportExpresswayNetworkTollAccessParams>
<isSupportControlCameraDetectAction>
  <!--ro, opt, bool, whether it supports controlling the patrol path of camera, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/ControlCameraDetectAction?format=json-->true
</isSupportControlCameraDetectAction>
<isSupportObjectDistanceAccessWay>
  <!--ro, opt, bool, whether it supports the method of getting the object distance, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceAccessWay?format=json-->true
</isSupportObjectDistanceAccessWay>
<isSupportCancelTemperatureRiseCalibration>
  <!--ro, opt, bool, whether it supports canceling temperature rise calibration, desc:PUT /ISAPI/Thermal/CancelTemperatureRiseCalibration?format=json--
>true
</isSupportCancelTemperatureRiseCalibration>
<isSupportRTCTimeTrigger>
  <!--ro, opt, bool, whether the device supports RTC time exception linkage, desc:GET/PUT /ISAPI/Event/RTCTimeTrigger?format=json-->true
</isSupportRTCTimeTrigger>
<isSupportChargingGunAbnormalEvent>
  <!--ro, opt, bool, whether it supports charging station inserted gun exception event, desc:eventType: ChargingGunAbnormalEvent-->true
</isSupportChargingGunAbnormalEvent>
<isSupportVideoChannelEnhance>
  <!--ro, opt, bool, whether it supports enhancement configuration of adding video channel, desc:GET/PUT
/ISAPI/ContentMgmt/InputProxy/VideoChannelEnhance?format=json-->true
</isSupportVideoChannelEnhance>
<isSupportDeleteObjectDistanceFocusInfo>
  <!--ro, opt, bool, whether it supports deleting the object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/DeleteObjectDistanceFocusInfo?format=json-->true
</isSupportDeleteObjectDistanceFocusInfo>
<isSupportGetStartTemperatureRiseCalibrationParams>
  <!--ro, opt, bool, whether the device supports getting parameter of starting temperature rise calibration, desc:GET
/ISAPI/Thermal/GetStartTemperatureRiseCalibrationParams?format=json-->true
</isSupportGetStartTemperatureRiseCalibrationParams>
<isSupportSearchAudioRecordPlan>
  <!--ro, opt, bool, whether the device supports searching all recording schedules, desc:POST /ISAPI/System/RecordAudio/SearchAudioRecordPlan?format=json-
>true
</isSupportSearchAudioRecordPlan>
<isSupportCleanerDetectionOverlapParams>
  <!--ro, opt, bool, whether the device supports cleaning liquid detection information overlay parameters, desc:GET/PUT
/ISAPI/System/Hardware/channels/<channelID>/CleanerDetectionOverlapParams?format=json-->true
</isSupportCleanerDetectionOverlapParams>
<isSupportWaterLevelVerticalAreaDisplayParams>
  <!--ro, opt, bool, whether it supports the display parameters of water level vertical view monitoring area, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/WaterLevelVerticalAreaDisplayParams?format=json-->true
</isSupportWaterLevelVerticalAreaDisplayParams>
<isSupportGetRecordStatusList>
  <!--ro, opt, bool, whether the device supports getting the recording status of all channels, desc:POST /ISAPI/ContentMgmt/search/GetRecordStatusList?
format=json-->true
</isSupportGetRecordStatusList>
<isSupportWiperClean>
  <!--ro, opt, bool, whether the device supports wiper cleaning event, desc:eventType: WiperClean-->true
</isSupportWiperClean>
<isSupportGetNATStatus>
  <!--ro, opt, bool, whether the device supports getting NAT status, desc:GET /ISAPI/System/Network/NetworkAdvancedCfg/GetNATStatus?format=json-->true
</isSupportGetNATStatus>
<isSupportRestoreSensitivityDefaultParams>
  <!--ro, opt, bool, whether the device supports restoring radar perimeter sensitivity parameters to the default, desc:PUT
/ISAPI/Smart/RadarPerimeterAlg/RestoreSensitivityDefaultParams?format=json-->true
</isSupportRestoreSensitivityDefaultParams>
<isSupportCarryPassengersPricingStatus>
  <!--ro, opt, bool, whether the device supports displaying the status change of meter, desc:eventType: CarryPassengersPricingStatus-->true
</isSupportCarryPassengersPricingStatus>
<isSupportCarryPassengersNoPricing>
  <!--ro, opt, bool, whether to support event of not using the meter when passengers are on board, desc:eventType: CarryPassengersNoPricing-->true
</isSupportCarryPassengersNoPricing>
<isSupportNPlusMultiple>
  <!--ro, opt, bool, whether the device supports N+M mode of host and standby device, desc:whether the device supports setting workMode to N+M. Related
API: PUT /ISAPI/System/Network/NPlusOne/capabilities?format=json-->true
</isSupportNPlusMultiple>
<ifPopUpNetworkCfgSetting>
  <!--ro, opt, bool, whether to pop up network configuration interface, desc:PUT/GET /ISAPI/System/Network/interfaces/<interfaceID>. true (if the default
IP address of the device is 192.168.0.100, the device will pop up the network configuration interface), false or node does not exist (the network
configuration interface will not pop up)-->true
</ifPopUpNetworkCfgSetting>
<AudioOutputChannelCfg>
  <!--ro, opt, object, support linking audio output channel with audio file-->
  <AudioOutputChannelIDList>
    <!--ro, opt, array, audio output channel list, subType:object-->
    <AudioOutputChannelID>
      <!--ro, req, int, audio output channel No.-->1
    </AudioOutputChannelID>
  </AudioOutputChannelIDList>
  </AudioOutputChannelCfg>

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</module></module>
<isSupportImportTemperatureRiseCalibrationFile>
  <!--ro, opt, bool, whether the device supports importing the temperature rise calibration file, desc:PUT
/ISAPI/Thermal/ImportTemperatureRiseCalibrationFile?format=json-->true
</isSupportImportTemperatureRiseCalibrationFile>
<isSupportGetAPPQRCODEPic>
  <!--ro, opt, bool, whether it supports getting QR code picture of the app accessed by the device, desc:GET /ISAPI/System/GetAPPQRCODEPic?format=json-->true
</isSupportGetAPPQRCODEPic>
<isSupportPersoonQueueRealTimeData>
  <!--ro, opt, bool, whether it supports real-time counting of the number of people in the queuing area, desc:GET
/ISAPI/Intelligent/personQueueDetection/channels/<channelID>/PersoonQueueRealTimeData?format=json-->true
</isSupportPersoonQueueRealTimeData>
<isSupportImageVignettingOptimizeParams>
  <!--ro, opt, bool, whether it supports image dark corner optimization parameters, desc:GET/PUT
/ISAPI/Streaming/channels/<channelID>/ImageVignettingOptimizeParams?format=json-->true
</isSupportImageVignettingOptimizeParams>
<isSupportStartMultispectralCalibration>
  <!--ro, opt, bool, whether it supports enabling multispectral calibration, desc:PUT
/ISAPI/System/SpectrumCalibration/channels/<channelID>/StartMultispectralCalibration?format=json-->true
</isSupportStartMultispectralCalibration>
<isSupportGetMultispectralCalibrationStatus>
  <!--ro, opt, bool, whether it supports getting the multispectral calibration status, desc:GET
/ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationStatus?format=json-->true
</isSupportGetMultispectralCalibrationStatus>
<isSupportStopRadarVideoTargetTracking>
  <!--ro, opt, bool, whether it supports stopping target tracking (radar), desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoTargetTracking?format=json-->true
</isSupportStopRadarVideoTargetTracking>
<isSupportGetRadarVideoTargetTrackingStatus>
  <!--ro, opt, bool, whether it supports getting the status of target tracking (radar), desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/GetRadarVideoTargetTrackingStatus?format=json-->true
</isSupportGetRadarVideoTargetTrackingStatus>
<isSupportStartRadarVideoTargetTracking>
  <!--ro, opt, bool, whether it supports starting target tracking (radar), desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StartRadarVideoTargetTracking?format=json-->true
</isSupportStartRadarVideoTargetTracking>
<isSupportBatteryChargeCtrl>
  <!--ro, opt, bool, whether it supports battery charging control parameters, desc:GET/PUT /ISAPI/System/BatteryChargeCtrl?format=json-->true
</isSupportBatteryChargeCtrl>
<isUpgradeAuxDevice opt="true,false">
  <!--ro, opt, string, whether it supports synchronous upgrade of auxiliary device, attr:opt(opt, string), desc:it represents support details of
isUpgradeAuxDevice in /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpgradeAuxDevice=
<isUpgradeAuxDevice>-->true
</isUpgradeAuxDevice>
<isSupportClearFile>
  <!--ro, opt, bool, whether it supports clearing files under the specified path of the device, desc:POST
/ISAPI/System/VideoQualityMeasurementCollection/ClearFile?format=json-->true
</isSupportClearFile>
<isSupportRestoreDefaultParamByChannel>
  <!--ro, opt, bool, whether it supports restoring default parameters of a single channel, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?
format=json&channelID=<channelID>-->true
</isSupportRestoreDefaultParamByChannel>
<isSupportGetMultispectralCalibrationParams>
  <!--ro, opt, bool, whether it supports getting multispectral calibration parameters, desc:GET
/ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationParams?format=json-->true
</isSupportGetMultispectralCalibrationParams>
<isSupportGetThermometryCalibrationData>
  <!--ro, opt, bool, whether it supports getting temperature calibration-related data, desc:GET /ISAPI/Thermal/GetThermometryCalibrationData?format=json-->true
</isSupportGetThermometryCalibrationData>
<isSupportImportEncryptedComponentFile>
  <!--ro, opt, bool, whether it supports importing encrypted component files, desc:POST
/ISAPI/System/EncryptionFileUpgradeMgr/ImportEncryptedComponentFile?format=json-->true
</isSupportImportEncryptedComponentFile>
<isSupportGetFrameMeanValue>
  <!--ro, opt, bool, whether it supports getting frame average range information, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetFrameMeanValue?format=json-->true
</isSupportGetFrameMeanValue>
<isSupportGetBaffleImage>
  <!--ro, opt, bool, whether it supports getting image data before and after the image is captured, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/GetBaffleImage?format=json-->true
</isSupportGetBaffleImage>
<isSupportThermometryGainMode>
  <!--ro, opt, bool, whether it supports the temperature measurement gain parameters, desc:GET/PUT /ISAPI/Thermal/ThermometryGainMode?format=json-->true
</isSupportThermometryGainMode>
<isSupportGetChannelEnhancePreOperationTips>
  <!--ro, opt, bool, whether it supports getting the prompt information before the enhancement configuration of adding the channel, desc:POST
/ISAPI/ContentMgmt/InputProxy/GetChannelEnhancePreOperationTips?format=json-->true
</isSupportGetChannelEnhancePreOperationTips>
<isSupportGetDeviceNetworkLinkStatus>
  <!--ro, opt, bool, whether it supports getting the network connection status of the device, desc:GET /ISAPI/System/GetDeviceNetworkLinkStatus?
format=json-->true
</isSupportGetDeviceNetworkLinkStatus>
<isSupportManualRefreshOnlineUserList>
  <!--ro, opt, bool, whether the web needs to display a refresh button, desc:used for web interface display, determine whether the device needs to display
the refresh button to refresh the List of online users via this field-->true
</isSupportManualRefreshOnlineUserList>
<isSupportVCAAlarmTrans>
  <!--ro, opt, bool, whether it supports uploading events related to abnormal behavior detection through the transmission of ISAPI messages, desc:for all
events related to abnormal behavior detection, refer to HCNetsdk enumeration of VCA_RULE_EVENT_TYPE_EX-->true
</isSupportVCAAlarmTrans>
<isSupportHumidityEvent>
  <!--ro, opt, bool, whether it supports humidity alarm events, desc:HumidityEvent-->true

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</isSupportHumidityEvent>
<isSupportDisplayThermometryFirePrevention>
  <!--ro, opt, bool, whether the web needs to display the temperature measurement function as
  temperature measurement & fire prevention,
  desc:if this field is true, the web interface display
  "Temperature Measurement & Fire Prevention", otherwise, it will be displayed as "Temperature Measurement". If this field is not supported, the web interface
  will display "Temperature Measurement"-->true
</isSupportDisplayThermometryFirePrevention>
<isSupportPreRemoteCheckEvent>
  <!--ro, opt, bool, whether it supports the collection of pre-verification events, desc:eventType:PreRemoteCheckEvent-->true
</isSupportPreRemoteCheckEvent>
<isSupportSearchExportDeviceStatusTaskProgress>
  <!--ro, opt, bool, whether it supports getting the progress of the task to export device information and status, desc:POST
  /ISAPI/System/SearchExportDeviceStatusTaskProgress?format=json-->true
</isSupportSearchExportDeviceStatusTaskProgress>
<isSupportAddExportDeviceStatusTask>
  <!--ro, opt, bool, whether it supports creating the task to export device information and status, desc:POST /ISAPI/System/AddExportDeviceStatusTask?
  format=json-->true
</isSupportAddExportDeviceStatusTask>
<isSupportInsulatorThermometryRuleParams>
  <!--ro, opt, bool, whether it supports the insulator temperature measurement rule parameters, desc:GET/PUT
  /ISAPI/Thermal/channels/<channelID>/InsulatorThermometryRuleParams?format=json-->true
</isSupportInsulatorThermometryRuleParams>
<isSupportInsulatorCruiseParams>
  <!--ro, opt, bool, whether it supports insulator patrol parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/InsulatorCruiseParams?format=json--
  >true
</isSupportInsulatorCruiseParams>
<isSupportModifyInsulatorDetectionRegion>
  <!--ro, opt, bool, whether it supports modifying the insulator group detection area, desc:PUT
  /ISAPI/Thermal/channels/<channelID>/ModifyInsulatorDetectionRegion?format=json-->true
</isSupportModifyInsulatorDetectionRegion>
<isSupportMoveInsulatorPosition>
  <!--ro, opt, bool, whether it supports movement to the specified insulator string position, desc:POST
  /ISAPI/Thermal/channels/<channelID>/MoveInsulatorPosition?format=json-->true
</isSupportMoveInsulatorPosition>
<isSupportDeleteInsulatorGroup>
  <!--ro, opt, bool, whether it supports deleting a specified insulator group, desc:POST /ISAPI/Thermal/channels/<channelID>/DeleteInsulatorGroup?
  format=json-->true
</isSupportDeleteInsulatorGroup>
<isSupportGenerateInsulatorRuleRegion>
  <!--ro, opt, bool, whether it supports generating a specified insulator string rule area, desc:POST
  /ISAPI/Thermal/channels/<channelID>/GenerateInsulatorRuleRegion?format=json-->true
</isSupportGenerateInsulatorRuleRegion>
<isSupportGetInsulatorAllName>
  <!--ro, opt, bool, whether it supports getting all the names of insulator groups, desc:GET /ISAPI/Thermal/channels/<channelID>/GetInsulatorAllName?
  format=json-->true
</isSupportGetInsulatorAllName>
<isSupportSearchUploadLog>
  <!--ro, opt, bool, whether it supports searching and uploading log files, desc:PUT /ISAPI/ContentMgmt/SearchUploadLog?format=json-->true
</isSupportSearchUploadLog>
<isSupportRainfallTelemeter>
  <!--ro, opt, bool, whether it supports remotely monitoring rainfall, desc:eventType:RainfallTelemeter-->true
</isSupportRainfallTelemeter>
<isSupportPosOverlayTest>
  <!--ro, opt, bool, whether it supports the test of enabling POS information overlay configuration, desc:POST
  /ISAPI/System/Video/inputs/channels/<channelID>/PosOverlayTest?format=json-->true
</isSupportPosOverlayTest>
<isSupportFullScreenPosOverlay>
  <!--ro, opt, bool, whether it supports full-screen POS information overlay parameters, desc:GET/PUT
  /ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlay?format=json-->true
</isSupportFullScreenPosOverlay>
<isSupportFullScreenPosOverlayCap>
  <!--ro, opt, bool, whether to get the dynamic capability of full-screen POS information overlay parameters, desc:GET
  /ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlayCap?format=json-->true
</isSupportFullScreenPosOverlayCap>
<isSupportVCAOptimizeDisplay>
  <!--ro, opt, bool, whether it supports the display of the intelligent resource usability page, desc:if the field is true, the user-friendly interface is
  displayed; otherwise, the previous one will be displayed; if the field is not supported, it is determined by the device type-->true
</isSupportVCAOptimizeDisplay>
<isSupportSearchInsulatorParams>
  <!--ro, opt, bool, whether it supports searching the parameters of a specified insulator group, desc:POST
  /ISAPI/Thermal/channels/<channelID>/SearchInsulatorParams?format=json-->true
</isSupportSearchInsulatorParams>
<isSupportSetInsulatorParams>
  <!--ro, opt, bool, whether it supports setting insulator group parameters, desc:PUT /ISAPI/Thermal/channels/<channelID>/SetInsulatorParams?format=json--
  >true
</isSupportSetInsulatorParams>
<isSupportInsulatorAbnormalAlarm>
  <!--ro, opt, bool, whether it supports insulator exception alarm, desc:eventType:InsulatorAbnormalAlarm-->true
</isSupportInsulatorAbnormalAlarm>
<isSupportGetFlashLifespanInfo>
  <!--ro, opt, bool, whether it supports getting the lifespan information of the built-in disk storage medium, desc:GET
  /ISAPI/System/GetFlashLifespanInfo?format=json-->true
</isSupportGetFlashLifespanInfo>
<isSupportDownloadAudioAnalysisFile>
  <!--ro, opt, bool, whether it supports downloading audio analysis files, desc:POST /ISAPI/Intelligent/SensitiveInfoAnalysis/DownloadAudioAnalysisFile?
  format=json-->true
</isSupportDownloadAudioAnalysisFile>
<isSupportCrowdDispersing>
  <!--ro, opt, bool, whether it supports crowd dispersing, desc:corresponding event: CrowdDispersing-->true
</isSupportCrowdDispersing>
<isSupportPeripheralAbnormalEvent>
  <!--ro, opt, bool, whether it supports peripheral fault event, desc:If it is not supported, this node will not be returned. Corresponding eventType:

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PeripheralAbnormalEvent-->true
</isSupportPeripheralAbnormalEvent>
<isSupportPeripheralAbnormalTrigger>
  <!--ro, opt, bool, whether it supports peripheral fault Linkage, desc:GET/PUT /ISAPI/Event/PeripheralAbnormalTrigger?format=json-->true
</isSupportPeripheralAbnormalTrigger>
<isSupportBodyCameraAbnormalEvent>
  <!--ro, opt, bool, whether it supports exception events of body cameras, desc:corresponding event: BodyCameraAbnormalEvent-->true
</isSupportBodyCameraAbnormalEvent>
<isSupportDeviceWorkModeAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports the event of enforcement device working mode asynchronous status, desc:corresponding event:
DeviceWorkModeAsyncStatusEvent-->true
</isSupportDeviceWorkModeAsyncStatusEvent>
<isSupportDeviceUserAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameter status event for law enforcement device users, desc:corresponding event:
DeviceUserAsyncStatusEvent-->true
</isSupportDeviceUserAsyncStatusEvent>
<isSupportVideoCodingAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports video encoding asynchronous parameter status event, desc:corresponding event: VideoCodingAsyncStatusEvent-->true
</isSupportVideoCodingAsyncStatusEvent>
<isSupportRecordAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports manual recording asynchronous parameter status event, desc:corresponding event: RecordAsyncStatusEvent-->true
</isSupportRecordAsyncStatusEvent>
<isSupportISUPAccessAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameter status event for ISUP protocol access, desc:corresponding event:
ISUPAccessAsyncStatusEvent-->true
</isSupportISUPAccessAsyncStatusEvent>
<isSupportWirelessConfigListAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameters status event for wireless configuration List, desc:corresponding event:
WirelessConfigListAsyncStatusEvent-->true
</isSupportWirelessConfigListAsyncStatusEvent>
<isSupportDeviceLocationAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameter status event for device Location, desc:corresponding event:
DeviceLocationAsyncStatusEvent-->true
</isSupportDeviceLocationAsyncStatusEvent>
<isSupportVideoCodingAsyncParams>
  <!--ro, opt, bool, whether it supports video encoding async parameters, desc:GET/PUT /ISAPI/Streaming/VideoCodingAsyncParams?format=json-->true
</isSupportVideoCodingAsyncParams>
<isSupportMoveShield>
  <!--ro, opt, bool, whether it supports movement to a specified shielded area, desc:POST /ISAPI/Intelligent/channels/<channelID>/MoveShield?format=json--
>true
</isSupportMoveShield>
<isSupportStopAutoGenerateShield>
  <!--ro, opt, bool, whether it supports stopping the automatic recognition of smart shielded area, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/StopAutoGenerateShield?format=json-->true
</isSupportStopAutoGenerateShield>
<isSupportStartGenerateParams>
  <!--ro, opt, bool, whether it supports starting generating smart shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/StartGenerateParams?
format=json-->true
</isSupportStartGenerateParams>
<isSupportIntelligentShieldAlgorithmsVersion>
  <!--ro, opt, bool, whether it supports getting the version information of the intelligent shielding area recognition algorithm library, desc:GET
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldAlgorithmsVersion?format=json-->true
</isSupportIntelligentShieldAlgorithmsVersion>
<isSupportConvertShield>
  <!--ro, opt, bool, whether it supports converting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ConvertShield?format=json-
-->true
</isSupportConvertShield>
<isSupportDeleteShield>
  <!--ro, opt, bool, whether it supports deleting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/DeleteShield?format=json--
>true
</isSupportDeleteShield>
<isSupportGetGenerateParams>
  <!--ro, opt, bool, whether it supports getting the parameters of generating smart shielded areas, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetGenerateParams?format=json-->true
</isSupportGetGenerateParams>
<isSupportManualAddShield>
  <!--ro, opt, bool, whether it supports manually adding shielded areas, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ManualAddShield?format=json--
>true
</isSupportManualAddShield>
<isSupportIntelligentShieldParams>
  <!--ro, opt, bool, whether it supports smart shielded area parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/IntelligentShieldParams?
format=json-->true
</isSupportIntelligentShieldParams>
<isSupportIntelligentShieldListParams>
  <!--ro, opt, bool, whether it supports the smart shielded area List parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldListParams?format=json-->true
</isSupportIntelligentShieldListParams>
<isSupportDeicingCfg>
  <!--ro, opt, bool, whether it supports Lens heating and de-icing control parameters, desc:GET/PUT /ISAPI/System/Hardware/DeicingCfg?format=json-->true
</isSupportDeicingCfg>
<isSupportInspectionOverdueEvent>
  <!--ro, opt, bool, whether it supports annual inspection overdue event for alcohol detection enforcement device, desc:corresponding event:
InspectionOverdueEvent-->true
</isSupportInspectionOverdueEvent>
<isSupportChannelsPanoramaCalibration>
  <!--ro, opt, bool, whether it supports batch configuring channel calibration distance, desc:GET/PUT /ISAPI/Panorama/channels/calibration/distance?
format=json-->true
</isSupportChannelsPanoramaCalibration>
<isSupportRadarPerimeterShieldOverlapParams>
  <!--ro, opt, bool, whether it supports radar perimeter overLay parameters, desc:GET/PUT
/ISAPI/Smart/RadarPerimeterAlg/Radar/<RadarID>/RadarPerimeterShieldOverlapParams?format=json-->true
</isSupportRadarPerimeterShieldOverlapParams>
<isSupportAlarmFilterInterval>
  <!--ro, opt, bool, whether it supports configuring alarm filter time interval, desc:corresponding to the "alarmFilterInterval" node in the POST

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/ISAPI/IoTGateway/AlarmRules/SetAlarmRules?format=json-->true
</isSupportAlarmFilterInterval>
<isSupportImportLicense>
  <!--ro, opt, bool, whether it supports importing the authorization file, desc:POST /ISAPI/System/SoftwareLicense/ImportLicense?format=json-->true
</isSupportImportLicense>
<isSupportShipStatisticsAnalysisParams>
  <!--ro, opt, bool, whether it supports ship traffic statistics parameters, desc:GET/PUT
/ISAPI/IntelLigent/channels/<channelID>/ShipStatisticsAnalysisParams?format=json-->true
</isSupportShipStatisticsAnalysisParams>
<isSupportShipStatisticsEvent>
  <!--ro, opt, bool, whether it supports ship traffic statistics event, desc:corresponding event: ShipStatisticsEvent-->true
</isSupportShipStatisticsEvent>
<isSupportPeopleVitalSignsDetection>
  <!--ro, opt, bool, whether it supports vital sign detection event, desc:eventType: PeopleVitalSignsDetection-->true
</isSupportPeopleVitalSignsDetection>
<isSupportPeopleVitalSignsStatistic>
  <!--ro, opt, bool, whether it supports vital sign statistics event, desc:eventType: PeopleVitalSignsStatistic-->true
</isSupportPeopleVitalSignsStatistic>
<isSupportPeopleBodyMoving>
  <!--ro, opt, bool, whether it supports human body movement detection event, desc:eventType: PeopleBodyMoving-->true
</isSupportPeopleBodyMoving>
<isSupportPeopleGetUp>
  <!--ro, opt, bool, whether it supports person get-up detection event, desc:eventType: PeopleGetUp (detected a person who got up from the bed during
posture detection)-->true
</isSupportPeopleGetUp>
<isSupportPersonLocationDetection>
  <!--ro, opt, bool, whether it supports person location detection event, desc:eventType: PersonLocationDetection-->true
</isSupportPersonLocationDetection>
<isSupportPeopleFallingDownRecovery>
  <!--ro, opt, bool, whether it supports person fall restoring event, desc:eventType: PeopleFallingDownRecovery-->true
</isSupportPeopleFallingDownRecovery>
<isSupportPersonRetentionEvent>
  <!--ro, opt, bool, whether it supports people overstay event, desc:eventType: PersonRetentionEvent-->true
</isSupportPersonRetentionEvent>
<isSupportPersonRetentionRecoveryEvent>
  <!--ro, opt, bool, whether it supports people overstay restoring event, desc:eventType: PersonRetentionRecoveryEvent-->true
</isSupportPersonRetentionRecoveryEvent>
<isSupportPeopleLongTimeLeaveEvent>
  <!--ro, opt, bool, whether it supports person long-time off-bed event, desc:eventType: PeopleLongTimeLeaveEvent-->true
</isSupportPeopleLongTimeLeaveEvent>
<isSupportPeopleLongTimeLeaveRecoveryEvent>
  <!--ro, opt, bool, whether it supports person long-time off-bed restoring event, desc:eventType: PeopleLongTimeLeaveRecoveryEvent-->true
</isSupportPeopleLongTimeLeaveRecoveryEvent>
<isSupportPeopleVitalSignsAbnormalEvent>
  <!--ro, opt, bool, whether it supports vital sign exception event, desc:eventType: PeopleVitalSignsAbnormalEvent-->true
</isSupportPeopleVitalSignsAbnormalEvent>
<isSupportPeopleVitalSignsRecoveryEvent>
  <!--ro, opt, bool, whether it supports vital sign exception restoring event, desc:eventType: PeopleVitalSignsRecoveryEvent-->true
</isSupportPeopleVitalSignsRecoveryEvent>
<isSupportPeopleVitalSignsExportTaskEvent>
  <!--ro, opt, bool, whether it supports vital sign data exporting task result event, desc:eventType: PeopleVitalSignsExportTaskEvent-->true
</isSupportPeopleVitalSignsExportTaskEvent>
<isSupportTOFCalibrate>
  <!--ro, opt, bool, whether it supports configuring TOF detection calibration, desc:PUT /ISAPI/System/Video/inputs/tamperDetection/TOFCalibrate?
format=json-->true
</isSupportTOFCalibrate>
<isSupportSearchBulkUpgradeChildDeviceTask>
  <!--ro, opt, bool, whether it supports searching for tasks of batch upgrading sub-devices, desc:/ISAPI/System/SearchBulkUpgradeChildDeviceTask?
format=json-->true
</isSupportSearchBulkUpgradeChildDeviceTask>
<isSupportModifyBulkUpgradeChildDeviceTask>
  <!--ro, opt, bool, whether it supports modifying the task of batch upgrading sub-devices, desc:/ISAPI/System/ModifyBulkUpgradeChildDeviceTask?
format=json-->true
</isSupportModifyBulkUpgradeChildDeviceTask>
<isSupportDeleteBulkUpgradeChildDeviceTask>
  <!--ro, opt, bool, whether it supports deleting tasks of batch upgrading sub-devices, desc:/ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json--
>true
</isSupportDeleteBulkUpgradeChildDeviceTask>
<isSupportAddEventTemplate>
  <!--ro, opt, bool, whether it supports adding event message templates, desc:POST /ISAPI/Event/notification/httpHosts/AddEventTemplate?format=json-->true
</isSupportAddEventTemplate>
<isSupportTestEventTemplate>
  <!--ro, opt, bool, whether it supports testing event message templates, desc:POST /ISAPI/Event/notification/httpHosts/TestEventTemplate?format=json--
>true
</isSupportTestEventTemplate>
<isSupportEventTemplateParams>
  <!--ro, opt, bool, whether it supports event message template parameters, desc:GET /ISAPI/Event/notification/httpHosts/EventTemplateParams?format=json--
>true
</isSupportEventTemplateParams>
<isSupportAddCombineEvents>
  <!--ro, opt, bool, whether it supports adding combined event, desc:POST /ISAPI/Event/CombineEvents/AddCombineEvents?format=json-->true
</isSupportAddCombineEvents>
<isSupportModifyCombineEventsParams>
  <!--ro, opt, bool, whether it supports modifying combined event parameters, desc:POST /ISAPI/Event/CombineEvents/ModifyCombineEventsParams?format=json--
>true
</isSupportModifyCombineEventsParams>
<isSupportSearchCombineEventsParams>
  <!--ro, opt, bool, whether it supports searching for combined event parameters, desc:POST /ISAPI/Event/CombineEvents/SearchCombineEventsParams?
format=json-->true
</isSupportSearchCombineEventsParams>
<isSupportDeleteCombineEventsList>
  <!--ro, opt, bool, whether it supports batch deleting combined event, desc:POST /ISAPI/Event/CombineEvents/DeleteCombineEventsList?format=json-->true
</isSupportDeleteCombineEventsList>

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<isSupportEventRuleAndTriggerParams>
  <!--ro, opt, bool, whether it supports configuring the rule linkage parameters of a specified event, desc:POST
  /ISAPI/Event/EventBatchRulesMgr/EventRuleAndTriggerParams?format=json-->true
</isSupportEventRuleAndTriggerParams>
<isSupportSearchEventRuleAndTriggerParams>
  <!--ro, opt, bool, whether it supports searching for the rule linkage parameters of a specified event, desc:POST
  /ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndTriggerParams?format=json-->true
</isSupportSearchEventRuleAndTriggerParams>
<isSupportCallMessageHandlingEvent>
  <!--ro, opt, bool, whether it supports the call message handling event, desc:eventType: CallMessageHandlingEvent-->true
</isSupportCallMessageHandlingEvent>
<isSupportResetBatchChild>
  <!--ro, opt, bool, whether it supports batch restoring sub-devices, desc:PUT /ISAPI/System/ResetBatchChild?format=json-->true
</isSupportResetBatchChild>
<isSupportSearchResetBatchChildStatus>
  <!--ro, opt, bool, whether it supports searching for the status of batch restoring sub-devices, desc:POST /ISAPI/System/SearchResetBatchChildStatus?
  format=json-->true
</isSupportSearchResetBatchChildStatus>
<isSupportSearchRebootBatchChildStatus>
  <!--ro, opt, bool, whether it supports searching for the status of batch rebooting sub-devices, desc:POST /ISAPI/System/SearchRebootBatchChildStatus?
  format=json-->true
</isSupportSearchRebootBatchChildStatus>
<isSupportRebootBatchChild>
  <!--ro, opt, bool, whether it supports batch restarting sub-devices, desc:PUT /ISAPI/System/RebootBatchChild?format=json-->true
</isSupportRebootBatchChild>
<isSupportFlowStatisticsTelemeter>
  <!--ro, opt, bool, whether it supports traffic statistics telemetry event, desc:eventType:FlowStatisticsTelemeter-->true
</isSupportFlowStatisticsTelemeter>
<isSupportModifyFlowStatisticsTelemeterUpload>
  <!--ro, opt, bool, whether it supports modifying the parameters of traffic statistics telemetry report strategy, desc:PUT
  /ISAPI/System/Network/NetworkFlowMonitor/ModifyFlowStatisticsTelemeterUpload?format=json-->true
</isSupportModifyFlowStatisticsTelemeterUpload>
<isSupportClearFlowStatisticsTelemeter>
  <!--ro, opt, bool, whether it supports clearing the flow statistics telemetry data, desc:PUT
  /ISAPI/System/Network/NetworkFlowMonitor/ClearFlowStatisticsTelemeter?format=json-->true
</isSupportClearFlowStatisticsTelemeter>
<isSupportFlowStatisticsTelemeterUpload>
  <!--ro, opt, bool, whether it supports the remote telemetry reporting strategy parameters for traffic statistics, desc:GET
  /ISAPI/System/Network/NetworkFlowMonitor/FlowStatisticsTelemeterUpload?format=json-->true
</isSupportFlowStatisticsTelemeterUpload>
<isSupportScreenGraphicsColorControl>
  <!--ro, opt, bool, whether it supports screen image color control, desc:POST /ISAPI/System/ScreenColorQualityDetection/ScreenGraphicsColorControl?
  format=json-->true
</isSupportScreenGraphicsColorControl>
<isSupportSearchScreenColorQualityDetectionResult>
  <!--ro, opt, bool, whether it supports searching for the screen image color quality detection results, desc:POST
  /ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionResult?format=json-->true
</isSupportSearchScreenColorQualityDetectionResult>
<isSupportSearchScreenColorQualityDetectionProcess>
  <!--ro, opt, bool, whether it supports searching for the screen image color quality detection progress, desc:POST
  /ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionProcess?format=json-->true
</isSupportSearchScreenColorQualityDetectionProcess>
<isSupportScreenColorQualityDetectionParams>
  <!--ro, opt, bool, whether it supports the screen image color quality detection parameters, desc:GET/PUT
  /ISAPI/System/ScreenColorQualityDetection/channels/<channelID>/ScreenColorQualityDetectionParams?format=json-->true
</isSupportScreenColorQualityDetectionParams>
<isSupportAddInsulatorGroup>
  <!--ro, opt, bool, whether it supports adding insulator list, desc:GET /ISAPI/Thermal/channels/<channelID>/AddInsulatorGroup?format=json-->true
</isSupportAddInsulatorGroup>
<isSupportSuperResolutionReboot>
  <!--ro, opt, bool, whether the device supports rebooting super resolution, desc:whether the device supports restart to complete the conversion: true
  (yes), false (no). Different devices use different chips. Some devices convert videos/images to ones with super resolution effect when the super resolution
  function is enabled, while others require a device restart before the conversion is completed-->true
</isSupportSuperResolutionReboot>
<isSupportStartAutoTest>
  <!--ro, opt, bool, whether it supports starting automatic test, desc:POST /ISAPI/System/AutoTest/StartAutoTest?format=json-->true
</isSupportStartAutoTest>
<isSupportGetUserManualLink>
  <!--ro, opt, bool, whether it supports getting the device user manual link, desc:GET /ISAPI/System/GetUserManualLink?format=json-->true
</isSupportGetUserManualLink>
<isSupportExportSignaturePublicKey>
  <!--ro, opt, bool, whether it supports exporting the signed public key, desc:GET /ISAPI/Security/ProtocolCertificate/ExportSignaturePublicKey?
  format=json-->true
</isSupportExportSignaturePublicKey>
<isSupportCipherCardRunningStatus>
  <!--ro, opt, bool, whether it supports the password card running status, desc:GET /ISAPI/System/CipherCardRunningStatus?format=json-->true
</isSupportCipherCardRunningStatus>
<isSupportBridgeDeflectionMonitoringCapture>
  <!--ro, opt, bool, whether it supports manual triggering of bridge deflection monitoring and capturing, desc:POST
  /ISAPI/IntelLigent/channels/<channelID>/BridgeDeflectionMonitoringCapture?format=json-->true
</isSupportBridgeDeflectionMonitoringCapture>
<isSupportAdaptiveVideoStreamParams>
  <!--ro, opt, bool, whether it supports adaptive video stream control parameters, desc:GET/PUT /ISAPI/ContentMgmt/InputProxy/AdaptiveVideoStreamParams?
  format=json-->true
</isSupportAdaptiveVideoStreamParams>
<isSupportSearchAudioDetectionTaskProcess>
  <!--ro, opt, bool, whether it supports searching for the progress of audio detection tasks, desc:POST
  /ISAPI/System/DeviceAudioDetection/SearchAudioDetectionTaskProcess?format=json-->true
</isSupportSearchAudioDetectionTaskProcess>
<isSupportSendAudioDetectionTask>
  <!--ro, opt, bool, whether it supports applying audio detection tasks, desc:POST /ISAPI/System/DeviceAudioDetection/SendAudioDetectionTask?format=json--
  >true
</isSupportSendAudioDetectionTask>
<isSupportMedicalImageMRI FmiredEvent>

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<!--ro, opt, bool, whether it supports medical image URL expiration event, desc:eventType:MedicalImageURLExpiredEvent-->true
</isSupportMedicalImageURLExpiredEvent>
<isSupportBridgeDeflectionMonitoringEvent>
  <!--ro, opt, bool, whether it supports bridge deflection monitoring event, desc:eventType:BridgeDeflectionMonitoringEvent-->true
</isSupportBridgeDeflectionMonitoringEvent>
<isSupportBridgeDeflectionTargetLossEvent>
  <!--ro, opt, bool, whether it supports bridge deflection target loss event, desc:eventType:BridgeDeflectionTargetLossEvent-->true
</isSupportBridgeDeflectionTargetLossEvent>
<isSupportNobodyDetectionRuleParams>
  <!--ro, opt, bool, whether it supports unmanned detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/NobodyDetectionRuleParams?
format=json-->true
</isSupportNobodyDetectionRuleParams>
<isSupportNobodyDetectionRegionShieldParams>
  <!--ro, opt, bool, whether it supports the shielding parameters for unmanned detection areas, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/NobodyDetectionRegionShieldParams?format=json-->true
</isSupportNobodyDetectionRegionShieldParams>
<isSupportReduceTemperatureFilterParams>
  <!--ro, opt, bool, whether it supports the cooling filter parameter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ReduceTemperatureFilterParams?
format=json-->true
</isSupportReduceTemperatureFilterParams>
<isSupportGetStreamOverlayLogDebugStatus>
  <!--ro, opt, bool, whether it supports getting the status of overlaying log debugging status on stream, desc:GET
/ISAPI/ContentMgmt/channels/<channelID>/GetStreamOverlayLogDebugStatus?format=json-->true
</isSupportGetStreamOverlayLogDebugStatus>
<isSupportStreamOverlayLogDebugParams>
  <!--ro, opt, bool, whether it supports the parameters of overlaying log debugging information on stream, desc:GET/PUT
/ISAPI/ContentMgmt/channels/<channelID>/StreamOverlayLogDebugParams?format=json-->true
</isSupportStreamOverlayLogDebugParams>
<isSupportBridgeDeflectionAbnormalRuleParams>
  <!--ro, opt, bool, whether it supports the bridge deflection exception detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionAbnormalRuleParams?format=json-->true
</isSupportBridgeDeflectionAbnormalRuleParams>
<isSupportBridgeDeflectionAbnormalEvent>
  <!--ro, opt, bool, whether it supports bridge deflection exception event, desc:eventType:BridgeDeflectionAbnormalEvent-->true
</isSupportBridgeDeflectionAbnormalEvent>
<isSupportRestoreBackup>
  <!--ro, opt, bool, whether it supports data restoring and backup, desc:POST /ISAPI/System/SystemSnapshot/RestoreBackup?format=json-->true
</isSupportRestoreBackup>
<isSupportDeleteBackupFile>
  <!--ro, opt, bool, whether it supports deleting backup files, desc:POST /ISAPI/System/SystemSnapshot/DeleteBackupFile?format=json-->true
</isSupportDeleteBackupFile>
<isSupportSearchRestoreBackupProgress>
  <!--ro, opt, bool, whether it supports searching for the progress of restoring backup files, desc:POST
/ISAPI/System/SystemSnapshot/SearchRestoreBackupProgress?format=json-->true
</isSupportSearchRestoreBackupProgress>
<isSupportGetBackupsFileList>
  <!--ro, opt, bool, whether it supports getting the list of backup files, desc:POST /ISAPI/System/SystemSnapshot/GetBackupsFileList?format=json-->true
</isSupportGetBackupsFileList>
<isSupportAutoBackupParams>
  <!--ro, opt, bool, whether it supports automatic backup parameters, desc:GET/PUT /ISAPI/System/SystemSnapshot/AutoBackupParams?format=json-->true
</isSupportAutoBackupParams>
<isSupportDeviceSoftwareRunningProgramRestart>
  <!--ro, opt, bool, whether it supports restarting the device software program, desc:POST /ISAPI/System/DeviceSoftwareRunningProgramRestart?format=json--
>true
</isSupportDeviceSoftwareRunningProgramRestart>
<isSupportSendScreenColorQualityDetectionRequest>
  <!--ro, opt, bool, whether it supports sending a screen color quality detection request, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SendScreenColorQualityDetectionRequest?format=json-->true
</isSupportSendScreenColorQualityDetectionRequest>
<isSupportTelephoningDetectionRuleParams>
  <!--ro, opt, bool, whether it supports phone call detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/telephoningDetect/TelephoningDetectionRuleParams?format=json-->true
</isSupportTelephoningDetectionRuleParams>
<isSupportTelephoningBehaviorAlarmEvent>
  <!--ro, opt, bool, whether it supports phone call alarm event, desc:eventType:TelephoningBehaviorAlarmEvent-->true
</isSupportTelephoningBehaviorAlarmEvent>
<isSupportTelephoningBehaviorRecoveryEvent>
  <!--ro, opt, bool, whether it supports phone call restoring event, desc:eventType:TelephoningBehaviorRecoveryEvent-->true
</isSupportTelephoningBehaviorRecoveryEvent>
<isSupportSleepDutyRecoveryEvent>
  <!--ro, opt, bool, whether the device supports absence or sleeping on duty restoring event, desc:eventType:SleepDutyRecoveryEvent-->true
</isSupportSleepDutyRecoveryEvent>
<isSupportGetBridgeDeflectionTargetOffsetResetStatus>
  <!--ro, opt, bool, whether it supports getting the zeroing result of bridge deflection target displacement, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetOffsetResetStatus?format=json-->true
</isSupportGetBridgeDeflectionTargetOffsetResetStatus>
<isSupportBridgeDeflectionTargetOffsetReset>
  <!--ro, opt, bool, whether it supports bridge deflection target displacement reset, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetOffsetReset?format=json-->true
</isSupportBridgeDeflectionTargetOffsetReset>
<isSupportBridgeDeflectionTargetLossRuleParams>
  <!--ro, opt, bool, whether it supports bridge deflection target loss rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetLossRuleParams?format=json-->true
</isSupportBridgeDeflectionTargetLossRuleParams>
<isSupportGetAnalogOutputParams>
  <!--ro, opt, bool, whether it supports getting all analog output parameters, desc:GET /ISAPI/System/AnalogOutput/GetAnalogOutputParams?format=json--
>true
</isSupportGetAnalogOutputParams>
<isSupportSystemReset>
  <!--ro, opt, bool, whether the device supports restoring system to default settings, desc:POST /ISAPI/System/SystemReset?format=json-->true
</isSupportSystemReset>
<isSupportFakePlateDetection>
  <!--ro, opt, bool, whether it supports fake license plate detection event, desc:corresponding event: FakePlateDetection-->true

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</isSupportFakePlateDetection>
<isSupportSafetyWorkwearDetectionEvent>
  <!--ro, opt, bool, whether it supports the event of wearing safety work clothes detection, desc:corresponding event: SafetyWorkwearDetectionEvent-->true
</isSupportSafetyWorkwearDetectionEvent>
<isSupportVehicleMaintenanceStatusEvent>
  <!--ro, opt, bool, whether it supports vehicle maintenance status events, desc:corresponding event: VehicleMaintenanceStatusEvent-->true
</isSupportVehicleMaintenanceStatusEvent>
<isSupportEventRetransmissionInfo>
  <!--ro, opt, bool, whether it supports reloading event information, desc:GET /ISAPI/System/EventRetransmissionInfo?format=json-->true
</isSupportEventRetransmissionInfo>
<isSupportGetEnvironmentCaptureStatus>
  <!--ro, opt, bool, whether it supports getting the environmental data collection status, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetEnvironmentCaptureStatus?format=json-->true
</isSupportGetEnvironmentCaptureStatus>
<isSupportEnvironmentCapture>
  <!--ro, opt, bool, whether it supports environmental data collection, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/EnvironmentCapture?format=json-->true
</isSupportEnvironmentCapture>
<isSupportSearchCPUDetailInfo>
  <!--ro, opt, bool, whether it supports searching for detailed CPU performance information, desc:POST /ISAPI/System/SearchCPUDetailInfo?format=json--
>true
</isSupportSearchCPUDetailInfo>
<isSupportPanelControlParams>
  <!--ro, opt, bool, whether it supports device hardware panel configuration parameters, desc:GET/PUT /ISAPI/System/PanelControlParams?format=json-->true
</isSupportPanelControlParams>
<isSupportSearchGoodsIncAndDecStatistics>
  <!--ro, opt, bool, whether it supports searching for cargo increase/decrease statistics, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/SearchGoodsIncAndDecStatistics?format=json-->true
</isSupportSearchGoodsIncAndDecStatistics>
<isSupportCustomAlarmHTTPParams>
  <!--ro, opt, bool, whether it supports custom alarm HTTP parameters, desc:GET/PUT /ISAPI/Event/notification/httpHosts/CustomAlarmHTTPParams?format=json-
>true
</isSupportCustomAlarmHTTPParams>
<isSupportTestCustomAlarmHTTP>
  <!--ro, opt, bool, whether it supports custom alarm HTTP request test, desc:PUT /ISAPI/Event/notification/httpHosts/TestCustomAlarmHTTP?format=json--
>true
</isSupportTestCustomAlarmHTTP>
<isSupportCustomAlarmHTTPLinkageParams>
  <!--ro, opt, bool, whether it supports customizing the alarm HTTP linkage parameters, desc:GET/PUT /ISAPI/Thermal/CustomAlarmHTTPLinkageParams?
format=json-->true
</isSupportCustomAlarmHTTPLinkageParams>
<isSupportAccumulationDetectionPeriodicReport>
  <!--ro, opt, bool, whether it supports scheduled reporting of pile detection events, desc:corresponding event: AccumulationDetectionPeriodicReport--
>true
</isSupportAccumulationDetectionPeriodicReport>
<isSupportImportDiskCompatibilityCertificate>
  <!--ro, opt, bool, whether it supports importing disk compatibility certificate, desc:PUT /ISAPI/Security/serverCertificate/certificate?type=
<type>&platformNo=<platformNo>, and type is DiskCompatibility-->true
</isSupportImportDiskCompatibilityCertificate>
<isSupportGetAccumulationDetectionAlgorithmicVersion>
  <!--ro, opt, bool, whether it supports getting the version of the stacking detection algorithm, desc:GET
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetAccumulationDetectionAlgorithmicVersion?format=json-->true
</isSupportGetAccumulationDetectionAlgorithmicVersion>
<isSupportSnapshotUpgradeDifferenceFile>
  <!--ro, opt, bool, whether it supports getting snapshot upgrade difference files, desc:GET /ISAPI/System/SystemSnapshot/SnapshotUpgradeDifferenceFile?
format=json-->true
</isSupportSnapshotUpgradeDifferenceFile>
<isSupportGroupRecoveryEvent>
  <!--ro, opt, bool, whether it supports people gathering restoring event-->true
</isSupportGroupRecoveryEvent>
<isSupportElectronicEyepieceParams>
  <!--ro, opt, bool, whether it supports electron microscope channel parameters, desc:GET/PUT /ISAPI/System/Video/ElectronicEyepieceParams?format=json--
>true
</isSupportElectronicEyepieceParams>
<isSupportSmokingDetectionRecoverEvent>
  <!--ro, opt, bool, whether it supports smoking detection restoring event, desc:fire alarm upload in progress, eventType: SmokingDetectionRecoverEvent--
>true
</isSupportSmokingDetectionRecoverEvent>
<isSupportDeviceRegionParams>
  <!--ro, opt, bool, whether it supports device area parameters, desc:GET/PUT /ISAPI/System/DeviceRegionParams?format=json-->true
</isSupportDeviceRegionParams>
<isSupportTemperatureEvent>
  <!--ro, opt, bool, whether it supports temperature alarm event, desc:TemperatureEvent-->true
</isSupportTemperatureEvent>
<isSupportTemperatureRecoverEvent>
  <!--ro, opt, bool, whether it supports temperature restoring event, desc:TemperatureRecoverEvent-->true
</isSupportTemperatureRecoverEvent>
<isSupportWebSSH>
  <!--ro, opt, bool, whether it supports WebSSH, desc:if the node is not returned, it indicates no-->true
</isSupportWebSSH>
<isSupportCreateDebugChannel>
  <!--ro, opt, bool, whether it supports creating WebSSH debugging channel, desc:if the node is not returned, it indicates no-->true
</isSupportCreateDebugChannel>
<isSupportExportDebugLog>
  <!--ro, opt, bool, whether it supports exporting the debugging log, desc:if the node is not returned, it indicates no-->true
</isSupportExportDebugLog>
<isSupportAccumulationDetectionEffectiveAreaCfgParams>
  <!--ro, opt, bool, whether it supports the configuration parameters of pile detection effective area, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/AccumulationDetectionEffectiveAreaCfgParams?format=json-->true
</isSupportAccumulationDetectionEffectiveAreaCfgParams>
<isSupportGetGoalDryWetBallTempCurve>
  <!--ro, opt, bool, whether it supports getting the target dry-wet bulb temperature curve, desc:POST
/ISAPI/System/TobaccoBakeMgr/GetGoalDryWetBallTempCurve?format=json-->true
</isSupportGetGoalDryWetBallTempCurve>

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</isSupportWetBulbTempCurve>
<isSupportBakeTobaccoRuleParams>
  <!--ro, opt, bool, whether it supports tobacco baking rule parameters, desc:GET/PUT /ISAPI/System/TobaccoBakeMgr/BakeTobaccoRuleParams?format=json-->true
>true
</isSupportBakeTobaccoRuleParams>
<isSupportStartBakeTobacco>
  <!--ro, opt, bool, whether it supports starting baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/StartBakeTobacco?format=json-->true
</isSupportStartBakeTobacco>
<isSupportEndBakeTobacco>
  <!--ro, opt, bool, whether it supports stopping baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/EndBakeTobacco?format=json-->true
</isSupportEndBakeTobacco>
<isSupportDryWetBallTempEventRuleParams>
  <!--ro, opt, bool, whether it supports the dry-wet bulb temperature reporting rule parameters, desc:GET/PUT
/ISAPI/System/TobaccoBakeMgr/DryWetBallTempEventRuleParams?format=json-->true
</isSupportDryWetBallTempEventRuleParams>
<isSupportDevAccessServerNetworkStatus>
  <!--ro, opt, bool, whether it supports device connection status with multi-access service network, desc:GET /ISAPI/System/DevAccessServerNetworkStatus?
format=json-->true
</isSupportDevAccessServerNetworkStatus>
<isSupportMaintenanceServiceAuthorization>
  <!--ro, opt, bool, whether it supports device maintenance service authorization information, desc:GET/PUT /ISAPI/System/MaintenanceServiceAuthorization?
format=json-->true
</isSupportMaintenanceServiceAuthorization>
<isSupportUploadIndividualCallSignalOperation>
  <!--ro, opt, bool, whether it supports uploading single-mode call signaling events, desc:UploadIndividualCallSignalOperation-->true
</isSupportUploadIndividualCallSignalOperation>
<isSupportUploadGroupCallSignalOperation>
  <!--ro, opt, bool, whether it supports uploading group call signaling event, desc:UploadGroupCallSignalOperation-->true
</isSupportUploadGroupCallSignalOperation>
<isSupportDrivingBehaviorVoiceReminder>
  <!--ro, opt, bool, whether it supports configuring intelligent voice reminders for driving behavior, desc:GET/PUT
/ISAPI/Event/DrivingBehaviorVoiceReminder?format=json-->true
</isSupportDrivingBehaviorVoiceReminder>
<isSupportImportMultiCameraCalibrationFile>
  <!--ro, opt, bool, whether it supports importing image calibration files of multiple cameras, desc:POST
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/ImportMultiCameraCalibrationFile?format=json-->true
</isSupportImportMultiCameraCalibrationFile>
<isSupportDeletePlayerFileList>
  <!--ro, opt, bool, whether it supports batch deleting player files, desc:POST /ISAPI/System/Video/DeletePlayerFileList?format=json-->true
</isSupportDeletePlayerFileList>
<isSupportPlayerInfoParams>
  <!--ro, opt, bool, whether it supports player information parameters, desc:GET /ISAPI/System/Video/PlayerInfoParams?format=json-->true
</isSupportPlayerInfoParams>
<isSupportBurnPlayerParams>
  <!--ro, opt, bool, whether it supports burning player parameters, desc:GET/PUT /ISAPI/System/Video/BurnPlayerParams?format=json-->true
</isSupportBurnPlayerParams>
<isSupportOneKeyVoiceAbnormalDetection>
  <!--ro, opt, bool, whether it supports quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/OneKeyVoiceAbnormalDetection?format=json-->true
</isSupportOneKeyVoiceAbnormalDetection>
<isSupportVoiceAbnormalDetectionRuleParams>
  <!--ro, opt, bool, whether it supports quick voice exception detection parameters, desc:GET/PUT
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/VoiceAbnormalDetectionRuleParams?format=json-->true
</isSupportVoiceAbnormalDetectionRuleParams>
<isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
  <!--ro, opt, bool, whether it supports searching for the progress of quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/SearchOneKeyVoiceAbnormalDetectionProcess?format=json-->true
</isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
<isSupportSearchOneKeyAudioDetection>
  <!--ro, opt, bool, whether it supports searching for the progress of quick audio detection, desc:POST
/ISAPI/Smart/AudioExceptionDetection/SearchOneKeyAudioDetection?format=json-->true
</isSupportSearchOneKeyAudioDetection>
<isSupportOneKeyAudioDetection>
  <!--ro, opt, bool, whether it supports quick audio detection, desc:POST /ISAPI/Smart/AudioExceptionDetection/channels/<channelID>/OneKeyAudioDetection?
format=json-->true
</isSupportOneKeyAudioDetection>
<isNotSupportScheduledRecordUpload>
  <!--ro, opt, bool, whether it supports the event of scheduled recording of channel, desc:if this node does not exist or returns false (default value),
it indicates for support; if it returns true, it indicates no-->>false
</isNotSupportScheduledRecordUpload>
<isSupportDevEnvironmentalCtrlMaterielInfoDetect>
  <!--ro, opt, bool, whether it supports the detection of equipment environmental control material information?, desc:POST
/ISAPI/System/DeviceModuleDiagnosis/DevEnvironmentalCtrlMaterielInfoDetect?format=json-->true
</isSupportDevEnvironmentalCtrlMaterielInfoDetect>
<isSupportWaterRescueDetectionRuleParams>
  <!--ro, opt, bool, whether it supports water rescue detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/WaterRescueDetectionRuleParams?format=json-->true
</isSupportWaterRescueDetectionRuleParams>
<isSupportWaterRescueDetectionEvent>
  <!--ro, opt, bool, whether it supports water rescue detection event, desc:corresponding event: WaterRescueDetectionEvent-->true
</isSupportWaterRescueDetectionEvent>
<isSupportGetActualDryWetBallTempCurve>
  <!--ro, opt, bool, whether it supports getting the actual wet-bulb temperature curve, desc:POST
/ISAPI/System/TobaccoBakeMgr/GetActualDryWetBallTempCurve?format=json-->true
</isSupportGetActualDryWetBallTempCurve>
<isSupportPerimeterAlgoTargetFilterParams>
  <!--ro, opt, bool, whether it supports target filtering parameters of perimeter algorithm, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/PerimeterAlgoTargetFilterParams?format=json-->true
</isSupportPerimeterAlgoTargetFilterParams>
<isSupportSendDeviceMaintenanceStatus>
  <!--ro, opt, bool, whether it supports applying device maintenance status, desc:PUT /ISAPI/System/SendDeviceMaintenanceStatus?format=json-->true
</isSupportSendDeviceMaintenanceStatus>
<videoEncodeChannel opt="1,2">
  <!--ro, opt, string, channel supporting video parameter display on the web, attr:opt{req, string}, desc:N/A-->1,2

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</videoEncodeChannel>
<isSupportLocalButtonRecordControl>
  <!--ro, opt, bool, whether it supports controlling local recording button on the device, desc:/ISAPI/System/LocalButtonRecord?format=json-->true
</isSupportLocalButtonRecordControl>
<isSupportLocalButtonSnapshotControl>
  <!--ro, opt, bool, whether it supports controlling local capture button on the device, desc:/ISAPI/System/LocalButtonSnapshot?format=json-->true
</isSupportLocalButtonSnapshotControl>
<isSupportSecurityDisclaimerParams>
  <!--ro, opt, bool, whether it supports the security disclaimer parameters, desc:GET/PUT /ISAPI/System/InformedConsentMgr/SecurityDisclaimerParams?format=json-->true
</isSupportSecurityDisclaimerParams>
<isSupportThermometryGatewayParams>
  <!--ro, opt, bool, whether it supports configuring parameters of the temperature measurement gateway, desc:GET/PUT /ISAPI/Thermal/ThermometryGatewayParams?format=json-->true
</isSupportThermometryGatewayParams>
<isSupportSearchRegionDefaultInfo>
  <!--ro, opt, bool, whether it supports searching for the default parameters for the region, desc:POST /ISAPI/System/SearchRegionDefaultInfo?format=json-->true
</isSupportSearchRegionDefaultInfo>
<isSupportAntiCollisionWarningEvent>
  <!--ro, opt, bool, whether it supports anti-collision pre-alarm event, desc:AntiCollisionWarningEvent-->true
</isSupportAntiCollisionWarningEvent>
<isSupportMovementWorkStatus>
  <!--ro, opt, bool, whether it supports gas cloud module working status, desc:GET /ISAPI/System/MovementMgr/MovementWorkStatus?format=json-->true
</isSupportMovementWorkStatus>
<isSupportMovementTimingPowerOffParams>
  <!--ro, opt, bool, whether it supports scheduled shutdown parameters of the gas cloud imaging module, desc:GET/PUT /ISAPI/System/MovementMgr/MovementTimingPowerOffParams?format=json-->true
</isSupportMovementTimingPowerOffParams>
<isSupportFireExtinguisherDetectionRuleParams>
  <!--ro, opt, bool, whether it supports fire extinguisher detection rule parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/FireExtinguisherDetectionRuleParams?format=json-->true
</isSupportFireExtinguisherDetectionRuleParams>
<isSupportFireExtinguisherRecoveryEvent>
  <!--ro, opt, bool, whether it supports fire extinguisher detection restoring event, desc:eventType:FireExtinguisherRecoveryEvent-->true
</isSupportFireExtinguisherRecoveryEvent>
<isSupportGasTankMonitoringRecoveryEvent>
  <!--ro, opt, bool, whether it supports gas tank detection restoring event, desc:eventType:GasTankMonitoringRecoveryEvent-->true
</isSupportGasTankMonitoringRecoveryEvent>
<isSupportSearchImportPlayerFileProcess>
  <!--ro, opt, bool, whether it supports searching for the progress of importing player files, desc:POST /ISAPI/System/Video/SearchImportPlayerFileProcess?format=json-->true
</isSupportSearchImportPlayerFileProcess>
<isSupportStatisticalDataFlashStorageParams>
  <!--ro, opt, bool, whether it supports data statistics for Flash storage parameters, desc:GET/PUT /ISAPI/System/dataAware/StatisticalDataFlashStorageParams?format=json-->true
</isSupportStatisticalDataFlashStorageParams>
<eventRecheckCap>
  <!--ro, opt, object, event review capability, desc:Low-end cameras depend on powerful NVRs, due to the limitation of computing power for secondary review of Low-end camera alarms to reduce false alarms-->
  <algorithmInfoList>
    <!--ro, opt, array, List of event detection algorithm information, subType:object, range:[1,5]-->
    <algorithmInfoItem>
      <!--ro, req, object, event detection algorithm element-->
      <algorithmType>
        <!--ro, req, enum, event detection algorithm type, subType:string, desc:"VMD", "regionEntrance", "regionExiting", "LineDetection" (Line crossing), "fieldDetection" (intrusion detection)-->VMD
      </algorithmType>
      <algorithmVersion>
        <!--ro, req, string, event detection algorithm version, desc:N/A-->1B
      </algorithmVersion>
    </algorithmInfoItem>
  </algorithmInfoList>
</eventRecheckCap>
<isSupportModifyWindowGroup>
  <!--ro, opt, bool, whether it supports modifying the specified window group of the scope, desc:PUT /ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/ModifyWindowGroup?format=json-->true
</isSupportModifyWindowGroup>
<isSupportCalibrateWindowGroup>
  <!--ro, opt, bool, whether it supports calibrating specified window group of the scope, desc:PUT /ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/CalibrateWindowGroup?format=json-->true
</isSupportCalibrateWindowGroup>
<isSupportAllWindowParams>
  <!--ro, opt, bool, whether it supports all scope window parameters, desc:GET /ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/AllWindowParams?format=json-->true
</isSupportAllWindowParams>
<isSupportWindowCalibrationStatus>
  <!--ro, opt, bool, whether it supports getting the scope window calibration status, desc:GET /ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowCalibrationStatus?format=json-->true
</isSupportWindowCalibrationStatus>
<isSupportOBUModuleInfo>
  <!--ro, opt, bool, whether it supports OBU module information, desc:GET /ISAPI/Traffic/ETC/OBUModuleInfo?format=json-->true
</isSupportOBUModuleInfo>
<isSupportExportOBUDetectionLogFile>
  <!--ro, opt, bool, whether it supports exporting the OBU Label detection module Logs, desc:POST /ISAPI/Traffic/ETC/ExportOBUDetectionLogFile?format=json-->true
</isSupportExportOBUDetectionLogFile>
<isSupportOBUDetectionReboot>
  <!--ro, opt, bool, whether it supports restarting OBU Label detection module, desc:PUT /ISAPI/Traffic/ETC/OBUDetectionReboot?format=json-->true
</isSupportOBUDetectionReboot>
<isSupportGetOBUREbootStatus>
  <!--ro, opt, bool, whether it supports getting the restart status of the OBU module, desc:GET /ISAPI/Traffic/ETC/GetOBUREbootStatus?format=json-->true
</isSupportGetOBUREbootStatus>

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<isSupportSearchOBUSStatisticsReport>
  <!--ro, opt, bool, whether it supports searching for the statistical data of OBU tag detection, desc:POST /ISAPI/Traffic/ETC/SearchOBUSStatisticsReport?
format=json-->true
</isSupportSearchOBUSStatisticsReport>
<isSupportSearchTargetEnhancementInfo>
  <!--ro, opt, bool, whether it supports searching for target prominence parameters, desc:POST
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/SearchTargetEnhancementInfo?format=json-->true
</isSupportSearchTargetEnhancementInfo>
<isSupportTargetEnhancementParams>
  <!--ro, opt, bool, whether it supports target prominence parameters, desc:GET/PUT
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/TargetEnhancementParams?format=json-->true
</isSupportTargetEnhancementParams>
<isSupportPicSearchByCoordinate>
  <!--ro, opt, bool, whether the device supports configuring AcuSearch parameters, desc:GET/PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json--
>true
</isSupportPicSearchByCoordinate>
<AcuSearchCap>
  <!--ro, opt, object, AcuSearch capability-->
  <targetType opt="face,human,vehicle">
    <!--ro, opt, enum, supported target types of AcuSearch, subType:string, attr:opt{req, string}, desc:"face", "human", "vehicle"-->face
  </targetType>
  <dataType opt="url,binary,modelData">
    <!--ro, opt, enum, supported image transmission format types, subType:string, attr:opt{req, string}, desc:"url", "binary", "modelData"-->url
  </dataType>
  <maxPicSize>
    <!--ro, opt, int, maximum size of image can be searched, range:[0,5], desc:unit: MB-->1
  </maxPicSize>
  <maxPicWidthSize>
    <!--ro, opt, int, maximum width of image can be searched, range:[0,4096]-->4096
  </maxPicWidthSize>
  <maxPicHeightSize>
    <!--ro, opt, int, maximum height of image can be searched, range:[0,4096]-->4096
  </maxPicHeightSize>
</AcuSearchCap>
<isSupportSendDevRemoteGUICtrl>
  <!--ro, opt, bool, whether it supports applying parameter settings of remote GUI control of devices, desc:POST
/ISAPI/System/RemoteCtrlDevGUI/SendDevRemoteGUICtrl?format=json-->true
</isSupportSendDevRemoteGUICtrl>
<isSupportStreamTemperatureParams>
  <!--ro, opt, bool, whether the device supports configuring stream temperature parameters of thermal imaging, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/StreamTemperatureParams?format=json-->true
</isSupportStreamTemperatureParams>
<moduleTypeSupportCap>
  <!--ro, opt, object, capability supported by device module-->
<isSupportMaintenanceModule>
  <!--ro, opt, bool, whether it supports maintenance module, desc:the wire products are embedded with the business module (businessModule) and
maintenance module (maintenanceModule), which supports independent access to EZVIZ cloud, etc.; at the same time, they support the cloud platform to send
maintenance instructions to the maintenance module-->true
</isSupportMaintenanceModule>
<moduleTypeSupportCap>
<isSupportSearchEventResources>
  <!--ro, opt, bool, whether it supports searching for event resource information, desc:POST /ISAPI/Event/EventBatchRulesMgr/SearchEventResources?
format=json-->true
</isSupportSearchEventResources>
<isSupportRestorePerimeterAlgoDefaultParam>
  <!--ro, opt, bool, whether it supports restoring parameters of perimeter algorithm, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/RestorePerimeterAlgoDefaultParam?format=json-->true
</isSupportRestorePerimeterAlgoDefaultParam>
<isSupportGetDeviceMaintenanceStatus>
  <!--ro, opt, bool, whether it supports getting device maintenance status, desc:GET /ISAPI/System/GetDeviceMaintenanceStatus?format=json-->true
</isSupportGetDeviceMaintenanceStatus>
<isSupportImportConfigurationFileByModel>
  <!--ro, opt, bool, whether it supports importing configuration files by device model, desc:POST /ISAPI/System/ImportConfigurationFileByModel?
format=json-->true
</isSupportImportConfigurationFileByModel>
<isSupportRadarVideoDetectionStorageParams>
  <!--ro, opt, bool, whether it supports the parameters of saving radar-assisted target detection result, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/RadarVideoDetectionStorageParams?format=json-->true
</isSupportRadarVideoDetectionStorageParams>
<isSupportBodyCameraMediaInfo>
  <!--ro, opt, bool, whether it supports body camera reporting media information, desc:eventType: BodyCameraMediaInfo-->true
</isSupportBodyCameraMediaInfo>
<isSupportRadarVideoPOSOVerlayParams>
  <!--ro, opt, bool, whether it supports smart radar-assisted POS information overlay parameters, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/RadarVideoPOSOVerlayParams?format=json-->true
</isSupportRadarVideoPOSOVerlayParams>
<isSupportDeleteNetworkCaptureFile>
  <!--ro, opt, bool, whether it supports deleting captured packet files, desc:POST /ISAPI/System/DeleteNetworkCaptureFile?format=json-->true
</isSupportDeleteNetworkCaptureFile>
<isSupportRecoverCustomEventTemplate>
  <!--ro, opt, bool, whether it supports restoring a custom event message template, desc:POST
/ISAPI/Event/notification/httpHosts/RecoverCustomEventTemplate?format=json-->true
</isSupportRecoverCustomEventTemplate>
<isSupportModCustomEventTemplateParams>
  <!--ro, opt, bool, whether it supports modifying a specified template of custom event message, desc:PUT
/ISAPI/Event/notification/httpHosts/ModCustomEventTemplateParams?format=json-->true
</isSupportModCustomEventTemplateParams>
<isSupportSearchAbnormalStreamInfo>
  <!--ro, opt, bool, whether it supports searching for streaming exception information, desc:POST /ISAPI/Streaming/SearchAbnormalStreamInfo?format=json--
>true
</isSupportSearchAbnormalStreamInfo>
<isSupportAbnormalStreamEvent>
  <!--ro, opt, bool, whether it supports real-time streaming exception report-->true
</isSupportAbnormalStreamEvent>

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<!--ro, opt, bool, whether it supports batch deleting captured packet files, desc:PUT /ISAPI/System/BatchDeleteNetworkCaptureFile?format=json-->true
</isSupportBatchDeleteNetworkCaptureFile>
<isSupportWindowRuleParams>
  <!--ro, opt, bool, whether it supports the scope window rule parameters, desc:GET/PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowRuleParams?format=json-->true
  </isSupportWindowRuleParams>
<isSupportGetMultiRemoteAlbumPicture>
  <!--ro, opt, bool, whether the device supports getting pictures from the remote album under specified path, desc:POST
/ISAPI/Thermal/GetMultiRemoteAlbumPicture?format=json-->true
  </isSupportGetMultiRemoteAlbumPicture>
<isSupportGetDeviceTokenInfo>
  <!--ro, opt, bool, whether the device supports getting the device token information, desc:/ISAPI/System/ImportConfigurationFileByURL?format=json
/ISAPI/System/DeviceAuth/GetDeviceTokenInfo?format=json-->true
  </isSupportGetDeviceTokenInfo>
<isSupportDisplayEnableCloudServicePrompt>
  <!--ro, opt, bool, whether it supports displaying the prompt for enabling cloud services, desc:false by default. When the device returns true, it
indicates that displaying a prompt of enabling cloud services is supported. Scenario example: After your device is connected to the Internet and the cloud
service is enabled, we will provide you with better cloud device management and maintenance upgrade services. Therefore, we will collect relevant device
information such as the device's IP address, device name, and device serial No. If you have any questions, you can contact customer service at any time--
>true
  </isSupportDisplayEnableCloudServicePrompt>
<isSupportFiredoorDetectionEvent>
  <!--ro, opt, bool, whether it supports fireproof door detection event, desc:eventType:FiredoorDetectionEvent-->true
  </isSupportFiredoorDetectionEvent>
<isSupportImportConfigurationFileByURL>
  <!--ro, opt, bool, whether it supports importing configuration file via URL, desc:POST /ISAPI/System/ImportConfigurationFileByURL?format=json-->true
  </isSupportImportConfigurationFileByURL>
<isSupportPlayCellphoneRecoveryEvent>
  <!--ro, opt, bool, whether it supports playing mobile restoring event, desc:eventType:PlayCellphoneRecoveryEvent-->true
  </isSupportPlayCellphoneRecoveryEvent>
<isSupportDetectorShelterAlarmRecovery>
  <!--ro, opt, bool, whether it supports detector occlusion detection restoring event, desc:eventType:DetectorShelterAlarmRecovery-->true
  </isSupportDetectorShelterAlarmRecovery>
<isSupportExportThermalMovementFile>
  <!--ro, opt, bool, whether it supports exporting the device temperature measuring module template, desc:POST /ISAPI/Thermal/ExportThermalMovementFile?
format=json-->true
  </isSupportExportThermalMovementFile>
<isSupportUPSAbnormalAutoShutDownParams>
  <!--ro, opt, bool, whether it supports parameters of UPS power supply exception and auto power off, desc:GET/PUT
/ISAPI/System/UPSAbnormalAutoShutDownParams?format=json-->true
  </isSupportUPSAbnormalAutoShutDownParams>
<isSupportThermalLensDriveParams>
  <!--ro, opt, bool, whether it supports configuring thermal lens driver parameters, desc:GET/PUT /ISAPI/Thermal/ThermalLensDriveParams?format=json-->true
  </isSupportThermalLensDriveParams>
<isSupportBatchDeleteUserInfoCfg>
  <!--ro, opt, bool, whether it supports batch deleting user information configurations, desc:PUT /ISAPI/Security/users/BatchDeleteUserInfoCfg?
format=json-->true
  </isSupportBatchDeleteUserInfoCfg>
<isSupportAddUserGroup>
  <!--ro, opt, bool, whether it supports adding user groups, desc:POST /ISAPI/System/UserGroupingMgr/AddUserGroup?format=json-->true
  </isSupportAddUserGroup>
<isSupportModifyUserGroup>
  <!--ro, opt, bool, whether it supports editing user groups, desc:PUT /ISAPI/System/UserGroupingMgr/ModifyUserGroup?format=json-->true
  </isSupportModifyUserGroup>
<isSupportSearchUserGroup>
  <!--ro, opt, bool, whether it supports searching user groups, desc:POST /ISAPI/System/UserGroupingMgr/SearchUserGroup?format=json-->true
  </isSupportSearchUserGroup>
<isSupportDeleteUserGroup>
  <!--ro, opt, bool, whether it supports deleting user groups, desc:PUT /ISAPI/System/UserGroupingMgr/DeleteUserGroup?format=json-->true
  </isSupportDeleteUserGroup>
<isSupportReadUpgradePackageData>
  <!--ro, opt, bool, whether it supports notifying the device to read upgrade package data, desc:PUT /ISAPI/System/SystemUpgrade/ReadUpgradePackageData?
format=json-->true
  </isSupportReadUpgradePackageData>
<isSupportWebDisplayThermometryDistanceTips>
  <!--ro, opt, bool, whether it supports displaying the prompt of temperature measurement distance on the web page, desc:the temperature measurement
distance not only affects the temperature measurement results, but also affects the accuracy of temperature measurement location-->true
  </isSupportWebDisplayThermometryDistanceTips>
<isSupportFlashLifespanProtectParams>
  <!--ro, opt, bool, whether it supports flash life protection parameter, desc:GET/PUT /ISAPI/System/FLASHMgr/FlashLifespanProtectParams?format=json--
>true
  </isSupportFlashLifespanProtectParams>
<isSupportIlluminanceTelemeter>
  <!--ro, opt, bool, whether it supports light intensity telemetry, desc:eventType:IlluminanceTelemeter-->true
  </isSupportIlluminanceTelemeter>
<isSupportWindDirectionInDegreesTelemeter>
  <!--ro, opt, bool, whether it supports wind direction telemetry, desc:eventType:WindDirectionInDegreesTelemeter-->true
  </isSupportWindDirectionInDegreesTelemeter>
<isSupportWindSpeedTelemeter>
  <!--ro, opt, bool, whether it supports wind telemetry, desc:eventType:WindSpeedTelemeter-->true
  </isSupportWindSpeedTelemeter>
<isSupportAirPressureTelemeter>
  <!--ro, opt, bool, whether it supports air pressure telemetry, desc:eventType:AirPressureTelemeter-->true
  </isSupportAirPressureTelemeter>
<isSupportBatteryPercentageTelemeter>
  <!--ro, opt, bool, whether it supports battery telemetry, desc:eventType:BatteryPercentageTelemeter-->true
  </isSupportBatteryPercentageTelemeter>
<isSupportTemperaturesTelemeter>
  <!--ro, opt, bool, whether it supports temperature telemetry, desc:eventType:TemperaturesTelemeter-->true
  </isSupportTemperaturesTelemeter>
<isSupportHumidityTelemeter>
  <!--ro, opt, bool, whether it supports humidity telemetry, desc:eventType:HumidityTelemeter-->true
  </isSupportHumidityTelemeter>

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</isSupportHumidityTelemeter>
<isSupportDipAngleTelemeter>
  <!--ro, opt, bool, whether it supports angle telemetry, desc:eventType:DipAngleTelemeter-->true
</isSupportDipAngleTelemeter>
<isSupportLocationEvent>
  <!--ro, opt, bool, whether it supports the device location information uploading event, desc:eventType:Location-->true
</isSupportLocationEvent>
<isSupportSoilMoistureTelemeter>
  <!--ro, opt, bool, whether it supports soil moisture telemetry, desc:eventType:SoilMoistureTelemeter-->true
</isSupportSoilMoistureTelemeter>
<isSupportLitterTelemeter>
  <!--ro, opt, bool, whether it supports litter telemetry, desc:eventType:LitterTelemeter-->true
</isSupportLitterTelemeter>
<isSupportSearchRealStreamStatus>
  <!--ro, opt, bool, whether it supports searching for the real-time streaming status, desc:POST /ISAPI/Streaming/SearchRealStreamStatus?format=json-->true
</isSupportSearchRealStreamStatus>
<isSupportSearchStreamBandwidthInfo>
  <!--ro, opt, bool, whether it supports searching for stream bandwidth status, desc:POST /ISAPI/Streaming/SearchStreamBandwidthInfo?format=json-->true
</isSupportSearchStreamBandwidthInfo>
<isSupportExportThermometryData>
  <!--ro, opt, bool, whether it supports exporting visible light temperature measurement data, desc:POST
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExportThermometryData?format=json-->true
</isSupportExportThermometryData>
<isSupportModifyClickToThermometry>
  <!--ro, opt, bool, whether it supports configuring parameters of visible light click-to-thermometry rules, desc:PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ModifyClickToThermometry?format=json-->true
</isSupportModifyClickToThermometry>
<isSupportExpertThermometryAlarmParams>
  <!--ro, opt, bool, whether it supports alarm parameters of temperature measurement in expert mode, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExpertThermometryAlarmParams?format=json-->true
</isSupportExpertThermometryAlarmParams>
<isSupportExpertThermometryRuleParams>
  <!--ro, opt, bool, whether it supports the rule parameters of temperature measurement in expert mode, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExpertThermometryRuleParams?format=json-->true
</isSupportExpertThermometryRuleParams>
<isSupportNormalThermometryRuleParams>
  <!--ro, opt, bool, whether it supports the rule parameters of temperature measurement in normal mode, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/NormalThermometryRuleParams?format=json-->true
</isSupportNormalThermometryRuleParams>
<isSupportModeParams>
  <!--ro, opt, bool, whether it supports mode parameters of visible light radiation temperature measurement, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ModeParams?format=json-->true
</isSupportModeParams>
<isSupportBasicParams>
  <!--ro, opt, bool, whether it supports basic parameters of visible light radiation temperature measurement, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/BasicParams?format=json-->true
</isSupportBasicParams>
<isSupportImportPlayerFile>
  <!--ro, opt, bool, whether it supports importing player files, desc:PUT /ISAPI/System/Video/ImportPlayerFile?format=json-->true
</isSupportImportPlayerFile>
<isSupportSearchQuickQRCode>
  <!--ro, opt, bool, whether it supports searching for quick configuration codes, desc:POST /ISAPI/System/QuickConfigQRCodeMgr/SearchQuickQRCode?
format=json-->true
</isSupportSearchQuickQRCode>
<isSupportQuickQRCodeEncryptionParams>
  <!--ro, opt, bool, whether it supports the encryption parameters of quick configuration codes, desc:PUT
/ISAPI/System/QuickConfigQRCodeMgr/QuickQRCodeEncryptionParams?format=json-->true
</isSupportQuickQRCodeEncryptionParams>
<isUnsynchronizedUpgradeKVMInput>
  <!--ro, opt, bool, whether it supports upgrading KVM input node synchronously, desc:1. For distributed main control, if this field is not returned or
returned as false, the KVM input node added to the main control will be upgraded. 2. If this field is returned as true, the KVM input node will not be
upgraded.-->true
</isUnsynchronizedUpgradeKVMInput>
<isSupportWaterRescueDetectionOverlapParams>
  <!--ro, opt, bool, whether it supports intelligent information overlay parameters of water rescue, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/WaterRescueDetectionOverlapParams?format=json-->true
</isSupportWaterRescueDetectionOverlapParams>
<isSupportSearchRecordsStatusList>
  <!--ro, opt, bool, whether it supports searching for the channel recording status, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStatusList?
format=json-->true
</isSupportSearchRecordStatusList>
<isSupportWiperSelfCheckParams>
  <!--ro, opt, bool, whether it supports the wiper self-checking parameters, desc:GET/PUT
/ISAPI/System/Hardware/channels/<channelID>/WiperSelfCheckParams?format=json-->true
</isSupportWiperSelfCheckParams>
<isSupportBlackoutNotification>
  <!--ro, opt, bool, whether it supports power-off notification, desc:PUT /ISAPI/System/channels/<channelID>/BlackoutNotification?format=json-->true
</isSupportBlackoutNotification>
<isSupportGotWaterFlowScene>
  <!--ro, opt, bool, whether it supports jumping to the specified water flow detection scene, desc:POST
/ISAPI/Event/channels/<channelID>/GotWaterFlowScene?format=json-->true
</isSupportGotWaterFlowScene>
<isSupportHVTVehicleDetectUploadChannel>
  <!--ro, opt, bool, whether it supports configuring the channel of uploading mixed-traffic detection information to the FTP server, desc:GET/PUT
/ISAPI/System/Network/ftp/HVTVehicleDetectUpload?format=json&channelID=<channelID>-->true
</isSupportHVTVehicleDetectUploadChannel>
<isSupportWaterFlowAdvancedParams>
  <!--ro, opt, bool, whether it supports advanced parameters of water flow detection, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/WaterFlowAdvancedParams?format=json-->true
</isSupportWaterFlowAdvancedParams>
<isSupportWaterFlowSceneList>
  <!--ro, opt, bool, whether it supports water flow detection scene list, desc:GET /ISAPI/Event/channels/<channelID>/WaterFlowSceneList?format=json-->true
</isSupportWaterFlowSceneList>

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</isSupportWaterFlowSceneRuleParams>
<!--ro, opt, bool, whether it supports water flow detection scene rule parameters, desc:PUT /ISAPI/Event/channels/<channelID>/WaterFlowSceneRuleParams?
format=json-->true
</isSupportWaterFlowSceneRuleParams>
<isSupportRainfallScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled rainfall information uploading event, desc:RainfallScheduledUpload-->true
</isSupportRainfallScheduledUpload>
<isSupportFlowVelocityScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled water flow speed uploading event, desc:FlowVelocityScheduledUpload-->true
</isSupportFlowVelocityScheduledUpload>
<isSupportFlowVelocityAlarm>
<!--ro, opt, bool, whether it supports water flow speed alarm event, desc:FlowVelocityAlarm-->true
</isSupportFlowVelocityAlarm>
<isSupportVolumeFlowRateScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled water flow rate uploading event, desc:VolumeFlowRateScheduledUpload-->true
</isSupportVolumeFlowRateScheduledUpload>
<isSupportVolumeFlowRateAlarm>
<!--ro, opt, bool, whether it supports water flow rate alarm event, desc:VolumeFlowRateAlarm-->true
</isSupportVolumeFlowRateAlarm>
<isSupportSecurityDisclaimerResultEvent>
<!--ro, opt, bool, whether it supports security disclaimer result uploading, desc:SecurityDisclaimerResultEvent-->true
</isSupportSecurityDisclaimerResultEvent>
<isSupportInfusionAlarmEvent>
<!--ro, opt, bool, whether it supports infusion alarm event, desc:InfusionAlarmEvent-->true
</isSupportInfusionAlarmEvent>
<isSupportBodyCameraBatteryTamperEvent>
<!--ro, opt, bool, whether it supports body camera battery tampering event, desc:eventType:BodyCameraBatteryTamperEvent-->true
</isSupportBodyCameraBatteryTamperEvent>
<isSupportShipSafeOperationDetectionEvent>
<!--ro, opt, bool, whether it supports ship operation safety event detection, desc:eventType:ShipSafeOperationDetectionEvent-->true
</isSupportShipSafeOperationDetectionEvent>
<isSupportTrafficLightStatusDetectionEvent>
<!--ro, opt, bool, whether it supports traffic light status detection event, desc:eventType:TrafficLightStatusDetectionEvent-->true
</isSupportTrafficLightStatusDetectionEvent>
<isSupportSoundPressureDetectionEvent>
<!--ro, opt, bool, whether it supports triggering overload sound pressure alarm, desc:eventType:SoundPressureDetectionEvent-->true
</isSupportSoundPressureDetectionEvent>
<isSupportInvigilatorBehaviorRuleParams>
<!--ro, opt, bool, whether it supports driving license exam proctoring behavior detection rule parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/InvigilatorBehaviorRuleParams?format=json-->true
</isSupportInvigilatorBehaviorRuleParams>
<isSupportInvigilatorBehaviorEvent>
<!--ro, opt, bool, whether it supports driving license exam proctoring behavior detection event, desc:eventType:InvigilatorBehaviorEvent-->true
</isSupportInvigilatorBehaviorEvent>
<isSupportPeopleSleepingDetectionReport>
<!--ro, opt, bool, whether it supports sleep detection report event, desc:eventType:PeopleSleepingDetectionReport-->true
</isSupportPeopleSleepingDetectionReport>
<isSupportPhysicalMatchMgr>
<!--ro, opt, bool, whether it supports sports competitions management, desc:GET /ISAPI/Intelligent/channels/<channelID>/PhysicalMatchMgr/capabilities?
format=json-->true
</isSupportPhysicalMatchMgr>
<isSupportSearchNetworkResources>
<!--ro, opt, bool, whether it supports searching for network resource information, desc:POST
/ISAPI/System/Network/NetworkAdvancedCfg/SearchNetworkResources?format=json-->true
</isSupportSearchNetworkResources>
<isSupportTriggerPeripheralAbnormalDetection>
<!--ro, opt, bool, whether it supports triggering the device to check component fault, desc:PUT /ISAPI/Event/TriggerPeripheralAbnormalDetection?
format=json-->true
</isSupportTriggerPeripheralAbnormalDetection>
<isSupportDeviceCabinetDoorOpenAlarm>
<!--ro, opt, bool, whether it support device chassis open alarm event, desc:eventType:DeviceCabinetDoorOpenAlarm-->true
</isSupportDeviceCabinetDoorOpenAlarm>
<isSupportSurplusPower>
<!--ro, opt, bool, whether it supports remaining battery information, desc:GET/PUT /ISAPI/System/SurplusPower?format=json-->true
</isSupportSurplusPower>
<isSupportPeripheralAbnormalScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled device component fault uploading event, desc:eventType:PeripheralAbnormalScheduledUpload-->true
</isSupportPeripheralAbnormalScheduledUpload>
<isSupportBandwidthLimitWarningParams>
<!--ro, opt, bool, whether it supports the bandwidth limit alarm parameters, desc:GET/PUT
/ISAPI/System/Network/NetworkFlowMonitor/BandwidthLimitWarningParams?format=json-->true
</isSupportBandwidthLimitWarningParams>
<isSupportRoadMaintSceneList>
<!--ro, opt, bool, whether it supports road maintenance scene List, desc:GET /ISAPI/Intelligent/channels/<channelID>/RoadMaintSceneList?format=json--
>true
</isSupportRoadMaintSceneList>
<isSupportSearchIntelligencePerformance>
<!--ro, opt, bool, whether it supports searching post-intelligence performance parameters, desc:POST
/ISAPI/System/VCAResource/SearchIntelligencePerformance?format=json-->true
</isSupportSearchIntelligencePerformance>
<isSupportRainfallRuleParams>
<!--ro, opt, bool, whether it supports rainfall detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/RainfallRuleParams?
format=json-->true
</isSupportRainfallRuleParams>
<isSupportRadarFlowmeterOverlapParams>
<!--ro, opt, bool, whether it supports radar detection information overlay parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/RadarFlowmeterOverlapParams?format=json-->true
</isSupportRadarFlowmeterOverlapParams>
<isSupportCurrentComStatus>
<!--ro, opt, bool, whether it supports getting the device communication status, desc:GET /ISAPI/System/CurrentComStatus?format=json-->true
</isSupportCurrentComStatus>
<isSupportRoadMaintAlgorithmVersion>
<!--ro, opt, bool, whether it supports road maintenance algorithm version, desc:GET /ISAPI/Intelligent/channels/<channelID>/RoadMaintAlgorithmVersion?

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format=json-->true
  </isSupportRoadMaintAlgorithmVersion>
  <isSupportRoadMaintOverlapParams>
    <!--ro, opt, bool, whether it supports road maintenance overlay parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/RoadMaintOverLapParams?
format=json-->true
  </isSupportRoadMaintOverlapParams>
  <isSupportActionExecuteStatus>
    <!--ro, opt, bool, whether it supports action execution status uploading, desc:eventType: ActionExecuteStatus-->true
  </isSupportActionExecuteStatus>
  <isSupportExportConfigurationFile>
    <!--ro, opt, bool, whether it supports exporting the configuration file, desc:GET /ISAPI/System/ExportConfigurationFile?format=json-->true
  </isSupportExportConfigurationFile>
  <isSupportMultiChannelStreamMergeParams>
    <!--ro, opt, bool, whether it supports the parameters of merging the encoding of multiple panoramic channels, desc:GET/PUT
/ISAPI/System/Video/inputs/MultiChannelStreamMergeParams?format=json-->true
  </isSupportMultiChannelStreamMergeParams>
  <isSupportAnalysisEngineCfgTest>
    <!--ro, opt, bool, whether it supports engine analysis parameters configuration restarting detection, desc:POST
/ISAPI/System/VCAResource/AnalysisEngineCfgTest?format=json-->true
  </isSupportAnalysisEngineCfgTest>
  <isSupportModifyPlaybackSpeed>
    <!--ro, opt, bool, whether it supports video speed playback of the streaming device, desc:POST
/ISAPI/Streaming/channels/<channelID>/ModifyPlaybackSpeed?format=json-->true
  </isSupportModifyPlaybackSpeed>
  <isSupportSeekToSegsPlayback>
    <!--ro, opt, bool, whether it supports dragging to the time period during streaming video playback, desc:POST
/ISAPI/Streaming/channels/<channelID>/SeekToSegsPlayback?format=json-->true
  </isSupportSeekToSegsPlayback>
  <isSupportIntersectionTrafficFlowStatisticsEvent>
    <!--ro, opt, bool, whether it supports event for counting motor vehicle traffic at an intersection, desc:eventType:
IntersectionTrafficFlowStatisticsEvent-->true
  </isSupportIntersectionTrafficFlowStatisticsEvent>
  <isSupportLensCoverAbnormalEvent>
    <!--ro, opt, bool, whether it supports lens cover abnormal event, desc:event type: LensCoverAbnormalEvent.-->true
  </isSupportLensCoverAbnormalEvent>
  <isSupportCrackMonitoringEvent>
    <!--ro, opt, bool, whether it supports geological cracks detection event, desc:event type: CrackMonitoringEvent-->true
  </isSupportCrackMonitoringEvent>
  <isSupportExportNetworkCaptureFile>
    <!--ro, opt, bool, whether it supports exporting captured packets, desc:POST /ISAPI/System/ExportNetworkCaptureFile?format=json-->true
  </isSupportExportNetworkCaptureFile>
  <isSupportGetNetworkCaptureStatus>
    <!--ro, opt, bool, whether it supports getting captured packet status, desc:POST /ISAPI/System/GetNetworkCaptureStatus?format=json-->true
  </isSupportGetNetworkCaptureStatus>
  <isSupportStartNetworkCapture>
    <!--ro, opt, bool, whether it supports starting capturing network packets, desc:POST /ISAPI/System/StartNetworkCapture?format=json-->true
  </isSupportStartNetworkCapture>
  <isSupportStopNetworkCapture>
    <!--ro, opt, bool, whether it supports stopping capturing network packets, desc:POST /ISAPI/System/StopNetworkCapture?format=json-->true
  </isSupportStopNetworkCapture>
  <isSupportObjectAccumulationAlgorithmParams>
    <!--ro, opt, bool, whether it supports stacking detection algorithm parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/ObjectAccumulationDetection/ObjectAccumulationAlgorithmParams?format=json-->true
  </isSupportObjectAccumulationAlgorithmParams>
  <isSupportGetTobaccoStageStatus>
    <!--ro, opt, bool, whether it supports getting tobacco baking status, desc:POST /ISAPI/System/TobaccoBakeMgr/GetTobaccoStageStatus?format=json-->true
  </isSupportGetTobaccoStageStatus>
  <isSupportMultipleIntelligentParams>
    <!--ro, opt, bool, whether it supports multi-intelligent mode parameters, desc:GET/PUT /ISAPI/System/VCAResource/MultipleIntelligentParams?format=json--
>true
  </isSupportMultipleIntelligentParams>
  <isSupportStartSystemEnvDetection>
    <!--ro, opt, bool, whether it supports starting system environment check, desc:POST /ISAPI/System/StartSystemEnvDetection?format=json-->true
  </isSupportStartSystemEnvDetection>
  <isSupportSearchSystemEnvDetectionProcess>
    <!--ro, opt, bool, whether it supports searching for system environment checking progress, desc:POST /ISAPI/System/SearchSystemEnvDetectionProcess?
format=json-->true
  </isSupportSearchSystemEnvDetectionProcess>
  <isSupportEVACInputParams>
    <!--ro, opt, bool, whether it supports EVAC input parameters, desc:GET/PUT /ISAPI/System/EVACInput/EVACInputParams?format=json-->true
  </isSupportEVACInputParams>
  <isSupportModifyT1TestParams>
    <!--ro, opt, bool, whether it editing T1 test parameters, desc:POST /ISAPI/System/AutoTest/ModifyT1TestParams?format=json-->true
  </isSupportModifyT1TestParams>
  <isSupportDetectionRegionRuleParams>
    <!--ro, opt, bool, whether it supports rule parameters of hazardous gas area scanning detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/DetectionRegionRuleParams?format=json-->true
  </isSupportDetectionRegionRuleParams>
  <isSupportFaceApproach>
    <!--ro, opt, bool, whether it supports face approaching event, desc:eventType:FaceApproach-->true
  </isSupportFaceApproach>
  <isSupportHardwareEnergySaving>
    <!--ro, opt, bool, whether it supports device hardware energy-saving parameters, desc:GET/PUT /ISAPI/System/consumptionMode/HardwareEnergySaving?
format=json-->true
  </isSupportHardwareEnergySaving>
  <isSupportDeviceHighTemperature>
    <!--ro, opt, bool, whether it supports device overheat event, desc:eventType:DeviceHighTemperature-->true
  </isSupportDeviceHighTemperature>
  <isSupportExportAudioAndVideoFile>
    <!--ro, opt, bool, whether it supports exporting audio and video files, desc:POST /ISAPI/System/ExportAudioAndVideoFile?format=json-->true
  </isSupportExportAudioAndVideoFile>
  <isSupportGetExportAudioAndVideoFileStatus>
    <!--ro, opt, bool, whether it supports getting the export status of video and audio files, desc:GET /ISAPI/System/GetExportAudioAndVideoFileStatus?
format=json-->true
  </isSupportGetExportAudioAndVideoFileStatus>

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format=json-->true
</isSupportGetExportAudioAndVideoFileStatus>
<isSupportStartExportAudioAndVideoFile>
  <!--ro, opt, bool, whether it supports starting exporting audio and video files, desc:PUT /ISAPI/System/StartExportAudioAndVideoFile?format=json-->true
</isSupportStartExportAudioAndVideoFile>
<isSupportStopExportAudioAndVideoFile>
  <!--ro, opt, bool, whether it supports stopping exporting audio and video files, desc:PUT /ISAPI/System/StopExportAudioAndVideoFile?format=json-->true
</isSupportStopExportAudioAndVideoFile>
<isSupportAsyncExportSearchEventResult>
  <!--ro, opt, bool, whether it supports exporting the results of searching for event-related videos asynchronously, desc:POST
/ISAPI/ContentMgmt/search/AsyncExportSearchEventResult?format=json-->true
</isSupportAsyncExportSearchEventResult>
<isRebootAfterUpgradeFailed>
  <!--ro, opt, bool, whether restarting device is required after upgrading failed, desc:if this field does not exist, the device will not be restarted
after upgrading failed-->true
</isRebootAfterUpgradeFailed>
<isSupportGetAsyncExportSearchEventResultProgress>
  <!--ro, opt, bool, whether it supports getting the async exporting progress of event videos search result, desc:POST
/ISAPI/ContentMgmt/search/GetAsyncExportSearchEventResultProgress?format=json-->true
</isSupportGetAsyncExportSearchEventResultProgress>
<isSupportManualTriggerWipe>
  <!--ro, opt, bool, whether it supports manually starting the wiper, desc:POST /ISAPI/System/Hardware/ManualTriggerWipe?format=json-->true
</isSupportManualTriggerWipe>
<isSupportWiperCleanMode>
  <!--ro, opt, bool, whether it supports wiper cleaning mode, desc:GET/PUT /ISAPI/System/Hardware/WiperCleanMode?format=json-->true
</isSupportWiperCleanMode>
<isSupportRadarWaterFlowRuleParams>
  <!--ro, opt, bool, whether it supports radar-assisted water flow detection rule parameters, desc:GET/PUT
/ISAPI/Event/Radar/<RadarID>/RadarWaterFlowRuleParams?format=json-->true
</isSupportRadarWaterFlowRuleParams>
<isSupportRadarWaterFlowAdvancedParams>
  <!--ro, opt, bool, whether it supports advanced parameters of radar-assisted water flow detection, desc:GET/PUT
/ISAPI/Event/Radar/<RadarID>/RadarWaterFlowAdvancedParams?format=json-->true
</isSupportRadarWaterFlowAdvancedParams>
<isSupportLaserSourceSwitch>
  <!--ro, opt, bool, whether it supports laser light source switching, desc:GET/PUT /ISAPI/System/channels/<channelID>/LaserSourceSwitch?format=json--
>true
</isSupportLaserSourceSwitch>
<isSupportLensStainConfigParams>
  <!--ro, opt, bool, whether it supports lens dirt detection configuration parameters, desc:GET/PUT /ISAPI/System/Hardware/LensStainConfigParams?
format=json-->true
</isSupportLensStainConfigParams>
<setupParamType opt="PTZLaser">
  <!--ro, opt, object, whether it supports construction parameters of device sub modules, attr:opt{req, string}, desc:whether /ISAPI/System/SetupParam?
type=<type> supports the optional parameter "type"-->
</setupParamType>
<isSupportLensStain>
  <!--ro, opt, bool, whether it supports lens stain event, desc:eventType:LensStain-->true
</isSupportLensStain>
<isSupportLensStainRecovered>
  <!--ro, opt, bool, whether it supports lens stain restoring event, desc:eventType:LensStainRecovered-->true
</isSupportLensStainRecovered>
<isSupportDeviceCircuitFault>
  <!--ro, opt, bool, whether it supports device circuit fault events, desc:eventType:DeviceCircuitFault-->true
</isSupportDeviceCircuitFault>
<isSupportDeviceCircuitRestored>
  <!--ro, opt, bool, whether it supports device circuit fault restoring event, desc:eventType:DeviceCircuitRestored-->true
</isSupportDeviceCircuitRestored>
<isSupportTurnOffBuiltInBattery>
  <!--ro, opt, bool, whether it supports disabling the built-in battery, desc:PUT /ISAPI/System/TurnOffBuiltInBattery?format=json-->true
</isSupportTurnOffBuiltInBattery>
<isSupportElevatorStartMove>
  <!--ro, opt, bool, whether it supports elevator starting running event, desc:eventType:ElevatorStartMove-->true
</isSupportElevatorStartMove>
<isSupportElevatorStopMove>
  <!--ro, opt, bool, whether it supports elevator stopping running event, desc:eventType:ElevatorStopMove-->true
</isSupportElevatorStopMove>
<isSupportAutoUpgradeCfg>
  <!--ro, opt, bool, whether it supports auto upgrade configuration, desc:GET/PUT /ISAPI/System/AutoUpgradeCfg?format=json-->true
</isSupportAutoUpgradeCfg>
<isSupportOpticalFiberRasterAlarm>
  <!--ro, opt, bool, whether it supports fiber and raster alarm event, desc:corresponding event: OpticalFiberRasterAlarm-->true
</isSupportOpticalFiberRasterAlarm>
<isSupportSensorRealTimeData>
  <!--ro, opt, bool, whether it supports real-time sensor data uploading event, desc:corresponding event: SensorRealTimeData-->true
</isSupportSensorRealTimeData>
<isSupportGNSSWorkStatus>
  <!--ro, opt, bool, whether it supports GNSS working status uploading event, desc:corresponding event: GNSSWorkStatus.-->true
</isSupportGNSSWorkStatus>
<isSupportDevOnPlatformInfoEvent>
  <!--ro, opt, bool, whether it supports on-platform device information request event, desc:corresponding event: DevOnPlatformInfoEvent-->true
</isSupportDevOnPlatformInfoEvent>
<isSupportUpdateScheduleEvent>
  <!--ro, opt, bool, whether it supports request event of updating the schedule, desc:corresponding event: UpdateScheduleEvent-->true
</isSupportUpdateScheduleEvent>
<isSupportIntelligentApply>
  <!--ro, opt, bool, whether it support intelligent mode application parameters, desc:GET/PUT /ISAPI/System/VCAResource/IntelligentApply?format=json--
>true
</isSupportIntelligentApply>
<isSupportMCUDevSerialWrite>
  <!--ro, opt, bool, whether it supports writing MCU device serial number, desc:POST /ISAPI/System/MCUDevSerialWrite?format=json-->true
</isSupportMCUDevSerialWrite>
<isSupportSambaServerParams>
  <!--ro, opt, bool, whether it supports for Samba server parameters, desc:GET/PUT /ISAPI/System/SambaServerManagement/SambaServerParams?format=json--

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>true
  </isSupportSambaServerParams>
  <isSupportToolingBoardConfigParams>
    <!--ro, opt, bool, whether it supports tooling plate configuration parameters, desc:PUT /ISAPI/System/AutoTest/ToolingBoardConfigParams?format=json-->
  >true
    </isSupportToolingBoardConfigParams>
    <isSupportAlumFlocObservationParams>
      <!--ro, opt, bool, whether it supports alum floc detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/AlumFlocObservation/channels/<channelID>/AlumFlocObservationParams?format=json-->true
    </isSupportAlumFlocObservationParams>
    <isDisplayRebootTips>
      <!--ro, opt, bool, whether to prompt users to restart the device when switching non-uniformity correction, desc:For devices with low performance,
switching non-uniformity correction will take effect only after restarting the device. The web should inform users of this. For devices with high
performance, restarting is not required. You can determine whether the web needs to prompt users to restart the device via this field. If this field does
not exist, the web needs to prompt by default.-->true
    </isDisplayRebootTips>
    <isSupportAIOPMPAndResChanged>
      <!--ro, opt, bool, whether the meaning of AIOP model package and resource interface is changed, desc:If the returned value is true, it indicates that
the meaning of the AIOP API has changed, that is, the AIOP API for model package and engine management have replaced that of algorithm package scheduling
and of the algorithm and engine management in algorithm application scheduling. For example,
/ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engine/<engineID>?format=json and /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engines?format=json will
return all the engines for scheduling the algorithm packages and applications, and /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management?format=json
will return the information of the AIOP model package, algorithm application scheduling, and algorithm package scheduling.-->true
    </isSupportAIOPMPAndResChanged>
    <isSupportRoadVisibilityAnalysisCfg>
      <!--ro, opt, bool, whether it supports road visibility detection analysis parameters, desc:GET/PUT
/ISAPI/Traffic/channels/<channelID>/RoadVisibilityAnalysisCfg?format=json-->true
    </isSupportRoadVisibilityAnalysisCfg>
    <isSupportHotStandbyParams>
      <!--ro, opt, bool, whether it supports hot spare parameters, desc:GET/PUT /ISAPI/System/MainSpareDeviceSettings/HotStandbyParams?format=json-->true
    </isSupportHotStandbyParams>
    <isSupportHotStandbySwitchParams>
      <!--ro, opt, bool, whether it supports hot spare switching, desc:GET/PUT /ISAPI/System/MainSpareDeviceSettings/HotStandbySwitchParams?format=json-->true
    </isSupportHotStandbySwitchParams>
    <isSupportManualControlHotStandby>
      <!--ro, opt, bool, whether it supports manually taking over hot spare, desc:POST /ISAPI/System/MainSpareDeviceSettings/ManualControlHotStandby?
format=json-->true
    </isSupportManualControlHotStandby>
    <isSupportGetHotStandbyStatus>
      <!--ro, opt, bool, whether it supports getting hot spare status, desc:GET /ISAPI/System/MainSpareDeviceSettings/GetHotStandbyStatus?format=json-->true
    </isSupportGetHotStandbyStatus>
    <isSupportRestoreStreamCollectionInfo>
      <!--ro, opt, bool, whether it supports confirmation for collecting the raw stream data without mosaic mask, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/RestoreStreamCollectionInfo?format=json-->true
    </isSupportRestoreStreamCollectionInfo>
    <isSupportFileNameFormat>
      <!--ro, opt, bool, whether it supports file name format, desc:GET/PUT /ISAPI/System/FileNameFormat?format=json-->true
    </isSupportFileNameFormat>
    <isSupportThermalInspectionParams>
      <!--ro, opt, bool, whether it supports thermal inspection parameters, desc:GET/PUT /ISAPI/Thermal/ThermalInspectionParams?format=json-->true
    </isSupportThermalInspectionParams>
    <isSupportThermalRecordParams>
      <!--ro, opt, bool, whether it supports configuring the recording parameters of specified thermal imaging channel, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/ThermalRecordParams?format=json-->true
    </isSupportThermalRecordParams>
    <isSupportThermalLensParams>
      <!--ro, opt, bool, whether it supports configuring thermal imaging lens parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ThermalLensParams?
format=json-->true
    </isSupportThermalLensParams>
    <isSupportThermalInfoDisplayParam>
      <!--ro, opt, bool, whether it supports the display parameters of temperature measurement information, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/ThermalInfoDisplayParam?format=json-->true
    </isSupportThermalInfoDisplayParam>
    <isSupportHotMapParam>
      <!--ro, opt, bool, whether it supports thermal image capturing parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/HotMapParam?format=json--
  >true
    </isSupportHotMapParam>
    <isSupportGetLaserRangingResult>
      <!--ro, opt, bool, whether it supports getting the Laser ranging results, desc:GET /ISAPI/System/GetLaserRangingResult?format=json-->true
    </isSupportGetLaserRangingResult>
    <isSupportLaserRangingOverlay>
      <!--ro, opt, bool, whether it supports parameters of overlaying laser ranging information, desc:GET/PUT /ISAPI/System/LaserRangingOverlay?format=json--
  >true
    </isSupportLaserRangingOverlay>
    <isSupportPersonQueueRegionRealTime>
      <!--ro, opt, bool, whether it supports real-time uploading of number of people in queue detection, desc:eventType:PersonQueueRegionRealTime-->true
    </isSupportPersonQueueRegionRealTime>
    <isSupportFullSpectrumAnalysis>
      <!--ro, opt, bool, whether it supports full spectrum analysis parameters, desc:GET /ISAPI/FullSpectrumAnalysis/capabilities?format=json-->true
    </isSupportFullSpectrumAnalysis>
    <isSupportColorDetectionActionDelayTimeParams>
      <!--ro, opt, bool, whether it supports the parameters of screen color quality detection action delay, desc:GET/PUT
/ISAPI/System/ScreenColorQualityDetection/ColorDetectionActionDelayTimeParams?format=json-->true
    </isSupportColorDetectionActionDelayTimeParams>
    <isSupportAudioDetectionActionDelayTimeParams>
      <!--ro, opt, bool, whether it supports the parameters of audio detection action delay, desc:GET/PUT
/ISAPI/System/DeviceAudioDetection/AudioDetectionActionDelayTimeParams?format=json-->true
    </isSupportAudioDetectionActionDelayTimeParams>
    <isSupportCustomVideoChannelStatusIconParams>
      <!--ro, opt, bool, whether it supports configuring the parameters of customizing the video channel status icon, desc:GET/PUT
/ISAPI/ContentMgmt/InputProxy/CustomVideoChannelStatusIconParams?format=json-->true
    </isSupportCustomVideoChannelStatusIconParams>
    <isSupportImportVideoChannelStatusIcon>

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<!--ro, opt, bool, whether it supports importing custom video channel status icons, desc:POST
/ISAPI/ContentMgmt/InputProxy/ImportVideoChannelStatusIcon?format=json-->true
</isSupportImportVideoChannelStatusIcon>
<isSupportDeleteVideoChannelStatusIcon>
<!--ro, opt, bool, whether it supports deleting custom video channel status icons, desc:POST /ISAPI/ContentMgmt/InputProxy/DeleteVideoChannelStatusIcon?
format=json-->true
</isSupportDeleteVideoChannelStatusIcon>
<isSupportSmokeScopeTelemeter>
<!--ro, opt, bool, whether it supports smoke density telemetry, desc:eventType:SmokeScopeTelemeter-->true
</isSupportSmokeScopeTelemeter>
<isSupportTempDataScheduledUpload>
<!--ro, opt, bool, whether it supports scheduled temperature measurement data uploading, desc:eventType:TempDataScheduledUpload-->true
</isSupportTempDataScheduledUpload>
<isSupportSafetyRopeDetectionRuleParams>
<!--ro, opt, bool, whether it supports safety rope detection rule parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/SafetyRopeDetectionRuleParams?format=json-->true
</isSupportSafetyRopeDetectionRuleParams>
<isSupportSafetyRopeDetectionEvent>
<!--ro, opt, bool, whether it supports safety rope detection event, desc:eventType: SafetyRopeDetectionEvent-->true
</isSupportSafetyRopeDetectionEvent>
<isSupportAddPositionShieldMask>
<!--ro, opt, bool, whether it supports adding the shielded area of a specified PTZ absolute position, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/AddPositionShieldMask?format=json-->true
</isSupportAddPositionShieldMask>
<isSupportPRPDUpload>
<!--ro, opt, bool, whether it supports uploading PRPD data, desc:eventType:PRPDUpload-->true
</isSupportPRPDUpload>
<isSupportAsynchronousUploadGNSSInitDataResult>
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  <!--ro, opt, bool-->true
</isSupporttiUpgrade>
<isSupportUpdateFineTuningSynonym>
  <!--ro, opt, bool-->true
</isSupportUpdateFineTuningSynonym>
<isSupportCommonGasParams>
  <!--ro, opt, bool-->true
</isSupportCommonGasParams>
<isSupportAddGasCombinationRuleTemplate>
  <!--ro, opt, bool-->true
</isSupportAddGasCombinationRuleTemplate>
<isSupportModifyGasCombinationRuleTemplate>
  <!--ro, opt, bool-->true
</isSupportModifyGasCombinationRuleTemplate>
<isSupportDeleteGasCombinationRuleTemplateList>
  <!--ro, opt, bool-->true
</isSupportDeleteGasCombinationRuleTemplateList>
<isSupportInOutConfigParams>
  <!--ro, opt, bool-->true
</isSupportInOutConfigParams>
<isSupportSeachInOutStatusInfo>
  <!--ro, opt, bool-->true
</isSupportSeachInOutStatusInfo>
<isSupportSearchGasCombinationRuleTemplateList>
  <!--ro, opt, bool-->true
</isSupportSearchGasCombinationRuleTemplateList>
<isSupportInputModuleParams>
  <!--ro, opt, bool-->true
</isSupportInputModuleParams>
<isSupportOutletDetectionParams>
  <!--ro, opt, bool-->true
</isSupportOutletDetectionParams>
<isSupportOutletConfigParams>
  <!--ro, opt, bool-->true
</isSupportOutletConfigParams>
<isSupportOutletTriggerParams>
  <!--ro, opt, bool-->true
</isSupportOutletTriggerParams>
<isSupportSearchOutletStatusInfo>
  <!--ro, opt, bool-->true
</isSupportSearchOutletStatusInfo>
<isSupportScenarioAwareTrigger>
  <!--ro, opt, bool-->true
</isSupportScenarioAwareTrigger>
<isSupportScenarioAwareOverlapParams>
  <!--ro, opt, bool-->true
</isSupportScenarioAwareOverlapParams>
<isSupportScenarioAwareRuleParams>
  <!--ro, opt, bool-->true
</isSupportScenarioAwareRuleParams>
<isSupportSeachInModuleStatusInfo>
  <!--ro, opt, bool-->true
</isSupportSeachInModuleStatusInfo>
<isSupportClippedSystemCapCtrl>
  <!--ro, opt, bool-->true
</isSupportClippedSystemCapCtrl>
<isSupportTestPicSearchByTextChannel>
  <!--ro, opt, bool-->true
```

```
</isSupportTestPicSearchByTextChannel>
<isSupportRedirectStorageUrlEvent>
  <!--ro, opt, bool-->true
</isSupportRedirectStorageUrlEvent>
<isSupportGetChallengeV3>
  <!--ro, opt, bool-->true
</isSupportGetChallengeV3>
<isSupportStartActivateV3>
  <!--ro, opt, bool-->true
</isSupportStartActivateV3>
<isSupportManualTriggerAlarm>
  <!--ro, opt, bool-->true
</isSupportManualTriggerAlarm>
<isSupportModifyInOutModuleParams>
  <!--ro, opt, bool-->true
</isSupportModifyInOutModuleParams>
<isSupportModifyInModuleParams>
  <!--ro, opt, bool-->true
</isSupportModifyInModuleParams>
<isSupportStopWebRTCPreview>
  <!--ro, opt, bool-->true
</isSupportStopWebRTCPreview>
<isSupportStartWebRTCPreview>
  <!--ro, opt, bool-->true
</isSupportStartWebRTCPreview>
<isSupportStartWebRTCPreviewPush>
  <!--ro, opt, bool-->true
</isSupportStartWebRTCPreviewPush>
<isSupportRestoreDebugLogCfg>
  <!--ro, opt, bool-->true
</isSupportRestoreDebugLogCfg>
<isSupportModifyTriggerParams>
  <!--ro, opt, bool-->true
</isSupportModifyTriggerParams>
<isSupportModifyDetectionParams>
  <!--ro, opt, bool-->true
</isSupportModifyDetectionParams>
<isSupportGetControllerStatus>
  <!--ro, opt, bool-->true
</isSupportGetControllerStatus>
<isSupportHotStandbyEvent>
  <!--ro, opt, bool-->true
</isSupportHotStandbyEvent>
<isSupportGetDeviceTamperClearAlarm>
  <!--ro, opt, bool-->true
</isSupportGetDeviceTamperClearAlarm>
<isSupportSendDeviceTamperClearAlarm>
  <!--ro, opt, bool-->true
</isSupportSendDeviceTamperClearAlarm>
<isSupportGetScenarioAwareRunStatus>
  <!--ro, opt, bool-->true
</isSupportGetScenarioAwareRunStatus>
<isSupportStartFullSpectrumCalibration>
  <!--ro, opt, bool-->true
</isSupportStartFullSpectrumCalibration>
<isSupportGetFullCalibrationResult>
  <!--ro, opt, bool-->true
</isSupportGetFullCalibrationResult>
<isSupportSendPointCalibrationCmd>
  <!--ro, opt, bool-->true
</isSupportSendPointCalibrationCmd>
<isSupportGetPointCalibrationResult>
  <!--ro, opt, bool-->true
</isSupportGetPointCalibrationResult>
<isSupportRestoreFullSpectrumModel>
  <!--ro, opt, bool-->true
</isSupportRestoreFullSpectrumModel>
<isSupportLectureSpeechTransParams>
  <!--ro, opt, bool-->true
</isSupportLectureSpeechTransParams>
<isSupportAutoCalibrateStatus>
  <!--ro, opt, bool-->true
</isSupportAutoCalibrateStatus>
<isSupportAlgorithmOrch>
  <!--ro, opt, bool-->true
</isSupportAlgorithmOrch>
<isSupportGroupDeviceRunningDataUpload>
  <!--ro, opt, bool-->true
</isSupportGroupDeviceRunningDataUpload>
<isSupportGroupDeviceRelatedComponentsStatus>
  <!--ro, opt, bool-->true
</isSupportGroupDeviceRelatedComponentsStatus>
<isSupportImageSecurityCheck>
  <!--ro, opt, bool-->true
</isSupportImageSecurityCheck>
<isSupportShipStatisticsRuleParams>
  <!--ro, opt, bool-->true
</isSupportShipStatisticsRuleParams>
<isSupportShipIllegalParkingRuleParams>
  <!--ro, opt, bool-->true
</isSupportShipIllegalParkingRuleParams>
<isSupportShipDetectionAlgVersion>
  <!--ro, opt, bool-->true
```

```
</isSupportShipDetectionAlgVersion>
<isSupportSearchSafeModeEntryRecord>
  <!--ro, opt, bool-->true
</isSupportSearchSafeModeEntryRecord>
<isSupportModifytDragCalibrationStatus>
  <!--ro, opt, bool-->true
</isSupportModifytDragCalibrationStatus>
<isSupportGunCalibrateMode>
  <!--ro, opt, bool-->true
</isSupportGunCalibrateMode>
<isSupportBirdMonitoringRuleParams>
  <!--ro, opt, bool-->true
</isSupportBirdMonitoringRuleParams>
<isSupportBirdMonitoringAdvancedParams>
  <!--ro, opt, bool-->true
</isSupportBirdMonitoringAdvancedParams>
<isSupportSystemBasicReset>
  <!--ro, opt, bool-->true
</isSupportSystemBasicReset>
<isSupportGetMNTServiceAuthFile>
  <!--ro, opt, bool-->true
</isSupportGetMNTServiceAuthFile>
<isSupportSearchAcuSeekFullLabel>
  <!--ro, opt, bool-->true
</isSupportSearchAcuSeekFullLabel>
<isSupportSearchAcuSeekType>
  <!--ro, opt, bool-->true
</isSupportSearchAcuSeekType>
<isSupportShipIllegalParking>
  <!--ro, opt, bool-->true
</isSupportShipIllegalParking>
<isSupportRegionalShipStatistics>
  <!--ro, opt, bool-->true
</isSupportRegionalShipStatistics>
<isSupportGetShootTablePositionRange>
  <!--ro, opt, bool-->true
</isSupportGetShootTablePositionRange>
<isSupportShipDetectionOverlapParams>
  <!--ro, opt, bool-->true
</isSupportShipDetectionOverlapParams>
<isSupportDryWetBallTempParams>
  <!--ro, opt, bool-->true
</isSupportDryWetBallTempParams>
<isSupportControllerCap>
  <!--ro, opt, bool-->true
</isSupportControllerCap>
<isSupportExportCrossData>
  <!--ro, opt, bool-->true
</isSupportExportCrossData>
<isSupportImportCrossData>
  <!--ro, opt, bool-->true
</isSupportImportCrossData>
<isSupportSearchRestrictedText>
  <!--ro, opt, bool-->true
</isSupportSearchRestrictedText>
<isSupportDeleteRestrictedText>
  <!--ro, opt, bool-->true
</isSupportDeleteRestrictedText>
<isSupportModifyRestrictedText>
  <!--ro, opt, bool-->true
</isSupportModifyRestrictedText>
<isSupportAddRestrictedText>
  <!--ro, opt, bool-->true
</isSupportAddRestrictedText>
<isSupportSwitchScheduleParams>
  <!--ro, opt, bool-->true
</isSupportSwitchScheduleParams>
<isSupportFlowVelocityPatrol>
  <!--ro, opt, bool-->true
</isSupportFlowVelocityPatrol>
<isSupportOutletOfflineShieldParams>
  <!--ro, opt, bool-->true
</isSupportOutletOfflineShieldParams>
<isSupportWaterFlowAlgorithmParams>
  <!--ro, opt, bool-->true
</isSupportWaterFlowAlgorithmParams>
<isSupportAudioRecordParams>
  <!--ro, opt, bool-->true
</isSupportAudioRecordParams>
<isSupportAudioRecordCodingParams>
  <!--ro, opt, bool-->true
</isSupportAudioRecordCodingParams>
<isSupportTECSelfCheckParams>
  <!--ro, opt, bool-->true
</isSupportTECSelfCheckParams>
<isSupportTECWorkException>
  <!--ro, opt, bool-->true
</isSupportTECWorkException>
<isSupportSearchWaterFlowAlarmData>
  <!--ro, opt, bool-->true
</isSupportSearchWaterFlowAlarmData>
<isSupportHumidityRecovery>
  <!--ro, opt, bool-->true
</isSupportHumidityRecovery>
```



```

</isSupportExpresswayDetection>
<isSupportControlOutlet>
  <!--ro, opt, bool-->true
</isSupportControlOutlet>
<isSupportExpresswayDetection>
  <!--ro, opt, bool-->true
</isSupportExpresswayDetection>
<isSupportEnhancement>
  <!--ro, opt, bool-->true
</isSupportEnhancement>
<isSupportPrivacyInfoCtrl>
  <!--ro, opt, bool-->true
</isSupportPrivacyInfoCtrl>
<isSupportSearchLocalAnalysisStatus>
  <!--ro, opt, bool-->true
</isSupportSearchLocalAnalysisStatus>
<isSupportScenarioAware>
  <!--ro, opt, bool-->true
</isSupportScenarioAware>
<isSupportSitAndReach>
  <!--ro, opt, bool-->true
</isSupportSitAndReach>
<isSupportDisplayDevModeSwitch>
  <!--ro, opt, bool-->true
</isSupportDisplayDevModeSwitch>
</DeviceCap>

```

9.1.9.4 Import device configuration file

Request URL

POST /ISAPI/System/configurationData?secretkey=<secretkey>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
deviceIndex	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	configurationFileAndEncryptionKeyMismatch	0x4000A3C8	--
6	Invalid Content	badDevType	0x60000010	--

9.1.9.5 Export device configuration file

Request URL

GET /ISAPI/System/configurationData?secretkey=<secretkey>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
mode	enum	--

Request Message

None

Response Message

Binary Data

statusCode	statusString	subStatusCode	errorCode	description
2	Device Busy	serviceUnavailable	0x20000002	--

9.1.9.6 Set restoring devices' default parameters

Request URL

PUT /ISAPI/System/factoryReset?mode=<mode>&childDevID=<devIndex>&loginPassword=<loginPassword>&module=<module>

Query Parameter

Parameter Name	Parameter Type	Description
mode	enum	--
devIndex	string	--
loginPassword	string	--
module	enum	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

9.1.9.7 Get the network service capability

Request URL

GET /ISAPI/System/Network/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportWired>
    <!--ro, opt, bool, whether the device supports wired network-->true
  </isSupportWired>
  <isSupportMobileNetwork>
    <!--ro, opt, bool, whether the device supports mobile network-->true
  </isSupportMobileNetwork>
  <isSupportWAN>
    <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
  </isSupportWAN>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportPNP>
    <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
  </isSupportPNP>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <isSupport28181>
    <!--ro, opt, bool, N/A-->true
  </isSupport28181>
  <SnmpCap>
    <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
</NetworkCap>
```

```

</SnmpCap>
<isSupportExtNetCfg>
  <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
</isSupportExtNetCfg>
<isSupportIPFilter>
  <!--ro, opt, bool, whether the device supports IP filtering-->true
</isSupportIPFilter>
<isSupportSSH>
  <!--ro, opt, bool, whether the device supports SSH-->true
</isSupportSSH>
<isSupportNetPreviewStrategy>
  <!--ro, opt, bool, whether the device supports network Live view strategy-->true
</isSupportNetPreviewStrategy>
<isSupportEZVIZ>
  <!--ro, opt, bool, whether the device supports EZ protocol-->true
</isSupportEZVIZ>
<isSupportEhome>
  <!--ro, opt, bool, whether the device supports EHome (ISUP) protocol-->true
</isSupportEhome>
<isSupportWirelessDial>
  <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
</isSupportWirelessDial>
<isSupportWirelessServer>
  <!--ro, opt, bool, whether the device supports wireless server-->true
</isSupportWirelessServer>
<isSupportWPS>
  <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
</isSupportWPS>
<isWirelessMutexWithWirelessServer>
  <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
</isWirelessMutexWithWirelessServer>
<isSupportMACFilter>
  <!--ro, opt, bool, whether the device supports physical address filtering-->true
</isSupportMACFilter>
<GB28181Cap>
  <!--ro, opt, object, N/A-->
  <isSupportGB28181Service>
    <!--ro, opt, bool, N/A-->true
  </isSupportGB28181Service>
</GB28181Cap>
<WPS>
  <!--ro, opt, object, Wi-Fi Protected Setup-->
  <NetworkInterfaceList size="2">
    <!--ro, opt, array, Internet API List, subType:object, attr:size(req, int)-->
    <NetworkInterface>
      <!--ro, opt, object, Internet API-->
      <id>
        <!--ro, req, string, NIC index-->test
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable or not-->true
      </enabled>
      <isSupportAutoConnect>
        <!--ro, opt, bool, whether the device supports automatic connection-->true
      </isSupportAutoConnect>
      <isSupportDevicePinCode>
        <!--ro, opt, bool, whether the device supports PIN code-->true
      </isSupportDevicePinCode>
      <isSupportDevicePinCodeUpdate>
        <!--ro, opt, bool, whether the device supports updating the PIN code-->true
      </isSupportDevicePinCodeUpdate>
      <ApPinCode>
        <!--ro, opt, object, PIN code of the access point-->
        <ssid min="1" max="10">
          <!--ro, opt, string, SSID (Service Set Identifier), attr:min(req, int),max(req, int)-->test
        </ssid>
        <pinCode min="1" max="10">
          <!--ro, opt, string, PIN code, attr:min(req, int),max(req, int)-->test
        </pinCode>
      </ApPinCode>
    </NetworkInterface>
  </NetworkInterfaceList>
</WPS>
<isSupportRFIDData>
  <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
</isSupportRFIDData>
<isSupportwifiProbeSSID>
  <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
</isSupportwifiProbeSSID>
<isSupportwifiProbe>
  <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
</isSupportwifiProbe>
<verificationCode min="1" max="10">
  <!--ro, opt, string, verification code, attr:min(req, int),max(req, int)-->test
</verificationCode>
<WPSCap>
  <!--ro, opt, object, Wi-Fi Protected Setup capability-->
  <isSupport>
    <!--ro, req, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupport>
  <isSupportAutoConnect>
    <!--ro, req, bool, whether the device supports automatic connection-->true
  </isSupportAutoConnect>
</WPSCap>

```

```

<!--ro, opt, object, network mode-->
<NetworkMode>
  <!--ro, opt, object, network mode-->
  <workMode>
    <!--ro, opt, enum, network mode, subType:string, desc:"close", "wifi" (wireless access), "wifiAp" (wireless access point)-->close
  </workMode>
</NetworkMode>
<VerificationCodeModification>
  <!--ro, opt, object, verification code editing-->
  <verificationCodeType opt="normal,empty">
    <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
  </verificationCodeType>
  <isSupportDeclarationURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
  </isSupportDeclarationURL>
  <isSupportPrivacyPolicyURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
  </isSupportPrivacyPolicyURL>
  <verificationCodeModify opt="true,false">
    <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{req, string}-->true
  </verificationCodeModify>
  <Hyperlinks>
    <!--ro, opt, object, hyperlink-->
    <declarationURL>
      <!--ro, opt, string, hyperlink to the Terms of Service-->test
    </declarationURL>
    <privacyPolicyURL>
      <!--ro, opt, string, hyperlink to the Privacy Policy-->test
    </privacyPolicyURL>
  </Hyperlinks>
  <isSupportVerificationCodeCheck>
    <!--ro, opt, bool, whether the device supports verifying the verification code-->true
  </isSupportVerificationCodeCheck>
  <isSupportOldVerificationCode>
    <!--ro, opt, bool, whether the device supports the EZ password configuration of old version (the password of old version consists of six characters)-->true
  </isSupportOldVerificationCode>
</VerificationCodeModification>
<isSupportIntegrate>
  <!--ro, opt, bool, whether the device supports access protocol configuration-->true
</isSupportIntegrate>
<isSupportPlatformAccess>
  <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP address filtering configuration (Ehome allowlist configuration))-->true
</isSupportPlatformAccess>
<isSupportIntelligentBoost>
  <!--ro, opt, bool, whether the device supports intelligent acceleration-->true
</isSupportIntelligentBoost>
<WifiRegionConfiguration>
  <!--ro, opt, object, Wi-Fi area configuration, desc:when both are not supported, the <WifiRegionConfiguration> node will not return-->
  <isSupportWifiRegion>
    <!--ro, opt, bool, whether the device supports Wi-Fi area configuration, desc:supported by overseas devices; true (support), this node is not returned (not support)-->true
  </isSupportWifiRegion>
  <isSupportWifiEnhancement>
    <!--ro, opt, bool, whether the device supports Wi-Fi enhancement, desc:N/A-->true
  </isSupportWifiEnhancement>
</WifiRegionConfiguration>
<isSupportResourceStatistics>
  <!--ro, opt, bool, whether the device supports getting the network resource information-->true
</isSupportResourceStatistics>
<isSupportBandwidthLimit>
  <!--ro, opt, bool, whether the device supports bandwidth limit-->true
</isSupportBandwidthLimit>
<isSupportPOEPortsDisableAdaptiveServer>
  <!--ro, opt, bool, whether the device supports the POE protocol-->true
</isSupportPOEPortsDisableAdaptiveServer>
<isSupportPOEConfiguration>
  <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
</isSupportPOEConfiguration>
<isSupportGetLinkSocketIP>
  <!--ro, opt, bool, whether the device supports getting the SocketIP of the current Link-->true
</isSupportGetLinkSocketIP>
<isSupportWebSocket>
  <!--ro, opt, bool, whether the device supports WebSocket-->true
</isSupportWebSocket>
<isSupportWebSockets>
  <!--ro, opt, bool, whether the device supports WebSockets-->true
</isSupportWebSockets>
<Adaption>
  <!--ro, opt, object, network self-adaption-->
  <streamType opt="0,1,2,3,4,5,7,8,9,10">
    <!--ro, opt, enum, stream type that supports network self-adaption during Live view, subType:int, attr:opt{req, string}, desc:stream types that support network self-adaption during Live view: 0 (main stream), 1 (sub-stream), 2 (third stream), 3 (fourth stream / virtual stream), 4 (fifth stream), 5 (sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10 (tenth stream), and so on-->0
  </streamType>
  <isSupportPlayback>
    <!--ro, opt, bool, whether the device supports playback-->true
  </isSupportPlayback>
</Adaption>
<isSupportVideoImgDB>
  <!--ro, opt, bool, whether the device supports image and video Library configuration-->true
</isSupportVideoImgDB>
<isSupportVideoImgLibService>

```

```

    <!--ro, opt, bool, whether the device supports image and video library service configuration-->true
</isSupportVideoImgLibService>
<isSupportWirelessWifi>
    <!--ro, opt, bool, whether the device supports Wi-Fi configuration-->true
</isSupportWirelessWifi>
<isSupportPrivatePlatomo>
    <!--ro, opt, bool, whether the device supports third party platform configuration-->true
</isSupportPrivatePlatomo>
<isSupportEventDataOverWebSocket opt="true,false">
    <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
</isSupportEventDataOverWebSocket>
<isSupportDynamicHostName>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
</isSupportDynamicHostName>
<isSupportTCPAcceleration>
    <!--ro, opt, bool, whether the device supports TCP acceleration-->true
</isSupportTCPAcceleration>
<isSupportFlowControl>
    <!--ro, opt, bool, whether the device supports traffic control-->true
</isSupportFlowControl>
<isSupportEZVIZTiming>
    <!--ro, opt, bool, whether the device supports EZ platform synchronization (enabledTiming of message returned by/ISAPI/System/Network/EZVIZ)-->true
</isSupportEZVIZTiming>
<isSupportTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring-->true
</isSupportTrafficMonitor>
<isSupportPingWatchdog>
    <!--ro, opt, bool, whether the device supports configuring the watchdog parameters-->true
</isSupportPingWatchdog>
<isSupportAutoFrequencyHop>
    <!--ro, opt, bool, whether the device supports auto frequency hopping-->true
</isSupportAutoFrequencyHop>
<isSupportTDMACfg>
    <!--ro, opt, bool, whether the device supports configuring TDMA-->true
</isSupportTDMACfg>
<isSupportDynamicChannel>
    <!--ro, opt, bool, whether the device supports dynamic channel selection-->true
</isSupportDynamicChannel>
<isSupportSIMCardSwitch>
    <!--ro, opt, bool, whether the device supports configuring scheduled SIM card switch, desc:related URI:
/ISAPI/System/Network/WirelessDial/SIMCardSwitch?format=json-->true
</isSupportSIMCardSwitch>
<isSupportDMSAuthInfo>
    <!--ro, opt, bool, whether the device supports DMS authentication information management-->true
</isSupportDMSAuthInfo>
<isSupportWiredandWirelessTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC)-->true
</isSupportWiredandWirelessTrafficMonitor>
<isSupport4G>
    <!--ro, opt, bool, whether the device supports 4G module-->true
</isSupport4G>
<isSupport4GConfig>
    <!--ro, opt, bool, whether the device supports 4G configuration-->true
</isSupport4GConfig>
<isSupportGRIDServer>
    <!--ro, opt, bool, N/A-->true
</isSupportGRIDServer>
<isSupportGRIDInfo>
    <!--ro, opt, bool, N/A-->true
</isSupportGRIDInfo>
<isSupportSipTransmit>
    <!--ro, opt, bool, whether the device supports configuring SIP transmission parameters, desc:GET|PUT /ISAPI/System/Network/SIP/transmit?format=json--
>true
</isSupportSipTransmit>
<isSupportSip>
    <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
</isSupportSip>
<isSupportWifiCascade>
    <!--ro, opt, bool, whether the device supports enabling automatic Wi-Fi cascading-->true
</isSupportWifiCascade>
<isSupportRelativeInfo>
    <!--ro, opt, bool, whether the device supports getting the information about the connected devices-->true
</isSupportRelativeInfo>
<isSupportANR>
    <!--ro, opt, bool, whether the device supports automatic network replenishment, desc:related URI: /ISAPI/System/Network/ANR-->true
</isSupportANR>
<isSupportEmailEncrypt>
    <!--ro, opt, bool, whether the device supports email encryption configuration-->true
</isSupportEmailEncrypt>
<isSupportPortFilter>
    <!--ro, opt, bool, whether the device supports port filtering-->true
</isSupportPortFilter>
<isSupportPingDeny>
    <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
</isSupportPingDeny>
<isSupportDOSDeny>
    <!--ro, opt, bool, whether the device supports configuring parameters to prevent the DoS attack-->true
</isSupportDOSDeny>
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether the device supports searching for the network status-->true
</isSupportNetworkStatus>
<isSupportEZVIZQRCode>
    <!--ro, opt, bool, whether the device supports getting the EZ QR code, desc:/ISAPI/System/Network/EZVIZ/QRCode?format=json-->true
</isSupportEZVIZQRCode>

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</isSupportEZVIZUnbind>
  <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
</isSupportEZVIZUnbind>
<isSupportPingTest>
  <!--ro, opt, bool, whether the device supports ping test, desc:related URI: /ISAPI/System/Network/pingtest/capabilities-->true
</isSupportPingTest>
<isSupportMOTCloudPlatform>
  <!--ro, opt, bool, whether the device supports configuring parameters for accessing the cloud platform service, desc:related URI:
/ISAPI/System/Network/Traffic/MOTCloudPlatform/capabilities?format=json-->true
</isSupportMOTCloudPlatform>
<isSupportSpectrumScan>
  <!--ro, opt, bool, whether the device supports frequency spectrum scanning, desc:related URI:
/ISAPI/System/Network/interface/<netCardID>/spectrumScan/capabilities?format=json-->true
</isSupportSpectrumScan>
<isSupportFRP>
  <!--ro, opt, bool, whether the device supports FRP (Fast Reverse Proxy), desc:related URI: /ISAPI/Syetem/Network/FRP?format=json-->true
</isSupportFRP>
<isSupport5GModule>
  <!--ro, opt, bool, whether the device supports 5G module configuration, desc:related URI: /ISAPI/System/Network/5GModule/capabilities?format=json-->true
</isSupport5GModule>
<isSupportWirelessDialRoute>
  <!--ro, opt, bool, whether the device supports configuring parameters of route forwarding rules of wireless dial-up, desc:related URI:
/ISAPI/System/Network/WirelessDia/route/capabilities?format=json-->true
</isSupportWirelessDialRoute>
<isSupportNPlusOne>
  <!--ro, opt, bool, whether the device supports N+1 protocol-->true
</isSupportNPlusOne>
<supportEnvironmentalStandards opt="HJ212">
  <!--ro, opt, string, environmental monitoring standard, attr:opt{req, string}, desc:N/A-->true
</supportEnvironmentalStandards>
<isSupportIntranetSegment>
  <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/intranetSegment/capabilities?
format=json-->true
</isSupportIntranetSegment>
<isSupportDHCPServer>
  <!--ro, opt, bool, whether the device supports DHCP service configuration, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/DHCPServer/capabilities?
format=json-->true
</isSupportDHCPServer>
<isSupportClusterIntercom>
  <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/clusterIntercom/capabilities?
format=json-->true
</isSupportClusterIntercom>
<isSupportHydrologicMonitoring>
  <!--ro, opt, bool, whether the device supports hydrologic monitoring data communication, desc:if this function is supported, this node will be returned
and its value is true; otherwise, this node will be returned and its value is false or not be returned. Related URI:
/ISAPI/System/Network/HydrologicMonitoring/capabilities?format=json-->true
</isSupportHydrologicMonitoring>
<isSupportGantryETCServer>
  <!--ro, opt, bool, N/A, desc:N/A-->true
</isSupportGantryETCServer>
<RoutePolicy>
  <!--ro, opt, object, policy routing-->
  <isSupportRoutePolicy>
    <!--ro, opt, bool, whether the device supports configuring the policy routing parameters-->true
  </isSupportRoutePolicy>
  <isRebootHint>
    <!--ro, opt, bool, whether to alarm to reboot after configuration-->true
  </isRebootHint>
</RoutePolicy>
<isWirelessDialMuxedWithWlanAndWifi>
  <!--ro, opt, bool, whether the wireless dial-up is mutually exclusive with the WLAN hotspot and Wi-Fi-->true
</isWirelessDialMuxedWithWlanAndWifi>
<isSupportDMZCfg>
  <!--ro, opt, bool, whether the device supports configuring DMZ host, desc:related URI: /ISAPI/System/Network/DMZCfg/capabilities?format=json-->true
</isSupportDMZCfg>
<isSupportSpeechTranscription>
  <!--ro, opt, bool, whether the device supports accessing the speech-to-text transcription platform, desc:if this function is supported, this node will
be returned and its value is true; otherwise, this node will be returned and its value is false or not be returned. Related URI:
/ISAPI/System/Network/speechTranscriptionPlatform/capabilities?format=json-->true
</isSupportSpeechTranscription>
<isSupportAsynPingTest>
  <!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and packet
loss rate of the network using the ping command for knowing about the network condition. Related URI: /ISAPI/System/Network/asynPingTest/capabilities?
format=json-->true
</isSupportAsynPingTest>
<isSupportBasebandCapture>
  <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and the
base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
</isSupportBasebandCapture>
<isSupportCellularNetworkSpeedTest>
  <!--ro, opt, bool, whether the device supports testing the connection speed between the device and the cellular network, desc:you can test the
connection speed of the network where the current device is. Related URI: /ISAPI/System/Network/cellularNetwork/speedTest/capabilities?format=json-->true
</isSupportCellularNetworkSpeedTest>
<isSupportCellularNetworkCellInfo>
  <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the cellular
network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&security=<security>&iv=<iv>-->true
</isSupportCellularNetworkCellInfo>
<isSupportMultiBandFrequencyScanning>
  <!--ro, opt, bool, whether the device supports scanning all frequency bands of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/multiBandFrequencyScanning/capabilities?format=json-->true

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</isSupportMultiBandFrequencyScanning>
<isSupportCellularNetworkLockBand>
  <!--ro, opt, bool, whether the device supports accessing the Locked frequency band of the cellular network, desc:related URI:
  /ISAPI/System/Network/cellularNetwork/LockBand/capabilities?format=json-->true
</isSupportCellularNetworkLockBand>
<isSupportRadioFrequencyFluctuation>
  <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
  /ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
</isSupportRadioFrequencyFluctuation>
<isSupportBasebandBlock>
  <!--ro, opt, bool, whether the device supports the unauthorized List of the base station, desc:related URI:
  /ISAPI/System/Network/cellularNetwork/basebandBlock/capabilities?format=json-->true
</isSupportBasebandBlock>
<isSupportWifiQuality>
  <!--ro, opt, bool, whether the device supports getting the device's Wi-Fi quality information, desc:related URI: /ISAPI/System/Network/wifiQuality?
  format=json-->true
</isSupportWifiQuality>
<isSupportWiredNetworkQuality>
  <!--ro, opt, bool, whether the device supports getting the wired network quality information, desc:related URI:
  /ISAPI/System/Network/wiredNetworkQuality?format=json-->true
</isSupportWiredNetworkQuality>
<isSupportQoSBehaviorCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS traffic behavior, desc:related URI: /ISAPI/System/Network/qos/BehaviorCfg/capabilities?
  format=json-->true
</isSupportQoSBehaviorCfg>
<isSupportQoSClassifierCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS traffic types, desc:related URI: /ISAPI/System/Network/qos/ClassifierCfg/capabilities?
  format=json-->true
</isSupportQoSClassifierCfg>
<isSupportQoSPolicyCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS policy, desc:related URI: /ISAPI/System/Network/qos/PolicyCfg/capabilities?format=json--
  >true
</isSupportQoSPolicyCfg>
<isSupportQoSPolicyVLANCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS policy VLAN, desc:related URI: /ISAPI/System/Network/qos/PolicyVLANCfg/capabilities?
  format=json-->true
</isSupportQoSPolicyVLANCfg>
<isSupportQoSPortRateLimiting>
  <!--ro, opt, bool, whether the device supports configuring port speed Limit, desc:related URI: /ISAPI/System/Network/qos/portRateLimiting/capabilities?
  format=json-->true
</isSupportQoSPortRateLimiting>
<isSupportQoSQueueAlgorithmCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS queue algorithm, desc:related URI:
  /ISAPI/System/Network/qos/queueAlgorithmCfg/capabilities?format=json-->true
</isSupportQoSQueueAlgorithmCfg>
<isSupportQoSPortPriorityCfg>
  <!--ro, opt, bool, whether the device supports QoS port priority configuration, desc:related URI:
  /ISAPI/System/Network/qos/portPriorityCfg/capabilities?format=json-->true
</isSupportQoSPortPriorityCfg>
<isSupportQoS802.1pPriorityMapping>
  <!--ro, opt, bool, whether the device supports DSCP and 802.1p mapping priority configuration, desc:related URI:
  /ISAPI/System/Network/qos/802.1pPriorityMapping/capabilities?format=json-->true
</isSupportQoS802.1pPriorityMapping>
<isSupportQoSSDCPEndPriorityMapping>
  <!--ro, opt, bool, whether the device supports DSCP mapping priority configuration, desc:related URI:
  /ISAPI/System/Network/qos/SDCPEndPriorityMapping/capabilities?format=json-->true
</isSupportQoSSDCPEndPriorityMapping>
<isSupportQoSLoaclPriorityMapping>
  <!--ro, opt, bool, whether the device supports 802.1p and Local mapping priority configuration, desc:related URI:
  /ISAPI/System/Network/qos/LoaclPriorityMapping/capabilities?format=json-->true
</isSupportQoSLoaclPriorityMapping>
<isSupportStopPingtest>
  <!--ro, opt, bool, whether to stop PingTest, desc:related URI: /ISAPI/System/Network/StopPingtest?format=json-->true
</isSupportStopPingtest>
<isSupportSNMPView>
  <!--ro, opt, bool, whether the device supports SNMP view, desc:related URI: /ISAPI/System/Network/SNMP/ViewCfg/capabilities?format=json-->true
</isSupportSNMPView>
<isSupportSNMPUserGroup>
  <!--ro, opt, bool, whether the device supports SNMP user groups, desc:related URI: /ISAPI/System/Network/SNMP/UserGroup/capabilities?format=json-->true
</isSupportSNMPUserGroup>
<isSupportSNMPBasicInfo>
  <!--ro, opt, bool, whether the device supports the basic information of SNMP, desc:related URI: /ISAPI/System/Network/SNMP/BasicInfo/capabilities?
  format=json-->true
</isSupportSNMPBasicInfo>
<isSupportRFIDTagCheckTest>
  <!--ro, opt, bool, whether the device supports the RFID tag disk storage performance test, desc:related URI:
  /ISAPI/System/Network/RFIDData/GetTagCheckTestInfo?format=json-->true
</isSupportRFIDTagCheckTest>
<isSupportQoSSDCPEndLoaclPriorityMapping>
  <!--ro, opt, bool, whether the device supports DSCP and Local mapping priority configuration, desc:related URI:
  /ISAPI/System/Network/qos/SDCPEndLoaclPriorityMapping/capabilities?format=json-->true
</isSupportQoSSDCPEndLoaclPriorityMapping>
<isSupportQoS802.1pto802.1pPriorityMapping>
  <!--ro, opt, bool, whether the device supports 802.1p mapping priority configuration, desc:related URI:
  /ISAPI/System/Network/qos/802.1pto802.1pPriorityMapping/capabilities?format=json-->true
</isSupportQoS802.1pto802.1pPriorityMapping>
<isSupportNFCAllowList>
  <!--ro, opt, bool, whether the device supports NFC network allowlist management, desc:related URI: /ISAPI/System/Network/NFC/allowList/capabilities?
  format=json-->true
</isSupportNFCAllowList>
<isSupportLinkTestStart>
  <!--ro, opt, bool, whether the device supports enabling network Link test, desc:related URI:
  /ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
</isSupportLinkTestStart>

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</isSupportLinkTestStart>
<isSupportLinkTestResult>
  <!--ro, opt, bool, whether the device supports getting the result of network Link test, desc:related URI:
  /ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Result?format=json-->true
</isSupportLinkTestResult>
<isSupportGetAsynTestEmailResult>
  <!--ro, opt, bool, whether the device supports getting email testing result asynchronously, desc:POST
  /ISAPI/System/Network/Email/GetAsynTestEmailResult?format=json-->true
</isSupportGetAsynTestEmailResult>
<isSupportTestAsynEmail>
  <!--ro, opt, bool, whether the device supports asynchronously testing email, desc:POST /ISAPI/System/Network/Email/TestAsynEmail?format=json-->true
</isSupportTestAsynEmail>
<isSupportModbusAccess>
  <!--ro, opt, bool, whether the device supports Modbus accessing parameters, desc:GET/PUT /ISAPI/System/Network/ModbusAccess?format=json-->true
</isSupportModbusAccess>
<isSupportNetworkModuleUpgrade>
  <!--ro, opt, bool, whether the device supports upgrading network module, desc:including importing upgrade package
  (/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
  (/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrading
  (/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json), get upgrade status
  (/ISAPI/System/Network/WirelessDial/GetUpgradeNetworkModuleStatus?format=json)-->true
</isSupportNetworkModuleUpgrade>
<isSupportNetworkModuleVersionInfo>
  <!--ro, opt, bool, whether the device supports getting the version information of network module, desc:GET
  /ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
</isSupportNetworkModuleVersionInfo>
<isSupportGreenShieldPlatform>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/GreenShieldPlatform/GreenShieldPlatform?format=json-->true
</isSupportGreenShieldPlatform>
<isSupportFusionUniversalApplicationPlatCfg>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/FusionUniversalApplicationPlatform/FusionUniversalApplicationPlatCfg?format=json-->true
</isSupportFusionUniversalApplicationPlatCfg>
<isSupportNPQControlStrategy>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/NPQMgr/NPQControlStrategy?format=json-->true
</isSupportNPQControlStrategy>
<isSupportNAT>
  <!--ro, opt, bool, whether NAT is supported, desc:GET/PUT /ISAPI/System/Network/nats-->true
</isSupportNAT>
<isSupportGetCurrentlyUsingNetworkCardInfo>
  <!--ro, opt, bool, whether the device supports getting the current NIC information, desc:GET
  /ISAPI/System/Network/WirelessDial/GetCurrentlyUsingNetworkCardInfo?format=json-->true
</isSupportGetCurrentlyUsingNetworkCardInfo>
<isSupportAutoRecampParams>
  <!--ro, opt, bool, whether the device supports parameters of auto re-camping on a regular basis, desc:GET/PUT
  /ISAPI/System/Network/WirelessDial/AutoRecampParams?format=json-->true
</isSupportAutoRecampParams>
<isSupportDSTPServerCfg>
  <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
  format=json-->true
</isSupportDSTPServerCfg>
<isSupportJTT1008ServerParams>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/JTT1008Access/JTT1008ServerParams?format=json-->true
</isSupportJTT1008ServerParams>
<isSupportImportNARIKeyFile>
  <!--ro, opt, bool, N/A, desc:PUT /ISAPI/System/Network/NAREncryptionKeyMgr/ImportNARIKeyFile?format=json-->true
</isSupportImportNARIKeyFile>
<isSupportExportNARIKeyFile>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/System/Network/NAREncryptionKeyMgr/ExportNARIKeyFile?format=json-->true
</isSupportExportNARIKeyFile>
<isSupportNAREncryptionParams>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/NAREncryptionKeyMgr/NAREncryptionParams?format=json-->true
</isSupportNAREncryptionParams>
<isSupportDSTPProtocolCap>
  <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolCap?format=json-->true
</isSupportDSTPProtocolCap>
<isSupportIPAdaptiveParams>
  <!--ro, opt, bool, whether the device supports self-adaptive IP address parameters, desc:GET/PUT /ISAPI/System/Network/IPAdaptiveParams?format=json--
  >true
</isSupportIPAdaptiveParams>
<isSupportPowerCenterPlatform>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/PowerCenterAccess/PowerCenterPlatform?format=json-->true
</isSupportPowerCenterPlatform>
<isSupportTestPowerCenterPlatformConnect>
  <!--ro, opt, bool, N/A, desc:POST /ISAPI/System/Network/PowerCenterAccess/TestPowerCenterPlatformConnect?format=json-->true
</isSupportTestPowerCenterPlatformConnect>
<isSupportModbusTestSubordinateStatus>
  <!--ro, opt, bool, whether the device supports Modbus testing the status of the specified client., desc:POST
  /ISAPI/System/Network/ModbusTestSubordinateStatus?format=json-->true
</isSupportModbusTestSubordinateStatus>
<isSupportGetModbusAndRegisterMap>
  <!--ro, opt, bool, whether the device supports get Modbus uploaded content and the mapping relation between the uploaded content and register quantity,
  desc:GET /ISAPI/System/Network/GetModbusAndRegisterMap?format=json-->true
</isSupportGetModbusAndRegisterMap>
<isSupportAddFRPDynamicAllocationMapPort>
  <!--ro, opt, bool, whether the device supports batch adding FRP dynamic allocation mapping ports, desc:POST
  /ISAPI/System/Network/NATPenetration/AddFRPDynamicAllocationMapPort?format=json-->true
</isSupportAddFRPDynamicAllocationMapPort>
<isSupportGetFRPDynamicAllocationMapPort>
  <!--ro, opt, bool, whether the device supports batch getting FRP dynamic allocation mapping ports, desc:GET
  /ISAPI/System/Network/NATPenetration/GetFRPDynamicAllocationMapPort?format=json-->true
</isSupportGetFRPDynamicAllocationMapPort>
<isSupportUSBNetworkCard>
  <!--ro, opt, bool, whether the device supports USB NIC-->true
</isSupportUSBNetworkCard>

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<isSupportOptimumOperatorParams>
  <!--ro, opt, bool, whether the device supports optimal carrier parameters, desc:GET/PUT /ISAPI/System/Network/WirelessDial/OptimumOperatorParams?
format=json-->true
</isSupportOptimumOperatorParams>
<isSupportERTCJoin>
  <!--ro, opt, bool, whether the device supports ERTC access, desc:PUT /ISAPI/System/Network/ERTCAccess/ERTCJoin?format=json-->true
</isSupportERTCJoin>
<isSupportERTCQuit>
  <!--ro, opt, bool, whether the device supports ERTC quitting, desc:PUT /ISAPI/System/Network/ERTCAccess/ERTCQuit?format=json-->true
</isSupportERTCQuit>
<isSupportLoRaCommunicationParams>
  <!--ro, opt, bool, whether the device supports LoRa protocol communication parameters, desc:GET/PUT
/ISAPI/System/Network/LoRAAccess/LoRaCommunicationParams?format=json-->true
</isSupportLoRaCommunicationParams>
<isSupportSetSIMCardSwitch>
  <!--ro, opt, bool, whether the device supports SIM NIC switch configuration, desc:PUT /ISAPI/System/Network/WirelessDial/SetSIMCardSwitch?format=json--
>true
</isSupportSetSIMCardSwitch>
<isSupportDialSMSAutoSwitchParams>
  <!--ro, opt, bool, whether the device supports configuring the NIC suto-switch parameter configuration, desc:GET/PUT
/ISAPI/System/Network/WirelessDial/DialSMSAutoSwitchParams?format=json, DialSMSAutoSwitchParams-->true
</isSupportDialSMSAutoSwitchParams>
<isSupportBatchModifyDeviceIPAddress>
  <!--ro, opt, bool, whether the device supports batch configuring device network addresses, desc:PUT /ISAPI/System/Network/BatchModifyDeviceIPAddress?
format=json-->true
</isSupportBatchModifyDeviceIPAddress>
<isSupportWirelessConfigListAsyncParams>
  <!--ro, opt, bool, whether the device supports wireless configuration list async parameters, desc:GET/PUT
/ISAPI/System/Network/WirelessConfigListAsyncParams?format=json-->true
</isSupportWirelessConfigListAsyncParams>
<isSupportRestartPaging>
  <!--ro, opt, bool, whether the device supports paging again, desc:PUT /ISAPI/System/Network/LoRAAccess/RestartPaging?format=json-->true
</isSupportRestartPaging>
<isSupportGetCurrentWirelessServerMode>
  <!--ro, opt, bool, whether the device supports getting the current network hotspot mode, desc:GET
/ISAPI/System/Network/WLANServer/GetCurrentWirelessServerMode?format=json-->true
</isSupportGetCurrentWirelessServerMode>
<isSupportNetworkInterfaceMaterielInfoDetect>
  <!--ro, opt, bool, whether the device supports NIC material information detection, desc:PUT /ISAPI/System/Network/NetworkInterfaceMaterielInfoDetect?
format=json-->true
</isSupportNetworkInterfaceMaterielInfoDetect>
<isSupportMQTTAccessCfg>
  <!--ro, opt, bool, whether the device supports MQTT access configuration, desc:GET/PUT /ISAPI/System/Network/MQTTAccess/MQTTAccessCfg?format=json-->true
</isSupportMQTTAccessCfg>
<isSupportModifySavedWirelessList>
  <!--ro, opt, bool, whether the device supports editing Wi-Fi list saved to specified network interface, desc:PUT
/ISAPI/System/Network/ModifySavedWirelessList?format=json-->true
</isSupportModifySavedWirelessList>
<isSupportGetSavedWirelessList>
  <!--ro, opt, bool, whether the device supports getting Wi-Fi list saved to specified network interface, desc:POST
/ISAPI/System/Network/GetSavedWirelessList?format=json-->true
</isSupportGetSavedWirelessList>
<isSupportWirelessCommunication433AccessParams>
  <!--ro, opt, bool, whether the device supports 433 protocol communication parameters, desc:GET/PUT
/ISAPI/System/Network/WirelessCommunication433Access/WirelessCommunication433AccessParams?format=json-->true
</isSupportWirelessCommunication433AccessParams>
<isSupportTSOverWebSocket>
  <!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
</isSupportTSOverWebSocket>
<isSupportSouthernPowerGridServer>
  <!--ro, opt, bool, N/A, desc:N/A-->true
</isSupportSouthernPowerGridServer>
<isSupportGeologicalHazardMonitoringCfg>
  <!--ro, opt, bool, whether the device supports geological hazard detection service parameters, desc:GET/PUT
/ISAPI/System/Network/GeologicalHazardMonitoring/GeologicalHazardMonitoringCfg?format=json-->true
</isSupportGeologicalHazardMonitoringCfg>
<isSupportFixedNetworkInterfaceCfg>
  <!--ro, opt, bool, whether the device supports network interface fixed address allocation parameters, desc:GET/PUT
/ISAPI/System/Network/FixedNetworkInterfaceCfg?format=json-->true
</isSupportFixedNetworkInterfaceCfg>
<isSupportSearchWiFiChannelDetectionState>
  <!--ro, opt, bool, whether the device supports searching for Wi-Fi channel detection status, desc:POST
/ISAPI/System/Network/SearchWiFiChannelDetectionState?format=json-->true
</isSupportSearchWiFiChannelDetectionState>
<isSupportStartWiFiChannelDetection>
  <!--ro, opt, bool, whether the device supports enabling Wi-Fi channel detection, desc:POST /ISAPI/System/Network/StartWiFiChannelDetection?format=json--
>true
</isSupportStartWiFiChannelDetection>
<isSupportStartManualChannelAdaptation>
  <!--ro, opt, bool, whether the device supports enabling manual configuration of channel self-adaption, desc:POST
/ISAPI/System/Network/StartManualChannelAdaptation?format=json-->true
</isSupportStartManualChannelAdaptation>
<isSupportSearchManualChannelAdaptationState>
  <!--ro, opt, bool, whether the device supports searching for manual channel configuration self-adaption status, desc:POST
/ISAPI/System/Network/SearchManualChannelAdaptationState?format=json-->true
</isSupportSearchManualChannelAdaptationState>
<isSupportLoRaWorkMode>
  <!--ro, opt, bool, whether the device supports LoRa module working mode, desc:GET/PUT /ISAPI/System/Network/LoRAAccess/LoRaWorkMode?format=json-->true
</isSupportLoRaWorkMode>
<isSupportPLCParams>
  <!--ro, opt, bool-->true
</isSupportPLCParams>
<isSupportWiFiRoamingConfigParams>

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    <!--ro, opt, bool-->true
</isSupportWiFiRoamingConfigParams>
<isSupportPLCNetworkingStatus>
    <!--ro, opt, bool-->true
</isSupportPLCNetworkingStatus>
<isSupportAddPLCDiagnosis>
    <!--ro, opt, bool-->true
</isSupportAddPLCDiagnosis>
<isSupportGetPLCDiagnosisProgress>
    <!--ro, opt, bool-->true
</isSupportGetPLCDiagnosisProgress>
<isSupportSearchOutlookAuthResult>
    <!--ro, opt, bool-->true
</isSupportSearchOutlookAuthResult>
<isSupportRequestOutlookAuth>
    <!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
    <isSupportRequestOutlookAuth>
    <isSupportOutlookEmailList>
        <!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
    </isSupportOutlookEmailList>
<isSupportNARIEncryptionPortParams>
    <!--ro, opt, bool-->true
</isSupportNARIEncryptionPortParams>
<isSupportAlphaPlatformParams>
    <!--ro, opt, bool-->true
</isSupportAlphaPlatformParams>
<isSupportSouthernUniverseIOTAccessParams>
    <!--ro, opt, bool-->true
</isSupportSouthernUniverseIOTAccessParams>
<isSupportModifyGeologicalHazardMonitoringParams>
    <!--ro, opt, bool-->true
</isSupportModifyGeologicalHazardMonitoringParams>
<isSupportGeologicalHazardMonitoringListParams>
    <!--ro, opt, bool-->true
</isSupportGeologicalHazardMonitoringListParams>
<isSupportServerCfg>
    <!--ro, opt, bool-->true
</isSupportServerCfg>
<isSupportDelimiterCfg>
    <!--ro, opt, bool-->true
</isSupportDelimiterCfg>
<isSupportNetworkSignalEnhancementParams>
    <!--ro, opt, bool-->true
</isSupportNetworkSignalEnhancementParams>
<isSupportISUPServiceParams>
    <!--ro, opt, bool-->true
</isSupportISUPServiceParams>
<isSupportWirelessDialSchedule>
    <!--ro, opt, bool-->true
</isSupportWirelessDialSchedule>
<isSupportISUPServiceParamsTest>
    <!--ro, opt, bool-->true
</isSupportISUPServiceParamsTest>
<isSupportVoiceTalkOverWebSocket>
    <!--ro, opt, bool-->true
</isSupportVoiceTalkOverWebSocket>
<isSupportPlaybackOverWebSocket>
    <!--ro, opt, bool-->true
</isSupportPlaybackOverWebSocket>
<isSupportALSOverWebSocket>
    <!--ro, opt, bool-->true
</isSupportALSOverWebSocket>
<isSupportPUSHCfg>
    <!--ro, opt, bool-->true
</isSupportPUSHCfg>
<RTDROverWebSocket>
    <!--ro, opt, object-->
    <isSupportGasSpectrumRealTimeData>
        <!--ro, opt, bool-->true
    </isSupportGasSpectrumRealTimeData>
    <isSupportDangerousGasSensorRealTimeData>
        <!--ro, opt, bool-->true
    </isSupportDangerousGasSensorRealTimeData>
    <isSupportComposeWindConditionRealTimeData>
        <!--ro, opt, bool-->true
    </isSupportComposeWindConditionRealTimeData>
    <isSupportMeteorologySensorRealTimeData>
        <!--ro, opt, bool-->true
    </isSupportMeteorologySensorRealTimeData>
</RTDROverWebSocket>
<isSupportWiFiAutoConnectParams>
    <!--ro, opt, bool-->true
</isSupportWiFiAutoConnectParams>
<isSupportAccessParams>
    <!--ro, opt, bool-->true
</isSupportAccessParams>
<isSupportStartCellNetSelfDiagnosis>
    <!--ro, opt, bool-->true
</isSupportStartCellNetSelfDiagnosis>
<isSupportCellNetSelfDiagnosisStatus>
    <!--ro, opt, bool-->true
</isSupportCellNetSelfDiagnosisStatus>

```

```

</isSupportMDNSParams>
<!--ro, opt, bool-->true
</isSupportMDNSParams>
<WebSocketPlaybackCap>
<!--ro, opt, object-->
<cmdType opt="speed,pause,resume,single,stop,seek,startExtracting,stopExtracting,transcoding">
<!--ro, req, string, attr:opt{opt, string}-->speed
</cmdType>
</WebSocketPlaybackCap>
<isSupportDevGetRTCCConfigurationOperationResult>
<!--ro, opt, bool-->true
</isSupportDevGetRTCCConfigurationOperationResult>
<isSupportAddICECandidate>
<!--ro, opt, bool-->true
</isSupportAddICECandidate>
<isSupportRTCCConfiguration>
<!--ro, opt, bool-->true
</isSupportRTCCConfiguration>
<isSupportOpenDataChannel>
<!--ro, opt, bool-->true
</isSupportOpenDataChannel>
<isSupportHTTPSecurityHeader>
<!--ro, opt, bool-->true
</isSupportHTTPSecurityHeader>
</NetworkCap>

```

9.1.9.8 Set SSH parameters

Request URL

PUT /ISAPI/System/Network/ssh?readerID=<readerID>&type=<type>&SOCChipID=<SOCChipID>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--
type	enum	--
SOCChipID	string	--
controllerID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<SSH xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
<!--wo, opt, object, attr:version{opt, string, protocolVersion}-->
<enabled>
<!--wo, req, bool, whether to enable the function-->true
</enabled>
<port>
<!--wo, opt, int-->22
</port>
</SSH>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional Error, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

9.1.9.9 Reboot device

Request URL

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
devIndex	string	Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json.
module	enum	The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera).
loginPassword	string	--
controllerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <errCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </errCode>
  <errDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </errDevSelfEx>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
2	Device Busy	upgrading	0x20000003	--
3	设备错误	applnUpgrade	0x3000001B	--
3	Device Error	formattingTheDisk	0x30008051	--
3	Device Error	upgradingThe4g5gModule	0x30008052	--
3	Device Error	rebootingTheDevice	0x30008053	--

9.1.10 Time Management

9.1.10.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message


```

<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite"
    (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization
    source), "GB28181" (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, Local time, range:[0,256], dep:and,{$.Time.timeMode,eq>manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq>manual},{$.Time.timeMode,eq,NTP}, desc:dayLight saving time configuration in
    time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently
    is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform--
    EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in
    GB28181List, reLated URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>

```

9.1.10.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite" (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization source), "GB28181" (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>

```

statusCode	statusString	subStatusCode
4	Invalid Operation	pleaseSyncToAReasonableTimeOtherwiseDeviceTimeExceptionMayOccurWhichMayCause
4	Invalid Operation	theTimeZonesLockedPleaseUnlockItThroughTheLocalGuiInterface
4	Invalid Operation	theTimeDoesNotComplyWithTheRules
4	Invalid Operation	theCurrentNtpTimingsIsValidTimingsIsNotAllowed
4	Invalid Operation	ntpTimelsInInvalidStatusProtection
6	Invalid Content	verifyingUseTimingError

9.1.10.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,QWIF,ALL(任何支持的校时方式都允许校时),NONE(不允许校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite" (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization source), "GB28181" (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:"Local" (Local time), "UTC" (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh:mm,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device's NTP service information with the camera, desc:related URI: /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
</Time>
```

```

<isSupportNTPService>
  <!--ro, opt, bool-->true
</isSupportNTPService>
<isSupportTimeZoneListInfo>
  <!--ro, opt, bool-->true
</isSupportTimeZoneListInfo>
<isSupportAutoEnterSummerTime>
  <!--ro, opt, bool-->true
</isSupportAutoEnterSummerTime>
<ethernetPort min="0" max="255">
  <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
</ethernetPort>
<IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/D
akar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Afr
ica/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,Americ
a/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,A
merica/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Tur
k,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,
America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,
America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monter
rey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Sal
em,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,Amer
ica/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bah
rain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta
,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyan
g,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlant
ic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbou
rne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe
/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Euro
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Amer
ica/Lower_Principes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Oyuzlorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apiia">
  <!--ro, opt, enum, IANA synchronization, subtype:string, attr:opt{opt, string}-->Africa/Abidjan
</IANA>
<isSupportSearchDeviceIANAList>
  <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
</isSupportSearchDeviceIANAList>
<windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Yekaterinburg Standard Time,Pakistan Standard Time,Oyuzlorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
  <!--ro, opt, enum, subtype:string, attr:opt{opt, string}-->Dateline Standard Time
</windowsZone>
<isSupportTimeStateParam>
  <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
</isSupportTimeStateParam>
<localTimeDST min="0" max="256">
  <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
</localTimeDST>
<isSupportStandard>
  <!--ro, opt, bool-->true
</isSupportStandard>
<isSupportPTPParams>
  <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
</isSupportPTPParams>
<isSupportTimeZoneLock>
  <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
</isSupportTimeZoneLock>
</Time>

```

9.1.10.4 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information List, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10000], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

9.1.10.5 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10000], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->>false
    </enabled>
  </NTPServer>
</NTPServerList>
```

9.1.10.6 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10000], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->>false
  </enabled>
</NTPServer>
```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

9.1.10.7 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10000], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->>false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

9.1.10.8 Get the configuration capability of a specific NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>

```

9.1.10.9 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information List, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>

```

9.1.11 Upgrade Management

9.1.11.1 Upgrade devices

Request URL

POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpdateAuxDevice=<isUpdateAuxDevice>&confirmEnabled=<confirmEnabled>&upgradeToken=<upgradeToken>

Query Parameter

Parameter Name	Parameter Type	Description
type	enum	device type 1.alarm device: "keypad" (keypad), "wirelessRecv" (wireless receiver module), "wiredZone" (wired zone module), "sirenIndoor" (indoor sounder), "sirenOutdoor" (outdoor sounder), "sirenAudio" (sounder two-way audio), "repeater" (repeater), "bluetoothModule" (bluetooth module), "single" (single output module), "wallSwitch" (wall mounted switch), "smartPlug" (smart plug), "detector" (detector), "transmitter" (transmitter peripheral), "remoteCtrl" (keyfob), "subModule" (sub module), "dispModule" (display module), "netReader" (network card reader), "faceModule" (face module), "touchScreenModule" (touch screen module), "temperatureModule" (temperature measurement module), "lockControlBoard" (lock control board), "networkZoneModule" (network zone module), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "electricLock" (electric lock), "heatingModule" (heating module), "RS485Module" (RS-485 module), "QRCodeModule" (QR code module), "electricGenie" (electric Genie), "zigbeeModule" (zigbee module), "PMRModule" (PMR module), "networkInputAndOutputModule" (network input and output module via network TAP) 2. access control device: "cardReader" (485 card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAAlgorithmProgram" (fingerprint algorithm program of card reader), "bluetoothModule" (bluetooth module), "MCU"-MCU upgrade, "temperatureModule" (temperature measurement module), "faceModule" (face module), "touchScreenModule" (touch screen module), "netReader" (network card reader), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "QRCodeModule" (QR code module) 3. intelligent cabinet: "lockControlBoard" (lock control board) 4. "ledReceiverCard"-receiving card; 5. "subsystem"-sub system; 6. radar-assisted device: "warningScreenFirmware"-firmware upgrade of pre-alarm screen, "warningScreenFont"-font library upgrade of pre-alarm screen, "warningScreenVoice"-audio upgrade of pre-alarm screen; 7. "electricGenie"-electric Genie
moduleAddress	string	module address is an exclusive node for Ax Hybrid example: keypad: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; relay: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; sounder: GET /ISAPI/SecurityCP/Configuration/wirelessSiren?format=json; output module: GET /ISAPI/SecurityCP/Configuration/outputModules?format=json; extension module: GET /ISAPI/SecurityCP/Configuration/extensionModule?format=json; zone: GET /ISAPI/SecurityCP/Configuration/zones?format=json
indexID	string	device No. 1. AX Hybrid Pro and wireless security control upgrade by ID; 2. access control devices upgrade by ID; 3. intelligent cabinet upgrade by IS
devIndex	string	--
isUpdateAuxDevice	string	--
confirmEnabled	string	--
upgradeToken	string	--

Request Message

Binary Data

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:it describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information, range:[0,1024], desc:this node is used for debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCo
2	Device Busy	noMemory	0x2000
2	Device Busy	upgrading	0x2000
2	Device Busy	networkError	0x2000
3	Device Error	badFlash	0x3000
3	Device Error	RFJammingDetected	0x3000
3	Device Error	devicesForwardedByTheRepeater	0x3000
3	Device Error	upgradeProgramExecutionError	0x3000
3	Device Error	pausingStorageRecordingServiceFailed	0x3000
3	Device Error	pausingStorageServiceFailed	0x3000
3	Device Error	hardwareCommunicationErrorOccurredDuringHardwareUpgradePleaseTryAgain	0x3000
3	Device Error	gettingDeviceFirmwareVersionFailedDuringHardwareUpgrade	0x3000
3	Device Error	unknownErrorOccurredDuringHardwareUpgrade	0x3000
3	Device Error	someHardwareComponentsFailedToUpgrade	0x3000
3	Device Error	failedToReadUpgradePackageData	0x3000
3	Device Error	upgradeServerError	0x3000
4	Invalid Operation	notSupport	0x4000
4	Invalid Operation	updateLangUnmatched	0x4000
4	Invalid Operation	writeFileError	0x4000
4	Invalid Operation	noMemory	0x4000
4	Invalid Operation	noFlash	0x4000
4	Invalid Operation	armedStatus	0x4000

4	Invalid Operation	registerMode	0x4000
4	Invalid Operation	upgradePackageTransferFailed	0x4000
4	Invalid Operation	DetectorOrPeripheralUpgrading	0x4000
4	Invalid Operation	pleaseStopAllHeopApplicationsBeforeUpgradingTheFirmware	0x4000
4	Invalid Operation	theEquipmentIsCurrentlyAlarmingAndCannotBeUpgraded	0x4000
4	Invalid Operation	0x4000A85A	0x4000
6	Invalid Content	connectSreverFail	0x6000
6	Invalid Content	badVersion	0x6000
6	Invalid Content	badDevType	0x6000
6	Invalid Content	badLanguage	0x6000
6	Invalid Content	OEMCodeMismatch	0x6000
6	Invalid Content	versionMismatch	0x6000
6	Invalid Content	noUpgradableFileFoundInTheSpecifiedPathDuringHardwareUpgrade	0x6000
6	Invalid Content	noUpgradableHardwareDeviceDetectedDuringHardwareUpgrade	0x6000
6	Invalid Content	verifyingTheBinFileFailedDuringHardwareUpgrade	0x6000
6	Invalid Content	upgradeFileDatalsIncompatible	0x6000
6	Invalid Content	httpRequestError	0x6000
6	Invalid Content	upgradeFileDoesNotExist	0x6000
6	Invalid Content	upgradetokenError	0x6000
7	Reboot Required	rebootRequired	0x7000
7	Reboot Required	autoReboot	0x7000

9.1.11.2 Get the device upgrade progress

Request URL

GET /ISAPI/System/upgradeStatus?format=json&type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	--

Request Message

None

Response Message

```
{
  "requestURL": "/ISAPI/Streaming/channels/1",
  /*ro, opt, string, request URL*/
  "statusCode": "test",
  /*ro, req, string, status code*/
  "statusString": "test",
  /*ro, req, string, status description*/
  "subStatusCode": "test",
  /*ro, req, string, sub status code*/
  "errorCode": 1,
  /*ro, opt, int, This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
  "errorMsg": "ok",
  /*ro, opt, string, This field is required when the value of statusCode is not 1. Detailed error description of a certain parameter can be provided*/
  "upgrading": "TRUE",
  /*ro, opt, string, whether the device is upgrading: "TRUE" (upgrading), "FALSE" (not in upgrading)*/
  "percent": 22,
  /*ro, opt, int, upgrade progress (% complete)*/
  "idList": [
    /*ro, opt, array, ID list, subType:object*/
    {
      "id": "test",
      /*ro, req, string, analysis unit ID*/
      "percent": 22,
      /*ro, opt, int, upgrade progress (% complete)*/
      "status": "test",
      /*ro, opt, string, "backingUp" (backing up upgrade)*/
    }
  ]
}
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	fixPackVersionTooLowForUpgrade	0x4000A1CF	--

9.1.11.3 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	noMemory	0x4000202A	--

9.1.12 User Management

9.1.12.1 Log in to the device by digest

Request URL

GET /ISAPI/Security/userCheck

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<userCheck xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <statusValue>
    <!--ro, req, enum, status code, subType:int, desc:"200"(succeeded), "401"(failed)-->200
  </statusValue>
  <statusString>
    <!--ro, opt, enum, status, subType:string, desc:"OK", "Unauthorized"-->OK
  </statusString>
  <isDefaultPassword>
    <!--ro, opt, bool-->true
  </isDefaultPassword>
  <isRiskPassword>
    <!--ro, opt, bool-->true
  </isRiskPassword>
  <isActivated>
    <!--ro, opt, bool-->true
  </isActivated>
  <residualValidity>
    <!--ro, opt, int-->1
  </residualValidity>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:"unlock", "Locked"-->unlock
  </lockStatus>
  <unlockTime>
    <!--ro, opt, int, unlocking remaining time, unit:s-->1
  </unlockTime>
  <retryLoginTime>
    <!--ro, opt, int, remaining login attempts-->1
  </retryLoginTime>
</userCheck>
```

statusCode	statusString	subStatusCode	errorCode	description
1	OK	needServicePlatformAccountAuthenticationAndAuthroization	0x10000010	--
4	Invalid Operation	devicesLocked	0x4000A3E9	--
6	Invalid Content	forbiddenIP	0x60000070	--

9.1.12.2 Get all users' permissions

Request URL

GET /ISAPI/Security/UserPermission

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UserPermissionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, List of user permissions, subType:object, attr:version(req, string, protocolVersion)-->
  <UserPermission>
    <!--ro, req, object, user permissions-->
    <id>
      <!--ro, req, string, index, range:[1,32]-->test
    </id>
    <userID>
      <!--ro, req, string, user ID, range:[1,32]-->test
    </userID>
    <userType>
      <!--ro, req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission),
"operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)--
>admin
    </userType>
    <localPermission>
      <!--ro, opt, object, Local permission node-->
      <backup>
        <!--ro, opt, bool, whether to enable the backup permission (of all channels), desc:true (enable), false (disable)-->true
      </backup>
      <record>
        <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
      </record>
      <playBack>
        <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
      </playBack>
      <preview>
        <!--ro, opt, bool, whether to enable the Live view permission (of all channels), desc:true (enable), false (disable)-->true
      </preview>
      <privacyMaskClose>
        <!--ro, opt, bool-->>false
      </privacyMaskClose>
      <videoChannelPermissionList>
        <!--ro, opt, object, permission configuration list for video channel-->
        <videoChannelPermission>
          <!--ro, opt, object, permission configuration for a single video channel-->
          <id>
            <!--ro, req, string, video channel ID, range:[1,32]-->test
          </id>
          <playBack>
            <!--ro, opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
          </playBack>
          <preview>
            <!--ro, opt, bool, whether to enable the Live view permission of the current channel, desc:true (enable), false (disable)-->true
          </preview>
          <record>
            <!--ro, opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
          </record>
          <backup>
            <!--ro, opt, bool, whether to enable the backup permission of the current channel, desc:true (enable), false (disable)-->true
          </backup>
          <playBackDoubleVerification>
            <!--ro, opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)--
>true
          </playBackDoubleVerification>
          <backupDoubleVerification>
```

```

    <!--ro, opt, bool, whether to enable double verification permission for backup of the current channel, desc:true (enable), false (disable)-->
>true
    </backupDoubleVerification>
    <privacyMaskClose>
    <!--ro, opt, bool-->>false
    </privacyMaskClose>
    </videoChannelPermission>
</videoChannelPermissionList>
<ptzControl>
    <!--ro, req, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
</ptzControl>
<ptzChannelPermissionList>
    <!--ro, opt, object, PTZ control permission List of the video channels-->
    <ptzChannelPermission>
    <!--ro, req, object, PTZ control permission of a specified video channel-->
    <id>
    <!--ro, req, string, video channel ID, range:[1,32]-->test
    </id>
    <ptzControl>
    <!--ro, opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
    </ptzControl>
    </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck>
    <!--ro, opt, bool, whether to enable permission for checking logs and device status, desc:true (enable), false (disable)-->true
</logOrStateCheck>
<parameterConfig>
    <!--ro, opt, bool, whether to enable the configuration permission for remote parameters, desc:true (enable), false (disable)-->true
</parameterConfig>
<restartOrShutdown>
    <!--ro, opt, bool, whether to enable the permission to reboot or power off, desc:true (enable), false (disable)-->true
</restartOrShutdown>
<upgrade>
    <!--ro, opt, bool, whether to enable the upgrade permission, desc:true (enable), false (disable)-->true
</upgrade>
<manageChannel>
    <!--ro, opt, bool, whether to enable the channel management permission, desc:true (enable), false (disable)-->true
</manageChannel>
<AIModelManagement>
    <!--ro, opt, bool, whether to enable permission for operation on the AI model packet, desc:true (enable), false (disable)-->true
</AIModelManagement>
<AITaskManagement>
    <!--ro, opt, bool, whether to enable permission for management and operation on AI tasks, desc:true (enable), false (disable)-->true
</AITaskManagement>
<subSysOrZoneArm>
    <!--ro, opt, bool, permission to arm partitions (areas) / zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
    <!--ro, opt, bool, permission to disarm partitions (areas) / zones-->true
</subSysOrZoneDisarm>
<operateOutput>
    <!--ro, opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
    <!--ro, opt, array, List of the partitions (areas) that support Linkage, subType:object, desc:N/A-->
    <subSystem>
    <!--ro, opt, int, the partition (area) that supports Linkage, range:[1,64]-->1
    </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
    <!--ro, opt, array, partition (area) List, subType:object, desc:partitions (areas) can be Linked when adding or configuring operators-->
    <subSystem>
    <!--ro, opt, int, partition (area), range:[1,64]-->1
    </subSystem>
</subSystemList>
<localFileMgr>
    <!--ro, opt, bool, whether the user has the permission of Local folder management, desc:it refers to the permission of adding, deleting, editing,
and getting local files-->true
</localFileMgr>
<LEDConfig>
    <!--ro, opt, bool, whether the user has the permission to edit LED configuration-->true
</LEDConfig>
<logMgr>
    <!--ro, opt, bool, whether the user has the permission of Log management-->true
</logMgr>
<TIPConfig>
    <!--ro, opt, bool, whether the user has the permission of TIP configuration-->true
</TIPConfig>
<securityFunctionCfg>
    <!--ro, opt, bool, whether the user has the permission of security check function configuration, desc:it is used for permission configuration for
security check devices-->true
</securityFunctionCfg>
<networkCfg>
    <!--ro, opt, bool, whether the user has the permission of network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
    <!--ro, opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a
device to access. Field: ISUP Access Management-->true
</ISUPAccess>
<timeMgr>
    <!--ro, opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone.
Field: Time Management-->true
</timeMgr>

```

```

<HDDMgr>
  <!--ro, opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
</HDDMgr>
<securitySystemMaintenance>
  <!--ro, opt, bool, permission for security inspection system maintenance, desc:the permission includes testing detector, ray source, and operation
  keyboard, and diagnosing system, maintaining records, and performing other tests on, e.g., main control board, I/O board, and ray source light-->true
</securitySystemMaintenance>
<securityCheckingPicExport>
  <!--ro, opt, bool, permission to export baggage security check pictures-->true
</securityCheckingPicExport>
<securityCheckingPicDelete>
  <!--ro, opt, bool, permission to delete baggage security check pictures-->true
</securityCheckingPicDelete>
<operationScheduleList>
  <!--ro, opt, array, operation schedule List, subType:object, dep:or,
  {$.UserPermissionList.UserPermission.LocalPermission.operationScheduleEnabled,eq,true}-->
  <operationSchedule>
    <!--ro, opt, object, operation schedule-->
    <operationScheduleEnabled>
      <!--ro, opt, bool, whether to enable operation schedule-->true
    </operationScheduleEnabled>
    <operationDetailItem>
      <!--ro, opt, enum, operation detailed item, subType:string, desc:"all", "arm" (arming), "disarm" (disarming), "operateOutput" (operating
      relays)-->arm
    </operationDetailItem>
    <timeRange>
      <!--ro, opt, object, time range-->
      <beginTime>
        <!--ro, opt, time, start time-->00:00:00
      </beginTime>
      <endTime>
        <!--ro, opt, time, end time-->00:00:00
      </endTime>
    </timeRange>
  </operationSchedule>
</operationScheduleList>
</localPermission>
<remotePermission>
  <!--ro, opt, object, remote permission node-->
  <record>
    <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
  </record>
  <playBack>
    <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
  </playBack>
  <preview>
    <!--ro, opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
  </preview>
  <privacyMaskClose>
    <!--ro, opt, bool-->false
  </privacyMaskClose>
  <videoChannelPermissionList>
    <!--ro, opt, object, permission configuration List for video channel-->
    <videoChannelPermission>
      <!--ro, opt, object, permission configuration for a single video channel-->
      <id>
        <!--ro, req, string, corresponding video channel ID, range:[1,32]-->test
      </id>
      <preview>
        <!--ro, opt, bool, whether to enable the live view permission of the current channel, desc:true (enable), false (disable)-->true
      </preview>
      <playBack>
        <!--ro, opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
      </playBack>
      <record>
        <!--ro, opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
      </record>
      <playBackDoubleVerification>
        <!--ro, opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)--
      >true
      </playBackDoubleVerification>
      <privacyMaskClose>
        <!--ro, opt, bool-->false
      </privacyMaskClose>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl>
    <!--ro, opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
  </ptzControl>
  <ptzChannelPermissionList>
    <!--ro, opt, object, PTZ control permission List of the video channels-->
    <ptzChannelPermission>
      <!--ro, opt, object, PTZ control permission of a specified video channel-->
      <id>
        <!--ro, req, string, video channel ID, range:[1,32], desc:/ISAPI/PTZCtrl/channels/ID-->test
      </id>
      <ptzControl>
        <!--ro, opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
      </ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck>
    <!--ro, opt, bool, whether to enable permission for checking Logs and device status-->true
  </logOrStateCheck>

```

```

<parameterConfig>
  <!--ro, opt, bool, whether to enable the permission for remote parameters setting-->true
</parameterConfig>
<restartOrShutdown>
  <!--ro, opt, bool, whether to enable the permission to reboot or power off-->true
</restartOrShutdown>
<upgrade>
  <!--ro, opt, bool, whether to enable the upgrade permission-->true
</upgrade>
<upgradeNotShow>
  <!--ro, opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure
compatibility, true (support), false or this node is not returned (not support)-->true
</upgradeNotShow>
<voiceTalk>
  <!--ro, opt, bool, whether to enable the two-way audio permission (of all channels)-->true
</voiceTalk>
<transparentChannel>
  <!--ro, opt, bool, whether to enable the transparent channel permission (of all channels), desc:namely, the serial port control-->true
</transparentChannel>
<controlLocalOut>
  <!--ro, opt, bool, whether to enable the local output control permission-->true
</controlLocalOut>
<manageChannel>
  <!--ro, opt, bool, whether to enable the channel management permission-->true
</manageChannel>
<alarmOutOrUpload>
  <!--ro, opt, bool, whether to enable permission for alarm uploading and output-->true
</alarmOutOrUpload>
<IRAID>
  <!--ro, opt, object, permission for iRAID quick configuration-->
  <rapidConfiguration>
    <!--ro, opt, bool, whether to enable permission for iRAID quick configuration-->true
  </rapidConfiguration>
  <reset>
    <!--ro, opt, bool, whether to enable permission for restoring iRAID configuration-->true
  </reset>
</IRAID>
<LEDConfig>
  <!--ro, opt, bool, whether it allows users to edit the LED configuration-->true
</LEDConfig>
<AIModelManagement>
  <!--ro, opt, bool, whether to enable permission for operation on the AI model packet-->true
</AIModelManagement>
<AITaskManagement>
  <!--ro, opt, bool, whether to enable permission for management and operation on AI tasks-->true
</AITaskManagement>
<materialReview>
  <!--ro, opt, bool, material checking permission, desc:information release-->true
</materialReview>
<programReview>
  <!--ro, opt, bool, program checking permission, desc:information release-->true
</programReview>
<uploadMaterial>
  <!--ro, opt, bool, upload material, desc:information release-->true
</uploadMaterial>
<addProgram>
  <!--ro, opt, bool, program editing permission, desc:add, delete, and edit-->true
</addProgram>
<addSchedule>
  <!--ro, opt, bool, schedule editing permission, desc:add, delete, and edit-->true
</addSchedule>
<programPlay>
  <!--ro, opt, bool, information playing permission-->true
</programPlay>
<terminalMgr>
  <!--ro, opt, bool, terminal management permission, desc:set terminal parameters such as backlight and Logo Light-->true
</terminalMgr>
<shopSmart>
  <!--ro, opt, bool, smart permission-->true
</shopSmart>
<materialReplace>
  <!--ro, opt, bool, permission to replace materials-->true
</materialReplace>
<scheduleRelease>
  <!--ro, opt, bool, permission to release schedules-->true
</scheduleRelease>
<scheduleReview>
  <!--ro, opt, bool, schedule checking-->true
</scheduleReview>
<addTerminal>
  <!--ro, opt, bool, terminal adding-->true
</addTerminal>
<terminalDispatch>
  <!--ro, opt, bool, permission to relocate terminals, desc:to relocate the terminals under this organization to lower-level organizations-->true
</terminalDispatch>
<addUser>
  <!--ro, opt, bool, the permission for editing users, desc:add, delete, and edit-->true
</addUser>
<addOrg>
  <!--ro, opt, bool, the permission for editing organizations, desc:add, delete, and edit-->true
</addOrg>
<logMgr>
  <!--ro, opt, bool, log management-->true

```

```

</logMgr>
<dataUpdate>
  <!--ro, opt, bool, data update, desc:calling data-->true
</dataUpdate>
<insertMessage>
  <!--ro, opt, bool, the permission for cutting texts-->true
</insertMessage>
<notReview>
  <!--ro, opt, bool, free-reviewing permission-->true
</notReview>
<MgrTerminal>
  <!--ro, opt, object, the managed terminals or terminal group-->
  <TerminalNoList>
    <!--ro, opt, object, terminal No. List-->
    <terminalNo>
      <!--ro, opt, int, terminal No., range:[1,10]-->1
    </terminalNo>
  </TerminalNoList>
  <TerminalGroupNoList>
    <!--ro, opt, object, terminal group List-->
    <terminalGroupNo>
      <!--ro, opt, int, terminal group No., range:[1,10]-->1
    </terminalGroupNo>
  </TerminalGroupNoList>
</MgrTerminal>
<subSysOrZoneArm>
  <!--ro, opt, bool, permission to arm partitions (areas)/zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--ro, opt, bool, permission to disarm partitions (areas)/zones-->true
</subSysOrZoneDisarm>
<subSysOrZoneClearArm>
  <!--ro, opt, bool, clear alarm permission of partition/alarm zone-->true
</subSysOrZoneClearArm>
<zoneBypass>
  <!--ro, opt, bool, permission to bypass zones-->true
</zoneBypass>
<zoneBypassRecover>
  <!--ro, opt, bool, permission to recover bypass of zones-->true
</zoneBypassRecover>
<operateOutput>
  <!--ro, opt, bool, permission to operate relays-->true
</operateOutput>
<subSystemList>
  <!--ro, opt, object, partition (area) List, desc:the user can link to the partition (area) when adding or configuring operator-->
  <subSystem>
    <!--ro, opt, int, partition (area), range:[1,64]-->1
  </subSystem>
</subSystemList>
<factoryReset>
  <!--ro, opt, bool, restore to default parameters-->true
</factoryReset>
<arm>
  <!--ro, opt, bool, arm-->true
</arm>
<disarm>
  <!--ro, opt, bool, disarm-->true
</disarm>
<restoreTamper>
  <!--ro, opt, bool, restore the tamper-proof setting, desc:see description in /ISAPI/Security/UserPermission/capabilities-->true
</restoreTamper>
<restoreConfirmedAlarm>
  <!--ro, opt, bool, restore the acknowledging alarm setting, desc:see description in /ISAPI/Security/UserPermission/capabilities-->true
</restoreConfirmedAlarm>
<LPListAudit>
  <!--ro, opt, bool, permission for blocklist and allowlist operations,
desc:users with this permission can import/export/edit/delete/search for blocklist and allowlist. The related URIs are as follows:
URI for searching for the license plate blocklist and allowlist: /ISAPI/Traffic/channels/<ID>/searchLPListAudit;
URI for importing/exporting the license plate blocklist and allowlist data: /ISAPI/Traffic/channels/<ID>/LicensePlateAuditData;
URI for editing the license plate List: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/record?format=json;
URI for deleting the license plate blocklist and allowlist: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/delete?format=json-->true
  </LPListAudit>
  <applicationWarehouse>
    <!--ro, opt, bool, APP Library management permission, desc:this node is not returned if it is not supported. With management permission, users can
perform operations such as importing, deleting and starting HEOP APP-->true
  </applicationWarehouse>
  <subSystemManagement>
    <!--ro, opt, bool, permission for partition (area) management, desc:all permissions for partition (area) management-->true
  </subSystemManagement>
  <userOfModifyKeypadPasswordList>
    <!--ro, opt, array, List of users allowed to change keypad password, subType:object-->
    <userOfModifyKeypadPassword>
      <!--ro, opt, object, users allowed to change keypad password-->
    </subUserType>
    <!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyKeypadPassword>
</userOfModifyKeypadPasswordList>
  <userOfModifyUserPermissionList>
    <!--ro, opt, array, List of users allowed to edit users permission, subType:object-->
    <userOfModifyUserPermission>
      <!--ro, opt, object, users allowed to edit users permission-->
    </subUserType>
  </userOfModifyUserPermission>

```

```

<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyUserPermission>
</userOfModifyUserPermissionList>
<userOfModifyPircamPermissionList>
<!--ro, opt, array, List of users who can edit the permission to receive pircam camera video or picture event, subType:object-->
<userOfModifyPircamPermission>
<!--ro, opt, object, user who can edit the permission to receive pircam camera video or picture event-->
<subUserType>
<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyPircamPermission>
</userOfModifyPircamPermissionList>
<userOfModifyIPCPermissionList>
<!--ro, opt, array, List of users who can edit the permission to receive network camera video or picture event, subType:object-->
<userOfModifyIPCPermission>
<!--ro, opt, object, user who can edit the permission to receive network camera video or picture event-->
<subUserType>
<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyIPCPermission>
</userOfModifyIPCPermissionList>
<userOfModifyUploadPermissionList>
<!--ro, opt, array, List of users who can edit the uploading permission, subType:object-->
<userOfModifyUploadPermission>
<!--ro, opt, object, user who can edit the uploading permission-->
<subUserType>
<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyUploadPermission>
</userOfModifyUploadPermissionList>
<falseAlarmMaterialLibMgr>
<!--ro, opt, bool, permission for false alarm library management-->true
</falseAlarmMaterialLibMgr>
<TIPConfig>
<!--ro, opt, bool, whether the user has the permission of TIP configuration-->true
</TIPConfig>
<networkCfg>
<!--ro, opt, bool, whether the user has the permission of network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
<!--ro, opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a
device to access. Field: ISUP Access Management-->true
</ISUPAccess>
<timeMgr>
<!--ro, opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone.
Field: Time Management-->true
</timeMgr>
<HDDMgr>
<!--ro, opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
</HDDMgr>
<statusUploadPermission>
<!--ro, opt, object, status event permission-->
<panelPowerStatus>
<!--ro, opt, bool, whether the user has the permission to receive security control panel's power status event, desc:the security control panel's
power status refers to battery status of AC power, storage battery power, etc. The corresponding CID event codes are: [1301,3301,1302,3302,1311,3311,1311]-->true
</panelPowerStatus>
<panelCommunicateStatus>
<!--ro, opt, bool, whether the user has the permission to receive security control panel's network status event, desc:the security control panel's
network status refers to mobile network, SIM card, WIFI, wired network, etc. The corresponding CID event codes are:
[1351,3351,1352,3352,1350,3350,1350,3350,1350,3350,1344,3344,1350,3350,1350,3350,1949,3949,1350,1948]-->true
</panelCommunicateStatus>
<zoneStatus>
<!--ro, opt, bool, whether the user has the permission to receive zone status event, desc:the zones refer to all zones linked to the user, and the
zone status refers to wireless detector's power, temperature, online status, etc., and peripheral module's connection status, sensor error status, etc. The
corresponding CID event codes are: [1306,3306,1570,3570,1144,3144,1380,3380,1158,3158,1159,3159]-->true
</zoneStatus>
<externalDeviceStatus>
<!--ro, opt, bool, whether the user has the permission to receive peripheral status event, desc:the peripherals refer to all peripherals linked to
the user, such as repeater, wireless repeater, expansion module AC power, wireless peripheral, wireless sounder, keypad, and card reader; the peripheral
status refers to the status of these devices. The corresponding CID event codes are:
[1312,3312,1319,3319,1312,3312,1384,3384,1301,3301,1311,3311,1301,3301,1381,3381,1384,3384,1381,3381,1381,3381,1384,3384,1384,3384,1501,3501,1501,
3501,1306,3306,1306,3306,1306,3306,1306,1306]-->true
</externalDeviceStatus>
</statusUploadPermission>
<eventUploadPermission>
<!--ro, opt, object, event uploading permission, desc:the uploading protocol of security control panel event is CID (event identifier: cidEvent).
For definition values of CID, refer to the CID nodes on PMP platform-->
<panicAlarm>
<!--ro, opt, bool, whether the user has the permission to receive emergency alarm event, desc:emergency alarm event is triggered by card, keyfob,
or keypad, and can be triggered in zones. The corresponding CID event codes are: [1120,3120]-->true
</panicAlarm>
<medicalAlarm>
<!--ro, opt, bool, whether the user has the permission to receive medical alarm event, desc:medical alarm event is triggered via card, keyfob, or
keypad by a person. The corresponding CID event codes are: [1100,3100]-->true
</medicalAlarm>
<fireAlarm>
<!--ro, opt, bool, whether the user has the permission to receive fire alarm event, desc:fire alarm event is uploaded when a fire occurs. The

```



```

corresponding CID event codes are: [1111,3111,1114,3114]-->true
</fireAlarm>
<smartAlarm>
  <!--ro, opt, bool, whether the user has the permission to receive intelligent alarm event, desc:intelligent alarm events include intrusion
detection alarm, line crossing alarm, fire source alarm, temperature alarm, motion detection alarm, and region entrance/exiting alarm. The corresponding CID
event codes are: [1131,3131,1131,3131,1112,3112,1158,3158,1159,3159,1158,3158,1159,3159,1131,3131,1134]-->true
</smartAlarm>
<panelTamper>
  <!--ro, opt, bool, whether the user has the permission to receive security control panel tampering event, desc:the security control panel
tampering event refers to the alarm event triggered when the security control panel is being tampered. The corresponding CID event codes are: [1137,3137]--
>true
</panelTamper>
<externalDeviceTamper>
  <!--ro, opt, bool, whether the user has the permission to receive peripheral tampering event, desc:the peripherals include extension module,
wireless repeater, wireless peripheral, and the peripheral tampering event refers to the alarm triggered when the Linked peripheral is being tampered. The
corresponding CID event codes are: [1144,3144,1144,3144,1144,3144,1144,3144]-->true
</externalDeviceTamper>
<zoneAlarm>
  <!--ro, opt, bool, whether the user has the permission to receive zone alarm event and zone tampering event, desc:the zone alarm events include
instant alarm, time out alarm, burglary alarm, 24-hour alarm, delay alarm, device moving alarm, detector blocking alarm, zone detector tampering alarm,
water leak alarm. The corresponding CID event codes are: [1120,3120,1100,3100,1111,3111,1114,3114,1162,3162]-->true
</zoneAlarm>
<panelOperation>
  <!--ro, opt, bool, whether the user has the permission to receive security control panel's operation event, desc:the security control panel's
operation events refer to the events triggered when some operations are performed on the security control panel, including disarming, away/stay/auto/key
zone arming/disarming, clearing alarm, pacing, forward/backward programming, control panel reset, entering incorrect password. The corresponding CID event
codes are: [1401,3401,3441,1403,3403,1409,3409,1121,1406,1452,1455,1454,3250,1865,1607,3607,1627,1628,1305,1461]-->true
</panelOperation>
<eventUploadPermission>
<FDLibPermissionList>
  <!--ro, opt, object, permission List of face picture Library control-->
<FDLibPermission>
  <!--ro, opt, object, permission of face picture Library control-->
  <FDID>
    <!--ro, req, string, face picture Library ID, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
  </FDID>
  <FDName>
    <!--ro, opt, string, face picture Library name, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
  </FDName>
  <FDLibControl>
    <!--ro, req, bool, whether tp enable the permission of face picture library control, desc:permission of face picture library control includes
adding/deleting/editing/getting the information in the Library-->true
  </FDLibControl>
</FDLibPermission>
<FDLibPermissionList>
<picSearch>
  <!--ro, opt, bool, whether to enable the permission of picture search, desc:true-enabled, false-disabled. See details in /ISAPI/ContentMgmt/search--
>true
</picSearch>
<pircamVideoAndPicture>
  <!--ro, opt, bool, whether to enable the permission to receive pircam camera video or picture event, desc:when an alarm is triggered, if this
permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or
picture. There is no corresponding CID event code-->true
</pircamVideoAndPicture>
<IPCVideoAndPicture>
  <!--ro, opt, bool, whether to enable the permission to receive network camera video or picture event, desc:when an alarm is triggered, if this
permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or
picture. There is no corresponding CID event code-->true
</IPCVideoAndPicture>
</remotePermission>
<passwordValidity>
  <!--ro, opt, int, password validity period, range:[0,365], desc:unit: day; value range: [0, 365]; only the administrator can edit-->1
</passwordValidity>
<subUserType>
  <!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</UserPermission>
</UserPermissionList>

```

9.1.12.3 Get the permission of a single user

Request URL

GET /ISAPI/Security/UserPermission/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<UserPermission xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user permissions, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, range:[1,32]-->test
  </id>
  <userID>
    <!--ro, req, string, user ID, range:[1,32]-->test
  </userID>
  <userType>
    <!--ro, req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission),
    "operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)--
  >admin
  </userType>
  <localPermission>
    <!--ro, opt, object, Local permission node-->
    <backup>
      <!--ro, opt, bool, whether to enable the backup permission (of all channels), desc:true (enable), false (disable)-->true
    </backup>
    <record>
      <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
    </record>
    <playBack>
      <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--ro, opt, bool, whether to enable the Live view permission (of all channels), desc:true (enable), false (disable)-->true
    </preview>
    <privacyMaskClose>
      <!--ro, opt, bool-->>false
    </privacyMaskClose>
    <videoChannelPermissionList>
      <!--ro, opt, object, permission configuration List for video channel-->
      <videoChannelPermission>
        <!--ro, opt, object, permission configuration for a single video channel-->
        <id>
          <!--ro, req, string, video channel ID, range:[1,32]-->test
        </id>
        <playBack>
          <!--ro, opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
        </playBack>
        <preview>
          <!--ro, opt, bool, whether to enable the Live view permission of the current channel, desc:true (enable), false (disable)-->true
        </preview>
        <record>
          <!--ro, opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
        </record>
        <backup>
          <!--ro, opt, bool, whether to enable the backup permission of the current channel, desc:true (enable), false (disable)-->true
        </backup>
        <playBackDoubleVerification>
          <!--ro, opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)--
        >true
        </playBackDoubleVerification>
        <backupDoubleVerification>
          <!--ro, opt, bool, whether to enable double verification permission for backup of the current channel, desc:true (enable), false (disable)-->true
        </backupDoubleVerification>
        <privacyMaskClose>
          <!--ro, opt, bool-->>false
        </privacyMaskClose>
        </videoChannelPermission>
      </videoChannelPermissionList>
      <ptzControl>
        <!--ro, req, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
      </ptzControl>
      <ptzChannelPermissionList>
        <!--ro, opt, object, PTZ control permission List of the video channels-->
        <ptzChannelPermission>
          <!--ro, req, object, PTZ control permission of a specified video channel-->
          <id>
            <!--ro, req, string, video channel ID, range:[1,32]-->test
          </id>
          <ptzControl>
            <!--ro, opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
          </ptzControl>
        </ptzChannelPermission>
      </ptzChannelPermissionList>
      <logOrStateCheck>
        <!--ro, opt, bool, whether to enable permission for checking logs and device status, desc:true (enable), false (disable)-->true
      </logOrStateCheck>
      <parameterConfig>
        <!--ro, opt, bool, whether to enable the configuration permission for remote parameters, desc:true (enable), false (disable)-->true
      </parameterConfig>
      <restartOrShutdown>
        <!--ro, opt, bool, whether to enable the permission to reboot or power off, desc:true (enable), false (disable)-->true
      </restartOrShutdown>
      <upgrade>
        <!--ro, opt, bool, whether to enable the upgrade permission, desc:true (enable), false (disable)-->true
      </upgrade>
      <upgradeNotShow>
        <!--ro, opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure

```

```

compatibility, true (support), false or this node is not returned (not support)-->true
</upgradeNotShow>
<manageChannel>
  <!--ro, opt, bool, whether to enable the channel management permission, desc:true (enable), false (disable)-->true
</manageChannel>
<AIModelManagement>
  <!--ro, opt, bool, whether to enable permission for operation on the AI model packet, desc:true (enable), false (disable)-->true
</AIModelManagement>
<AITaskManagement>
  <!--ro, opt, bool, whether to enable permission for management and operation on the AI model packet, desc:true (enable), false (disable)-->true
</AITaskManagement>
<subSysOrZoneArm>
  <!--ro, opt, bool, permission to arm partitions (areas) / zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--ro, opt, bool, permission to disarm partitions (areas) / zones-->true
</subSysOrZoneDisarm>
<operateOutput>
  <!--ro, opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
  <!--ro, opt, array, List of the partitions (areas) that support Linkage, subType:object, desc:N/A-->
  <subSystem>
    <!--ro, opt, int, the partition (area) that supports Linkage, range:[1,64]-->1
  </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
  <!--ro, opt, array, partition (area) List, subType:object, desc:partitions (areas) can be Linked when adding or configuring operators N/A-->
  <subSystem>
    <!--ro, opt, int, partition (area), range:[1,64]-->1
  </subSystem>
</subSystemList>
<localFileMgr>
  <!--ro, opt, bool, permission for Local folder management, desc:it refers to the permission of adding, deleting, editing, and getting Local files-->true
</localFileMgr>
<LEDConfig>
  <!--ro, opt, bool, permission to edit LED configuration-->true
</LEDConfig>
<logMgr>
  <!--ro, opt, bool, permission for Log management-->true
</logMgr>
<TIPConfig>
  <!--ro, opt, bool, permission for TIP configuration-->true
</TIPConfig>
<securityFunctionCfg>
  <!--ro, opt, bool, permission for security check function configuration, desc:it is used for permission configuration for security check devices-->true
</securityFunctionCfg>
<networkCfg>
  <!--ro, opt, bool, permission for network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
  <!--ro, opt, bool, permission for ISUP access configuration, desc:it refers to the ISUP configuration permission for a device to access. Field: ISUP
Access Management-->true
</ISUPAccess>
<timeMgr>
  <!--ro, opt, bool, permission for time management, desc:it refers to the permission of managing system time and time zone. Field: Time Management-->true
</timeMgr>
<HDDMgr>
  <!--ro, opt, bool, permission for HDD management, desc:field: HDD Management-->true
</HDDMgr>
<securitySystemMaintenance>
  <!--ro, opt, bool, permission for security inspection system maintenance, desc:the permission includes testing detector, ray source, and operation
keyboard, and diagnosing system, maintaining records, and performing other tests on, e.g., main control board, I/O board, and ray source Light-->true
</securitySystemMaintenance>
<securityCheckingPicExport>
  <!--ro, opt, bool, permission to export baggage security check pictures-->true
</securityCheckingPicExport>
<securityCheckingPicDelete>
  <!--ro, opt, bool, permission to delete baggage security check pictures-->true
</securityCheckingPicDelete>
<operationScheduleList>
  <!--ro, opt, array, operation schedule List, subType:object, dep:or,
{$.UserPermissionList.UserPermission.LocalPermission.operationScheduleEnabled,eq,true}-->
  <operationSchedule>
    <!--ro, opt, object, operation schedule-->
    <operationScheduleEnabled>
      <!--ro, opt, bool, whether to enable operation schedule-->true
    </operationScheduleEnabled>
    <operationDetailItem>
      <!--ro, opt, enum, operation detailed item, subType:string, desc:"all", "arm" (arming), "disarm" (disarming), "operateOutput" (operating relays)-->
    </operationDetailItem>
  </operationSchedule>
</operationScheduleList>
<timeRange>
  <!--ro, opt, object, time range-->
  <beginTime>
    <!--ro, opt, time, start time-->00:00:00
  </beginTime>
  <endTime>
    <!--ro, opt, time, end time-->00:00:00
  </endTime>
</timeRange>

```

```

</operationSchedule>
</operationScheduleList>
</localPermission>
<remotePermission>
  <!--ro, opt, object, remote permission node-->
  <record>
    <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
  </record>
  <playBack>
    <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
  </playBack>
  <preview>
    <!--ro, opt, bool, whether to enable the Live view permission (of all channels), desc:true (enable), false (disable)-->true
  </preview>
  <privacyMaskClose>
    <!--ro, opt, bool-->false
  </privacyMaskClose>
  <videoChannelPermissionList>
    <!--ro, opt, object, permission configuration List for video channel-->
    <videoChannelPermission>
      <!--ro, opt, object, permission configuration for a single video channel-->
      <id>
        <!--ro, req, string, corresponding video channel ID, range:[1,32]-->test
      </id>
      <preview>
        <!--ro, opt, bool, whether to enable the Live view permission of the current channel, desc:true (enable), false (disable)-->true
      </preview>
      <playBack>
        <!--ro, opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
      </playBack>
      <record>
        <!--ro, opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
      </record>
      <playBackDoubleVerification>
        <!--ro, opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)--
>true
      </playBackDoubleVerification>
      <privacyMaskClose>
        <!--ro, opt, bool-->false
      </privacyMaskClose>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl>
    <!--ro, opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
  </ptzControl>
  <ptzChannelPermissionList>
    <!--ro, opt, object, PTZ control permission List of the video channels-->
    <ptzChannelPermission>
      <!--ro, opt, object, PTZ control permission of a specified video channel-->
      <id>
        <!--ro, req, string, video channel ID, range:[1,32], desc:/ISAPI/PTZCtrl/channels/ID-->test
      </id>
      <ptzControl>
        <!--ro, opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
      </ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck>
    <!--ro, opt, bool, whether to enable permission for checking Logs and device status-->true
  </logOrStateCheck>
  <parameterConfig>
    <!--ro, opt, bool, whether to enable the configuration permission for remote parameters-->true
  </parameterConfig>
  <restartOrShutdown>
    <!--ro, opt, bool, whether to enable the permission to reboot or power off-->true
  </restartOrShutdown>
  <upgrade>
    <!--ro, opt, bool, whether to enable the upgrade permission-->true
  </upgrade>
  <upgradeNotShow>
    <!--ro, opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure
compatibility, true (support), false or this node is not returned (not support)-->true
  </upgradeNotShow>
  <voiceTalk>
    <!--ro, opt, bool, whether to enable the two-way audio permission (of all channels)-->true
  </voiceTalk>
  <transParentChannel>
    <!--ro, opt, bool, whether to enable the transparent channel permission (of all channels), desc:the serial port control-->true
  </transParentChannel>
  <contorlLocalOut>
    <!--ro, opt, bool, whether to enable the Local output control permission-->true
  </contorlLocalOut>
  <manageChannel>
    <!--ro, opt, bool, whether to enable the channel management permission-->true
  </manageChannel>
  <alarmOutOrUpload>
    <!--ro, opt, bool, whether to enable permission for alarm uploading and output-->true
  </alarmOutOrUpload>
  <IRAIID>
    <!--ro, opt, object, permission for Iraid quick configuration-->
    <rapidConfiguration>
      <!--ro, opt, bool, whether to enable permission for Iraid quick configuration-->true
    </rapidConfiguration>

```

```

<reset>
  <!--ro, opt, bool, whether to enable permission for restoring Iraid configuration-->true
</reset>
</IRAID>
<LEDConfig>
  <!--ro, opt, bool, whether it allows users to edit the LED configuration-->true
</LEDConfig>
<AIModelManagement>
  <!--ro, opt, bool, whether to enable permission for operation on the AI model packet-->true
</AIModelManagement>
<AITaskManagement>
  <!--ro, opt, bool, whether to enable permission for management and operation on AI tasks-->true
</AITaskManagement>
<materialReview>
  <!--ro, opt, bool, material checking permission, desc:information release-->true
</materialReview>
<programReview>
  <!--ro, opt, bool, program checking permission, desc:information release-->true
</programReview>
<uploadMaterial>
  <!--ro, opt, bool, upload material, desc:information release-->true
</uploadMaterial>
<addProgram>
  <!--ro, opt, bool, program editing permission, desc:add, delete, and edit-->true
</addProgram>
<addSchedule>
  <!--ro, opt, bool, schedule editing permission, desc:add, delete, and edit-->true
</addSchedule>
<programPlay>
  <!--ro, opt, bool, information playing permission-->true
</programPlay>
<terminalMgr>
  <!--ro, opt, bool, terminal management permission, desc:set terminal parameters such as backLight and Logo Light-->true
</terminalMgr>
<shopSmart>
  <!--ro, opt, bool, permission for shop smart-->true
</shopSmart>
<materialReplace>
  <!--ro, opt, bool, permission to replace materials-->true
</materialReplace>
<scheduleRelease>
  <!--ro, opt, bool, permission to release schedules-->true
</scheduleRelease>
<scheduleReview>
  <!--ro, opt, bool, schedule checking-->true
</scheduleReview>
<addTerminal>
  <!--ro, opt, bool, terminal adding-->true
</addTerminal>
<terminalDispatch>
  <!--ro, opt, bool, permission to relocate terminals, desc:to relocate the terminals under this organization to lower-level organizations-->true
</terminalDispatch>
<addUser>
  <!--ro, opt, bool, the permission for editing users, desc:add, delete, and edit-->true
</addUser>
<addOrg>
  <!--ro, opt, bool, the permission for editing organizations, desc:add, delete, and edit-->true
</addOrg>
<logMgr>
  <!--ro, opt, bool, Log management-->true
</logMgr>
<dataUpdate>
  <!--ro, opt, bool, data update, desc:calling data-->true
</dataUpdate>
<insertMessage>
  <!--ro, opt, bool, the permission for cutting texts-->true
</insertMessage>
<notReview>
  <!--ro, opt, bool, free-reviewing permission-->true
</notReview>
<MgrTerminal>
  <!--ro, opt, object, the managed terminals or terminal group-->
  <TerminalNoList size="10">
    <!--ro, opt, object, terminal No. List, attr:size{req, int}-->
    <terminalNo>
      <!--ro, opt, int, terminal No., range:[1,10]-->1
    </terminalNo>
  </TerminalNoList>
  <TerminalGroupNoList size="10">
    <!--ro, opt, object, terminal group List, attr:size{req, int}-->
    <terminalGroupNo>
      <!--ro, opt, int, terminal group No., range:[1,10]-->1
    </terminalGroupNo>
  </TerminalGroupNoList>
</MgrTerminal>
<subSysOrZoneArm>
  <!--ro, opt, bool, permission to arm partitions (areas)/zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--ro, opt, bool, permission to disarm partitions (areas)/zones-->true
</subSysOrZoneDisarm>
<subSysOrZoneClearArm>
  <!--ro, opt, bool, permission to clear alarms triggered in partitions (areas)/zones-->true
</subSysOrZoneClearArm>

```

```

</zoneBypass>
<!--ro, opt, bool, permission to bypass zones-->true
</zoneBypass>
<zoneBypassRecover>
<!--ro, opt, bool, permission to recover bypass of zones-->true
</zoneBypassRecover>
<operateOutput>
<!--ro, opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
<!--ro, opt, object, List of the partitions (areas) that support Linkage-->
<subSystem>
<!--ro, opt, int, the partition (area) that supports Linkage, range:[1,64]-->1
</subSystem>
</supportLinkageSubSystemList>
<subSystemList>
<!--ro, opt, object, partition (area) List, desc:users can link partitions (areas) when adding or configuring operators-->
<subSystem>
<!--ro, opt, int, partition (area), range:[1,64]-->1
</subSystem>
</subSystemList>
<factoryReset>
<!--ro, opt, bool, restore to default parameters-->true
</factoryReset>
<arm>
<!--ro, opt, bool, arm-->true
</arm>
<disarm>
<!--ro, opt, bool, disarm-->true
</disarm>
<accessControl>
<!--ro, opt, bool, access control-->true
</accessControl>
<restoreTamper>
<!--ro, opt, bool, restore the tamper-proof setting, desc:refer to /ISAPI/Security/UserPermission/capabilities-->true
</restoreTamper>
<restoreConfirmedAlarm>
<!--ro, opt, bool, restore the acknowledging alarm setting, desc:refer to /ISAPI/Security/UserPermission/capabilities-->true
</restoreConfirmedAlarm>
<LPListAudit>
<!--ro, opt, bool, permission for blacklist and allowlist operations,
desc:users with this permission can import/export/edit/delete/search for blacklist and allowlist. The related URIs are as follows:
URI for searching for the license plate blacklist and allowlist: /ISAPI/Traffic/channels/<ID>/searchLPListAudit;
URI for importing/exporting the license plate blacklist and allowlist data: /ISAPI/Traffic/channels/<ID>/LicensePlateAuditData;
URI for editing the license plate list: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/record?format=json;
URI for deleting the license plate blacklist and allowlist: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/delete?format=json-->true
</LPListAudit>
<applicationWarehouse>
<!--ro, opt, bool, APP Library management permission, desc:this node is not returned if it is not supported. With management permission, users can
perform operations such as importing, deleting and starting HEOP APP-->true
</applicationWarehouse>
<subSystemManagement>
<!--ro, opt, bool, permission to manage partitions (areas), desc:all permissions for partition (area) management-->true
</subSystemManagement>
<userOfModifyKeypadPasswordList>
<!--ro, opt, array, List of users allowed to change keypad password, subType:object-->
<userOfModifyKeypadPassword>
<!--ro, opt, object, users allowed to change keypad password-->
<subUserType>
<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyKeypadPassword>
</userOfModifyKeypadPasswordList>
<userOfModifyUserPermissionList>
<!--ro, opt, array, List of users allowed to edit user permissions, subType:object-->
<userOfModifyUserPermission>
<!--ro, opt, object, users allowed to edit user permissions-->
<subUserType>
<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyUserPermission>
</userOfModifyUserPermissionList>
<userOfModifyPircamPermissionList>
<!--ro, opt, array, List of users who can edit the permission to receive pircam camera video or picture event, subType:object-->
<userOfModifyPircamPermission>
<!--ro, opt, object, user who can edit the permission to receive pircam camera video or picture event-->
<subUserType>
<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyPircamPermission>
</userOfModifyPircamPermissionList>
<userOfModifyIPCPermissionList>
<!--ro, opt, array, List of users who can edit the permission to receive network camera video or picture event, subType:object-->
<userOfModifyIPCPermission>
<!--ro, opt, object, user who can edit the permission to receive network camera video or picture event-->
<subUserType>
<!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cloudOperator", "LocalOperator"-->cloudAdmin
</subUserType>
</userOfModifyIPCPermission>

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</userOfModifyIPCPermissionList>
<userOfModifyUploadPermissionList>
  <!--ro, opt, array, list of users who can edit the uploading permission, subType:object-->
  <userOfModifyUploadPermission>
    <!--ro, opt, object, user who can edit the uploading permission-->
    <subUserType>
      <!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
      "cloudOperator", "LocalOperator"-->cloudAdmin
    </subUserType>
  </userOfModifyUploadPermission>
</userOfModifyUploadPermissionList>
<falseAlarmMaterialLibMgr>
  <!--ro, opt, bool, permission for false alarm library management-->true
</falseAlarmMaterialLibMgr>
<TIPConfig>
  <!--ro, opt, bool, whether the user has the permission of TIP configuration-->true
</TIPConfig>
<networkCfg>
  <!--ro, opt, bool, whether the user has the permission of network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
  <!--ro, opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a device
  to access. Field: ISUP Access Management-->true
</ISUPAccess>
<timeMgr>
  <!--ro, opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone.
  Field: Time Management-->true
</timeMgr>
<HDDMgr>
  <!--ro, opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
</HDDMgr>
<statusUploadPermission>
  <!--ro, opt, object, status event permission-->
  <panelPowerStatus>
    <!--ro, opt, bool, whether the user has the permission to receive security control panel's power status event, desc:the security control panel's
    power status refers to battery status of AC power, storage battery power, etc. The corresponding CID event codes are: [1301,3301,1302,3302,1311,3311,1311]--
    >true
  </panelPowerStatus>
  <panelCommunicateStatus>
    <!--ro, opt, bool, whether the user has the permission to receive security control panel's network status event, desc:the security control panel's
    network status refers to mobile network, SIM card, WIFI, wired network, etc. The corresponding CID event codes are:
    [1351,3351,1352,3352,1350,3350,1350,3350,1350,3350,1344,3344,1350,3350,1350,3350,1949,3949,1350,1948]-->true
  </panelCommunicateStatus>
  <zoneStatus>
    <!--ro, opt, bool, whether the user has the permission to receive zone status event, desc:the zones refer to all zones linked to the user, and the
    zone status refers to wireless detector's power, temperature, online status, etc., and peripheral module's connection status, sensor error status, etc. The
    corresponding CID event codes are: [1306,3306,1570,3570,1144,3144,1380,3380,1158,3158,1159,3159]-->true
  </zoneStatus>
  <externalDeviceStatus>
    <!--ro, opt, bool, whether the user has the permission to receive peripheral status event, desc:the peripherals refer to all peripherals linked to
    the user, such as repeater, wireless repeater, expansion module AC power, wireless peripheral, wireless sounder, keypad, and card reader; the peripheral
    status refers to the status of these devices. The corresponding CID event codes are:
    [1312,3312,1319,3319,1312,3312,1384,3384,1301,3301,1311,3311,1301,3301,1381,3381,1384,3384,1381,3381,1381,3381,1384,3384,1384,3384,1501,3501,1501,
    3501,1306,3306,1306,3306,1306,3306,1306,3306]-->true
  </externalDeviceStatus>
  </statusUploadPermission>
  <eventUploadPermission>
    <!--ro, opt, object, event uploading permission, desc:the uploading protocol of security control panel event is CID (event identifier: cidEvent). For
    definition values of CID, refer to the CID nodes on PMP platform-->
    <panicAlarm>
      <!--ro, opt, bool, whether the user has the permission to receive emergency alarm event, desc:emergency alarm event is triggered by card, keyfob, or
      keypad, and can be triggered in zones. The corresponding CID event codes are: [1120,3120]-->true
    </panicAlarm>
    <medicalAlarm>
      <!--ro, opt, bool, whether the user has the permission to receive medical alarm event, desc:medical alarm event is triggered via card, keyfob, or
      keypad by a person. The corresponding CID event codes are: [1100,3100]-->true
    </medicalAlarm>
    <fireAlarm>
      <!--ro, opt, bool, whether the user has the permission to receive fire alarm event, desc:fire alarm event is uploaded when a fire occurs. The
      corresponding CID event codes are: [1111,3111,1114,3114]-->true
    </fireAlarm>
    <smartAlarm>
      <!--ro, opt, bool, whether the user can have the permission to receive intelligent alarm event, desc:intelligent alarm events include intrusion
      detection alarm, line crossing alarm, fire source alarm, temperature alarm, motion detection alarm, and region entrance/exiting alarm. The corresponding CID
      event codes are: [1131,3131,1131,3131,1112,3112,1158,3158,1159,3159,1158,3158,1159,3159,1131,3131,1134]-->true
    </smartAlarm>
    <panelTamper>
      <!--ro, opt, bool, whether the user has the permission to receive security control panel tampering event, desc:the security control panel tampering
      event refers to the alarm event triggered when the security control panel is being tampered. The corresponding CID event codes are: [1137,3137]-->true
    </panelTamper>
    <externalDeviceTamper>
      <!--ro, opt, bool, whether the user has the permission to receive peripheral tampering event, desc:the peripherals include extension module,
      wireless repeater, wireless peripheral, and the peripheral tampering event refers to the alarm triggered when the linked peripheral is being tampered. The
      corresponding CID event codes are: [1144,3144,1144,3144,1144,3144,1144,3144]-->true
    </externalDeviceTamper>
    <zoneAlarm>
      <!--ro, opt, bool, whether the user has the permission to receive zone alarm event and zone tampering event, desc:the zone alarm events include
      instant alarm, time out alarm, burglary alarm, 24-hour alarm, delay alarm, device moving alarm, detector blocking alarm, zone detector tampering alarm,
      water leak alarm. The corresponding CID event codes are: [1120,3120,1100,3100,1111,3111,1114,3114,1162,3162]-->true
    </zoneAlarm>
    <panelOperation>
      <!--ro, opt, bool, whether the user has the permission to receive security control panel's operation event, desc:the security control panel's
      operation events refer to the events triggered when some operations are performed on the security control panel, including disarming, away/stay/auto/key
      zone arming/disarming, cloning alarm, panic, forward/backward programming, control panel reset, entering incorrect password. The corresponding CID event

```



```

zone arming/disarming, clearing alarm, panic, firmware/backup programming, control panel reset, entering incorrect password. The corresponding CID event codes are: [1401,3401,3441,1403,3403,1409,3409,1121,1406,1452,1455,1454,3250,1865,1607,3607,1627,1628,1305,1461]-->true
</panelOperation>
</eventUploadPermission>
<FDLibPermissionList>
  <!--ro, opt, object, permission list of face picture library control-->
  <FDLibPermission>
    <!--ro, opt, object, permission of face picture Library control-->
    <FDID>
      <!--ro, req, string, face picture Library ID, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
    </FDID>
    <FDName>
      <!--ro, opt, string, face picture Library name, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
    </FDName>
    <FDLibControl>
      <!--ro, req, bool, whether to enable the permission of face picture Library control, desc:permission of face picture library control includes adding/deleting/editing/getting the information in the Library-->true
    </FDLibControl>
  </FDLibPermission>
</FDLibPermissionList>
<picSearch>
  <!--ro, opt, bool, whether to enable the permission of picture search, desc:true-enabled, false-disabled. See details in /ISAPI/ContentMgmt/search-->true
</picSearch>
<pircamVideoAndPicture>
  <!--ro, opt, bool, whether to enable the permission to receive pircam camera video or picture event, desc:when an alarm is triggered, if this permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or picture. There is no corresponding CID event code-->true
</pircamVideoAndPicture>
<IPCVideoAndPicture>
  <!--ro, opt, bool, whether to enable the permission to receive network camera video or picture event, desc:when an alarm is triggered, if this permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or picture. There is no corresponding CID event code-->true
</IPCVideoAndPicture>
</remotePermission>
<passwordValidity>
  <!--ro, opt, int, password validity period, range:[0,365], desc:unit: day; value range: [0, 365]; only the administrator can edit-->1
</passwordValidity>
<subUserType>
  <!--ro, opt, enum, type of sub-users, subType:string, desc:"cCloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller", "cCloudOperator", "LocalOperator"-->cCloudAdmin
</subUserType>
<internalParametersAdjustment>
  <!--ro, opt, bool-->true
</internalParametersAdjustment>
</UserPermission>

```

9.1.12.4 Set permissions for a single user

Request URL

PUT /ISAPI/Security/UserPermission/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<UserPermission xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user permissions, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, range:[1,32]-->test
  </id>
  <userID>
    <!--req, string, user ID, range:[1,32]-->test
  </userID>
  <userType>
    <!--req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission), "operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)-->admin
  </userType>
  <localPermission>
    <!--opt, object, local permission node-->
    <backup>
      <!--opt, bool, whether to enable the backup permission (of all channels), desc:true (enable), false (disable)-->true
    </backup>
    <record>
      <!--opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
    </record>
    <playBack>
      <!--opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
    </preview>
  </localPermission>

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<!--opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
</preview>
<privacyMaskClose>
  <!--opt, bool-->false
</privacyMaskClose>
<videoChannelPermissionList>
  <!--opt, object, permission configuration List for video channel-->
  <videoChannelPermission>
    <!--opt, object, permission configuration for a single video channel-->
    <id>
      <!--req, string, video channel ID, range:[1,32]-->test
    </id>
    <playBack>
      <!--opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--opt, bool, whether to enable the Live view permission of the current channel, desc:true (enable), false (disable)-->true
    </preview>
    <record>
      <!--opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
    </record>
    <backup>
      <!--opt, bool, whether to enable the backup permission of the current channel, desc:true (enable), false (disable)-->true
    </backup>
    <playBackDoubleVerification>
      <!--opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)-->true
    </playBackDoubleVerification>
    <backupDoubleVerification>
      <!--opt, bool, whether to enable double verification permission for backup of the current channel, desc:true (enable), false (disable)-->true
    </backupDoubleVerification>
    <privacyMaskClose>
      <!--opt, bool-->false
    </privacyMaskClose>
  </videoChannelPermission>
</videoChannelPermissionList>
<ptzControl>
  <!--req, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
</ptzControl>
<ptzChannelPermissionList>
  <!--opt, object, PTZ control permission List of the video channels-->
  <ptzChannelPermission>
    <!--opt, object, PTZ control permission of a specified video channel-->
    <id>
      <!--req, string, video channel ID, range:[1,32]-->test
    </id>
    <ptzControl>
      <!--opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
    </ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck>
  <!--opt, bool, whether to enable permission for checking logs and device status, desc:true (enable), false (disable)-->true
</logOrStateCheck>
<parameterConfig>
  <!--opt, bool, whether to enable the permission for remote parameters setting, desc:true (enable), false (disable)-->true
</parameterConfig>
<restartOrShutdown>
  <!--opt, bool, whether to enable the permission to reboot or power off, desc:true (enable), false (disable)-->true
</restartOrShutdown>
<upgrade>
  <!--opt, bool, whether to enable the upgrade permission, desc:true (enable), false (disable)-->true
</upgrade>
<upgradeNotShow>
  <!--opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure
compatibility, true (support), false or this node is not returned (not support)-->true
</upgradeNotShow>
<manageChannel>
  <!--opt, bool, whether to enable the channel management permission, desc:true (enable), false (disable)-->true
</manageChannel>
<AIModelManagement>
  <!--opt, bool, whether to enable permission for operation on the AI model packet, desc:true (enable), false (disable)-->true
</AIModelManagement>
<AITaskManagement>
  <!--opt, bool, whether to enable permission for management and operation on AI tasks, desc:true (enable), false (disable)-->true
</AITaskManagement>
<subSysOrZoneArm>
  <!--opt, bool, permission to arm partitions (areas) / zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--opt, bool, permission to disarm partitions (areas) / zones-->true
</subSysOrZoneDisarm>
<operateOutput>
  <!--opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
  <!--opt, array, List of the partitions (areas) that support Linkage, subType:object, desc:N/A-->
  <subSystem>
    <!--opt, int, the partition (area) that supports Linkage, range:[1,64]-->1
  </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
  <!--opt, array, partition (area) List, subType:object, desc:partitions (areas) can be linked when adding or configuring operators N/A-->
  <subSystem>
    <!--opt, int, partition (area), range:[1,64]-->1
  </subSystem>
</subSystemList>

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    </subSystem>
  </subSystemList>
  <localFileMgr>
    <!--opt, bool, whether the user has the permission of Local folder management, desc:it refers to the permission of adding, deleting, editing, and
getting local files-->true
  </localFileMgr>
  <LEDConfig>
    <!--opt, bool, whether the user has the permission to edit LED configuration-->true
  </LEDConfig>
  <logMgr>
    <!--opt, bool, whether the user has the Log management permission-->true
  </logMgr>
  <TIPConfig>
    <!--opt, bool, whether the user has the permission of TIP configuration-->true
  </TIPConfig>
  <securityFunctionCfg>
    <!--opt, bool, whether the user has the permission of security check function configuration, desc:it is used for permission configuration for security
check devices-->true
  </securityFunctionCfg>
  <networkCfg>
    <!--opt, bool, whether the user has the permission of network configuration, desc:including wireLess and wired network-->true
  </networkCfg>
  <ISUPAccess>
    <!--opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a device to
access. Field: ISUP Access Management-->true
  </ISUPAccess>
  <timeMgr>
    <!--opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone. Field:
Time Management-->true
  </timeMgr>
  <HDDMgr>
    <!--opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
  </HDDMgr>
  <securitySystemMaintenance>
    <!--opt, bool, permission for security inspection system maintenance, desc:the permission includes testing detector, ray source, and operation
keyboard, and diagnosing system, maintaining records, and performing other tests on, e.g., main control board, I/O board, and ray source Light-->true
  </securitySystemMaintenance>
  <securityCheckingPicExport>
    <!--opt, bool, permission to export baggage security check pictures-->true
  </securityCheckingPicExport>
  <securityCheckingPicDelete>
    <!--opt, bool, permission to delete baggage security check pictures-->true
  </securityCheckingPicDelete>
  <operationScheduleList>
    <!--opt, array, operation schedule List, subType:object, dep:or,
{$.UserPermissionList.UserPermission.LocalPermission.operationScheduleEnabled,eq,true}-->
  <operationSchedule>
    <!--opt, object, operation schedule-->
    <operationScheduleEnabled>
      <!--opt, bool, whether to enable operation schedule-->true
    </operationScheduleEnabled>
    <operationDetailItem>
      <!--opt, enum, operation detailed item, subType:string, desc:"all", "arm" (arming), "disarm" (disarming), "operateOutput" (operating relays)-->arm
    </operationDetailItem>
    <timeRange>
      <!--opt, object, time range-->
      <beginTime>
        <!--opt, time, start time-->00:00:00
      </beginTime>
      <endTime>
        <!--opt, time, end time-->00:00:00
      </endTime>
    </timeRange>
  </operationSchedule>
</operationScheduleList>
</localPermission>
<remotePermission>
  <!--opt, object, remote permission node-->
  <record>
    <!--opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
  </record>
  <playBack>
    <!--opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
  </playBack>
  <preview>
    <!--opt, bool, whether to enable the Live view permission (of all channels), desc:true (enable), false (disable)-->true
  </preview>
  <privacyMaskClose>
    <!--opt, bool-->false
  </privacyMaskClose>
  <videoChannelPermissionList>
    <!--opt, object, List of configuring permission for video channel-->
    <videoChannelPermission>
      <!--opt, object, configure the permission for a single video channel-->
      <id>
        <!--req, string, video channel ID, range:[1,32]-->test
      </id>
      <preview>
        <!--opt, bool, whether to enable the Live view permission of the current channel, desc:true (enable), false (disable)-->true
      </preview>
      <playBack>
        <!--opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
      </playBack>
    </videoChannelPermission>
  </videoChannelPermissionList>

```

```

    <record>
      <!--opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
    </record>
    <playBackDoubleVerification>
      <!--opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)-->true
    </playBackDoubleVerification>
    <privacyMaskClose>
      <!--opt, bool-->false
    </privacyMaskClose>
  </videoChannelPermission>
</videoChannelPermissionList>
<ptzControl>
  <!--opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
</ptzControl>
<ptzChannelPermissionList>
  <!--opt, object, PTZ control permission list of the video channels-->
  <ptzChannelPermission>
    <!--opt, object, PTZ control permission of a specified video channel-->
    <id>
      <!--req, string, video channel ID, range:[1,32], desc:/ISAPI/PTZCtrl/channels/ID-->test
    </id>
    <ptzControl>
      <!--opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
    </ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck>
  <!--opt, bool, whether to enable permission for checking logs and device status-->true
</logOrStateCheck>
<parameterConfig>
  <!--opt, bool, whether to enable the permission for remote parameters setting-->true
</parameterConfig>
<restartOrShutdown>
  <!--opt, bool, whether to enable the permission to reboot or power off-->true
</restartOrShutdown>
<upgrade>
  <!--opt, bool, whether to enable the upgrade permission-->true
</upgrade>
<upgradeNotShow>
  <!--opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure
compatibility, true (support), false or this node is not returned (not support)-->true
</upgradeNotShow>
<voiceTalk>
  <!--opt, bool, whether to enable the two-way audio permission (of all channels)-->true
</voiceTalk>
<transparentChannel>
  <!--opt, bool, whether to enable the transparent channel, desc:namely, the serial port control-->true
</transparentChannel>
<controlLocalOut>
  <!--opt, bool, whether to enable the local output control permission-->true
</controlLocalOut>
<manageChannel>
  <!--opt, bool, whether to enable the channel management permission-->true
</manageChannel>
<alarmOutOrUpload>
  <!--opt, bool, whether to enable permission for alarm uploading and output-->true
</alarmOutOrUpload>
<iRAID>
  <!--opt, object, permission for iRAID quick configuration-->
  <rapidConfiguration>
    <!--opt, bool, whether to enable permission for iRAID quick configuration-->true
  </rapidConfiguration>
  <reset>
    <!--opt, bool, whether to enable permission for restoring iRAID configuration-->true
  </reset>
</iRAID>
<LEDConfig>
  <!--opt, bool, whether it allows the user to edit the LED configuration-->true
</LEDConfig>
<AIModelManagement>
  <!--opt, bool, whether to enable permission for operation on the AI model packet-->true
</AIModelManagement>
<AITaskManagement>
  <!--opt, bool, whether to enable permission for management and operation on AI tasks-->true
</AITaskManagement>
<materialReview>
  <!--opt, bool, material reviewing permission, desc:information release-->true
</materialReview>
<programReview>
  <!--opt, bool, program reviewing permission, desc:information release-->true
</programReview>
<uploadMaterial>
  <!--opt, bool, material upload, desc:information release-->true
</uploadMaterial>
<addProgram>
  <!--opt, bool, program editing permission, desc:add, delete, and edit-->true
</addProgram>
<addSchedule>
  <!--opt, bool, schedule editing permission, desc:add, delete, and edit-->true
</addSchedule>
<programPlay>
  <!--opt, bool, information playing permission-->true
</programPlay>
<terminalMgr>

```

```

    <!--opt, bool, terminal management permission, desc:set terminal parameters such as backlight and Logo Light-->true
</terminalMgr>
<shopSmart>
    <!--opt, bool, permission for shop smart-->true
</shopSmart>
<materialReplace>
    <!--opt, bool, permission to replace materials-->true
</materialReplace>
<scheduleRelease>
    <!--opt, bool, permission for releasing schedule-->true
</scheduleRelease>
<scheduleReview>
    <!--opt, bool, schedule checking-->true
</scheduleReview>
<addTerminal>
    <!--opt, bool, terminal adding-->true
</addTerminal>
<terminalDispatch>
    <!--opt, bool, permission to relocate terminals, desc:to relocate the terminals under this organization to lower-level organizations-->true
</terminalDispatch>
<addUser>
    <!--opt, bool, the permission for editing users, desc:add, delete, and edit-->true
</addUser>
<addOrg>
    <!--opt, bool, the permission for editing organizations, desc:add, delete, and edit-->true
</addOrg>
<logMgr>
    <!--opt, bool, Log management-->true
</logMgr>
<dataUpdate>
    <!--opt, bool, data update, desc:calling data-->true
</dataUpdate>
<insertMessage>
    <!--opt, bool, the permission for cutting texts-->true
</insertMessage>
<notReview>
    <!--opt, bool, the permission for free-review-->true
</notReview>
<MgrTerminal>
    <!--opt, object, the managed terminals or terminal group-->
    <TerminalNoList size="10">
        <!--opt, object, terminal No. List, attr:size{req, int}-->
        <terminalNo>
            <!--opt, int, terminal No., range:[1,10]-->1
        </terminalNo>
    </TerminalNoList>
    <TerminalGroupNoList size="10">
        <!--opt, object, terminal group List, attr:size{req, int}-->
        <terminalGroupNo>
            <!--opt, int, terminal group No., range:[1,10]-->1
        </terminalGroupNo>
    </TerminalGroupNoList>
</MgrTerminal>
<subSysOrZoneArm>
    <!--opt, bool, permission to arm partitions (areas) / zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
    <!--opt, bool, permission to disarm partitions (areas) / zones-->true
</subSysOrZoneDisarm>
<subSysOrZoneClearArm>
    <!--opt, bool, permission to clear alarms triggered in partitions (areas) / zones-->true
</subSysOrZoneClearArm>
<zoneBypass>
    <!--opt, bool, permission to bypass zones-->true
</zoneBypass>
<zoneBypassRecover>
    <!--opt, bool, permission to recover bypass of zones-->true
</zoneBypassRecover>
<operateOutput>
    <!--opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
    <!--opt, object, list of the partitions (areas) that support linkage-->
    <subSystem>
        <!--opt, int, the partition (area) that supports linkage, range:[1,64]-->1
    </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
    <!--opt, object, partition (area) List, desc:partitions (areas) can be Linked when adding or configuring operators-->
    <subSystem>
        <!--opt, int, partition (area), range:[1,64]-->1
    </subSystem>
</subSystemList>
<factoryReset>
    <!--opt, bool, restore to default parameters-->true
</factoryReset>
<arm>
    <!--opt, bool, arm-->true
</arm>
<disarm>
    <!--opt, bool, disarm-->true
</disarm>
<accessControl>

```

```

    <!--opt, bool, access control-->true
  </accessControl>
  <restoreTamper>
    <!--opt, bool, restore the tamper-proof setting, desc:for usage details, refer to /ISAPI/Security/UserPermission/capabilities-->true
  </restoreTamper>
  <restoreConfirmedAlarm>
    <!--opt, bool, restore the acknowledging alarm setting, desc:for usage details, refer to /ISAPI/Security/UserPermission/capabilities-->true
  </restoreConfirmedAlarm>
  <LPListAudit>
    <!--opt, bool, permission for blacklist and allowList operations,
    desc:users with this permission can import/export/edit/delete/search for blacklist and allowList. The related URIs are as follows:
    URI for searching for the license plate blacklist and allowList: /ISAPI/Traffic/channels/<ID>/searchLPListAudit;
    URI for importing/exporting the license plate blacklist and allowList data: /ISAPI/Traffic/channels/<ID>/LicensePlateAuditData;
    URI for editing the license plate list: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/record?format=json;
    URI for deleting the license plate blacklist and allowList: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/delete?format=json-->true
  </LPListAudit>
  <applicationWarehouse>
    <!--opt, bool, APP Library management permission, desc:with management permission, users can perform operations such as importing, deleting and
    starting HEOP APP-->true
  </applicationWarehouse>
  <subSystemManagement>
    <!--opt, bool, permission to manage partitions (areas), desc:all permissions for partition (area) management-->true
  </subSystemManagement>
  <userOfModifyKeypadPasswordList>
    <!--opt, array, List of users allowed to change keypad password, subType:object-->
  <userOfModifyKeypadPassword>
    <!--opt, object, users allowed to change keypad password-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyKeypadPassword>
  <userOfModifyKeypadPasswordList>
  <userOfModifyUserPermissionList>
    <!--opt, array, List of users allowed to edit user permissions, subType:object-->
  <userOfModifyUserPermission>
    <!--opt, object, users allowed to edit users permission-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyUserPermission>
  <userOfModifyUserPermissionList>
  <userOfModifyPircamPermissionList>
    <!--opt, array, List of users who can edit the permission to receive pircam camera video or picture event, subType:object-->
  <userOfModifyPircamPermission>
    <!--opt, object, user who can edit the permission to receive pircam camera video or picture event-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyPircamPermission>
  <userOfModifyPircamPermissionList>
  <userOfModifyIPCPermissionList>
    <!--opt, array, List of users who can edit the permission to receive network camera video or picture event, subType:object-->
  <userOfModifyIPCPermission>
    <!--opt, object, user who can edit the permission to receive network camera video or picture event-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyIPCPermission>
  <userOfModifyIPCPermissionList>
  <userOfModifyUploadPermissionList>
    <!--opt, array, List of users who can edit the uploading permission, subType:object-->
  <userOfModifyUploadPermission>
    <!--opt, object, user who can edit the uploading permission-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyUploadPermission>
  <userOfModifyUploadPermissionList>
  <falseAlarmMaterialLibMgr>
    <!--opt, bool, permission for false alarm library management-->true
  </falseAlarmMaterialLibMgr>
  <TIPConfig>
    <!--opt, bool, whether the user has the permission of TIP configuration-->true
  </TIPConfig>
  <networkCfg>
    <!--opt, bool, whether the user has the permission of network configuration, desc:including wireless and wired network-->true
  </networkCfg>
  <ISUPAccess>
    <!--opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a device to
    access. Field: ISUP Access Management-->true
  </ISUPAccess>
  <timeMgr>
    <!--opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone. Field:
    Time Management-->true
  </timeMgr>
  <HDDMgr>
    <!--opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
  </HDDMgr>
  <statusUnloadPermission>

```

```

<!--opt, object, status event permission-->
<panelPowerStatus>
  <!--opt, bool, whether the user has the permission to receive security control panel's power status event, desc:the security control panel's power status refers to battery status of AC power, storage battery power, etc. The corresponding CID event codes are: [1301,3301,1302,3302,1311,3311,1311]-->true
</panelPowerStatus>
<panelCommunicateStatus>
  <!--opt, bool, whether the user has the permission to receive security control panel's network status event, desc:the security control panel's network status refers to mobile network, SIM card, WIFI, wired network, etc. The corresponding CID event codes are: [1351,3351,1352,3352,1350,3350,1350,3350,1350,3350,1344,3344,1350,3350,1350,3350,1949,3949,1350,1948]-->true
</panelCommunicateStatus>
<zoneStatus>
  <!--opt, bool, whether the user has the permission to receive zone status event, desc:the zones refer to all zones linked to the user, and the zone status refers to wireless detector's power, temperature, online status, etc., and peripheral module's connection status, sensor error status, etc. The corresponding CID event codes are: [1306,3306,1570,3570,1144,3144,1380,3380,1158,3158,1159,3159]-->true
</zoneStatus>
<externalDeviceStatus>
  <!--opt, bool, whether the user has the permission to receive peripheral status event, desc:the peripherals refer to all peripherals linked to the user, such as repeater, wireless repeater, expansion module AC power, wireless peripheral, wireless sounder, keypad, and card reader; the peripheral status refers to the status of these devices. The corresponding CID event codes are: [1312,3312,1319,3319,1312,3312,1384,3384,1301,3301,1311,3311,1301,3301,1381,3381,1384,3384,1381,3381,1381,3381,1381,3381,1384,3384,1384,3384,1501,3501,1501,3501,1306,3306,1306,3306,1306,3306]-->true
</externalDeviceStatus>
</statusUploadPermission>
<eventUploadPermission>
  <!--opt, object, event uploading permission, desc:the uploading protocol of security control panel event is CID (event identifier: cidEvent). For definition values of CID, refer to the CID nodes on PMP platform-->
<panicAlarm>
  <!--opt, bool, whether the user has the permission to receive emergency alarm event, desc:emergency alarm event is triggered by card, keyfob, or keypad, and can be triggered in zones. The corresponding CID event codes are: [1120,3120]-->true
</panicAlarm>
<medicalAlarm>
  <!--opt, bool, whether the user has the permission to receive medical alarm event, desc:medical alarm event is triggered via card, keyfob, or keypad by a person. The corresponding CID event codes are: [1100,3100]-->true
</medicalAlarm>
<fireAlarm>
  <!--opt, bool, whether the user has the permission to receive fire alarm event, desc:fire alarm event is uploaded when a fire occurs. The corresponding CID event codes are: [1111,3111,1114,3114]-->true
</fireAlarm>
<smartAlarm>
  <!--opt, bool, whether the user has the permission to receive intelligent alarm event, desc:intelligent alarm events include intrusion detection alarm, line crossing alarm, fire source alarm, temperature alarm, motion detection alarm, and region entrance/exiting alarm. The corresponding CID event codes are: [1131,3131,1131,3131,1112,3112,1158,3158,1159,3159,1158,3158,1159,3159,1131,3131,1134]-->true
</smartAlarm>
<panelTamper>
  <!--opt, bool, whether the user has the permission to receive security control panel tampering event, desc:the security control panel tampering event refers to the alarm event triggered when the security control panel is being tampered. The corresponding CID event codes are: [1137,3137]-->true
</panelTamper>
<externalDeviceTamper>
  <!--opt, bool, whether the user has the permission to receive peripheral tampering event, desc:the peripherals include extension module, wireless repeater, wireless peripheral, and the peripheral tampering event refers to the alarm triggered when the linked peripheral is being tampered. The corresponding CID event codes are: [1144,3144,1144,3144,1144,3144,1144,3144]-->true
</externalDeviceTamper>
<zoneAlarm>
  <!--opt, bool, whether the user has the permission to receive zone alarm event and zone tampering event, desc:the zone alarm events include instant alarm, time out alarm, burglary alarm, 24-hour alarm, delay alarm, device moving alarm, detector blocking alarm, zone detector tampering alarm, water leak alarm. The corresponding CID event codes are: [1120,3120,1100,3100,1111,3111,1114,3114,1162,3162]-->true
</zoneAlarm>
<panelOperation>
  <!--opt, bool, whether the user has the permission to receive security control panel's operation event, desc:the security control panel's operation events refer to the events triggered when some operations are performed on the security control panel, including disarming, away/stay/auto/key zone arming/disarming, clearing alarm, pacing, forward/backward programming, control panel reset, entering incorrect password. The corresponding CID event codes are: [1401,3401,3441,1403,3403,1409,3409,1121,1406,1452,1455,1454,3250,1865,1607,3607,1627,1628,1305,1461]-->true
</panelOperation>
</eventUploadPermission>
<FDLibPermissionList>
  <!--ro, opt, object, permission list of face picture library control-->
<FDLibPermission>
  <!--ro, opt, object, permission of face picture library control-->
  <FDID>
    <!--ro, req, string, face picture library ID, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
  </FDID>
  <FDName>
    <!--ro, opt, string, face picture library name, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
  </FDName>
  <FDLibControl>
    <!--ro, req, bool, whether to enable the permission of face picture library control, desc:permission of face picture library control includes adding/deleting/editing/getting the information in the library-->true
  </FDLibControl>
</FDLibPermission>
</FDLibPermissionList>
<picSearch>
  <!--ro, opt, bool, whether to enable the permission of picture search, desc:true-enabled, false-disabled. See details in /ISAPI/ContentMgmt/search-->true
</picSearch>
<pircamVideoAndPicture>
  <!--opt, bool, whether to enable the permission to receive pircam camera video or picture event, desc:when an alarm is triggered, if this permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or picture. There is no corresponding CID event code-->true
</pircamVideoAndPicture>
<IPCVideoAndPicture>
  <!--opt, bool, whether to enable the permission to receive network camera video or picture event, desc:when an alarm is triggered, if this permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or picture. There is no corresponding CID event code-->true

```



```

</IPCVideoAndPicture>
<internalParametersAdjustment>
  <!--opt, bool-->true
</internalParametersAdjustment>
<subSystemMode>
  <!--opt, enum, subType:string-->assigned
</subSystemMode>
</remotePermission>
<passwordValidity>
  <!--opt, int, password validity period, range:[0,365], desc:unit: day; value range: [0, 365]; only the administrator can edit-->1
</passwordValidity>
</UserPermission>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode
4	Invalid Operation	privacyProtectionFunctionConflictsWithAiByNvrDisableAiByNvrOfTheChannel	0x4000A5I
4	Invalid Operation	noMoreChannelsCanBeUsedToEnableTheOriginalStreamForPrivacyProtection	0x4000A6.

9.1.12.5 Get the configuration capability of a specific user

Request URL

GET /ISAPI/Security/users/<indexID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	user (index) ID (it indicates the single ID in the array)

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user, attr:version{req, string, protocolVersion}-->
  <id min="1" max="200">
    <!--ro, req, int, user parameter No., range:[1,200], attr:min{req, int},max{req, int}-->1
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable user, attr:opt{req, string}-->true
  </enabled>
  <userName min="1" max="32">
    <!--ro, req, string, user name, range:[1,32], attr:min{req, int},max{req, int}-->test
  </userName>
  <userNickName>

```

```

<!--ro, opt, string, user's nickname, range:[1,32], desc:userNickName is the user's nickname on the Hik-Connect client, and the user can modify it. The
web client can get userNickName via the device-->test
</userNickName>
<phoneNum min="1" max="11">
<!--ro, opt, string, phone number, range:[1,11], attr:min{req, int},max{req, int}, desc:it can replace userName for Login. The sensitive information
should be encrypted-->test
</phoneNum>
<emailAddress min="1" max="32">
<!--ro, opt, string, email address, range:[1,32], attr:min{req, int},max{req, int}, desc:it can replace userName for Login. The sensitive information
should be encrypted-->test
</emailAddress>
<oldPassword min="1" max="16">
<!--ro, opt, string, old password before the current logging-in, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-
-->null
</oldPassword>
<password min="1" max="16">
<!--ro, req, string, current password, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
</password>
<duressPasswordMode opt="auto,custom">
<!--ro, opt, string, range:[1,16], attr:opt{req, string}-->null
</duressPasswordMode>
<duressPassword min="1" max="16">
<!--ro, opt, string, duress password, which is used for special password which is under duress, the sensitive information should be encrypted, range:
[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
</duressPassword>
<keypadPassword min="1" max="16">
<!--ro, opt, string, keypad password, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
</keypadPassword>
<singleKeypadEnable opt="true,false">
<!--ro, opt, bool, whether to enable the one-time keypad password, attr:opt{req, string}-->true
</singleKeypadEnable>
<invalidKeypadPassword opt="true,false">
<!--ro, opt, bool, whether the current keypad password is invalid, attr:opt{req, string}, desc:it is valid after setting singleKeypadEnable-->true
</invalidKeypadPassword>
<userOperateType opt="1,2,3">
<!--ro, opt, enum, user type, subType:int, attr:opt{req, string}, desc:operator accounts shared from the cloud can log in to the cloud, the Web Client,
the Local device, operator accounts created on the cloud and LAN can only log in to Local device 1 (network user), 2 (keypad user), 3 (network user and
keypad user)-->1
</userOperateType>
<bondIpAddressList>
<!--ro, opt, array, List of socket IP address that bound to the user, subType:object-->
<bondIpAddress>
<!--ro, opt, object, socket IP address that bound to the user-->
<id>
<!--ro, req, int, bounded IP address No., range:[1,16]-->1
</id>
<ipAddress>
<!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
</ipAddress>
<ipv6Address>
<!--ro, opt, string, bounded IPv6 address, range:[1,32]-->test
</ipv6Address>
</bondIpAddress>
</bondIpAddressList>
<bondMacAddressList>
<!--ro, opt, array, List of MAC address that bound to the user, subType:object-->
<bondMacAddress>
<!--ro, opt, object, MAC address that bound to the user-->
<id>
<!--ro, req, int, bounded MAC address No., range:[1,16]-->1
</id>
<macAddress>
<!--ro, opt, string, bounded MAC address, range:[1,32]-->test
</macAddress>
</bondMacAddress>
</bondMacAddressList>
<userLevel opt="Administrator,Operator,Viewer,installer,manufacturer,single,installerAdmin,installEmployee,Employee,DepartmentManager">
<!--ro, opt, enum, user type, subType:string, attr:opt{req, string}, desc:"Administrator", "Operator", "Viewer" (viewer, which is not applied on devices
currently), "installer", "manufacturer", "single" (single user, which is not applied on devices currently), "installerAdmin" (installation admin),
"installEmployee" (installation employee), "Employee" (for attendance), "DepartmentManager" (department manager, for attendance)-->Administrator
</userLevel>
<attribute>
<!--ro, opt, object, user's special attributes-->
<inherent>
<!--ro, opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
</inherent>
</attribute>
<belongUserId>
<!--ro, opt, int, upper-level user ID, range:[0,16], desc:0 (no upper-level user)-->1
</belongUserId>
<belongOrgNo>
<!--ro, opt, int, organization No., range:[1,16]-->1
</belongOrgNo>
<cardList>
<!--ro, opt, object, List of cards that belong to the user-->
<card>
<!--ro, opt, object, card-->
<id>
<!--ro, opt, int, No., range:[1,16]-->1
</id>
<name>
<!--ro, opt, string, name, range:[1,32]-->test
</name>

```

```

</card>
</cardList>
<remoteCtrlList>
  <!--ro, opt, array, List of keyfobs that belong to the user, subType:object-->
  <remoteCtrl>
    <!--ro, opt, object, keyfob-->
    <id>
      <!--ro, opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
  </remoteCtrl>
</remoteCtrlList>
<userNo min="501" max="null">
  <!--ro, opt, int, displayed No. corresponding to the user ID, attr:min{req, int},max{req, int},
  desc:the No. is used on the App and the Web Client and reporting alarms to ARC. It is valid to all user types. The format of the No. is 500 + id adding
  userNo for supporting distinguishing users from user names on some ARC
  scenario: department. residents in one department shall share a security control panel and ARC service. Only one resident shall be the administrator. for
  this scenario, after adding residents as users on the ARC, user ID of the server and App users shall be linked, for contacting the corresponding user when
  problems occur. All users shall know their numbers on the ARC after creating their App accounts-->1
  <userNo>
  <adminType opt="Cloud,LAN">
    <!--ro, opt, enum, admin type, subType:string, attr:opt{req, string}, desc:for hybrid security control panel, it is used for distinguishing whether the
    administrator account is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account
    created on the cloud), "LAN" (user account created on LAN)-->Cloud
  </adminType>
  <installerType opt="Cloud,LAN">
    <!--ro, opt, enum, installer type, subType:string, attr:opt{req, string}, desc:for hybrid security control panel, it is used for distinguishing whether
    the installer account is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account
    created on the cloud), "LAN" (user account created on LAN)-->Cloud
  </installerType>
  <PasswordPeriodPrompt>
    <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
    expired shall be prompted during logging-in. password changing is not required in using-->
    <passwordPeriodType opt="permanent,byDate" def="permanent">
      <!--ro, req, enum, notification type of the password validity period, subType:string, attr:opt{req, string},def{req, string}, desc:"permanent" (take
      effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take
      effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
    </passwordPeriodType>
    <passwordPeriodbyDate min="1" max="1200" def="90">
      <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:h, dep:and,
      {$:User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}, attr:min{req, int},max{req, int},def{req, int}-->90
    </passwordPeriodbyDate>
  </PasswordPeriodPrompt>
  <userActivationStatus opt="true,false">
    <!--ro, opt, bool, user activation status, attr:opt{req, string}, desc:true (activated; default value), false (inactivated)-->true
  </userActivationStatus>
  <invalidDuressPassword opt="true,false">
    <!--ro, opt, bool, whether current duress password is invalid, attr:opt{req, string}-->true
  </invalidDuressPassword>
  <userID min="1" max="64">
    <!--ro, opt, string, user No., range:[1,64], attr:min{req, int},max{req, int}, desc:UUID (unique ID of the user)-->test
  </userID>
  <passwordType opt="simple,complex">
    <!--ro, opt, enum, password type, subType:string, attr:opt{req, string}, desc:simple, "complex"-->simple
  </passwordType>
  <temporaryLocalOperator>
    <!--ro, opt, object, information of temporary Local operator, desc:the temporary Local operator can only log in on Local keypad-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the permission of temporary Local operator, attr:opt{req, string}-->true
    </enabled>
    <scheduledOperationList size="1">
      <!--ro, opt, array, validity period List of the permission of temporary Local operator, subType:object, attr:size{req, int}-->
      <scheduledOperation>
        <!--ro, opt, object, validity period of the permission of temporary Local operator-->
        <beginTime>
          <!--ro, req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
        </endTime>
      </scheduledOperation>
    </scheduledOperationList>
    <maxValidTimeSpan min="1" max="720">
      <!--ro, opt, int, maximum valid span for one time, range:[1,720], step:1, unit:h, attr:min{req, int},max{req, int}-->1
    </maxValidTimeSpan>
  </temporaryLocalOperator>
  <IssuerDN opt="true,false">
    <!--ro, opt, bool, whether it has a bound DN, attr:opt{req, string}, desc:a bound DN refers to a bound server directory.The process of the client
    sending a request to establish session with the LDAP server is called binding.When binding, the client needs to specify the access account so as to access
    the directory information on the server.The user searches for contents according to the bound directory.This node will be used only when logging in to the
    device (GB35114) via pin (UKey)-->true
  </IssuerDN>
  <isSupportModifyAdminUserName>
    <!--ro, opt, bool, support modifying the user name of administrator, desc:if this node is not returned or its value is false, the function is not
    supported. This node will be returned when getting the capability of configuring the administrator information parameters-->true
  </isSupportModifyAdminUserName>
  <userGroupID min="0" max="3" def="0">
    <!--ro, opt, int, user group ID, range:[0,3], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist or its value is 0, it
    indicates that there are no groups. As for the security inspection, the upper-level software supports dividing users into several groups and delivering the
    group IDs to them, which is managed by the device.-->0
  </userGroupID>
  <isSupportLocalBackup opt="true,false">

```

```

<!--ro, opt, bool, whether it supports configuring the weak password, attr:opt{req, string}, desc:according to the requirements of the security red
line, weak password is not allowed to configure in the new version-->true
</isSupportWeakPassword>
<isSupportKeyWord opt="true,false">
  <!--ro, opt, bool, whether it supports the key word in the password, attr:opt{req, string}, desc:true (support the key word in the password), false (not
support the key word in the password)-->true
</isSupportKeyWord>
<facePicture>
  <!--ro, opt, object-->
  <filePathType opt="URL,multipart">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->multipart
  </filePathType>
  <filePath min="0" max="262000">
    <!--ro, opt, string, range:[0,262000], attr:min{req, int},max{req, int}-->facePicture
  </filePath>
</facePicture>
</User>

```

9.1.12.6 Get information about a single user

Request URL

GET /ISAPI/Security/users/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, user parameter No.-->1
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable user-->true
  </enabled>
  <userName>
    <!--ro, req, string, user name, range:[1,32]-->test
  </userName>
  <phoneNum>
    <!--ro, opt, string, phone number, range:[1,11], desc:the sensitive information should be encrypted-->test
  </phoneNum>
  <emailAddress>
    <!--ro, opt, string, Email Address, range:[1,32], desc:the sensitive information should be encrypted-->test
  </emailAddress>
  <duressPassword>
    <!--ro, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
  </duressPassword>
  <keypadPassword>
    <!--ro, opt, string, Keypad Password, range:[1,16], desc:the sensitive information should be encrypted. entering password is required when the security
control panel is armed or disarmed via keypad-->test
  </keypadPassword>
  <singleKeypadEnable>
    <!--ro, opt, bool, whether to enable the keypad password for one time-->true
  </singleKeypadEnable>
  <invalidkeypadPassword>
    <!--ro, opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
  </invalidkeypadPassword>
  <userOperateType>
    <!--ro, opt, enum, user type, subType:int,
desc:1 (if userOperateType is 1 or none, password is valid. if method is POST, password is required. otherwise this node is optional)
2 (if userOperateType is 2, keypadPassword is valid. if method is POST, keypadPassword is required. otherwise this node is optional)
3 (if userOperateType is 3,both passwordkeypad and Password is valid. if method is POST, password and keypadPassword is required. otherwise this node is
optional) 1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
  </userOperateType>
  <bondIpAddressList>
    <!--ro, opt, object, List of socket IP address that bound to the user-->
    <bondIpAddress>
      <!--ro, opt, object, socket IP address that bound to the user-->
      <id>
        <!--ro, req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
      </id>
      <ipAddress>
        <!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>

```

```

    <!--ro, opt, string, bounded IPv6 address, range:[1,128]-->test
    </ipV6Address>
  </bondIpAddress>
</bondIpAddressList>
<bondMacAddressList>
  <!--ro, opt, object, List of mac address that bound to the user-->
  <bondMacAddress>
    <!--ro, opt, object, mac address that bound to the user-->
    <id>
      <!--ro, req, int, bounded mac address No., range:[1,16], desc:It starts from 1-->1
    </id>
    <macAddress>
      <!--ro, opt, string, bounded mac address, range:[1,32]-->test
    </macAddress>
  </bondMacAddress>
</bondMacAddressList>
</bondMacAddressList>
<userLevel>
  <!--ro, opt, enum, user level, subType:string, desc:"Administrator", "Operator", "Viewer" (viewer, which is not applied on devices currently),
  "installer", "manufacturer", "single" (single user, which is not applied on devices currently), "installerAdmin" (installation admin), "installEmployee"
  (installation employee), "Employee" (for attendance), "DepartmentManager" (department manager, for attendance)-->Administrator
  </userLevel>
<attribute>
  <!--ro, opt, object, user's special attributes-->
  <inherent>
    <!--ro, opt, bool, whether it can be deleted, desc:true: the user cannot be deleted. false / the node is invalid: the user can be deleted-->true
  </inherent>
</attribute>
<cardList>
  <!--ro, opt, object, List of cards that belong to the user-->
  <card>
    <!--ro, opt, object, card-->
    <id>
      <!--ro, opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
  </card>
</cardList>
<remoteCtrlList>
  <!--ro, opt, object, List of keyfobs that belong to the user-->
  <remoteCtrl>
    <!--ro, opt, object, keyfob-->
    <id>
      <!--ro, opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
  </remoteCtrl>
</remoteCtrlList>
<userNo>
  <!--ro, opt, int, displayed No. corresponding to the user ID, desc:the No. is used on the App and the Web Client and reporting alarms to ARC. it is
  valid to all user types. the format of the No. is 500 + id-->501
  </userNo>
  <adminType>
    <!--ro, opt, enum, admin type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the administrator account
    is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
    (user account created on LAN)-->Cloud
  </adminType>
  <installerType>
    <!--ro, opt, enum, installer type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the installer account
    is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
    (user account created on LAN)-->Cloud
  </installerType>
  <PasswordPeriodPrompt>
    <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
    expired shall be prompted during logging-in, but the expired password does not affect the usage-->
    <passwordPeriodType>
      <!--ro, req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take
      effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password
      expired shall be prompted in time, but password changing is not required in using-->permanent
    </passwordPeriodType>
    <passwordPeriodbyDate>
      <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
      {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
    </passwordPeriodbyDate>
  </PasswordPeriodPrompt>
  <userActivationStatus>
    <!--ro, opt, bool, user activation status, desc:true (activated; default value), false (inactivated)-->true
  </userActivationStatus>
  <invalidDuressPassword>
    <!--ro, opt, bool, whether current duress password is invalid, desc:if the duress password is not configured, the value is true-->true
  </invalidDuressPassword>
  <passwordType>
    <!--ro, opt, enum, subType:string-->simple
  </passwordType>
  <temporaryLocalOperator>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </temporaryLocalOperator>
  <scheduledOperationList>
    <!--ro, opt, array, subType:object, dep:or,{$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->

```

```

<scheduledOperation>
  <!--ro, opt, object-->
  <beginTime>
    <!--ro, req, datetime-->1970-01-01T00:00:00+08:00
  </beginTime>
  <endTime>
    <!--ro, req, datetime-->1970-01-01T00:00:00+08:00
  </endTime>
</scheduledOperation>
</scheduledOperationList>
<validPeriodStatus>
  <!--ro, opt, enum, subType:string-->notReached
</validPeriodStatus>
<remainingValidTime>
  <!--ro, opt, int, range:[0,720], step:1, unit:h-->1
</remainingValidTime>
</temporaryLocalOperator>
<IssuerDN>
  <!--ro, opt, bool-->>true
</IssuerDN>
</User>

```

9.1.12.7 Set information for a single user

Request URL

PUT /ISAPI/Security/users/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, user parameter No., range:[1,16], desc:supported range: [1,16]-->1
  </id>
  <enabled>
    <!--opt, bool, whether to enable user-->true
  </enabled>
  <userName>
    <!--req, string, user name, range:[1,32]-->test
  </userName>
  <phoneNumber>
    <!--wo, opt, string, phone number, range:[1,11], desc:the sensitive information should be encrypted-->test
  </phoneNumber>
  <emailAddress>
    <!--wo, opt, string, email address, range:[1,32], desc:the sensitive information should be encrypted-->test
  </emailAddress>
  <password>
    <!--wo, opt, string, password, range:[1,16], desc:the sensitive information should be encrypted-->test
  </password>
  <duressPasswordMode>
    <!--wo, opt, enum, subType:object-->auto
  </duressPasswordMode>
  <duressPassword>
    <!--wo, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
  </duressPassword>
  <keypadPassword>
    <!--wo, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the security control panel is armed or disarmed via keypad-->test
  </keypadPassword>
  <invalidKeypadPassword>
    <!--opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
  </invalidKeypadPassword>
  <bondIpAddressList>
    <!--opt, object, list of socket IP address that bound to the user-->
    <bondIpAddress>
      <!--opt, object, socket IP address that bound to the user-->
      <id>
        <!--req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
      </id>
      <ipAddress>
        <!--opt, string, bounded IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, bounded IPv6 address, range:[1,32]-->test
      </ipv6Address>
    </bondIpAddress>
  </bondIpAddressList>
  <bondMacAddressList>

```

```

<!--opt, object, List of MAC address that bound to the user-->
<bondMacAddress>
  <!--opt, object, MAC address that bound to the user-->
  <id>
    <!--req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
  </id>
  <macAddress>
    <!--opt, string, bounded MAC address, range:[1,32]-->test
  </macAddress>
</bondMacAddress>
</bondMacAddressList>
<attribute>
  <!--opt, object, user's special attributes-->
  <inherent>
    <!--opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
  </inherent>
</attribute>
<loginPassword>
  <!--req, string, confirm password, range:[1,16], desc:confirming password is required when adding and editing user information. Encrypting the sensitive information is required when the confirm password is transmitted via HTTP-->King726014
</loginPassword>
<cardList>
  <!--opt, object, List of cards that belong to the user-->
  <card>
    <!--opt, object, card-->
    <id>
      <!--opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--opt, string, name, range:[1,32]-->test
    </name>
  </card>
</cardList>
<remoteCtrlList>
  <!--opt, object, List of keyfobs that belong to the user-->
  <remoteCtrl>
    <!--opt, object, keyfob-->
    <id>
      <!--opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--opt, string, name, range:[1,32]-->test
    </name>
  </remoteCtrl>
</remoteCtrlList>
<PasswordPeriodPrompt>
  <!--opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password expired shall be prompted during Logging-in, but the expired password does not affect the usage-->
  <passwordPeriodType>
    <!--req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
  </passwordPeriodType>
  <passwordPeriodbyDate>
    <!--opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
    {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
  </passwordPeriodbyDate>
</PasswordPeriodPrompt>
<temporaryLocalOperator>
  <!--opt, object, information of temporary Local operator, desc:the temporary Local operator can only Log in on Local keypad-->
  <scheduledOperationList>
    <!--opt, array, validity period list of the permission of temporary Local operator, subType:object, dep:and,
    {$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
    <scheduledOperation>
      <!--opt, object, validity period of the permission of temporary Local operator-->
      <beginTime>
        <!--req, dateTime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, dateTime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
      </endTime>
    </scheduledOperation>
  </scheduledOperationList>
</temporaryLocalOperator>
<IssuerDN>
  <!--opt, bool, whether it has a bound DN, desc:a bound DN refers to a bound server directory. The process of the client sending a request to establish session with the LDAP server is called binding. When binding, the client needs to specify the access account so as to access the directory information on the server. The user searches for contents according to the bound directory. N/A-->true
</IssuerDN>
<userGroupID>
  <!--opt, int, user group ID, range:[0,3], desc:if this node does not exist or its value is 0, it indicates that there are no groups. As for the security inspection, the upper-level software supports dividing users into several groups and delivering the group IDs to them, which is managed by the device.-->0
</userGroupID>
<facePicture>
  <!--opt, object-->
  <filePathType>
    <!--req, enum, subType:string-->multipart
  </filePathType>
  <filePath>
    <!--opt, string, range:[0,262000]-->test
  </filePath>
</facePicture>
</User>

```


Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
User	[Message content]	application/xml	--	--	--
facePicture	[Binary picture data]	image/jpeg	facePicture	facePicture.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"Lock", "unlock". It is returned when verifying LoginPassword failed-->lock
  </lockStatus>
  <resLockTime>
    <!--ro, opt, int, range:[0,1800], unit:s, dep:or,{$.ResponseStatus.LockStatus,eq,Lock}-->1
  </resLockTime>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying LoginPassword failed-->1
  </retryTimes>
  <lockType>
    <!--ro, opt, enum, subType:string, dep:or,{$.ResponseStatus.LockStatus,eq,Lock}-->lockIP
  </lockType>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	exDevNotExist	0x40008016	--
4	Invalid Operation	keypadPasswordConflict	0x4000A33F	--

9.1.12.8 Get information about all users

Request URL

GET /ISAPI/Security/users

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<UserList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

```
<!--ro, req, array, user List, subType:object, attr:version{req, string, protocolVersion}-->
```

```
<User>
```

```
<!--ro, opt, object, user-->
```

```
<id>
```

```
<!--ro, req, int, user parameter No., range:[1,200]-->1
```

```
</id>
```

```
<enabled>
```

```
<!--ro, opt, bool, whether to enable user-->true
```

```
</enabled>
```

```
<userName>
```

```
<!--ro, req, string, user name, range:[1,32]-->test
```

```
</userName>
```

```
<userNickName>
```

```
<!--ro, opt, string, user's nickname, range:[1,32], desc:userNickName is the user's nickname on the Hik-Connect client, and the user can modify it.
```

The web client can get userNickName via the device-->test

```
</userNickName>
```

```
<phoneNumber>
```

```
<!--ro, opt, string, phone number, range:[1,13], desc:the sensitive information should be encrypted-->test
```

```
</phoneNumber>
```

```
<emailAddress>
```

```
<!--ro, opt, string, email address, range:[1,32], desc:the sensitive information should be encrypted-->test
```

```
</emailAddress>
```

```
<duressPassword>
```

```
<!--ro, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
```

```
</duressPassword>
```

```
<keypadPassword>
```

<!--ro, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the security control panel is armed or disarmed via keypad-->test

```
</keypadPassword>
```

```
<singleKeypadEnable>
```

```
<!--ro, opt, bool, whether to enable single keypad operation-->true
```

```
</singleKeypadEnable>
```

```
<invalidKeypadPassword>
```

```
<!--ro, opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
```

```
</invalidKeypadPassword>
```

```
<userOperateType>
```

```
<!--ro, opt, enum, user operation type, subType:int,
```

desc:1 (if userOperateType is 1 or none, password is valid, if method is POST, password is required. otherwise this node is optional)

2 (if userOperateType is 2, keypadPassword is valid. if method is POST, keypadPassword is required. otherwise this node is optional)

3 (if userOperateType is 3, both passwordKeypad and Password is valid. if method is POST, password and keypadPassword is required. otherwise this node is optional) 1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1

```
</userOperateType>
```

```
<bondIpAddressList>
```

```
<!--ro, opt, object, List of socket IP address that bound to the user-->
```

```
<bondIpAddress>
```

```
<!--ro, opt, object, socket IP address that bound to the user-->
```

```
<id>
```

```
<!--ro, req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
```

```
</id>
```

```
<ipAddress>
```

```
<!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
```

```
</ipAddress>
```

```
<ipv6Address>
```

```
<!--ro, opt, string, bounded IPv6 address, range:[1,128]-->test
```

```
</ipv6Address>
```

```
</bondIpAddress>
```

```
</bondIpAddressList>
```

```
<bondMacAddressList>
```

```
<!--ro, opt, object, List of MAC address that bound to the user-->
```

```
<bondMacAddress>
```

```
<!--ro, opt, object, MAC address that bound to the user-->
```

```
<id>
```

```
<!--ro, req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
```

```
</id>
```

```
<macAddress>
```

```
<!--ro, opt, string, bounded MAC address, range:[1,32]-->test
```

```
</macAddress>
```

```
</bondMacAddress>
```

```
</bondMacAddressList>
```

```
<userLevel>
```

<!--ro, opt, enum, user Level, subType:string, desc:"Administrator", "Operator", "Viewer", "installer", "manufacturer", "single" (single user), "installerAdmin" (installation admin), "installEmployee" (installation employee), "Employee", "DepartmentManager"-->Administrator

```
</userLevel>
```

```
<attribute>
```

```

<!--ro, opt, object, user's special attributes-->
<inherent>
  <!--ro, opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
</inherent>
</attribute>
<cardList>
  <!--ro, opt, object, List of cards that belong to the user-->
  <card>
    <!--ro, opt, object, card-->
    <id>
      <!--ro, opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
  </card>
</cardList>
<remoteCtrlList>
  <!--ro, opt, object, List of keyfobs that belong to the user-->
  <remoteCtrl>
    <!--ro, opt, object, keyfob-->
    <id>
      <!--ro, opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
  </remoteCtrl>
</remoteCtrlList>
<userNo>
  <!--ro, opt, int, displayed No. corresponding to the user ID, desc:the No. is used on the App and the Web Client and reporting alarms to ARC. it is valid to all user types. the format of the No. is 500 + id-->501
</userNo>
<adminType>
  <!--ro, opt, enum, admin type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the administrator account is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN" (user account created on LAN)-->Cloud
</adminType>
<installerType>
  <!--ro, opt, enum, installer type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the installer account is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN" (user account created on LAN)-->Cloud
</installerType>
<passwordPeriodPrompt>
  <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password expired shall be prompted during logging-in, but the expired password does not affect the usage-->
  <passwordPeriodType>
    <!--ro, req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
  </passwordPeriodType>
  <passwordPeriodbyDate>
    <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
    {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
  </passwordPeriodbyDate>
</PasswordPeriodPrompt>
<userActivationStatus>
  <!--ro, opt, bool, user activation status, desc:true (activated; default value), false (inactivated)-->true
</userActivationStatus>
<invalidDuressPassword>
  <!--ro, opt, bool, whether current duress password is invalid, desc:if the duress password is not configured, the value is true-->true
</invalidDuressPassword>
<userID>
  <!--ro, opt, string, user No., range:[1,64], desc:UUID (unique ID of the user)-->test
</userID>
<passwordType>
  <!--ro, opt, enum, password type, subType:string,
  desc:1. Attendance check devices support setting simple password and complex password. The password is used for device activation and login.
  2. The password should contain 6 digits. "simple", "complex"-->simple
</passwordType>
<temporaryLocalOperator>
  <!--ro, opt, object, information of temporary Local operator, desc:the temporary Local operator can only Log in on Local keypad-->
  <enabled>
    <!--ro, req, bool, whether to enable the permission of temporary Local operator-->true
  </enabled>
  <scheduledOperationList>
    <!--ro, opt, array, validity period List of the permission of temporary Local operator, subType:object, dep:or,
    {$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
    <scheduledOperation>
      <!--ro, opt, object, validity period of the permission of temporary Local operator-->
      <beginTime>
        <!--ro, req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
      </endTime>
    </scheduledOperation>
  </scheduledOperationList>
  <validPeriodStatus>
    <!--ro, opt, enum, validity period status, subType:string, desc:"notReached" (the validity period has not started), "available" (within the validity period), "expired"-->notReached
  </validPeriodStatus>
  <remainingValidTime>
    <!--ro, opt, int, remaining validity period, range:[0, 3600], unit:秒, dep:and,
    {$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
  </remainingValidTime>

```

```
<!--ro, opt, int, remaining validity period, range:[0,120], step:1, unit:n-->1
</remainingValidTime>
</temporaryLocalOperator>
<userGroupID>
  <!--ro, opt, int, user group ID, range:[0,3], desc:if this node does not exist or its value is 0, it indicates that there are no groups. As for the
  security inspection, the upper-level software supports dividing users into several groups and delivering the group IDs to them, which is managed by the
  device.-->0
  </userGroupID>
</User>
</UserList>
```

9.1.12.9 Set information about all users

Request URL

PUT /ISAPI/Security/users

Query Parameter

None

Request Message

hieu.tu@tacorp.vn
CNG TY C PHN U T CNG NGH THIN N

```

<?xml version="1.0" encoding="UTF-8"?>

<UserList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user list, attr:version{req, string, protocolVersion}-->
  <User>
    <!--opt, object, user-->
    <id>
      <!--req, int, user parameter No., range:[1,64]-->1
    </id>
    <enabled>
      <!--opt, bool, whether to enable user-->true
    </enabled>
    <userName>
      <!--req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--wo, req, string, password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </password>
    <keypadPassword>
      <!--wo, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the security control panel is armed or disarmed via keypad-->test
    </keypadPassword>
    <userOperateType>
      <!--opt, enum, user type, subType:int, desc:1 (network user), 2 (keypad user), 3 (network and keypad user). When userOperateType is 1 or none, the node password is valid and it is required when the method is POST. When userOperateType is 2, the node keypadPassword is valid and it is required when the method is POST. When userOperateType is 3, both nodes password and keypadPassword are valid and they are required when the method is POST.-->1
    </userOperateType>
    <bondIpAddressList>
      <!--opt, object, List of socket IP address that bound to the user-->
      <bondIpAddress>
        <!--opt, object, socket IP address that bound to the user-->
        <id>
          <!--req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <ipAddress>
          <!--opt, string, bounded IPv4 address, range:[1,32]-->test
        </ipAddress>
        <ipv6Address>
          <!--opt, string, bounded IPv6 address, range:[1,32]-->test
        </ipv6Address>
      </bondIpAddress>
    </bondIpAddressList>
    <bondMacAddressList>
      <!--opt, object, List of MAC address that bound to the user-->
      <bondMacAddress>
        <!--opt, object, MAC address that bound to the user-->
        <id>
          <!--req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <macAddress>
          <!--opt, string, bounded MAC address, range:[1,32]-->test
        </macAddress>
      </bondMacAddress>
    </bondMacAddressList>
    <userLevel>
      <!--opt, enum, user Level, subType:string, desc:"Administrator", "Operator", "Viewer", "installer", "manufacturer", "single" (single user), "installerAdmin" (installation admin), "installEmployee" (installation employee)-->Administrator
    </userLevel>
    <attribute>
      <!--opt, object, user's special attributes-->
      <inherent>
        <!--opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false/the node is invalid (the user can be deleted)-->true
      </inherent>
    </attribute>
    <loginPassword>
      <!--req, string, confirm password, range:[1,16], desc:confirming password is required when adding and editing user information. Encrypting the sensitive information is required when the confirm password is transmitted via HTTP-->King726014
    </loginPassword>
    <PasswordPeriodPrompt>
      <!--opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password expired shall be prompted during logging-in. password changing is not required in using-->
      <passwordPeriodType>
        <!--req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
      </passwordPeriodType>
      <passwordPeriodbyDate>
        <!--opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and, {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
      </passwordPeriodbyDate>
    </PasswordPeriodPrompt>
    <userID>
      <!--opt, string, user No., range:[1,64], desc:UUID (unique ID of the user)-->test
    </userID>
  </User>
</UserList>

```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"Lock", "unlock". It is returned when verifying LoginPassword failed-->lock
  </lockStatus>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying LoginPassword failed-->1
  </retryTimes>
</ResponseStatus>
```

9.2 Network Configuration

9.2.1 HTTP/HTTPS Access Management

9.2.1.1 Get the certificate type parameters of web service

Request URL

GET /ISAPI/Security/webCertificate

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WebCertificate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, web service certificate type configuration, attr:version{req, string, protocolVersion}-->
  <certificateType>
    <!--ro, req, enum, authentication type, subType:string, desc:"basic", "digest", "digest/basic"-->basic
  </certificateType>
  <securityAlgorithm>
    <!--ro, opt, object, security algorithm, desc:this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:string, desc:"MD5", "SHA256", "MD5/SHA256"-->MD5/SHA256
    </algorithmType>
  </securityAlgorithm>
</WebCertificate>
```

9.2.1.2 Set the certificate type parameters of web service

Request URL

PUT /ISAPI/Security/webCertificate

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WebCertificate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <CertificateType>
    <!--req, enum, authentication type, subType:string, desc:"basic", "digest", "digest/baisc"-->basic
  </CertificateType>
  <SecurityAlgorithm>
    <!--opt, object, security algorithm, desc:this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType>
      <!--opt, enum, algorithm type, subType:string, desc:"MD5", "SHA256", "MD5/SHA256"-->MD5/SHA256
    </algorithmType>
  </SecurityAlgorithm>
</WebCertificate>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, req, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024], desc:the detailed information of custom error returned by device applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

statusCode	statusString	subStatusCode
4	Invalid Operation	switchingFailedForAdminUserPleaseGoToUserPageAndReconfigurePasswordForTheFollow

9.2.2 IP Address Management

9.2.2.1 Get capability of a specific network interface

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	NIC ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkInterface xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, network, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <IPAddress>
    <!--ro, req, object, IP address-->
    <ipVersion opt="v4,v6,dual">
      <!--ro, req, enum, IP address version, subType:string, attr:opt{req, string}, desc:"v4" (IPv4), "v6" (IPv6), "dual" (IPv4 and IPv6)-->v4
    </ipVersion>
  </IPAddress>
</NetworkInterface>
```



```

</ipversion>
<addressingType opt="static,dynamic,apiipa">
  <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"static", "dynamic", "apiipa" (automatic private IP addressing)-->static
</addressingType>
<ipAddress>
  <!--ro, opt, string, IP address-->test
</ipAddress>
<subnetMask>
  <!--ro, opt, string, sub-net mask-->test
</subnetMask>
<ipv6Address>
  <!--ro, opt, string, IPv6 address-->test
</ipv6Address>
<bitMask>
  <!--ro, opt, string, bitmask-->test
</bitMask>
<DefaultGateway>
  <!--ro, opt, object, default gateway-->
  <ipAddress>
    <!--ro, opt, string, IP address-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address-->test
  </ipv6Address>
</DefaultGateway>
<PrimaryDNS>
  <!--ro, opt, object-->
  <ipAddress>
    <!--ro, opt, string, IP address-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address-->test
  </ipv6Address>
</PrimaryDNS>
<SecondaryDNS>
  <!--ro, opt, object, secondary DNS-->
  <ipAddress>
    <!--ro, opt, string, IP address-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address-->test
  </ipv6Address>
</SecondaryDNS>
<DNSEnable opt="true,false">
  <!--ro, opt, bool, whether to enable automatic DNS assignment, attr:opt{req, string}-->true
</DNSEnable>
<IPv6Mode>
  <!--ro, opt, object, IPv6 mode-->
  <ipV6AddressingType opt="ra,manual,dhcp">
    <!--ro, opt, enum, IPv6 address type, subType:string, attr:opt{req, string}, desc:"ra" (router advertisement), "manual" (manual assignment), "dhcp"
(DHCP server assignment)-->ra
  </ipV6AddressingType>
  <ipV6AddressList>
    <!--ro, opt, array, IPv6 address list, subType:object-->
    <v6Address>
      <!--ro, opt, object, IPv6 address-->
      <id min="1" max="10">
        <!--ro, opt, string, ID, attr:min{req, int},max{req, int}-->test
      </id>
      <type opt="ra,manual,dhcp">
        <!--ro, opt, string, type, attr:opt{req, string}-->test
      </type>
      <address min="1" max="10">
        <!--ro, opt, string, address, attr:min{req, int},max{req, int}-->test
      </address>
      <bitMask min="1" max="10">
        <!--ro, opt, int, bitmask, attr:min{req, int},max{req, int}-->1
      </bitMask>
    </v6Address>
  </ipV6AddressList>
</IPv6Mode>
<DoubleNetRoute>
  <!--ro, opt, object, dual WAN router-->
  <ipAddress min="1" max="10">
    <!--ro, opt, string, IP address, attr:min{req, int},max{req, int}-->test
  </ipAddress>
  <subnetMask min="1" max="10">
    <!--ro, opt, string, subnet mask, attr:min{req, int},max{req, int}-->test
  </subnetMask>
</DoubleNetRoute>
<LANConfigMode opt="manual,auto">
  <!--ro, opt, string, attr:opt{req, string}-->test
</LANConfigMode>
<DoubleNetRouteList size="3">
  <!--ro, opt, object, attr:size{req, int}, desc:in dual network isolation mode, up to four router configurations are supported; the address of the
first router needs to be configured via DoubleNetRoute, and the additional expansion routes can be configured through this node-->
  <DoubleNetRoute>
    <!--ro, opt, object, dual WAN router-->
    <ipAddress min="1" max="10">
      <!--ro, opt, string, IP address, attr:min{req, int},max{req, int}-->test
    </ipAddress>
    <subnetMask min="1" max="10">
      <!--ro, opt, string, subnet mask, attr:min{req, int},max{req, int}-->test
    </subnetMask>
  </DoubleNetRoute>

```

```

    </DoubleNetRoute>
  </DoubleNetRouteList>
  <ipv6Enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </ipv6Enabled>
</IPAddress>
<Wireless>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <wirelessNetworkMode opt="infrastructure,adhoc">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->infrastructure
  </wirelessNetworkMode>
  <channel opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,auto">
    <!--ro, opt, enum, channel 1, subType:string, attr:opt{req, string}-->auto
  </channel>
  <ssid min="1" max="10">
    <!--ro, opt, string, SSID, attr:min{req, int},max{req, int}-->test
  </ssid>
  <wmmEnabled>
    <!--ro, opt, bool-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--ro, opt, object, wireless network security-->
    <securityMode opt="disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise,WPA3">
      <!--ro, opt, enum, security mode, subType:string, attr:opt{req, string}-->disable
    </securityMode>
    <WEP>
      <!--ro, opt, object, WEP-->
      <authenticationType opt="open,sharedkey,auto">
        <!--ro, req, enum, Identity Authentication Type, subType:string, attr:opt{req, string}-->open
      </authenticationType>
      <defaultTransmitKeyIndex min="1" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength opt="64,128">
        <!--ro, opt, enum, subType:int, attr:opt{req, string}-->64
      </wepKeyLength>
      <EncryptionKeyList>
        <!--ro, opt, array, subType:object-->
        <encryptionKey>
          <!--ro, opt, string, encryption key-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--ro, opt, object, WPA-->
      <algorithmType opt="TKIP,AES,TKIP/AES">
        <!--ro, req, enum, algorithm type, subType:string, attr:opt{req, string}, desc:algorithm type-->TKIP
      </algorithmType>
      <sharedKey>
        <!--ro, req, string-->test
      </sharedKey>
      <wpaKeyLength min="8" max="64">
        <!--ro, req, int, range:[8,64], attr:min{req, int},max{req, int}-->8
      </wpaKeyLength>
    </WPA>
    <support64bitKey opt="WPA-personal,WPA2-personal">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->WPA-personal
    </support64bitKey>
  </WirelessSecurity>
</AccessPointList>
  <!--ro, opt, array, subType:object-->
  <InterfaceDisplay>
    <!--ro, opt, object-->
    <TableElementList>
      <!--ro, opt, array, subType:object-->
      <TableElement opt="SSID,workingMode,securityMode,channel,signalStrength,speed,connectionStatus">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->securityMode
      </TableElement>
    </TableElementList>
  </InterfaceDisplay>
</AccessPointList>
<isSupportConnectStatus>
  <!--ro, opt, bool-->true
</isSupportConnectStatus>
<workScene opt="computerRoom,monitorTerminal,centralTerminal">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->computerRoom
</workScene>
<protocol opt="802.11ac">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->802.11ac
</protocol>
<hideSsid>
  <!--ro, opt, bool-->true
</hideSsid>
<ChannelConfig>
  <!--ro, opt, object, channel parameters-->
  <transmitPower opt="9,12,15,18,21,24,27">
    <!--ro, opt, enum, power Load, subType:int, attr:opt{req, string}-->9
  </transmitPower>
  <ChannelCountryList>
    <!--ro, opt, array, subType:object-->

```

```

<Country>
  <!--ro, opt, object, country-->
  <countryID>
    <!--ro, req, int, country code-->1
  </countryID>
  <ChannelList>
    <!--ro, opt, object, channel list-->
    <Node>
      <!--ro, opt, object, ID-->
      <width opt="auto,20,40,80">
        <!--ro, req, enum, bandwidth, subType:string, attr:opt{req, string}-->auto
      </width>
      <channel>
        <!--ro, req, int, channel 1-->1
      </channel>
    </Node>
  </ChannelList>
</Country>
</ChannelCountryList>
</ChannelConfig>
<rate opt="2.4GHz,5GHz,auto">
  <!--ro, opt, enum, frequency, subType:string, attr:opt{req, string}-->auto
</rate>
<dialUpGroupNo min="1" max="10">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</dialUpGroupNo>
<BaseNoisevalue min="1" max="10">
  <!--ro, opt, int, background noise value, attr:min{req, int},max{req, int}-->1
</BaseNoisevalue>
</Wireless>
<Discovery>
  <!--ro, opt, object, explore-->
  <UPnP>
    <!--ro, req, object, UPnP-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
  </UPnP>
  <Zeroconf>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
  </Zeroconf>
</Discovery>
<Link>
  <!--ro, opt, object, connected-->
  <MACAddress>
    <!--ro, req, string, MAC address-->test
  </MACAddress>
  <autoNegotiation opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </autoNegotiation>
  <speed opt="0,10,100,1000,2500">
    <!--ro, req, int, "10,100,1000,10000", attr:opt{req, string}, desc:"10,100,1000,10000"-->1
  </speed>
  <duplex opt="half,full">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->half
  </duplex>
  <MTU min="1" max="1500">
    <!--ro, req, int, maximum transmission unit, attr:min{req, int},max{req, int}-->1
  </MTU>
  <linkMode opt="optical,electrical">
    <!--ro, opt, enum, connection mode, subType:string, attr:opt{req, string}, desc:"optical", "electrical"-->optical
  </linkMode>
  <workMode opt="1,2,3">
    <!--ro, opt, int, operating mode, attr:opt{req, string}-->1
  </workMode>
</Link>
<defaultConnection opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</defaultConnection>
<ActiveMulticast>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <streamID opt="main">
    <!--ro, req, string, stream ID, attr:opt{req, string}-->test
  </streamID>
  <ipV4Address min="1" max="10">
    <!--ro, opt, string, IPv4 address, attr:min{req, int},max{req, int}-->test
  </ipV4Address>
  <ipV6Address min="1" max="10">
    <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}-->test
  </ipV6Address>
  <port min="1" max="10">
    <!--ro, opt, int, port No., attr:min{req, int},max{req, int}-->1
  </port>
</ActiveMulticast>
<macAddress min="1" max="10">
  <!--ro, opt, string, MAC address, attr:min{req, int},max{req, int}-->test
</macAddress>
<EthernetPortList size="4">

```

```

<!--ro, opt, array, network interface information, subType:object, attr:size{req, int}-->
<EthernetPort>
  <!--ro, opt, object, network interface-->
  <id min="1" max="4">
    <!--ro, req, int, ID, attr:min{req, int},max{req, int}-->1
  </id>
  <MACAddress>
    <!--ro, req, string, MAC address-->test
  </MACAddress>
  <status opt="connected,disconnect">
    <!--ro, opt, enum, status, subType:string, attr:opt{req, string}, desc:status-->connected
  </status>
  <speed opt="10,100,1000,10000">
    <!--ro, req, enum, speed, subType:int, attr:opt{req, string}, desc:"10,100,1000,10000"-->10
  </speed>
</EthernetPort>
</EthernetPortList>
<Extensions>
  <!--ro, opt, object, extension information-->
  <NetworkCardType opt="normal,internal,external,main,wired">
    <!--ro, opt, enum, NIC type, subType:string, attr:opt{req, string}-->normal
  </NetworkCardType>
</Extensions>
<interfaceName min="1" max="10">
  <!--ro, opt, string, NIC name, attr:min{req, int},max{req, int}-->test
</interfaceName>
<LANConfigMode opt="manual,auto">
  <!--ro, opt, string, attr:opt{req, string}-->test
</LANConfigMode>
<enabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</enabled>
<isSupportSerialNumberInURL>
  <!--ro, opt, bool, whether URL supports containing serialNumber-->true
</isSupportSerialNumberInURL>
<softStorageinterFaceEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</softStorageinterFaceEnabled>
<dualControlNICList size="2">
  <!--ro, opt, array, subType:object, attr:size{req, int}-->
  <dualControlNIC>
    <!--ro, opt, object-->
    <role opt="controllerA,controllerB">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->controllerA
    </role>
    <ipV4Address>
      <!--ro, opt, ipv4-->10.255.255.0
    </ipV4Address>
  </dualControlNIC>
</dualControlNICList>
</NetworkInterface>

```

statusCode	statusString	subStatusCode	errorCode	des
4	Invalid Operation	theCurrentPoeModeDoesNotSupportNetworkParameterConfiguration	0x4000A214	--

9.2.2.2 Set parameters of a specific network interface

Request URL

PUT /ISAPI/System/Network/interfaces/<interfacID>?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
interfacID	string	--
serialNumber	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<NetworkInterface xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, IP address parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, NIC index-->test
  </id>
  <IPAddress>
    <!--req, object, IP address parameters-->

```

```

<ipVersion>
  <!--req, enum, IP address version, subType:string, desc:"v4" (IPv4), "v6" (IPv6), "dual" (IPv4 and IPv6)-->v4
</ipVersion>
<addressingType>
  <!--req, enum, address type, subType:string, desc:"static", "dynamic", "apiipa" (automatic private IP addressing)-->static
</addressingType>
<ipAddress>
  <!--opt, string, IPv4 address-->test
</ipAddress>
<subnetMask>
  <!--opt, string, IPv4 subnet mask-->test
</subnetMask>
<ipv6Address>
  <!--opt, string, IPv6 address-->test
</ipv6Address>
<bitMask>
  <!--opt, string, IPv6 subnet mask-->test
</bitMask>
<DefaultGateway>
  <!--opt, object, default gateway-->
  <ipAddress>
    <!--opt, string, IPv4 address-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address-->test
  </ipv6Address>
</DefaultGateway>
<PrimaryDNS>
  <!--opt, object, preferred DNS-->
  <ipAddress>
    <!--opt, string, IPv4 address-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address-->test
  </ipv6Address>
</PrimaryDNS>
<SecondaryDNS>
  <!--opt, object, alternate DNS-->
  <ipAddress>
    <!--opt, string, IPv4 address-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address-->test
  </ipv6Address>
</SecondaryDNS>
<DNSEnable>
  <!--opt, bool-->true
</DNSEnable>
<IPv6Mode>
  <!--opt, object, IPv6 mode-->
  <ipV6AddressingType>
    <!--opt, enum, IPv6 address type, subType:string, desc:"ra" (router advertisement), "manual" (manual assignment), "dhcp" (DHCP server assignment)-->ra
  </ipV6AddressingType>
  <ipV6AddressList>
    <!--opt, array, IPv6 address List, subType:object-->
    <v6Address>
      <!--opt, object, IPv6 address-->
      <id>
        <!--opt, string, IPv6 address index-->test
      </id>
      <type>
        <!--opt, enum, IPv6 address type, subType:string, desc:"ra" (router advertisement), "manual" (manual assignment), "dhcp" (DHCP server assignment)-->ra
      </type>
      <address>
        <!--opt, string, IPv6 address-->test
      </address>
      <bitMask>
        <!--opt, string, IPv6 subnet mask-->test
      </bitMask>
    </v6Address>
  </ipV6AddressList>
</IPv6Mode>
<DoubleNetRoute>
  <!--opt, object, dual WAN router-->
  <ipAddress>
    <!--opt, string, IP address-->test
  </ipAddress>
  <subnetMask>
    <!--opt, string, sub-net mask-->test
  </subnetMask>
</DoubleNetRoute>
<LANConfigMode>
  <!--opt, enum, subType:string-->manual
</LANConfigMode>
<DoubleNetRouteList>
  <!--opt, array, subType:object, desc:in dual network isolation mode, up to four router configurations are supported; the address of the first router needs to be configured via DoubleNetRoute, and the additional expansion routes can be configured through this node-->
  <DoubleNetRoute>
    <!--opt, object, dual WAN router-->
    <ipAddress>
      <!--opt, string, IP address-->test
    </ipAddress>
  </DoubleNetRoute>

```

```

</ipsecEsp>
<subnetMask>
  <!--opt, string, subnet mask-->test
</subnetMask>
</DoubleNetRoute>
</DoubleNetRouteList>
<ipv6Enabled>
  <!--opt, bool-->>true
</ipv6Enabled>
</IPAddress>
<Wireless>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <wirelessNetworkMode>
    <!--opt, enum, subType:string-->infrastructure
  </wirelessNetworkMode>
  <channel>
    <!--opt, string-->test
  </channel>
  <ssid>
    <!--opt, string, SSID (Service Set Identifier)-->test
  </ssid>
  <wmmEnabled>
    <!--opt, bool, enable WMM (Wi-Fi multimedia)-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--opt, object, wireless network security-->
    <securityMode>
      <!--opt, enum, security mode, subType:string-->disable
    </securityMode>
    <WEP>
      <!--opt, object, dep:or,{$.NetworkInterface.Wireless.WirelessSecurity.securityMode,eq,WEP}-->
      <authenticationType>
        <!--req, enum, authentication type, subType:string, desc:authentication type-->open
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--req, int, default number of transmitted keys-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength>
        <!--opt, int, key length, range:[64,128]-->1
      </wepKeyLength>
      <EncryptionKeyList>
        <!--opt, array, subType:object-->
        <encryptionKey>
          <!--req, string, Secret key-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--opt, object, WPA-personal, dep:or,{$.NetworkInterface.Wireless.WirelessSecurity.securityMode,eq,WPA}-->
      <algorithmType>
        <!--req, enum, algorithm type, subType:string, desc:algorithm type-->TKIP
      </algorithmType>
      <sharedKey>
        <!--req, string, shared key authentication-->test
      </sharedKey>
      <wpaKeyLength>
        <!--req, int, key length, range:[8,63]-->1
      </wpaKeyLength>
    </WPA>
  </WirelessSecurity>
  <workScene>
    <!--opt, enum, subType:string-->computerRoom
  </workScene>
  <protocol>
    <!--opt, enum, protocol type, subType:string, desc:protocol type-->802.11ac
  </protocol>
  <protocolRealTime>
    <!--opt, string, real-time mode-->test
  </protocolRealTime>
  <hideSsid>
    <!--opt, bool, hidden SSID-->true
  </hideSsid>
  <ChannelConfig>
    <!--opt, object, channel configuration-->
    <width>
      <!--opt, enum, bandwidth, subType:string-->auto
    </width>
    <autoWidth>
      <!--ro, opt, string, bandwidth-->test
    </autoWidth>
    <channel>
      <!--opt, string-->test
    </channel>
    <autoChannel>
      <!--ro, opt, string, automatic channel-->test
    </autoChannel>
    <transmitPower>
      <!--opt, enum, power load, subType:int-->9
    </transmitPower>
    <transmitPowerRealTime>
      <!--opt, int, real-time power-->1
    </transmitPowerRealTime>
  </ChannelConfig>
</Wireless>

```

```

</transmitPowerRealTime>
<countryID>
  <!--opt, int, country code-->1
</countryID>
</ChannelConfig>
<rate>
  <!--opt, enum, frequency, subType:string-->2.4GHz
</rate>
<dialUpGroupNo>
  <!--opt, string, SSID dial group No.-->test
</dialUpGroupNo>
<baseNoisevalue>
  <!--ro, opt, int, background noise value, step:1-->1
</baseNoisevalue>
</Wireless>
<Discovery>
  <!--opt, object, the network discovery configuration-->
  <UPnP>
    <!--req, object, UPnP-->
    <enabled>
      <!--req, bool, whether to enable UPnP-->true
    </enabled>
  </UPnP>
  <Zeroconf>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
  </Zeroconf>
</Discovery>
<Link>
  <!--opt, object, NIC connection information-->
  <MACAddress>
    <!--req, string, MAC address-->test
  </MACAddress>
  <autoNegotiation>
    <!--req, bool, whether to enable the auto-negotiation-->true
  </autoNegotiation>
  <speed>
    <!--req, enum, "10,100,1000,10000", subType:int, desc:"10,100,1000,10000"-->10
  </speed>
  <duplex>
    <!--req, enum, duplex mode, subType:string-->half
  </duplex>
  <MTU>
    <!--req, int, MTU-->1
  </MTU>
  <linkMode>
    <!--opt, enum, connection mode, subType:string, desc:"optical", "electrical"-->optical
  </linkMode>
</Link>
<defaultConnection>
  <!--opt, bool, whether it is the default network connection, desc:whether it is the default network connection-->true
</defaultConnection>
<ActiveMulticast>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <streamID>
    <!--req, enum, stream type, subType:string-->main
  </streamID>
  <ipV4Address>
    <!--opt, string, IPV4 address-->test
  </ipV4Address>
  <ipV6Address>
    <!--opt, string, IPV6 address-->test
  </ipV6Address>
  <port>
    <!--opt, int, port No.-->1
  </port>
</ActiveMulticast>
<macAddress>
  <!--opt, string, MAC address-->test
</macAddress>
<EthernetPortList>
  <!--opt, array, ethernet port information list, subType:object-->
  <EthernetPort>
    <!--opt, object, ethernet port information-->
    <id>
      <!--req, int, port ID, range:[1,4]-->1
    </id>
    <MACAddress>
      <!--req, string, MAC address-->test
    </MACAddress>
    <status>
      <!--opt, enum, connection status, subType:string, desc:disconnect-->connected
    </status>
    <speed>
      <!--req, enum, "10,100,1000,10000", subType:string, desc:"10,100,1000,10000"-->10
    </speed>
  </EthernetPort>
</EthernetPortList>

```



```

<extensions>
  <!--opt, object, extension information-->
  <NetworkCardType>
    <!--opt, enum, NIC type, subType:string-->normal
  </NetworkCardType>
</Extensions>
<interfaceName>
  <!--opt, string, NIC name-->test
</interfaceName>
<enabled>
  <!--opt, bool-->true
</enabled>
<softStorageinterfaceEnabled>
  <!--ro, opt, bool-->true
</softStorageinterfaceEnabled>
<dualControlNICList>
  <!--ro, opt, array, subType:object, range:[1,2]-->
  <dualControlNIC>
    <!--ro, opt, object-->
    <role>
      <!--ro, req, enum, subType:string-->controllerA
    </role>
    <ipV4Address>
      <!--ro, opt, string-->10.255.255.0
    </ipV4Address>
  </dualControlNIC>
</dualControlNICList>
</NetworkInterface>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK", "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:read-only,describe the error reason in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

statusCode	statusString	subStatusCode
4	Invalid Operation	wifiIsOpenedUnableToConnectWiredNetwork
4	Invalid Operation	theNetworkCardHasNotEnabledTheIpv6Protocol
4	Invalid Operation	theIpAddressConflictsWithThatOfTheNetworkCamera
4	Invalid Operation	synchronizingNetworkConfigurationInformationWithThePeerControllerFailedCheckIfTheP
4	Invalid Operation	afterUnbindingTheIpAddressConflictsWithThatOfSingleNetworkInterfaceOrBond
4	Invalid Operation	theIpAddressConflictsWithTheIpAddressOfSingleNetworkInterfaceOrBond

4	Invalid Operation	ipAddressesConflictAmongMultipleNicsOnTheDevice
4	Invalid Operation	invalidIpAddressAndSubnetMask
4	Invalid Operation	pleaseEnterCompleteSubnetMask
4	Invalid Operation	ipAddressesConflictOnTwoNics
4	Invalid Operation	settingServiceIpAddressFailed
4	Invalid Operation	theIpAddressesOfTwoControllersShouldBeDifferent
6	Invalid Content	badIPv6Address
6	Invalid Content	conflictIPv6Address
6	Invalid Content	badNetMask
6	Invalid Content	netGateInvalid
6	Invalid Content	theIpAddressIsInconsistentWithTheGatewayAddressNetworkSegment
6	Invalid Content	invalidDns1
6	Invalid Content	invalidDns2
6	Invalid Content	theIpAddressOfTheGatewayCannotStartWith127
6	Invalid Content	theIpAddressCannotStartWith127

9.2.2.3 Get the information of a specific network interface.

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>?serialNumber=<serialNumber>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--
serialNumber	string	--
controllerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkInterface xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, IP address parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, NIC index-->test
  </id>
  <IPAddress>
    <!--ro, req, object, IP address parameters-->
    <ipVersion>
      <!--ro, req, enum, IP address version, subType:string-->v4
    </ipVersion>
    <addressingType>
      <!--ro, req, enum, address type, subType:string-->static
    </addressingType>
    <ipAddress>
      <!--ro, opt, string, IPv4 address-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, IPv4 subnet mask-->test
    </subnetMask>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address-->test
    </ipv6Address>
    <bitMask>
      <!--ro, opt, string, IPv6 subnet mask-->test
    </bitMask>
    <DefaultGateway>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </DefaultGateway>
    <PrimaryDNS>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </PrimaryDNS>
    <SecondaryDNS>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </SecondaryDNS>
    <DNSEnable>
      <!--ro, opt, bool-->true
    </DNSEnable>
    <IPv6Mode>
      <!--ro, opt, object-->
      <ipV6AddressingType>
        <!--ro, opt, enum, subType:string-->ra
      </ipV6AddressingType>
      <ipV6AddressList>
        <!--ro, opt, array, subType:object-->
        <v6Address>
          <!--ro, opt, object, IPv6 address-->
          <id>
            <!--ro, opt, string-->test
          </id>
          <type>
            <!--ro, opt, enum, subType:string-->ra
          </type>
          <address>
            <!--ro, opt, string, IPv6 address-->test
          </address>
          <bitMask>
            <!--ro, opt, string, IPv6 subnet mask-->test
          </bitMask>
        </v6Address>
      </ipV6AddressList>
    </IPv6Mode>
    <DoubleNetRoute>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IP address-->test
      </ipAddress>
      <subnetMask>
        <!--ro, opt, string, subnet mask-->test
      </subnetMask>
    </DoubleNetRoute>
  </IPAddress>

```

```

<LANConfigMode>
  <!--ro, opt, enum, subType:string-->manual
</LANConfigMode>
<DoubleNetRouteList>
  <!--ro, opt, array, subType:object-->
  <DoubleNetRoute>
    <!--ro, opt, object-->
    <ipAddress>
      <!--ro, opt, string, IP address-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, subnet mask-->test
    </subnetMask>
  </DoubleNetRoute>
</DoubleNetRouteList>
</IPAddress>
<Wireless>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <wirelessNetworkMode>
    <!--ro, opt, enum, subType:string-->infrastructure
  </wirelessNetworkMode>
  <channel>
    <!--ro, opt, string-->test
  </channel>
  <ssid>
    <!--ro, opt, string, SSID (Service Set Identifier)-->test
  </ssid>
  <wmmEnabled>
    <!--ro, opt, bool-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--ro, opt, object-->
    <securityMode>
      <!--ro, opt, enum, subType:string-->disable
    </securityMode>
    <WEP>
      <!--ro, opt, object, dep:or,{$.NetworkInterface.Wireless.WireLessSecurity.securityMode,eq,WEP}-->
      <authenticationType>
        <!--ro, req, enum, authentication type, subType:string-->open
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--ro, req, int-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength>
        <!--ro, opt, int, range:[64,128]-->1
      </wepKeyLength>
      <EncryptionKeyList>
        <!--ro, opt, array, subType:object-->
        <encryptionKey>
          <!--ro, req, string, access key-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--ro, opt, object, dep:or,{$.NetworkInterface.Wireless.WireLessSecurity.securityMode,eq,WPA}-->
      <algorithmType>
        <!--ro, req, enum, algorithm type, subType:string-->TKIP
      </algorithmType>
      <sharedKey>
        <!--ro, req, string-->test
      </sharedKey>
      <wpaKeyLength>
        <!--ro, req, int, range:[8,63]-->1
      </wpaKeyLength>
    </WPA>
  </WirelessSecurity>
</workScene>
<workScene>
  <!--ro, opt, enum, subType:string-->computerRoom
</workScene>
<protocol>
  <!--ro, opt, enum, protocol type, subType:string-->802.11ac
</protocol>
<protocolRealTime>
  <!--ro, opt, string, real-time mode-->test
</protocolRealTime>
<hideSsid>
  <!--ro, opt, bool-->true
</hideSsid>
<ChannelConfig>
  <!--ro, opt, object-->
  <width>
    <!--ro, opt, enum, subType:string-->auto
  </width>
  <autoWidth>
    <!--ro, opt, string-->test
  </autoWidth>
  <channel>
    <!--ro, opt, string-->test
  </channel>
  <autoChannel>

```

```

    <!--ro, opt, string-->test
  </autoChannel>
  <transmitPower>
    <!--ro, opt, enum, subType:int-->9
  </transmitPower>
  <transmitPowerRealTime>
    <!--ro, opt, int-->1
  </transmitPowerRealTime>
  <countryID>
    <!--ro, opt, int-->1
  </countryID>
</ChannelConfig>
<rate>
  <!--ro, opt, enum, subType:string-->2.4GHz
</rate>
<dialUpGroupNo>
  <!--ro, opt, string-->test
</dialUpGroupNo>
<baseNoisevalue>
  <!--ro, opt, int, step:1-->1
</baseNoisevalue>
</Wireless>
<Discovery>
  <!--ro, opt, object-->
  <UPnP>
    <!--ro, req, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </UPnP>
  <Zeroconf>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </Zeroconf>
</Discovery>
<Link>
  <!--ro, opt, object-->
  <MACAddress>
    <!--ro, req, string, MAC address-->test
  </MACAddress>
  <autoNegotiation>
    <!--ro, req, bool-->true
  </autoNegotiation>
  <speed>
    <!--ro, req, enum, "10,100,1000,10000", subType:int, desc:"10,100,1000,10000"-->10
  </speed>
  <duplex>
    <!--ro, req, enum, subType:string-->half
  </duplex>
  <MTU>
    <!--ro, req, int, MTU-->1
  </MTU>
  <linkMode>
    <!--ro, opt, enum, connect mode, subType:string-->optical
  </linkMode>
</Link>
<defaultConnection>
  <!--ro, opt, bool, wether to enable default network connection-->true
</defaultConnection>
<ActiveMulticast>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <streamID>
    <!--ro, req, enum, subType:string-->main
  </streamID>
  <ipV4Address>
    <!--ro, opt, string, IPv4 address-->test
  </ipV4Address>
  <ipV6Address>
    <!--ro, opt, string, IPv6 address-->test
  </ipV6Address>
  <port>
    <!--ro, opt, int, port No.-->1
  </port>
</ActiveMulticast>
<macAddress>
  <!--ro, opt, string, MAC Address-->test
</macAddress>
<EthernetPortList>
  <!--ro, opt, array, network interface information list, subType:object-->
  <EthernetPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, range:[1,4]-->1
    </id>
    <MACAddress>
      <!--ro, req, string, MAC Address-->test
    </MACAddress>
    <status>
      <!--ro, opt, enum, connection status, subType:string, desc:disconnect-->connected
    </status>
  </EthernetPort>
</EthernetPortList>

```

```

</status>
<speed>
  <!--ro, req, enum, speed, subType:string, desc:"10,100,1000,10000"-->10
</speed>
</EthernetPort>
</EthernetPortList>
<Extensions>
  <!--ro, opt, object, extended information-->
  <NetworkCardType>
    <!--ro, opt, enum, subType:string-->normal
  </NetworkCardType>
</Extensions>
<interfaceName>
  <!--ro, opt, string-->test
</interfaceName>
</NetworkInterface>

```

9.2.2.4 Get the configuration capability of multiple network interfaces

Request URL

GET /ISAPI/System/Network/interfaces/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<NetworkInterfaceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, device NIC capability set, subType:object, attr:version{req, string}-->
  <NetworkInterface>
    <!--ro, opt, object, NIC-->
    <id>
      <!--ro, req, string, NIC index-->test
    </id>
    <IPAddress>
      <!--ro, req, object, IP address-->
      <ipVersion opt="v4,v6,dual">
        <!--ro, req, string, IP address version, attr:opt{req, string}-->test
      </ipVersion>
      <addressingType opt="static,dynamic,apiipa">
        <!--ro, req, string, address type, attr:opt{req, string}-->test
      </addressingType>
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->test
      </ipAddress>
      <subnetMask>
        <!--ro, opt, string, IPv4 subnet mask-->test
      </subnetMask>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
      <bitMask>
        <!--ro, opt, string, IPv6 subnet mask-->test
      </bitMask>
      <DefaultGateway>
        <!--ro, opt, object, default gateway-->
        <ipAddress>
          <!--ro, opt, string, IPv4 address-->test
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, IPv6 address-->test
        </ipv6Address>
      </DefaultGateway>
      <PrimaryDNS>
        <!--ro, opt, object, preferred DNS-->
        <ipAddress>
          <!--ro, opt, string, IPv4 address-->test
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, IPv6 address-->test
        </ipv6Address>
      </PrimaryDNS>
      <SecondaryDNS>
        <!--ro, opt, object, alternate DNS-->
        <ipAddress>
          <!--ro, opt, string, IPv4 address-->test
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, IPv6 address-->test
        </ipv6Address>
      </SecondaryDNS>
    </NetworkInterface>
  </NetworkInterfaceList>

```

```

</secondaryuns>
<DNSEnable opt="true,false">
  <!--ro, opt, bool, whether to enable automatic DNS assignment, attr:opt{req, string}-->true
</DNSEnable>
<IPv6Mode>
  <!--ro, opt, object, IPv6 mode-->
  <ipV6AddressingType opt="ra,manual,dhcp">
    <!--ro, opt, string, IPv6 address type, attr:opt{req, string}-->test
  </ipV6AddressingType>
  <ipV6AddressList>
    <!--ro, opt, array, IPv6 address list, subType:object-->
    <v6Address>
      <!--ro, opt, object, IPv6 address-->
      <id min="1" max="10">
        <!--ro, opt, string, IPv6 address index, attr:min{req, int},max{req, int}-->test
      </id>
      <type opt="ra,manual,dhcp">
        <!--ro, opt, string, IPv6 address type, attr:opt{req, string}-->test
      </type>
      <address min="1" max="10">
        <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}-->test
      </address>
      <bitMask min="1" max="10">
        <!--ro, opt, string, IPv6 subnet mask, attr:min{req, int},max{req, int}-->test
      </bitMask>
    </v6Address>
  </ipV6AddressList>
</IPv6Mode>
<DoubleNetRoute>
  <!--ro, opt, object, dual WAN router-->
  <ipAddress min="1" max="10">
    <!--ro, opt, string, IP address, attr:min{req, int},max{req, int}-->test
  </ipAddress>
  <subnetMask min="1" max="10">
    <!--ro, opt, string, subnet mask, attr:min{req, int},max{req, int}-->test
  </subnetMask>
</DoubleNetRoute>
</IPAddress>
<Wireless>
  <!--ro, opt, object, wireless-->
  <enabled>
    <!--ro, req, bool, whether to enable wireless-->true
  </enabled>
  <wirelessNetworkMode opt="infrastructure,adhoc">
    <!--ro, opt, string, wireless network mode, attr:opt{req, string}-->test
  </wirelessNetworkMode>
  <channel opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,auto">
    <!--ro, opt, string, channel index, attr:opt{req, string}-->test
  </channel>
  <ssid min="1" max="10">
    <!--ro, opt, string, SSID (Service Set Identifier), attr:min{req, int},max{req, int}-->test
  </ssid>
  <wmmEnabled>
    <!--ro, opt, bool, enable WMM (Wi-Fi multimedia)-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--ro, opt, object, wireless network security-->
    <securityMode opt="disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise,WPA3">
      <!--ro, opt, string, security mode, attr:opt{req, string}-->test
    </securityMode>
    <WEP>
      <!--ro, opt, object, WEP (Wired Equivalent Privacy)-->
      <authenticationType opt="open,sharedkey,auto">
        <!--ro, req, string, authentication type, attr:opt{req, string}-->test
      </authenticationType>
      <defaultTransmitKeyIndex min="1" max="10">
        <!--ro, req, int, default number of transmitted keys, attr:min{req, int},max{req, int}-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength opt="64,128">
        <!--ro, opt, int, key length, attr:opt{req, string}-->1
      </wepKeyLength>
      <EncryptionKeyList>
        <!--ro, opt, array, secret key list, subType:object-->
        <encryptionKey>
          <!--ro, opt, string, secret key-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--ro, opt, object, WPA-personal-->
      <algorithmType opt="TKIP,AES,TKIP/AES">
        <!--ro, req, string, algorithm type, attr:opt{req, string}-->test
      </algorithmType>
      <sharedKey>
        <!--ro, req, string, shared key authentication-->test
      </sharedKey>
      <wpaKeyLength min="8" max="64">
        <!--ro, req, int, key length, attr:min{req, int},max{req, int}-->1
      </wpaKeyLength>
    </WPA>
    <support64bitKey opt="WPA-personal,WPA2-personal">
      <!--ro, opt, string, encryption type supporting 64-byte secret key, attr:opt{req, string}-->test
    </support64bitKey>
  </WirelessSecurity>

```



```

<AccessPointList>
  <!--ro, opt, array, access point List, subType:object-->
  <InterfaceDisplay>
    <!--ro, opt, object, interface display-->
    <TableElementList>
      <!--ro, opt, array, list of table elements to be displayed, subType:object-->
      <TableElement opt="SSID,workingMode,securityMode,channel,signalStrength,speed,connectionStatus">
        <!--ro, opt, string, table elements to be displayed, attr:opt{req, string}-->test
      </TableElement>
    </TableElementList>
  </InterfaceDisplay>
</AccessPointList>
<isSupportConnectStatus>
  <!--ro, opt, bool, whether it supports Wi-Fi connection status-->true
</isSupportConnectStatus>
<workScene opt="computerRoom,monitorTerminal,centralTerminal">
  <!--ro, opt, string, working scene mode, attr:opt{req, string}-->test
</workScene>
<protocol opt="802.11ac">
  <!--ro, opt, string, protocol mode, attr:opt{req, string}-->test
</protocol>
<hideSsid>
  <!--ro, opt, bool, hidden SSID-->true
</hideSsid>
<ChannelConfig>
  <!--ro, opt, object, channel configuration-->
  <transmitPower opt="9,12,15,18,21,24,27">
    <!--ro, opt, int, transmission power, attr:opt{req, string}-->1
  </transmitPower>
  <ChannelCountryList>
    <!--ro, opt, array, country/region list of channels, subType:object-->
    <Country>
      <!--ro, opt, object, country/region-->
      <countryID>
        <!--ro, req, int, country code-->1
      </countryID>
    </Country>
    <ChannelList>
      <!--ro, opt, array, channel list, subType:object-->
      <Node>
        <!--ro, opt, object, node-->
        <width opt="auto,20,40,80">
          <!--ro, req, string, bandwidth, attr:opt{req, string}-->test
        </width>
        <channel>
          <!--ro, req, string, channel-->test
        </channel>
      </Node>
    </ChannelList>
  </Country>
</ChannelCountryList>
</ChannelConfig>
<rate opt="2.4GHz,5GHz,auto">
  <!--ro, opt, string, frequency, attr:opt{req, string}-->test
</rate>
<dialUpGroupNo min="1" max="10">
  <!--ro, opt, string, SSID dial group No., attr:min{req, int},max{req, int}-->test
</dialUpGroupNo>
<BaseNoisevalue min="1" max="10">
  <!--ro, opt, int, background noise value, attr:min{req, int},max{req, int}-->1
</BaseNoisevalue>
</Wireless>
<Discovery>
  <!--ro, opt, object, search result-->
  <UPnP>
    <!--ro, req, object, UPnP-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable UPnP, attr:opt{req, string}-->true
    </enabled>
  </UPnP>
  <Zeroconf>
    <!--ro, opt, object, Zeroconf protocol-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable Zeroconf protocol, attr:opt{req, string}-->true
    </enabled>
  </Zeroconf>
</Discovery>
<Link>
  <!--ro, opt, object, connection information-->
  <MACAddress>
    <!--ro, req, string, MAC address-->test
  </MACAddress>
  <autoNegotiation opt="true,false">
    <!--ro, req, bool, auto-negotiation, attr:opt{req, string}-->true
  </autoNegotiation>
  <speed opt="10,100,1000,2500">
    <!--ro, req, int, rate, attr:opt{req, string}-->1
  </speed>
  <duplex opt="half,full">
    <!--ro, req, string, data communication mode, attr:opt{req, string}-->test
  </duplex>
  <MTU min="1" max="1500">
    <!--ro, req, int, maximum transmission unit, attr:min{req, int},max{req, int}-->1
  </MTU>

```

```

<linkMode opt="optical,electrical">
  <!--ro, opt, string, connect mode, attr:opt{req, string}-->test
</linkMode>
<workMode opt="1,2,3">
  <!--ro, opt, int, working mode, attr:opt{req, string}, desc:1 (10M half duplex), 2 (10M full duplex), 3 (100M half duplex), 4 (100M full duplex), 5
(adaptive), 6 (1G full duplex)-->1
</workMode>
</Link>
<defaultConnection opt="true,false">
  <!--ro, opt, bool, whether it is default network connection, attr:opt{req, string}-->true
</defaultConnection>
<ActiveMulticast>
  <!--ro, opt, object, active multicast-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable or not, attr:opt{req, string}-->true
  </enabled>
  <streamID opt="main">
    <!--ro, req, string, stream type, attr:opt{req, string}-->test
  </streamID>
  <ipV4Address min="1" max="10">
    <!--ro, opt, string, IPv4 address, attr:min{req, int},max{req, int}-->test
  </ipV4Address>
  <ipV6Address min="1" max="10">
    <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}-->test
  </ipV6Address>
  <port min="1" max="10">
    <!--ro, opt, int, port No., attr:min{req, int},max{req, int}-->1
  </port>
</ActiveMulticast>
<macAddress min="1" max="10">
  <!--ro, opt, string, MAC Address, attr:min{req, int},max{req, int}-->test
</macAddress>
<EthernetPortList size="4">
  <!--ro, opt, array, Ethernet port information List, subType:object, attr:size{req, int}-->
  <EthernetPort>
    <!--ro, opt, object, Ethernet port information-->
    <id min="1" max="4">
      <!--ro, req, int, Ethernet port index, attr:min{req, int},max{req, int}-->1
    </id>
    <MACAddress>
      <!--ro, req, string, MAC Address-->test
    </MACAddress>
    <status opt="connected,disconnect">
      <!--ro, opt, string, connection status, attr:opt{req, string}-->test
    </status>
    <speed opt="10,100,1000,10000">
      <!--ro, req, int, rate, attr:opt{req, string}-->1
    </speed>
  </EthernetPort>
</EthernetPortList>
<Extensions>
  <!--ro, opt, object, extended information-->
  <NetworkCardType opt="normal,internal,external,main,internalDirect">
    <!--ro, opt, string, NIC type, attr:opt{req, string}-->test
  </NetworkCardType>
</Extensions>
<interfaceName min="1" max="10">
  <!--ro, opt, string, name of the network interfaces, attr:min{req, int},max{req, int}-->test
</interfaceName>
<enabled opt="true,false">
  <!--ro, opt, bool, enable NIC, attr:opt{req, string}-->true
</enabled>
</NetworkInterface>
</NetworkInterfaceList>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	pleaseWaitWhileTheDevicesInitializing	0x4000A375	--

9.2.2.5 Get parameters of all network interfaces

Request URL

GET /ISAPI/System/Network/interfaces?controllerID= <controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
controllerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<NetworkInterfaceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

```
<!--ro, req, array, IP address parameter List, subType:object, attr:version{req, string, protocolVersion}-->
```

```
<NetworkInterface>
```

```
<!--ro, req, object, IP address parameters-->
```

```
<id>
```

```
<!--ro, req, string, NIC index-->test
```

```
</id>
```

```
<IPAddress>
```

```
<!--ro, req, object, IP address parameters-->
```

```
<ipVersion>
```

```
<!--ro, req, enum, IP address version, subType:string, desc:"v4" (IPv4), "v6" (IPv6), "dual" (IPv4 and IPv6)-->v4
```

```
</ipVersion>
```

```
<addressingType>
```

```
<!--ro, req, enum, address type, subType:string, desc:"static", "dynamic", "apiipa" (automatic private IP addressing)-->static
```

```
</addressingType>
```

```
<ipAddress>
```

```
<!--ro, opt, string, IPV4 address-->test
```

```
</ipAddress>
```

```
<subnetMask>
```

```
<!--ro, opt, string, IPv4 subnet mask-->test
```

```
</subnetMask>
```

```
<ipv6Address>
```

```
<!--ro, opt, string, IPv6 address-->test
```

```
</ipv6Address>
```

```
<bitMask>
```

```
<!--ro, opt, string, IPv6 subnet mask-->test
```

```
</bitMask>
```

```
<DefaultGateway>
```

```
<!--ro, opt, object, default gateway-->
```

```
<ipAddress>
```

```
<!--ro, opt, string, IPV4 address-->test
```

```
</ipAddress>
```

```
<ipv6Address>
```

```
<!--ro, opt, string, IPv6 address-->test
```

```
</ipv6Address>
```

```
</DefaultGateway>
```

```
<PrimaryDNS>
```

```
<!--ro, opt, object, preferred DNS-->
```

```
<ipAddress>
```

```
<!--ro, opt, string, IPV4 address-->test
```

```
</ipAddress>
```

```
<ipv6Address>
```

```
<!--ro, opt, string, IPv6 address-->test
```

```
</ipv6Address>
```

```
</PrimaryDNS>
```

```
<SecondaryDNS>
```

```
<!--ro, opt, object, alternate DNS-->
```

```
<ipAddress>
```

```
<!--ro, opt, string, IPV4 address-->test
```

```
</ipAddress>
```

```
<ipv6Address>
```

```
<!--ro, opt, string, IPv6 address-->test
```

```
</ipv6Address>
```

```
</SecondaryDNS>
```

```
<DNSEnable>
```

```
<!--ro, opt, bool, DNS automatic allocation-->>true
```

```
</DNSEnable>
```

```
<IPv6Mode>
```

```
<!--ro, opt, object, IPv6 mode-->
```

```
<ipV6AddressingType>
```

```
<!--ro, opt, enum, IPv6 address type, subType:string, desc:"ra" (router advertisement), "manual" (manual assignment), "dhcp" (DHCP server assignment)-->ra
```

```
</ipV6AddressingType>
```

```
<ipV6AddressList>
```

```
<!--ro, opt, array, IPv6 address List, subType:object-->
```

```
<v6Address>
```

```
<!--ro, opt, object, IPV6 Address-->
```

```
<id>
```

```
<!--ro, opt, string, IPv6 address index-->test
```

```
</id>
```

```
<type>
```

```
<!--ro, opt, enum, IPv6 address type, subType:string, desc:"ra" (router advertisement), "manual" (manual assignment), "dhcp" (DHCP server assignment)-->ra
```

```
</type>
```

```
<address>
```

```
<!--ro, opt, string, IPV6 Address-->test
```

```
</address>
```

```
<bitMask>
```

```
<!--ro, opt, string, IPv6 subnet mask-->test
```

```
</bitMask>
```

```
</v6Address>
```

```
</ipV6AddressList>
```

```
</IPv6Mode>
```

```
<DoubleNetRoute>
```

```
<!--ro, opt, object, dual WAN router-->
```

```
<ipAddress>
```

```
<!--ro, opt, string, IP address-->test
```

```
</ipAddress>
```

```
<subnetMask>
```

```
<!--ro, opt, string, subnet mask-->test
```

```

    <!--ro, opt, string, subnet mask-->test
  </subnetMask>
</DoubleNetRoute>
<LANConfigMode>
  <!--ro, opt, enum, subType:string-->manual
</LANConfigMode>
<DoubleNetRouteList>
  <!--ro, opt, array, subType:object, desc:in dual network isolation mode, up to four router configurations are supported; the address of the first
  router needs to be configured via DoubleNetRoute, and the additional expansion routes can be configured through this node-->
  <DoubleNetRoute>
    <!--ro, opt, object, dual WAN router-->
    <ipAddress>
      <!--ro, opt, string, IP address-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, subnet mask-->test
    </subnetMask>
  </DoubleNetRoute>
</DoubleNetRouteList>
<ipv6Enabled>
  <!--ro, opt, bool-->true
</ipv6Enabled>
</IPAddress>
<Wireless>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <wirelessNetworkMode>
    <!--ro, opt, enum, subType:string-->infrastructure
  </wirelessNetworkMode>
  <channel>
    <!--ro, opt, string-->test
  </channel>
  <ssid>
    <!--ro, opt, string, SSID (Service Set Identifier)-->test
  </ssid>
  <wmmEnabled>
    <!--ro, opt, bool, enable WMM (Wi-Fi multimedia)-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--ro, opt, object, wireless network security-->
    <securityMode>
      <!--ro, opt, enum, security mode, subType:string-->disable
    </securityMode>
    <WEP>
      <!--ro, opt, object, dep:or,{$.NetworkInterface.Wireless.WirelessSecurity.securityMode,eq,WEP}-->
      <authenticationType>
        <!--ro, req, enum, the authentication type, subType:string-->open
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--ro, req, int, default number of transmitted keys-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength>
        <!--ro, opt, int, key Length, range:[64,128]-->1
      </wepKeyLength>
      <EncryptionKeyList>
        <!--ro, opt, array, subType:object-->
        <encryptionKey>
          <!--ro, req, string, Secret key-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--ro, opt, object, WPA-personal, dep:or,{$.NetworkInterface.Wireless.WirelessSecurity.securityMode,eq,WPA}-->
      <algorithmType>
        <!--ro, req, enum, algorithm type, subType:string-->TKIP
      </algorithmType>
      <sharedKey>
        <!--ro, req, string, shared key authentication-->test
      </sharedKey>
      <wpaKeyLength>
        <!--ro, req, int, key Length, range:[8,63]-->1
      </wpaKeyLength>
    </WPA>
  </WirelessSecurity>
  <workScene>
    <!--ro, opt, enum, subType:string-->computerRoom
  </workScene>
  <protocol>
    <!--ro, opt, enum, protocol type, subType:string-->802.11ac
  </protocol>
  <protocolRealTime>
    <!--ro, opt, string, real-time mode-->test
  </protocolRealTime>
  <hideSsid>
    <!--ro, opt, bool, hidden SSID-->true
  </hideSsid>
  <ChannelConfig>
    <!--ro, opt, object, channel configuration-->
    <width>
      <!--ro, opt, enum, bandwidth, subType:string-->auto
    </width>
    <autoWidth>

```

```

        <!--ro, opt, string, bandwidth-->test
    </autowidth>
    <channel>
        <!--ro, opt, string, channel-->test
    </channel>
    <autoChannel>
        <!--ro, opt, string, automatic channel-->test
    </autoChannel>
    <transmitPower>
        <!--ro, opt, enum, power Load, subType:int-->9
    </transmitPower>
    <transmitPowerRealTime>
        <!--ro, opt, int, real-time power-->1
    </transmitPowerRealTime>
    <countryID>
        <!--ro, opt, int, country code-->1
    </countryID>
</ChannelConfig>
<rate>
    <!--ro, opt, enum, frequency, subType:string-->2.4GHz
</rate>
<dialUpGroupNo>
    <!--ro, opt, string, SSID dial group No.-->test
</dialUpGroupNo>
<baseNoisevalue>
    <!--ro, opt, int, background noise value, step:1-->1
</baseNoisevalue>
</Wireless>
<Discovery>
    <!--ro, opt, object, the network discovery configuration-->
    <UPnP>
        <!--ro, req, object, UPnP-->
        <enabled>
            <!--ro, req, bool, whether to enable UPnP-->true
        </enabled>
    </UPnP>
    <Zeroconf>
        <!--ro, opt, object-->
        <enabled>
            <!--ro, req, bool-->true
        </enabled>
    </Zeroconf>
</Discovery>
<Link>
    <!--ro, opt, object, NIC connection information-->
    <MACAddress>
        <!--ro, req, string, MAC address-->test
    </MACAddress>
    <autoNegotiation>
        <!--ro, req, bool, whether to enable the auto-negotiation-->true
    </autoNegotiation>
    <speed>
        <!--ro, req, enum, Link speed, subType:int-->10
    </speed>
    <duplex>
        <!--ro, req, enum, duplex mode, subType:string-->half
    </duplex>
    <MTU>
        <!--ro, req, int, MTU-->1
    </MTU>
    <linkMode>
        <!--ro, opt, enum, connection mode, subType:string, desc:"optical", "eElectrical"-->optical
    </linkMode>
</Link>
<defaultConnection>
    <!--ro, opt, bool-->true
</defaultConnection>
<ActiveMulticast>
    <!--ro, opt, object, active multicast-->
    <enabled>
        <!--ro, req, bool, whether to enable-->true
    </enabled>
    <streamID>
        <!--ro, req, enum, stream type, subType:string-->main
    </streamID>
    <ipv4Address>
        <!--ro, opt, string, IPV4 address-->test
    </ipv4Address>
    <ipv6Address>
        <!--ro, opt, string, IPV6 address-->test
    </ipv6Address>
    <port>
        <!--ro, opt, int, port No.-->1
    </port>
</ActiveMulticast>
<macAddress>
    <!--ro, opt, string, MAC address-->test
</macAddress>
<EthernetPortList>
    <!--ro, opt, array, subType:object-->
    <EthernetPort>
        <!--ro, opt, object-->
        <id>

```

```

    <!--ro, req, int, port ID, range:[1,4]-->1
  </id>
  <MACAddress>
    <!--ro, req, string, MAC address-->test
  </MACAddress>
  <status>
    <!--ro, opt, enum, connection status, subType:string-->connected
  </status>
  <speed>
    <!--ro, req, enum, speed, subType:string-->10
  </speed>
</EthernetPort>
</EthernetPortList>
<Extensions>
  <!--ro, opt, object, extension information-->
  <NetworkCardType>
    <!--ro, opt, enum, NIC type, subType:string-->normal
  </NetworkCardType>
</Extensions>
<interfaceName>
  <!--ro, opt, string, NIC name-->test
</interfaceName>
<enabled>
  <!--ro, opt, bool-->true
</enabled>
<softStorageinterfaceEnabled>
  <!--ro, opt, bool-->true
</softStorageinterfaceEnabled>
<dualControlNICList>
  <!--ro, opt, array, subType:object, range:[1,2]-->
  <dualControlNIC>
    <!--ro, opt, object-->
    <role>
      <!--ro, req, enum, subType:string-->controllerA
    </role>
    <ipV4Address>
      <!--ro, opt, string-->10.255.255.0
    </ipV4Address>
  </dualControlNIC>
</dualControlNICList>
</NetworkInterface>
</NetworkInterfaceList>

```

9.2.3 Network Advanced Settings

9.2.3.1 Get the configuration capability of a NAT

Request URL

GET /ISAPI/System/Network/nats/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<NetAddrTransCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, NetAddrTransCap, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, NAT module No.-->1
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable NAT, attr:opt{req, string}-->true
  </enabled>
  <configMode opt="manual,auto">
    <!--ro, req, enum, NAT configuration mode, subType:string, attr:opt{req, string}, desc:; "manual", "auto"-->auto
  </configMode>
  <OuterIPAddress>
    <!--ro, req, object, external IP address-->
    <ipVersion opt="v4,v6">
      <!--ro, req, string, IP address version, attr:opt{ro, req, const}-->v4
    </ipVersion>
    <ipv4Addr>
      <!--ro, opt, string, IPv4 address-->test
    </ipv4Addr>
    <subnetMask>
      <!--ro, opt, string, subnetMask, desc:subnet mask for IPv4 address-->test
    </subnetMask>
    <ipv6Addr>
      <!--ro, opt, string, IPv6 address-->test
    </ipv6Addr>
    <bitMask>
      <!--ro, opt, int, bitMask, desc:bitmask for IPv6 address-->1
    </bitMask>
    <DefaultGateway>
      <!--ro, req, object, gateway-->
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->10.10.10.10
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </DefaultGateway>
  </OuterIPAddress>
  <InnerIPAddress>
    <!--ro, req, object, internal IP address-->
    <ipVersion opt="v4,v6">
      <!--ro, req, string, IP address version, attr:opt{ro, req, const}-->test
    </ipVersion>
    <ipv4Addr>
      <!--ro, opt, string, IPv4 address-->10.10.10.10
    </ipv4Addr>
    <subnetMask>
      <!--ro, opt, string, subnetMask, desc:subnet mask for IPv4 address-->test
    </subnetMask>
    <ipv6Addr>
      <!--ro, opt, string, IPv6 address-->test
    </ipv6Addr>
    <bitMask>
      <!--ro, opt, int, bitMask, desc:bitmask for IPv6 address-->1
    </bitMask>
  </InnerIPAddress>
</NetAddrTransCap>

```

9.2.4 Network Limitation Management

9.2.4.1 Set the IP address filter

Request URL

PUT /ISAPI/System/Network/ipFilter

Query Parameter

None

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>

<IPFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, the character size indicates the max. number of supported IP address. See XML_IPFilterAddressList for details, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <permissionType>
    <!--opt, enum, "deny,allow", subType:string, desc:"deny", "allow"-->deny
  </permissionType>
  <IPFilterAddressList size="48">
    <!--opt, array, subType:object, attr:size{req, int}-->
    <IPFilterAddress>
      <!--opt, object-->
      <id>
        <!--req, string, No.-->test
      </id>
      <permissionType>
        <!--opt, enum, "deny", "allow", subType:string, desc:"deny", "allow"-->deny
      </permissionType>
      <addressFilterType>
        <!--req, enum, address filtering type, subType:string, desc:address filtering type-->mask
      </addressFilterType>
      <AddressRange>
        <!--opt, object, dep:and,{$.IPFilter.IPFilterAddressList[*].IPFilterAddress.addressFilterType,eq,range}-->
        <startIPAddress>
          <!--opt, string-->test
        </startIPAddress>
        <endIPAddress>
          <!--opt, string-->test
        </endIPAddress>
        <startIPv6Address>
          <!--opt, string-->test
        </startIPv6Address>
        <endIPv6Address>
          <!--opt, string-->test
        </endIPv6Address>
      </AddressRange>
      <AddressMask>
        <!--opt, object, dep:and,{$.IPFilter.IPFilterAddressList[*].IPFilterAddress.addressFilterType,eq,mask}-->
        <ipAddress>
          <!--opt, string, IPv4 address-->test
        </ipAddress>
        <ipv6Address>
          <!--opt, string, IPv6 address-->test
        </ipv6Address>
        <bitMask>
          <!--req, string-->test
        </bitMask>
      </AddressMask>
    </IPFilterAddress>
  </IPFilterAddressList>
  <localIPCheckEnabled>
    <!--opt, bool-->false
  </localIPCheckEnabled>
</IPFilter>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	theFilterTypeAllowlistBlocklistIsRequired	0x4000A2A8	--
4	Invalid Operation	theAllowlistAndBlocklistContainSameIpAddress	0x4000A2B1	--
4	Invalid Operation	theSameIpAddressExistsInTheAllowlist	0x4000A6CC	--

9.2.4.2 Get the IP filter configuration parameters

Request URL

GET /ISAPI/System/Network/ipFilter

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IPFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the character size indicates the max. number of supported IP address. See XML_IPFilterAddressList for details, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <permissionType>
    <!--ro, opt, enum, "deny,allow", subType:string, desc:"deny", "allow"-->deny
  </permissionType>
  <IPFilterAddressList size="48">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <IPFilterAddress>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <permissionType>
        <!--ro, opt, enum, subType:string, desc:"deny", "allow"-->deny
      </permissionType>
      <addressFilterType>
        <!--ro, req, enum, subType:string-->mask
      </addressFilterType>
      <AddressRange>
        <!--ro, opt, object, dep:and,{$.IPFilter.IPFilterAddressList[*].IPFilterAddress.addressFilterType,eq,range}-->
        <startIPAddress>
          <!--ro, opt, string-->test
        </startIPAddress>
        <endIPAddress>
          <!--ro, opt, string-->test
        </endIPAddress>
        <startIPv6Address>
          <!--ro, opt, string-->test
        </startIPv6Address>
        <endIPv6Address>
          <!--ro, opt, string-->test
        </endIPv6Address>
      </AddressRange>
      <AddressMask>
        <!--ro, opt, object, dep:and,{$.IPFilter.IPFilterAddressList[*].IPFilterAddress.addressFilterType,eq,mask}-->
        <ipAddress>
          <!--ro, opt, string-->test
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, IPv6 address-->test
        </ipv6Address>
        <bitMask>
          <!--ro, req, string-->test
        </bitMask>
      </AddressMask>
    </IPFilterAddress>
  </IPFilterAddressList>
  <localIPCheckEnabled>
    <!--ro, opt, bool-->>false
  </localIPCheckEnabled>
</IPFilter>
```

9.2.4.3 Get the capability for configuring IP address filter

Request URL

GET /ISAPI/System/Network/ipFilter/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IPFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, IP address filter, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <permissionType opt="deny,allow">
    <!--ro, opt, string, IP address filter status: "deny,allow". If this node is configured,it will overwrite value of all <permissionType> in
<IPFilterAddress>, attr:opt{req, string}-->test
  </permissionType>
  <IPFilterAddressList size="10">
    <!--ro, opt, object, the attribute "size" refers to the maximum number of IP addresses supported by the device. If the attribute "size" is not
returned,the default supported number of IP addresses is 48, attr:size{req, int}-->
    <IPFilterAddress>
      <!--ro, opt, object-->
      <id min="1" max="10">
        <!--ro, req, int, ID, attr:min{req, int},max{req, int}-->1
      </id>
      <permissionType opt="deny,allow">
        <!--ro, opt, string, IP address filter status: "deny,allow". If this node is configured,it will overwrite value of all <permissionType> in
<IPFilterAddress>, attr:opt{req, string}-->test
      </permissionType>
      <addressFilterType opt="mask,range">
        <!--ro, req, string, ro,filter type: "mask"-single address,"range"-address range, attr:opt{req, string}-->test
      </addressFilterType>
      <AddressRange>
        <!--ro, opt, object, this node is valid when <addressFilterType> contains "range"-->
        <startIPAddress min="1" max="10">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </startIPAddress>
        <endIPAddress min="1" max="10">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </endIPAddress>
        <startIPv6Address min="1" max="10">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </startIPv6Address>
        <endIPv6Address min="1" max="10">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </endIPv6Address>
      </AddressRange>
      <AddressMask>
        <!--ro, opt, object, it is valid when <addressFilterType> contains "mask"-->
        <ipAddress min="1" max="10">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </ipAddress>
        <ipv6Address min="1" max="10">
          <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}-->test
        </ipv6Address>
        <bitMask min="1" max="10">
          <!--ro, req, string, attr:min{req, int},max{req, int}-->test
        </bitMask>
        <bitMaskIPv6 min="1" max="10">
          <!--ro, req, string, IPv6 prefix length, attr:min{req, int},max{req, int}-->test
        </bitMaskIPv6>
      </AddressMask>
      <describeMsg min="1" max="10">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
      </describeMsg>
    </IPFilterAddress>
  </IPFilterAddressList>
  <localIPCheckEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->>false
  </localIPCheckEnabled>
</IPFilter>
```

9.2.5 SIP Access Management

9.2.5.1 Set the parameters of a piece of SIP (Session Initiation Protocol) information

Request URL

Query Parameter

Parameter Name	Parameter Type	Description
SIPServerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SIPInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <videoID>
    <!--req, string-->test
  </videoID>
  <AlarmInList>
    <!--opt, array, subType:object-->
    <AlarmIn>
      <!--opt, object, alarm input-->
      <id>
        <!--req, string, alarm input No.-->test
      </id>
      <alarmInID>
        <!--req, string, alarm input encoding ID-->test
      </alarmInID>
    </AlarmIn>
  </AlarmInList>
  <VideoInputList>
    <!--opt, array, subType:object-->
    <VideoInput>
      <!--opt, object-->
      <id>
        <!--req, string, channel No.-->test
      </id>
      <videoInputID>
        <!--req, string, video channel encoding ID-->test
      </videoInputID>
    </VideoInput>
  </VideoInputList>
  <VideoOutputList>
    <!--opt, array, subType:object-->
    <VideoOutput>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <name>
        <!--opt, string, name-->test
      </name>
      <videoOutputID>
        <!--req, string-->test
      </videoOutputID>
    </VideoOutput>
  </VideoOutputList>
  <AudioOutputList>
    <!--opt, array, subType:object-->
    <AudioOutput>
      <!--opt, object, audio output-->
      <id>
        <!--req, string-->test
      </id>
      <audioOutputID>
        <!--req, string, audio output channel encoding ID-->test
      </audioOutputID>
    </AudioOutput>
  </AudioOutputList>
  <platformNo>
    <!--opt, int, platform No.-->1
  </platformNo>
</SIPInfo>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

9.2.5.2 Get the parameters of a specific SIP (Session Initiation Protocol) server

Request URL

GET /ISAPI/System/Network/SIP/<SIPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
SIPServerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SIPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, SIP server, attr:version{req, float, protocolVersion}-->
  <id>
    <!--ro, req, string, SIP server ID, range:[1,2]-->test
  </id>
  <localPort>
    <!--ro, opt, int, Local port No., range:[1,65535]-->1
  </localPort>
  <streamID>
    <!--ro, opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
  </streamID>
  <Standard>
    <!--ro, opt, object, standard protocol-->
    <registerStatus>
      <!--ro, req, bool, registration status, desc:registration status: "false"-unregistered, "true"-registered-->true
    </registerStatus>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <registrar>
      <!--ro, req, string, address, range:[1,128]-->test
    </registrar>
    <registrarPort>
      <!--ro, req, int, port No., range:[1,65535]-->1
    </registrarPort>
    <proxy>
      <!--ro, req, string, agent address, range:[1,128]-->test
    </proxy>
    <proxyPort>
      <!--ro, req, int, proxy port, range:[1,65535]-->1
    </proxyPort>
    <displayName>
      <!--ro, req, string, display name, range:[1,64]-->test
    </displayName>
    <userName>
      <!--ro, req, string, user name, range:[1,64]-->test
    </userName>
    <authID>
      <!--ro, req, string, user authentication ID, range:[1,64]-->test
    </authID>
    <expires>
      <!--ro, req, int, registration expiration, range:[3600,100000]-->1
    </expires>
    <transportType>
      <!--ro, opt, enum, SIP service registration method, subType:string, desc:"UDP", "TCP", "TLS"-->UDP
    </transportType>
  </Standard>
</SIPServer>
```

```

</transportType>
<udpRegEnabled>
  <!--ro, opt, bool, whether to enable UDP registration-->true
</udpRegEnabled>
<udpRegPort>
  <!--ro, opt, int, UDP registration port No., range:[1,65535]-->1
</udpRegPort>
<tcpRegEnabled>
  <!--ro, opt, bool, whether to enable TCP registration-->true
</tcpRegEnabled>
<tcpRegPort>
  <!--ro, opt, int, TCP registration port No., range:[1,65535]-->1
</tcpRegPort>
<tlsRegEnabled>
  <!--ro, opt, bool, whether to enable TLS registration-->true
</tlsRegEnabled>
<tlsRegPort>
  <!--ro, opt, int, TLS registration port No., range:[1,65535]-->1
</tlsRegPort>
<addressingFormatType>
  <!--ro, opt, enum, address type, subType:string, desc:if this node is not applied, it is registration by IP by default; "ipAddress" (IP address),
"hostname" (domain name)-->ipAddress
</addressingFormatType>
<hostName>
  <!--ro, opt, string, domain name, range:[1,64], desc:it is valid when addressingFormatType is hostName-->test
</hostName>
<networkMode>
  <!--ro, opt, enum, network mode, subType:int, desc:1-wired NIC 1 (default), 2-wired NIC 2, 3- wireLess network-->1
</networkMode>
<targetUserName>
  <!--ro, opt, string, username of the called party, range:[1,32], desc:English Letters, numbers, special characters, and Chinese characters are
allowed-->test
</targetUserName>
</Standard>
<GB28181>
  <!--ro, opt, object, single GB platform registration parameter; exclusive with <GB28181List>-->
  <registerStatus>
    <!--ro, req, bool, registration status-->true
  </registerStatus>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <registrar>
    <!--ro, req, string, address, range:[1,128]-->test
  </registrar>
  <registrarPort>
    <!--ro, req, int, port No., range:[1,65535]-->1
  </registrarPort>
  <serverId>
    <!--ro, req, string, SIP server ID, range:[1,64]-->test
  </serverId>
  <serverDomain>
    <!--ro, req, int, SIP server domain, range:[1,128]-->1
  </serverDomain>
  <userName>
    <!--ro, req, string, SIP user name, range:[1,64]-->test
  </userName>
  <authID>
    <!--ro, req, string, SIP authentication ID, range:[1,64]-->test
  </authID>
  <expires>
    <!--ro, req, int, registration expiration, range:[3600,100000], unit:s-->1
  </expires>
  <liveTime>
    <!--ro, req, int, effective time, range:[30,100000], unit:s-->1
  </liveTime>
  <heartbeatTime>
    <!--ro, req, int, heartbeat interval, range:[5,255], unit:s-->1
  </heartbeatTime>
  <heartbeatCount>
    <!--ro, req, int, maximum times of heartbeat timeout, range:[3,255]-->1
  </heartbeatCount>
  <transportType>
    <!--ro, req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
  </transportType>
  <streamType>
    <!--ro, opt, enum, stream type, subType:string, desc:"main" (main stream),"sub" (sub stream)-->main
  </streamType>
  <tcpConnectType>
    <!--ro, opt, enum, TCP connection type, subType:string, desc:"active", "passive"-->active
  </tcpConnectType>
  <registerInterval>
    <!--ro, req, int, registration period, range:[60,600], desc:unit: second-->60
  </registerInterval>
  <protocolVersion>
    <!--ro, req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
  </protocolVersion>
  <autoAllocEncodeChanID>
    <!--ro, opt, bool, whether to allocate encoding channel ID automatically-->true
  </autoAllocEncodeChanID>
  <autoAllocDecodeChanID>
    <!--ro, opt, bool, whether to allocate decoding channel ID automatically-->true
  </autoAllocDecodeChanID>
  <autoAllocAlarmInputChanID>

```

```

    <!--ro, opt, bool, whether to allocate alarm input channel ID automatically-->true
  </autoAllocAlarmInputChanID>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2]-->1
  </platformNo>
  <GBEncryptEnabled>
    <!--ro, req, bool, whether to enable domestic password-->true
  </GBEncryptEnabled>
  <exchangeCertificateEnabled>
    <!--ro, opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default-->true
  </exchangeCertificateEnabled>
  <offlineCode>
    <!--ro, opt, string, device status code returned during offline registration; PUT method is not supported for this node, desc:this node is returned
only when registerStatus is "false"-->0xFFFFFFFF
  </offlineCode>
  <unidirectionEnabled>
    <!--ro, opt, bool, whether to allow GB35114 one way authentication, desc:this node is valid when GBEncryptEnabled is "true"; one-way authentication is
authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->>false
  </unidirectionEnabled>
  <voiceBroadcastEnabled>
    <!--ro, opt, bool, whether to enable TCP voice broadcast protocol-->true
  </voiceBroadcastEnabled>
</GB28181>
<GB28181List>
  <!--ro, opt, array, subType:object-->
  <GB28181>
    <!--ro, opt, object, single GB platform registration parameter; exclusive with <GB28181List-->
    <registerStatus>
      <!--ro, req, bool, registration status-->true
    </registerStatus>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <registrar>
      <!--ro, req, string, address, range:[1,128]-->test
    </registrar>
    <registrarPort>
      <!--ro, req, int, port No., range:[1,65535]-->1
    </registrarPort>
    <serverId>
      <!--ro, req, string, SIP server ID, range:[1,64]-->test
    </serverId>
    <serverDomain>
      <!--ro, req, int, SIP server domain, range:[1,128]-->1
    </serverDomain>
    <userName>
      <!--ro, req, string, SIP user name, range:[1,64]-->test
    </userName>
    <authID>
      <!--ro, req, string, SIP authentication ID, range:[1,64]-->test
    </authID>
    <expires>
      <!--ro, req, int, registration expiration, range:[3600,100000], unit:s-->1
    </expires>
    <liveTime>
      <!--ro, req, int, effective time, range:[30,100000], unit:s-->1
    </liveTime>
    <heartbeatTime>
      <!--ro, req, int, heartbeat interval, range:[5,255], unit:s-->1
    </heartbeatTime>
    <heartbeatCount>
      <!--ro, req, int, maximum times of heartbeat timeout, range:[3,255]-->1
    </heartbeatCount>
    <transportType>
      <!--ro, req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
    </transportType>
    <registerInterval>
      <!--ro, req, int, registration period, range:[60,600], desc:unit: second-->60
    </registerInterval>
    <protocolVersion>
      <!--ro, req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
    </protocolVersion>
    <streamID>
      <!--ro, opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
    </streamID>
    <platformNo>
      <!--ro, opt, int, platform No., range:[1,2]-->1
    </platformNo>
    <GBEncryptEnabled>
      <!--ro, req, bool, whether to enable domestic password-->true
    </GBEncryptEnabled>
    <exchangeCertificateEnabled>
      <!--ro, opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default--
>true
    </exchangeCertificateEnabled>
    <offlineCode>
      <!--ro, opt, string, device status code returned during offline registration; PUT method is not supported for this node, desc:this node is returned
only when registerStatus is "false"-->0xFFFFFFFF
    </offlineCode>
    <localPort>
      <!--ro, req, int, Local SIP port, range:[1,65535]-->1
    </localPort>
    <unidirectionEnabled>

```



```

    <!--ro, opt, bool, whether to allow GB35114 one way authentication, desc:this node is valid when GBEncryptEnabled is "true"; one-way authentication
    is authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->false
    </unidirectionEnabled>
    <voiceBroadcastEnabled>
    <!--ro, opt, bool, whether to enable TCP voice broadcast protocol-->true
    </voiceBroadcastEnabled>
    </GB28181>
    </GB28181List>
    <channelID>
    <!--ro, opt, string, channel No., range:[1,2048], desc:channel No.-->1
    </channelID>
    </SIPServer>

```

9.2.5.3 Set the parameters of a specific SIP (Session Initiation Protocol) server

Request URL

PUT /ISAPI/System/Network/SIP/<SIPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
SIPServerID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<SIPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, SIP server, attr:version{req, float, protocolVersion}-->
  <id>
    <!--req, string, SIP server ID, range:[1,2]-->test
  </id>
  <localPort>
    <!--opt, int, Local port No., range:[1,65535]-->1
  </localPort>
  <streamID>
    <!--opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
  </streamID>
  <Standard>
    <!--opt, object, standard protocol-->
    <registerStatus>
      <!--req, bool, registration status, desc:registration status: "false"-unregistered,"true"-registered-->true
    </registerStatus>
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <registrar>
      <!--req, string, address, range:[1,128]-->test
    </registrar>
    <registrarPort>
      <!--req, int, port No., range:[1,65535]-->1
    </registrarPort>
    <proxy>
      <!--req, string, proxy address, range:[1,128]-->test
    </proxy>
    <proxyPort>
      <!--req, int, proxy port, range:[1,65535]-->1
    </proxyPort>
    <displayName>
      <!--req, string, display name, range:[1,64]-->test
    </displayName>
    <userName>
      <!--req, string, user name, range:[1,64]-->test
    </userName>
    <authID>
      <!--req, string, user authentication ID, range:[1,64]-->test
    </authID>
    <password>
      <!--req, string, password, range:[1,32]-->test
    </password>
    <expires>
      <!--req, int, registration expiration, range:[3600,100000]-->1
    </expires>
    <transportType>
      <!--opt, enum, SIP service registration method, subType:string, desc:"UDP", "TCP", "TLS"-->UDP
    </transportType>
    <udpRegEnabled>
      <!--opt, bool, whether to enable UDP registration-->true
    </udpRegEnabled>
    <udpRegPort>
      <!--opt, int, UDP registration port No., range:[1,65535]-->1
    </udpRegPort>
    <tcpRegEnabled>
      <!--opt, bool, whether to enable TCP registration-->true
    </tcpRegEnabled>

```

```

<tcpRegPort>
  <!--opt, int, TCP registration port No., range:[1,65535]-->1
</tcpRegPort>
<tlsRegEnabled>
  <!--opt, bool, whether to enable TLS registration-->true
</tlsRegEnabled>
<tlsRegPort>
  <!--opt, int, TLS registration port No., range:[1,65535]-->1
</tlsRegPort>
<addressingFormatType>
  <!--opt, enum, address type, subType:string, desc:if this node is not applied, it is registration by IP by default; "ipAddress" (IP address),
"hostname" (domain name)-->ipAddress
</addressingFormatType>
<hostname>
  <!--opt, string, domain name, range:[1,64], desc:it is valid when addressingFormatType is hostname-->test
</hostname>
<networkMode>
  <!--opt, enum, network mode, subType:int, desc:1-wired NIC 1 (default), 2-wired NIC 2, 3- wireless network-->1
</networkMode>
<targetUserName>
  <!--opt, string, username of the called party, range:[1,32], desc:English Letters, numbers, special characters, and Chinese characters are allowed--
>test
</targetUserName>
</Standard>
<GB28181>
  <!--opt, object, single GB platform registration parameter; exclusive with <GB28181List-->
<registerStatus>
  <!--req, bool, registration status-->true
</registerStatus>
<enabled>
  <!--req, bool, whether to enable-->true
</enabled>
<registrar>
  <!--req, string, address, range:[1,128]-->test
</registrar>
<registrarPort>
  <!--req, int, port No., range:[1,65535]-->1
</registrarPort>
<serverId>
  <!--req, string, SIP server ID, range:[1,64]-->test
</serverId>
<serverDomain>
  <!--req, int, SIP server domain, range:[1,128]-->1
</serverDomain>
<userName>
  <!--req, string, SIP user name, range:[1,64]-->test
</userName>
<authID>
  <!--req, string, SIP authentication ID, range:[1,64]-->test
</authID>
<password>
  <!--req, string, password; encryption required for sensitive information, range:[1,32]-->test
</password>
<expires>
  <!--req, int, registration expiration, range:[3600,100000], unit:s-->1
</expires>
<liveTime>
  <!--req, int, effective time, range:[30,100000], unit:s-->1
</liveTime>
<heartbeatTime>
  <!--req, int, heartbeat interval, range:[5,255], unit:s-->1
</heartbeatTime>
<heartbeatCount>
  <!--req, int, maximum times of heartbeat timeout, range:[3,255]-->1
</heartbeatCount>
<transportType>
  <!--req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
</transportType>
<streamType>
  <!--opt, enum, stream type, subType:string, desc:"main" (main stream),"sub" (sub stream)-->main
</streamType>
<tcpConnectType>
  <!--opt, enum, TCP connection type, subType:string, desc:"active", "passive"-->active
</tcpConnectType>
<registerInterval>
  <!--req, int, registration period, range:[60,600], desc:unit: second-->60
</registerInterval>
<protocolVersion>
  <!--req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
</protocolVersion>
<autoAllocEncodeChanID>
  <!--opt, bool, whether to allocate encoding channel ID automatically-->true
</autoAllocEncodeChanID>
<autoAllocDecodeChanID>
  <!--opt, bool, whether to allocate decoding channel ID automatically-->true
</autoAllocDecodeChanID>
<autoAllocAlarmInputChanID>
  <!--opt, bool, whether to allocate alarm input channel ID automatically-->true
</autoAllocAlarmInputChanID>
<platformNo>
  <!--opt, int, platform No., range:[1,2]-->1
</platformNo>
<GBEncryptEnabled>
  <!--req, bool, whether to enable domestic password-->true

```

```

</GBEncryptEnabled>
<exchangeCertificateEnabled>
  <!--opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default-->true
</exchangeCertificateEnabled>
<unidirectionEnabled>
  <!--opt, bool, whether to allow GB35114 one way authentication, desc:this node is valid when GBEncryptEnabled is "true"; one-way authentication is authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->>false
</unidirectionEnabled>
<voiceBroadcastEnabled>
  <!--opt, bool, whether to enable TCP voice broadcast protocol-->true
</voiceBroadcastEnabled>
</GB28181>
<GB28181List>
  <!--opt, array, subType:object-->
<GB28181>
  <!--opt, object, single GB platform registration parameter; exclusive with <GB28181List-->
  <registerStatus>
    <!--req, bool, registration status-->true
  </registerStatus>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <registrar>
    <!--req, string, address, range:[1,128]-->test
  </registrar>
  <registrarPort>
    <!--req, int, port No., range:[1,65535]-->1
  </registrarPort>
  <serverId>
    <!--req, string, SIP server ID, range:[1,64]-->test
  </serverId>
  <serverDomain>
    <!--req, int, SIP server domain, range:[1,128]-->1
  </serverDomain>
  <userName>
    <!--req, string, SIP user name, range:[1,64]-->test
  </userName>
  <authID>
    <!--req, string, SIP authentication ID, range:[1,64]-->test
  </authID>
  <password>
    <!--req, string, password; encryption required for sensitive information, range:[1,32]-->test
  </password>
  <expires>
    <!--req, int, registration expiration, range:[3600,100000], unit:s-->1
  </expires>
  <liveTime>
    <!--req, int, effective time, range:[30,100000], unit:s-->1
  </liveTime>
  <heartbeatTime>
    <!--req, int, heartbeat interval, range:[5,255], unit:s-->1
  </heartbeatTime>
  <heartbeatCount>
    <!--req, int, maximum times of heartbeat timeout, range:[3,255]-->1
  </heartbeatCount>
  <transportType>
    <!--req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
  </transportType>
  <registerInterval>
    <!--req, int, registration period, range:[60,600], desc:unit: second-->60
  </registerInterval>
  <protocolVersion>
    <!--req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
  </protocolVersion>
  <streamID>
    <!--opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
  </streamID>
  <platformNo>
    <!--opt, int, platform No., range:[1,2]-->1
  </platformNo>
  <GBEncryptEnabled>
    <!--req, bool, whether to enable domestic password-->true
  </GBEncryptEnabled>
  <exchangeCertificateEnabled>
    <!--opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default-->true
  </exchangeCertificateEnabled>
  <localPort>
    <!--req, int, Local SIP port, range:[1,65535]-->1
  </localPort>
  <unidirectionEnabled>
    <!--opt, bool, whether to allow GB35114 one way authentication, desc:this node is valid when GBEncryptEnabled is "true"; one-way authentication is authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->>false
  </unidirectionEnabled>
  <voiceBroadcastEnabled>
    <!--opt, bool, whether to enable TCP voice broadcast protocol-->true
  </voiceBroadcastEnabled>
</GB28181>
</GB28181List>
<channelID>
  <!--opt, string, channel No., range:[1,2048], desc:this node is returned when isSupportChannelSIP is supported in /ISAPI/System/Network/SIP/capabilities-->1
</channelID>
</SIPServer>

```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

9.2.5.4 Get the capability of configuring GB28181 server

Request URL

GET /ISAPI/System/Network/SIP/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SIPServerListCap xmlns="http://www.isapi.org/ver20/XMLSchema" size="10" version="2.0">
  <!--ro, req, object, List of SIP service capability parameters, attr:size{req, int, protocolVersion},version{req, string}-->
  <SIPServer>
    <!--ro, req, object, SIP service-->
    <id min="1" max="2">
      <!--ro, req, int, ID, range:[1,2], attr:min{req, int},max{req, int}-->1
    </id>
    <localPort min="1" max="65535">
      <!--ro, req, int, Local port, range:[1,65535], attr:min{req, int},max{req, int}-->1
    </localPort>
    <streamID opt="1,2,3">
      <!--ro, req, int, stream type ID, range:[1,3], attr:opt{req, string}, desc:1 (main stream), 2 (sub stream)-->1
    </streamID>
    <Standard>
      <!--ro, opt, object, standard-->
      <registerStatus opt="true,false">
        <!--ro, opt, bool, registration status, attr:opt{req, string}-->true
      </registerStatus>
      <enabled opt="true,false">
        <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
      </enabled>
      <registrar min="1" max="128">
        <!--ro, opt, string, registration IP, range:[1,128], attr:min{req, int},max{req, int}-->test
      </registrar>
      <registrarPort min="1" max="65535">
        <!--ro, opt, int, registration port, range:[1,65535], attr:min{req, int},max{req, int}-->1
      </registrarPort>
      <proxy min="1" max="128">
        <!--ro, opt, string, proxy IP, range:[1,128], attr:min{req, int},max{req, int}-->test
      </proxy>
      <proxyPort min="1" max="65535">
        <!--ro, opt, int, proxy port, range:[1,65535], attr:min{req, int},max{req, int}-->1
      </proxyPort>
      <displayName min="1" max="64">
        <!--ro, opt, string, name display, range:[1,64], attr:min{req, int},max{req, int}-->test
      </displayName>
      <userName min="1" max="64">
        <!--ro, opt, string, user name, range:[1,64], attr:min{req, int},max{req, int}-->test
      </userName>
      <authID min="1" max="64">
        <!--ro, opt, string, authentication ID, range:[1,64], attr:min{req, int},max{req, int}-->test
      </authID>
      <password min="1" max="32">
        <!--ro, opt, string, password, range:[1,32], attr:min{req, int},max{req, int}-->null
      </password>
```

```

<expires min="3600" max="100000">
  <!--ro, opt, int, validity, range:[3600,100000], attr:min{req, int},max{req, int, unit:s}-->1
</expires>
<transportType opt="UDP,TCP,TLS">
  <!--ro, opt, string, SIP service registration method, attr:opt{req, string}, desc:"UDP", "TCP", "TLS"; this node is exclusive with protocolType in
SipTransmit, and it is not recommended to keep both in one device-->UDP
</transportType>
<udpRegEnabled opt="true,false">
  <!--ro, opt, bool, whether to enable UDP registration, attr:opt{req, string}-->true
</udpRegEnabled>
<udpRegPort min="1" max="65535">
  <!--ro, opt, int, UDP registration port No., range:[1,65535], attr:min{req, int},max{req, int}-->1
</udpRegPort>
<tcpRegEnabled opt="true,false">
  <!--ro, opt, bool, whether to enable TCP registration, attr:opt{req, string}-->true
</tcpRegEnabled>
<tcpRegPort min="1" max="65535">
  <!--ro, opt, int, TCP registration port No., range:[1,65535], attr:min{req, int},max{req, int}-->1
</tcpRegPort>
<tlsRegEnabled opt="true,false">
  <!--ro, opt, bool, whether to enable TLS registration, attr:opt{req, string}-->true
</tlsRegEnabled>
<tlsRegPort min="1" max="65535">
  <!--ro, opt, int, TLS registration port No., range:[1,65535], attr:min{req, int},max{req, int}-->1
</tlsRegPort>
<addressingFormatType opt="ipAddress,hostName">
  <!--ro, opt, enum, address type, subType:string, attr:opt{req, string}, desc:if this node is not applied, it is registration by IP by default;
"ipAddress" (IP address), "hostname" (domain name)-->ipAddress
</addressingFormatType>
<hostName min="1" max="64">
  <!--ro, opt, string, domain name, attr:min{req, int},max{req, int}-->test
</hostName>
<networkMode opt="1,2,3">
  <!--ro, opt, enum, network mode, subType:int, attr:opt{req, string}, desc:1-wired NIC 1, 2-wired NIC 2, 3- wireless network-->1
</networkMode>
<targetUserName min="1" max="32">
  <!--ro, opt, string, username of the called party, attr:min{req, int},max{req, int}-->test
</targetUserName>
</Standard>
<GB28181>
  <!--ro, opt, object, GB-->
  <registerStatus opt="true,false">
    <!--ro, opt, bool, registration status, attr:opt{req, string}-->true
  </registerStatus>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <registrar min="1" max="128">
    <!--ro, opt, string, registration IP, range:[1,128], attr:min{req, int},max{req, int}-->test
  </registrar>
  <registrarPort min="1" max="65535">
    <!--ro, opt, int, registration port, range:[1,65535], attr:min{req, int},max{req, int}-->1
  </registrarPort>
  <serverId min="1" max="64">
    <!--ro, opt, string, service ID, range:[1,64], attr:min{req, int},max{req, int}-->test
  </serverId>
  <serverDomain min="1" max="128">
    <!--ro, opt, string, service domain name, range:[1,128], attr:min{req, int},max{req, int}-->test
  </serverDomain>
  <userName min="1" max="64">
    <!--ro, opt, string, user name, range:[1,64], attr:min{req, int},max{req, int}-->test
  </userName>
  <authID min="1" max="64">
    <!--ro, opt, string, authentication ID, range:[1,64], attr:min{req, int},max{req, int}-->test
  </authID>
  <password min="1" max="32">
    <!--ro, opt, string, password, range:[3600,100000], attr:min{req, int},max{req, int}-->null
  </password>
  <expires min="3600" max="100000">
    <!--ro, opt, int, validity, range:[3600,100000], attr:min{req, int},max{req, int, unit:s}-->1
  </expires>
  <liveTime min="30" max="100000">
    <!--ro, opt, int, Lifespan, range:[30,100000], attr:min{req, int},max{req, int, unit:s}-->1
  </liveTime>
  <heartbeatTime min="5" max="255">
    <!--ro, opt, int, Poling Rate, range:[5,255], attr:min{req, int},max{req, int, unit:s}-->1
  </heartbeatTime>
  <heartbeatCount min="3" max="255">
    <!--ro, opt, int, heartbeat times, range:[3,255], attr:min{req, int},max{req, int}-->1
  </heartbeatCount>
  <transportType opt="UDP,TCP,TLS">
    <!--ro, opt, string, SIP service registration method, attr:opt{req, string}, desc:"UDP", "TCP", "TLS"-->UDP
  </transportType>
  <streamType opt="sub,main">
    <!--ro, opt, string, stream type, attr:opt{req, string}, desc:"main", "sub"-->main
  </streamType>
  <tcpConnectType opt="active,passive">
    <!--ro, opt, string, TCP connection type, attr:opt{req, string}, desc:"active", "passive"; this node is valid when transportType is "TCP"-->active
  </tcpConnectType>
  <registerInterval min="60" max="600">
    <!--ro, opt, int, registration interval, range:[60,600], unit:s, attr:min{req, int},max{req, int, unit:s}-->1
  </registerInterval>
  <GBEncryptEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable GB encryption, attr:opt{req, string}-->

```

```

    <!--ro, req, bool, whether to enable domestic password, attr:opt{req, string}-->true
  </GBEncryptEnabled>
  <unidirectionEnabled opt="true,false" def="false">
    <!--ro, opt, bool, whether to allow GB35114 one way authentication, attr:opt{req, string},def{req, string}, desc:this node is valid when
    GBEncryptEnabled is "true"; one-way authentication is authenticating the client on the platform, while two-way authentication is mutual authentication
    between the client and platform-->false
  </unidirectionEnabled>
  <offlineCode min="0" max="16">
    <!--ro, opt, string, device status code returned during offline registration; PUT method is not supported for this node, attr:min{opt,
    string},max{opt, string}, desc:this node is returned only when registerStatus is "false"-->0xFFFFFFFF
  </offlineCode>
  <voiceBroadcastEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable TCP voice broadcast protocol, attr:opt{req, string}-->true
  </voiceBroadcastEnabled>
</GB28181>
<isSupportSIPMultiInfo>
  <!--ro, opt, bool, whether it supports configuring information list of multiple SIP servers-->true
</isSupportSIPMultiInfo>
<isSupportChannelSIP>
  <!--ro, opt, bool, whether it supports configuring SIP by channel-->true
</isSupportChannelSIP>
<isSupportSIPInfoCap>
  <!--ro, opt, bool-->true
</isSupportSIPInfoCap>
</SIPServer>
</SIPServerListCap>

```

9.2.5.5 Get the parameter list of a specific SIP (Session Initiation Protocol) server

Request URL

PUT /ISAPI/System/Network/SIP

Query Parameter

None

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<SIPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, SIP server List, subType:object, attr:version{req, string, protocolVersion}-->
  <SIPServer>
    <!--req, object, SIP server-->
    <id>
      <!--req, string, SIP server ID, range:[1,2]-->test
    </id>
    <localPort>
      <!--opt, int, Local port No., range:[1,65535]-->1
    </localPort>
    <streamID>
      <!--opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
    </streamID>
    <Standard>
      <!--opt, object, standard protocol-->
      <registerStatus>
        <!--req, bool, registration status, desc:registration status: "false"-unregistered,"true"-registered-->true
      </registerStatus>
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <registrar>
        <!--req, string, address, range:[1,128]-->test
      </registrar>
      <registrarPort>
        <!--req, int, port No., range:[1,65535]-->1
      </registrarPort>
      <proxy>
        <!--req, string, proxy address, range:[1,128]-->test
      </proxy>
      <proxyPort>
        <!--req, int, proxy port, range:[1,65535]-->1
      </proxyPort>
      <displayName>
        <!--req, string, displayed name, range:[1,64]-->test
      </displayName>
      <userName>
        <!--req, string, user name, range:[1,64]-->test
      </userName>
      <authID>
        <!--req, string, user authentication ID, range:[1,64]-->test
      </authID>
      <password>
        <!--req, string, password, range:[1,32]-->test
      </password>
      <expires>
        <!--req, int, registration expiration, range:[3600,100000]-->1
      </expires>

```

```

<transportType>
  <!--opt, enum, SIP service registration method, subType:string, desc:"UDP", "TCP", "TLS"-->UDP
</transportType>
<udpRegEnabled>
  <!--opt, bool, whether to enable UDP registration-->true
</udpRegEnabled>
<udpRegPort>
  <!--opt, int, UDP registration port No., range:[1,65535]-->1
</udpRegPort>
<tcpRegEnabled>
  <!--opt, bool, whether to enable TCP registration-->true
</tcpRegEnabled>
<tcpRegPort>
  <!--opt, int, TCP registration port No., range:[1,65535]-->1
</tcpRegPort>
<tlsRegEnabled>
  <!--opt, bool, whether to enable TLS registration-->true
</tlsRegEnabled>
<tlsRegPort>
  <!--opt, int, TLS registration port No., range:[1,65535]-->1
</tlsRegPort>
<addressingFormatType>
  <!--opt, enum, address type, subType:string, desc:if this node is not applied, it is registration by IP by default; "ipAddress" (IP address),
"hostname" (domain name)-->ipAddress
</addressingFormatType>
<hostName>
  <!--opt, string, domain name, range:[1,64], desc:it is valid when addressingFormatType is hostName-->test
</hostName>
<networkMode>
  <!--opt, enum, network mode, subType:int, desc:1-wired NIC 1 (default), 2-wired NIC 2, 3- wireless network-->1
</networkMode>
<targetUserName>
  <!--opt, string, username of the called party, range:[1,32], desc:English letters, numbers, special characters, and Chinese characters are allowed--
>test
</targetUserName>
</Standard>
<GB28181>
  <!--opt, object, single GB platform registration parameter; exclusive with <GB28181List-->
  <registerStatus>
    <!--req, bool, registration status-->true
  </registerStatus>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <registrar>
    <!--req, string, address, range:[1,128]-->test
  </registrar>
  <registrarPort>
    <!--req, int, port No., range:[1,65535]-->1
  </registrarPort>
  <serverId>
    <!--req, string, SIP server ID, range:[1,64]-->test
  </serverId>
  <serverDomain>
    <!--req, int, SIP server domain, range:[1,128]-->1
  </serverDomain>
  <userName>
    <!--req, string, SIP user name, range:[1,64]-->test
  </userName>
  <authID>
    <!--req, string, SIP authentication ID, range:[1,64]-->test
  </authID>
  <password>
    <!--req, string, password; encryption required for sensitive information, range:[1,32]-->test
  </password>
  <expires>
    <!--req, int, registration expiration, range:[3600,100000], unit:s-->1
  </expires>
  <liveTime>
    <!--req, int, effective time, range:[30,100000], unit:s-->1
  </liveTime>
  <heartbeatTime>
    <!--req, int, heartbeat interval, range:[5,255], unit:s-->1
  </heartbeatTime>
  <heartbeatCount>
    <!--req, int, maximum times of heartbeat timeout, range:[3,255]-->1
  </heartbeatCount>
  <transportType>
    <!--req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
  </transportType>
  <streamType>
    <!--opt, enum, stream type, subType:string, desc:"main" (main stream),"sub" (sub stream)-->main
  </streamType>
  <tcpConnectType>
    <!--opt, enum, TCP connection type, subType:string, desc:"active", "passive"-->active
  </tcpConnectType>
  <registerInterval>
    <!--req, int, registration period, range:[60,600], desc:unit: second-->60
  </registerInterval>
  <protocolVersion>
    <!--req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
  </protocolVersion>
  <autoAllocEncodeChanID>
    <!--opt, bool, whether to allocate encoding channel ID automatically-->true

```

```

</autoAllocEncodeChanID>
<autoAllocDecodeChanID>
  <!--opt, bool, whether to allocate decoding channel ID automatically-->true
</autoAllocDecodeChanID>
<autoAllocAlarmInputChanID>
  <!--opt, bool, whether to allocate alarm input channel ID automatically-->true
</autoAllocAlarmInputChanID>
<platformNo>
  <!--opt, int, platform No., range:[1,2]-->1
</platformNo>
<GBEncryptEnabled>
  <!--req, bool, whether to enable domestic password-->true
</GBEncryptEnabled>
<exchangeCertificateEnabled>
  <!--opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default-->true
</exchangeCertificateEnabled>
<unidirectionEnabled>
  <!--opt, bool, whether to allow GB35114 one way authentication, desc:this node is valid when GBEncryptEnabled is "true"; one-way authentication is authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->>false
</unidirectionEnabled>
<voiceBroadcastEnabled>
  <!--opt, bool, whether to enable TCP voice broadcast protocol-->true
</voiceBroadcastEnabled>
</GB28181>
<GB28181List>
  <!--opt, array, subType:object-->
<GB28181>
  <!--opt, object, single GB platform registration parameter; exclusive with <GB28181List-->
  <registerStatus>
    <!--req, bool, registration status-->true
  </registerStatus>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <registrar>
    <!--req, string, address, range:[1,128]-->test
  </registrar>
  <registrarPort>
    <!--req, int, port No., range:[1,65535]-->1
  </registrarPort>
  <serverID>
    <!--req, string, SIP server ID, range:[1,64]-->test
  </serverID>
  <serverDomain>
    <!--req, int, SIP server domain, range:[1,128]-->1
  </serverDomain>
  <userName>
    <!--req, string, SIP user name, range:[1,64]-->test
  </userName>
  <authID>
    <!--req, string, SIP authentication ID, range:[1,64]-->test
  </authID>
  <password>
    <!--req, string, password; encryption required for sensitive information, range:[1,32]-->test
  </password>
  <expires>
    <!--req, int, registration expiration, range:[3600,100000], unit:s-->1
  </expires>
  <liveTime>
    <!--req, int, effective time, range:[30,100000], unit:s-->1
  </liveTime>
  <heartbeatTime>
    <!--req, int, heartbeat interval, range:[5,255], unit:s-->1
  </heartbeatTime>
  <heartbeatCount>
    <!--req, int, maximum times of heartbeat timeout, range:[3,255]-->1
  </heartbeatCount>
  <transportType>
    <!--req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
  </transportType>
  <registerInterval>
    <!--req, int, registration period, range:[60,600], desc:unit: second-->60
  </registerInterval>
  <protocolVersion>
    <!--req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
  </protocolVersion>
  <streamID>
    <!--opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
  </streamID>
  <platformNo>
    <!--opt, int, platform No., range:[1,2]-->1
  </platformNo>
  <GBEncryptEnabled>
    <!--req, bool, whether to enable domestic password-->true
  </GBEncryptEnabled>
  <exchangeCertificateEnabled>
    <!--opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default-->true
  </exchangeCertificateEnabled>
  <localPort>
    <!--req, int, local SIP port, range:[1,65535]-->1
  </localPort>
  <unidirectionEnabled>
    <!--opt, bool, whether to allow GB35114 one way authentication, desc:this node is valid when GBEncryptEnabled is "true"; one-way authentication is

```



```

authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->false
</unidirectionEnabled>
<voiceBroadcastEnabled>
  <!--opt, bool, whether to enable TCP voice broadcast protocol-->true
</voiceBroadcastEnabled>
</GB28181>
</GB28181List>
<channelID>
  <!--opt, string, channel No., range:[1,2048], desc:this node is returned when isSupportChannelSIP is supported in
/ISAPI/System/Network/SIP/capabilities-->1
</channelID>
</SIPServer>
</SIPServerList>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode
4	操作错误	serviceGB28181AndEZVIZExclusive
4	Invalid Operation	thelupServiceAndgbT28181ServiceAreMutuallyExclusiveShutDownOneOfThemAndTryAg
4	Invalid Operation	videoEncryptionIsEnabledForHikConnectPleaseTurnOffVideoEncryptionAndTryAgain

9.2.5.6 Get the parameters of a specific SIP (Session Initiation Protocol) server

Request URL

GET /ISAPI/System/Network/SIP

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<SIPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, SIP server List, subType:object, attr:version{req, string, protocolVersion}-->
  <SIPServer>
    <!--ro, req, object, SIP server-->
    <id>
      <!--ro, req, string, SIP server ID, range:[1,2]-->test
    </id>
    <localPort>
      <!--ro, opt, int, local port No., range:[1,65535]-->1
    </localPort>
    <streamID>
      <!--ro, opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
    </streamID>
    <Standard>
      <!--ro, opt, object, standard protocol-->
    </Standard>
    <registerStatus>

```

```

    <!--ro, req, bool, registration status, desc:false (unregistered), true (registered)-->true
</registerStatus>
<enabled>
    <!--ro, req, bool, whether to enable-->true
</enabled>
<registrar>
    <!--ro, req, string, address, range:[1,128]-->test
</registrar>
<registrarPort>
    <!--ro, req, int, port No., range:[1,65535]-->1
</registrarPort>
<proxy>
    <!--ro, req, string, proxy address, range:[1,128]-->test
</proxy>
<proxyPort>
    <!--ro, req, int, proxy port, range:[1,65535]-->1
</proxyPort>
<displayName>
    <!--ro, req, string, displayed name, range:[1,64]-->test
</displayName>
<userName>
    <!--ro, req, string, user name, range:[1,64]-->test
</userName>
<authID>
    <!--ro, req, string, user authentication ID, range:[1,64]-->test
</authID>
<expires>
    <!--ro, req, int, registration expiration, range:[3600,100000]-->1
</expires>
<transportType>
    <!--ro, opt, enum, SIP service registration method, subType:string, desc:"UDP", "TCP", "TLS"-->UDP
</transportType>
<udpRegEnabled>
    <!--ro, opt, bool, whether to enable UDP registration-->true
</udpRegEnabled>
<udpRegPort>
    <!--ro, opt, int, UDP registration port No., range:[1,65535]-->1
</udpRegPort>
<tcpRegEnabled>
    <!--ro, opt, bool, whether to enable TCP registration-->true
</tcpRegEnabled>
<tcpRegPort>
    <!--ro, opt, int, TCP registration port No., range:[1,65535]-->1
</tcpRegPort>
<tlsRegEnabled>
    <!--ro, opt, bool, whether to enable TLS registration-->true
</tlsRegEnabled>
<tlsRegPort>
    <!--ro, opt, int, TLS registration port No., range:[1,65535]-->1
</tlsRegPort>
<addressingFormatType>
    <!--ro, opt, enum, address type, subType:string, desc:if this node is not applied, it is registration by IP by default; "ipAddress" (IP address),
"hostname" (domain name)-->ipAddress
</addressingFormatType>
<hostName>
    <!--ro, opt, string, domain name, range:[1,64], desc:it is valid when addressingFormatType is hostName-->test
</hostName>
<networkMode>
    <!--ro, opt, enum, network mode, subType:int, desc:1-wired NIC 1 (default), 2-wired NIC 2, 3- wireless network-->1
</networkMode>
<targetUserName>
    <!--ro, opt, string, username of the called party, range:[1,32], desc:English Letters, numbers, special characters, and Chinese characters are
allowed-->test
</targetUserName>
</Standard>
<GB28181>
    <!--ro, opt, object, single GB platform registration parameter; exclusive with <GB28181List>-->
<registerStatus>
    <!--ro, req, bool, registration status-->true
</registerStatus>
<enabled>
    <!--ro, req, bool, whether to enable-->true
</enabled>
<registrar>
    <!--ro, req, string, address, range:[1,128]-->test
</registrar>
<registrarPort>
    <!--ro, req, int, port No., range:[1,65535]-->1
</registrarPort>
<serverId>
    <!--ro, req, string, SIP server ID, range:[1,64]-->test
</serverId>
<serverDomain>
    <!--ro, req, int, SIP server domain, range:[1,128]-->1
</serverDomain>
<userName>
    <!--ro, req, string, SIP user name, range:[1,64]-->test
</userName>
<authID>
    <!--ro, req, string, SIP authentication ID, range:[1,64]-->test
</authID>
<expires>
    <!--ro, req, int, registration expiration, range:[3600,100000], unit:s-->1
</expires>

```

```

<expires>
<liveTime>
  <!--ro, req, int, effective time, range:[30,100000], unit:s-->1
</liveTime>
<heartbeatTime>
  <!--ro, req, int, heartbeat interval, range:[5,255], unit:s-->1
</heartbeatTime>
<heartbeatCount>
  <!--ro, req, int, maximum times of heartbeat timeout, range:[3,255]-->1
</heartbeatCount>
<transportType>
  <!--ro, req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
</transportType>
<streamType>
  <!--ro, opt, enum, stream type, subType:string, desc:"main" (main stream),"sub" (sub stream)-->main
</streamType>
<tcpConnectType>
  <!--ro, opt, enum, TCP connection type, subType:string, desc:"active", "passive"-->active
</tcpConnectType>
<registerInterval>
  <!--ro, req, int, registration period, range:[60,600], desc:unit: second-->60
</registerInterval>
<protocolVersion>
  <!--ro, req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
</protocolVersion>
<autoAllocEncodeChanID>
  <!--ro, opt, bool, whether to allocate encoding channel ID automatically-->true
</autoAllocEncodeChanID>
<autoAllocDecodeChanID>
  <!--ro, opt, bool, whether to allocate decoding channel ID automatically-->true
</autoAllocDecodeChanID>
<autoAllocAlarmInputChanID>
  <!--ro, opt, bool, whether to allocate alarm input channel ID automatically-->true
</autoAllocAlarmInputChanID>
<platformNo>
  <!--ro, opt, int, platform No., range:[1,2]-->1
</platformNo>
<GBEncryptEnabled>
  <!--ro, req, bool, whether to enable domestic password-->true
</GBEncryptEnabled>
<exchangeCertificateEnabled>
  <!--ro, opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default--
>true
</exchangeCertificateEnabled>
<offlineCode>
  <!--ro, opt, string, device status code returned during offline registration; PUT method is not supported for this node, range:[1,128], desc:this
node is returned only when registerStatus is "false"-->0xFFFFFFFF
</offlineCode>
<unidirectionEnabled>
  <!--ro, opt, bool, this node is valid when GBEncryptEnabled is "true"; one-way authentication is authenticating the client on the platform, while
two-way authentication is mutual authentication between the client and platform, desc:this node is valid when GBEncryptEnabled is "true"; one-way
authentication is authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->>false
</unidirectionEnabled>
<voiceBroadcastEnabled>
  <!--ro, opt, bool-->true
</voiceBroadcastEnabled>
</GB28181>
<GB28181List>
  <!--ro, opt, array, subType:object-->
<GB28181>
  <!--ro, opt, object, single GB platform registration parameter; exclusive with <GB28181List-->
  <registerStatus>
    <!--ro, req, bool, registration status-->true
  </registerStatus>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <registrar>
    <!--ro, req, string, address, range:[1,128]-->test
  </registrar>
  <registrarPort>
    <!--ro, req, int, port No., range:[1,65535]-->1
  </registrarPort>
  <serverId>
    <!--ro, req, string, SIP server ID, range:[1,64]-->test
  </serverId>
  <serverDomain>
    <!--ro, req, int, SIP server domain, range:[1,128]-->1
  </serverDomain>
  <userName>
    <!--ro, req, string, SIP user name, range:[1,64]-->test
  </userName>
  <authID>
    <!--ro, req, string, SIP authentication ID, range:[1,64]-->test
  </authID>
  <expires>
    <!--ro, req, int, registration expiration, range:[3600,100000], unit:s-->1
  </expires>
  <liveTime>
    <!--ro, req, int, effective time, range:[30,100000], unit:s-->1
  </liveTime>
  <heartbeatTime>
    <!--ro, req, int, heartbeat interval, range:[5,255], unit:s-->1
  </heartbeatTime>

```

```

<heartbeatCount>
  <!--ro, req, int, maximum times of heartbeat timeout, range:[3,255]-->1
</heartbeatCount>
<transportType>
  <!--ro, req, enum, transport protocol, subType:string, desc:"UDP" (UDP protocol), "TCP" (TCP Protocol)-->UDP
</transportType>
<registerInterval>
  <!--ro, req, int, registration period, range:[60,600], desc:unit: second-->60
</registerInterval>
<protocolVersion>
  <!--ro, req, enum, protocol version, subType:string, desc:"GB/T28181-2011", "GB/T28181-2016"-->GB/T28181-2011
</protocolVersion>
<streamID>
  <!--ro, opt, enum, stream type, subType:string, desc:1 (main stream), 2 (sub-stream), 3 (third stream)-->1
</streamID>
<platformNo>
  <!--ro, opt, int, platform No., range:[1,2]-->1
</platformNo>
<GBEncryptEnabled>
  <!--ro, req, bool, whether to enable domestic password-->true
</GBEncryptEnabled>
<exchangeCertificateEnabled>
  <!--ro, opt, bool, whether to enable certificate exchange; this node is displayed when domestic password is enabled; it is disabled by default--
>true
</exchangeCertificateEnabled>
<offlineCode>
  <!--ro, opt, string, device status code returned during offline registration; PUT method is not supported for this node, range:[1,128], desc:this
node is returned only when registerStatus is "false"-->0xFFFFFFFF
</offlineCode>
<localPort>
  <!--ro, req, int, Local SIP port, range:[1,65535]-->1
</localPort>
<unidirectionEnabled>
  <!--ro, opt, bool, whether to allow GB35114 one way authentication, desc:this node is valid when GBEncryptEnabled is "true"; one-way
authentication is authenticating the client on the platform, while two-way authentication is mutual authentication between the client and platform-->false
</unidirectionEnabled>
<voiceBroadcastEnabled>
  <!--ro, opt, bool-->true
</voiceBroadcastEnabled>
</GB28181>
</GB28181List>
<channelID>
  <!--ro, opt, string, channel No., range:[1,2048], desc:channel No.-->1
</channelID>
</SIPServer>
</SIPServerList>

```

9.3 Information Security

9.3.1 Protocol Certificate

9.3.1.1 Get device certificate information in a batch

Request URL

GET /ISAPI/Security/deviceCertificate/certificates?format=json&type= <type>

Query Parameter

Parameter Name	Parameter Type	Description
type	enum	--

Request Message

None

Response Message

```

{
  "DeviceCertificates": {
    /*ro, req, object*/
    "CertificateInfoList": {
      /*ro, req, object*/
      "CertificateInfo": [
        /*ro, req, array, subType:object*/
        {
          "id": 1,
          /*ro, req, int, step:1*/
          "issuerDN": "test",
          /*ro, req, string*/
          "subjectDN": "test",
          /*ro, req, string*/
          "startDate": "1970-01-01+08:00",
          /*ro, req, date, start date of the certificate effective period, desc:it is accurate to day*/
          "endDate": "1970-01-01+08:00",
          /*ro, req, date, desc:it is accurate to day*/
          "type": "SecurityLog",
          /*ro, opt, enum, subType:string*/
          "platformNo": 1,
          /*ro, opt, int*/
          "certificatesOwner": "NVR",
          /*ro, opt, enum, subType:string*/
          "status": "normal",
          /*ro, opt, enum, certificate status, subType:string*/
          "customID": "abcd1234"
          /*ro, opt, string, user-defined certificate ID*/
        }
      ]
    }
  }
}

```

9.4 Video (General)

9.4.1 Video Channel Management

9.4.1.1 Get access protocols supported by the device

Request URL

GET /ISAPI/ContentMgmt/sourceSupport?send=<send>&mfgr=<mfgr>

Query Parameter

Parameter Name	Parameter Type	Description
send	string	Value range: "head", "top", "middle", "bottom", when set to "head". The device only returns the total number of supported protocols, corresponding to the mediaDeviceListSize field; when set to "top", "middle" or "bottom", the device returns the first 1/3, middle 1/3, and last 1/3 of the supported protocol information respectively.
mfgr	string	Search for specified types of protocols and it supports searching for multiple types simultaneously (GET /ISAPI/ContentMgmt/souceSupport?mfgr=HIKVISION&mfgr=ONVIF). Value range: "HIKVISION", "ONVIF", "GB28181", "ClearView", "ISAPI".

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<CMSourceSupport xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <mediaDeviceListSize>
    <!--ro, req, int-->1
  </mediaDeviceListSize>
  <supportedMediaDeviceSourceList>
    <!--ro, opt, array, subType:object-->
    <MediaDeviceSource>
      <!--ro, opt, object-->
      <MediaDeviceMfr>
        <!--ro, req, enum, subType:string-->
      </MediaDeviceMfr>
      <MediaDeviceMgmtMode>
        <!--ro, req, string-->test
      </MediaDeviceMgmtMode>
      <MediaDeviceModel>
        <!--ro, req, string-->test
      </MediaDeviceModel>
    </MediaDeviceSource>
  </supportedMediaDeviceSourceList>
</CMSourceSupport>

```

9.5 Data Intelligent Analysis

9.5.1 Device Information Management

9.5.1.1 Edit the device information

Request URL

PUT /ISAPI/System/deviceInfo?format=json

Query Parameter

None

Request Message

```

{
  "deviceName": "test",
  /*ro, opt, string, device name*/
  "deviceId": "48353836-3033-3734-3837-2857beed15a",
  /*ro, opt, string, device ID*/
  "deviceDescription": "IPCamera",
  /*ro, opt, string, device description*/
  "deviceLocation": "test",
  /*ro, opt, string, actual location of the device*/
  "systemContact": "test",
  /*ro, opt, string, contact information of the device*/
  "model": "DS-8632N-I8",
  /*ro, opt, string, device model*/
  "serialNumber": "test",
  /*ro, opt, string, device serial No.*/
  "macAddress": "test",
  /*ro, opt, string, device MAC address*/
  "firmwareVersion": "test",
  /*ro, opt, string, */
  "firmwareReleasedDate": "test",
  /*ro, opt, string, */
  "bootVersion": "test",
  /*ro, opt, string, */
  "bootReleasedDate": "test",
  /*ro, opt, string, */
  "hardwareVersion": "test",
  /*ro, opt, string, */
  "encoderVersion": "test",
  /*ro, opt, string, */
  "encoderReleasedDate": "test",
  /*ro, opt, string, */
  "decoderVersion": "test",
  /*ro, opt, string, */
  "decoderReleasedDate": "test",
  /*ro, opt, string, */
  "deviceType": "FA",
  /*ro, opt, enum, device type, subType:string, desc:"IPCamera", "IPDome", "DVR", "HybridNVR", "NVR", "DVS", "IPZoom", "FA", "FS", "FS", "PURE", "HAWK",
  "Blade", "HMCP", "BladePS", "DMbehavior"*/
  "isManageNode": true,
  /*ro, opt, bool, */
  "telecontrolID": 4,
  /*ro, opt, int, keyfob control ID*/
  "supportBeep": true,
  /*ro, opt, bool, whether it supports buzzer*/
  "language": "English",
  /*ro, opt, enum, , subType:string*/
  "illageCharacterList": [
  /*ro, opt, array, , subType:object*/
    {
      "illageCharacter": "&"
      /*ro, opt, string, */
    }
  ],
  "operatingSystem": "Centos Linux",
  /*ro, opt, string, operating system*/
  "manufacturer": "test"
  /*ro, opt, string, manufacturer information, desc:it is used for OEM devices*/
}

```

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "OK",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}

```

9.5.1.2 Get the device information

Request URL

GET /ISAPI/System/deviceInfo?format=json&childDevID=<devIndex>

Query Parameter

Parameter Name	Parameter Type	Description
devIndex	string	--

Request Message

None

Response Message

```
{
  "deviceName": "test",
  /*ro, opt, string, device name*/
  "deviceID": "48353836-3033-3734-3837-2857beed15a",
  /*ro, opt, string, device ID*/
  "deviceDescription": "IPCamera",
  /*ro, opt, string, device description*/
  "deviceLocation": "test",
  /*ro, opt, string, device location*/
  "systemContact": "test",
  /*ro, opt, string, system contact*/
  "model": "DS-8632N-I8",
  /*ro, opt, string, device model*/
  "serialNumber": "test",
  /*ro, opt, string, device serial No.*/
  "macAddress": "test",
  /*ro, opt, string, device MAC address*/
  "firmwareVersion": "test",
  /*ro, opt, string, device firmware version No.*/
  "firmwareReleasedDate": "test",
  /*ro, opt, string, compile date of the device firmware version*/
  "bootVersion": "test",
  /*ro, opt, string, startup BOOT version No.*/
  "bootReleasedDate": "test",
  /*ro, opt, string, compile date of the startup BOOT version*/
  "hardwareVersion": "test",
  /*ro, opt, string, device hardware version No.*/
  "encoderVersion": "test",
  /*ro, opt, string, device encoder version No.*/
  "encoderReleasedDate": "test",
  /*ro, opt, string, compile date of the device encoder version*/
  "decoderVersion": "test",
  /*ro, opt, string, device decoder version No.*/
  "decoderReleasedDate": "test",
  /*ro, opt, string, compile date of the device decoder version*/
  "deviceType": "FA",
  /*ro, opt, enum, device type, subType:string, desc:"IPCamera"-network camera, "IPDome"-network speed dome, "DVR"-digital video recorder, "HybridNVR"-
  hybrid network video recorder, "NVR"-network video recorder, "DVS"-digital video server, "IPZoom"-positioning system, "FA"-face picture real-time comparison
  alarm server, "FS"-face picture static data server, "FD"-capture detection server, "PURE"-pure analysis server, "DM"-Full Analytics Server, "BladePS"-
  Security Blade, "Blade"-DeepinMind vehicle picture analysis server (Blade series), "HAWK"-DeepinMind video structural analysis server (Falcon series),
  "HMCP"-hybrid model comparison server, "lightFace"-Intelligent Fusion Server, "DMbehavior"-abnormal event detection device*/
  "isManageNode": true,
  /*ro, opt, bool, whether it is a management node*/
  "telecontrolID": 4,
  /*ro, opt, int, device ID for keyfob*/
  "supportBeep": true,
  /*ro, opt, bool, whether to support buzzer*/
  "language": "English",
  /*ro, opt, enum, supported language, subType:string, desc:"English", "SimChinese"-Simplified Chinese*/
  "illageCharacterList": [
  /*ro, opt, array, list of illegal characters, subType:object*/
    {
      "illageCharacter": "&"
      /*ro, opt, string, illegal character not supported by the device*/
    }
  ],
  "operatingSystem": "Centos Linux",
  /*ro, opt, string, operating system*/
  "manufacturer": "test",
  /*ro, opt, string, manufacturer information, desc:which is used to distinguish OEM device*/
  "certificateImported": true,
  /*ro, opt, bool, whether the device authorization certificate is imported, desc:if it is not imported, you need to call
  /ISAPI/SDT/Management/certificateImport?format=json to import it*/
  "oneKeySceneEventCombination": {
  /*ro, opt, object, */
    "combinationID": 1,
    /*ro, req, int, */
    "combinationName": "监所"
    /*ro, req, string, , range:[1,64]*/
  }
}
```

9.6 Digital Signage

9.6.1 Information Release

9.6.1.1 Get the information release capability

Request URL

GET /ISAPI/Publish/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PublishServerCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, information release capability, attr:version{req, string, protocolVersion}-->
  <maxStaticMaterialNum>
    <!--ro, req, int, the number of supported static materials-->1
  </maxStaticMaterialNum>
  <maxDynamicMaterialNum>
    <!--ro, opt, int, the number of supported dynamic materials-->1
  </maxDynamicMaterialNum>
  <maxPictureNum>
    <!--ro, opt, int, the number of supported static pictures-->0
  </maxPictureNum>
  <maxVideoNum>
    <!--ro, opt, int, the number of supported static videos-->0
  </maxVideoNum>
  <maxVideoSize>
    <!--ro, opt, int, the total size of supported static video files, step:1, unit:MB-->1
  </maxVideoSize>
  <maxDocumentNum>
    <!--ro, opt, int, the number of supported static texts-->0
  </maxDocumentNum>
  <maxProgramNum>
    <!--ro, req, int, the number of supported programs-->1
  </maxProgramNum>
  <programTypeLimit>
    <!--ro, opt, bool, whether the program has limitations,
    desc:1. true: program number is limited by program type; if the value of maxProgramNum is 2 and the supported programs are pictures and welcome words, it
    indicates the device only supports 1 picture and 1 piece of welcome words.
    2. false: program number is not limited by program type; if the value of maxProgramNum is 2 and the supported programs are pictures and welcome words, it
    indicates the device supports 2 programs (the 2 programs can both be pictures).
    3. this node is only available for access control devices.
    4. true: only 1 of each program type is supported-->true
  </programTypeLimit>
  <maxScheduleNum>
    <!--ro, req, int, the number of supported schedules-->1
  </maxScheduleNum>
  <maxTerminalNum>
    <!--ro, opt, int, the number of supported terminals-->1
  </maxTerminalNum>
  <maxTerminalGroupNum>
    <!--ro, opt, int, the number of supported terminal groups-->1
  </maxTerminalGroupNum>
  <maxSchedulePlanNum>
    <!--ro, opt, int, the number of supported releasing schedules-->1
  </maxSchedulePlanNum>
  <maxOrgNum>
    <!--ro, opt, int, the number of supported organizations-->1
  </maxOrgNum>
  <maxPackLength>
    <!--ro, opt, int, the size of supported package, unit:KB-->1
  </maxPackLength>
  <isSupportMaterialMgr>
    <!--ro, req, bool, whether it supports material management-->true
  </isSupportMaterialMgr>
  <isSupportProgramMgr>
    <!--ro, req, bool, whether it supports program management-->true
  </isSupportProgramMgr>
  <isSupportScheduleMgr>
    <!--ro, req, bool, whether it supports schedule management-->true
  </isSupportScheduleMgr>
  <isSupportTerminalMgr>
    <!--ro, opt, bool, whether it supports terminal management-->true
  </isSupportTerminalMgr>
  <isSupportSwitchPlan>
    <!--ro, opt, bool, whether it supports startup or shutdown schedule-->true
  </isSupportSwitchPlan>
  <isSupportVolumePlan>
    <!--ro, opt, bool, whether it supports volume adjustment schedule-->true
  </isSupportVolumePlan>
  <isSupportServerAddr>
    <!--ro, opt, bool, whether it supports configuring terminal registration server address-->true
  </isSupportServerAddr>
  <isSupportBusinessIntelligence>
    <!--ro, opt, bool, whether it supports business intelligence functions-->true
  </isSupportBusinessIntelligence>

```

```
</isSupportBusinessIntelligence>
<isSupportTerminalPreview>
  <!--ro, opt, bool, whether it supports live view on the terminal-->true
</isSupportTerminalPreview>
<isSupportTemplateMgr>
  <!--ro, opt, bool, whether it supports template management-->true
</isSupportTemplateMgr>
<isSupportGerDataTrans>
  <!--ro, opt, bool, whether it supports data transmission-->true
</isSupportGerDataTrans>
<isSupportThridPartyFile>
  <!--ro, opt, bool, whether it supports uploading the third-party file-->true
</isSupportThridPartyFile>
<isSupportXmlUserMgr>
  <!--ro, opt, bool, whether it supports XML user management-->true
</isSupportXmlUserMgr>
<isSupportUserPermissionCfg>
  <!--ro, opt, bool, whether it supports user permission configuration-->true
</isSupportUserPermissionCfg>
<isSupportOrgMgr>
  <!--ro, opt, bool, whether it supports template management-->true
</isSupportOrgMgr>
<isSupportRetransmitPack>
  <!--ro, opt, bool, whether it supports re-uploading the lost packet-->true
</isSupportRetransmitPack>
<isSupportWeatherFactory>
  <!--ro, opt, bool, whether it supports configuring the server's weather manufacture information-->true
</isSupportWeatherFactory>
<isSupportScheduleExport>
  <!--ro, opt, bool, whether it supports exporting schedules-->true
</isSupportScheduleExport>
<isSupportInputPlan>
  <!--ro, opt, bool, whether it supports auto-switching schedule configuration-->true
</isSupportInputPlan>
<isSupportInsertCharacter>
  <!--ro, opt, bool, whether it supports cut-in message-->true
</isSupportInsertCharacter>
<isSupportHdInit>
  <!--ro, opt, bool, whether it supports disk initialization-->true
</isSupportHdInit>
<isSupportSSH>
  <!--ro, opt, bool, whether it supports SSH configuration-->true
</isSupportSSH>
<isSupportTerminalAdbDebug>
  <!--ro, opt, bool, whether it supports terminal ADB debugging configuration-->true
</isSupportTerminalAdbDebug>
<isSupportBackupData>
  <!--ro, opt, bool, whether it supports data backup-->true
</isSupportBackupData>
<isSupportServerTimeZone>
  <!--ro, opt, bool, whether it supports server time zone configuration-->true
</isSupportServerTimeZone>
<isSupportTerminalTimeZone>
  <!--ro, opt, bool, whether it supports terminal area configuration-->true
</isSupportTerminalTimeZone>
<isSupportTerminalDiscovery>
  <!--ro, opt, bool, whether it supports terminal discovery-->true
</isSupportTerminalDiscovery>
<isSupportScheduleUtcTime>
  <!--ro, opt, bool, whether it supports UTC time plan-->true
</isSupportScheduleUtcTime>
<isSupportAdditionModule>
  <!--ro, opt, bool, whether it supports additional components-->true
</isSupportAdditionModule>
<isSupportTerminalMgrCap>
  <!--ro, opt, bool, whether it supports getting terminal management capability set-->true
</isSupportTerminalMgrCap>
<isSupportMaterialThumbnailDownload>
  <!--ro, opt, bool, whether it supports downloading material thumbnail-->true
</isSupportMaterialThumbnailDownload>
<isSupportProgramThumbnailUpload>
  <!--ro, opt, bool, whether it supports uploading program thumbnail-->true
</isSupportProgramThumbnailUpload>
<isSupportProgramThumbnailDownload>
  <!--ro, opt, bool, whether it supports downloading program thumbnail-->true
</isSupportProgramThumbnailDownload>
<isSupportRePublish>
  <!--ro, opt, bool, whether it supports re-releasing-->true
</isSupportRePublish>
<isSupportTemplateThumbnailUpload>
  <!--ro, opt, bool, whether it supports uploading template thumbnail-->true
</isSupportTemplateThumbnailUpload>
<isSupportTemplateThumbnailDownload>
  <!--ro, opt, bool, whether it supports downloading template thumbnail-->true
</isSupportTemplateThumbnailDownload>
<isSupportSchedulePlanIDSearch>
  <!--ro, opt, bool, whether it supports searching for a single schedule of program schedule-->true
</isSupportSchedulePlanIDSearch>
<isSupportTemplateBaseInfoSearch>
  <!--ro, opt, bool, whether it supports template message-->true
</isSupportTemplateBaseInfoSearch>
<maxDelSchedulePlanNum>
  <!--ro, opt, int, the maximum number of deleting releasing program schedules in a batch-->1
```

```

</maxDelSchedulePlanNum>
<maxDelMaterialNum>
  <!--ro, opt, int, the maximum number of deleting materials in a batch-->1
</maxDelMaterialNum>
<maxDelProgramNum>
  <!--ro, req, int, the maximum number of deleting programs in a batch-->1
</maxDelProgramNum>
<maxDelTerminalNum>
  <!--ro, opt, int, the maximum number of deleting terminals in a batch-->1
</maxDelTerminalNum>
<isSupportUniversalISUPKey>
  <!--ro, opt, bool, whether it supports ISUP general secret key configuration-->true
</isSupportUniversalISUPKey>
<isSupportSubPackageUpload>
  <!--ro, opt, bool, whether it supports uploading materials in package, desc:related URI: /ISAPI/Publish/MaterialMgr/material/<ID>/upload?format=json-->true
</isSupportSubPackageUpload>
<isSupportTerminalStdProto>
  <!--ro, opt, bool, whether the terminal supports standard protocol, desc:after the standardization of terminal protocol, this field is used for the
distinguishing and compatibility of private protocol-->true
</isSupportTerminalStdProto>
<isSupportInsertProgram>
  <!--ro, opt, bool, whether it supports terminal cut-in program-->true
</isSupportInsertProgram>
<isSupportInsertMaterial>
  <!--ro, opt, bool, whether it supports terminal cut-in materials-->true
</isSupportInsertMaterial>
<isSupportMaterialMgrAsync>
  <!--ro, opt, bool, whether it supports material asynchronous applying, desc:related URI: /ISAPI/Publish/MaterialMgr/material/asyncDownload?format=json-->true
</isSupportMaterialMgrAsync>
<isSupportMaterialBatchAdd>
  <!--ro, opt, bool, whether it supports adding materials in a batch, desc:related URI: /ISAPI/Publish/MaterialMgr/material/batchAdd-->true
</isSupportMaterialBatchAdd>
<isSupportScheduleDefaultParam>
  <!--ro, opt, bool, whether it supports default parameter configuration, desc:related URI: /ISAPI/Publish/ScheduleMgr/defaultParam/capabilities?
format=json-->true
</isSupportScheduleDefaultParam>
<isSupportWeatherInfo>
  <!--ro, opt, bool, whether it supports applying city weather data, desc:related URI: /ISAPI/Publish/WeatherInfo?format=json-->true
</isSupportWeatherInfo>
<isSupportCapturePicture>
  <!--ro, opt, bool, whether it supports capture parameter configuration, desc:related URI: /ISAPI/Publish/GetDataMgr/capturePicture/capabilities?
format=json-->true
</isSupportCapturePicture>
<isSupportUpdateDataSource>
  <!--ro, opt, bool, whether it supports source data replacement, desc:related URI: /ISAPI/Publish/Terminal/UpdateDataSource?format=json-->true
</isSupportUpdateDataSource>
<isSupportPageCfg>
  <!--ro, opt, bool, whether it supports page configuration, desc:related URI: /ISAPI/Publish/Terminal/UpdateDataSource?format=json-->true
</isSupportPageCfg>
<isSupportOPSCfg>
  <!--ro, opt, bool, whether it supports OPS parameter configuration, desc:related URI: /ISAPI/Publish/OPSCfg/capabilities?format=json-->true
</isSupportOPSCfg>
<isSupportScreenDisplayCfg>
  <!--ro, opt, bool, whether it supports screen display parameter configuration, desc:related URI: /ISAPI/Publish/OPSCfg/capabilities?format=json-->true
</isSupportScreenDisplayCfg>
<maxMainInterfaceProgramNum>
  <!--ro, opt, int, the number of supported programs on the home page, desc:the home page program cannot be added via schedule and time configuration is
not required, in order to distinguish it in the capability set, this field is added to indicate the number of supported programs on the home page-->true
</maxMainInterfaceProgramNum>
<isSupportScreenGroupDevice>
  <!--ro, opt, bool, whether it supports video wall device configuration, desc:related URI: /ISAPI/Publish/ScreenGroupDevice/capabilities?format=json-->true
</isSupportScreenGroupDevice>
<isSupportScreenGroupMaintenance>
  <!--ro, opt, bool, whether it supports video wall maintenance, desc:related URI: /ISAPI/Publish/ScreenGroupMaintenance/capabilities?format=json-->true
</isSupportScreenGroupMaintenance>
<isSupportScreenGroupControl>
  <!--ro, opt, bool, whether it supports video wall control, desc:related URI: /ISAPI/Publish/ScreenGroupControl/capabilities?format=json-->true
</isSupportScreenGroupControl>
<maxIdleScreenProgramNum>
  <!--ro, opt, int, the number of standby screen programs, desc:the standby screen program cannot be added via schedule and time configuration is not
required, in order to distinguish it in the capability set, this field is added to indicate the number of supported standby screen programs on the home
page-->true
</maxIdleScreenProgramNum>
<isSupportSearchPlaylist>
  <!--ro, opt, bool, whether it supports searching for the playlist, desc:related URI: /ISAPI/Publish/PlayMgr/SearchPlaylist?format=json-->true
</isSupportSearchPlaylist>
<isSupportQuickSnap>
  <!--ro, opt, bool, whether it supports built-in camera capture, desc:related URI: /ISAPI/Publish/TerminalMgr/bindIPC/<ID>-->true
</isSupportQuickSnap>
<isSupportCustomTextShow>
  <!--ro, opt, bool, whether it supports custom text display, desc:related URI: /ISAPI/Publish/TerminalMgr/CustomText/capabilities?format=json-->true
</isSupportCustomTextShow>
<isSupportEZVIZAdvertisement>
  <!--ro, opt, bool, whether it supports EZ advertisement service, desc:related URI: /ISAPI/Publish/EZVIZAdvertisement/capabilities?format=json-->true
</isSupportEZVIZAdvertisement>
<isSupportReleaseScheduleProgress>
  <!--ro, opt, bool, whether it supports uploading schedule release progress, desc:related event: ReleaseScheduleProgress-->true
</isSupportReleaseScheduleProgress>
<isSupportInsertPlayProgress>
  <!--ro, opt, bool, whether it supports uploading cut-in releasing progress, desc:related event: InsertPlayProgress-->true
</isSupportInsertPlayProgress>

```

```

<!--ro, opt, bool, whether it supports uploading cut-off releasing progress, desc:related event: InsertPlayProgress-->true
</isSupportInsertPlayProgress>
<isSupportReleaseSequel>
  <!--ro, opt, bool, whether it supports ANR (Automatic Network Replenishment)-->true
</isSupportReleaseSequel>
<isSupportScreenSaverCfg>
  <!--ro, opt, bool, whether it supports configuring screen saver, desc:related URI: GET/PUT /ISAPI/Publish/DisplayScreen/ScreenSaverCfg?format=json-->true
</isSupportScreenSaverCfg>
<isSupportUploadScreenSaverPicture>
  <!--ro, opt, bool, whether it supports uploading screen picture, desc:related URI: POST /ISAPI/Publish/DisplayScreen/UploadScreenSaverPicture?format=json-->true
</isSupportUploadScreenSaverPicture>
<isSupportGetScreenSaverPicList>
  <!--ro, opt, bool, whether it supports getting screen picture list, desc:related URI: GET /ISAPI/Publish/DisplayScreen/GetScreenSaverPicList?format=json-->true
</isSupportGetScreenSaverPicList>
<isSupportAddBackgroundPicInfo>
  <!--ro, opt, bool, whether it supports adding background picture information, desc:related URI: POST /ISAPI/Publish/UI/AddBackgroundPicInfo?format=json-->true
</isSupportAddBackgroundPicInfo>
<isSupportSearchBackgroundPicInfo>
  <!--ro, opt, bool, whether it supports searching for background picture information, desc:related URI: POST /ISAPI/Publish/UI/SearchBackgroundPicInfo?format=json-->true
</isSupportSearchBackgroundPicInfo>
<isSupportCurrentBackgroundPic>
  <!--ro, opt, bool, whether it supports configuring the current background picture, desc:related URI: GET/PUT /ISAPI/Publish/UI/CurrentBackgroundPic?format=json-->true
</isSupportCurrentBackgroundPic>
<isSupportScreenLockPlanParams>
  <!--ro, opt, bool, whether it supports lock screen schedule parameters, desc:related URI: GET/PUT /ISAPI/Publish/DisplayScreen/ScreenLockPlanParams?format=json-->true
</isSupportScreenLockPlanParams>
<isSupportDeleteBackgroundPicInfo>
  <!--ro, opt, bool, whether it supports deleting background picture information, desc:related URI: POST /ISAPI/Publish/UI/DeleteBackgroundPicInfo?format=json-->true
</isSupportDeleteBackgroundPicInfo>
<isSupportScreenLockPlanStatus>
  <!--ro, opt, bool, whether it supports uploading lock screen schedule status, desc:related event: ScreenLockPlanStatus-->true
</isSupportScreenLockPlanStatus>
<isSupportVoiceBroadcastParams>
  <!--ro, opt, bool, whether it supports voice prompt parameters, desc:related URI: GET/PUT /ISAPI/Publish/InformationReleaseVoiceBroadcast/VoiceBroadcastParams?format=json-->true
</isSupportVoiceBroadcastParams>
<isSupportSendBroadcastInfo>
  <!--ro, opt, bool, whether it supports applying broadcast content, desc:related URI: POST /ISAPI/Publish/InformationReleaseVoiceBroadcast/SendBroadcastInfo?format=json-->true
</isSupportSendBroadcastInfo>
<isSupportScreenStateParams>
  <!--ro, opt, bool, whether it supports display screen status parameters, desc:related URI: GET/PUT /ISAPI/Publish/DisplayScreen/ScreenStateParams?format=json-->true
</isSupportScreenStateParams>
<isSupportProgramSearch>
  <!--ro, opt, bool, whether it supports searching for programs, desc:related URI: POST /ISAPI/Publish/ProgramMgr/programSearch-->true
</isSupportProgramSearch>
<maxNormalProgramNum>
  <!--ro, opt, int, the number of supported standard programs-->true
</maxNormalProgramNum>
<maxScreensaverProgramNum>
  <!--ro, opt, int, the number of supported screen savers-->true
</maxScreensaverProgramNum>
<maxCharacterProgramNum>
  <!--ro, opt, int, the number of supported welcome words-->true
</maxCharacterProgramNum>
<maxTxtProgramNum>
  <!--ro, opt, int, the number of supported texts-->true
</maxTxtProgramNum>
<maxConferenceProgramNum>
  <!--ro, opt, int, the number of supported conferences-->true
</maxConferenceProgramNum>
<maxAdvertiseProgramNum>
  <!--ro, opt, int, the number of supported advertisements-->true
</maxAdvertiseProgramNum>
<isSupportScreenNoDisplayTimeParams>
  <!--ro, opt, bool, whether it supports time parameters of auto locking the screen after no operations, desc:related URI: GET/PUT /ISAPI/Publish/DisplayScreen/ScreenNoDisplayTimeParams?format=json-->true
</isSupportScreenNoDisplayTimeParams>
<isSupportOPSPassword>
  <!--ro, opt, bool, whether it supports OPS password, desc:related URI: PUT /ISAPI/Publish/OPSCfg/OPSPassword?format=json-->true
</isSupportOPSPassword>
<isSupportModifyMaterialAccessToken>
  <!--ro, opt, bool, whether it supports editing access token of materials, desc:related URI: PUT /ISAPI/Publish/MaterialMgr/ModifyMaterialAccessToken?format=json-->true
</isSupportModifyMaterialAccessToken>
<isSupportGetPublishServiceRecentSendInfoRecord>
  <!--ro, opt, bool, whether it supports getting notification records of digital signage server, desc:related URI: POST /ISAPI/Publish/GetPublishServiceRecentSendInfoRecord?format=json-->true
</isSupportGetPublishServiceRecentSendInfoRecord>
<isNotSupportMaterialSearch>
  <!--ro, opt, bool-->true
</isNotSupportMaterialSearch>
<isSupportExportCustomPageConfigFile>
  <!--ro, opt, bool-->true
</isSupportExportCustomPageConfigFile>

```

```
<!--ro, opt, bool-->true
</isSupportCustomPageLinkRule>
<isSupportCustomPage>
  <!--ro, opt, bool-->true
</isSupportCustomPage>
<isSupportScheduleInfo>
  <!--ro, opt, bool-->true
</isSupportScheduleInfo>
<isSupportReleaseSchedule>
  <!--ro, opt, bool-->true
</isSupportReleaseSchedule>
<isSupportGetReleaseScheduleProgress>
  <!--ro, opt, bool-->true
</isSupportGetReleaseScheduleProgress>
<isSupportCancelReleaseSchedule>
  <!--ro, opt, bool-->true
</isSupportCancelReleaseSchedule>
<isSupportCancelInsertPlay>
  <!--ro, opt, bool-->true
</isSupportCancelInsertPlay>
<isSupportInsertPlay>
  <!--ro, opt, bool-->true
</isSupportInsertPlay>
<isSupportInsertPlayControl>
  <!--ro, opt, bool-->true
</isSupportInsertPlayControl>
<isSupportScreenShot>
  <!--ro, opt, bool-->true
</isSupportScreenShot>
<isSupportGetScreenShotResult>
  <!--ro, opt, bool-->true
</isSupportGetScreenShotResult>
<isSupportGetInsertPlayProgress>
  <!--ro, opt, bool-->true
</isSupportGetInsertPlayProgress>
<isSupportMaterialInfo>
  <!--ro, opt, bool-->true
</isSupportMaterialInfo>
<isSupportPageItem>
  <!--ro, opt, bool-->true
</isSupportPageItem>
<isSupportTouchBrightScreen>
  <!--ro, opt, bool-->true
</isSupportTouchBrightScreen>
<isSupportScreenOnDurationTime>
  <!--ro, opt, bool-->true
</isSupportScreenOnDurationTime>
<isSupportCustomPageSupportParams>
  <!--ro, opt, bool-->true
</isSupportCustomPageSupportParams>
<isSupportSearchCustomPageControlTemplate>
  <!--ro, opt, bool-->true
</isSupportSearchCustomPageControlTemplate>
<isSupportSearchCustomPageControlInfo>
  <!--ro, opt, bool-->true
</isSupportSearchCustomPageControlInfo>
<isSupportGetCustomPageThumbnailsList>
  <!--ro, opt, bool-->true
</isSupportGetCustomPageThumbnailsList>
<isSupportAddImportCustomPageFileTask>
  <!--ro, opt, bool-->true
</isSupportAddImportCustomPageFileTask>
<isSupportSearchImportCustomPageFileTaskProgress>
  <!--ro, opt, bool-->true
</isSupportSearchImportCustomPageFileTaskProgress>
<isSupportDeleteAllMaterial>
  <!--ro, opt, bool-->true
</isSupportDeleteAllMaterial>
<isSupportGetPageFont>
  <!--ro, opt, bool-->true
</isSupportGetPageFont>
<isSupportImportPageFont>
  <!--ro, opt, bool-->true
</isSupportImportPageFont>
<isSupportCloudLockScreenParams>
  <!--ro, opt, bool-->true
</isSupportCloudLockScreenParams>
<isSupportCloudLockScreenTimeParams>
  <!--ro, opt, bool-->true
</isSupportCloudLockScreenTimeParams>
<isSupportMainInterfaceModeParams>
  <!--ro, opt, bool-->true
</isSupportMainInterfaceModeParams>
<isSupportSearchMainInterfaceThumbnail>
  <!--ro, opt, bool-->true
</isSupportSearchMainInterfaceThumbnail>
<isSupportCustomStandbyPicMgr>
  <!--ro, opt, bool-->true
</isSupportCustomStandbyPicMgr>
<isSupportStandbyPicParams>
  <!--ro, opt, bool-->true
</isSupportStandbyPicParams>
```

```

<!--ro, opt, bool-->true
</isSupportStandbyPicDisplayParams>
<isSupportGetCustomPageFileSupportParams>
  <!--ro, opt, bool-->true
</isSupportGetCustomPageFileSupportParams>
<isSupportSendRestoreDefaultPageTask>
  <!--ro, opt, bool-->true
</isSupportSendRestoreDefaultPageTask>
<isSupportSearchRestoreDefaultPageTaskStatus>
  <!--ro, opt, bool-->true
</isSupportSearchRestoreDefaultPageTaskStatus>
</PublishServerCap>

```

9.7 Decoding and Video Wall

9.7.1 Board Management

9.7.1.1 Set parameters of a specified sub-board

Request URL

PUT /ISAPI/System/Board/<BoardID>/config

Query Parameter

Parameter Name	Parameter Type	Description
BoardID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<InputBoardCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, sub-board parameters, attr:version{req, string, protocolVersion}-->
  <slotNo>
    <!--req, int, slot No.-->1
  </slotNo>
  <fullFrameEnable>
    <!--req, bool-->true
  </fullFrameEnable>
</InputBoardCfg>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

9.7.1.2 Get sub-board capability

Request URL

GET /ISAPI/System/Board/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AllBoardCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <BoardCapList>
    <!--ro, req, array, subType:object-->
    <BoardCap>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, No.-->test
      </id>
      <isSupportBasicParam opt="ture,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </isSupportBasicParam>
      <isSupportSvrParam opt="ture,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </isSupportSvrParam>
      <isSupportNetwork opt="ture,false">
        <!--ro, opt, bool, whether it supports network parameter configuration, attr:opt{req, string}-->true
      </isSupportNetwork>
      <isSupportNetAddrTrans opt="true,false">
        <!--ro, opt, bool, whether it supports NAT configuration, attr:opt{req, string}-->true
      </isSupportNetAddrTrans>
      <isSupportStatus opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </isSupportStatus>
      <isSupportNetworkArea opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </isSupportNetworkArea>
      <isSupportBatchUpgrade opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </isSupportBatchUpgrade>
      <isSupportSubSysConfigFileImport opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </isSupportSubSysConfigFileImport>
    </BoardCap>
  </BoardCapList>
</AllBoardCap>
```

9.7.1.3 Set parameters of all sub-boards

Request URL

PUT /ISAPI/System/Board/config

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<InputBoardCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <InputBoardCfg>
    <!--opt, object-->
    <slotNo>
      <!--req, int, slot No.-->1
    </slotNo>
    <fullFrameEnable>
      <!--req, bool-->true
    </fullFrameEnable>
  </InputBoardCfg>
</InputBoardCfgList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	uhdOutputBoardPort2IsInTheWall	0x4000A1A2	--
4	Invalid Operation	uhdInputBoardSource2IsInTheWall	0x4000A1A3	--
4	Invalid Operation	uhdInputBoardSource2IsSplicingSource	0x4000A1A4	--
4	Invalid Operation	inStandardModeSource1IsNotAllowedToDragMoreThanOne	0x4000A1A5	--
4	Invalid Operation	oneDragMultipleEncodingResourcesAreInsufficient	0x4000A1A6	--

9.7.1.4 Get capability of the status of all sub-boards

Request URL

GET /ISAPI/System/Board/status/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BoardStatusCapList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <BoardStatusCap>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, No.-->1
    </id>
    <boardModle min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </boardModle>
    <hardVersion min="1" max="10">
      <!--ro, opt, string, hardware version, attr:min{req, int},max{req, int}-->test
    </hardVersion>
    <SubSysStatusList>
      <!--ro, req, array, subType:object-->
      <SubSysStatus>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, No.-->1
        </id>
        <type opt="DI2,D02,UHDI,YI2">
          <!--ro, opt, string, type, attr:opt{req, string}-->test
        </type>
        <isOnline opt="true,false">
```



```

    <!--ro, opt, bool, online status, attr:opt{req, string}-->true
</isOnline>
<syncStatus opt="normal,abnormal">
    <!--ro, opt, string, synchronization status, attr:opt{req, string}-->test
</syncStatus>
<ModuleList>
    <!--ro, opt, array, subType:object-->
    <Module>
        <!--ro, opt, object-->
        <id>
            <!--ro, req, int, No.-->1
        </id>
        <modDescription opt="CPU1,CPU2,CPU3">
            <!--ro, opt, string, attr:opt{req, string}-->test
        </modDescription>
        <MCU>
            <!--ro, opt, object-->
            <mcuUtilization min="1" max="10">
                <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
            </mcuUtilization>
        </MCU>
        <PhysiMem>
            <!--ro, opt, object-->
            <physiMemUsage min="1" max="10">
                <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
            </physiMemUsage>
            <physiMemAvailable min="1" max="10">
                <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
            </physiMemAvailable>
        </PhysiMem>
        <DispMem>
            <!--ro, opt, object-->
            <dispMemUsage min="1" max="10">
                <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
            </dispMemUsage>
            <dispMemAvailable min="1" max="10">
                <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
            </dispMemAvailable>
        </DispMem>
        <CacheMem>
            <!--ro, opt, object, cache-->
            <cacheMemUsage min="1" max="10">
                <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
            </cacheMemUsage>
            <cacheMemAvailable min="1" max="10">
                <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
            </cacheMemAvailable>
        </CacheMem>
        <temperature min="1" max="10">
            <!--ro, opt, float, temperature, attr:min{req, int},max{req, int}-->0.000
        </temperature>
    </Module>
</ModuleList>
<NetPortStatusList size="10">
    <!--ro, opt, array, network port status list, subType:object, attr:size{req, int}-->
    <NetPortStatus>
        <!--ro, opt, object-->
        <id min="1" max="10">
            <!--ro, opt, int, No., attr:min{req, int},max{req, int}-->1
        </id>
        <netPortDescription opt="ctrl,data1,data2">
            <!--ro, opt, string, attr:opt{req, string}-->test
        </netPortDescription>
        <linkStatus opt="connected,disconnected">
            <!--ro, opt, string, connection status, attr:opt{req, string}-->test
        </linkStatus>
        <upSpeed min="1" max="10">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
        </upSpeed>
        <downSpeed min="1" max="10">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
        </downSpeed>
    </NetPortStatus>
</NetPortStatusList>
<decodingResourcesNum min="1" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</decodingResourcesNum>
<decodingResourcesUsageRate min="1" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</decodingResourcesUsageRate>
<decodingResourcesAvailable min="1" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</decodingResourcesAvailable>
</SubSysStatus>
<SubSysStatusList>
    <customBoardModle min="1" max="10">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </customBoardModle>
</BoardStatusCap>
</BoardStatusCapList>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	notSupport	0x40000001	--

9.7.2 Decoding Management

9.7.2.1 Get decoding device status

Request URL

GET /ISAPI/DisplayDev/decoingDevice/status?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "DevCaseStatus": {
    /*ro, opt, object, cabinet status*/
    "height": "3U",
    /*ro, opt, string, cabinet height, desc:example: 3U*/
    "row": 1,
    /*ro, opt, int, number of cabinet rows, range:[1,64]*/
    "col": 1
    /*ro, opt, int, number of cabinet columns, range:[1,64]*/
  },
  "MainBoardStatusList": [
    /*ro, opt, array, main board status list, subType:object*/
    {
      "ID": 1,
      /*ro, opt, int, sequence number*/
      "NetworkInterfaceList": [
        /*ro, opt, array, network interface list, subType:object*/
        {
          "ID": 1,
          /*ro, opt, int, network interface No.*/
          "type": "kilomega",
          /*ro, opt, enum, network interface type, subType:string, desc:"kilomega" (gigabit Ethernet), "optical" (optical port)*/
          "ipV4Address": "test",
          /*ro, opt, string, IPv4*/
          "ipV6Address": "test",
          /*ro, opt, string, IPv6*/
          "portNo": 8100,
          /*ro, opt, int, port No.*/
          "MACAddress": "test",
          /*ro, opt, string, MAC address*/
          "ethernetPortStatus": "Connected",
          /*ro, opt, enum, the network interface status, subType:string, desc:"connected", "notconnect" (not connected)*/
          "sendingRate": 0.0,
          /*ro, opt, float, sending rate, range:[0,1099511627776], unit:bps*/
          "receivingRate": 0.0,
          /*ro, opt, float, receiving rate, range:[0,1099511627776], unit:bps*/
          "lineRate": 0.0,
          /*ro, opt, float, line rate, range:[0,1099511627776], unit:bps, desc:NIC bandwidth*/
          "row": 1,
          /*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
          "col": 2
          /*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
        }
      ],
      "temperature": 0.0,
      /*ro, opt, float, temperature*/
      "status": "normal",
      /*ro, opt, enum, status, subType:string, desc:"normal", "alarm", "notPower" (not powered on), "notInsert" (no board/module inserted)*/
      "alarmType": "temp",
      /*ro, opt, enum, alarm type, subType:string, desc:it is valid when the status==alarm. "temp" (temperature), "voltage", "network", "chassis" (cabinet)*/
      "subAlarmType": "upLimit",
      /*ro, opt, enum, sub alarm type, subType:string, desc:"upLimit" (exceeded the upper limit), "downLimit" (exceeded the lower limit),
      "netDisconnect" (network disconnected), "netConflict" (network conflict), "illegallogin" (illegal login), "powerNotCon" (not powered up), "fanNotCon" (no fan installed)*/
      "alarmLevel": "general",
      /*ro, opt, enum, alarm level, subType:string, desc:"general", "serious", "deadly"*/
      "exceptionList": [
        /*ro, opt, array, exception list, subType:object*/
        {
          "alarmType": "temp",
          /*ro, opt, enum, exception type, subType:string, desc:it is valid when the status==alarm. "temp" (temperature), "voltage", "network",
          "security", "chassis" (cabinet), "decodingResources" (decoding resource), "networkSource" (network source), "selfDefine" (custom), "transferStreamException" (stream forwarding exception)*/
          "exceptionInfo": "undefined",

```

```

        /*ro, opt, string, exception information, range:[1,256]*/
        "errorCode": 1073784640,
        /*ro, opt, enum, error code, subType:int, desc:error code*/
        "customeInfo": "undefined"
        /*ro, opt, string, custom information, range:[1,256]*/
    }
},
"CPUUtilization": 0,
/*ro, opt, int, CPU usage rate, range:[0,100]*/
"memoryUtilization": 0,
/*ro, opt, int, percentage of the used memory, range:[0,100]*/
"audioInputList": [
/*ro, opt, array, audio input list, subType:object*/
{
    "ID": 1,
    /*ro, req, int, sequence number*/
    "status": "connected",
    /*ro, req, enum, connection status, subType:string, desc:"connected", "notconnect" (not connected)*/
    "row": 1,
    /*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
    "col": 2
    /*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
}
],
"audioOutputList": [
/*ro, opt, array, audio output list, subType:object*/
{
    "ID": 1,
    /*ro, req, int, sequence number*/
    "status": "connected",
    /*ro, req, enum, connection status, subType:string, desc:"connected", "notconnect" (not connected)*/
    "row": 1,
    /*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
    "col": 2
    /*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
}
],
"serialPortList": [
/*ro, opt, array, serial port list, subType:object*/
{
    "ID": 1,
    /*ro, req, int, sequence number*/
    "serialPortType": "232",
    /*ro, req, enum, serial port type, subType:string, desc:serial port type*/
    "status": "connected",
    /*ro, req, enum, connection status, subType:string, desc:"connected", "notconnect" (not connected)*/
    "row": 1,
    /*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
    "col": 2
    /*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
}
],
"genlockPortList": [
/*ro, opt, array, genlock interface list, subType:object, desc:genlock interface list*/
{
    "ID": 1,
    /*ro, req, int, sequence number*/
    "status": "connected",
    /*ro, req, enum, connection status, subType:string, desc:"connected", "notconnect" (not connected)*/
    "row": 1,
    /*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
    "col": 2
    /*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
}
],
"USBInterfaceList": [
/*ro, opt, array, USB interface list, subType:object*/
{
    "ID": 1,
    /*ro, opt, int, USB port serial No.*/
    "status": "connected",
    /*ro, opt, enum, subType:string*/
    "row": 1,
    /*ro, opt, int, the row number where USB located, range:[1,64]*/
    "col": 2
    /*ro, opt, int, the column number where USB located, range:[1,64]*/
}
],
"row": 1,
/*ro, opt, int, the row number where main board located, range:[1,64]*/
"col": 2,
/*ro, opt, int, the column number where main board located, range:[1,64]*/
"rowCover": 1,
/*ro, opt, int, number of rows occupied by the main board*/
"colCover": 1,
/*ro, opt, int, number of columns occupied by the main board*/
"runTime": 1
/*ro, opt, int, main board running duration, range:[0,100000000], unit:s*/
}
],
"SwitchBoardStatusList": [
/*ro, opt, array, switch board status list, subType:object*/
{

```

```

        "ID": 1,
        /*ro, opt, int, sequence number*/
        "serialNumber": "test",
        /*ro, opt, string, sub-board serial No., range:[9,48]*/
        "CPUUtilization": 0,
        /*ro, opt, int, CPU usage rate, range:[0,100]*/
        "NetworkInterfaceList": [
        /*ro, opt, array, network interface list, subType:object*/
        {
            "ID": 1,
            /*ro, opt, int, network interface No.*/
            "ipV4Address": "test",
            /*ro, opt, string, IPv4*/
            "ipV6Address": "test",
            /*ro, opt, string, IPv6*/
            "portNo": 8100,
            /*ro, opt, int, port No.*/
            "MACAddress": "undefined",
            /*ro, opt, string, MAC address*/
            "ethernetPortStatus": "connected",
            /*ro, opt, enum, the network interface status, subType:string, desc:"connected", "notconnect" (not connected)*/
            "sendingRate": 0.0,
            /*ro, opt, float, sending rate, range:[0,1099511627776], unit:bps*/
            "receivingRate": 0.0,
            /*ro, opt, float, receiving rate, range:[0,1099511627776], unit:bps*/
            "row": 1,
            /*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
            "col": 2
            /*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
        }
    ],
    "backplaneTemperature": 0.0,
    /*ro, opt, float, backplane temperature*/
    "temperature": 0.0,
    /*ro, opt, float, switch board temperature*/
    "status": "normal",
    /*ro, opt, enum, status, subType:string, desc:"normal", "alarm", "notPower" (not powered on), "notInsert" (no board/module inserted)*/
    "alarmType": "temp",
    /*ro, opt, enum, alarm type, subType:string, desc:it is valid when the status=alarm. "temp" (temperature), "voltage", "network", "chassis"
(cabinet)*/
    "subAlarmType": "upLimit",
    /*ro, opt, enum, sub alarm type, subType:string, desc:"upLimit" (exceeded the upper limit), "downLimit" (exceeded the lower limit),
"netDisconnect" (network disconnected), "netConflict" (network conflict), "illegallogin" (illegal login), "powerNotCon" (not powered up), "fanNotCon" (no
fan installed), "NATConflict" (NAT conflict)*/
    "alarmLevel": "general",
    /*ro, opt, enum, alarm level, subType:string, desc:"general", "serious", "deadly"*/
    "row": 1,
    /*ro, opt, int, row No. where switch board located, range:[1,64]*/
    "col": 2,
    /*ro, opt, int, column No. where switch board located, range:[1,64]*/
    "rowCover": 1,
    /*ro, opt, int, number of rows occupied by the switch board*/
    "colCover": 1
    /*ro, opt, int, number of columns occupied by the switch board*/
}
],
"SubBoardStatusList": [
/*ro, opt, array, sub board status list, subType:object*/
{
    "ID": 1,
    /*ro, opt, int, sub-board No.*/
    "subSystemID": 1,
    /*ro, opt, int, partition No.*/
    "boardModel": "test",
    /*ro, opt, string, sub-board model, range:[1,32]*/
    "subBoardType": "input",
    /*ro, opt, enum, sub-board type, subType:string, desc:"input" (input board), "output" (output board), "decode" (decoding board), "LEDSendCard"
(LED sending card), "decodeOutput" (decoding output board), "encodeInput" (encoding input board), "inputOutputMixed" (input and output board), "preBoard"
(preview board), "LEDOpticalSendCard" (LED optical port sending card), "monitorBoard" (preview board)*/
    "serialNumber": "test",
    /*ro, opt, string, sub-board serial No., range:[9,48]*/
    "firmwareVersion": "V5.4.20",
    /*ro, opt, string, firmware version No., range:[1,16], desc:example: V5.4.20*/
    "firmwareReleaseDate": "build 20200508",
    /*ro, opt, string, version date, range:[1,32], desc:build 20200508*/
    "CPUUtilization": 0,
    /*ro, opt, int, CPU usage rate, range:[0,100]*/
    "memoryUtilization": 0,
    /*ro, opt, int, percentage of the used memory, range:[0,100]*/
    "SubBoardInterfaceList": [
    /*ro, opt, array, sub-board interface list, subType:object*/
    {
        "ID": 1,
        /*ro, opt, int, sequence number*/
        "subBoardInterfaceType": "HDMI",
        /*ro, opt, enum, sub-board interface type, subType:string, desc:"HDMI", "DVI", "VGA", "DP", "HDMI4K", "DP4K", "SDI"*/
        "signalStatus": "signal",
        /*ro, opt, enum, signal source status, subType:string, desc:"signal" (with signal input), "noSignal" (no signal), "abnormal"*/
        "decodeWallStatus": "decoding",
        /*ro, opt, enum, status of decoding and displaying on video wall, subType:string, desc:"decoding", "noDecoding" (no decoding)*/
        "outputPortLinkStatus": "connected",
        /*ro, opt, enum, output port wiring status, subType:string, desc:"connected", "notconnect" (not connected)*/
        "subBoardInterfaceMode": "input",
        /*ro, opt, enum, sub board interface mode. subType:string, desc:"input", "output". It is valid when "subBoardType" is

```

```

"inputOutputMixed", and used to distinguish the interface is output or input*/
"row": 1,
/*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
"col": 2
/*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
}
],
"NetworkInterfaceList": [
/*ro, opt, array, network group interface list, subType:object*/
{
  "ID": 1,
  /*ro, opt, int, group ID*/
  "type": "kilomega",
  /*ro, opt, enum, network interface type, subType:string, desc:"kilomega" (gigabit Ethernet), "optical" (optical port)*/
  "netPortList": [
  /*ro, opt, array, network interface list, subType:object*/
  {
    "ID": 1,
    /*ro, opt, int, network interface No.*/
    "ipV4Address": "test",
    /*ro, opt, string, IPv4*/
    "ipV6Address": "test",
    /*ro, opt, string, IPv6*/
    "portNo": 8100,
    /*ro, opt, int, port No.*/
    "MACAddress": "undefined",
    /*ro, opt, string, MAC address*/
    "ethernetPortStatus": "connected",
    /*ro, opt, enum, the network interface status, subType:string, desc:"connected", "notconnect" (not connected)*/
    "sendingRate": 0.0,
    /*ro, opt, float, sending rate, range:[0,1099511627776], unit:bps*/
    "receivingRate": 0.0,
    /*ro, opt, float, receiving rate, range:[0,1099511627776], unit:bps*/
    "lineRate": 0.0,
    /*ro, opt, float, line rate, range:[0,1099511627776], unit:bps, desc:NIC bandwidth*/
    "row": 1,
    /*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
    "col": 2
    /*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
  }
]
}
],
"decodingResourcesNum": 1,
/*ro, opt, int, decoding resource channels*/
"decodingResourcesUsageRate": 1,
/*ro, opt, int, decoding resource usage*/
"decodingResourcesAvailable": 1,
/*ro, opt, int, decoding resource available channels*/
"temperature": 0.0,
/*ro, opt, float, temperature*/
"status": "normal",
/*ro, opt, enum, status, subType:string, desc:"normal", "alarm", "notPower" (not powered on), "notInsert" (no board/module inserted)*/
"alarmType": "temp",
/*ro, opt, enum, alarm type, subType:string, desc:it is valid when the status==alarm. "temp" (temperature), "voltage", "network", "security",
"chassis" (cabinet), "decodingResources" (decoding resource), "networkSource" (network source)*/
"subAlarmType": "upLimit",
/*ro, opt, enum, sub alarm type, subType:string, desc:"upLimit" (exceeded the upper limit), "downLimit" (exceeded the lower limit),
"netDisconnect" (network disconnected), "netConflict" (network conflict), "illegalLogin" (illegal login), "powerNotCon" (not powered up), "fanNotCon" (no
fan installed)*/
"errorCode": 1,
/*ro, opt, int, error code information*/
"alarmLevel": "general",
/*ro, opt, enum, alarm level, subType:string, desc:"general", "serious", "deadly"*/
"exceptionList": [
/*ro, opt, array, exception list, subType:object*/
{
  "alarmType": "temp",
  /*ro, opt, enum, exception type, subType:string, desc:it is valid when the status==alarm. "temp" (temperature), "voltage", "network",
"security", "chassis" (cabinet), "decodingResources" (decoding resource), "networkSource" (network source), "selfDefine" (custom), "transferStreamException"
(stream forwarding exception)*/
  "exceptionInfo": "undefined",
  /*ro, opt, string, exception information, range:[1,256]*/
  "errorCode": 1073784640,
  /*ro, opt, enum, error code information, subType:int*/
  "customeInfo": "undefined"
  /*ro, opt, string, custom information, range:[1,256]*/
}
]
},
"isMainSubMixed": false,
/*ro, opt, bool, whether it is mixed main/sub board, desc:defaults to non-mixed sub board*/
"occupySlotNum": 1,
/*ro, opt, int, number of occupied slots, desc:defaults to one when this field is absent*/
"row": 1,
/*ro, opt, int, row number where the sub board located, range:[1,64]*/
"col": 2,
/*ro, opt, int, column number where the sub board located, range:[1,64]*/
"rowCover": 1,
/*ro, opt, int, number of rows occupied by the sub board*/
"colCover": 1,
/*ro, opt, int, number of columns occupied by the sub board*/
"runTime": 1,
/*ro, opt, int, sub board running duration, range:[0,100000000], unit:s*/

```

```

"audioInputList": [
/*ro, opt, array, audio input list, subType:object*/
{
  "ID": 1,
/*ro, req, int, sequence number*/
  "status": "connected",
/*ro, req, enum, connection status, subType:string, desc:"connected", "notconnect" (not connected)*/
  "row": 1,
/*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
  "col": 2
/*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
}
],
"audioOutputList": [
/*ro, opt, array, audio output list, subType:object*/
{
  "ID": 1,
/*ro, req, int, sequence number*/
  "status": "connected",
/*ro, req, enum, connection status, subType:string, desc:"connected", "notconnect" (not connected)*/
  "row": 1,
/*ro, opt, int, the row number of the physical interface on current board, range:[1,64]*/
  "col": 2
/*ro, opt, int, the column number of the physical interface on current board, range:[1,64]*/
}
]
},
"BackplaneStatusList": [
/*ro, opt, array, backplane status list, subType:object, desc:backplane is a independent PCB board in the cabinet*/
{
  "ID": 1,
/*ro, opt, int, sequence number*/
  "backplaneTemperature": 0.0
/*ro, opt, float, backplane temperature*/
}
]
}

```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
	[Message content]	application/json	--	--	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```

--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value

```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

statusCode	statusString	subStatusCode	errorC
2	Device Busy	noMemory	0x200
3	Device Error	insufficientDecRes	0x300
3	Device Error	theDevicelsOffline	0x300
3	Device Error	cpuUsagelsTooHigh	0x300
3	Device Error	maximumTemperatureOfDeviceReached	0x300
3	Device Error	gpustatusexcept	0x300
3	Device Error	fpgastatusexcept	0x300

3	Device Error	dspstatusexcept	0x300
3	Device Error	localsourcestreamsnotreceivedunicastdata	0x300
3	Device Error	abnormalUplinkBandwidth	0x300
3	Device Error	abnormalDownlinkBandwidth	0x300
3	Device Error	subsystemClockSynchronizationFailure	0x300
4	Invalid Operation	USBNotExist	0x400
4	Invalid Operation	ModifiedIpCannotBeDuplicate	0x400
4	Invalid Operation	networkSourceStreamsNotReceived	0x400
4	Invalid Operation	networkSourceRtspStreamingFailed	0x400
4	Invalid Operation	networkSourceRtpPacketsNotReceivedContinuously	0x400
4	Invalid Operation	tooManyErrorStreamsInTheForwardedStreams	0x400
4	Invalid Operation	encodingTypesOfForwardedStreamsMismatch	0x400
4	Invalid Operation	decodingNetworkSourceStreamsFailedEncodingTypeException	0x400
4	Invalid Operation	tooMuchErrorDataWhenDecodingNetworkSourceStreams	0x400
4	Invalid Operation	iFrameIntervalTooHighWhenDecodingNetworkSourceStreams	0x400
4	Invalid Operation	timestampDifferenceTooLargeWhenDecodingNetworkSourceStreams	0x400
4	Invalid Operation	localSourceStreamsDisplayedOnTheVideoWallButMulticastDataNotReceived	0x400
4	Invalid Operation	packetLossExistsInMulticastDataReceivedFromLocalSourceStreams	0x400
4	Invalid Operation	shallowDecodingFailedAbnormalDataReceivedFromLocalSourceStreams	0x400
4	Invalid Operation	theSignalSourceProcessingByWindowDivisionApplicationAndThatByDspMismatch	0x400
4	Invalid Operation	noDeviceConnectedToVideoOutputInterface	0x400
4	Invalid Operation	deviceInternalErrorOccurredInRtspStreaming	0x400
4	Invalid Operation	deviceOptionsRequestException	0x400
4	Invalid	peerEndOptionsResponseException	0x400

4	Invalid Operation	peerEndOptionsResponseException	0x4000
4	Invalid Operation	deviceDescribeRequestException	0x4000
4	Invalid Operation	peerEndDescribeResponseException	0x4000
4	Invalid Operation	peerEndAuthenticationExceptionOccurredInRtspStreaming	0x4000
4	Invalid Operation	peerEndMediaInformationErrorOccurredInRtspStreaming	0x4000
4	Invalid Operation	networkExceptionOccurredInRtspStreaming	0x4000
4	Invalid Operation	deviceSetupRequestException	0x4000
4	Invalid Operation	peerEndSetupResponseException	0x4000
4	Invalid Operation	devicePlayRequestException	0x4000
4	Invalid Operation	peerEndPlayResponseException	0x4000
4	Invalid Operation	thePacketOrderIsNotContinuousInStream	0x4000
4	Invalid Operation	streamingIsNotContinuous	0x4000
4	Invalid Operation	streamDataIsAccumulated	0x4000
4	Invalid Operation	encodingFormatIsNotSupported	0x4000
4	Invalid Operation	streamEncapsulationFormatCannotBeRecognized	0x4000
4	Invalid Operation	encryptionKeysNotConfiguredForTheEncryptedStream	0x4000
4	Invalid Operation	streamingIsNotContinuousInForwarding	0x4000
4	Invalid Operation	streamDataInForwardingIsAccumulated	0x4000
4	Invalid Operation	encodingFormatOfStreamForwardingIsNotSupported	0x4000
4	Invalid Operation	streamEncapsulationFormatOfStreamForwardingCannotBeRecognized	0x4000
4	Invalid Operation	encryptionKeyOfStreamForwardingIsNotConfiguredForTheEncryptedStream	0x4000
4	Invalid Operation	streamForwardingTimeout	0x4000

4	Invalid Operation	unstableFrameRateOfStreamForwarding	0x400
4	Invalid Operation	highNetworkUsage	0x400
4	Invalid Operation	fanStatusAbnormal	0x400
4	Invalid Operation	highMemoryUsage	0x400
6	Invalid Content	insufficientNetworkBandwidth	0x600

9.7.2.2 Get network pre-monitor parameters of a video wall

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/nPreMonitor

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	Video wall ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallNPreMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, network pre-monitoring parameters of a video wall, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, video wall No.-->1
  </id>
  <resolution opt="1080*720P,1920*1080P,...">
    <!--ro, req, string, pre-monitoring signal output resolution, attr:opt{req, string}, desc:it is obtained from capability set-->test
  </resolution>
  <frameRate opt="15,20,25,...">
    <!--ro, req, int, frame rate, attr:opt{req, string}-->1
  </frameRate>
  <bitRateType opt="CBR,VBR">
    <!--ro, req, string, bitrate type, attr:opt{req, string}, desc:including CBR (Constant Bit Rate) and VBR (Variable Bit Rate).-->test
  </bitRateType>
  <bitRate opt="8,16,32,...">
    <!--ro, opt, int, bitrate, unit:kpbs 码率, 包括8、16、32等, 单位: kpbs, 定码率时有效, attr:opt{req, string}-->1
  </bitRate>
  <maxBitRate opt="128,256,...">
    <!--ro, opt, int, max. bitrate, unit:kpbs 最大码率, 128、256等, 单位: kpbs, 变码率时有效, attr:opt{req, string}-->1
  </maxBitRate>
  <intervalFrameI>
    <!--ro, req, int, I frame interval, unit:ms-->1
  </intervalFrameI>
</WallNPreMonitor>
```

9.7.2.3 Set network pre-monitor parameters of a video wall

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/nPreMonitor

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallNPreMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, network pre-monitoring parameters of a video wall, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, video wall No.-->1
  </id>
  <resolution opt="1080*720P,1920*1080P,...">
    <!--req, string, pre-monitoring signal output resolution, attr:opt{req, string}, desc:it is obtained from capability set-->test
  </resolution>
  <frameRate opt="15,20,25,...">
    <!--req, int, frame rate, attr:opt{req, string}-->1
  </frameRate>
  <bitRateType opt="CBR,VBR">
    <!--req, string, bitrate type, attr:opt{req, string}, desc:including CBR (Constant Bit Rate) and VBR (Variable Bit Rate).-->test
  </bitRateType>
  <bitRate opt="8,16,32,...">
    <!--opt, int, bitrate, unit:kbps 码率, 包括8、16、32等, 单位: kbps, 定码率时有效, attr:opt{req, string}-->1
  </bitRate>
  <maxBitRate opt="128,256,...">
    <!--opt, int, max. bitrate, unit:kbps 最大码率, 128、256等, 单位: kbps, 变码率时有效, attr:opt{req, string}-->1
  </maxBitRate>
  <intervalFrameI>
    <!--req, int, I frame interval, unit:ms-->1
  </intervalFrameI>
</WallNPreMonitor>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.2.4 Get sub window configuration capability

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/param/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubWindowParamCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sub window configuration capability, attr:version{req, string, protocolVersion}-->
  <enabledAudio opt="true,fasle">
    <!--ro, opt, bool, whether to turn on the audio of sub window, attr:opt{req, string}-->true
  </enabledAudio>
  <rotateAngle min="0" max="270">
    <!--ro, opt, int, rotate angle, attr:min{req, int},max{req, int}-->1
  </rotateAngle>
  <borderEnabled opt="true,fasle">
    <!--ro, opt, bool, whether to enable border light, attr:opt{req, string}, desc:true=yes (the boarder light is always on), false=no (the boarder Light is always off)-->true
  </borderEnabled>
  <borderWidth min="0" max="100">
    <!--ro, opt, int, border width, attr:min{req, int},max{req, int}-->1
  </borderWidth>
  <borderColor opt="red,orange,yellow,green,blue,purple,black">
    <!--ro, opt, string, border color, attr:opt{req, string}, desc:red by default-->red
  </borderColor>
  <flashEnabled opt="true,fasle">
    <!--ro, opt, bool, whether to enable flash of the border light, attr:opt{req, string}, desc:true=yes, false=no (the light will always be bright); this node is valid only when the value of borderEnabled is "true"-->true
  </flashEnabled>
  <flashDurationTime min="0" max="100">
    <!--ro, opt, int, border flash time duration, attr:min{req, int},max{req, int}-->1
  </flashDurationTime>
  <flashOnTime min="0" max="100">
    <!--ro, opt, int, duration time of the light keeping bright in one flash, unit: s, attr:min{req, int},max{req, int}-->1
  </flashOnTime>
  <flashOffTime min="0" max="100">
    <!--ro, opt, int, duration time of the light keeping dark in one flash, unit: s, attr:min{req, int},max{req, int}-->1
  </flashOffTime>
  <rolateOSD opt="true,fasle">
    <!--ro, opt, bool, whether to relate to OSD overLay, attr:opt{req, string}-->true
  </rolateOSD>
</SubWindowParamCap>
```

9.7.2.5 Start dynamic decoding

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/start

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
    quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode
3	Device Error	localsourcestreamsnotreceivedunicastdata	0x30008025
3	Device Error	localsourcestreamsnotreceivedunicastdata	0x30008026
4	Invalid Operation	thePacketOrderIsNotContinuousInStream	0x4000A3FA
4	Invalid Operation	streamingIsNotContinuous	0x4000A3FB
4	Invalid Operation	streamDataIsAccumulated	0x4000A3FC
4	Invalid Operation	encodingFormatIsNotSupported	0x4000A3FD
4	Invalid Operation	streamEncapsulationFormatCannotBeRecognized	0x4000A3FE
4	Invalid Operation	encryptionKeysNotConfiguredForTheEncryptedStream	0x4000A3FF
4	Invalid Operation	unstableDecodingFrameRate	0x4000A400
4	Invalid Operation	decodingFirmwareException	0x4000A401
4	Invalid Operation	inputChipException	0x4000A402
4	Invalid Operation	frontInputSignal5vIsNotRecognized	0x4000A403
4	Invalid Operation	unstableInputSignal	0x4000A404
4	Invalid Operation	inputResolutionIsNotSupported	0x4000A405

4	Invalid Operation	inputInterfaceProtocolMismatch	0x4000A406
4	Invalid Operation	inputColorSpaceMismatch	0x4000A407
4	Invalid Operation	samplingOfTheTwoMergedInputsExistsDeviation	0x4000A408
4	Invalid Operation	inputProtocolInteractionException	0x4000A409
4	Invalid Operation	abnormalChipOfOutput	0x4000A40A
4	Invalid Operation	theHotPlugFrequencyBetweenHdmiOutputAndPostScreenIsTooHigh	0x4000A40B
4	Invalid Operation	outputModeMismatch	0x4000A40C
4	Invalid Operation	unstableOutputSignal	0x4000A40D
4	Invalid Operation	outputProtocolInteractionException	0x4000A40E
4	Invalid Operation	thePacketOrderIsNotContinuousInStreamForwarding	0x4000A40F
4	Invalid Operation	streamingIsNotContinuousInForwarding	0x4000A410
4	Invalid Operation	streamDataInForwardingIsAccumulated	0x4000A411
4	Invalid Operation	encodingFormatOfStreamForwardingIsNotSupported	0x4000A412
4	Invalid Operation	streamEncapsulationFormatOfStreamForwardingCannotBeRecognized	0x4000A413
4	Invalid Operation	encryptionKeyOfStreamForwardingIsNotConfiguredForTheEncryptedStream	0x4000A414
4	Invalid Operation	streamForwardingTimeout	0x4000A415
4	Invalid Operation	unstableFrameRateOfStreamForwarding	0x4000A416
4	Invalid Operation	theNumberOfEncryptedStreamsExceedsTheLimit	0x4000A44A
4	Invalid Operation	thereShouldBeTwoOrMoreBoxesForDecoding	0x4000A44B

9.7.2.6 Get decoding status of all sub windows of a specific window

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/status

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubWinStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sub window decoding status, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, window ID-->1
  </id>
  <isLinked>
    <!--ro, opt, bool, connection status-->true
  </isLinked>
  <isDecoding>
    <!--ro, opt, bool, decoding status-->true
  </isDecoding>
  <isDecodingEnabled>
    <!--ro, opt, bool, decoding switch status-->true
  </isDecodingEnabled>
  <imageWidth>
    <!--ro, opt, int, image width-->1
  </imageWidth>
  <imageHeight>
    <!--ro, opt, int, image height-->1
  </imageHeight>
  <videoFPS>
    <!--ro, opt, int, video frame rate-->1
  </videoFPS>
  <audioFPS>
    <!--ro, opt, int, audio frame rate-->1
  </audioFPS>
  <videoTotal>
    <!--ro, opt, int, total decoded video frames-->1
  </videoTotal>
  <decodedAudioFrames>
    <!--ro, opt, int, total decoded audio frames-->1
  </decodedAudioFrames>
  <streamRate>
    <!--ro, opt, int, stream transmission rate-->1
  </streamRate>
  <videoType>
    <!--ro, opt, string, encoding type-->test
  </videoType>
  <packetType>
    <!--ro, opt, string, encapsulation format-->test
  </packetType>
  <wndDecodeType>
    <!--ro, opt, enum, window decoding type, subType:string, desc:"dynamic" (dynamic decoding), "cycle" (auto-switch decoding)-->dynamic
  </wndDecodeType>
  <SubWindowParam>
    <!--ro, opt, object, sub window signal parameter-->
    <signalMode>
      <!--ro, opt, enum, signal mode, subType:string, desc:"videoInput" (video input), "streamId" (stream ID), "streamSetting" (stream settings),
      "alarmLinkage" (alarm linkage)-->video input
    </signalMode>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <inputChannelID>
      <!--ro, opt, string, input channel ID, dep:and,{$.SubWindow.SubWindowParam.signalMode,eq,video input}-->test
    </inputChannelID>
    <streamingChannelID>
      <!--ro, opt, string, stream channel ID-->test
    </streamingChannelID>
    <StreamInput>
      <!--ro, opt, object, network stream information-->
      <streamInputMode>
        <!--ro, req, enum, stream type, subType:string, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
      </streamInputMode>
      <StreamInputRealtime>
        <!--ro, opt, object, real-time stream information-->
        <durationInUnit>
          <!--ro, opt, int, decoding display interval of real-time stream, unit:s-->1
        </durationInUnit>
        <StreamRealtimeUnitList>
          <!--ro, opt, array, real-time stream list, subType:object-->
          <StreamRealtimeUnit>
```

```

<!--ro, opt, object, single real-time stream unit-->
<streamType>
<!--ro, req, enum, stream type, subType:string, desc:"in URL", "by ddns", "by domain"-->in URL
</streamType>
<StreamInURL>
<!--ro, opt, object, URL stream, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
<URL>
<!--ro, req, string, URL address-->test
</URL>
</StreamInURL>
<StreamByDdns>
<!--ro, opt, object, DDNS stream, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
<DdnsServerInfo>
<!--ro, opt, object, DDNS server information-->
<domain>
<!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
</domain>
<port>
<!--ro, req, int, port number-->80
</port>
<ddnsType>
<!--ro, req, enum, DDNS type, subType:string, desc:"hiDdns", "noIp"-->hiDdns
</ddnsType>
<username>
<!--ro, req, string, user name-->test
</username>
<password>
<!--ro, req, string, password-->test
</password>
</DdnsServerInfo>
<EncodeDevInfo>
<!--ro, opt, object, information of device in playback-->
<domain>
<!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
</domain>
<port>
<!--ro, req, int, port number-->1
</port>
<transmitProtocol>
<!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
>tcp

</transmitProtocol>
<protocol>
<!--ro, req, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
</protocol>
<username>
<!--ro, req, string, user name-->test
</username>
<password>
<!--ro, req, string, password-->test
</password>
<channelMode>
<!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType>
<!--ro, req, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
</channelType>
<channelZero>
<!--ro, opt, int, channel-zero-->1
</channelZero>
<channelNormal>
<!--ro, opt, int, normal channel-->1
</channelNormal>
<channelStreaming>
<!--ro, opt, int, stream channel-->1
</channelStreaming>
<channelDistributed>
<!--ro, opt, int, distributed channel-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
<!--ro, opt, object, media gateway information-->
<enabled>
<!--ro, req, enum, whether to enable the function, subType:string, desc:true (enable); false (disable);-->>true
</enabled>
<domain>
<!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
</domain>
<port>
<!--ro, req, int, port number-->1
</port>
<transmitProtocol>
<!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
>tcp

</transmitProtocol>
</MediaGatewayInfo>
<StreamByDdns>
<StreamByDomain>
<!--ro, opt, object, domain stream-->
<EncodeDevInfo>
<!--ro, opt, object, information of device in playback-->
<domain>

```

```

<!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
</domain>
<port>
  <!--ro, req, int, port number-->1
</port>
<transmitProtocol>
  <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "multicast" (multicast protocol)--
>tcp

</transmitProtocol>
<protocol>
  <!--ro, req, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
</protocol>
<username>
  <!--ro, req, string, user name-->test
</username>
<password>
  <!--ro, req, string, password-->test
</password>
<channelMode>
  <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType>
  <!--ro, req, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
</channelType>
<channelZero>
  <!--ro, opt, int, channel-zero-->1
</channelZero>
<channelNormal>
  <!--ro, opt, int, normal channel-->1
</channelNormal>
<channelStreaming>
  <!--ro, opt, int, stream channel-->1
</channelStreaming>
<channelDistributed>
  <!--ro, opt, int, distributed channel-->1
</channelDistributed>
<EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object, media gateway information-->
  <enabled>
    <!--ro, req, enum, whether to enable the function, subType:string, desc:true (enable); false (disable)-->true
  </enabled>
  <domain>
    <!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
  </domain>
  <port>
    <!--ro, req, int, port number-->1
  </port>
  <transmitProtocol>
    <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "multicast" (multicast protocol)--
>tcp

  </transmitProtocol>
  </MediaGatewayInfo>
  </StreamByDomain>
  </StreamRealtimeUnit>
  </StreamRealtimeUnitList>
  </StreamInputRealtime>
  <StreamInputPlayback>
    <!--ro, opt, object, playback stream information-->
    <playbackMode>
      <!--ro, opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file name
    </playbackMode>
    <EncodeDevInfo>
      <!--ro, opt, object, information of device in playback-->
      <domain>
        <!--ro, req, enum, domain name, subType:string, desc:domain name-->ipv4
      </domain>
      <port>
        <!--ro, req, int, port number-->1
      </port>
      <transmitProtocol>
        <!--ro, req, enum, transport protocol, subType:string, desc:transport protocol-->tcp
      </transmitProtocol>
      <protocol>
        <!--ro, req, enum, protocol, subType:string, desc:protocol-->DAHUA
      </protocol>
      <username>
        <!--ro, req, string, user name-->test
      </username>
      <password>
        <!--ro, req, string, password-->test
      </password>
      <channelMode>
        <!--ro, req, enum, channel mode, subType:string, desc:channel mode-->normal
      </channelMode>
      <channelType>
        <!--ro, req, enum, channel type, subType:string, desc:channel type-->main
      </channelType>
      <channelZero>
        <!--ro, opt, int, channel-zero-->1
      </channelZero>
      <channelNormal>
        <!--ro, opt, int, normal channel-->1
      </channelNormal>

```



```

</channelNormal>
<channelStreaming>
  <!--ro, opt, int, stream channel-->1
</channelStreaming>
<channelDistributed>
  <!--ro, opt, int, distributed channel-->1
</channelDistributed>
</EncodeDevInfo>
<fileName>
  <!--ro, opt, string, file name-->test
</fileName>
<TimeRange>
  <!--ro, opt, object, time range-->
  <beginTime>
    <!--ro, opt, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--ro, opt, time, end time-->00:00:00+08:00
  </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--ro, opt, enum, true (enable); false (disable);, subType:string, desc:true (enable); false (disable);-->>true
</streamEncryptEnable>
<streamPassword>
  <!--ro, opt, string, stream password, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}, desc:sensitive information encryption. It is valid when the value of streamEncryptEnable is true-->test
</streamPassword>
</StreamInput>
<subWindowMode>
  <!--ro, opt, enum, sub-window mode, subType:string, desc:"normal" (normal window), "polling" (auto-switching window)-->normal
</subWindowMode>
</SubWindowParam>
<isAlarmDecoding>
  <!--ro, opt, bool, alarm status, desc:when the device's alarm output is enabled, the client will keep getting the device's alarm status and check whether an alarm occurred via this field-->true
</isAlarmDecoding>
<isPollPending>
  <!--ro, opt, bool, auto-switch suspending status, desc:when the current window is in auto-switch mode, the web client needs to test the auto-switch suspending status-->true
</isPollPending>
</SubWinStatus>

```

9.7.2.7 Stop dynamic decoding

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/stop

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, describe the error reason in detail, desc:describe the error reason in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.2.8 Get decoding status of all sub-windows of all windows

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/status

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallWindowStatusList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, decoding status List, attr:version{req, string, protocolVersion}-->
  <WallWindowStatus>
    <!--ro, opt, object, decoding status-->
    <id>
      <!--ro, req, string, window ID-->test
    </id>
    <windowMode>
      <!--ro, req, int, 1-->0
    </windowMode>
    <SubWinStatusList>
      <!--ro, opt, object, sub window status List-->
      <SubWinStatus>
        <!--ro, opt, object, sub window decoding status-->
        <id>
          <!--ro, req, int, sub-window number-->1
        </id>
        <isLinked>
          <!--ro, opt, bool, connection status-->true
        </isLinked>
        <isDecoding>
          <!--ro, opt, bool, decoding status-->true
        </isDecoding>
        <isDecodingEnabled>
          <!--ro, opt, bool, decoding switch status-->true
        </isDecodingEnabled>
        <imageWidth>
          <!--ro, opt, int, image width-->1
        </imageWidth>
        <imageHeight>
          <!--ro, opt, int, image height-->1
        </imageHeight>
        <videoFPS>
          <!--ro, opt, int, video frame rate-->1
        </videoFPS>
      </SubWinStatus>
    </SubWinStatusList>
  </WallWindowStatus>
</WallWindowStatusList>
```

```

<audioFPS>
  <!--ro, opt, int, audio frame rate-->1
</audioFPS>
<videoTotal>
  <!--ro, opt, int, total decoded video frames-->1
</videoTotal>
<decodedAudioFrames>
  <!--ro, opt, int, total decoded audio frames-->1
</decodedAudioFrames>
<streamRate>
  <!--ro, opt, int, stream transmission rate-->1
</streamRate>
<videoType>
  <!--ro, opt, string, encoding type-->test
</videoType>
<packetType>
  <!--ro, opt, string, encapsulation format-->test
</packetType>
<wndDecodeType>
  <!--ro, opt, enum, window decoding type, subType:string, desc:"dynamic" (dynamic decoding), "cycle" (auto-switch decoding)-->dynamic
</wndDecodeType>
<SubWindowParam>
  <!--ro, opt, object, sub-window signal parameter-->
  <signalMode>
    <!--ro, opt, enum, signal mode, subType:string, desc:"videoInput" (video input), "streamId" (stream ID), "streamSetting" (stream settings),
    "alarmLinkage" (alarm linkage)-->video input
  </signalMode>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID-->1
  </videoInputChannelID>
  <inputChannelID>
    <!--ro, opt, string, input channel ID, dep:and,{$.SubWindow.SubWindowParam.signalMode,eq,video input}-->test
  </inputChannelID>
  <streamingChannelID>
    <!--ro, opt, string, stream channel ID-->test
  </streamingChannelID>
  <StreamInput>
    <!--ro, opt, object, network stream information-->
    <streamInputMode>
      <!--ro, req, enum, stream type, subType:string, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
    </streamInputMode>
    <StreamInputRealtime>
      <!--ro, opt, object, real-time stream information-->
      <durationInUnit>
        <!--ro, opt, int, decoding display interval of real-time stream, unit:s-->1
      </durationInUnit>
      <StreamRealtimeUnitList>
        <!--ro, opt, array, real-time stream list, subType:object-->
        <StreamRealtimeUnit>
          <!--ro, opt, object, single real-time stream unit-->
          <streamType>
            <!--ro, req, enum, stream type, subType:string, desc:"in URL", "by ddns", "by domain"-->in URL
          </streamType>
          <StreamInURL>
            <!--ro, opt, object, URL stream, dep:and,
            {$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
            <URL>
              <!--ro, req, string, URL address-->test
            </URL>
          </StreamInURL>
          <StreamByDdns>
            <!--ro, opt, object, DDNS stream, dep:and,
            {$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
            <DdnsServerInfo>
              <!--ro, opt, object, DDNS server information-->
              <domain>
                <!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
              </domain>
              <port>
                <!--ro, req, int, port No.-->80
              </port>
              <ddnsType>
                <!--ro, req, enum, DDNS type, subType:string, desc:"hiDdns", "noIp"-->hiDdns
              </ddnsType>
              <username>
                <!--ro, req, string, user name.-->test
              </username>
              <password>
                <!--ro, req, string, password-->test
              </password>
            </DdnsServerInfo>
            <EncodeDevInfo>
              <!--ro, opt, object, information of device in playback-->
              <domain>
                <!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
              </domain>
              <port>
                <!--ro, req, int, port No.-->1
              </port>
              <transmitProtocol>
                <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast
                protocol)-->tcp
              </transmitProtocol>
            </EncodeDevInfo>
          </StreamByDdns>
        </StreamRealtimeUnit>
      </StreamRealtimeUnitList>
    </StreamInput>
  </streamingChannelID>
  <inputChannelID>
    <!--ro, opt, string, input channel ID, dep:and,{$.SubWindow.SubWindowParam.signalMode,eq,video input}-->test
  </inputChannelID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID-->1
  </videoInputChannelID>
  <signalMode>
    <!--ro, opt, enum, signal mode, subType:string, desc:"videoInput" (video input), "streamId" (stream ID), "streamSetting" (stream settings),
    "alarmLinkage" (alarm linkage)-->video input
  </signalMode>
  <wndDecodeType>
    <!--ro, opt, enum, window decoding type, subType:string, desc:"dynamic" (dynamic decoding), "cycle" (auto-switch decoding)-->dynamic
  </wndDecodeType>
  <packetType>
    <!--ro, opt, string, encapsulation format-->test
  </packetType>
  <videoType>
    <!--ro, opt, string, encoding type-->test
  </videoType>
  <streamRate>
    <!--ro, opt, int, stream transmission rate-->1
  </streamRate>
  <decodedAudioFrames>
    <!--ro, opt, int, total decoded audio frames-->1
  </decodedAudioFrames>
  <videoTotal>
    <!--ro, opt, int, total decoded video frames-->1
  </videoTotal>
  <audioFPS>
    <!--ro, opt, int, audio frame rate-->1
  </audioFPS>
</SubWindowParam>

```

```

    <!--ro, req, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
  </protocol>
  <username>
    <!--ro, req, string, user name.-->test
  </username>
  <password>
    <!--ro, req, string, password-->test
  </password>
  <channelMode>
    <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
  </channelMode>
  <channelType>
    <!--ro, req, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
  </channelType>
  <channelZero>
    <!--ro, opt, int, channel-zero-->1
  </channelZero>
  <channelNormal>
    <!--ro, opt, int, normal channel-->1
  </channelNormal>
  <channelStreaming>
    <!--ro, opt, int, stream channel-->1
  </channelStreaming>
  <channelDistributed>
    <!--ro, opt, int, distributed channel-->1
  </channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object, media gateway information-->
  <enabled>
    <!--ro, req, enum, whether to enable, subType:string, desc:true (enable), false (disable)-->true
  </enabled>
  <domain>
    <!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
  </domain>
  <port>
    <!--ro, req, int, port No.-->1
  </port>
  <transmitProtocol>
    <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast
protocol)-->tcp
  </transmitProtocol>
</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--ro, opt, object, domain stream-->
  <EncodeDevInfo>
    <!--ro, opt, object, information of device in playback-->
    <domain>
      <!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->1
    </port>
    <transmitProtocol>
      <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast
protocol)-->tcp
    </transmitProtocol>
  <protocol>
    <!--ro, req, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
  </protocol>
  <username>
    <!--ro, req, string, user name.-->test
  </username>
  <password>
    <!--ro, req, string, password-->test
  </password>
  <channelMode>
    <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
  </channelMode>
  <channelType>
    <!--ro, req, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
  </channelType>
  <channelZero>
    <!--ro, opt, int, channel-zero-->1
  </channelZero>
  <channelNormal>
    <!--ro, opt, int, normal channel-->1
  </channelNormal>
  <channelStreaming>
    <!--ro, opt, int, stream channel-->1
  </channelStreaming>
  <channelDistributed>
    <!--ro, opt, int, distributed channel-->1
  </channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object, media gateway information-->
  <enabled>
    <!--ro, req, enum, whether to enable, subType:string, desc:true (enable), false (disable)-->true
  </enabled>
  <domain>
    <!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4

```

```

        </domain>
        <port>
            <!--ro, req, int, port No.-->1
        </port>
        <transmitProtocol>
            <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast
protocol)-->tcp
        </transmitProtocol>
        </MediaGatewayInfo>
        </StreamByDomain>
        </StreamRealtimeUnit>
        </StreamRealtimeUnitList>
        </StreamInputRealtime>
        <StreamInputPlayback>
            <!--ro, opt, object, playback stream information-->
            <playbackMode>
                <!--ro, opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file
name
            </playbackMode>
            <EncodeDevInfo>
                <!--ro, opt, object, information of device in playback-->
                <domain>
                    <!--ro, req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
                </domain>
                <port>
                    <!--ro, req, int, port No.-->1
                </port>
                <transmitProtocol>
                    <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
>tcp
                </transmitProtocol>
                <protocol>
                    <!--ro, req, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
                </protocol>
                <username>
                    <!--ro, req, string, user name.-->test
                </username>
                <password>
                    <!--ro, req, string, password-->test
                </password>
                <channelMode>
                    <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
                </channelMode>
                <channelType>
                    <!--ro, req, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
                </channelType>
                <channelZero>
                    <!--ro, opt, int, channel-zero-->1
                </channelZero>
                <channelNormal>
                    <!--ro, opt, int, normal channel-->1
                </channelNormal>
                <channelStreaming>
                    <!--ro, opt, int, stream channel-->1
                </channelStreaming>
                <channelDistributed>
                    <!--ro, opt, int, distributed channel-->1
                </channelDistributed>
            </EncodeDevInfo>
            <fileName>
                <!--ro, opt, string, file name-->test
            </fileName>
            <TimeRange>
                <!--ro, opt, object, time range-->
                <beginTime>
                    <!--ro, opt, time, start time-->00:00:00+08:00
                </beginTime>
                <endTime>
                    <!--ro, opt, time, end time-->00:00:00+08:00
                </endTime>
            </TimeRange>
            </StreamInputPlayback>
            <streamEncryptEnable>
                <!--ro, opt, enum, whether to enable stream encryption, subType:string, desc:true (enable), false (disable)-->true
            </streamEncryptEnable>
            <streamPassword>
                <!--ro, opt, string, stream password, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}, desc:sensitive information
encryption. It is valid when the value of streamEncryptEnable is true-->test
            </streamPassword>
            </StreamInput>
            <subWindowMode>
                <!--ro, opt, enum, sub-window mode, subType:string, desc:"normal" (normal window), "polling" (auto-switching window)-->normal
            </subWindowMode>
            </SubWindowParam>
            <isAlarmDecoding>
                <!--ro, opt, bool, alarm status, desc:when the device's alarm output is enabled, the client will keep getting the device's alarm status and check
whether an alarm occurred via this field-->true
            </isAlarmDecoding>
            <isPollPending>
                <!--ro, opt, bool, auto-switch suspending status, desc:when the current window is in auto-switch mode, the web client needs to test the auto-
switch suspending status-->true
            </isPollPending>
            </SubWinStatus>
            </SubWinStatusList>

```

```
</WallWindowStatus>
</WallWindowStatusList>
```

9.7.2.9 Get sub-board stream exporting configurations

Request URL

GET /ISAPI/DisplayDev/VideoWall/DecodeMgr/BoardStreamExportCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "enabled": "true"
  /*opt, boolean, whether to enable exporting sub-board stream*/
}
```

9.7.2.10 Set sub-board stream exporting configurations

Request URL

PUT /ISAPI/DisplayDev/VideoWall/DecodeMgr/BoardStreamExportCfg?format=json

Query Parameter

None

Request Message

```
{
  "enabled": "true"
  /*opt, boolean, whether to enable exporting sub-board stream*/
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "OK",
  /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status code is not "ok" (success)*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok",
  /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
  "MErrCode": "0x00000000",
  /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
  "MErrDevSelfEx": "0x00000000",
  /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

9.7.2.11 Get capability of default decoding delay parameters

Request URL

GET /ISAPI/DisplayDev/VideoWall/DecodeMgr/DefaultDecodeDelayParams/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "DefaultDecodeDelayParamsCap": {
    /*opt, object, capability of default decoding delay parameters, desc:corresponding API: GET or PUT
    /ISAPI/DisplayDev/VideoWall/DecodeMgr/DefaultDecodeDelayParams?format=json*/
    "defaultDecodeDelayParam": {
      /*req, object, value range of default decoding delay parameter*/
      "@opt": [],
      /*opt, array, enumeration of default decoding delay parameter, subType:string, range:[1,5]*/
      "@def": "mostRealTime"
      /*opt, string, default value*/
    }
  }
}

```

9.7.2.12 Get default decoding delay parameters

Request URL

GET /ISAPI/DisplayDev/VideoWall/DecodeMgr/DefaultDecodeDelayParams?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "defaultDecodeDelayParam": "mostRealTime"
  /*req, enum, default decoding delay parameters, subType:string, desc:"mostRealTime" (most real-time), "moreRealTime" (relatively real-time), "balance"
  (balancing), "moreFluency" (relatively fluent), "mostFluency" (most fluent)*/
}

```

9.7.2.13 Set default decoding delay parameters

Request URL

PUT /ISAPI/DisplayDev/VideoWall/DecodeMgr/DefaultDecodeDelayParams?format=json

Query Parameter

None

Request Message

```

{
  "defaultDecodeDelayParam": "mostRealTime"
  /*req, enum, default decoding delay parameters, subType:string, desc:"mostRealTime" (most real-time), "moreRealTime" (relatively real-time), "balance"
  (balancing), "moreFluency" (relatively fluent), "mostFluency" (most fluent)*/
}

```

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "OK",
  /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status code is not "ok" (success)*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok",
  /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
  "MErrCode": "0x00000000",
  /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
  "MErrDevSelfEx": "0x00000000"
  /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}

```

9.7.2.14 Get network pre-monitoring parameters of all video walls

Request URL

GET /ISAPI/DisplayDev/VideoWall/nPreMonitor

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallNPreMonitorList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, network pre-monitoring parameter list of all video walls, attr:version{req, string, protocolVersion}-->
  <WallNPreMonitor version="2.0">
    <!--ro, opt, object, network pre-monitoring parameters of a video wall, attr:version{req, string}-->
    <id>
      <!--ro, req, int, video wall No.-->1
    </id>
    <resolution>
      <!--ro, req, string, pre-monitoring signal output resolution, desc:it is obtained from capability set-->test
    </resolution>
    <frameRate>
      <!--ro, req, int, frame rate-->1
    </frameRate>
    <bitRateType>
      <!--ro, req, string, bitrate type, desc:including CBR (Constant Bit Rate) and VBR (Variable Bit Rate).-->test
    </bitRateType>
    <bitRate>
      <!--ro, opt, int, bitrate, unit:kpbs 码率, 包括8、16、32等, 单位: kpbs, 定码率时有效-->1
    </bitRate>
    <maxBitRate>
      <!--ro, opt, int, max. bitrate, unit:kpbs 最大码率, 128、256等, 单位: kpbs, 变码率时有效-->1
    </maxBitRate>
    <intervalFrameI>
      <!--ro, req, int, I frame interval, unit:ms-->1
    </intervalFrameI>
  </WallNPreMonitor>
</WallNPreMonitorList>
```

9.7.2.15 Set network pre-monitoring parameters of all video walls

Request URL

PUT /ISAPI/DisplayDev/VideoWall/nPreMonitor

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallNPreMonitorList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, network pre-monitoring parameter list of all video walls, attr:version{req, string, protocolVersion}-->
  <WallNPreMonitor version="2.0">
    <!--opt, object, network pre-monitoring parameters of a video wall, attr:version{req, string}-->
    <id>
      <!--req, int, video wall No.-->1
    </id>
    <resolution>
      <!--req, string, pre-monitoring signal output resolution, desc:it is obtained from capability set-->test
    </resolution>
    <frameRate>
      <!--req, int, frame rate-->1
    </frameRate>
    <bitRateType>
      <!--req, string, bitrate type, desc:including CBR (Constant Bit Rate) and VBR (Variable Bit Rate).-->test
    </bitRateType>
    <bitRate>
      <!--opt, int, bitrate, unit:kpbs 码率, 包括8、16、32等, 单位: kpbs, 定码率时有效-->1
    </bitRate>
    <maxBitRate>
      <!--opt, int, max. bitrate, unit:kpbs 最大码率, 128、256等, 单位: kpbs, 变码率时有效-->1
    </maxBitRate>
    <intervalFrameI>
      <!--req, int, I frame interval, unit:ms-->1
    </intervalFrameI>
  </WallNPreMonitor>
</WallNPreMonitorList>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
    quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.2.16 Get capability of network pre-monitoring parameters of video wall

Request URL

GET /ISAPI/DisplayDev/VideoWall/nPreMonitor/capabilities.

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallNPreMonitorListCap xmlns="http://www.isapi.org/ver20/XMLSchema" size="10" version="2.0">
  <!--ro, opt, object, network pre-monitoring parameter List of all video walls, attr:size{req, int, protocolVersion},version{req, string}-->
  <WallNPreMonitor>
    <!--ro, req, object, network pre-monitoring parameters of a video wall-->
    <id min="1" max="10">
      <!--ro, req, int, window ID, attr:min{req, int},max{req, int}-->1
    </id>
    <resolution opt="1080*720P,1920*1080P,...">
      <!--ro, req, string, resolution, attr:opt{req, string}, desc:1080*720P, 1920*1080P-->1080*720P
    </resolution>
    <frameRate opt="15,20,25,...">
      <!--ro, req, int, frame rate, attr:opt{req, string}, desc:15, 20, 25, ...-->15
    </frameRate>
    <bitRateType opt="CBR,VBR">
      <!--ro, req, string, bit rate type, attr:opt{req, string}, desc:CBR, VBR-->CBR
    </bitRateType>
    <bitRate opt="8,16,32,...">
      <!--ro, opt, int, bit rate, unit:kpbs, attr:opt{req, string}, desc:unit: kpbs-->8
    </bitRate>
    <maxBitRate opt="128,256,...">
      <!--ro, opt, int, maximum bit rate, unit:kpbs, attr:opt{req, string}-->1
    </maxBitRate>
    <intervalFrameI min="1" max="10">
      <!--ro, req, int, interval, unit:millisecond, attr:min{req, int},max{req, int}-->1
    </intervalFrameI>
    <maxScreenRow>
      <!--ro, opt, int, the row and column identification is mainly related to stream scaling-->1
    </maxScreenRow>
    <maxScreenColumn>
      <!--ro, opt, int, the row and column identification is mainly related to stream scaling-->1
    </maxScreenColumn>
    <maxScreenCount>
      <!--ro, opt, int, maximum number of screens that support video wall image displaying-->1
    </maxScreenCount>
  </WallNPreMonitor>
</WallNPreMonitorListCap>
```

9.7.3 Output Channel Management

9.7.3.1 Get the audio output channels' parameters

Request URL

GET /ISAPI/DisplayDev/Audio/outputs/channels

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<AudioOutputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <AudioOutputChannel>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <channelName>
      <!--ro, opt, string-->test
    </channelName>
    <sourceType>
      <!--ro, req, enum, subType:string-->audioInput
    </sourceType>
    <AudioWithInput>
      <!--ro, opt, object-->
      <AudioInputBindID>
        <!--ro, opt, int-->0
      </AudioInputBindID>
    </AudioWithInput>
    <AudioWithSubWin>
      <!--ro, opt, object-->
      <wallID>
        <!--ro, opt, int-->0
      </wallID>
      <windowID>
        <!--ro, opt, int-->0
      </windowID>
      <subWinID>
        <!--ro, opt, int-->0
      </subWinID>
    </AudioWithSubWin>
  </AudioOutputChannel>
</AudioOutputChannelList>

```

9.7.3.2 Set parameters of all audio output channels

Request URL

PUT /ISAPI/DisplayDev/Audio/outputs/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioOutputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <AudioOutputChannel>
    <!--opt, object, the audio output channels' parameters-->
    <id>
      <!--req, int-->0
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <channelName>
      <!--opt, string-->test
    </channelName>
    <sourceType>
      <!--req, enum, subType:string-->audioInput
    </sourceType>
    <AudioWithInput>
      <!--opt, object-->
      <AudioInputBindID>
        <!--opt, int-->0
      </AudioInputBindID>
    </AudioWithInput>
    <AudioWithSubWin>
      <!--opt, object-->
      <wallID>
        <!--opt, int-->0
      </wallID>
      <windowID>
        <!--opt, int-->0
      </windowID>
      <subWinID>
        <!--opt, int-->0
      </subWinID>
    </AudioWithSubWin>
  </AudioOutputChannel>
</AudioOutputChannelList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	audioCannotBeEnabledInTheWindow	0x4000A3C4	--

9.7.3.3 Set parameters of all video outputs

Request URL

PUT /ISAPI/DisplayDev/Video/outputs/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoOutputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video output channel List, attr:version{req, string, protocolVersion}-->
  <VideoOutputChannel>
    <!--opt, object, video output channel-->
```

```

<!--opt, object, video output channel-->
<id>
  <!--req, string, channel ID-->test
</id>
<portType>
  <!--req, enum, port type, subType:string, desc:"VGA", "CVBS", "HDMI", "Spot", "SDI", "DVI", "TVI", "LEDSendCard"-->VGA
</portType>
<OutputResolution>
  <!--opt, object, decoding output resolution-->
  <resolution>
    <!--opt, enum, resolution, subType:string,
desc:3840*2160@30HZ;3840*2160@30HZ;1920*1080@60HZ;1920*1080@60HZ;1920*1080@50HZ;1280*720@50HZ;1280*720@50HZ;1280*720@60HZ;1280*720@60HZ;1024*
768@60HZ;1024*768@60HZ;-->3840*2160@30HZ
  </resolution>
  <imageWidth>
    <!--opt, int, resolution (width)-->0
  </imageWidth>
  <imageHeight>
    <!--opt, int, resolution (height)-->0
  </imageHeight>
</OutputResolution>
<PortInBoard>
  <!--opt, object-->
  <boardID>
    <!--opt, int-->0
  </boardID>
  <portID>
    <!--opt, int-->0
  </portID>
  <boardType>
    <!--opt, enum, subType:string-->netra
  </boardType>
</PortInBoard>
<deviceID>
  <!--opt, int-->0
</deviceID>
<useEDIDResolution>
  <!--opt, bool-->true
</useEDIDResolution>
<outputMode>
  <!--opt, enum, subType:string-->HDMI
</outputMode>
<outputType>
  <!--opt, enum, subType:string-->main
</outputType>
<timeSequenceMode>
  <!--opt, enum, subType:string-->standard
</timeSequenceMode>
<CustomMode>
  <!--opt, object, custom mode-->
  <timeSequenceWidth>
    <!--opt, int-->0
  </timeSequenceWidth>
  <timeSequenceHeight>
    <!--opt, int-->0
  </timeSequenceHeight>
  <timeSequenceFrequency>
    <!--opt, int-->0
  </timeSequenceFrequency>
</CustomMode>
<AdvanceMode>
  <!--opt, object-->
  <timeSequenceWidth>
    <!--opt, int-->0
  </timeSequenceWidth>
  <timeSequenceHeight>
    <!--opt, int-->0
  </timeSequenceHeight>
  <timeSequenceFrequency>
    <!--opt, int-->0
  </timeSequenceFrequency>
  <horizontalFront>
    <!--opt, int-->1
  </horizontalFront>
  <horizontalBack>
    <!--opt, int-->1
  </horizontalBack>
  <horizontalSyncSignal>
    <!--opt, int-->1
  </horizontalSyncSignal>
  <verticalFront>
    <!--opt, int-->1
  </verticalFront>
  <verticalBack>
    <!--opt, int-->1
  </verticalBack>
  <verticalSyncSignal>
    <!--opt, int-->1
  </verticalSyncSignal>
</AdvanceMode>
<name>
  <!--opt, string, decoding output name-->test
</name>
<outputPortEnabled>

```

```

    <!--opt, bool-->true
  </outputPortEnabled>
  <LEDSendCardResolutionEnabled>
    <!--opt, bool-->true
  </LEDSendCardResolutionEnabled>
  <outputPortAccessType>
    <!--opt, enum, subType:string-->display
  </outputPortAccessType>
  <outputBackgroundType>
    <!--opt, enum, subType:string-->default
  </outputBackgroundType>
  <outputBackgroundRGBColor>
    <!--opt, int, dep:and,{$.VideoOutputChannelList.VideoOutputChannel.outputBackgroundType,eq,solidColor}-->1
  </outputBackgroundRGBColor>
  <HDMIEnabled>
    <!--opt, bool-->true
  </HDMIEnabled>
</VideoOutputChannel>
</VideoOutputChannelList>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details; desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

9.7.3.4 Get basic parameters of all video outputs

Request URL

GET /ISAPI/DisplayDev/Video/outputs/channels

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoOutputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, float, protocolVersion}-->
  <VideoOutputChannel>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <portType>
      <!--ro, req, enum, subType:string-->VGA
    </portType>
    <OutputResolution>
      <!--ro, opt, object-->
      <resolution>
        <!--ro, opt, enum, subType:string-->3840*2160@30HZ
      </resolution>
      <imageWidth>
        <!--ro, opt, int-->0
      </imageWidth>
      <imageHeight>
        <!--ro, opt, int-->0
      </imageHeight>
    </OutputResolution>
    <PortInBoard>
      <!--ro, opt, object-->
      <boardID>
        <!--ro, opt, int-->0
      </boardID>
    </PortInBoard>
  </VideoOutputChannel>

```

```

</boardID>
<portID>
  <!--ro, opt, int-->0
</portID>
<boardType>
  <!--ro, opt, enum, subType:string-->netra
</boardType>
</PortInBoard>
<deviceID>
  <!--ro, opt, int-->0
</deviceID>
<useEDIDResolution>
  <!--ro, opt, bool-->true
</useEDIDResolution>
<outputMode>
  <!--ro, opt, enum, subType:string-->HDMI
</outputMode>
<outputType>
  <!--ro, opt, enum, subType:string-->main
</outputType>
<timeSequenceMode>
  <!--ro, opt, enum, subType:string-->standard
</timeSequenceMode>
<CustomMode>
  <!--ro, opt, object-->
  <timeSequenceWidth>
    <!--ro, opt, int-->0
  </timeSequenceWidth>
  <timeSequenceHeight>
    <!--ro, opt, int-->0
  </timeSequenceHeight>
  <timeSequenceFrequency>
    <!--ro, opt, int-->0
  </timeSequenceFrequency>
</CustomMode>
<AdvanceMode>
  <!--ro, opt, object-->
  <timeSequenceWidth>
    <!--ro, opt, int-->0
  </timeSequenceWidth>
  <timeSequenceHeight>
    <!--ro, opt, int-->0
  </timeSequenceHeight>
  <timeSequenceFrequency>
    <!--ro, opt, int-->0
  </timeSequenceFrequency>
  <horizontalFront>
    <!--ro, opt, int-->1
  </horizontalFront>
  <horizontalBack>
    <!--ro, opt, int-->1
  </horizontalBack>
  <horizontalSyncSignal>
    <!--ro, opt, int-->1
  </horizontalSyncSignal>
  <verticalFront>
    <!--ro, opt, int-->1
  </verticalFront>
  <verticalBack>
    <!--ro, opt, int-->1
  </verticalBack>
  <verticalSyncSignal>
    <!--ro, opt, int-->1
  </verticalSyncSignal>
</AdvanceMode>
<name>
  <!--ro, opt, string-->test
</name>
<outputPortEnabled>
  <!--ro, opt, bool-->true
</outputPortEnabled>
<LEDSendCardResolutionEnabled>
  <!--ro, opt, bool-->true
</LEDSendCardResolutionEnabled>
<outputPortAccessType>
  <!--ro, opt, enum, subType:string-->display
</outputPortAccessType>
<outputBackgroundType>
  <!--ro, opt, enum, subType:string-->default
</outputBackgroundType>
<outputBackgroundRGBColor>
  <!--ro, opt, int, dep:and,{$.VideoOutputChannelList.VideoOutputChannel.outputBackgroundType,eq,solidColor}-->1
</outputBackgroundRGBColor>
<HDMIIDialEnabled>
  <!--ro, opt, bool-->true
</HDMIIDialEnabled>
</VideoOutputChannel>
</VideoOutputChannelList>

```

statusCode	statusString	subStatusCode	errorCode	description
3	Device Error	outputClockException	0x30008060	--
4	Invalid Operation	theOutputCablesNotConnected	0x4000A54D	--
4	Invalid Operation	outputResolutionMismatch	0x4000A54E	--
4	Invalid Operation	exceptionOfOutput10GigabitLink	0x4000A54F	--

9.7.3.5 Get parameters of a specific video output

Request URL

GET /ISAPI/DisplayDev/Video/outputs/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOutputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video output channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, output ID-->test
  </id>
  <portType>
    <!--ro, req, enum, port type, subType:string, desc:"VGA,CVBS,HDMI,Spot,SDI,DVI,TVI,etc."-->VGA
  </portType>
  <OutputResolution>
    <!--ro, opt, object, decoding output resolution-->
    <resolution>
      <!--ro, opt, enum, resolution, subType:string, desc:"1280*720@50HZ...",resolution name-->3840*2160@30HZ
    </resolution>
    <imageWidth>
      <!--ro, opt, int, resolution (width)-->0
    </imageWidth>
    <imageHeight>
      <!--ro, opt, int, resolution (height)-->0
    </imageHeight>
  </OutputResolution>
  <PortInBoard>
    <!--ro, opt, object-->
    <boardID>
      <!--ro, opt, int-->0
    </boardID>
    <portID>
      <!--ro, opt, int-->0
    </portID>
    <boardType>
      <!--ro, opt, enum, subType:string-->netra
    </boardType>
    <ipAddress>
      <!--ro, opt, string, range:[0,32]-->test
    </ipAddress>
    <port>
      <!--ro, opt, int-->0
    </port>
  </PortInBoard>
  <deviceID>
    <!--ro, opt, int-->0
  </deviceID>
  <useEDIDResolution>
    <!--ro, opt, bool-->true
  </useEDIDResolution>
  <outputMode>
    <!--ro, opt, enum, output mode, subType:string, desc:"HDMI", "DVI"-->HDMI
  </outputMode>
  <outputType>
    <!--ro, opt, enum, subType:string-->main
  </outputType>
  <timeSequenceMode>
    <!--ro, opt, enum, subType:string-->standard
  </timeSequenceMode>
  <CustomMode>
```



```

<!--ro, opt, object-->
<timeSequenceWidth>
  <!--ro, opt, int-->0
</timeSequenceWidth>
<timeSequenceHeight>
  <!--ro, opt, int-->0
</timeSequenceHeight>
<timeSequenceFrequency>
  <!--ro, opt, int-->0
</timeSequenceFrequency>
</CustomMode>
<AdvanceMode>
  <!--ro, opt, object-->
  <timeSequenceWidth>
    <!--ro, opt, int-->0
  </timeSequenceWidth>
  <timeSequenceHeight>
    <!--ro, opt, int-->0
  </timeSequenceHeight>
  <timeSequenceFrequency>
    <!--ro, opt, int-->0
  </timeSequenceFrequency>
  <horizontalFront>
    <!--ro, opt, int-->1
  </horizontalFront>
  <horizontalBack>
    <!--ro, opt, int-->1
  </horizontalBack>
  <horizontalSyncSignal>
    <!--ro, opt, int-->1
  </horizontalSyncSignal>
  <verticalFront>
    <!--ro, opt, int-->1
  </verticalFront>
  <verticalBack>
    <!--ro, opt, int-->1
  </verticalBack>
  <verticalSyncSignal>
    <!--ro, opt, int-->1
  </verticalSyncSignal>
</AdvanceMode>
<name>
  <!--ro, opt, string, decoding output name-->test
</name>
<outputPortEnabled>
  <!--ro, opt, bool-->true
</outputPortEnabled>
<LEDSendCardResolutionEnabled>
  <!--ro, opt, bool-->true
</LEDSendCardResolutionEnabled>
<outputPortAccessType>
  <!--ro, opt, enum, subType:string-->display
</outputPortAccessType>
<outputBackgroundType>
  <!--ro, opt, enum, subType:string-->default
</outputBackgroundType>
<outputBackgroundRGBColor>
  <!--ro, opt, int, dep:and,{$.VideoOutputChannelList.VideoOutputChannel.outputBackgroundType,eq,solidColor}-->1
</outputBackgroundRGBColor>
<HDMI DialEnabled>
  <!--ro, opt, bool-->true
</HDMI DialEnabled>
<enabled>
  <!--ro, opt, bool-->true
</enabled>
<outputResolutionAutoEnabled>
  <!--ro, opt, bool-->true
</outputResolutionAutoEnabled>
<LEDSendCardSerialNumber>
  <!--ro, opt, string, range:[9,128], dep:and,{$.VideoOutputChannelList[*].VideoOutputChannel.portType,eq,LEDSendCard}-->test
</LEDSendCardSerialNumber>
<outputPortAccessStatus>
  <!--ro, opt, enum, subType:string-->normal
</outputPortAccessStatus>
<abnormalReason>
  <!--ro, opt, string, range:[0,256], dep:and,{$.VideoOutputChannel.outputPortAccessStatus,eq,abnormal}-->test
</abnormalReason>
</VideoOutputChannel>

```

9.7.3.6 Set parameters of a specific video output

Request URL

PUT /ISAPI/DisplayDev/Video/outputs/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoOutputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video output channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel ID-->test
  </id>
  <portType>
    <!--req, enum, port type, subType:string, desc:"VGA", "CVBS", "HDMI", "Spot", "SDI", "DVI", "TVI", "LEDSendCard"-->VGA
  </portType>
  <OutputResolution>
    <!--opt, object, decoding output resolution-->
    <resolution>
      <!--opt, enum, resolution, subType:string, desc:resolution-->3840*2160@30HZ
    </resolution>
    <imageWidth>
      <!--opt, int, resolution (width)-->0
    </imageWidth>
    <imageHeight>
      <!--opt, int, resolution (height)-->0
    </imageHeight>
  </OutputResolution>
  <PortInBoard>
    <!--opt, object-->
    <boardID>
      <!--opt, int, sub-board ID-->0
    </boardID>
    <portID>
      <!--opt, int, slot ID of the sub-board-->0
    </portID>
    <boardType>
      <!--opt, enum, subType:string-->netra
    </boardType>
  </PortInBoard>
  <deviceID>
    <!--opt, int, device ID-->0
  </deviceID>
  <useEDIDResolution>
    <!--opt, bool-->true
  </useEDIDResolution>
  <outputMode>
    <!--opt, enum, subType:string-->HDMI
  </outputMode>
  <outputType>
    <!--opt, enum, subType:string-->main
  </outputType>
  <timeSequenceMode>
    <!--opt, enum, subType:string-->standard
  </timeSequenceMode>
  <CustomMode>
    <!--opt, object, custom mode-->
    <timeSequenceWidth>
      <!--opt, int-->0
    </timeSequenceWidth>
    <timeSequenceHeight>
      <!--opt, int-->0
    </timeSequenceHeight>
    <timeSequenceFrequency>
      <!--opt, int-->0
    </timeSequenceFrequency>
  </CustomMode>
  <AdvanceMode>
    <!--opt, object, advanced mode-->
    <timeSequenceWidth>
      <!--opt, int-->0
    </timeSequenceWidth>
    <timeSequenceHeight>
      <!--opt, int-->0
    </timeSequenceHeight>
    <timeSequenceFrequency>
      <!--opt, int-->0
    </timeSequenceFrequency>
    <horizontalFront>
      <!--opt, int-->1
    </horizontalFront>
    <horizontalBack>
      <!--opt, int-->1
    </horizontalBack>
    <horizontalSyncSignal>
      <!--opt, int-->1
    </horizontalSyncSignal>
    <verticalFront>
      <!--opt, int-->1
    </verticalFront>
  </AdvanceMode>
</VideoOutputChannel>
```

```

<verticalBack>
  <!--opt, int-->1
</verticalBack>
<verticalSyncSignal>
  <!--opt, int-->1
</verticalSyncSignal>
</AdvanceMode>
<name>
  <!--opt, string, decoding output name-->test
</name>
<outputPortEnabled>
  <!--opt, bool-->true
</outputPortEnabled>
<LEDSendCardResolutionEnabled>
  <!--opt, bool-->true
</LEDSendCardResolutionEnabled>
<outputPortAccessType>
  <!--opt, enum, subType:string-->display
</outputPortAccessType>
<outputBackgroundType>
  <!--opt, enum, subType:string-->default
</outputBackgroundType>
<outputBackgroundRGBColor>
  <!--opt, int, dep:and,{$.VideoOutputChannelList.VideoOutputChannel.outputBackgroundType,eq,solidColor}-->1
</outputBackgroundRGBColor>
<HDMI DialEnabled>
  <!--opt, bool-->true
</HDMI DialEnabled>
</VideoOutputChannel>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	desc
4	Invalid Operation	configuringDvilsNotSupportedBy4kDevice	0x4000A312	--
4	Invalid Operation	theNameOfOutputChannelExists	0x4000A330	--
4	Invalid Operation	theOutputSupportsOnlyTheLcdModeRatherThanLedMode	0x4000A553	--
4	Invalid Operation	outputPortsInTheSameSlotAreConnectedToInconsistentWallNumbers	0x4000A79E	--
4	Invalid Operation	theOutputPortDoesNotSupportTheCurrentBitDepth	0x4000A7C3	--

9.7.3.7 Get the capability of a specific video output

Request URL

GET /ISAPI/DisplayDev/Video/outputs/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoOutputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video output channel, attr:version{req, string, protocolVersion}-->
  <id min="0" max="10">
    <!--ro, opt, string, channel ID, attr:min{req, int},max{req, int}-->test
  </id>
  <portType opt=" VGA,CVBS,HDMI,Spot,SDI,DVI,TVI,LEDSendCardResolutionEnabled,HDMI4K">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </portType>
  <needRebot>
    <!--ro, opt, bool, "true,false"-->true
  </needRebot>
  <OutputResolution>
    <!--ro, opt, object-->
    <resolution
      opt="1920*1080@60HZ,1280*720@50HZ,3840*2160@30HZ,1920*1080@50HZ,1280*720@60HZ,1024*768@60HZ,BNC_MODE_PAL_25HZ,BNC_MODE_NTSC_30HZ,1920*1200@60HZ,3840*2160@60
      HZ,1600*1200@60HZ">
      <!--ro, opt, string, resolution, attr:opt{req, string}-->test
    </resolution>
    <imageWidth min="0" max="10">
      <!--ro, opt, int, resolution (width), attr:min{req, int},max{req, int}-->1
    </imageWidth>
    <imageHeight min="0" max="10">
      <!--ro, opt, int, resolution (height), attr:min{req, int},max{req, int}-->1
    </imageHeight>
  </OutputResolution>
  <PortInBoard>
    <!--ro, opt, object-->
    <boardID min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </boardID>
    <portID min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </portID>
    <boardType opt="netra">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </boardType>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </ipAddress>
    <port min="0" max="65535">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </port>
  </PortInBoard>
  <deviceID min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </deviceID>
  <useEDIDResolution opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </useEDIDResolution>
  <outputMode opt="HDMI,DVI,AUTO">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </outputMode>
  <outputType opt="main,sub,alone">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </outputType>
  <timeSequenceMode opt="standard,custom,advance">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </timeSequenceMode>
  <CustomMode>
    <!--ro, opt, object-->
    <timeSequenceWidth min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </timeSequenceWidth>
    <timeSequenceHeight min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </timeSequenceHeight>
    <timeSequenceFrequency min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </timeSequenceFrequency>
  </CustomMode>
  <AdvanceMode>
    <!--ro, opt, object-->
    <timeSequenceWidth min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </timeSequenceWidth>
    <timeSequenceHeight min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </timeSequenceHeight>
    <timeSequenceFrequency min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </timeSequenceFrequency>
  </AdvanceMode>
</VideoOutputChannel>
```

```

</timeSequenceFrequency>
<timeSequenceFrequency min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
</timeSequenceFrequency>
<horizontalFront min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</horizontalFront>
<horizontalBack min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</horizontalBack>
<horizontalSyncSignal min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</horizontalSyncSignal>
<verticalFront min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</verticalFront>
<verticalBack min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</verticalBack>
<verticalSyncSignal min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</verticalSyncSignal>
</AdvanceMode>
<name min="1" max="64">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</name>
<outputPortEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</outputPortEnabled>
<LEDSendCardResolutionEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</LEDSendCardResolutionEnabled>
<outputBackgroundType opt="default,solidColor">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->default
</outputBackgroundType>
<outputBackgroundRGBColor>
  <!--ro, opt, int, dep:and,{$.VideoOutputChannelList.VideoOutputChannel.outputBackgroundType,eq,solidColor}-->1
</outputBackgroundRGBColor>
<HDMI DialEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</HDMI DialEnabled>
<outputAlignUnit>
  <!--ro, opt, object-->
  <width>
    <!--ro, req, int, range:[1,10000]-->1
  </width>
  <height>
    <!--ro, req, int, range:[1,10000]-->1
  </height>
</outputAlignUnit>
<enabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</enabled>
<outputResolutionAutoEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</outputResolutionAutoEnabled>
<LEDSendCardSerialNumber min="9" max="128">
  <!--ro, opt, string, dep:and,{$.VideoOutputChannelList[*].VideoOutputChannel.portType,eq,LEDSendCard}, attr:min{req, int},max{req, int}-->test&gt;
</LEDSendCardSerialNumber>
</VideoOutputChannel>

```

9.7.3.8 Set parameters of all video output channels

Request URL

PUT /ISAPI/DisplayDev/Video/outputs/channels/all

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoOutputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video output channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, channel ID-->0
  </id>
  <portType opt="VGA,CVBS,HDMI,Spot,SDI,DVI">
    <!--opt, string, port type, attr:opt{req, string}-->test
  </portType>
  <OutputResolution>
    <!--opt, object, output resolution-->
    <resolution opt="1920*1080@60HZ,1280*720@50HZ,3840x2160@30HZ">
      <!--opt, string, resolution, attr:opt{req, string}-->test
    </resolution>
    <imageWidth min="0" max="10">
      <!--opt, int, resolution (width), attr:min{req, int},max{req, int}-->1
    </imageWidth>
    <imageHeight min="0" max="10">
      <!--opt, int, resolution (height), attr:min{req, int},max{req, int}-->1
    </imageHeight>
  </OutputResolution>
</VideoOutputChannel>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device-Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

9.7.3.9 Get the configuration capability of all video output channels

Request URL

GET /ISAPI/DisplayDev/Video/outputs/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoOutputsCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <videoOutputPortNums>
    <!--ro, opt, int, the number of output ports-->0
  </videoOutputPortNums>
  <isSupportMultiOutputType>
    <!--ro, opt, bool-->true
  </isSupportMultiOutputType>
  <isSupportMultiResolution>
    <!--ro, opt, bool-->true
  </isSupportMultiResolution>
  <isSupportColorSetting>
    <!--ro, opt, bool, ro-->true
  </isSupportColorSetting>
  <isSupportWidthHeightSetting>
    <!--ro, opt, bool-->true
  </isSupportWidthHeightSetting>
  <isSupportOutputIdentity>
    <!--ro, opt, bool-->true
  </isSupportOutputIdentity>
  <OutputResolutionCapList>
    <!--ro, opt, array, subType:object-->
    <OutputResolutionCap>
      <!--ro, opt, object-->
      <resolution>
        <!--ro, opt, string, resolution-->test
      </resolution>
      <imageWidth min="0" max="4096">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
      </imageWidth>
      <imageHeight min="0" max="4096">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
      </imageHeight>
    </OutputResolutionCap>
  </OutputResolutionCapList>
  <isSupportEDIDResolution>
    <!--ro, opt, bool-->true
  </isSupportEDIDResolution>
  <isNeedWallResolutionUnanimous>
    <!--ro, opt, bool-->true
  </isNeedWallResolutionUnanimous>
  <outputPortAccessType opt="display,projector,LCD,LED,bigScreen,conferenceTablet,recordHost,universal">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->display
  </outputPortAccessType>
  <outputBackgroundType opt="default,solidColor">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->default
  </outputBackgroundType>
  <outputBackgroundRGBColor>
    <!--ro, opt, int, dep:and,{$.VideoOutputChannelList.VideoOutputChannel.outputBackgroundType,eq,solidColor}-->1
  </outputBackgroundRGBColor>
</VideoOutputsCap>
```

9.7.4 Signal Source Management

9.7.4.1 Get the audio capabilities

Request URL

GET /ISAPI/DisplayDev/Audio/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <audioInputsPortNums>
    <!--ro, opt, int-->0
  </audioInputsPortNums>
  <audioOutputsPortNums>
    <!--ro, opt, int-->0
  </audioOutputsPortNums>
  <audioOutputMaxNum>
    <!--ro, opt, int-->0
  </audioOutputMaxNum>
</AudioCap>
```

9.7.4.2 Get capability set of adding signal source group

Request URL

GET /ISAPI/DisplayDev/SignalSource/AddSignalSourceGroup/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "AddSignalSourceGroupCap": {
    /*opt, object, capability of adding signal source group, desc:input parameter capability of corresponding function (URL: POST
    /ISAPI/DisplayDev/SignalSource/AddSignalSourceGroup?format=json)*/
    "groupName": {
      /*req, object, group name*/
      "@def": "undefined",
      /*opt, string, default value*/
      "@min": 1,
      /*req, int, minimum length*/
      "@max": 64
      /*req, int, maximum length*/
    },
    "signalSourceList": {
      /*opt, object, signal source list*/
      "@size": 128,
      /*opt, int, the maximum number of array elements*/
      "signalSourceID": {
        /*opt, object, signal source ID range*/
        "@def": 1,
        /*opt, int, default value*/
        "@min": 1,
        /*req, int, minimum value*/
        "@max": 2147483647
        /*req, int, maximum value*/
      }
    }
  }
}
```

9.7.4.3 Get signal source groups

Request URL

POST /ISAPI/DisplayDev/SignalSource/GetSignalSourceGroup?format=json

Query Parameter

None

Request Message

None

Response Message


```

{
  "signalSourceGroupList": [
    /*opt, array, signal source group list, subType:object, range:[1,32]*/
    {
      "groupID": 1,
      /*req, int, signal source group ID, range:[1,32]*/
      "groupName": "undefined",
      /*req, string, group name, range:[1,64]*/
      "signalSourceList": [
        /*opt, array, signal source list, subType:object, range:[0,128]*/
        {
          "signalSourceID": 1
          /*opt, int, signal source ID, range:[1,2147483647]*/
        }
      ]
    }
  ]
}

```

9.7.4.4 Get capability of editing signal source group

Request URL

GET /ISAPI/DisplayDev/SignalSource/ModifySignalSourceGroup/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "ModifySignalSourceGroupCap": {
    /*opt, object, capability of editing signal source group, desc:input parameter capability of corresponding function (URL: POST
    /ISAPI/DisplayDev/SignalSource/ModifySignalSourceGroup?format=json)*/
    "groupID": {
      /*req, object, value range of signal source group ID*/
      "@def": 1,
      /*opt, int, default value*/
      "@min": 1,
      /*req, int, minimum value*/
      "@max": 32
      /*req, int, maximum value*/
    },
    "groupName": {
      /*req, object, group name*/
      "@def": "undefined",
      /*opt, string, default value*/
      "@min": 1,
      /*req, int, minimum length*/
      "@max": 64
      /*req, int, maximum length*/
    },
    "signalSourceList": {
      /*opt, object, signal source list*/
      "@size": 128,
      /*opt, int, the maximum number of array elements*/
      "signalSourceID": {
        /*opt, object, signal source ID range*/
        "@def": 1,
        /*opt, int, default value*/
        "@min": 1,
        /*req, int, minimum value*/
        "@max": 2147483647
        /*req, int, maximum value*/
      }
    }
  }
}

```

9.7.4.5 Get capability of no signal parameters of signal source

Request URL

GET /ISAPI/DisplayDev/SignalSource/SignalSourceNoSignalParams/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "SignalSourceNoSignalParamsCap": {
    /*opt, object, capability of no signal parameters of signal source, desc:corresponding API: GET or PUT
    /ISAPI/DisplayDev/SignalSource/SignalSourceNoSignalParams?format=json*/
    "noSignalScreenMode": {
      /*req, object, value range of no signal image type of signal source*/
      "@opt": [],
      /*opt, array, enumeration of no signal image type of signal source, subType:string, range:[1,4]*/
      "@def": "default"
      /*opt, string, default value*/
    },
    "customFile": {
      /*opt, object, custom picture file*/
      "filePathType": {
        /*req, object, range of file storage path type, desc:"simpleStorage" (simple storage protocol): filePath is storageID; "URL": such as
        "http://storage.com/1.jpg", the file can be directly accessed; "localPath" (device local storage): the file is stored on the device, filepath is a path
        generated by device. The platform obtains files from the device through a special file transfer link. "binary" (direct transmission): filePath is the binary
        data of the file. A base64 transcoding is needed when transferring, and the size of message after transcoding should be no more than 256 KB. Message after
        transcoding larger than 256 KB is not suitable to this method; "multipart" (form format): supported only under HTTP/HTTPS protocols, filePath is form name*/
        "@opt": [],
        /*opt, array, enumeration list of file storage path types, subType:string, range:[1,5]*/
        "@def": "URL"
        /*opt, string, default value*/
      },
      "filePath": {
        /*req, object, range of file storage path type, desc:different filePathType values suggest different ways of file access*/
        "@def": "",
        /*opt, string, default value*/
        "@min": 0,
        /*req, int, minimum length*/
        "@max": 30720
        /*req, int, maximum length*/
      }
    }
  }
}
```

9.7.4.6 Get no signal parameters of signal source

Request URL

GET /ISAPI/DisplayDev/SignalSource/SignalSourceNoSignalParams?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "noSignalScreenMode": "default",
  /*req, enum, no signal image type of signal source, subType:string, desc:"default", "custom", "retainLastFrame" (save the last frame), "noSignal" (no
  video signal)*/
  "customFile": {
    /*opt, object, custom picture file, dep:and,{noSignalScreenMode,eq,custom}*/
    "filePathType": "URL",
    /*req, enum, file path type, subType:string, desc:"simpleStorage" (simple storage protocol): filePath is storageID; "URL": such as
    "http://storage.com/1.jpg", the file can be directly accessed; "localPath" (device local storage): the file is stored on the device, filepath is a path
    generated by device. The platform obtains files from the device through a special file transfer link. "binary" (direct transmission): filePath is the binary
    data of the file. The files should be converted via base64 and the size of the converted message should be no more than 256 KB. Otherwise, other types of
    picture storage path are recommended "multipart" (form format): supported only under HTTP/HTTPS protocols, filePath is form name. "URL", "binary" (data
    encoded via base64), "localPath" (device local storage), "simpleStorage" (simple storage protocol), "multipart" (from format)*/
    "filePath": ""
    /*req, string, file path, range:[0,30720], desc:different filePathType values suggest different ways of file access*/
  }
}
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	theImageBitDepthIsIncorrect	0x4000A621	--

9.7.4.7 Get video capabilities

Request URL

GET /ISAPI/DisplayDev/Video/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, video capability, attr:version{req, string, protocolVersion}-->
  <VideoInputsCap version="2.0">
    <!--ro, opt, object, video input capability, attr:version{req, string}-->
    <videoInputPortNums>
      <!--ro, opt, int, number of video input ports-->0
    </videoInputPortNums>
    <isSupportColorSetting>
      <!--ro, opt, bool, whether the device supports color settings-->true
    </isSupportColorSetting>
    <isSupportPostionSetting>
      <!--ro, opt, bool, whether the device supports location settings-->true
    </isSupportPostionSetting>
    <isSupportCutoffSetting>
      <!--ro, opt, bool, whether the device supports cropping-->true
    </isSupportCutoffSetting>
    <isSupportPictureCapture>
      <!--ro, opt, bool, whether the device supports capturing pictures-->true
    </isSupportPictureCapture>
    <isSupportText>
      <!--ro, opt, bool, whether the device supports text-->true
    </isSupportText>
    <encodeOSDInputSignalType opt="HDMI,DP">
      <!--ro, opt, string, input signal type, attr:opt{req, string}-->test
    </encodeOSDInputSignalType>
    <SupportSelfdefineResolution>
      <!--ro, opt, object, custom resolution supported by DP (display port) input source-->
      <signalType opt="DVI,VGA,HDMI,DP">
        <!--ro, req, string, signal type, attr:opt{req, string}-->test
      </signalType>
      <supportedDPResolution opt="640*480@60,800*600@60">
        <!--ro, opt, string, custom resolution supported by DP (display port) input source, attr:opt{req, string}-->test
      </supportedDPResolution>
    </SupportSelfdefineResolution>
    <isSupportEDIDResolution>
      <!--ro, opt, bool, whether the device supports EDID resolution-->true
    </isSupportEDIDResolution>
    <isSupportJoinSignalCfg>
      <!--ro, opt, bool, whether the device supports signal source splicing-->true
    </isSupportJoinSignalCfg>
    <SupportAudioCfg>
      <!--ro, opt, object, audio settings-->
      <signalType opt="DVI,VGA,HDMI,DP">
        <!--ro, req, string, supported encoding embedded audio / independent audio, attr:opt{req, string}-->test
      </signalType>
    </SupportAudioCfg>
  </VideoInputsCap>
  <VideoOutputsCap version="2.0">
    <!--ro, opt, object, video output capability, attr:version{req, string}-->
    <videoOutputPortNums>
      <!--ro, opt, int, the number of output ports-->0
    </videoOutputPortNums>
    <isSupportMultiOutputType>
      <!--ro, opt, bool, whether the device supports multiple output types-->true
    </isSupportMultiOutputType>
    <isSupportMultiResolution>
      <!--ro, opt, bool, whether the device supports multi-resolution-->true
    </isSupportMultiResolution>
    <isSupportColorSetting>
      <!--ro, opt, bool, whether the device supports output port color configuration-->true
    </isSupportColorSetting>
    <isSupportWidthHeightSetting>
      <!--ro, opt, bool, whether the device supports LED resolution configuration-->true
    </isSupportWidthHeightSetting>
    <isSupportOutputIdentity>
      <!--ro, opt, bool, whether the device supports output port customization-->true
    </isSupportOutputIdentity>
    <OutputResolutionCapList>
      <!--ro, opt, array, output port resolution capability set list, subType:object-->
      <OutputResolutionCap>
        <!--ro, opt, object, output port resolution capability set-->
        <resolution>
          <!--ro, opt, string, resolution-->test
        </resolution>
        <imageWidth min="0" max="10">

```

```

        <!--ro, opt, int, resolution (width), attr:min{req, int},max{req, int}-->0
    </imageWidth>
    <imageHeight min="0" max="10">
        <!--ro, opt, int, resolution (height), attr:min{req, int},max{req, int}-->0
    </imageHeight>
    </OutputResolutionCap>
</OutputResolutionCapList>
<isSupportedEDIDResolution>
    <!--ro, opt, bool, whether the device supports LED resolution-->true
</isSupportedEDIDResolution>
</VideoOutputsCap>
<VideoStreamingCap version="2.0">
    <!--ro, opt, object, video stream capability, attr:version{req, string}-->
    <streamingNums>
        <!--ro, opt, int, stream count-->0
    </streamingNums>
    <isSupportURL>
        <!--ro, opt, bool, whether the device supports URL address-->true
    </isSupportURL>
    <isSupportDDNS>
        <!--ro, opt, bool, whether the device supports DDNS address-->true
    </isSupportDDNS>
    <isSupportIPAddress>
        <!--ro, opt, bool, whether the device supports IP address-->true
    </isSupportIPAddress>
    <isSupportDistributedIPSignal opt="true,false">
        <!--ro, opt, bool, whether the device supports adding distributed network source, attr:opt{req, string}-->true
    </isSupportDistributedIPSignal>
    <isSupportAddBatch>
        <!--ro, opt, bool, whether the device supports batch adding-->true
    </isSupportAddBatch>
    <isSupportStreamChanSearch>
        <!--ro, opt, bool, whether the device supports getting stream channel information-->true
    </isSupportStreamChanSearch>
    <isSupportEncryptStream>
        <!--ro, opt, bool, whether the device supports stream encryption-->true
    </isSupportEncryptStream>
    <isSupportLoopEncryptStream>
        <!--ro, opt, bool, whether the device supports decoding encrypted streams in auto-switch-->true
    </isSupportLoopEncryptStream>
</VideoStreamingCap>
<isSupportClarityConfig>
    <!--ro, opt, bool, whether it supports definition configuration when displaying videos on the video wall-->true
</isSupportClarityConfig>
</VideoCap>

```

9.7.4.8 Get parameters of all video input channels

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoInputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
    <!--ro, req, array, signal source channel list, subType:object, attr:version{req, string, protocolVersion}-->
    <VideoInputChannel>
        <!--ro, req, object, signal source-->
        <id>
            <!--ro, req, int, Local signal source ID-->1
        </id>
        <name>
            <!--ro, opt, string, signal source name-->test
        </name>
        <portType>
            <!--ro, opt, enum, signal source type, subType:string, desc:"SDI,OPT,VGA,HDMI,YpbPr,Matrix,Joint,DVI"-->SDI
        </portType>
        <useSelfdefineResolution>
            <!--ro, opt, bool, custom resolution-->true
        </useSelfdefineResolution>
        <selfdefineResolutionNo>
            <!--ro, opt, int, custom resolution ID, desc:custom resolution ID-->1
        </selfdefineResolutionNo>
        <imageWidth>
            <!--ro, opt, int, resolution (width)-->1
        </imageWidth>
        <imageHeight>
            <!--ro, opt, int, resolution (height)-->1
        </imageHeight>
    </VideoInputChannel>

```

```

<!--ro, opt, int-->1
</FPS>
<PortInBoard>
  <!--ro, opt, object-->
  <boardID>
    <!--ro, opt, int, sub-board ID-->1
  </boardID>
  <portID>
    <!--ro, opt, int, slot ID of the sub-board-->1
  </portID>
  <ipv4Address>
    <!--ro, opt, string, ipv4-->192.168.1.1
  </ipv4Address>
  <ipv6Address>
    <!--ro, opt, string, ipv6-->fe80::4ba5:790e
  </ipv6Address>
  <port>
    <!--ro, opt, int, port No., range:[0,65535]-->1
  </port>
  <trackNum>
    <!--ro, opt, object-->
    <x>
      <!--ro, req, int-->1
    </x>
    <y>
      <!--ro, req, int-->1
    </y>
  </trackNum>
</PortInBoard>
<deviceID>
  <!--ro, opt, int, device ID-->1
</deviceID>
<JointPort>
  <!--ro, opt, object-->
  <horizontalCount>
    <!--ro, opt, int, the number of horizontal signal sources-->1
  </horizontalCount>
  <verticalCount>
    <!--ro, opt, int, the number of vertical signal sources-->1
  </verticalCount>
</JointPort>
<RelateScreenServer>
  <!--ro, opt, bool, whether the input is related to a screen server-->true
</RelateScreenServer>
<signalStatus>
  <!--ro, opt, enum, signal source status, subType:string, desc:"signal" (with signal input), "noSignal" (no signal), "abnormal"-->signal
</signalStatus>
<decodeWallStatus>
  <!--ro, opt, enum, status of decoding and displaying on video wall, subType:string, desc:"decoding", "noDecoding" (no decoding)-->decoding
</decodeWallStatus>
<jointMatrixEnabled>
  <!--ro, opt, bool-->false
</jointMatrixEnabled>
<videoCodecType>
  <!--ro, opt, enum, video encoding type, subType:string, desc:"H.264", "H.265"-->H.264
</videoCodecType>
<SmartCodec>
  <!--ro, opt, object, smart codec-->
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
</SmartCodec>
<audioCompressionType>
  <!--ro, opt, enum, audio encoding type, subType:string, desc:audio encoding type-->G.711alaw
</audioCompressionType>
<inputPortEnabled>
  <!--ro, opt, bool-->true
</inputPortEnabled>
<timeSequenceResolution>
  <!--ro, opt, enum, video timing, subType:string, desc:1024*768@60HZ, 720P@50HZ, 720P@60HZ#720P@60HZ, 1366*768@60HZ#1366, 1440*900@60HZ, 1280*1024@60HZ, 1680*1050@60, 1080P@50HZ, 1080P@60HZ, 3840*2160@30HZ-->1024*768@60HZ
</timeSequenceResolution>
<inputContent>
  <!--ro, opt, enum, input content, subType:string, desc:"default", "broadcastVideo" (broadcast video), "wirelessMirroring" (wireless screen mirroring)-->default
</inputContent>
<OptimumResolution>
  <!--ro, opt, object, resolution self-adaption-->
  <enabled>
    <!--ro, opt, bool, whether to enable-->true
  </enabled>
  <imageWidth>
    <!--ro, opt, int, resolution (width), range:[0,4096]-->0
  </imageWidth>
  <imageHeight>
    <!--ro, opt, int, resolution (height), range:[0,4096]-->0
  </imageHeight>
</OptimumResolution>
<inputPortAccessType>
  <!--ro, opt, enum, input port access type, subType:string, desc:input port access type-->computer
</inputPortAccessType>
<streamType>

```

```

    <!--ro, opt, enum, stream type, subType:int, desc:1-main stream, 2-sub stream-->1
  </streamType>
  <bitRateType>
    <!--ro, opt, enum, video bitrate type, subType:string, desc:"CBR" (constant bit rate), "VBR" (variable bit rate)-->CBR
  </bitRateType>
  <intervalFrame>
    <!--ro, opt, int, I frame interval, range:[1,400]-->25
  </intervalFrame>
  <picQuality>
    <!--ro, opt, enum, picture quality, subType:int, desc:1 (Lowest), 2 (Lower), 3 (Low), 4 (medium), 5 (higher), 6 (highest)-->1
  </picQuality>
  <resolution>
    <!--ro, opt, enum, resolution, subType:string, desc:352*240#CIF (352*240), 352*288#CIF (352*288), 176*120#QCIF (176*120), 176*144#QCIF (176*144),
    704*288#CIF (704*288), 704*240#CIF (704*240), 528*384#DCIF (528*384), 528*320#DCIF (528*320), 704*576#FCIF (704*576), 704*480#FCIF (704*480), 960*576#HD1
    (960*576), 960*480#HD1 (960*480), 1280*720#720P (1280*720), 1280*960#XVGA (1280*960), 1600*1200#UXGA (1600*1200), 1080P#1080P (1920*1080), 1080I#1080I
    (1920*1080), 1024*768#XGA (1024*768), 1360*768#XGA (1360*768), 1366*768#XGA (1366*768), 1280*1024#SXGA (1280*1024), 1400*1050#SXGA (1400*1050),
    1440*900#SXGA(1440*900), 1680*1050#SXGA+(1680*1050)-->352*240
  </resolution>
  <videoType>
    <!--ro, opt, enum, video type, subType:string, desc:video type-->0
  </videoType>
  <videoBitRate>
    <!--ro, opt, enum, maximum bit rate, subType:int, desc:maximum bit rate-->2
  </videoBitRate>
  <customBitRate>
    <!--ro, opt, int, custom bit rate, range:[128,16384], desc:from 128 to 16384 kbps-->128
  </customBitRate>
  <frameType>
    <!--ro, opt, enum, frame type, subType:string, desc:P (P-frame)-->P
  </frameType>
  <frameRate>
    <!--ro, opt, enum, video frame rate, subType:string, desc:full (full frame rate), 1/16, 1/8, 1/4, 1/2, 1, 2, 4, 6, 8, 10, 12, 15, 16, 18, 20, 22, 25,
    30-->25
  </frameRate>
  <audioBitRate>
    <!--ro, opt, enum, audio bitrate, subType:int, desc:64 (64 kbps)-->64
  </audioBitRate>
  <audioSamplingRate>
    <!--ro, opt, enum, audio sampling rate, subType:int, desc:unit: Hz. "8000" (8 kHz), "16000" (16 kHz), "32000" (32 kHz), "44100" (44.1 kHz), "48000"
    (48 kHz)-->8000
  </audioSamplingRate>
  <audioCodecType>
    <!--ro, opt, enum, audio encoding type, subType:string, desc:audio encoding type-->G.711alaw
  </audioCodecType>
  <audioVolume>
    <!--ro, opt, int, volume level, range:[0,100]-->1
  </audioVolume>
  <localStreamDeepEncodeRatioType>
    <!--ro, opt, enum, deep compression rate type of local source, subType:string, desc:"Low", "middle", "high"-->low
  </localStreamDeepEncodeRatioType>
  <localStreamLightEncodeRatioType>
    <!--ro, opt, enum, shallow compression rate type of local source, subType:string, desc:"Low", "middle", "high"-->low
  </localStreamLightEncodeRatioType>
  <abnormalReason>
    <!--ro, opt, string, cause of exception, range:[0,256], dep:and,{$.VideoInputChannel.signalStatus,eq,abnormal}-->test
  </abnormalReason>
  <isSupportSingleJoin>
    <!--ro, opt, bool-->true
  </isSupportSingleJoin>
  <signalInfoList>
    <!--ro, opt, array, list of spliced signal source information, subType:object, range:[0,128]-->
  <signalInfo>
    <!--ro, req, object, signal source information-->
    <signalNo>
      <!--ro, opt, int, signal source No., range:[1,1024]-->0
    </signalNo>
    <name>
      <!--ro, opt, string, signal source name, range:[0,32]-->test
    </name>
    <hostname>
      <!--ro, opt, enum, address type, subType:string, desc:"ipv4Address" (IPv4 address), "ipv6Address" (IPv6 address)-->ipv4Address
    </hostname>
    <ipv4Address>
      <!--ro, opt, string, ipv4, dep:and,{$.VideoInputChannelList[*].VideoInputChannel.SignalInfoList[*].SignalInfo.hostname,eq,ipv4Address}--
    >192.168.1.1
    </ipv4Address>
    <ipv6Address>
      <!--ro, opt, string, ipv6, dep:and,{$.VideoInputChannelList[*].VideoInputChannel.SignalInfoList[*].SignalInfo.hostname,eq,ipv6Address}--
    >fe80::4ba5:790e
    </ipv6Address>
    <port>
      <!--ro, opt, int, port No., range:[0,65535]-->1
    </port>
    <trackNum>
      <!--ro, opt, object-->
      <x>
        <!--ro, req, int-->1
      </x>
      <y>
        <!--ro, req, int-->1
      </y>
    </trackNum>
  </signalInfo>
</SignalInfoList>

```

```

</signalResolutionType>
<signalResolutionType>
  <!--ro, opt, enum, subType:string-->HD
</signalResolutionType>
<bitDepth>
  <!--ro, opt, enum, the number of bits used for each color channel (such as red, green, and blue). 8 (8-bit), 10 (10-bit), subType:int, desc:the number of bits used for each color channel (such as red, green, and blue). 8 (8-bit), 10 (10-bit)-->8
</bitDepth>
<sharedSignalEnabled>
  <!--ro, opt, bool-->false
</sharedSignalEnabled>
</VideoInputChannel>
</VideoInputChannelList>

```

statusCode	statusString	subStatusCode	errorCode	description
3	Device Error	theExceptionReasonIsEmpty	0x30008054	--
3	Device Error	unknownAccessStatus	0x30008055	--
3	Device Error	gettingMaintenanceInformationFailed	0x30008056	--
3	Device Error	theSubSystemIsOffline	0x30008057	--
3	Device Error	incorrectSubSystemType	0x30008058	--
3	Device Error	theInputInternalLinksUnstable	0x30008059	--
3	Device Error	exceptionOfInput10GigabitLink	0x3000805A	--
3	Device Error	theSplicingChannelsNotConfigured	0x3000805B	--
3	Device Error	gettingTheInputSyncStatusFailed	0x3000805C	--
3	Device Error	theSplicingChannelsNotDisplayedOnVideoWall	0x3000805D	--
3	Device Error	gettingTheSyncStatusOfTheSplicingChannelFailed	0x3000805E	--
3	Device Error	theSplicingChannelImagesNotSynced	0x3000805F	--
4	Invalid Operation	unknownExceptionReason	0x4000A54B	--
4	Invalid Operation	inputFrameRatelsNotSupported	0x4000A54C	--

9.7.4.9 Set parameters of all video input channels

Request URL

PUT /ISAPI/DisplayDev/Video/inputs/channels

Query Parameter

None

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<VideoInputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <VideoInputChannel>
    <!--req, object, signal source-->
    <id>
      <!--req, int, Local signal source ID-->1
    </id>
    <name>
      <!--opt, string, signal source name-->test
    </name>
    <portType>
      <!--ro, opt, enum, signal source type, subType:string, desc:"SDI", "OPT", "VGA", "HDMI", "YpbPr", "Matrix", "Joint", "DVI", "TVI", "innerCamera", "HDMI/Hdbase-T", "Hdbase-T"-->SDI
    </portType>
    <useSelfdefineResolution>
      <!--opt, bool, custom resolution-->true
    </useSelfdefineResolution>
    <selfdefineResolutionNo>
      <!--opt, int, custom resolution ID, desc:custom resolution ID-->1
    </selfdefineResolutionNo>
    <imageWidth>

```

```

    <!--opt, int, image width-->1
</imageWidth>
<imageHeight>
    <!--opt, int, image height-->1
</imageHeight>
<FPS>
    <!--opt, int-->1
</FPS>
<PortInBoard>
    <!--opt, object-->
    <boardID>
        <!--opt, int, sub-board ID-->1
    </boardID>
    <portID>
        <!--opt, int, slot ID of the sub-board-->1
    </portID>
</PortInBoard>
<deviceID>
    <!--opt, int, device ID-->1
</deviceID>
<JointPort>
    <!--opt, object, read-only, it is required only when portType is "Joint"-->
    <horizontalCount>
        <!--opt, int, the number of horizontal signal sources-->1
    </horizontalCount>
    <verticalCount>
        <!--opt, int, the number of vertical signal sources-->1
    </verticalCount>
</JointPort>
<RelateScreenServer>
    <!--req, bool, whether the input has been linked to a screen server-->true
</RelateScreenServer>
<signalStatus>
    <!--opt, enum, subType:string-->signal
</signalStatus>
<decodeWallStatus>
    <!--opt, enum, subType:string-->decoding
</decodeWallStatus>
<jointMatrixEnabled>
    <!--opt, bool-->false
</jointMatrixEnabled>
<videoCodecType>
    <!--opt, enum, subType:string-->H.264
</videoCodecType>
<SmartCodec>
    <!--opt, object-->
    <enabled>
        <!--opt, bool-->true
    </enabled>
</SmartCodec>
<audioCompressionType>
    <!--opt, enum, subType:string-->G.711alaw
</audioCompressionType>
<inputPortEnabled>
    <!--opt, bool-->true
</inputPortEnabled>
<timeSequenceResolution>
    <!--opt, enum, subType:string-->1024*768@60Hz
</timeSequenceResolution>
<inputContent>
    <!--opt, enum, subType:string-->default
</inputContent>
<OptimumResolution>
    <!--opt, object, optimum resolution-->
    <enabled>
        <!--opt, bool-->true
    </enabled>
    <imageWidth>
        <!--opt, int, image width, range:[0,4096]-->0
    </imageWidth>
    <imageHeight>
        <!--opt, int, image height, range:[0,4096]-->0
    </imageHeight>
</OptimumResolution>
<inputPortAccessType>
    <!--opt, enum, subType:string-->computer
</inputPortAccessType>
<streamType>
    <!--opt, enum, subType:int-->1
</streamType>
<bitRateType>
    <!--opt, enum, subType:string-->CBR
</bitRateType>
<intervalFrame>
    <!--opt, int, range:[1,400]-->25
</intervalFrame>
<picQuality>
    <!--opt, enum, subType:int-->1
</picQuality>
<resolution>
    <!--opt, enum, subType:string-->352*240
</resolution>
<videoType>
    <!--opt, enum, subType:string-->0

```



```

<!--opt, enum, subType:string-->
</videoType>
<videoBitRate>
<!--opt, enum, subType:int-->2
</videoBitRate>
<customBitRate>
<!--opt, int, range:[128,16384]-->128
</customBitRate>
</VideoInputChannel>
</VideoInputChannelList>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK", "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:read-only,describe the error reason in detail-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode
4	Invalid Operation	signalSourceHasBeenConfiguredForCroppingAndDoesNotSupportEncodingParameterCon
4	Invalid Operation	numberOfAddedSignalSourcesExceededLimit
4	Invalid Operation	theResolutionIsNotSupportedBy2kDevice
4	Invalid Operation	theNameOfSignalSourceExists

9.7.4.10 Set parameters of a specified signal source

Request URL

PUT /ISAPI/DisplayDev/Video/inputs/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoInputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, signal source, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, local signal source ID-->1
  </id>
  <name>
    <!--opt, string, signal source name-->test
  </name>
  <portType>
    <!--ro, opt, enum, signal source type, subType:string, desc:"SDI,OPT,VGA,HDMI,YpbPr,Matrix,Joint,DVI,TVI"-->SDI
  </portType>
  <useSelfdefineResolution>
    <!--opt, bool, custom resolution-->true
  </useSelfdefineResolution>

```

```

<selfdefineResolutionNo>
  <!--opt, int, custom resolution ID, desc:custom resolution ID-->1
</selfdefineResolutionNo>
<imageWidth>
  <!--opt, int, resolution (width)-->1
</imageWidth>
<imageHeight>
  <!--opt, int, resolution (height)-->1
</imageHeight>
<FPS>
  <!--opt, int-->1
</FPS>
<PortInBoard>
  <!--opt, object-->
  <boardID>
    <!--opt, int, sub-board ID, which starts from 1-->1
  </boardID>
  <portID>
    <!--opt, int, Slot ID of the sub-board, which starts from 1-->1
  </portID>
</PortInBoard>
<deviceID>
  <!--opt, int, device ID-->1
</deviceID>
<JointPort>
  <!--opt, object, it is required only when portType is "Joint"-->
  <horizontalCount>
    <!--opt, int, the number of horizontal signal sources-->1
  </horizontalCount>
  <verticalCount>
    <!--opt, int, the number of vertical signal sources-->1
  </verticalCount>
</JointPort>
<RelateScreenServer>
  <!--req, bool, whether the input related a screen server-->true
</RelateScreenServer>
<signalStatus>
  <!--opt, enum, subType:string-->signal
</signalStatus>
<decodeWallStatus>
  <!--opt, enum, subType:string-->decoding
</decodeWallStatus>
<jointMatrixEnabled>
  <!--opt, bool-->false
</jointMatrixEnabled>
<videoCodecType>
  <!--opt, enum, subType:string-->H.264
</videoCodecType>
<SmartCodec>
  <!--opt, object-->
  <enabled>
    <!--opt, bool-->true
  </enabled>
</SmartCodec>
<audioCompressionType>
  <!--opt, enum, subType:string-->G.711alaw
</audioCompressionType>
<inputPortEnabled>
  <!--opt, bool-->true
</inputPortEnabled>
<timeSequenceResolution>
  <!--opt, enum, subType:string-->1024*768@60HZ
</timeSequenceResolution>
<inputContent>
  <!--opt, enum, subType:string-->default
</inputContent>
<OptimumResolution>
  <!--opt, object-->
  <enabled>
    <!--opt, bool-->true
  </enabled>
  <imageWidth>
    <!--opt, int, resolution (width), range:[0,4096]-->0
  </imageWidth>
  <imageHeight>
    <!--opt, int, resolution (height), range:[0,4096]-->0
  </imageHeight>
</OptimumResolution>
<inputPortAccessType>
  <!--opt, enum, subType:string-->computer
</inputPortAccessType>
<streamType>
  <!--opt, enum, subType:int-->1
</streamType>
<bitRateType>
  <!--opt, enum, subType:string-->CBR
</bitRateType>
<intervalFrame>
  <!--opt, int, range:[1,400]-->25
</intervalFrame>
<picQuality>
  <!--opt, enum, subType:int-->1
</picQuality>

```

```

<resolution>
  <!--opt, enum, subType:string-->352*240
</resolution>
<videoType>
  <!--opt, enum, subType:string-->0
</videoType>
<videoBitRate>
  <!--opt, enum, subType:int-->2
</videoBitRate>
<customBitRate>
  <!--opt, int, range:[128,16384]-->128
</customBitRate>
</VideoInputChannel>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode
4	Invalid Operation	signalSourceHasBeenConfiguredWithAUserDefinedResolutionAndDoesNotSupportEncodi
4	Invalid Operation	theResolutionExceededLoadCapacity
4	Invalid Operation	operationNotSupportedTheCurrentSignalSourceHasBeenSpliced
4	Invalid Operation	failedToSetTheResolutionWhichIsNotSupportedByTheDevice
4	Invalid Operation	displayingAVideoWallImagelsEnabledOnAClientDisableItFirst

9.7.4.11 Get parameters of a specific signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoInputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->

```

```

<!--ro, req, object, desc:resolution, req, string, procedure:-->
<id>
<!--ro, req, int, local signal source ID-->1
</id>
<name>
<!--ro, opt, string, signal source name-->test
</name>
<portType>
<!--ro, opt, enum, signal source type, subType:string, desc:"SDI, VGA, HDMI, YpbPr, Matrix, Joint..."-->SDI
</portType>
<useSelfdefineResolution>
<!--ro, opt, bool, custom resolution-->true
</useSelfdefineResolution>
<selfdefineResolutionNo>
<!--ro, opt, int, custom resolution ID, desc:custom resolution ID-->1
</selfdefineResolutionNo>
<imageWidth>
<!--ro, opt, int-->1
</imageWidth>
<imageHeight>
<!--ro, opt, int-->1
</imageHeight>
<FPS>
<!--ro, opt, int, [TBC]-->1
</FPS>
<PortInBoard>
<!--ro, opt, object-->
<boardID>
<!--ro, opt, int, sub-board ID, which starts from 1-->1
</boardID>
<portID>
<!--ro, opt, int, Slot ID of the sub-board, which starts from 1-->1
</portID>
<ipv4Address>
<!--ro, opt, string-->192.168.1.1
</ipv4Address>
<ipv6Address>
<!--ro, opt, string-->fe80::4ba5:790e
</ipv6Address>
</PortInBoard>
<deviceID>
<!--ro, opt, int, device ID-->1
</deviceID>
<JointPort>
<!--ro, opt, object, it is required only when portType is "Joint"-->
<horizontalCount>
<!--ro, opt, int, the number of horizontal signal sources-->1
</horizontalCount>
<verticalCount>
<!--ro, opt, int, the number of vertical signal sources-->1
</verticalCount>
</JointPort>
<RelateScreenServer>
<!--ro, opt, bool, whether the input related a screen server-->true
</RelateScreenServer>
<signalStatus>
<!--ro, opt, enum, subType:string-->signal
</signalStatus>
<decodeWallStatus>
<!--ro, opt, enum, subType:string-->decoding
</decodeWallStatus>
<jointMatrixEnabled>
<!--ro, opt, bool-->false
</jointMatrixEnabled>
<videoCodecType>
<!--ro, opt, enum, video encoding type, subType:string, desc:video encoding type-->H.264
</videoCodecType>
<SmartCodec>
<!--ro, opt, object-->
<enabled>
<!--ro, opt, bool-->true
</enabled>
</SmartCodec>
<audioCompressionType>
<!--ro, opt, enum, audio encoding type, subType:string, desc:audio encoding type-->G.711alaw
</audioCompressionType>
<inputPortEnabled>
<!--ro, opt, bool-->true
</inputPortEnabled>
<timeSequenceResolution>
<!--ro, opt, enum, subType:string-->1024*768@60HZ
</timeSequenceResolution>
<inputContent>
<!--ro, opt, enum, subType:string-->default
</inputContent>
<OptimumResolution>
<!--ro, opt, object-->
<enabled>
<!--ro, opt, bool, whether to enable the function-->true
</enabled>
<imageWidth>
<!--ro, opt, int, range:[0,4096]-->0
</imageWidth>
<imageHeight>

```

```

    <!--ro, opt, int, range:[0,4096]-->0
  </imageHeight>
</OptimumResolution>
<inputPortAccessType>
  <!--ro, opt, enum, subType:string-->computer
</inputPortAccessType>
<streamType>
  <!--ro, opt, enum, subType:int-->1
</streamType>
<bitRateType>
  <!--ro, opt, enum, subType:string-->CBR
</bitRateType>
<intervalFrame>
  <!--ro, opt, int, range:[1,400]-->25
</intervalFrame>
<picQuality>
  <!--ro, opt, enum, subType:int-->1
</picQuality>
<resolution>
  <!--ro, opt, enum, subType:string-->352*240
</resolution>
<videoType>
  <!--ro, opt, enum, subType:string-->0
</videoType>
<videoBitRate>
  <!--ro, opt, enum, subType:int-->2
</videoBitRate>
<customBitRate>
  <!--ro, opt, int, range:[128,16384]-->128
</customBitRate>
<frameType>
  <!--ro, opt, enum, subType:string-->P
</frameType>
<frameRate>
  <!--ro, opt, enum, subType:string-->25
</frameRate>
<audioBitRate>
  <!--ro, opt, enum, subType:int-->64
</audioBitRate>
<audioSamplingRate>
  <!--ro, opt, enum, subType:int-->8000
</audioSamplingRate>
<audioCodecType>
  <!--ro, opt, enum, subType:string-->G.711alaw
</audioCodecType>
<audioVolume>
  <!--ro, opt, int, range:[0,100]-->1
</audioVolume>
<localStreamDeepEncodeRatioType>
  <!--ro, opt, enum, subType:string-->low
</localStreamDeepEncodeRatioType>
<localStreamLightEncodeRatioType>
  <!--ro, opt, enum, subType:string-->low
</localStreamLightEncodeRatioType>
<abnormalReason>
  <!--ro, opt, string, range:[0,256], dep:and,{$.VideoInputChannel.signalStatus,eq,abnormal}-->test
</abnormalReason>
<isSupportSingleJoin>
  <!--ro, opt, bool-->true
</isSupportSingleJoin>
<SignalInfoList>
  <!--ro, opt, array, subType:object, range:[0,128]-->
  <SignalInfo>
    <!--ro, req, object-->
    <signalNo>
      <!--ro, opt, int, range:[1,1024]-->0
    </signalNo>
    <name>
      <!--ro, opt, string, range:[0,32]-->test
    </name>
  </SignalInfo>
</SignalInfoList>
<signalResolutionType>
  <!--ro, opt, enum, subType:string-->HD
</signalResolutionType>
</VideoInputChannel>

```

9.7.4.12 Get color parameters of a specific signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, color configuration, attr:version{req, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness, range:[0,100]-->1
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast, range:[0,100]-->1
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, range:[0,100]-->1
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, tint, range:[0,100]-->1
  </hueLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->1
  </sharpnessLevel>
</Color>
```

9.7.4.13 Set color parameters of a specified signal source

Request URL

POT /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, color configuration, attr:version{req, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness, range:[0,100]-->1
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast, range:[0,100]-->1
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, range:[0,100]-->1
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue, range:[0,100]-->1
  </hueLevel>
  <sharpnessLevel>
    <!--opt, int, sharpness, range:[0,100]-->1
  </sharpnessLevel>
</Color>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

9.7.4.14 Get the color configuration capability of a signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/color/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, color configuration capability, attr:version{req, string, protocolVersion}-->
  <brightnessLevel min="0" max="100">
    <!--ro, opt, int, brightness, attr:min{req, int},max{req, int}-->1
  </brightnessLevel>
  <contrastLevel min="0" max="100">
    <!--ro, opt, int, contrast, attr:min{req, int},max{req, int}-->1
  </contrastLevel>
  <saturationLevel min="0" max="100">
    <!--ro, opt, int, saturation, attr:min{req, int},max{req, int}-->1
  </saturationLevel>
  <hueLevel min="0" max="100">
    <!--ro, opt, int, hue, attr:min{req, int},max{req, int}-->1
  </hueLevel>
  <sharpnessLevel min="0" max="100">
    <!--ro, opt, int, sharpness, attr:min{req, int},max{req, int}-->1
  </sharpnessLevel>
</Color>
```

9.7.4.15 Get picture cropping parameters of a specific signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/cutOff

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<InputCutOff xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <leftCutOff>
    <!--ro, opt, int, range:[0,30]-->1
  </leftCutOff>
  <rightCutOff>
    <!--ro, opt, int, range:[0,30]-->1
  </rightCutOff>
  <topCutOff>
    <!--ro, opt, int, range:[0,30]-->1
  </topCutOff>
  <bottomCutOff>
    <!--ro, opt, int, range:[0,30]-->1
  </bottomCutOff>
</InputCutOff>
```

9.7.4.16 Set picture cropping parameters of a specified signal source

Request URL

PUT /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/cutOff

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<InputCutOff xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture cropping parameters, attr:version{req, string, protocolVersion}-->
  <leftCutOff>
    <!--opt, int, left cropping, range:[0,30]-->1
  </leftCutOff>
  <rightCutOff>
    <!--opt, int, right cropping, range:[0,30]-->1
  </rightCutOff>
  <topCutOff>
    <!--opt, int, up cropping, range:[0,30]-->1
  </topCutOff>
  <bottomCutOff>
    <!--opt, int, down cropping, range:[0,30]-->1
  </bottomCutOff>
</InputCutOff>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

9.7.4.17 Get the capability of configuring picture cropping parameters of a signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/cutOff/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<InputCutOff xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, image cropping configuration, attr:version{req, string, protocolVersion}-->
  <leftCutOff min="0" max="30">
    <!--ro, opt, int, left cropping, attr:min{req, int},max{req, int}-->1
  </leftCutOff>
  <rightCutOff min="0" max="30">
    <!--ro, opt, int, right cropping, attr:min{req, int},max{req, int}-->1
  </rightCutOff>
  <topCutOff min="0" max="30">
    <!--ro, opt, int, top cropping, attr:min{req, int},max{req, int}-->1
  </topCutOff>
  <bottomCutOff min="0" max="30">
    <!--ro, opt, int, bottom cropping, attr:min{req, int},max{req, int}-->1
  </bottomCutOff>
  <leftCutOffMultiple>
    <!--ro, opt, int, left cropping ratio-->16
  </leftCutOffMultiple>
  <rightCutOffMultiple>
    <!--ro, opt, int, right cropping ratio-->16
  </rightCutOffMultiple>
  <topCutOffMultiple>
    <!--ro, opt, int, top cropping ratio-->4
  </topCutOffMultiple>
  <bottomCutOffMultiple>
    <!--ro, opt, int, bottom cropping ratio-->4
  </bottomCutOffMultiple>
  <displayCutOffIndicatorLine>
    <!--ro, opt, bool, whether to display the cropping line-->true
  </displayCutOffIndicatorLine>
</InputCutOff>
```

9.7.4.18 Get captured pictures

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/picture

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

Binary Data

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	insufficientResourcesForStitchedSourcePreview	0x4000A828	--

9.7.4.19 Get the capability of configuring image position adjustment parameters of a signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/position/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<InputPosition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, image position adjustment parameters, attr:version{req, string, protocolVersion}-->
  <horizontal min="-30" max="30">
    <!--ro, opt, int, horizontal, attr:min{req, int},max{req, int}-->1
  </horizontal>
  <vertical min="-30" max="30">
    <!--ro, opt, int, vertical, attr:min{req, int},max{req, int}-->1
  </vertical>
  <adjustmentUnit opt="0,8,16,24,32">
    <!--ro, opt, int, tuning unit, attr:opt{req, string}-->0
  </adjustmentUnit>
</InputPosition>
```

9.7.4.20 Set the custom resolution of a specified signal source

Request URL

PUT /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/resolution

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Resolution xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, resolution, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, signal source ID-->0
  </id>
  <resolutionName>
    <!--opt, string, configuration name-->test
  </resolutionName>
  <imageWidth>
    <!--req, int, image width-->0
  </imageWidth>
  <imageHeight>
    <!--req, int, image height-->0
  </imageHeight>
  <refreshRate>
    <!--opt, int-->0
  </refreshRate>
  <colorDepth>
    <!--opt, enum, subType:string-->test
  </colorDepth>
  <scanType>
    <!--opt, enum, subType:string-->intervalScan
  </scanType>
</Resolution>
```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	theResolutionDoesNotMeetTheAlignmentConstraints	0x4000A7D0	--

9.7.4.21 Get the capability of customizing the resolution of a specified signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/resolution/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Resolution xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, resolution settings, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, signal source ID-->0
  </id>
  <resolutionName min="0" max="10">
    <!--ro, opt, string, signal source name, attr:min{req, int},max{req, int}-->test
  </resolutionName>
  <imageWidth min="0" max="10">
    <!--ro, req, int, image width, attr:min{req, int},max{req, int}-->0
  </imageWidth>
  <imageHeight min="0" max="10">
    <!--ro, req, int, image height, attr:min{req, int},max{req, int}-->0
  </imageHeight>
  <refreshRate opt="25,30,60">
    <!--ro, opt, int, refresh rate, attr:opt{req, string}-->0
  </refreshRate>
  <colorDepth opt="8,16,32">
    <!--ro, opt, enum, color, subType:int, attr:opt{req, string}, desc:32, 16, 8-->32
  </colorDepth>
  <scanType opt="progressiveScan,intervalScan">
    <!--ro, opt, scanning type, subType:string, attr:opt{req, string}, desc:"progressiveScan" (progressive scan), "intervalScan" (interLaced scan)-->progressiveScan
  </scanType>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable resolution configuration, attr:opt{opt, string}-->true
  </enabled>
  <imageWidthMultiple opt="2,4,8">
    <!--ro, opt, int, image width multiplier, attr:opt{req, string}-->2
  </imageWidthMultiple>
  <imageHeightMultiple opt="2,4,8">
    <!--ro, opt, int, image height multiplier, attr:opt{req, string}-->2
  </imageHeightMultiple>
  <resolutionMin opt="800*600@30Hz,800*600@60Hz">
    <!--ro, opt, string, min. resolution range, attr:opt{req, string}, desc:obsolete, and replaced by field resolutionCapList-->800*600@30Hz
  </resolutionMin>
  <resolutionMax opt="3840*2160@30Hz,1920*1200@60Hz">
    <!--ro, opt, string, max. resolution range, attr:opt{req, string}, desc:obsolete, and replaced by field resolutionCapList-->3840*2160@30Hz
  </resolutionMax>
  <resolutionCapList>
    <!--ro, opt, array, resolution capability List, subType:object-->
    <resolutionCap>
      <!--ro, opt, object, resolution capability-->
      <refreshRate opt="25,30,60">
        <!--ro, opt, int, frame rate, attr:opt{req, string}-->0
      </refreshRate>
      <resolutionWidth min="800" max="3840">
        <!--ro, opt, string, resolution (width), attr:min{req, int},max{req, int}-->test
      </resolutionWidth>
      <resolutionHeight min="600" max="2160">
        <!--ro, opt, string, resolution (height), attr:min{req, int},max{req, int}-->test
      </resolutionHeight>
      <resolutionArea min="480000" max="100000000">
        <!--ro, opt, string, resolution area range, attr:min{req, int},max{req, int}, desc:resolution area range-->test
      </resolutionArea>
    </resolutionCap>
  </resolutionCapList>
</Resolution>
```

9.7.4.22 Get the OSD configuration capability of a signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/<channelID>/text/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SignalSourceTextList xmlns="http://www.isapi.org/ver20/XMLSchema" size="10" version="2.0">
  <!--ro, opt, array, configuration list, subType:object, attr:size{req, int, protocolVersion},version{req, string}-->
  <SignalSourceText version="2.0">
    <!--ro, opt, object, signal source OSD configuration, attr:version{req, string}-->
    <id>
      <!--ro, req, int, text number-->0
    </id>
    <enable>
      <!--ro, req, bool, whether to enable-->true
    </enable>
    <fontSize opt="1">
      <!--ro, req, int, font size, attr:opt{opt, int}, desc:font size-->0
    </fontSize>
    <backgroundMode opt="transparent">
      <!--ro, req, enum, background mode, subType:string, attr:opt{opt, string}, desc:transparent, coverage-->transparent
    </backgroundMode>
    <positionX min="0" max="100">
      <!--ro, req, int, relative x-coordinate of input source, range:[0,1920], attr:min{opt, int},max{opt, int}, desc:the origin is the upper-left corner of the screen-->0
    </positionX>
    <positionY min="0" max="100">
      <!--ro, req, int, relative y-coordinate of input source, range:[0,1920], attr:min{opt, int},max{opt, int}, desc:the origin is the upper-left corner of the screen-->0
    </positionY>
    <ForegroundColor>
      <!--ro, req, object, text foreground color-->
      <RGB>
        <!--ro, req, int, RGB value-->0
      </RGB>
    </ForegroundColor>
    <BackgroundColor>
      <!--ro, req, object, text background color-->
      <RGB>
        <!--ro, req, int, RGB value-->0
      </RGB>
    </BackgroundColor>
    <textContent>
      <!--ro, req, string, text content-->test
    </textContent>
    <alignment opt="customize">
      <!--ro, opt, enum, alignment mode, subType:string, attr:opt{req, string}, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (Left align), "GB", "adsorption" (adsorption mode)-->alignRight
    </alignment>
  </SignalSourceText>
</SignalSourceTextList>
```

9.7.4.23 Get the video input capability

Request URL

GET /ISAPI/DisplayDev/Video/inputs/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoInputsCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video input capability, attr:version{req, string, protocolVersion}-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->0
  </videoInputPortNums>
  <isSupportColorSetting>
    <!--ro, opt, bool, whether the device supports color settings-->true
  </isSupportColorSetting>
  <isSupportPostionSetting>
    <!--ro, opt, bool, whether the device supports location settings-->true
  </isSupportPostionSetting>
  <isSupportCutOffSetting>
    <!--ro, opt, bool, whether the device supports cropping-->true
  </isSupportCutOffSetting>
  <isSupportPictureCapture>
    <!--ro, opt, bool, whether the device supports capturing pictures-->true
  </isSupportPictureCapture>
  <isSupportText>
    <!--ro, opt, bool, whether the device supports text-->true
  </isSupportText>
  <encodeOSDInputSignalType opt="HDMI,DP">
```

```

<!--ro, opt, string, input signal type, attr:opt{req, string}-->test
</encodeOSDInputSignalType>
<SupportSelfdefineResolution>
<!--ro, opt, object, custom resolution supported by DP (display port) input source-->
<signalType opt="DVI,VGA,HDMI,DP">
<!--ro, req, string, signal type, attr:opt{req, string}-->test
</signalType>
<supportedDPResolution opt="640*480@60,800*600@60">
<!--ro, opt, string, supported DP resolution, attr:opt{req, string}-->test
</supportedDPResolution>
</SupportSelfdefineResolution>
<isSupportEDIDResolution>
<!--ro, opt, bool, whether it supports edid resolution-->true
</isSupportEDIDResolution>
<isSupportJoinSignalCfg>
<!--ro, opt, bool, whether the device supports signal source splicing-->true
</isSupportJoinSignalCfg>
<SupportAudioCfg>
<!--ro, opt, object, audio settings-->
<signalType opt="DVI,VGA,HDMI,DP">
<!--ro, req, string, supported encoding embedded audio / independent audio, attr:opt{req, string}-->test
</signalType>
</SupportAudioCfg>
<inputPortEnabled opt="true,false">
<!--ro, opt, bool, whether to enable input port, attr:opt{req, string}-->true
</inputPortEnabled>
<timeSequenceResolution
opt="1024*768@60Hz,720P@50Hz,720P@60Hz#720P@60Hz,1366*768@60Hz#1366,1440*900@60Hz,1280*1024@60Hz,1680*1050@60,1080P@50Hz,1080P@60Hz,3840*2160@30Hz">
<!--ro, opt, enum, video timing, subType:string, attr:opt{req, string}, desc:1024*768@60Hz, 720P@50Hz, 720P@60Hz#720P@60Hz, 1366*768@60Hz#1366,
1440*900@60Hz, 1280*1024@60Hz, 1680*1050@60, 1080P@50Hz, 1080P@60Hz, 3840*2160@30Hz-->1024*768@60Hz
</timeSequenceResolution>
<inputContent opt="default,broadcastVideo,wirelessMirroring">
<!--ro, opt, enum, input content, subType:string, attr:opt{req, string}, desc:"default", "broadcastVideo" (broadcast video), "wirelessMirroring"
(wireless screen mirroring)-->default
</inputContent>
<OptimumResolution>
<!--ro, opt, object, resolution self-adaption-->
<enabled opt="true,false">
<!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
</enabled>
<ResolutionList>
<!--ro, opt, object, resolution, desc:LED sending cards support any resolution with custom width and height-->
<imageWidth min="0" max="4096">
<!--ro, opt, int, resolution (width), attr:min{req, int},max{req, int}-->0
</imageWidth>
<imageHeight min="0" max="4096">
<!--ro, opt, int, resolution (height), attr:min{req, int},max{req, int}-->0
</imageHeight>
</ResolutionList>
<OptimumResolution>
<inputPortAccessType opt="computer,notebook,mobilePhone,tablet,wirelessMirroring,highCamera,recordHost,broadcastVideo,universal">
<!--ro, opt, enum, input port access type, subType:string, attr:opt{req, string}, desc:"computer", "notebook", "mobilePhone" (cellphone), "tablet",
"wirelessMirroring" (wireless dongle), "highCamera" (document camera), "recordHost" (video recorder), "broadcastVideo" (broadcast)-->computer
</inputPortAccessType>
<isSupportVideoInputSubtrack>
<!--ro, opt, bool, whether the device supports splitting an ultra HD input signal into multiple tracks,
desc:ISAPI/DisplayDev/Video/inputs/subtrack/capabilities?format=json-->true
</isSupportVideoInputSubtrack>
<streamType opt="1,2">
<!--ro, opt, int, stream type, attr:opt{req, string}-->1
</streamType>
<bitRateType opt="CBR,VBR">
<!--ro, opt, string, video bitrate type, attr:opt{req, string}-->CBR
</bitRateType>
<intervalFrame>
<!--ro, opt, int, I frame interval, range:[1,400]-->25
</intervalFrame>
<picQuality opt="1,2,3,4,5,6">
<!--ro, opt, int, picture quality, attr:opt{req, string}-->1
</picQuality>
<resolution
opt="352*240,352*288,176*120,176*144,704*288,704*240,528*384,528*320,704*576,704*480,960*576,960*480,1280*720,1280*960,1600*1200,1080P,1080I,1024*768,1360*7
68,1366*768,1280*1024,1400*1050,1440*900,1680*1050">
<!--ro, opt, enum, resolution, subType:string, attr:opt{req, string}, desc:352*240#CIF (352*240), 352*288#CIF (352*288), 176*120#QCIF (176*120),
176*144#QCIF (176*144), 704*288#CIF (704*288), 704*240#CIF (704*240), 528*384#DCIF (528*384), 528*320#DCIF (528*320), 704*576#CIF (704*576), 704*480#CIF
(704*480), 960*576#HD1 (960*576), 960*480#HD1 (960*480), 1280*720#720P (1280*720), 1280*960#XGA (1280*960), 1600*1200#UXGA (1600*1200), 1080P#1080P
(1920*1080), 1080I#1080I (1920*1080), 1024*768#XGA (1024*768), 1360*768#XGA (1360*768), 1366*768#XGA (1366*768), 1280*1024#SXGA (1280*1024),
1400*1050#SXGA (1400*1050), 1440*900#SXGA(1440*900), 1680*1050#SXGA+(1680*1050)-->352*240
</resolution>
<videoType opt="0,1,2">
<!--ro, opt, int, video type, attr:opt{req, string}-->1
</videoType>
<videoBitRate opt="2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23">
<!--ro, opt, int, maximum bit rate, attr:opt{req, string}-->1
</videoBitRate>
<customBitRate>
<!--ro, opt, int, custom bit rate, range:[128,16384], desc:from 128 to 16384 kbps-->128
</customBitRate>
<frameType opt="P">
<!--ro, opt, enum, frame type, subType:string, attr:opt{req, string}, desc:P (P-frame)-->P
</frameType>
<frameRate opt="full,1/16,1/8,1/4,1/2,1,2,4,6,8,10,12,15,16,18,20,22,25,30">
<!--ro, opt, enum, video frame rate, subType:string, attr:opt{req, string}, desc:full (full frame rate), 1/16, 1/8, 1/4, 1/2, 1, 2, 4, 6, 8, 10, 12, 15,
16, 18, 20, 22, 25, 30-->25

```

```

</frameRate>
<audioBitRate opt="64">
  <!--ro, opt, enum, audio bitrate, subType:int, attr:opt{req, int}, desc:64 (64 kbps)-->64
</audioBitRate>
<audioSamplingRate opt="8000,16000,32000,44100,48000">
  <!--ro, opt, enum, audio sampling rate, subType:int, attr:opt{req, string}, desc:unit: Hz. "8000" (8 kHz), "16000" (16 kHz), "32000" (32 kHz), "44100"
(44.1 kHz), "48000" (48 kHz)-->8000
</audioSamplingRate>
<audioCodecType opt="G.711alaw,G.711ulaw,G.722.1,AAC">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->G.711alaw
</audioCodecType>
<audioVolume min="0" max="100">
  <!--ro, opt, int, volume level, range:[0,100], attr:min{req, int},max{req, int}-->1
</audioVolume>
<audioCompressionType opt="G.711alaw,G.711ulaw,G.722.1,AAC">
  <!--ro, opt, enum, audio encoding type, subType:string, attr:opt{req, string}, desc:G.711alaw, G.711ulaw, G.722.1-->G.711alaw
</audioCompressionType>
<isSupportVideoPreview>
  <!--ro, opt, bool, whether the device supports video live view of signal source, desc:get live streams via RTSP in pull mode-->true
</isSupportVideoPreview>
<isSupportSubStreamVideoPreview>
  <!--ro, opt, bool, whether the device supports sub stream live view of signal source, desc:get live streams via RTSP in pull mode-->true
</isSupportSubStreamVideoPreview>
<localStreamDeepEncodeRatioType opt="low,middle,high">
  <!--ro, opt, enum, deep compression rate type of local source, subType:string, attr:opt{req, string}, desc:"Low", "middle", "high"-->low
</localStreamDeepEncodeRatioType>
<localStreamLightEncodeRatioType opt="low,middle,high">
  <!--ro, opt, enum, shallow compression rate type of local source, subType:string, attr:opt{req, string}, desc:"Low", "middle", "high"-->low
</localStreamLightEncodeRatioType>
<name min="1" max="32">
  <!--ro, opt, string, signal source name, attr:min{req, int},max{req, int}-->1
</name>
<isSupportJoinSignalResolution>
  <!--ro, opt, bool, whether the device supports custom resolution configuration of splicing source (fusion source), desc:call API
(/ISAPI/DisplayDev/Video/inputs/channels/<channelID>/resolution) to configure custom resolution of splicing source (fusion source), the <channelID>
indicates the signal source ID after fusion-->true
</isSupportJoinSignalResolution>
<bitDepth opt="8,10">
  <!--ro, opt, enum, bit depth, subType:int, attr:opt{req, string}, desc:the number of bits used for each color channel (such as red, green, and blue). 8
(8-bit), 10 (10-bit)-->8
</bitDepth>
</VideoInputsCap>

```

9.7.4.24 Get splicing configuration of all signal resources

Request URL

GET /ISAPI/DisplayDev/Video/inputs/joinSignal

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<JoinSignalList xmlns="http://www.isapi.org/ver20/XMLSchema" size="10" version="2.0">
  <!--ro, opt, array, splicing configuration of all signal resources, subType:object, attr:size{req, int, protocolVersion},version{req, string}-->
  <JoinSignal>
    <!--ro, opt, object, signal source splicing-->
    <id>
      <!--ro, req, int, splicing No.-->0
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <joinName>
      <!--ro, req, string, joint source name-->test
    </joinName>
    <camMode>
      <!--ro, req, int, joint source type-->0
    </camMode>
    <rows>
      <!--ro, req, int, number of spliced rows-->0
    </rows>
    <columns>
      <!--ro, req, int, number of spliced columns-->0
    </columns>
    <SignalList>
      <!--ro, req, object, signal source list designed for splicing-->
      <signalNo>
        <!--ro, opt, int, signal source No.-->0
      </signalNo>
    </SignalList>
    <joinSignalNo>
      <!--ro, req, int, spliced signal source No. (read-only)-->1
    </joinSignalNo>
    <SignalInfoList>
      <!--ro, opt, array, list of spliced signal source information, subType:object-->
      <SignalInfo>
        <!--ro, req, object, signal source information-->
        <signalNo>
          <!--ro, opt, int, signal source No.-->0
        </signalNo>
        <name>
          <!--ro, opt, string, signal source name-->test
        </name>
        <portType>
          <!--ro, opt, enum, signal source type, subType:string, desc:SDI#SDI, OPT#OPT, VGA#VGA, HDMI#HDMI, YpbPr#YpbPr, Matrix#Matrix, Joint#Joint,
          DVI#DVI, TVI#TVI, innerCamera#built-in camera, HDMI/Hdbase-T#HDMI and Hdbase-T simultaneous output, Hdbase-T#Hdbase-T, DP#DP-->SDI
        </portType>
        <hostname>
          <!--ro, opt, enum, address type, subType:string, desc:"ipv4Address" (IPv4 address), "ipv6Address" (IPv6 address)-->ipv4Address
        </hostname>
        <ipv4Address>
          <!--ro, opt, string, ipv4, dep:and,{$.JoinSignalList[*].JoinSignal.SignalInfoList[*].SignalInfo.hostname,eq,ipv4Address}-->192.168.1.1
        </ipv4Address>
        <ipv6Address>
          <!--ro, opt, string, ipv6, dep:and,{$.JoinSignalList[*].JoinSignal.SignalInfoList[*].SignalInfo.hostname,eq,ipv6Address}-->fe80::4ba5:790e
        </ipv6Address>
        <port>
          <!--ro, opt, int, port No., range:[0,65535]-->1
        </port>
        <trackNum>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int-->1
          </x>
          <y>
            <!--ro, req, int-->1
          </y>
        </trackNum>
        <signalStatus>
          <!--ro, opt, enum, signal source status, subType:string, desc:"signal" (with signal input), "noSignal" (no signal), "abnormal"-->signal
        </signalStatus>
      </SignalInfo>
    </SignalInfoList>
  </JoinSignal>
</JoinSignalList>

```

9.7.4.25 Set jointing parameters of a specified signal source

Request URL

PUT /ISAPI/DisplayDev/Video/inputs/joinSignal/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<JoinSignal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, signal source jointing parameters, attr:version{req, string, protocolVersion}-->
  <id min="0" max="10">
    <!--req, int, splicing No., attr:min{req, int},max{req, int}-->0
  </id>
  <enabled opt="true,false">
    <!--req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <joinName max="10">
    <!--req, string, joint source name, attr:max{req, int}-->test
  </joinName>
  <camMode>
    <!--req, int, joint source type-->0
  </camMode>
  <rows min="0" max="100">
    <!--req, int, number of spliced rows, attr:min{req, int},max{req, int}-->0
  </rows>
  <columns min="0" max="100">
    <!--req, int, number of spliced columns, attr:min{req, int},max{req, int}-->0
  </columns>
  <SignalList size="10">
    <!--req, object, signal source List designed for splicing, attr:size{req, int}-->
    <signalNo>
      <!--opt, int, signal source No.-->0
    </signalNo>
  </SignalList>
  <joinSignalNo>
    <!--req, int, spliced signal source No. (read-only)-->1
  </joinSignalNo>
</JoinSignal>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode
4	Invalid Operation	splicingIsNotAllowedForSignalSourcesThatHaveBeenDisplayedOnVideoWall	0x4000A164
4	Invalid Operation	deletingSignalSourcesThatHaveBeenDisplayedOnVideoWallIsNotAllowed	0x4000A165
4	Invalid Operation	numberOfSplicedSignalSourcesExceededLimit	0x4000A166
4	Invalid Operation	uhdAndHdChannelsDoNotSupportMixedSplicing	0x4000A7B9
4	Invalid Operation	lowLatencyModeProhibitsSplicingSourcesIntoThePrimaryStream	0x4000A80C
4	Invalid Operation	deletionOfTheSplicedPrimarySourcesIsDisabledInLowLatencyMode	0x4000A80D
4	Invalid Operation	sourceLayerInSyncLockCannotBeStitched	0x4000A823
4	Invalid Operation	stitchedSourcesInSyncLockCannotBeDeleted	0x4000A824

9.7.4.26 Get splicing parameters of a signal source

Request URL

GET /ISAPI/DisplayDev/Video/inputs/joinSignal/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<JoinSignal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, signal source splicing, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, splicing No.-->0
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <joinName>
    <!--ro, req, string, joint source name-->test
  </joinName>
  <camMode>
    <!--ro, req, int, joint source type-->0
  </camMode>
  <rows>
    <!--ro, req, int, number of spliced rows-->0
  </rows>
  <columns>
    <!--ro, req, int, number of spliced columns-->0
  </columns>
  <SignalList>
    <!--ro, req, object, signal source List designed for splicing-->
    <signalNo>
      <!--ro, opt, int, signal source No.-->0
    </signalNo>
  </SignalList>
  <joinSignalNo>
    <!--ro, req, int, spliced signal source No. (read-only)-->1
  </joinSignalNo>
  <SignalInfoList>
    <!--ro, opt, array, List of spliced signal source information, subType:object-->
    <SignalInfo>
      <!--ro, req, object, signal source information-->
      <signalNo>
        <!--ro, opt, int, signal source No.-->0
      </signalNo>
      <name>
        <!--ro, opt, string, signal source name-->test
      </name>
      <portType>
        <!--ro, opt, enum, signal source type, subType:string, desc:SDI#SDI, OPT#OPT, VGA#VGA, HDMI#HDMI, YpbPr#YpbPr, Matrix#Matrix, Joint#Joint, DVI#DVI,
        TVI#TVI, innerCamera#built-in camera, HDMI/HDBase-T#HDMI and HDBase-T simultaneous output, HDBase-T#HDBase-T, DP#DP-->SDI
      </portType>
      <hostname>
        <!--ro, opt, enum, address type, subType:string, desc:"ipv4Address" (IPv4 address), "ipv6Address" (IPv6 address)-->ipv4Address
      </hostname>
      <ipv4Address>
        <!--ro, opt, string, IPv4 address, dep:and,{$.JoinSignal.SignalInfoList[*].SignalInfo.hostname,eq,ipv4Address}-->192.168.1.1
      </ipv4Address>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address, dep:and,{$.JoinSignal.SignalInfoList[*].SignalInfo.hostname,eq,ipv6Address}-->fe80::4ba5:790e
      </ipv6Address>
      <port>
        <!--ro, opt, int, port No., range:[0,65535]-->1
      </port>
      <trackNum>
        <!--ro, opt, object-->
        <x>
          <!--ro, req, int-->1
        </x>
        <y>
          <!--ro, req, int-->1
        </y>
      </trackNum>
      <signalStatus>
        <!--ro, opt, enum, signal source status, subType:string, desc:"signal" (with signal input), "noSignal" (no signal), "abnormal"-->signal
      </signalStatus>
    </SignalInfo>
  </SignalInfoList>
</JoinSignal>

```

9.7.4.27 Get signal source splicing capability

Request URL

GET /ISAPI/DisplayDev/Video/inputs/joinSignal/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<JoinSignalList xmlns="http://www.isapi.org/ver20/XMLSchema" size="10" version="2.0">
  <!--ro, opt, object, capability of signal source splicing configuration, attr:size{req, int, protocolVersion},version{req, string}-->
  <JoinSignal version="2.0">
    <!--ro, opt, object, signal source splicing, attr:version{req, string}-->
    <id min="0" max="10">
      <!--ro, req, int, splicing No., attr:min{req, int},max{req, int}-->0
    </id>
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <joinName max="10">
      <!--ro, req, string, joint source name, attr:max{req, int}-->test
    </joinName>
    <camMode>
      <!--ro, req, int, joint source type-->0
    </camMode>
    <rows min="0" max="100">
      <!--ro, req, int, number of spliced rows, attr:min{req, int},max{req, int}-->0
    </rows>
    <columns min="0" max="100">
      <!--ro, req, int, number of spliced columns, attr:min{req, int},max{req, int}-->0
    </columns>
    <SignalList size="10">
      <!--ro, opt, object, signal source list designed for splicing, attr:size{req, int}-->
      <signalNo>
        <!--ro, opt, int, signal source No.-->0
      </signalNo>
    </SignalList>
    <joinSignalNo>
      <!--ro, opt, int, spliced signal source No. (read-only)-->1
    </joinSignalNo>
  </JoinSignal>
</JoinSignalList>
```

9.7.4.28 Get all video streams' parameters

Request URL

GET /ISAPI/DisplayDev/Video/streaming/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<StreamInputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, all video stream configurations, attr:version{req, string, protocolVersion}-->
  <StreamInputChannel version="2.0">
    <!--ro, opt, object, video stream input channel, attr:version{req, string}-->
    <id>
      <!--ro, req, string, stream ID-->test
    </id>
    <name>
      <!--ro, opt, string, stream name-->test
    </name>
    <group>
      <!--ro, opt, string, stream group-->test
    </group>
    <startDecoding>
      <!--ro, req, bool-->true
    </startDecoding>
    <StreamInput>
      <!--ro, opt, object, video stream input-->
      <streamInputMode>
        <!--ro, req, enum, stream type, subType:string, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
      </streamInputMode>
      <StreamInputRealtime>
        <!--ro, opt, object, real-time stream information-->
        <durationInUnit>
          <!--ro, opt, int, decoding display interval of real-time stream, unit:s-->1
        </durationInUnit>
        <StreamRealtimeUnitList>
          <!--ro, opt, object, real-time stream list-->
          <StreamRealtimeUnit>
            <!--ro, opt, object, single real-time stream unit-->
            <streamType>
              <!--ro, req, enum, stream type, subType:string, desc:"in URL", "by dns", "by domain"-->by domain
            </streamType>
          </StreamRealtimeUnit>
        </StreamRealtimeUnitList>
      </StreamInputRealtime>
    </StreamInput>
  </StreamInputChannel>
</StreamInputChannelList>
```

```

<StreamInURL>
  <!--ro, opt, object, URL stream-->
  <URL>
    <!--ro, req, string, URL-->test
  </URL>
</StreamInURL>
<StreamByDdns>
  <!--ro, opt, object, DDNS-->
  <DdnsServerInfo>
    <!--ro, req, object, DDNS server information-->
    <domain>
      <!--ro, req, string, domain name-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <ddnsType>
      <!--ro, req, enum, DDNS type, subType:string, desc:"hiDdns", "noIp"-->hiDdns
    </ddnsType>
    <username>
      <!--ro, req, string, user name.-->test
    </username>
    <password>
      <!--ro, req, string, password-->test
    </password>
  </DdnsServerInfo>
  <EncodeDevInfo>
    <!--ro, req, object, encoding device information-->
    <domain>
      <!--ro, req, string, domain name-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <transmitProtocol>
      <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
    </transmitProtocol>
    <protocol>
      <!--ro, opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
    </protocol>
    <username>
      <!--ro, req, string, user name.-->test
    </username>
    <password>
      <!--ro, req, string, password-->test
    </password>
    <channelMode>
      <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType>
      <!--ro, opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
    </channelType>
    <channelZero>
      <!--ro, opt, int, channelMode is zero-->1
    </channelZero>
    <channelNormal>
      <!--ro, opt, int, channelMode is normal-->1
    </channelNormal>
    <channelStreaming>
      <!--ro, opt, string, channelMode is streaming-->test
    </channelStreaming>
    <channelDistributed>
      <!--ro, opt, int, channelMode is distributed-->1
    </channelDistributed>
  </EncodeDevInfo>
  <MediaGatewayInfo>
    <!--ro, opt, object, media gateway information-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <domain>
      <!--ro, req, string, domain name-->test
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <transmitProtocol>
      <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
    </transmitProtocol>
  </MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--ro, opt, object, streamType is by domain-->
  <EncodeDevInfo>
    <!--ro, req, object, device information-->
    <domain>
      <!--ro, req, string, domain name-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <transmitProtocol>

```

```

<!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
>tcp
</transmitProtocol>
<protocol>
<!--ro, opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
</protocol>
<username>
<!--ro, req, string, user name.-->test
</username>
<password>
<!--ro, req, string, password-->test
</password>
<channelMode>
<!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType>
<!--ro, opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
</channelType>
<channelZero>
<!--ro, opt, int, channelMode is zero-->1
</channelZero>
<channelNormal>
<!--ro, opt, int, channelMode is normal-->1
</channelNormal>
<channelStreaming>
<!--ro, opt, string, channelMode is streaming-->test
</channelStreaming>
<channelDistributed>
<!--ro, opt, int, channelMode is distributed-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
<!--ro, opt, object, media gateway information-->
<enabled>
<!--ro, req, bool, whether to enable-->true
</enabled>
<domain>
<!--ro, req, string, domain name-->ipv4
</domain>
<port>
<!--ro, req, int, port No.-->0
</port>
<transmitProtocol>
<!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
>tcp
</transmitProtocol>
</MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
<!--ro, opt, object, playback stream information-->
<playbackMode>
<!--ro, opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file name
</playbackMode>
<EncodeDevInfo>
<!--ro, opt, object, information of device in playback-->
<domain>
<!--ro, req, string, domain name-->ipv4
</domain>
<port>
<!--ro, req, int, port No.-->0
</port>
<transmitProtocol>
<!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
</transmitProtocol>
<protocol>
<!--ro, opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
</protocol>
<username>
<!--ro, opt, string, user name.-->test
</username>
<password>
<!--ro, opt, string, password-->test
</password>
<channelMode>
<!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType>
<!--ro, opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
</channelType>
<channelZero>
<!--ro, opt, int, zero-->0
</channelZero>
<channelNormal>
<!--ro, opt, int, normal-->0
</channelNormal>
<channelStreaming>
<!--ro, opt, string, streaming-->test
</channelStreaming>
<channelDistributed>
<!--ro, opt, int, distributed-->1

```

```

</channelDistributed>
</EncodeDevInfo>
<fileName>
  <!--ro, opt, string, file name-->test
</fileName>
<timeRange>
  <!--ro, opt, object, time range-->
  <beginTime>
    <!--ro, opt, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--ro, opt, time, end time-->00:00:00+08:00
  </endTime>
</timeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--ro, opt, bool, whether to enable stream encryption-->true
</streamEncryptEnable>
<streamPassword>
  <!--ro, opt, string, stream password-->test
</streamPassword>
</StreamInput>
</StreamInputChannel>
</StreamInputChannelList>

```

9.7.4.29 Set all video stream parameters

Request URL

PUT /ISAPI/DisplayDev/Video/streaming/channels

Query Parameter

None

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<StreamInputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, all video stream configurations, attr:version{req, string, protocolVersion}-->
  <StreamInputChannel version="2.0">
    <!--opt, object, video stream input channel, attr:version{req, string}-->
    <id>
      <!--req, string, stream ID-->test
    </id>
    <name>
      <!--opt, string, stream name-->test
    </name>
    <group>
      <!--opt, string, stream group-->test
    </group>
    <startDecoding>
      <!--req, bool, whether to start decoding-->true
    </startDecoding>
    <StreamInput>
      <!--opt, object, video stream input-->
      <streamInputMode>
        <!--req, enum, stream type, subType:string, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
      </streamInputMode>
      <StreamInputRealtime>
        <!--opt, object, real-time stream information-->
        <durationInUnit>
          <!--opt, int, decoding display interval of real-time stream, unit:s-->1
        </durationInUnit>
        <StreamRealtimeUnitList>
          <!--opt, object, real-time stream List-->
          <StreamRealtimeUnit>
            <!--opt, object, single real-time stream unit-->
            <streamType>
              <!--req, enum, stream type, subType:string, desc:"in URL", "by ddns", "by domain"-->by domain
            </streamType>
            <StreamInURL>
              <!--opt, object, URL stream-->
              <URL>
                <!--req, string, URL-->test
              </URL>
            </StreamInURL>
            <StreamByDdns>
              <!--opt, object, DDNS-->
              <DdnsServerInfo>
                <!--req, object, DDNS server information-->
                <domain>
                  <!--req, string, domain name-->ipv4
                </domain>
                <port>
                  <!--req, int, port No.-->0
                </port>
                <ddnsType>

```

```

        <!--req, enum, DDNS type, subType:string, desc:"hiDdns", "noIp"-->hiDdns
    </ddnsType>
    <username>
        <!--req, string, user name-->test
    </username>
    <password>
        <!--req, string, password-->test
    </password>
</DdnsServerInfo>
<EncodeDevInfo>
    <!--req, object, encoding device information-->
    <domain>
        <!--req, string, domain name-->ipv4
    </domain>
    <port>
        <!--req, int, port No.-->0
    </port>
    <transmitProtocol>
        <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
    </transmitProtocol>
    <protocol>
        <!--opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
    </protocol>
    <username>
        <!--req, string, user name-->test
    </username>
    <password>
        <!--req, string, password-->test
    </password>
    <channelMode>
        <!--req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType>
        <!--opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
    </channelType>
    <channelZero>
        <!--opt, int, channelMode is zero-->1
    </channelZero>
    <channelNormal>
        <!--opt, int, channelMode is normal-->1
    </channelNormal>
    <channelStreaming>
        <!--opt, string, channelMode is streaming-->test
    </channelStreaming>
    <channelDistributed>
        <!--opt, int, channelMode is distributed-->1
    </channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
    <!--opt, object, media gateway information-->
    <enabled>
        <!--req, bool, whether to enable-->true
    </enabled>
    <domain>
        <!--req, string, domain name-->test
    </domain>
    <port>
        <!--req, int, port No.-->0
    </port>
    <transmitProtocol>
        <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
    </transmitProtocol>
</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
    <!--opt, object, streamType is by domain-->
    <EncodeDevInfo>
        <!--req, object, device information-->
        <domain>
            <!--req, string, domain name-->ipv4
        </domain>
        <port>
            <!--req, int, port No.-->0
        </port>
        <transmitProtocol>
            <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
        </transmitProtocol>
        <protocol>
            <!--opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
        </protocol>
        <username>
            <!--req, string, user name-->test
        </username>
        <password>
            <!--req, string, password-->test
        </password>
        <channelMode>
            <!--req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
        </channelMode>
        <channelType>
            <!--opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
        </channelType>
        <channelZero>
            <!--opt, int, channelMode is zero-->1
        </channelZero>
        <channelNormal>
            <!--opt, int, channelMode is normal-->1
        </channelNormal>
        <channelStreaming>
            <!--opt, string, channelMode is streaming-->test
        </channelStreaming>
        <channelDistributed>
            <!--opt, int, channelMode is distributed-->1
        </channelDistributed>
    </EncodeDevInfo>
    <MediaGatewayInfo>
        <!--opt, object, media gateway information-->
        <enabled>
            <!--req, bool, whether to enable-->true
        </enabled>
        <domain>
            <!--req, string, domain name-->test
        </domain>
        <port>
            <!--req, int, port No.-->0
        </port>
        <transmitProtocol>
            <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
        </transmitProtocol>
    </MediaGatewayInfo>
</StreamByDomain>

```



```

<!--opt, int, channelMode is zero-->1
</channelZero>
<channelNormal>
  <!--opt, int, channelMode is normal-->1
</channelNormal>
<channelStreaming>
  <!--opt, string, channelMode is streaming-->test
</channelStreaming>
<channelDistributed>
  <!--opt, int, channelMode is distributed-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--opt, object, media gateway information-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <domain>
    <!--req, string, domain name-->ipv4
  </domain>
  <port>
    <!--req, int, port No.-->0
  </port>
  <transmitProtocol>
    <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
  </transmitProtocol>
</MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
  <!--opt, object, playback stream information-->
  <playbackMode>
    <!--opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file name
  </playbackMode>
  <EncodeDevInfo>
    <!--opt, object, information of device in playback-->
    <domain>
      <!--req, string, domain name-->ipv4
    </domain>
    <port>
      <!--req, int, port No.-->0
    </port>
    <transmitProtocol>
      <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
    </transmitProtocol>
    <protocol>
      <!--opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
    </protocol>
    <username>
      <!--opt, string, user name-->test
    </username>
    <password>
      <!--opt, string, password-->test
    </password>
    <channelMode>
      <!--req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType>
      <!--opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
    </channelType>
    <channelZero>
      <!--opt, int, zero-->0
    </channelZero>
    <channelNormal>
      <!--opt, int, normal-->0
    </channelNormal>
    <channelStreaming>
      <!--opt, string, streaming-->test
    </channelStreaming>
    <channelDistributed>
      <!--opt, int, distributed-->1
    </channelDistributed>
  </EncodeDevInfo>
  <fileName>
    <!--opt, string, file name-->test
  </fileName>
  <TimeRange>
    <!--opt, object, time range-->
    <beginTime>
      <!--opt, time, start time-->00:00:00+08:00
    </beginTime>
    <endTime>
      <!--opt, time, end time-->00:00:00+08:00
    </endTime>
  </TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--opt, bool, whether to enable stream encryption-->true
</streamEncryptEnable>
<streamPassword>
  <!--opt, string, stream password-->test
</streamPassword>

```

```
</StreamInput>
</StreamInputChannel>
</StreamInputChannelList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK", "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:read-only,describe the error reason in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXMLFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.4.30 Set parameters of a specific video stream

Request URL

PUT /ISAPI/DisplayDev/Video/streaming/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<StreamInputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video stream input channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, stream ID-->test
  </id>
  <name>
    <!--opt, string, stream name-->test
  </name>
  <group>
    <!--opt, string, stream group-->test
  </group>
  <startDecoding>
    <!--req, bool, whether to start decoding-->true
  </startDecoding>
  <StreamInput>
    <!--opt, object, video stream input-->
    <streamInputMode>
      <!--req, enum, stream type, subType:string, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
    </streamInputMode>
    <StreamInputRealtime>
      <!--opt, object, real-time stream information-->
      <durationInUnit>
        <!--opt, int, decoding display interval of real-time stream, unit:s-->1
      </durationInUnit>
      <StreamRealtimeUnitList>
        <!--opt, object, real-time stream list-->
        <StreamRealtimeUnit>
          <!--opt, object, single real-time stream unit-->
          <streamType>
            <!--req, enum, stream type, subType:string, desc:"in URL", "by ddns", "by domain"-->by domain
          </streamType>
          <StreamInURL>
```

```

<!--opt, object, URL stream-->
<URL>
<!--req, string, URL-->test
</URL>
</StreamInURL>
<StreamByDdns>
<!--opt, object, DDNS-->
<DdnsServerInfo>
<!--req, object, DDNS server information-->
<domain>
<!--req, string, domain name-->ipv4
</domain>
<port>
<!--req, int, port-->0
</port>
<ddnsType>
<!--req, enum, DDNS type, subType:string, desc:"hiDdns", "noIp"-->hiDdns
</ddnsType>
<username>
<!--req, string, user name-->test
</username>
<password>
<!--req, string, password-->test
</password>
</DdnsServerInfo>
<EncodeDevInfo>
<!--req, object, encoding device information-->
<domain>
<!--req, string, domain name-->ipv4
</domain>
<port>
<!--req, int, port-->0
</port>
<transmitProtocol>
<!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
</transmitProtocol>
<protocol>
<!--opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
</protocol>
<username>
<!--req, string, user name-->test
</username>
<password>
<!--req, string, password-->test
</password>
<channelMode>
<!--req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType>
<!--opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
</channelType>
<channelZero>
<!--opt, int, channelMode is zero-->1
</channelZero>
<channelNormal>
<!--opt, int, channelMode is normal-->1
</channelNormal>
<channelStreaming>
<!--opt, string, channelMode is streaming-->test
</channelStreaming>
<channelDistributed>
<!--opt, int, channelMode is distributed-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
<!--opt, object, media gateway information-->
<enabled>
<!--req, bool, whether to enable-->true
</enabled>
<domain>
<!--req, string, domain name-->test
</domain>
<port>
<!--req, int, port-->0
</port>
<transmitProtocol>
<!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
</transmitProtocol>
</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
<!--opt, object, streamType is by domain-->
<EncodeDevInfo>
<!--req, object, device information-->
<domain>
<!--req, string, domain name-->ipv4
</domain>
<port>
<!--req, int, port-->0
</port>
<transmitProtocol>
<!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
</transmitProtocol>
</EncodeDevInfo>

```

```

<!--opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
</protocol>
<username>
  <!--req, string, user name-->test
</username>
<password>
  <!--req, string, password-->test
</password>
<channelMode>
  <!--req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType>
  <!--opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
</channelType>
<channelZero>
  <!--opt, int, channelMode is zero-->1
</channelZero>
<channelNormal>
  <!--opt, int, channelMode is normal-->1
</channelNormal>
<channelStreaming>
  <!--opt, string, channelMode is streaming-->test
</channelStreaming>
<channelDistributed>
  <!--opt, int, channelMode is distributed-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--opt, object, media gateway information-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <domain>
    <!--req, string, domain name-->ipv4
  </domain>
  <port>
    <!--req, int, port-->0
  </port>
  <transmitProtocol>
    <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
  </transmitProtocol>
</MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
  <!--opt, object, playback stream information-->
  <playbackMode>
    <!--opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file name
  </playbackMode>
</StreamInputPlayback>
<EncodeDevInfo>
  <!--opt, object, information of device in playback-->
  <domain>
    <!--req, string, domain name-->ipv4
  </domain>
  <port>
    <!--req, int, port-->0
  </port>
  <transmitProtocol>
    <!--req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
  </transmitProtocol>
</protocol>
  <!--opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
</protocol>
<username>
  <!--opt, string, user name-->test
</username>
<password>
  <!--opt, string, password-->test
</password>
<channelMode>
  <!--req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType>
  <!--opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
</channelType>
<channelZero>
  <!--opt, int, zero-->0
</channelZero>
<channelNormal>
  <!--opt, int, normal-->0
</channelNormal>
<channelStreaming>
  <!--opt, string, streaming-->test
</channelStreaming>
<channelDistributed>
  <!--opt, int, distributed-->1
</channelDistributed>
</EncodeDevInfo>
<fileName>
  <!--opt, string, file name-->test
</fileName>

```

```

<TimeRange>
  <!--opt, object, time range-->
  <beginTime>
    <!--opt, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--opt, time, end time-->00:00:00+08:00
  </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--opt, bool, whether to enable stream encryption-->true
</streamEncryptEnable>
<streamPassword>
  <!--opt, string, stream password-->test
</streamPassword>
</StreamInput>
</StreamInputChannel>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXMLFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

9.7.4.31 Delete parameters of a specific video stream

Request URL

DELETE /ISAPI/DisplayDev/Video/streaming/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.4.32 Get parameters of a specified video stream

Request URL

GET /ISAPI/DisplayDev/Video/streaming/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<StreamInputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video stream input channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, stream ID-->test
  </id>
  <name>
    <!--ro, opt, string, stream name-->test
  </name>
  <group>
    <!--ro, opt, string, stream group-->test
  </group>
  <startDecoding>
    <!--ro, req, bool, whether to start decoding-->true
  </startDecoding>
  <StreamInput>
    <!--ro, opt, object, video stream input-->
    <streamInputMode>
      <!--ro, req, enum, stream type, subType:string, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
    </streamInputMode>
    <StreamInputRealtime>
      <!--ro, opt, object, real-time stream information-->
      <durationInUnit>
        <!--ro, opt, int, decoding display interval of real-time stream, unit:s-->1
      </durationInUnit>
      <StreamRealtimeUnitList>
        <!--ro, opt, object, real-time stream list-->
        <StreamRealtimeUnit>
          <!--ro, opt, object, single real-time stream unit-->
          <streamType>
            <!--ro, req, enum, stream type, subType:string, desc:"in URL", "by ddns", "by domain"-->by domain
          </streamType>
          <StreamInURL>
            <!--ro, opt, object, URL stream-->
            <URL>
              <!--ro, req, string, URL-->test
            </URL>
          </StreamInURL>
        </StreamRealtimeUnit>
      </StreamRealtimeUnitList>
    </StreamInputRealtime>
  </StreamInput>
</StreamInputChannel>
```

```

</URL>
</StreamInURL>
<StreamByDdns>
  <!--ro, opt, object, DDNS-->
  <DdnsServerInfo>
    <!--ro, req, object, DDNS server information-->
    <domain>
      <!--ro, req, string, domain name-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <ddnsType>
      <!--ro, req, enum, DDNS type, subType:string, desc:"hiDdns", "noIp"-->hiDdns
    </ddnsType>
    <username>
      <!--ro, req, string, user name-->test
    </username>
    <password>
      <!--ro, req, string, password-->test
    </password>
  </DdnsServerInfo>
  <EncodeDevInfo>
    <!--ro, req, object, encoding device information-->
    <domain>
      <!--ro, req, string, domain name-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <transmitProtocol>
      <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
    </transmitProtocol>
    <protocol>
      <!--ro, opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
    </protocol>
    <username>
      <!--ro, req, string, user name-->test
    </username>
    <password>
      <!--ro, req, string, password-->test
    </password>
    <channelMode>
      <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType>
      <!--ro, opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
    </channelType>
    <channelZero>
      <!--ro, opt, int, channelMode is zero-->1
    </channelZero>
    <channelNormal>
      <!--ro, opt, int, channelMode is normal-->1
    </channelNormal>
    <channelStreaming>
      <!--ro, opt, string, channelMode is streaming-->test
    </channelStreaming>
    <channelDistributed>
      <!--ro, opt, int, channelMode is distributed-->1
    </channelDistributed>
  </EncodeDevInfo>
  <MediaGatewayInfo>
    <!--ro, opt, object, media gateway information-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <domain>
      <!--ro, req, string, domain name-->test
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <transmitProtocol>
      <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
    </transmitProtocol>
  </MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--ro, opt, object, streamType is by domain-->
  <EncodeDevInfo>
    <!--ro, req, object, device information-->
    <domain>
      <!--ro, req, string, domain name-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <transmitProtocol>
      <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
    </transmitProtocol>
    <protocol>

```

```

    <!--ro, opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
  </protocol>
  <username>
    <!--ro, req, string, user name-->test
  </username>
  <password>
    <!--ro, req, string, password-->test
  </password>
  <channelMode>
    <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
  </channelMode>
  <channelType>
    <!--ro, opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
  </channelType>
  <channelZero>
    <!--ro, opt, int, channelMode is zero-->1
  </channelZero>
  <channelNormal>
    <!--ro, opt, int, channelMode is normal-->1
  </channelNormal>
  <channelStreaming>
    <!--ro, opt, string, channelMode is streaming-->test
  </channelStreaming>
  <channelDistributed>
    <!--ro, opt, int, channelMode is distributed-->1
  </channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object, media gateway information-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <domain>
    <!--ro, req, string, domain name-->ipv4
  </domain>
  <port>
    <!--ro, req, int, port No.-->0
  </port>
  <transmitProtocol>
    <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)--
    </transmitProtocol>
  </MediaGatewayInfo>
  <StreamByDomain>
  </StreamByDomain>
  <StreamRealtimeUnit>
  </StreamRealtimeUnit>
  <StreamInputRealtime>
  </StreamInputRealtime>
  <StreamInputPlayback>
    <!--ro, opt, object, playback stream information-->
    <playbackMode>
      <!--ro, opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file name
    </playbackMode>
  </StreamInputPlayback>
  <EncodeDevInfo>
    <!--ro, opt, object, information of device in playback-->
    <domain>
      <!--ro, req, string, domain name-->ipv4
    </domain>
    <port>
      <!--ro, req, int, port No.-->0
    </port>
    <transmitProtocol>
      <!--ro, req, enum, transport protocol, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)-->tcp
    </transmitProtocol>
    <protocol>
      <!--ro, opt, enum, protocol, subType:string, desc:"DAHUA"-->DAHUA
    </protocol>
    <username>
      <!--ro, opt, string, user name-->test
    </username>
    <password>
      <!--ro, opt, string, password-->test
    </password>
    <channelMode>
      <!--ro, req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType>
      <!--ro, opt, enum, channel type, subType:string, desc:"main" (main stream), "sub" (sub stream), "third" (third stream)-->main
    </channelType>
    <channelZero>
      <!--ro, opt, int, channelMode is zero-->0
    </channelZero>
    <channelNormal>
      <!--ro, opt, int, channelMode is normal-->0
    </channelNormal>
    <channelStreaming>
      <!--ro, opt, string, channelMode is streaming-->test
    </channelStreaming>
    <channelDistributed>
      <!--ro, opt, int, channelMode is distributed-->1
    </channelDistributed>
  </EncodeDevInfo>
  <fileName>
    <!--ro, opt, string, file name-->test
  </fileName>

```



```

</fileName>
<TimeRange>
  <!--ro, opt, object, time range-->
  <beginTime>
    <!--ro, opt, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--ro, opt, time, end time-->00:00:00+08:00
  </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--ro, opt, bool, whether to enable stream encryption-->true
</streamEncryptEnable>
<streamPassword>
  <!--ro, opt, string, stream password-->test
</streamPassword>
</StreamInput>
</StreamInputChannel>

```

9.7.4.33 Get video stream capability

Request URL

GET /ISAPI/DisplayDev/Video/streaming/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoStreamingCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video stream capability, attr:version{req, string, protocolVersion}-->
  <streamingNums>
    <!--ro, opt, int, the number of video streams-->0
  </streamingNums>
  <isSupportURL>
    <!--ro, opt, bool, whether it supports URLs-->true
  </isSupportURL>
  <isSupportDDNS>
    <!--ro, opt, bool, whether it supports DDNS address-->true
  </isSupportDDNS>
  <isSupportIPAddress>
    <!--ro, opt, bool, whether it supports IP address-->true
  </isSupportIPAddress>
  <isSupportDistributedIPSignal opt="true,false">
    <!--ro, opt, bool, whether it supports adding distributed network sources, attr:opt{req, string}-->true
  </isSupportDistributedIPSignal>
  <isSupportAddBatch>
    <!--ro, opt, bool, whether it supports batch adding network signal sources-->true
  </isSupportAddBatch>
  <isSupportStreamChanSearch>
    <!--ro, opt, bool-->true
  </isSupportStreamChanSearch>
  <isSupportEncryptStream>
    <!--ro, opt, bool, whether it supports encrypting channel stream-->true
  </isSupportEncryptStream>
  <isSupportLoopEncryptStream>
    <!--ro, opt, bool, whether it supports configuration for stream encryption of auto-switch decoding-->true
  </isSupportLoopEncryptStream>
  <id min="1" max="100">
    <!--ro, opt, string, stream ID, attr:min{req, int},max{req, int}-->test
  </id>
  <name min="1" max="32">
    <!--ro, opt, string, stream name, attr:min{req, int},max{req, int}-->test
  </name>
  <group min="1" max="32">
    <!--ro, opt, string, stream group, attr:min{req, int},max{req, int}-->test
  </group>
  <StreamInput>
    <!--ro, opt, object, network stream information-->
    <streamInputMode opt="realtime,playback">
      <!--ro, req, enum, stream type, subType:string, attr:opt{req, string}, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
    </streamInputMode>
    <StreamInputRealtime>
      <!--ro, opt, object, real-time stream information-->
      <durationInUnit min="1" max="10">
        <!--ro, opt, int, decoding display interval of real-time stream, unit:s, attr:min{req, int},max{req, int}-->1
      </durationInUnit>
      <StreamRealtimeUnitList>
        <!--ro, opt, object, real-time stream List-->
        <StreamRealtimeUnit>

```

```

<!--ro, opt, object, single real-time stream unit-->
<streamType opt="in URL,by ddns,by domain">
  <!--ro, opt, enum, stream type, subType:string, attr:opt{req, string}, desc:"in URL", "by ddns", "by domain"-->in URL
</streamType>
<StreamInURL>
  <!--ro, opt, object, URL stream, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL)-->
  <URL min="1" max="1024">
    <!--ro, opt, string, URL address, attr:min{req, int},max{req, int)-->test
  </URL>
</StreamInURL>
<StreamByDdns>
  <!--ro, opt, object, DDNS stream, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns)-->
  <DdnsServerInfo>
    <!--ro, opt, object, DDNS server information-->
    <domain opt="ipv4,ipv6,domain">
      <!--ro, opt, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port min="1" max="10">
      <!--ro, opt, int, port No., attr:min{req, int},max{req, int)-->80
    </port>
    <ddnsType opt="hiDdns,noIp,...">
      <!--ro, opt, string, DDNS type, attr:opt{req, string)-->test
    </ddnsType>
    <username min="1" max="32">
      <!--ro, opt, string, user name., attr:min{req, int},max{req, int)-->test
    </username>
    <password min="1" max="16">
      <!--ro, opt, string, password, attr:min{req, int},max{req, int)-->test
    </password>
  </DdnsServerInfo>
  <EncodeDevInfo>
    <!--ro, opt, object, information of device in playback-->
    <domain opt="ipv4,ipv6,domain">
      <!--ro, opt, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port min="1" max="65535">
      <!--ro, opt, int, port No., attr:min{req, int},max{req, int)-->1
    </port>
    <transmitProtocol opt="tcp,udp,mcast">
      <!--ro, opt, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
    </transmitProtocol>
    <protocol opt="HIKVISION,DAHUA,...">
      <!--ro, opt, enum, protocol, subType:string, attr:opt{req, string)-->DAHUA
    </protocol>
    <username min="1" max="32">
      <!--ro, opt, string, user name., attr:min{req, int},max{req, int)-->test
    </username>
    <password min="1" max="16">
      <!--ro, opt, string, password, attr:min{req, int},max{req, int)-->test
    </password>
    <channelMode opt="normal,zero,streaming,distributed">
      <!--ro, opt, enum, channel mode, subType:string, attr:opt{req, string}, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType opt="main,sub,third">
      <!--ro, opt, enum, channel type, subType:string, attr:opt{req, string}, desc:"main" (main stream),"sub" (sub stream), "third" (third
stream)-->main
    </channelType>
    <channelZero min="1" max="1024">
      <!--ro, opt, int, channel-zero, attr:min{req, int},max{req, int)-->1
    </channelZero>
    <channelNormal min="1" max="1024">
      <!--ro, opt, int, normal channel, attr:min{req, int},max{req, int)-->1
    </channelNormal>
    <channelStreaming min="1" max="1024">
      <!--ro, opt, int, stream channel, attr:min{req, int},max{req, int)-->1
    </channelStreaming>
    <channelDistributed min="1" max="1024">
      <!--ro, opt, int, distributed channel, attr:min{req, int},max{req, int)-->1
    </channelDistributed>
  </EncodeDevInfo>
  <MediaGatewayInfo>
    <!--ro, opt, object, media gateway information-->
    <enabled opt="true,false">
      <!--ro, opt, enum, whether to enable, subType:string, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
    <domain opt="ipv4,ipv6,domain">
      <!--ro, opt, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port min="1" max="65535">
      <!--ro, opt, int, port No., attr:min{req, int},max{req, int)-->1
    </port>
    <transmitProtocol opt="tcp,udp,mcast">
      <!--ro, opt, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
    </transmitProtocol>
  </MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--ro, opt, object, domain stream-->
  <EncodeDevInfo>

```

```

<!--ro, opt, object, information of device in playback-->
<domain opt="ipv4,ipv6,domain">
  <!--ro, opt, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
</domain>
<port min="1" max="65535">
  <!--ro, opt, int, port No., attr:min{req, int},max{req, int}-->1
</port>
<transmitProtocol opt="tcp,udp,mcast">
  <!--ro, opt, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
</transmitProtocol>
<protocol opt="HIKVISION,DAHUA,...">
  <!--ro, opt, enum, protocol, subType:string, attr:opt{req, string}-->DAHUA
</protocol>
<username min="1" max="32">
  <!--ro, opt, string, user name., attr:min{req, int},max{req, int}-->test
</username>
<password min="1" max="16">
  <!--ro, opt, string, password, attr:min{req, int},max{req, int}-->test
</password>
<channelMode opt="normal,zero,streaming,distributed">
  <!--ro, opt, enum, channel mode, subType:string, attr:opt{req, string}, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelType opt="main,sub,third">
  <!--ro, opt, enum, channel type, subType:string, attr:opt{req, string}, desc:"main" (main stream),"sub" (sub stream), "third" (third
stream)-->main
</channelType>
<channelZero min="1" max="1024">
  <!--ro, opt, int, channel-zero, attr:min{req, int},max{req, int}-->1
</channelZero>
<channelNormal min="1" max="1024">
  <!--ro, opt, int, normal channel, attr:min{req, int},max{req, int}-->1
</channelNormal>
<channelStreaming min="1" max="1024">
  <!--ro, opt, int, stream channel, attr:min{req, int},max{req, int}-->1
</channelStreaming>
<channelDistributed min="1" max="1024">
  <!--ro, opt, int, distributed channel, attr:min{req, int},max{req, int}-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object, media gateway information-->
  <enabled opt="true,false">
    <!--ro, opt, enum, whether to enable, subType:string, attr:opt{req, string}, desc:true (enable), false (disable)-->true
  </enabled>
  <domain opt="ipv4,ipv6,domain">
    <!--ro, opt, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
  </domain>
  <port min="1" max="65535">
    <!--ro, opt, int, port No., attr:min{req, int},max{req, int}-->1
  </port>
  <transmitProtocol opt="tcp,udp,mcast">
    <!--ro, opt, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
  </transmitProtocol>
  </MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
  <!--ro, opt, object, playback stream information-->
  <playbackMode>
    <!--ro, opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file name
  </playbackMode>
  <EncodeDevInfo>
    <!--ro, opt, object, information of device in playback-->
    <domain opt="ipv4,ipv6,domain">
      <!--ro, opt, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port min="1" max="10">
      <!--ro, opt, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <transmitProtocol opt="tcp,udp,mcast">
      <!--ro, opt, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
    </transmitProtocol>
    <protocol opt="HIKVISION,DAHUA,...">
      <!--ro, opt, enum, protocol, subType:string, attr:opt{req, string}-->DAHUA
    </protocol>
    <username min="1" max="32">
      <!--ro, opt, string, user name., attr:min{req, int},max{req, int}-->test
    </username>
    <password min="1" max="16">
      <!--ro, opt, string, password, attr:min{req, int},max{req, int}-->test
    </password>
    <channelMode opt="normal,zero,streaming,distributed">
      <!--ro, opt, enum, channel mode, subType:string, attr:opt{req, string}, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType opt="main,sub,third">
      <!--ro, opt, enum, channel type, subType:string, attr:opt{req, string}, desc:"main" (main stream),"sub" (sub stream), "third" (third stream)--
>main
    </channelType>
    <channelZero min="1" max="10">

```

```

    <!--ro, opt, int, channel-zero, attr:min{req, int},max{req, int}-->1
  </channelZero>
  <channelNormal min="1" max="1024">
    <!--ro, opt, int, normal channel, attr:min{req, int},max{req, int}-->1
  </channelNormal>
  <channelStreaming min="1" max="1240">
    <!--ro, opt, int, stream channel, attr:min{req, int},max{req, int}-->1
  </channelStreaming>
  <channelDistributed min="1" max="1024">
    <!--ro, opt, int, distributed channel, attr:min{req, int},max{req, int}-->1
  </channelDistributed>
</EncodeDevInfo>
<fileName min="1" max="128">
  <!--ro, opt, string, file name, attr:min{req, int},max{req, int}-->test
</fileName>
<TimeRange>
  <!--ro, opt, object, time range-->
  <beginTime>
    <!--ro, opt, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--ro, opt, time, end time-->00:00:00+08:00
  </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable opt="true,false">
  <!--ro, opt, enum, whether to enable stream encryption, subType:string, attr:opt{req, string}, desc:true (enable), false (disable)-->true
</streamEncryptEnable>
<streamPassword min="1" max="16">
  <!--ro, opt, string, stream password, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}, attr:min{req, int},max{req, int},
desc:sensitive information encryption. It is valid when the value of streamEncryptEnable is true-->test
</streamPassword>
</StreamInput>
</VideoStreamingCap>

```

9.7.4.34 Get capability of searching for network input source parameters

Request URL

GET /ISAPI/DisplayDev/Video/streaming/channels/search/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<StreamInChanSearchCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of network input source parameters, attr:version{req, string, protocolVersion}-->
  <searchID min="0" max="10">
    <!--ro, req, string, search ID, attr:min{req, int},max{req, int}-->test
  </searchID>
  <maxMatchResults min="0" max="10">
    <!--ro, req, int, maximum number of returned results per search, attr:min{req, int},max{req, int}-->0
  </maxMatchResults>
  <maxConcurrentSearches min="0" max="10">
    <!--ro, req, int, maximum number of concurrent searches, attr:min{req, int},max{req, int}-->0
  </maxConcurrentSearches>
</StreamInChanSearchCap>

```

9.7.5 Video Wall Management

9.7.5.1 Get the capability of video wall controller

Request URL

GET /ISAPI/DisplayDev/capabilities

Query Parameter

None

Request Message

None

Response Message

<?xml version="1.0" encoding="UTF-8"?>

```
<DisplayCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the capability of video wall controller, attr:version{req, string, protocolVersion}-->
  <isSupportScreenCtrl>
    <!--ro, opt, bool-->true
  </isSupportScreenCtrl>
  <VideoCap>
    <!--ro, opt, object-->
    <VideoInputsCap>
      <!--ro, opt, object-->
      <videoInputPortNums>
        <!--ro, opt, int-->0
      </videoInputPortNums>
      <isSupportColorSetting>
        <!--ro, opt, bool-->true
      </isSupportColorSetting>
      <isSupportPostionSetting>
        <!--ro, opt, bool-->true
      </isSupportPostionSetting>
      <isSupportCutOffSetting>
        <!--ro, opt, bool-->true
      </isSupportCutOffSetting>
      <isSupportPictureCapture>
        <!--ro, opt, bool, whether the device supports capturing pictures-->true
      </isSupportPictureCapture>
      <isSupportText>
        <!--ro, opt, bool-->true
      </isSupportText>
      <encodeOSDInputSignalType opt="HDMI,DP">
        <!--ro, opt, string, attr:opt{req, string}-->test
      </encodeOSDInputSignalType>
      <SupportSelfdefineResolution>
        <!--ro, opt, object-->
        <signalType opt="DVI,VGA,HDMI,DP">
          <!--ro, req, string, attr:opt{req, string}-->test
        </signalType>
        <supportedDPResolution opt="640*480@60,800*600@60">
          <!--ro, opt, string, attr:opt{req, string}-->test
        </supportedDPResolution>
      </SupportSelfdefineResolution>
      <isSupportEDIDResolution>
        <!--ro, opt, bool-->true
      </isSupportEDIDResolution>
      <isSupportJoinSignalCfg>
        <!--ro, opt, bool-->true
      </isSupportJoinSignalCfg>
      <SupportAudioCfg>
        <!--ro, opt, object, audio settings-->
        <signalType opt="DVI,VGA,HDMI,DP">
          <!--ro, req, string, attr:opt{req, string}-->test
        </signalType>
      </SupportAudioCfg>
    </VideoInputsCap>
    <VideoOutputsCap>
      <!--ro, opt, object-->
      <videoOutputPortNums>
        <!--ro, opt, int-->0
      </videoOutputPortNums>
      <isSupportMultiOutputType>
        <!--ro, opt, bool-->true
      </isSupportMultiOutputType>
      <isSupportMultiResolution>
        <!--ro, opt, bool-->true
      </isSupportMultiResolution>
      <isSupportColorSetting>
        <!--ro, opt, bool-->true
      </isSupportColorSetting>
      <isSupportWidthHeightSetting>
        <!--ro, opt, bool-->true
      </isSupportWidthHeightSetting>
      <isSupportOutputIdentity>
        <!--ro, opt, bool-->true
      </isSupportOutputIdentity>
      <OutputResolutionCapList>
        <!--ro, opt, array, subType:object-->
        <OutputResolutionCap>
          <!--ro, opt, object-->
          <resolution>
            <!--ro, opt, string, resolution name-->test
          </resolution>
          <imageWidth min="0" max="10">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
          </imageWidth>
          <imageHeight min="0" max="10">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
          </imageHeight>
        </OutputResolutionCap>
      </OutputResolutionCapList>
      <isSupportEDIDResolution>
        <!--ro, opt, bool-->true
      </isSupportEDIDResolution>
    </VideoOutputsCap>
  </VideoCap>
</DisplayCap>
```

```

</VideoOutputsCap>
<VideoStreamingCap>
  <!--ro, opt, object-->
  <streamingNums>
    <!--ro, opt, int-->0
  </streamingNums>
  <isSupportURL>
    <!--ro, opt, bool-->true
  </isSupportURL>
  <isSupportDDNS>
    <!--ro, opt, bool-->true
  </isSupportDDNS>
  <isSupportIPAddress>
    <!--ro, opt, bool-->true
  </isSupportIPAddress>
  <isSupportDistributedIPSignal opt="true,false">
    <!--ro, opt, bool, attr::opt{req, string}-->true
  </isSupportDistributedIPSignal>
  <isSupportAddBatch>
    <!--ro, opt, bool-->true
  </isSupportAddBatch>
  <isSupportStreamChanSearch>
    <!--ro, opt, bool-->true
  </isSupportStreamChanSearch>
  <isSupportEncryptStream>
    <!--ro, opt, bool-->true
  </isSupportEncryptStream>
  <isSupportLoopEncryptStream>
    <!--ro, opt, bool-->true
  </isSupportLoopEncryptStream>
</VideoStreamingCap>
<isSupportClarityConfig>
  <!--ro, opt, bool-->true
</isSupportClarityConfig>
</VideoCap>
<AudioCap>
  <!--ro, opt, object-->
  <audioInputsPortNums>
    <!--ro, opt, int-->0
  </audioInputsPortNums>
  <audioOutputsPortNums>
    <!--ro, opt, int-->0
  </audioOutputsPortNums>
</AudioCap>
<VideoWallCap>
  <!--ro, opt, object-->
  <maxWallNums>
    <!--ro, opt, int-->0
  </maxWallNums>
  <maxWindowNums>
    <!--ro, opt, int-->0
  </maxWindowNums>
  <baseOutputSize>
    <!--ro, opt, int-->1920
  </baseOutputSize>
  <isSupportBaseMap>
    <!--ro, opt, bool-->true
  </isSupportBaseMap>
  <isSupportVirtualLED>
    <!--ro, opt, bool-->true
  </isSupportVirtualLED>
  <isSupportPlan>
    <!--ro, opt, bool-->true
  </isSupportPlan>
  <isSupportScene>
    <!--ro, opt, bool-->true
  </isSupportScene>
  <isSupportRoam>
    <!--ro, opt, bool-->true
  </isSupportRoam>
  <isSupportAutoSwitchMainSub>
    <!--ro, opt, bool-->true
  </isSupportAutoSwitchMainSub>
  <isSupportGlobalCtrl>
    <!--ro, opt, bool-->true
  </isSupportGlobalCtrl>
<BaseMapCap>
  <!--ro, opt, object-->
  <baseMapOnWallNums>
    <!--ro, opt, int-->1
  </baseMapOnWallNums>
</BaseMapCap>
<VirtualLEDCap>
  <!--ro, opt, object-->
  <virtualLEDNums>
    <!--ro, opt, int-->1
  </virtualLEDNums>
  <virtualLEDFirstHeightList>
    <!--ro, opt, object-->
    <virtualLEDFirstHeight>
      <!--ro, opt, object-->
      <resolution>
        <!--ro, opt, string, resolution-->test
      </resolution>
    </virtualLEDFirstHeight>
  </virtualLEDFirstHeightList>

```

```

</virtualLEDHeight>
<!--ro, opt, int-->1
</virtualLEDHeight>
</virtualLEDFirstHeight>
</virtualLEDFirstHeightList>
<isSupportFontRestore>
<!--ro, opt, bool-->true
</isSupportFontRestore>
<fontTypeDllMaxSize>
<!--ro, opt, int-->1
</fontTypeDllMaxSize>
</VirtualLEDCap>
<PlanCap>
<!--ro, opt, object-->
<maxPlanNums>
<!--ro, opt, int-->1
</maxPlanNums>
<isSupportBaseMapCycleSwitch>
<!--ro, opt, bool-->true
</isSupportBaseMapCycleSwitch>
</PlanCap>
<SceneCap>
<!--ro, opt, object-->
<maxSceneNums>
<!--ro, opt, int-->0
</maxSceneNums>
<isSupportSceneCopy>
<!--ro, opt, bool-->true
</isSupportSceneCopy>
</SceneCap>
<InputBoardCfgList>
<!--ro, opt, object-->
<InputBoardCfg>
<!--ro, opt, object-->
<slotNo>
<!--ro, req, int-->0
</slotNo>
<fullFrameEnable>
<!--ro, req, bool-->true
</fullFrameEnable>
</InputBoardCfg>
</InputBoardCfgList>
<ledShowMode opt="normal,tiled">
<!--ro, opt, string, attr:opt{req, string}-->test
</ledShowMode>
<isSupportAreaMirror opt="true,false">
<!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportAreaMirror>
<supportLEDResolutionVoutType opt="hdmi,dvi,sdi">
<!--ro, opt, string, attr:opt{req, string}-->test
</supportLEDResolutionVoutType>
<SupportSubStreamAutoSwitch>
<!--ro, opt, object-->
<subWndWidth min="0" max="10">
<!--ro, req, int, attr:min{req, int},max{req, int}-->0
</subWndWidth>
<subWndHeight min="0" max="10">
<!--ro, req, int, attr:min{req, int},max{req, int}-->0
</subWndHeight>
</SupportSubStreamAutoSwitch>
<wndStaticMode opt="blackScreen,lastFrame">
<!--ro, opt, string, attr:opt{req, string}-->test
</wndStaticMode>
<streamFailedMode opt="noSignal,lastFrame,linkException,clearScreen">
<!--ro, opt, string, attr:opt{req, string}-->test
</streamFailedMode>
<isSupportVcaDec>
<!--ro, opt, bool-->true
</isSupportVcaDec>
<isSupportDisplayWinNo>
<!--ro, opt, bool-->true
</isSupportDisplayWinNo>
<isSupportVideoWallsMirror opt="true,false">
<!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportVideoWallsMirror>
<isSupportWallNPreMonitor opt="true">
<!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportWallNPreMonitor>
<isSupportWallLPreMonitor opt="true">
<!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportWallLPreMonitor>
<isSupportAudioMatrix opt="true">
<!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportAudioMatrix>
</VideoWallCap>
<DistriDevCap>
<!--ro, opt, object-->
<maxDistriDevNum>
<!--ro, req, int-->0
</maxDistriDevNum>
<isSupportAddDistriDev opt="true">
<!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportAddDistriDev>

```

```

<isSupportDelDistriDev opt="true">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportDelDistriDev>
<isSupportOutputModeCfg opt="true">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportOutputModeCfg>
</DistriDevCap>
<SIPServerCap>
  <!--ro, opt, object-->
  <isSupportEnDecodeSeparateCfg>
    <!--ro, opt, bool, "true"-->true
  </isSupportEnDecodeSeparateCfg>
</SIPServerCap>
<SwitchBoardCap>
  <!--ro, opt, object-->
  <isSptSwthBdInfo>
    <!--ro, opt, bool-->true
  </isSptSwthBdInfo>
  <isSptSwthBdWorkMode>
    <!--ro, opt, bool-->true
  </isSptSwthBdWorkMode>
  <isSptProperty>
    <!--ro, opt, bool-->true
  </isSptProperty>
  <isSptPolymerize>
    <!--ro, opt, bool-->true
  </isSptPolymerize>
  <isSptStack>
    <!--ro, opt, bool-->true
  </isSptStack>
</SwitchBoardCap>
<isSupportAnalogSwitchMode>
  <!--ro, opt, bool-->true
</isSupportAnalogSwitchMode>
<isSupportCtrlDev opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportCtrlDev>
<isSupportWorkMode opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportWorkMode>
<isSupportsyncLineStatus opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportsyncLineStatus>
<isSupportOpenSourceCert opt="true">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportOpenSourceCert>
<isSupportCustomBoardModle opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportCustomBoardModle>
<isSupportFontLibraryImport opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportFontLibraryImport>
<isSupportFontLibraryCfg opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportFontLibraryCfg>
<maxFontLibraryNum min="1" max="3">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
</maxFontLibraryNum>
<isSupportInternalIpAddress opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportInternalIpAddress>
<isSupportDecoingDeviceStatus opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportDecoingDeviceStatus>
</DisplayCap>

```

9.7.5.2 Get parameters of all video walls

Request URL

GET /ISAPI/DisplayDev/VideoWall

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoWallList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, video wall List, subType:object, attr:version{req, string, protocolVersion}-->
  <VideoWall>
    <!--ro, opt, object, video wall-->
    <id>

```



```

<!--ro, opt, int, video wall No.-->1
</id>
<name>
  <!--ro, opt, string, video wall name-->test
</name>
<backgroundColor>
  <!--ro, opt, enum, background color, subType:string, desc:"black", "blue", "white"-->black
</backgroundColor>
<autoSwitchMainSub>
  <!--ro, opt, bool, whether to enable auto switching main/sub-stream for divided windows-->true
</autoSwitchMainSub>
<WallOutputList>
  <!--ro, opt, array, subType:object-->
  <WallOutput>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <outputID>
      <!--ro, req, int-->1
    </outputID>
    <Rect>
      <!--ro, opt, object-->
      <Coordinate>
        <!--ro, opt, object-->
        <x>
          <!--ro, opt, int, range:[0,1920]-->1
        </x>
        <y>
          <!--ro, opt, int, range:[0,1920]-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, opt, int, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, opt, int, range:[0,1920]-->1
      </height>
    </Rect>
  </WallOutput>
</WallOutputList>
<WallWindowList>
  <!--ro, opt, array, subType:object-->
  <WallWindow>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <wndOperateMode>
      <!--ro, opt, enum, subType:string-->uniformCoordinate
    </wndOperateMode>
    <Rect>
      <!--ro, opt, object-->
      <Coordinate>
        <!--ro, opt, object-->
        <x>
          <!--ro, opt, int, range:[0,1920]-->1
        </x>
        <y>
          <!--ro, opt, int, range:[0,1920]-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, opt, int, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, opt, int, range:[0,1920]-->1
      </height>
    </Rect>
    <ResolutionRect>
      <!--ro, opt, object-->
      <Coordinate>
        <!--ro, opt, object-->
        <x>
          <!--ro, opt, int-->1
        </x>
        <y>
          <!--ro, opt, int-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, opt, int, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, opt, int, range:[0,1920]-->1
      </height>
    </ResolutionRect>
    <layerIdx>
      <!--ro, opt, int-->1
    </layerIdx>
    <displayWinNo>
      <!--ro, opt, enum, subType:object-->true
    </displayWinNo>

```

```

<windowMode>
  <!--ro, opt, enum, subType:int-->1
</windowMode>
<wndShowMode>
  <!--ro, opt, enum, subType:string-->subWndMode
</wndShowMode>
<amplifyingSubWndNo>
  <!--ro, opt, int-->1
</amplifyingSubWndNo>
<SubWindowList>
  <!--ro, opt, array, subType:object-->
  <SubWindow>
    <!--ro, req, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <SubWindowParam>
      <!--ro, opt, object-->
      <signalMode>
        <!--ro, opt, enum, subType:string-->video input
      </signalMode>
      <videoInputChannelID>
        <!--ro, opt, string-->1
      </videoInputChannelID>
      <streamingChannelID>
        <!--ro, opt, string-->test
      </streamingChannelID>
      <StreamInput>
        <!--ro, opt, object-->
        <streamInputMode>
          <!--ro, req, enum, subType:string-->realtime
        </streamInputMode>
        <StreamInputRealtime>
          <!--ro, opt, object-->
          <durationInUnit>
            <!--ro, opt, int, unit:s-->1
          </durationInUnit>
          <StreamRealtimeUnitList>
            <!--ro, opt, object-->
            <StreamRealtimeUnit>
              <!--ro, opt, object-->
              <streamType>
                <!--ro, req, enum, subType:string-->in URL
              </streamType>
              <StreamInURL>
                <!--ro, opt, object, dep:and,
                {$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
                <URL>
                  <!--ro, req, string-->test
                </URL>
              </StreamInURL>
              <StreamByDdns>
                <!--ro, opt, object, dep:and,
                {$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
                <DdnsServerInfo>
                  <!--ro, opt, object-->
                  <domain>
                    <!--ro, req, enum, subType:string-->ipv4
                  </domain>
                  <port>
                    <!--ro, req, int-->80
                  </port>
                  <ddnsType>
                    <!--ro, req, string-->test
                  </ddnsType>
                  <username>
                    <!--ro, req, string-->test
                  </username>
                  <password>
                    <!--ro, req, string-->test
                  </password>
                </DdnsServerInfo>
                <EncodeDevInfo>
                  <!--ro, opt, object-->
                  <domain>
                    <!--ro, req, enum, subType:string-->ipv4
                  </domain>
                  <port>
                    <!--ro, req, int-->1
                  </port>
                  <transmitProtocol>
                    <!--ro, req, enum, subType:string-->tcp
                  </transmitProtocol>
                  <protocol>
                    <!--ro, req, enum, subType:string-->test
                  </protocol>
                  <username>
                    <!--ro, req, string-->test
                  </username>
                  <password>
                    <!--ro, req, string-->test
                  </password>
                  <channelMode>
                    <!--ro, req, enum, subType:string-->normal

```

```

<!--ro, req, enum, subType:string-->main
</channelMode>
<channelType>
<!--ro, req, enum, subType:string-->main
</channelType>
<channelZero>
<!--ro, opt, int-->1
</channelZero>
<channelNormal>
<!--ro, opt, int-->1
</channelNormal>
<channelStreaming>
<!--ro, opt, int-->1
</channelStreaming>
<channelDistributed>
<!--ro, opt, int-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
<!--ro, opt, object-->
<enabled>
<!--ro, req, enum, subType:string-->true
</enabled>
<domain>
<!--ro, req, enum, subType:string-->ipv4
</domain>
<port>
<!--ro, req, int-->1
</port>
<transmitProtocol>
<!--ro, req, enum, subType:string-->tcp
</transmitProtocol>
</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
<!--ro, opt, object-->
<EncodeDevInfo>
<!--ro, opt, object-->
<domain>
<!--ro, req, enum, subType:string-->ipv4
</domain>
<port>
<!--ro, req, int-->1
</port>
<transmitProtocol>
<!--ro, req, enum, subType:string-->tcp
</transmitProtocol>
<protocol>
<!--ro, req, enum, subType:string-->test
</protocol>
<username>
<!--ro, req, string, range:[1,32]-->test
</username>
<password>
<!--ro, req, string-->test
</password>
<channelMode>
<!--ro, req, enum, subType:string-->normal
</channelMode>
<channelType>
<!--ro, req, enum, subType:string-->main
</channelType>
<channelZero>
<!--ro, opt, int-->1
</channelZero>
<channelNormal>
<!--ro, opt, int-->1
</channelNormal>
<channelStreaming>
<!--ro, opt, int-->1
</channelStreaming>
<channelDistributed>
<!--ro, opt, int-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
<!--ro, opt, object-->
<enabled>
<!--ro, req, enum, subType:string-->true
</enabled>
<domain>
<!--ro, req, enum, subType:string-->ipv4
</domain>
<port>
<!--ro, req, int-->1
</port>
<transmitProtocol>
<!--ro, req, enum, subType:string-->tcp
</transmitProtocol>
</MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>

```

```

<StreamInputPlayback>
  <!--ro, opt, object-->
  <playbackMode>
    <!--ro, opt, enum, subType:string-->file name
  </playbackMode>
  <EncodeDevInfo>
    <!--ro, opt, object-->
    <domain>
      <!--ro, req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--ro, req, int-->1
    </port>
    <transmitProtocol>
      <!--ro, req, enum, subType:string-->tcp
    </transmitProtocol>
    <protocol>
      <!--ro, req, enum, subType:string-->test
    </protocol>
    <username>
      <!--ro, req, string-->test
    </username>
    <password>
      <!--ro, req, string-->test
    </password>
    <channelMode>
      <!--ro, req, enum, subType:string-->normal
    </channelMode>
    <channelType>
      <!--ro, req, enum, subType:string-->main
    </channelType>
    <channelZero>
      <!--ro, opt, int-->1
    </channelZero>
    <channelNormal>
      <!--ro, opt, int-->1
    </channelNormal>
    <channelStreaming>
      <!--ro, opt, int-->1
    </channelStreaming>
    <channelDistributed>
      <!--ro, opt, int-->1
    </channelDistributed>
  </EncodeDevInfo>
  <fileName>
    <!--ro, opt, string-->test
  </fileName>
  <TimeRange>
    <!--ro, opt, object-->
    <beginTime>
      <!--ro, opt, time-->00:00:00+08:00
    </beginTime>
    <endTime>
      <!--ro, opt, time-->00:00:00+08:00
    </endTime>
  </TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--ro, opt, enum, subType:string-->true
</streamEncryptEnable>
<streamPassword>
  <!--ro, opt, string, dep:and,{$.Subwindow.SubwindowParam.StreamInput.streamEncryptEnable,eq,true}-->test
</streamPassword>
</StreamInput>
<subWindowMode>
  <!--ro, opt, enum, subType:string-->normal
</subWindowMode>
</SubWindowParam>
</SubWindow>
</SubWindowList>
<wndType>
  <!--ro, opt, enum, subType:string-->signalSource
</wndType>
<zoomEnabled>
  <!--ro, opt, bool-->true
</zoomEnabled>
<audioEnabled>
  <!--ro, opt, bool-->true
</audioEnabled>
<wndLockKeep>
  <!--ro, opt, bool-->true
</wndLockKeep>
<Graphic>
  <!--ro, opt, object-->
  <windowRegisterID>
    <!--ro, opt, int-->0
  </windowRegisterID>
</Graphic>
<Subtitle>
  <!--ro, opt, object-->
  <moveDirection>
    <!--ro, opt, enum, subType:string-->left
  </moveDirection>

```

```

        <move>peea>
        <!--ro, opt, enum, subType:string-->fast
    </moveSpeed>
</Subtitle>
</WallWindow>
</WallWindowList>
<ledShowMode>
    <!--ro, opt, enum, subType:string-->normal
</ledShowMode>
<SubStreamAutoSwitch>
    <!--ro, opt, object-->
    <subWndWidth>
        <!--ro, req, int-->1
    </subWndWidth>
    <subWndHeight>
        <!--ro, req, int-->1
    </subWndHeight>
</SubStreamAutoSwitch>
<wndStaticMode>
    <!--ro, opt, enum, subType:object-->blackScreen
</wndStaticMode>
<streamFailedMode>
    <!--ro, opt, enum, display mode when window streaming failed, subType:object, desc:"noSignal"-no network video signal, "LastFrame"-Last frame image--
>noSignal
    </streamFailedMode>
</VideoWall>
</VideoWallList>

```

9.7.5.3 Set parameters of a specific video wall

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoWall xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
    <!--opt, object, Video Wall, attr:version{req,string,protocolVersion}-->
    <id>
        <!--opt, int, video wall No.-->1
    </id>
    <name>
        <!--opt, string, Smart Wall Name-->test
    </name>
    <backgroundColor>
        <!--opt, enum, background color, subType:string, desc:"black,blue,white"-->black
    </backgroundColor>
    <autoSwitchMainSub>
        <!--opt, bool, whether to enable auto switching main/sub-stream for divided windows-->true
    </autoSwitchMainSub>
    <WallOutputList>
        <!--opt, array, subType:object-->
        <WallOutput>
            <!--opt, object-->
            <id>
                <!--opt, int-->1
            </id>
            <outputID>
                <!--req, int-->1
            </outputID>
            <Rect>
                <!--opt, object-->
                <Coordinate>
                    <!--opt, object-->
                    <x>
                        <!--opt, int, range:[0,1920]-->1
                    </x>
                    <y>
                        <!--opt, int, range:[0,1920]-->1
                    </y>
                </Coordinate>
                <width>
                    <!--opt, int, range:[0,1920]-->1
                </width>
                <height>
                    <!--opt, int, range:[0,1920]-->1
                </height>
            </Rect>
        </WallOutput>
    </WallOutputList>

```

```

</wallOutputList>
<WallWindowList>
  <!--opt, array, subType:object-->
  <WallWindow>
    <!--opt, object-->
    <id>
      <!--opt, int-->1
    </id>
    <wndOperateMode>
      <!--opt, enum, subType:string-->uniformCoordinate
    </wndOperateMode>
    <Rect>
      <!--opt, object-->
      <Coordinate>
        <!--opt, object-->
        <x>
          <!--opt, int, range:[0,1920]-->1
        </x>
        <y>
          <!--opt, int, range:[0,1920]-->1
        </y>
      </Coordinate>
      <width>
        <!--opt, int, range:[0,1920]-->1
      </width>
      <height>
        <!--opt, int, range:[0,1920]-->1
      </height>
    </Rect>
    <ResolutionRect>
      <!--opt, object-->
      <Coordinate>
        <!--opt, object-->
        <x>
          <!--opt, int-->1
        </x>
        <y>
          <!--opt, int-->1
        </y>
      </Coordinate>
      <width>
        <!--opt, int, range:[0,1920]-->1
      </width>
      <height>
        <!--opt, int, range:[0,1920]-->1
      </height>
    </ResolutionRect>
    <layerIdx>
      <!--opt, int-->1
    </layerIdx>
    <displayWinNo>
      <!--opt, enum, subType:object-->true
    </displayWinNo>
    <windowMode>
      <!--opt, enum, subType:int-->1
    </windowMode>
    <wndShowMode>
      <!--opt, enum, subType:string-->subWndMode
    </wndShowMode>
    <amplifyingSubWindNo>
      <!--opt, int-->1
    </amplifyingSubWindNo>
    <SubWindowList>
      <!--opt, array, subType:object-->
      <SubWindow>
        <!--req, object-->
        <id>
          <!--opt, int-->1
        </id>
        <SubWindowParam>
          <!--opt, object-->
          <signalMode>
            <!--opt, enum, subType:string-->video input
          </signalMode>
          <videoInputChannelID>
            <!--opt, string-->1
          </videoInputChannelID>
          <streamingChannelID>
            <!--opt, string-->test
          </streamingChannelID>
          <StreamInput>
            <!--opt, object-->
            <streamInputMode>
              <!--req, enum, subType:string-->realtime
            </streamInputMode>
            <StreamInputRealtime>
              <!--opt, object-->
              <durationInUnit>
                <!--opt, int, unit:s-->1
              </durationInUnit>
              <StreamRealtimeUnitList>
                <!--opt, object-->
                <StreamRealtimeUnit>
                  <!--opt, object-->

```

```

<streamType>
  <!--req, enum, subType:string-->in URL
</streamType>
<StreamInURL>
  <!--opt, object, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
  <URL>
    <!--req, string-->test
  </URL>
</StreamInURL>
<StreamByDdns>
  <!--opt, object, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
  <DdnsServerInfo>
    <!--opt, object-->
    <domain>
      <!--req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--req, int-->80
    </port>
    <ddnsType>
      <!--req, string-->test
    </ddnsType>
    <username>
      <!--req, string-->test
    </username>
    <password>
      <!--req, string-->test
    </password>
  </DdnsServerInfo>
  <EncodeDevInfo>
    <!--opt, object-->
    <domain>
      <!--req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--req, int-->1
    </port>
    <transmitProtocol>
      <!--req, enum, subType:string-->tcp
    </transmitProtocol>
    <protocol>
      <!--req, enum, subType:string-->
    </protocol>
    <username>
      <!--req, string-->test
    </username>
    <password>
      <!--req, string-->test
    </password>
    <channelMode>
      <!--req, enum, subType:string-->normal
    </channelMode>
    <channelType>
      <!--req, enum, subType:string-->main
    </channelType>
    <channelZero>
      <!--opt, int-->1
    </channelZero>
    <channelNormal>
      <!--opt, int-->1
    </channelNormal>
    <channelStreaming>
      <!--opt, int-->1
    </channelStreaming>
    <channelDistributed>
      <!--opt, int-->1
    </channelDistributed>
  </EncodeDevInfo>
  <MediaGatewayInfo>
    <!--opt, object-->
    <enabled>
      <!--req, enum, subType:string-->true
    </enabled>
    <domain>
      <!--req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--req, int-->1
    </port>
    <transmitProtocol>
      <!--req, enum, subType:string-->tcp
    </transmitProtocol>
  </MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--opt, object-->
  <EncodeDevInfo>
    <!--opt, object-->
    <domain>
      <!--req, enum, subType:string-->ipv4
    </domain>

```

```

    <port>
      <!--req, int-->1
    </port>
    <transmitProtocol>
      <!--req, enum, subType:string-->tcp
    </transmitProtocol>
    <protocol>
      <!--req, enum, subType:string-->
    </protocol>
    <username>
      <!--req, string, range:[1,32]-->test
    </username>
    <password>
      <!--req, string-->test
    </password>
    <channelMode>
      <!--req, enum, subType:string-->normal
    </channelMode>
    <channelType>
      <!--req, enum, subType:string-->main
    </channelType>
    <channelZero>
      <!--opt, int-->1
    </channelZero>
    <channelNormal>
      <!--opt, int-->1
    </channelNormal>
    <channelStreaming>
      <!--opt, int-->1
    </channelStreaming>
    <channelDistributed>
      <!--opt, int-->1
    </channelDistributed>
  </EncodeDevInfo>
  <MediaGatewayInfo>
    <!--opt, object-->
    <enabled>
      <!--req, enum, subType:string-->true
    </enabled>
    <domain>
      <!--req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--req, int-->1
    </port>
    <transmitProtocol>
      <!--req, enum, subType:string-->tcp
    </transmitProtocol>
  </MediaGatewayInfo>
  <StreamByDomain>
  </StreamByDomain>
  <StreamRealtimeUnit>
  </StreamRealtimeUnit>
  <StreamRealtimeUnitList>
  </StreamRealtimeUnitList>
  <StreamInputRealtime>
  </StreamInputRealtime>
  <StreamInputPlayback>
    <!--opt, object-->
    <playbackMode>
      <!--opt, enum, subType:string-->file name
    </playbackMode>
  </StreamInputPlayback>
  <EncodeDevInfo>
    <!--opt, object-->
    <domain>
      <!--req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--req, int-->1
    </port>
    <transmitProtocol>
      <!--req, enum, subType:string-->tcp
    </transmitProtocol>
    <protocol>
      <!--req, enum, subType:string-->
    </protocol>
    <username>
      <!--req, string-->test
    </username>
    <password>
      <!--req, string-->test
    </password>
    <channelMode>
      <!--req, enum, subType:string-->normal
    </channelMode>
    <channelType>
      <!--req, enum, subType:string-->main
    </channelType>
    <channelZero>
      <!--opt, int-->1
    </channelZero>
    <channelNormal>
      <!--opt, int-->1
    </channelNormal>
    <channelStreaming>
      <!--opt, int-->1
    </channelStreaming>
    <channelDistributed>

```



```

        <!--opt, int-->1
    </channelDistributed>
</EncodeDevInfo>
<fileName>
    <!--opt, string-->test
</fileName>
<TimeRange>
    <!--opt, object-->
    <beginTime>
        <!--opt, time-->00:00:00+08:00
    </beginTime>
    <endTime>
        <!--opt, time-->00:00:00+08:00
    </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
    <!--opt, enum, subType:string-->true
</streamEncryptEnable>
<streamPassword>
    <!--opt, string, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}-->test
</streamPassword>
</StreamInput>
<subWindowMode>
    <!--opt, enum, subType:string-->normal
</subWindowMode>
</SubWindowParam>
</SubWindow>
</SubWindowList>
<wndType>
    <!--opt, enum, subType:string-->signalSource
</wndType>
<zoomEnabled>
    <!--opt, bool-->true
</zoomEnabled>
<audioEnabled>
    <!--opt, bool-->true
</audioEnabled>
<wndLockKeep>
    <!--opt, bool-->true
</wndLockKeep>
<Graphic>
    <!--opt, object-->
    <windowRegisterID>
        <!--opt, int-->0
    </windowRegisterID>
</Graphic>
<Subtitle>
    <!--opt, object-->
    <moveDirection>
        <!--opt, enum, subType:string-->left
    </moveDirection>
    <moveSpeed>
        <!--opt, enum, subType:string-->fast
    </moveSpeed>
</Subtitle>
</WallWindow>
</WallWindowList>
<ledShowMode>
    <!--opt, enum, subType:string-->normal
</ledShowMode>
<SubStreamAutoSwitch>
    <!--opt, object-->
    <subWndWidth>
        <!--req, int-->1
    </subWndWidth>
    <subWndHeight>
        <!--req, int-->1
    </subWndHeight>
</SubStreamAutoSwitch>
<wndStaticMode>
    <!--opt, enum, subType:object-->blackScreen
</wndStaticMode>
<streamFailedMode>
    <!--opt, enum, display mode when window streamed failed, subType:object, desc:noSignal-no network video signal, LastFrame-last frame image-->noSignal
</streamFailedMode>
</VideoWall>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

9.7.5.4 Get parameters of a specific video wall

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoWall xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video wall, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, video wall No.-->1
  </id>
  <name>
    <!--ro, opt, string, smart wall name-->test
  </name>
  <backgroundColor>
    <!--ro, opt, enum, "black,blue,white", subType:string, desc:"black, blue, white"-->black
  </backgroundColor>
  <autoSwitchMainSub>
    <!--ro, opt, bool, whether to enable auto switching main/sub-stream for divided windows; if enabled,no matter the decoding device is configured with
    main-stream or sub-stream when dynamic decoding,divided windows take sub-stream to decode and display live view on video wall-->true
  </autoSwitchMainSub>
  <ledShowMode>
    <!--ro, opt, enum, subType:string-->normal
  </ledShowMode>
  <SubStreamAutoSwitch>
    <!--ro, opt, object-->
    <subWndWidth>
      <!--ro, req, int-->1
    </subWndWidth>
    <subWndHeight>
      <!--ro, req, int-->1
    </subWndHeight>
  </SubStreamAutoSwitch>
  <wndStaticMode>
    <!--ro, opt, enum, subType:string-->blackScreen
  </wndStaticMode>
  <streamFailedMode>
    <!--ro, opt, enum, display mode when window streamed failed:, subType:string, desc:"noSignal"-no network video signal, "LastFrame"-last frame image--
  >noSignal
  </streamFailedMode>
  <zeroLatencyModeEnabled>
    <!--ro, opt, bool-->true
  </zeroLatencyModeEnabled>
  <streamTransportType>
    <!--ro, opt, enum, subType:string-->unicast
  </streamTransportType>
  <wallBackMode>
    <!--ro, opt, enum, subType:string-->color
  </wallBackMode>
  <backgroundRGBColor>
    <!--ro, opt, object-->
    <RGB>
      <!--ro, req, string-->FF00FF
    </RGB>
  </backgroundRGBColor>
  <sceneSwitchDelayEnabled>
    <!--ro, opt, bool-->true
  </sceneSwitchDelayEnabled>
  <sceneSwitchDelay>
    <!--ro, opt, int, range:[1,604800], unit:s, dep:and,{$.VideoWallList[*].VideoWall.sceneSwitchDelayEnabled,eq,true}-->1
  </sceneSwitchDelay>
  <windowsMoveDelayEnabled>
    <!--ro, opt, bool-->true
  </windowsMoveDelayEnabled>
  <windowsMoveDelay>
    <!--ro, opt, int, range:[1,604800], unit:s, dep:and,{$.VideoWallList[*].VideoWall.windowsMoveDelayEnabled,eq,true}-->1
  </windowsMoveDelay>
  <localStreamDeepEncodeType>
    <!--ro, opt, enum, subType:string-->normal
  </localStreamDeepEncodeType>
  <deviceAbnormalDisplayEnabled>
    <!--ro, opt, bool-->true
  </deviceAbnormalDisplayEnabled>
  <netStreamAbnormalDisplayEnabled>
    <!--ro, opt, bool-->false
  </netStreamAbnormalDisplayEnabled>
  <localStreamListDeepEncodeEnabled>
    <!--ro, opt, bool-->true
  </localStreamListDeepEncodeEnabled>
  <wallBindOutputStatus>
    <!--ro, opt, enum, subType:string-->bound
  </wallBindOutputStatus>
</VideoWall>

```

9.7.5.5 Get linked screen parameters of all outputs

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/outputs

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallOutputList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <WallOutput>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <outputID>
      <!--ro, req, int-->16842753
    </outputID>
    <Rect>
      <!--ro, req, object-->
      <Coordinate>
        <!--ro, req, object-->
        <x>
          <!--ro, opt, int, range:[0,30720], step:1920-->1
        </x>
        <y>
          <!--ro, opt, int, range:[0,30720], step:1920-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, opt, int, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, opt, int, range:[0,1920]-->1
      </height>
    </Rect>
    <outputWinNum>
      <!--ro, opt, int-->0
    </outputWinNum>
  </WallOutput>
</WallOutputList>
```

9.7.5.6 Get video wall capabilities

Request URL

GET /ISAPI/DisplayDev/VideoWall/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoWallCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, video wall capabilities, attr:version{req, string, protocolVersion}-->
  <maxWallNums>
    <!--ro, opt, int, the maximum number of video walls-->0
  </maxWallNums>
  <maxWindowNums>
    <!--ro, opt, int, the maximum number of windows-->0
  </maxWindowNums>
  <baseOutputSize>
    <!--ro, opt, int, output port coordinates, on which the array calculations of window and output port are based, desc:generally , it is 1920-->1920
  </baseOutputSize>
  <isSupportBaseMap>
    <!--ro, opt, bool, whether it supports base map-->true
  </isSupportBaseMap>
  <isSupportVirtualLED>
    <!--ro, opt, bool, whether it supports virtual LED-->true
  </isSupportVirtualLED>
  <isSupportPlan>
    <!--ro, opt, bool, whether it supports plan-->true
  </isSupportPlan>
  <isSupportScene>
```

```

<!--ro, opt, bool, whether it supports scene-->true
</isSupportScene>
<isSupportRoam>
  <!--ro, opt, bool, whether it supports roaming-->true
</isSupportRoam>
<isSupportAutoSwitchMainSub>
  <!--ro, opt, bool, whether it supports auto-switching main/sub-stream-->true
</isSupportAutoSwitchMainSub>
<BaseMapCap>
  <!--ro, opt, object-->
  <baseMapOnWallNums>
    <!--ro, opt, int-->1
  </baseMapOnWallNums>
</BaseMapCap>
<VirtualLEDCap>
  <!--ro, opt, object-->
  <virtualLEDNums>
    <!--ro, opt, int-->1
  </virtualLEDNums>
  <virtualLEDFirstHeightList>
    <!--ro, opt, object-->
    <virtualLEDFirstHeight>
      <!--ro, opt, object-->
      <resolution>
        <!--ro, opt, string-->test
      </resolution>
      <virtualLEDHeight>
        <!--ro, opt, int-->1
      </virtualLEDHeight>
    </virtualLEDFirstHeight>
  </virtualLEDFirstHeightList>
  <isSupportFontRestore>
    <!--ro, opt, bool-->true
  </isSupportFontRestore>
  <fontTypeDllMaxSize>
    <!--ro, opt, int-->1
  </fontTypeDllMaxSize>
</VirtualLEDCap>
<PlanCap>
  <!--ro, opt, object-->
  <maxPlanNums>
    <!--ro, opt, int-->1
  </maxPlanNums>
  <isSupportBaseMapCycleSwitch>
    <!--ro, opt, bool-->true
  </isSupportBaseMapCycleSwitch>
</PlanCap>
<SceneCap>
  <!--ro, opt, object-->
  <maxSceneNums>
    <!--ro, opt, int-->0
  </maxSceneNums>
  <isSupportSceneCopy>
    <!--ro, opt, bool-->true
  </isSupportSceneCopy>
</SceneCap>
<InputBoardCfgList>
  <!--ro, opt, object, output board configuration information-->
  <InputBoardCfg>
    <!--ro, opt, object-->
    <slotNo>
      <!--ro, req, int, slot No.-->0
    </slotNo>
    <fullFrameEnable>
      <!--ro, req, bool, slot ID-->true
    </fullFrameEnable>
  </InputBoardCfg>
</InputBoardCfgList>
<ledShowMode opt="normal,tiled">
  <!--ro, opt, string, attr:opt{req, string}-->test
</ledShowMode>
<isSupportAreaMirror opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportAreaMirror>
<supportLEDResolutionVoutType opt="hmi,dvi,sdi">
  <!--ro, opt, string, attr:opt{req, string}-->test
</supportLEDResolutionVoutType>
<SupportSubStreamAutoSwitch>
  <!--ro, opt, object-->
  <subWndWidth min="0" max="10">
    <!--ro, req, int, attr:min{req, int},max{req, int}-->0
  </subWndWidth>
  <subWndHeight min="0" max="10">
    <!--ro, req, int, attr:min{req, int},max{req, int}-->0
  </subWndHeight>
</SupportSubStreamAutoSwitch>
<wndStaticMode opt="blackScreen,lastFrame">
  <!--ro, opt, string, attr:opt{req, string}-->test
</wndStaticMode>
<streamFailedMode opt="noSignal,lastFrame,linkException,clearScreen">
  <!--ro, opt, string, attr:opt{req, string}-->test
</streamFailedMode>
<isSupportVcaDec>

```

```

<!--ro, opt, bool-->true
</isSupportVcaDec>
<isSupportDisplayWinNo>
  <!--ro, opt, bool, whether it supports window No. configuration-->true
</isSupportDisplayWinNo>
<isSupportVideoWallsMirror opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportVideoWallsMirror>
<isSupportWallNPreMonitor opt="true">
  <!--ro, opt, bool, whether it supports network echo, attr:opt{req, string}-->true
</isSupportWallNPreMonitor>
<isSupportWallPreMonitor opt="true">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportWallPreMonitor>
<isSupportAudioMatrix opt="true">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportAudioMatrix>
<isSupportAudioMatrix opt="true">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportAudioMatrix>
<isSupportWindowsLock opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportWindowsLock>
<isSupportVideoWallPreEdit opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportVideoWallPreEdit>
<isSupportAlarmDecode>
  <!--ro, opt, bool-->true
</isSupportAlarmDecode>
<isSupportClearAlarmDecode>
  <!--ro, opt, bool-->true
</isSupportClearAlarmDecode>
<wndType opt="signalSource,graphic,subtitle">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->signalSource
</wndType>
<zoomEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</zoomEnabled>
<audioEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</audioEnabled>
<supportAudioPortType opt="HDMI">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->HDMI
</supportAudioPortType>
<wndLockKeep opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</wndLockKeep>
<isSupportWindowsMoveUp>
  <!--ro, opt, bool-->true
</isSupportWindowsMoveUp>
<isSupportWindowsMoveDown>
  <!--ro, opt, bool-->true
</isSupportWindowsMoveDown>
<SignalSource>
  <!--ro, opt, object-->
  <maxWindowNum>
    <!--ro, opt, int-->0
  </maxWindowNum>
</SignalSource>
<Graphic>
  <!--ro, opt, object-->
  <maxWindowNum>
    <!--ro, opt, int-->0
  </maxWindowNum>
  <graphicWindowRegisterID opt="1,2">
    <!--ro, opt, int, attr:opt{req, string}-->0
  </graphicWindowRegisterID>
</Graphic>
<Subtitle>
  <!--ro, opt, object-->
  <maxWindowNum>
    <!--ro, opt, int-->0
  </maxWindowNum>
  <moveDirection opt="left">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->left
  </moveDirection>
  <moveSpeed opt="fast,slow,static">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->fast
  </moveSpeed>
</Subtitle>
<VideoWallRect>
  <!--ro, opt, object, desc:the origin is the upper-Left corner of the screen-->
  <width>
    <!--ro, opt, int, range:[0,1920]-->1
  </width>
  <height>
    <!--ro, opt, int, range:[0,1920]-->1
  </height>
</VideoWallRect>
<isSupportBindAudioOut opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</isSupportBindAudioOut>
</VideoWallCap>

```

9.7.6 Video Wall Plan Management

9.7.6.1 Add a plan

Request URL

POST /ISAPI/DisplayDev/VideoWall/<videoWallID>/plan

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallPlan xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, range:[1,8]-->1
  </id>
  <name>
    <!--opt, string, range:[1,32]-->test
  </name>
  <ActTimeDetail>
    <!--opt, object-->
    <actTimeMode>
      <!--opt, enum, "at once,on day,weekly", subType:string-->at once
    </actTimeMode>
    <OnDayTime>
      <!--opt, object, actTimeMode is "on day"-->
      <datetime>
        <!--opt, datetime, ISO 8601 time-->1970-01-01T00:00:00+08:00
      </datetime>
    </OnDayTime>
    <WeeklyTime>
      <!--opt, object, actTimeMode is "weekly"-->
      <TimeBlockList>
        <!--req, object-->
        <TimeBlock>
          <!--opt, object-->
          <dayOfWeek>
            <!--opt, int, ISO 8601 weekday number,1=Monday,...-->1
          </dayOfWeek>
          <beginTime>
            <!--req, time, ISO 8601 time-->00:00:00+08:00
          </beginTime>
        </TimeBlock>
      </TimeBlockList>
    </WeeklyTime>
  </ActTimeDetail>
  <PlanDetailList>
    <!--opt, object-->
    <PlanDetail>
      <!--opt, object-->
      <operationType>
        <!--req, enum, operations about plan: "activateScene"-switch scenes,"closeScreen"-close screens,"openScreen"-open screens,"switchBaseMap"-switch background pictures, subType:string-->activateScene
      </operationType>
      <sceneID>
        <!--opt, int, scene ID-->1
      </sceneID>
      <duration>
        <!--req, int, time interval between two plans, unit: second-->1
      </duration>
      <baseMapType>
        <!--opt, enum, background picture type: baseMap,UHD, subType:string-->baseMap
      </baseMapType>
      <baseMapWindNo>
        <!--opt, int, ID of the background picture window-->1
      </baseMapWindNo>
      <baseMapNo>
        <!--opt, int, background picture ID-->1
      </baseMapNo>
    </PlanDetail>
  </PlanDetailList>
  <actCount>
    <!--opt, int-->1
  </actCount>
</WallPlan>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	thePlanNameAlreadyExists	0x4000A30D	--
4	Invalid Operation	numberOfScenesInThisPlanReachedLimit	0x4000A47C	--

9.7.6.2 Get configuration capability of a specific plan

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/plan/<planTemplateID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
planTemplateID	string	--

Request Message

None

Response Message


```

<?xml version="1.0" encoding="UTF-8"?>

<WallPlan xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id min="1" max="8">
    <!--ro, req, int, range:[1,8], attr:min{req, int},max{req, int}-->1
  </id>
  <name min="1" max="32">
    <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}-->test
  </name>
  <ActTimeDetail>
    <!--ro, opt, object-->
    <actTimeMode opt=" at once,on day,weekly">
      <!--ro, opt, string, "at once, on day, weekly", attr:opt{req, string}-->test
    </actTimeMode>
    <OnDayTime>
      <!--ro, opt, object, actTimeMode is "on day"-->
      <datetime>
        <!--ro, opt, datetime, ISO 8601 time-->1970-01-01T00:00+08:00
      </datetime>
    </OnDayTime>
    <WeeklyTime>
      <!--ro, opt, object, actTimeMode is "weekly"-->
      <TimeBlockList>
        <!--ro, req, object-->
        <TimeBlock>
          <!--ro, opt, object-->
          <dayOfWeek>
            <!--ro, opt, int, ISO 8601 weekday number, 1=Monday,...-->1
          </dayOfWeek>
          <beginTime>
            <!--ro, req, time, ISO 8601 time-->00:00+08:00
          </beginTime>
          <cycleSum min="1" max="128">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
          </cycleSum>
        </TimeBlock>
      </TimeBlockList>
    </WeeklyTime>
  </ActTimeDetail>
  <PlanDetailList>
    <!--ro, opt, object-->
    <PlanDetail>
      <!--ro, opt, object-->
      <operationType>
        <!--ro, req, enum, operations about plan: "activateScene"-switch scenes, "closeScreen"-close screens, "openScreen"-open screens, "switchBaseMap"-switch background pictures, subType:string, desc:operations about plan: "activateScene"-switch scenes, "closeScreen"-close screens, "openScreen"-open screens, "switchBaseMap"-switch background pictures-->activateScene
      </operationType>
      <sceneID min="1" max="128">
        <!--ro, opt, int, scene ID, attr:min{req, int},max{req, int}-->1
      </sceneID>
      <duration min="1" max="86400">
        <!--ro, req, int, time interval between two plans, unit: second, attr:min{req, int},max{req, int}-->1
      </duration>
      <baseMapType>
        <!--ro, opt, enum, background picture type: baseMap, UHD, subType:string, desc:background picture type: baseMap, UHD-->baseMap
      </baseMapType>
      <baseMapWindNo>
        <!--ro, opt, int, ID of the background picture window-->1
      </baseMapWindNo>
      <baseMapNo>
        <!--ro, opt, int, background picture ID-->1
      </baseMapNo>
    </PlanDetail>
  </PlanDetailList>
  <actCount>
    <!--ro, opt, int-->1
  </actCount>
</WallPlan>

```

9.7.6.3 Get plan configuration capability

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/plan/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlanCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <maxPlanNums>
    <!--ro, opt, int, the maximum number of plans-->1
  </maxPlanNums>
  <isSupportBaseMapCycleSwitch>
    <!--ro, opt, bool-->true
  </isSupportBaseMapCycleSwitch>
</PlanCap>
```

9.7.6.4 Get the current plan

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/plan/isRunning

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RunningPlan xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <planID>
    <!--ro, req, int-->1
  </planID>
</RunningPlan>
```

9.7.7 Video Wall Scene Management

9.7.7.1 Get all scenes' parameters

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallSceneList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <WallScene>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, string-->test
    </id>
    <name>
      <!--ro, opt, string-->test
    </name>
  </WallScene>
</WallSceneList>
```

9.7.7.2 Set parameters of a specific scene

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
SID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallScene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, string-->test
  </id>
  <name>
    <!--opt, string-->test
  </name>
</WallScene>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

9.7.7.3 Switch to a specific scene

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>/activate

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
SID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	inSceneSwitchingPleaseDoNotOperate	0x4000A1AB	--

9.7.7.4 Save the current scene

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>/saveData

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
SID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<Request xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, it is a placeholder, the body payload can be empty, attr:version{req, string, protocolVersion}-->
</Request>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
    quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

9.7.7.5 Get scene configuration capability

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SceneCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability, attr:version{req, string, protocolVersion}-->
  <maxSceneNums>
    <!--ro, opt, int, maximum number of scenes-->1
  </maxSceneNums>
  <isSupportSceneCopy>
    <!--ro, opt, bool, whether scene copying is supported, desc:/ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>/copy-->true
  </isSupportSceneCopy>
  <isSupportSceneInfo>
    <!--ro, opt, bool, whether scene information is supported, desc:/ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/<SID>/sceneInfo?format=json-->true
  </isSupportSceneInfo>
  <isSupportSceneImport>
    <!--ro, opt, bool, whether it supports scene import, desc:/ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/import?format=json-->true
  </isSupportSceneImport>
  <isSupportSceneExport>
    <!--ro, opt, bool, whether it supports scene export, desc:/ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/export?format=json-->true
  </isSupportSceneExport>
  <isSupportSaveSceneLogo>
    <!--ro, opt, bool, whether the device supports saving the scene with decoding Logo-->true
  </isSupportSaveSceneLogo>
  <isSupportSaveSceneAudio>
    <!--ro, opt, bool, whether the device supports saving the scene with audio-->true
  </isSupportSaveSceneAudio>
  <isSupportSaveSceneVirLed>
    <!--ro, opt, bool, whether the device supports saving the scene with virtual LED-->true
  </isSupportSaveSceneVirLed>
  <isSupportSaveSceneSmartDec>
    <!--ro, opt, bool, whether the device supports saving the scene with smart decoding-->true
  </isSupportSaveSceneSmartDec>
  <isSupportSaveSceneBaseMap>
    <!--ro, opt, bool, whether the device supports saving the scene with background-->true
  </isSupportSaveSceneBaseMap>
  <isSupportSaveSceneDecOsd>
    <!--ro, opt, bool, whether the device supports saving the scene with decoding OSD-->true
  </isSupportSaveSceneDecOsd>
  <isSupportSaveSceneDecDelay>
    <!--ro, opt, bool, whether the device supports saving the scene with decoding delay-->true
  </isSupportSaveSceneDecDelay>
  <name min="1" max="10">
    <!--ro, opt, string, scene name, attr:min{req, int},max{req, int}-->test
  </name>
</SceneCap>
```

9.7.7.6 Get the current scene

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/scene/isRunning

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RunningScene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, current scene information, attr:version{req, string, protocolVersion}-->
  <sceneID>
    <!--ro, req, int, current scene ID-->1
  </sceneID>
</RunningScene>
```

9.7.7.7 Get scene control parameters capability

Request URL

GET /ISAPI/DisplayDev/VideoWallScene/SceneControlParams/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "SceneControlParamsCap": {
    /*opt, object, scene control parameters capability, desc:corresponding URL: GET or PUT /ISAPI/DisplayDev/VideoWallScene/SceneControlParams?format=json*/
    "switchSubtitlesEnabled": {
      /*req, object, value range of whether to enable subtitle switching, desc:by default, subtitles remain unchanged as scene changes*/
      "@opt": [],
      /*req, array, enumeration list of whether to enable background switching, subType:boolean, range:[1,2]*/
      "@def": "false"
      /*opt, boolean, default value*/
    },
    "switchPictureEnabled": {
      /*opt, object, whether to enable background switching, desc:by default, background remains unchanged as background changes*/
      "@opt": [],
      /*req, array, whether to enable background switching, subType:boolean, range:[1,2]*/
      "@def": "false"
      /*opt, boolean, default value*/
    }
  }
}
```

9.7.7.8 Get scene control parameters

Request URL

GET /ISAPI/DisplayDev/VideoWallScene/SceneControlParams?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "switchSubtitlesEnabled": "false",
  /*req, boolean, whether to enable subtitle changing, desc:by default, subtitles remain unchanged as scene changes*/
  "switchPictureEnabled": "false"
  /*opt, boolean, whether to enable background changing, desc:by default, background remains unchanged as background changes*/
}
```

9.7.7.9 Set scene control parameters

Request URL

PUT /ISAPI/DisplayDev/VideoWallScene/SceneControlParams?format=json

Query Parameter

None

Request Message

```
{
  "switchSubtitlesEnabled": "false",
  /*req, boolean, whether to enable subtitle switching, desc:by default, subtitles remain unchanged as scene changes*/
  "switchPictureEnabled": "false"
  /*opt, boolean, whether to enable background changing, desc:by default, background remains unchanged as background changes*/
}
```

Response Message

```

{
  "statusCode": 1,
  /*opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "OK",
  /*opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
  "subStatusCode": "ok",
  /*opt, string, sub status code, range:[1,64], desc:this field is required when the status code is not "ok" (success)*/
  "errorCode": 1,
  /*opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok",
  /*opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
  "MErrCode": "0x00000000",
  /*opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
  "MErrDevSelfEx": "0x00000000"
  /*opt, string, error codes categorized by functional modules, desc:N/A*/
}

```

9.7.8 Video Wall Screen Management

9.7.8.1 Close all screens

Request URL

PUT /ISAPI/DisplayDev/ScreenCtrl/closeAll

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```

<ScreenCtrl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, undefined, attr:version{req, string, protocolVersion}-->
  <VideoWallID>
    <!--opt, int, undefined-->1
  </VideoWallID>
  <OutputID>
    <!--opt, int, undefined-->1
  </OutputID>
</ScreenCtrl>

```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>

```

9.7.9 Video Wall Text Management

9.7.9.1 Set parameters of all virtual LEDs

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VirtualLEDOnWallList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <VirtualLEDOnWall version="2.0">
    <!--opt, object, Virtual LED Settings, attr:version{req, string}-->
    <id>
      <!--opt, int, No.-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <wndOperateMode>
      <!--opt, enum, subType:string-->uniformCoordinate
    </wndOperateMode>
    <Rect>
      <!--opt, object-->
      <Coordinate>
        <!--req, object-->
        <x>
          <!--req, int, X-coordinate, range:[0,1920]-->1
        </x>
        <y>
          <!--req, int, Y-coordinate, range:[0,1920]-->1
        </y>
      </Coordinate>
      <width>
        <!--req, int, width, range:[0,1920]-->1
      </width>
      <height>
        <!--req, int, height, range:[0,1920]-->1
      </height>
    </Rect>
    <Coordinate>
      <!--opt, object, desc:the origin is the upper-left corner of the screen-->
      <x>
        <!--req, int, X-coordinate, range:[0,1920]-->0
      </x>
      <y>
        <!--req, int, Y-coordinate, range:[0,1920]-->0
      </y>
    </Coordinate>
    <ResolutionRect>
      <!--opt, object-->
      <Coordinate>
        <!--req, object-->
        <x>
          <!--req, int, X-coordinate-->1
        </x>
        <y>
          <!--req, int, Y-coordinate-->1
        </y>
      </Coordinate>
      <width>
        <!--req, int, width-->1
      </width>
      <height>
        <!--req, int, height-->1
      </height>
    </ResolutionRect>
    <text>
      <!--opt, string-->test
    </text>
    <foregroundColor>
      <!--opt, object, Foreground Color-->
      <RGB>
        <!--req, int-->1
      </RGB>
    </foregroundColor>
    <backgroundColor>
      <!--opt, object, Background Color-->
      <RGB>
        <!--req, int-->1
      </RGB>
    </backgroundColor>
    <transparencyMode>
      <!--opt, enum, subType:string-->opaque
    </transparencyMode>
    <moveDirection>
      <!--opt, enum, Moving Direction, subType:string-->left to right
    </moveDirection>
    <moveSpeed>
      <!--opt, int, Moving Mode-->1
    </moveSpeed>
  </VirtualLEDOnWall>
</VirtualLEDOnWallList>
```

```

<moveSpeed>
  <!--opt, enum, Moving Mode, subType:int-->1
</moveSpeed>
<ledType>
  <!--opt, enum, subType:string-->text
</ledType>
<ledDirection>
  <!--opt, enum, subType:string-->horizontal
</ledDirection>
<timeType>
  <!--opt, enum, time type, subType:string-->dateTime
</timeType>
<dateFormat>
  <!--opt, enum, Date Format, subType:string-->yyyy-mm-dd
</dateFormat>
<timeFormat>
  <!--opt, enum, time display format, subType:string-->h:mm:ss
</timeFormat>
<hourFormat>
  <!--opt, enum, subType:string-->12h
</hourFormat>
<amFormat>
  <!--opt, enum, subType:string-->am
</amFormat>
<pmFormat>
  <!--opt, enum, subType:string-->pm
</pmFormat>
<alignmentX>
  <!--opt, enum, subType:string-->left
</alignmentX>
<alignmentY>
  <!--opt, enum, subType:string-->up
</alignmentY>
<fontType>
  <!--opt, enum, subType:string-->simSun
</fontType>
<fontSize>
  <!--opt, enum, font size, subType:string-->1times
</fontSize>
<fontDisplayMode>
  <!--opt, enum, subType:int-->0
</fontDisplayMode>
<fontBold>
  <!--opt, bool-->true
</fontBold>
</VirtualLEDOnWall>
</VirtualLEDOnWallList>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version(ro, req, string, protocolVersion)-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	theWidthOfTheTextExceededLimit	0x4000A2B6	--
4	Invalid Operation	theHeightOfTheTextExceededLimit	0x4000A2B7	--
4	Invalid Operation	currentFontIsNotSupportedBySomeTextsOfTheSubtitles	0x4000A459	--

9.7.9.2 Get parameters of all virtual LEDs

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VirtualLEDOnWallList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version(req, string, protocolVersion)-->
  <VirtualLEDOnWall>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <enabled>
      <!--ro, req, bool-->>true
    </enabled>
    <wndOperateMode>
      <!--ro, opt, enum, subType:string-->uniformCoordinate
    </wndOperateMode>
    <Rect>
      <!--ro, opt, object-->
      <Coordinate>
        <!--ro, req, object-->
        <x>
          <!--ro, req, int, range:[0,1920]-->1
        </x>
        <y>
          <!--ro, req, int, range:[0,1920]-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, req, int, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, req, int, range:[0,1920]-->1
      </height>
    </Rect>
    <Coordinate>
      <!--ro, opt, object-->
      <x>
        <!--ro, req, int, range:[0,1920]-->0
      </x>
      <y>
        <!--ro, req, int, range:[0,1920]-->0
      </y>
    </Coordinate>
    <ResolutionRect>
      <!--ro, opt, object-->
      <Coordinate>
        <!--ro, req, object-->
        <x>
          <!--ro, req, int-->1
        </x>
        <y>
          <!--ro, req, int-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, req, int-->1
      </width>
      <height>
        <!--ro, req, int-->1
      </height>
    </ResolutionRect>
    <text>
      <!--ro, opt, string-->test
    </text>
    <foregroundColor>
      <!--ro, opt, object-->
      <RGB>
        <!--ro, req, string-->ff00ff
      </RGB>
    </foregroundColor>
    <backgroundColor>
      <!--ro, opt, object-->
      <RGB>
        <!--ro, req, string-->ff00ff
      </RGB>
    </backgroundColor>
    <transparencyMode>
      <!--ro, opt, enum, subType:string-->opaque
    </transparencyMode>
  </VirtualLEDOnWall>
</VirtualLEDOnWallList>
```

```

</transparencyMode>
<moveDirection>
  <!--ro, opt, enum, subType:string-->left to right
</moveDirection>
<moveSpeed>
  <!--ro, opt, int-->1
</moveSpeed>
<moveMode>
  <!--ro, opt, enum, subType:string-->normal
</moveMode>
<ledType>
  <!--ro, opt, enum, subType:string-->text
</ledType>
<ledDirection>
  <!--ro, opt, enum, subType:string-->horizontal
</ledDirection>
<timeType>
  <!--ro, opt, enum, subType:string-->dateTime
</timeType>
<showWeekInTime>
  <!--ro, opt, bool-->true
</showWeekInTime>
<dateFormat>
  <!--ro, opt, enum, subType:string-->yyyy-mm-dd
</dateFormat>
<timeFormat>
  <!--ro, opt, enum, subType:string-->h:mm:ss
</timeFormat>
<hourFormat>
  <!--ro, opt, enum, subType:string-->12h
</hourFormat>
<amFormat>
  <!--ro, opt, enum, subType:string-->am
</amFormat>
<pmFormat>
  <!--ro, opt, enum, subType:string-->pm
</pmFormat>
<clockLineFeedEnabled>
  <!--ro, opt, bool-->true
</clockLineFeedEnabled>
<alignmentX>
  <!--ro, opt, enum, subType:string-->left
</alignmentX>
<alignmentY>
  <!--ro, opt, enum, subType:string-->up
</alignmentY>
<fontType>
  <!--ro, opt, enum, subType:string-->simSun
</fontType>
<fontSize>
  <!--ro, opt, enum, subType:string-->1times
</fontSize>
<fontDisplayMode>
  <!--ro, opt, enum, subType:int-->0
</fontDisplayMode>
<fontBold>
  <!--ro, opt, bool-->true
</fontBold>
<backgroundPicType>
  <!--ro, opt, enum, subType:int-->0
</backgroundPicType>
<clockParam>
  <!--ro, opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,clock}-->
  <clockType>
    <!--ro, opt, enum, subType:string-->dialClock
  </clockType>
  <clockStyle>
    <!--ro, opt, enum, subType:string-->style1
  </clockStyle>
</clockParam>
<weatherParam>
  <!--ro, opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,weather}-->
  <weatherCondition>
    <!--ro, opt, enum, subType:string-->1
  </weatherCondition>
  <temperature>
    <!--ro, opt, int, range:[-100,100], unit: °C-->26
  </temperature>
  <humidity>
    <!--ro, opt, int, range:[0,100]-->50
  </humidity>
  <city>
    <!--ro, opt, string, range:[1,64]-->hangzhou
  </city>
  <province>
    <!--ro, opt, string, range:[1,16]-->zhejiang
  </province>
  <weatherOverview>
    <!--ro, opt, string, range:[1,128]-->test
  </weatherOverview>
  <airQuality>
    <!--ro, opt, string, range:[1,16]-->test
  </airQuality>
</weatherParam>

```

```

<forecastList>
  <!--ro, opt, array, subType:object, range:[1,5]-->
  <forecast>
    <!--ro, opt, object-->
    <highestTemperature>
      <!--ro, opt, int, range:[-100,100], unit:℃-->28
    </highestTemperature>
    <lowestTemperature>
      <!--ro, opt, int, range:[-100,100], unit:℃-->12
    </lowestTemperature>
    <weatherCondition>
      <!--ro, opt, enum, subType:string-->1
    </weatherCondition>
  </forecast>
</forecastList>
</VirtualLEDOnWall>
</VirtualLEDOnWallList>

```

9.7.9.3 Add all virtual LEDs

Request URL

POST /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VirtualLEDOnWall xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, virtual LED configuration, attr:version{req, string, protocolVersion}, desc:virtual LED configuration-->
  <id>
    <!--opt, int, No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <wndOperateMode>
    <!--opt, enum, subType:string-->uniformCoordinate
  </wndOperateMode>
  <Rect>
    <!--opt, object-->
    <Coordinate>
      <!--req, object-->
      <x>
        <!--req, int, X-coordinate, range:[0,1920]-->1
      </x>
      <y>
        <!--req, int, Y-coordinate, range:[0,1920]-->1
      </y>
    </Coordinate>
    <width>
      <!--req, int, range:[0,1920]-->1
    </width>
    <height>
      <!--req, int, range:[0,1920]-->1
    </height>
  </Rect>
  <Coordinate>
    <!--opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
    <x>
      <!--req, int, X-coordinate, range:[0,1920]-->0
    </x>
    <y>
      <!--req, int, Y-coordinate, range:[0,1920]-->0
    </y>
  </Coordinate>
  <ResolutionRect>
    <!--opt, object-->
    <Coordinate>
      <!--req, object-->
      <x>
        <!--req, int, X-coordinate, desc:X-coordinate-->1
      </x>
      <y>
        <!--req, int, Y-coordinate, desc:Y-coordinate-->1
      </y>
    </Coordinate>
    <width>
      <!--req, int-->1
    </width>
    <height>

```

```

        <!--req, int-->1
    </height>
</ResolutionRect>
<text>
    <!--opt, string-->test
</text>
<foregroundColor>
    <!--ro, opt, object-->
    <RGB>
        <!--ro, req, string-->ff00ff
    </RGB>
</foregroundColor>
<backgroundColor>
    <!--ro, opt, object-->
    <RGB>
        <!--ro, req, string-->ff00ff
    </RGB>
</backgroundColor>
<transparencyMode>
    <!--opt, enum, subType:string-->opaque
</transparencyMode>
<moveDirection>
    <!--opt, enum, incorrect moving direction, subType:string, desc:incorrect moving direction-->left to right
</moveDirection>
<moveSpeed>
    <!--opt, int, incorrect moving speed-->1
</moveSpeed>
<moveMode>
    <!--opt, enum, incorrect moving mode, subType:string, desc:incorrect moving mode-->normal
</moveMode>
<ledType>
    <!--opt, enum, subType:string-->text
</ledType>
<ledDirection>
    <!--opt, enum, subType:string-->horizontal
</ledDirection>
<timeType>
    <!--opt, enum, subType:string-->dateTime
</timeType>
<showWeekInTime>
    <!--ro, opt, bool-->true
</showWeekInTime>
<dateFormat>
    <!--opt, enum, subType:string-->yyyy-mm-dd
</dateFormat>
<timeFormat>
    <!--opt, enum, subType:string-->h:mm:ss
</timeFormat>
<hourFormat>
    <!--opt, enum, subType:string-->12h
</hourFormat>
<amFormat>
    <!--opt, enum, subType:string-->am
</amFormat>
<pmFormat>
    <!--opt, enum, subType:string-->pm
</pmFormat>
<clockLineFeedEnabled>
    <!--opt, bool-->true
</clockLineFeedEnabled>
<alignmentX>
    <!--opt, enum, subType:string-->left
</alignmentX>
<alignmentY>
    <!--opt, enum, subType:string-->up
</alignmentY>
<fontType>
    <!--opt, enum, subType:string-->simSun
</fontType>
<fontSize>
    <!--opt, enum, subType:string-->1times
</fontSize>
<fontDisplayMode>
    <!--opt, enum, subType:int-->0
</fontDisplayMode>
<fontBold>
    <!--opt, bool-->true
</fontBold>
<backgroundPicType>
    <!--opt, enum, subType:int-->0
</backgroundPicType>
<clockParam>
    <!--opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,clock}-->
    <clockType>
        <!--opt, enum, subType:string-->dialClock
    </clockType>
    <clockStyle>
        <!--opt, enum, subType:string-->style1
    </clockStyle>
</clockParam>
<weatherParam>
    <!--opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,weather}-->
    <weatherCondition>

```

```

    <!--opt, enum, subType:string-->1
  </weatherCondition>
  <temperature>
    <!--opt, int, range:[-100,100], unit: °C-->26
  </temperature>
  <humidity>
    <!--opt, int, range:[0,100]-->50
  </humidity>
  <city>
    <!--opt, string, range:[1,64]-->hangzhou
  </city>
  <province>
    <!--opt, string, range:[1,16]-->zhejiang
  </province>
  <weatherOverview>
    <!--opt, string, range:[1,128]-->test
  </weatherOverview>
  <airQuality>
    <!--opt, string, range:[1,16]-->test
  </airQuality>
</weatherParam>
<forecastList>
  <!--opt, array, subType:object, range:[1,5]-->
  <forecast>
    <!--opt, object-->
    <highestTemperature>
      <!--opt, int, range:[-100,100], unit: °C-->28
    </highestTemperature>
    <lowestTemperature>
      <!--opt, int, range:[-100,100], unit: °C-->12
    </lowestTemperature>
    <weatherCondition>
      <!--opt, enum, subType:string-->1
    </weatherCondition>
  </forecast>
</forecastList>
</VirtualLEDOnWall>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	pleaseEnterTheSubtitle	0x4000A3D3	--

9.7.9.4 Get parameters of a specified LED

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED/<SubtitlesID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	Window ID
SubtitlesID	string	Subtitle ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VirtualLEDOnWall xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int-->1
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <wndOperateMode>
    <!--ro, opt, enum, subType:string-->uniformCoordinate
  </wndOperateMode>
  <Rect>
    <!--ro, opt, object-->
    <Coordinate>
      <!--ro, req, object-->
      <x>
        <!--ro, req, int, range:[0,1920]-->1
      </x>
      <y>
        <!--ro, req, int, range:[0,1920]-->1
      </y>
    </Coordinate>
    <width>
      <!--ro, req, int, range:[0,1920]-->1
    </width>
    <height>
      <!--ro, req, int, range:[0,1920]-->1
    </height>
  </Rect>
  <Coordinate>
    <!--ro, opt, object-->
    <x>
      <!--ro, req, int, range:[0,1920]-->0
    </x>
    <y>
      <!--ro, req, int, range:[0,1920]-->0
    </y>
  </Coordinate>
  <ResolutionRect>
    <!--ro, opt, object-->
    <Coordinate>
      <!--ro, req, object-->
      <x>
        <!--ro, req, int-->1
      </x>
      <y>
        <!--ro, req, int-->1
      </y>
    </Coordinate>
    <width>
      <!--ro, req, int-->1
    </width>
    <height>
      <!--ro, req, int-->1
    </height>
  </ResolutionRect>
  <text>
    <!--ro, opt, string-->test
  </text>
  <foregroundColor>
    <!--ro, opt, object-->
    <RGB>
      <!--ro, req, string-->ff00ff
    </RGB>
  </foregroundColor>
  <backgroundColor>
    <!--ro, opt, object-->
    <RGB>
      <!--ro, req, string-->ff00ff
    </RGB>
  </backgroundColor>
  <transparencyMode>
    <!--ro, opt, enum, subType:string-->opaque
  </transparencyMode>
  <moveDirection>
```



```
<!--ro, opt, enum, subType:string-->left to right
</moveDirection>
<moveSpeed>
<!--ro, opt, int-->1
</moveSpeed>
<moveMode>
<!--ro, opt, enum, subType:string-->normal
</moveMode>
<ledType>
<!--ro, opt, enum, subType:string-->text
</ledType>
<ledDirection>
<!--ro, opt, enum, subType:string-->horizontal
</ledDirection>
<timeType>
<!--ro, opt, enum, subType:string-->dateTime
</timeType>
<showWeekInTime>
<!--ro, opt, bool-->true
</showWeekInTime>
<dateFormat>
<!--ro, opt, enum, subType:string-->yyyy-mm-dd
</dateFormat>
<timeFormat>
<!--ro, opt, enum, subType:string-->h:mm:ss
</timeFormat>
<hourFormat>
<!--ro, opt, enum, subType:string-->12h
</hourFormat>
<amFormat>
<!--ro, opt, enum, subType:string-->am
</amFormat>
<pmFormat>
<!--ro, opt, enum, subType:string-->pm
</pmFormat>
<clockLineFeedEnabled>
<!--ro, opt, bool-->true
</clockLineFeedEnabled>
<alignmentX>
<!--ro, opt, enum, subType:string-->left
</alignmentX>
<alignmentY>
<!--ro, opt, enum, subType:string-->up
</alignmentY>
<fontType>
<!--ro, opt, enum, subType:string-->simSun
</fontType>
<fontSize>
<!--ro, opt, enum, subType:string-->1times
</fontSize>
<fontDisplayMode>
<!--ro, opt, enum, subType:int-->0
</fontDisplayMode>
<fontBold>
<!--ro, opt, bool-->true
</fontBold>
<backgroundPicType>
<!--ro, opt, enum, subType:int-->0
</backgroundPicType>
<clockParam>
<!--ro, opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,clock}-->
<clockType>
<!--ro, opt, enum, subType:string-->dialClock
</clockType>
<clockStyle>
<!--ro, opt, enum, subType:string-->style1
</clockStyle>
</clockParam>
<weatherParam>
<!--ro, opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,weather}-->
<weatherCondition>
<!--ro, opt, enum, subType:string-->1
</weatherCondition>
<temperature>
<!--ro, opt, int, range:[-100,100], unit:℃-->26
</temperature>
<humidity>
<!--ro, opt, int, range:[0,100]-->50
</humidity>
<city>
<!--ro, opt, string, range:[1,64]-->hangzhou
</city>
<province>
<!--ro, opt, string, range:[1,16]-->zhejiang
</province>
<weatherOverview>
<!--ro, opt, string, range:[1,128]-->test
</weatherOverview>
<airQuality>
<!--ro, opt, string, range:[1,16]-->test
</airQuality>
</weatherParam>
<forecastList>
```

```

<!--ro, opt, array, subType:object, range:[1,5]-->
<forecast>
  <!--ro, opt, object-->
  <highestTemperature>
    <!--ro, opt, int, range:[-100,100], unit:℃-->28
  </highestTemperature>
  <lowestTemperature>
    <!--ro, opt, int, range:[-100,100], unit:℃-->12
  </lowestTemperature>
  <weatherCondition>
    <!--ro, opt, enum, subType:string-->1
  </weatherCondition>
</forecast>
</forecastList>
<wordSpace>
  <!--ro, opt, int-->1
</wordSpace>
</VirtualLEDOnWall>

```

9.7.9.5 Delete a specific virtual LED

Request URL

DELETE /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED/<SubtitlesID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	Window ID
SubtitlesID	string	Subtitle ID

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version(ro, req, string, protocolVersion)-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

9.7.9.6 Set parameters of a specific virtual LED

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED/<SubtitlesID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
SubtitlesID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VirtualLEDOnWall xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, virtual LED settings, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <wndOperateMode>
    <!--opt, enum, subType:string-->uniformCoordinate
  </wndOperateMode>
  <Rect>
    <!--opt, object-->
    <Coordinate>
      <!--req, object-->
      <x>
        <!--req, int, X-coordinate, range:[0,1920]-->1
      </x>
      <y>
        <!--req, int, Y-coordinate, range:[0,1920]-->1
      </y>
    </Coordinate>
    <width>
      <!--req, int, width, range:[0,1920]-->1
    </width>
    <height>
      <!--req, int, height, range:[0,1920]-->1
    </height>
  </Rect>
  <Coordinate>
    <!--opt, object, desc:the origin is the upper-left corner of the screen-->
    <x>
      <!--req, int, X-coordinate, range:[0,1920]-->0
    </x>
    <y>
      <!--req, int, Y-coordinate, range:[0,1920]-->0
    </y>
  </Coordinate>
  <ResolutionRect>
    <!--opt, object-->
    <Coordinate>
      <!--req, object-->
      <x>
        <!--req, int, X-coordinate-->1
      </x>
      <y>
        <!--req, int, Y-coordinate-->1
      </y>
    </Coordinate>
    <width>
      <!--req, int, width-->1
    </width>
    <height>
      <!--req, int, height-->1
    </height>
  </ResolutionRect>
  <text>
    <!--opt, string, text-->test
  </text>
  <foregroundColor>
    <!--opt, object, foreground color-->
    <RGB>
      <!--req, int-->1
    </RGB>
  </foregroundColor>
  <backgroundColor>
    <!--opt, object, background color-->
    <RGB>
      <!--req, int-->1
    </RGB>
  </backgroundColor>
  <transparencyMode>
    <!--opt, enum, subType:string-->opaque
  </transparencyMode>
  <moveDirection>
    <!--opt, enum, moving direction, subType:string-->left to right
  </moveDirection>
  <moveSpeed>
    <!--opt, enum, moving speed, subType:int-->1
  </moveSpeed>
</VirtualLEDOnWall>
```

```

</moveSpeed>
<moveMode>
  <!--opt, enum, moving mode, subType:string-->normal
</moveMode>
<ledType>
  <!--opt, enum, subType:string-->text
</ledType>
<ledDirection>
  <!--opt, enum, subType:string-->horizontal
</ledDirection>
<timeType>
  <!--opt, enum, time type, subType:string-->dateTime
</timeType>
<dateFormat>
  <!--opt, enum, date display format, subType:string-->yyyy-mm-dd
</dateFormat>
<timeFormat>
  <!--opt, enum, time display format, subType:string-->h:mm:ss
</timeFormat>
<hourFormat>
  <!--opt, enum, subType:string-->12h
</hourFormat>
<amFormat>
  <!--opt, enum, subType:string-->am
</amFormat>
<pmFormat>
  <!--opt, enum, subType:string-->pm
</pmFormat>
<alignmentX>
  <!--opt, enum, subType:string-->left
</alignmentX>
<alignmentY>
  <!--opt, enum, subType:string-->up
</alignmentY>
<fontType>
  <!--opt, enum, font type, subType:string-->simSun
</fontType>
<fontSize>
  <!--opt, enum, font size, subType:string-->1times
</fontSize>
<fontDisplayMode>
  <!--opt, enum, subType:int-->0
</fontDisplayMode>
<fontBold>
  <!--opt, bool-->true
</fontBold>
</VirtualLEDOnWall>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version(ro, req, string, protocolVersion)-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	descriptio
4	Invalid Operation	theNumberOfVirtualLedsInASingleOutputPortExceedsTheLimit	0x4000A073	--
4	Invalid Operation	virtualLedWindowAreaOverlap	0x4000A074	--
4	Invalid Operation	virtualLedCannotContainBothLedAndLcdOutputs	0x4000A075	--
4	Invalid Operation	theNumberOfVirtualLedsForASingleWallClockExceedsTheLimit	0x4000A076	--
4	Invalid Operation	theLengthOfTheTextExceededLimit	0x4000A2A4	--
4	Invalid Operation	theNumberOfVerticalTextsExceededLimit	0x4000A2A5	--
4	Invalid Operation	theWidthOfTheTextExceededLimit	0x4000A2B6	--
4	Invalid Operation	theHeightOfTheTextExceededLimit	0x4000A2B7	--
4	Invalid Operation	currentFontIsNotSupportedBySomeTextsOfTheSubtitles	0x4000A459	--

9.7.9.7 Get the virtual LED configuration capability

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED/<SubtitlesID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	Window ID
SubtitlesID	string	Subtitle ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VirtualLEDOnWall xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int-->1
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <wndOperateMode opt="uniformCoordinate,resolutionCoordinate">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->uniformCoordinate
  </wndOperateMode>
  <Rect>
    <!--ro, opt, object-->
    <Coordinate>
      <!--ro, req, object-->
      <x min="0" max="1920">
        <!--ro, req, int, range:[0,1920], attr:min{req, int},max{req, int}-->1
      </x>
      <y min="0" max="1920">
        <!--ro, req, int, range:[0,1920], attr:min{req, int},max{req, int}-->1
      </y>
    </Coordinate>
  </Rect>

```

```

</Coordinate>
<width min="0" max="1920">
  <!--ro, req, int, range:[0,1920], attr:min{req, int},max{req, int}-->1
</width>
<height min="0" max="1920">
  <!--ro, req, int, range:[0,1920], attr:min{req, int},max{req, int}-->1
</height>
</Rect>
<Coordinate>
  <!--ro, opt, object-->
  <x min="0" max="1920">
    <!--ro, req, int, range:[0,1920], attr:min{req, int},max{req, int}-->0
  </x>
  <y min="0" max="1920">
    <!--ro, req, int, range:[0,1920], attr:min{req, int},max{req, int}-->0
  </y>
</Coordinate>
<ResolutionRect>
  <!--ro, opt, object-->
  <Coordinate>
    <!--ro, req, object-->
    <x min="0" max="1920">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </x>
    <y min="0" max="1920">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </y>
  </Coordinate>
  <width min="0" max="1920">
    <!--ro, req, int, attr:min{req, int},max{req, int}-->1
  </width>
  <height min="0" max="1920">
    <!--ro, req, int, attr:min{req, int},max{req, int}-->1
  </height>
</ResolutionRect>
<text max="1024">
  <!--ro, opt, string, attr:max{req, int}-->test
</text>
<foregroundColor>
  <!--ro, opt, object-->
  <RGB>
    <!--ro, req, int-->1
  </RGB>
</foregroundColor>
<backgroundColor>
  <!--ro, opt, object-->
  <RGB>
    <!--ro, req, int-->1
  </RGB>
</backgroundColor>
<transparencyMode opt="opaque,half-transparent,transparent">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->opaque
</transparencyMode>
<moveDirection opt="left to right,right to left,top to bottom,bottom to top">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->left to right
</moveDirection>
<moveSpeed min="1" max="4">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</moveSpeed>
<moveMode opt="normal,smooth,static">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
</moveMode>
<ledType opt="text,time,clock,weather">
  <!--ro, opt, string, attr:opt{req, string}-->test
</ledType>
<ledDirection opt="horizontal,vertical">
  <!--ro, opt, string, attr:opt{req, string}-->test
</ledDirection>
<timeType opt="timeOnly,dateTime">
  <!--ro, opt, string, attr:opt{req, string}-->test
</timeType>
<showWeekInTime opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</showWeekInTime>
<dateFormat opt="yyyy-mm-dd,mm-dd-yyyy,dd-mm-yyyy,xxxx\y00\bx0D,x0\bx0xxxxY,x0Dx0\bx0xxY,yyyy/mm/dd,yy/mm/dd">
  <!--ro, opt, string, attr:opt{req, string}-->test
</dateFormat>
<timeFormat opt="h:mm:ss,hh:mm:ss,hh:mm,hh:mm:ssS,hh:mmM">
  <!--ro, opt, string, attr:opt{req, string}-->test
</timeFormat>
<hourFormat opt="12h,24h">
  <!--ro, opt, string, attr:opt{req, string}-->test
</hourFormat>
<amFormat opt="am,morning">
  <!--ro, opt, string, attr:opt{req, string}-->test
</amFormat>
<pmFormat opt="pm,afternoon">
  <!--ro, opt, string, attr:opt{req, string}-->test
</pmFormat>
<clockLineFeedEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</clockLineFeedEnabled>
<alignmentX opt="left,middle,right">
  <!--ro, opt, string, attr:opt{req, string}-->test

```

```

</alignmentX>
<alignmentY opt="up,middle,down">
  <!--ro, opt, string, attr:opt{req, string}-->test
</alignmentY>
<fontType opt="simSun,simHei,kaiTi,default,custom,LiSu,custom2,custom3">
  <!--ro, opt, string, attr:opt{req, string}-->test
</fontType>
<fontSize opt="0.5times,0.75times,1times,1.25times,1.5times,1.75times,2times,2.5times,3times,3.5times,4times,6times,8times">
  <!--ro, opt, string, attr:opt{req, string}-->test
</fontSize>
<fontDisplayMode opt="0,1,2">
  <!--ro, opt, string, attr:opt{req, string}-->test
</fontDisplayMode>
<fontBold opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</fontBold>
<backgroundPicType opt="0,1,2,3">
  <!--ro, opt, string, attr:opt{req, string}-->test
</backgroundPicType>
<clockParam>
  <!--ro, opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,clock}-->
  <clockType opt="dialClock,digitalClock">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->dialClock
  </clockType>
  <clockStyle opt="style1,style2,style3">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->style1
  </clockStyle>
</clockParam>
<weatherParam>
  <!--ro, opt, object, dep:and,{$.VirtualLEDOnWall.LedType,eq,weather}-->
  <weatherCondition
opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,5
4,55,56,57,58,59,60,61">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->1
  </weatherCondition>
  <temperature min="-100" max="100">
    <!--ro, opt, int, range:[-100,100], unit:℃, attr:min{req, int},max{req, int}-->26
  </temperature>
  <humidity min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->50
  </humidity>
  <city min="1" max="64">
    <!--ro, opt, string, range:[1,64], attr:min{req, int},max{req, int}-->hangzhou
  </city>
  <province min="1" max="16">
    <!--ro, opt, string, range:[1,16], attr:min{req, int},max{req, int}-->zhejiang
  </province>
  <weatherOverview min="1" max="128">
    <!--ro, opt, string, range:[1,128], attr:min{req, int},max{req, int}-->test
  </weatherOverview>
  <airQuality min="1" max="16">
    <!--ro, opt, string, range:[1,16], attr:min{req, int},max{req, int}-->test
  </airQuality>
</weatherParam>
<forecastList min="1" max="5">
  <!--ro, opt, array, subType:object, range:[1,5], attr:min{req, int},max{req, int}-->
  <forecast>
    <!--ro, opt, object-->
    <highestTemperature min="-100" max="100">
      <!--ro, opt, int, range:[-100,100], unit:℃, attr:min{req, int},max{req, int}-->28
    </highestTemperature>
    <lowestTemperature min="-100" max="100">
      <!--ro, opt, int, range:[-100,100], unit:℃, attr:min{req, int},max{req, int}-->12
    </lowestTemperature>
    <weatherCondition
opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,5
4,55,56,57,58,59,60,61">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->1
    </weatherCondition>
  </forecast>
</forecastList>
<wordSpace>
  <!--ro, opt, int-->0
</wordSpace>
</VirtualLEDOnWall>

```

9.7.9.8 Get configuration capability of all virtual LEDs

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/virtualLED/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VirtualLEDCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <virtualLEDNums>
    <!--ro, opt, int, number of virtual LEDs-->1
  </virtualLEDNums>
  <virtualLEDFirstHeightList>
    <!--ro, opt, object-->
    <virtualLEDFirstHeight>
      <!--ro, opt, object, height of virtual LED-->
      <resolution>
        <!--ro, opt, string, resolution-->test
      </resolution>
      <virtualLEDHeight>
        <!--ro, opt, int, height of virtual LED-->1
      </virtualLEDHeight>
    </virtualLEDFirstHeight>
  </virtualLEDFirstHeightList>
  <isSupportFontRestore>
    <!--ro, opt, bool-->true
  </isSupportFontRestore>
  <fontTypeDllMaxSize>
    <!--ro, opt, int-->1
  </fontTypeDllMaxSize>
  <perWallClockSubtitlesMaxNum>
    <!--ro, opt, int-->1
  </perWallClockSubtitlesMaxNum>
  <perWallDynamicSubtitlesMaxNum>
    <!--ro, opt, int-->1
  </perWallDynamicSubtitlesMaxNum>
  <perWallVerticalSubtitlesMaxNum>
    <!--ro, opt, int-->1
  </perWallVerticalSubtitlesMaxNum>
  <dynamicSubtitlesMaxNum>
    <!--ro, opt, int-->1
  </dynamicSubtitlesMaxNum>
  <Rect>
    <!--ro, opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
    <MaxVertSubtitleWindowSize>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,65535]-->1
      </width>
      <height>
        <!--ro, req, int, height, range:[0,65535]-->1
      </height>
    </MaxVertSubtitleWindowSize>
    <MaxHoriSubtitleWindowSize>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,65535]-->1
      </width>
      <height>
        <!--ro, req, int, height, range:[0,65535]-->1
      </height>
    </MaxHoriSubtitleWindowSize>
    <MinVertSubtitleWindowSize>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1920]-->1
      </height>
    </MinVertSubtitleWindowSize>
    <MinHoriSubtitleWindowSize>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1920]-->1
      </height>
    </MinHoriSubtitleWindowSize>
  </Rect>
  <ResolutionRect>
    <!--ro, opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
    <MaxVertSubtitleWindowSize>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,131072]-->1
      </width>
      <height>
        <!--ro, req, int, height, range:[0,131072]-->1
      </height>
    </MaxVertSubtitleWindowSize>
```



```

</MaxVertSubtitleWindowSize>
<MaxHoriSubtitleWindowSize>
  <!--ro, opt, object-->
  <width>
    <!--ro, req, int, width, range:[0,131072]-->1
  </width>
  <height>
    <!--ro, req, int, height, range:[0,131072]-->1
  </height>
</MaxHoriSubtitleWindowSize>
<MinVertSubtitleWindowSize>
  <!--ro, opt, object-->
  <width>
    <!--ro, req, int, width, range:[0,1920]-->1
  </width>
  <height>
    <!--ro, req, int, height, range:[0,1920]-->1
  </height>
</MinVertSubtitleWindowSize>
<MinHoriSubtitleWindowSize>
  <!--ro, opt, object-->
  <width>
    <!--ro, req, int, width, range:[0,1920]-->1
  </width>
  <height>
    <!--ro, req, int, height, range:[0,1920]-->1
  </height>
</MinHoriSubtitleWindowSize>
</ResolutionRect>
<isSupportShowWeekInClockSubtitles>
  <!--ro, opt, bool-->true
</isSupportShowWeekInClockSubtitles>
</VirtualLEDcap>

```

9.7.10 Video Wall Wallpaper Management

9.7.10.1 Get configuration capability of background picture window

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/baseMap/<mapFileID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
mapFileID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BaseMapOnWall xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of background picture window, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, background picture window No.-->1
  </id>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <baseMapType opt="picture,video">
    <!--ro, opt, string, background picture type, attr:opt{req, string}-->test
  </baseMapType>
  <baseMapID>
    <!--ro, opt, int, background picture No.-->1
  </baseMapID>
  <wndOperateMode opt="uniformCoordinate,resolutionCoordinate">
    <!--ro, opt, enum, unified coordinates, subType:string, attr:opt{req, string}, desc:uniformCoordinate; resolutionCoordinate;-->uniformCoordinate
  </wndOperateMode>
  <Rect>
    <!--ro, opt, object-->
    <Coordinate>
      <!--ro, opt, object, coordinates-->
      <x min="0" max="1920">
        <!--ro, opt, int, X-coordinate, range:[0,1920], attr:min{req, int},max{req, int}-->1
      </x>
      <y min="0" max="1920">
        <!--ro, opt, int, y-coordinate, range:[0,1920], attr:min{req, int},max{req, int}-->1
      </y>
    </Coordinate>
    <width min="0" max="1920">
      <!--ro, opt, int, width, range:[0,1920], attr:min{req, int},max{req, int}-->1
    </width>
    <height min="0" max="1920">
      <!--ro, opt, int, length, range:[0,1920], attr:min{req, int},max{req, int}-->1
    </height>
  </Rect>
  <Coordinate>
    <!--ro, opt, object, origin coordinates, desc:the origin is the upper-left corner of the screen-->
    <x min="0" max="1920">
      <!--ro, opt, int, X-coordinate, range:[0,1920], attr:min{req, int},max{req, int}-->1
    </x>
    <y min="0" max="1920">
      <!--ro, opt, int, Y-coordinate, range:[0,1920], attr:min{req, int},max{req, int}-->1
    </y>
  </Coordinate>
  <ResolutionRect>
    <!--ro, opt, object, resolution coordinate, desc:resolution coordinate-->
    <Coordinate>
      <!--ro, opt, object, coordinates-->
      <x min="0" max="1920">
        <!--ro, opt, int, X-coordinate, attr:min{req, int},max{req, int}, desc:X-coordinate-->1
      </x>
      <y min="0" max="1920">
        <!--ro, opt, int, Y-coordinate, attr:min{req, int},max{req, int}, desc:Y-coordinate-->1
      </y>
    </Coordinate>
    <width min="0" max="1920">
      <!--ro, opt, int, width, attr:min{req, int},max{req, int}, desc:width-->1
    </width>
    <height min="0" max="1920">
      <!--ro, opt, int, height, attr:min{req, int},max{req, int}, desc:height-->1
    </height>
  </ResolutionRect>
</BaseMapOnWall>
```

9.7.10.2 Get the capability of all background pictures

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/baseMap/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BaseMapOnWallCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <baseMapOnWallNums>
    <!--ro, opt, int-->1
  </baseMapOnWallNums>
</BaseMapOnWallCap>
```

9.7.10.3 Set parameters of all background pictures

Request URL

PUT /ISAPI/DisplayDev/VideoWall/baseMap

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BaseMapList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configuration of all background pictures, attr:version{req, string, protocolVersion}-->
  <BaseMap>
    <!--opt, object, configuration of a single background picture-->
    <id>
      <!--opt, string, background picture ID-->test
    </id>
    <name>
      <!--opt, string, background picture name-->test
    </name>
    <fileType>
      <!--opt, string, picture type-->test
    </fileType>
    <imageWidth>
      <!--opt, int, picture width-->1
    </imageWidth>
    <imageHeight>
      <!--opt, int, picture height-->1
    </imageHeight>
  </BaseMap>
</BaseMapList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->test
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details-->test
  </subStatusCode>
</ResponseStatus>
```

9.7.10.4 Delete a specific background picture

Request URL

DELETE /ISAPI/DisplayDev/VideoWall/baseMap/<mapFileID>

Query Parameter

Parameter Name	Parameter Type	Description
mapFileID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.10.5 Set parameters of a specific background picture

Request URL

PUT /ISAPI/DisplayDev/VideoWall/baseMap/<mapFileID>

Query Parameter

Parameter Name	Parameter Type	Description
mapFileID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BaseMap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, string, No. of background picture-->test
  </id>
  <name>
    <!--opt, string, background picture name-->test
  </name>
  <fileType>
    <!--opt, string, picture type-->test
  </fileType>
  <imageWidth>
    <!--opt, int, picture width-->1
  </imageWidth>
  <imageHeight>
    <!--opt, int, picture height-->1
  </imageHeight>
</BaseMap>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

9.7.10.6 Get parameters of a background picture

Request URL

GET /ISAPI/DisplayDev/VideoWall/baseMap/<mapFileID>

Query Parameter

Parameter Name	Parameter Type	Description
mapFileID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BaseMap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, string-->test
  </id>
  <name>
    <!--ro, opt, string-->test
  </name>
  <fileType>
    <!--ro, opt, string-->test
  </fileType>
  <imageWidth>
    <!--ro, opt, int-->1
  </imageWidth>
  <imageHeight>
    <!--ro, opt, int-->1
  </imageHeight>
</BaseMap>
```

9.7.10.7 Get the background picture configuration capability

Request URL

GET /ISAPI/DisplayDev/VideoWall/baseMap/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BaseMapCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <baseMapNums>
    <!--ro, opt, int-->1
  </baseMapNums>
  <supportFileType>
    <!--ro, opt, enum, subType:string-->JPEG
  </supportFileType>
  <maxFileSize>
    <!--ro, opt, int-->1
  </maxFileSize>
  <maxImageWidth>
    <!--ro, opt, int-->1
  </maxImageWidth>
  <maxImageHeight>
    <!--ro, opt, int-->1
  </maxImageHeight>
  <BaseMapAlignUnit>
    <!--ro, opt, object-->
    <width opt="0,2,4,8,...">
      <!--ro, req, int, attr:opt{req, string}-->1
    </width>
    <height opt="0,2,4,8,...">
      <!--ro, req, int, attr:opt{req, string}-->1
    </height>
  </BaseMapAlignUnit>
  <isSupportBaseMapCircle opt="true">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportBaseMapCircle>
  <imageWidth min="0" max="16384">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </imageWidth>
  <imageHeight min="0" max="8192">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </imageHeight>
</BaseMapCap>
```

9.7.10.8 Get configuration of all background pictures

Request URL

GET /ISAPI/DisplayDev/VideoWall/baseMap?isGetBaseMapFile=<isGetBaseMapFile>

Query Parameter

Parameter Name	Parameter Type	Description
isGetBaseMapFile	enum	The background picture will not be obtained by default.

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<BaseMapList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configuration of all background pictures, attr:version{req, string, protocolVersion}-->
  <BaseMap>
    <!--ro, opt, object, configuration of a single background picture-->
    <id>
      <!--ro, opt, string, No. of background picture-->test
    </id>
    <name>
      <!--ro, opt, string, background picture name-->test
    </name>
    <fileType>
      <!--ro, opt, string, picture type-->test
    </fileType>
    <imageWidth>
      <!--ro, opt, int, picture width-->1
    </imageWidth>
    <imageHeight>
      <!--ro, opt, int, picture height-->1
    </imageHeight>
    <baseMapFile>
      <!--ro, opt, object-->
      <filePathType>
        <!--ro, opt, enum, subType:string-->multipart
      </filePathType>
      <filePath>
        <!--ro, opt, string, range:[0,1024]-->test
      </filePath>
    </baseMapFile>
  </BaseMap>
</BaseMapList>

```

9.7.11 Video Wall Window Management

9.7.11.1 Get LED or LCD areas

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/ledArea

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LedAreaList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, LED area list, attr:version{req, string, protocolVersion}-->
  <LedArea>
    <!--ro, opt, object, LED area-->
    <id>
      <!--ro, req, int, LED area ID-->1
    </id>
    <areaType>
      <!--ro, opt, enum, area type, subType:string, desc:area type-->LED
    </areaType>
    <Rect>
      <!--ro, opt, object, LED area, desc:LED area-->
      <Coordinate>
        <!--ro, opt, object, coordinates-->
        <x>
          <!--ro, opt, int, X-coordinate, range:[0,1920]-->1
        </x>
        <y>
          <!--ro, opt, int, Y-coordinate, range:[0,1920]-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, opt, int, width, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, opt, int, length, range:[0,1920]-->1
      </height>
    </Rect>
    <OutputChanList>
      <!--ro, req, object, output port list-->
      <id>
        <!--ro, req, int, LED area ID-->1
      </id>
    </OutputChanList>
  </LedArea>
</LedAreaList>
```

9.7.11.2 Get all windows' parameters

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallWindowList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <WallWindow>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <wndOperateMode>
      <!--ro, opt, enum, subType:string-->uniformCoordinate
    </wndOperateMode>
    <Rect>
      <!--ro, opt, object-->
      <Coordinate>
        <!--ro, opt, object-->
        <x>
          <!--ro, opt, int, range:[0,1920]-->1
        </x>
        <y>
          <!--ro, opt, int, range:[0,1920]-->1
        </y>
      </Coordinate>
      <width>
        <!--ro, opt, int, range:[0,1920]-->1
      </width>
      <height>
        <!--ro, opt, int, range:[0,1920]-->1
      </height>
    </Rect>
  </WallWindow>
</WallWindowList>
```



```

</height>
</Rect>
<Coordinate>
  <!--ro, opt, object-->
  <x>
    <!--ro, opt, int, range:[0,1920]-->1
  </x>
  <y>
    <!--ro, opt, int, range:[0,1920]-->1
  </y>
</Coordinate>
<ResolutionRect>
  <!--ro, opt, object-->
  <Coordinate>
    <!--ro, opt, object-->
    <x>
      <!--ro, opt, int-->1
    </x>
    <y>
      <!--ro, opt, int-->1
    </y>
  </Coordinate>
  <width>
    <!--ro, opt, int-->1
  </width>
  <height>
    <!--ro, opt, int-->1
  </height>
</ResolutionRect>
<layerIdx>
  <!--ro, opt, int-->1
</layerIdx>
<displayWinNo>
  <!--ro, opt, bool-->true
</displayWinNo>
<windowMode>
  <!--ro, opt, enum, subType:int-->1
</windowMode>
<wndShowMode>
  <!--ro, opt, enum, subType:string-->subWndMode
</wndShowMode>
<amplifyingSubWndNo>
  <!--ro, opt, int-->1
</amplifyingSubWndNo>
<wndTopKeep>
  <!--ro, opt, bool-->true
</wndTopKeep>
<wndOpenKeep>
  <!--ro, opt, bool-->true
</wndOpenKeep>
<SubWindowList>
  <!--ro, opt, object-->
  <SubWindow>
    <!--ro, opt, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <SubWindowParam>
      <!--ro, opt, object-->
      <signalMode>
        <!--ro, opt, enum, subType:string-->video input
      </signalMode>
      <videoInputChannelID>
        <!--ro, opt, string-->16842753
      </videoInputChannelID>
      <streamingChannelID>
        <!--ro, opt, string-->test
      </streamingChannelID>
      <StreamInput>
        <!--ro, opt, object-->
        <streamInputMode>
          <!--ro, req, enum, subType:string-->realtime
        </streamInputMode>
        <StreamInputRealtime>
          <!--ro, opt, object-->
          <durationInUnit>
            <!--ro, opt, int, unit:s-->1
          </durationInUnit>
          <StreamRealtimeUnitList>
            <!--ro, opt, object-->
            <StreamRealtimeUnit>
              <!--ro, opt, object-->
              <streamType>
                <!--ro, req, enum, subType:string-->in URL
              </streamType>
              <StreamInURL>
                <!--ro, opt, object, dep:and,
                {$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
              </URL>
              <!--ro, req, string-->test
            </URL>
          </StreamInURL>
          <StreamByDdns>
            <!--ro, opt, object, dep:and,

```

```
{$.SubWindow.SubWindowParam.StreamInput.Realtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
```

```
<DdnsServerInfo>
  <!--ro, opt, object-->
  <domain>
    <!--ro, req, enum, subType:string-->ipv4
  </domain>
  <port>
    <!--ro, req, int-->80
  </port>
  <ddnsType>
    <!--ro, req, string-->test
  </ddnsType>
  <username>
    <!--ro, req, string-->test
  </username>
  <password>
    <!--ro, req, string-->test
  </password>
</DdnsServerInfo>
<EncodeDevInfo>
  <!--ro, opt, object-->
  <domain>
    <!--ro, req, enum, subType:string-->ipv4
  </domain>
  <port>
    <!--ro, req, int-->1
  </port>
  <transmitProtocol>
    <!--ro, req, enum, subType:string-->tcp
  </transmitProtocol>
  <protocol>
    <!--ro, req, enum, subType:string-->DAHUA
  </protocol>
  <username>
    <!--ro, req, string-->test
  </username>
  <password>
    <!--ro, req, string-->test
  </password>
  <channelMode>
    <!--ro, req, enum, subType:string-->normal
  </channelMode>
  <channelType>
    <!--ro, req, enum, subType:string-->main
  </channelType>
  <channelZero>
    <!--ro, opt, int-->1
  </channelZero>
  <channelNormal>
    <!--ro, opt, int-->1
  </channelNormal>
  <channelStreaming>
    <!--ro, opt, int-->1
  </channelStreaming>
  <channelDistributed>
    <!--ro, opt, int-->1
  </channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <domain>
    <!--ro, req, enum, subType:string-->ipv4
  </domain>
  <port>
    <!--ro, req, int-->1
  </port>
  <transmitProtocol>
    <!--ro, req, enum, subType:string-->tcp
  </transmitProtocol>
</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--ro, opt, object-->
  <EncodeDevInfo>
    <!--ro, opt, object-->
    <domain>
      <!--ro, req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--ro, req, int-->1
    </port>
    <transmitProtocol>
      <!--ro, req, enum, subType:string-->tcp
    </transmitProtocol>
    <protocol>
      <!--ro, req, enum, subType:string-->DAHUA
    </protocol>
    <username>
      <!--ro, req, string, range:[1,32]-->test
    </username>
```

```

    <password>
      <!--ro, req, string-->test
    </password>
    <channelMode>
      <!--ro, req, enum, subType:string-->normal
    </channelMode>
    <channelType>
      <!--ro, req, enum, subType:string-->main
    </channelType>
    <channelZero>
      <!--ro, opt, int-->1
    </channelZero>
    <channelNormal>
      <!--ro, opt, int-->1
    </channelNormal>
    <channelStreaming>
      <!--ro, opt, int-->1
    </channelStreaming>
    <channelDistributed>
      <!--ro, opt, int-->1
    </channelDistributed>
  </EncodeDevInfo>
  <MediaGatewayInfo>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, enum, subType:string-->true
    </enabled>
    <domain>
      <!--ro, req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--ro, req, int-->1
    </port>
    <transmitProtocol>
      <!--ro, req, enum, subType:string-->tcp
    </transmitProtocol>
  </MediaGatewayInfo>
  <StreamByDomain>
  </StreamByDomain>
  <StreamRealtimeUnit>
  </StreamRealtimeUnit>
  <StreamRealtimeUnitList>
  </StreamRealtimeUnitList>
  <StreamInputRealtime>
  </StreamInputRealtime>
  <StreamInputPlayback>
    <!--ro, opt, object-->
    <playbackMode>
      <!--ro, opt, enum, subType:string-->file name
    </playbackMode>
  </StreamInputPlayback>
  <EncodeDevInfo>
    <!--ro, opt, object-->
    <domain>
      <!--ro, req, enum, subType:string-->ipv4
    </domain>
    <port>
      <!--ro, req, int-->1
    </port>
    <transmitProtocol>
      <!--ro, req, enum, subType:string-->tcp
    </transmitProtocol>
    <protocol>
      <!--ro, req, enum, subType:string-->DAHUA
    </protocol>
    <username>
      <!--ro, req, string-->test
    </username>
    <password>
      <!--ro, req, string-->test
    </password>
    <channelMode>
      <!--ro, req, enum, subType:string-->normal
    </channelMode>
    <channelType>
      <!--ro, req, enum, subType:string-->main
    </channelType>
    <channelZero>
      <!--ro, opt, int-->1
    </channelZero>
    <channelNormal>
      <!--ro, opt, int-->1
    </channelNormal>
    <channelStreaming>
      <!--ro, opt, int-->1
    </channelStreaming>
    <channelDistributed>
      <!--ro, opt, int-->1
    </channelDistributed>
  </EncodeDevInfo>
  <fileName>
    <!--ro, opt, string-->test
  </fileName>
  <TimeRange>
    <!--ro, opt, object-->
    <beginTime>
      <!--ro, opt, time-->00:00:00+08:00
    </beginTime>
    <endTime>

```

```

</StreamRange>
<!--ro, opt, time-->00:00:00+08:00
</endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
<!--ro, opt, bool-->>true
</streamEncryptEnable>
<streamPassword>
<!--ro, opt, string, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}-->test
</streamPassword>
</StreamInput>
<pictureFormat>
<!--ro, opt, enum, subType:string-->BMP
</pictureFormat>
<signalSourceName>
<!--ro, opt, string-->test
</signalSourceName>
</SubWindowParam>
</SubWindow>
</SubWindowList>
<wndType>
<!--ro, opt, enum, subType:string-->signalSource
</wndType>
<zoomEnabled>
<!--ro, opt, bool-->>true
</zoomEnabled>
<audioEnabled>
<!--ro, opt, bool-->true
</audioEnabled>
<wndLockKeep>
<!--ro, opt, bool-->true
</wndLockKeep>
<Graphic>
<!--ro, opt, object-->
<windowRegisterID>
<!--ro, opt, int-->0
</windowRegisterID>
</Graphic>
<Subtitle>
<!--ro, opt, object-->
<moveDirection>
<!--ro, opt, enum, subType:string-->left
</moveDirection>
<moveSpeed>
<!--ro, opt, enum, subType:string-->fast
</moveSpeed>
</Subtitle>
</WallWindow>
</WallWindowList>

```

9.7.11.3 Delete all windows

Request URL

DELETE /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.11.4 Add a window

Request URL

POST /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallWindow xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, window configuration capability of the video wall, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, ID-->1
  </id>
  <wndOperateMode>
    <!--opt, enum, "uniformCoordinate,resolutionCoordinate", subType:string, desc:"uniformCoordinate, resolutionCoordinate"-->uniformCoordinate
  </wndOperateMode>
  <Rect>
    <!--opt, object, normalized coordinates, desc:normalized coordinates-->
    <Coordinate>
      <!--opt, object, coordinates-->
      <x>
        <!--opt, int, X-coordinate, range:[0,1920]-->1
      </x>
      <y>
        <!--opt, int, Y-coordinate, range:[0,1920]-->1
      </y>
    </Coordinate>
    <width>
      <!--opt, int, width, range:[0,1920]-->1
    </width>
    <height>
      <!--opt, int, Length, range:[0,1920]-->1
    </height>
  </Rect>
  <Coordinate>
    <!--opt, object, original point coordinates, desc:the origin is the upper-left corner of the screen-->
    <x>
      <!--opt, int, X-coordinate, range:[0,1920]-->1
    </x>
    <y>
      <!--opt, int, Y-coordinate, range:[0,1920]-->1
    </y>
  </Coordinate>
  <ResolutionRect>
    <!--opt, object, resolution coordinate, desc:resolution coordinate-->
    <Coordinate>
      <!--opt, object, coordinates-->
      <x>
        <!--opt, int, X-coordinate, desc:X-coordinate-->1
      </x>
      <y>
        <!--opt, int, Y-coordinate, desc:Y-coordinate-->1
      </y>
    </Coordinate>
  </ResolutionRect>
</WallWindow>
```

```

    </x>
  <y>
    <!--opt, int, Y-coordinate, desc:Y-coordinate-->1
  </y>
</Coordinate>
<width>
  <!--opt, int, width, desc:width-->1
</width>
<height>
  <!--opt, int, height, desc:height-->1
</height>
</ResolutionRect>
<layerIdx>
  <!--opt, int-->1
</layerIdx>
<displayWinNo>
  <!--opt, enum, subType:object-->true
</displayWinNo>
<windowMode>
  <!--opt, enum, "1,4,9,16", subType:int, desc:"1, 4, 9, 16"-->1
</windowMode>
<wndShowMode>
  <!--opt, enum, fullScreenMode", subType:string, desc:"subWndMode", "fullScreenMode"-->subWndMode
</wndShowMode>
<amplifyingSubWndNo>
  <!--opt, int-->1
</amplifyingSubWndNo>
<wndTopKeep>
  <!--opt, enum, "true,false", subType:bool, desc:true (enable), false (disable)-->true
</wndTopKeep>
<wndOpenKeep>
  <!--opt, enum, "true,false", subType:bool, desc:true (enable), false (disable)-->true
</wndOpenKeep>
<SubWindowList>
  <!--opt, object, sub-window List-->
  <SubWindow>
    <!--opt, object, sub-window-->
    <id>
      <!--opt, int, sub-window No.-->1
    </id>
    <SubWindowParam>
      <!--opt, object-->
      <signalMode>
        <!--opt, enum, subType:string-->video input
      </signalMode>
      <videoInputChannelID>
        <!--opt, string, video input channel ID, desc:video input channel ID-->16842753
      </videoInputChannelID>
      <streamingChannelID>
        <!--opt, string, streaming channel ID-->test
      </streamingChannelID>
      <StreamInput>
        <!--opt, object, network stream information-->
        <streamInputMode>
          <!--req, enum, stream type, subType:string, desc:stream type-->realtime
        </streamInputMode>
        <StreamInputRealtime>
          <!--opt, object, real-time stream information-->
          <durationInUnit>
            <!--opt, int, unit:s-->1
          </durationInUnit>
          <StreamRealtimeUnitList>
            <!--opt, object-->
            <StreamRealtimeUnit>
              <!--opt, object-->
              <streamType>
                <!--req, enum, stream type, subType:string, desc:stream type-->in URL
              </streamType>
              <StreamInURL>
                <!--opt, object, dep:and,
                <!--req, string, URL address-->test
              </URL>
            </StreamInURL>
            <StreamByDdns>
              <!--opt, object, dep:and,
              <!--req, string, ddns type-->test
            </ddnsType>
            <username>
              <!--req, string, user name-->test
            </username>
            <password>

```

```

        <!--req, string, password-->test
    </password>
</DdnsServerInfo>
<EncodeDevInfo>
    <!--opt, object-->
    <domain>
        <!--req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port>
        <!--req, int, port-->1
    </port>
    <transmitProtocol>
        <!--req, enum, transport protocol, subType:string, desc:"tcp", "udp", "mcast"-->tcp
    </transmitProtocol>
    <protocol>
        <!--req, enum, protocol, subType:string, desc:protocol-->DAHUA
    </protocol>
    <username>
        <!--req, string, user name-->test
    </username>
    <password>
        <!--req, string, password-->test
    </password>
    <channelMode>
        <!--req, enum, channel mode, subType:string, desc:channel mode-->normal
    </channelMode>
    <channelType>
        <!--req, enum, channel type, subType:string, desc:channel type-->main
    </channelType>
    <channelZero>
        <!--opt, int-->1
    </channelZero>
    <channelNormal>
        <!--opt, int-->1
    </channelNormal>
    <channelStreaming>
        <!--opt, int, streaming channel-->1
    </channelStreaming>
    <channelDistributed>
        <!--opt, int-->1
    </channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
    <!--opt, object-->
    <enabled>
        <!--req, enum, whether to enable, subType:string, desc:true (enable), false (disable)-->true
    </enabled>
    <domain>
        <!--req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port>
        <!--req, int, port-->1
    </port>
    <transmitProtocol>
        <!--req, enum, transport protocol, subType:string, desc:"tcp", "udp", "mcast"-->tcp
    </transmitProtocol>
</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
    <!--opt, object-->
    <EncodeDevInfo>
        <!--opt, object-->
        <domain>
            <!--req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
        </domain>
        <port>
            <!--req, int, port-->1
        </port>
        <transmitProtocol>
            <!--req, enum, transport protocol, subType:string, desc:"tcp", "udp", "mcast"-->tcp
        </transmitProtocol>
        <protocol>
            <!--req, enum, protocol, subType:string, desc:protocol-->DAHUA
        </protocol>
        <username>
            <!--req, string, user name, range:[1,32]-->test
        </username>
        <password>
            <!--req, string, password-->test
        </password>
        <channelMode>
            <!--req, enum, channel mode, subType:string, desc:channel mode-->normal
        </channelMode>
        <channelType>
            <!--req, enum, channel type, subType:string, desc:channel type-->main
        </channelType>
        <channelZero>
            <!--opt, int-->1
        </channelZero>
        <channelNormal>
            <!--opt, int-->1
        </channelNormal>
        <channelStreaming>
            <!--opt, int, streaming channel-->1
        </channelStreaming>
    </EncodeDevInfo>
    <MediaGatewayInfo>
        <!--opt, object-->
        <enabled>
            <!--req, enum, whether to enable, subType:string, desc:true (enable), false (disable)-->true
        </enabled>
        <domain>
            <!--req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
        </domain>
        <port>
            <!--req, int, port-->1
        </port>
        <transmitProtocol>
            <!--req, enum, transport protocol, subType:string, desc:"tcp", "udp", "mcast"-->tcp
        </transmitProtocol>
    </MediaGatewayInfo>
</StreamByDomain>

```

```

        <!--opt, int, streaming channel-->1
    </channelStreaming>
    <channelDistributed>
        <!--opt, int-->1
    </channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
    <!--opt, object-->
    <enabled>
        <!--req, enum, whether to enable, subType:string, desc:true (enable), false (disable)-->true
    </enabled>
    <domain>
        <!--req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port>
        <!--req, int, port-->1
    </port>
    <transmitProtocol>
        <!--req, enum, transport protocol, subType:string, desc:"tcp", "udp", "mcast"-->tcp
    </transmitProtocol>
</MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
    <!--opt, object-->
    <playbackMode>
        <!--opt, enum, playback mode, subType:string, desc:playback mode-->file name
    </playbackMode>
</EncodeDevInfo>
    <!--opt, object-->
    <domain>
        <!--req, enum, domain name, subType:string, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port>
        <!--req, int, port-->1
    </port>
    <transmitProtocol>
        <!--req, enum, transport protocol, subType:string, desc:"tcp", "udp", "mcast"-->tcp
    </transmitProtocol>
    <protocol>
        <!--req, enum, protocol, subType:string, desc:protocol-->DAHUA
    </protocol>
    <username>
        <!--req, string, user name-->test
    </username>
    <password>
        <!--req, string, password-->test
    </password>
    <channelMode>
        <!--req, enum, channel mode, subType:string, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelType>
        <!--req, enum, channel type, subType:string, desc:channel type-->main
    </channelType>
    <channelZero>
        <!--opt, int-->1
    </channelZero>
    <channelNormal>
        <!--opt, int, normal channel-->1
    </channelNormal>
    <channelStreaming>
        <!--opt, int, streaming channel-->1
    </channelStreaming>
    <channelDistributed>
        <!--opt, int, distributed channel-->1
    </channelDistributed>
</EncodeDevInfo>
<fileName>
    <!--opt, string, file name-->test
</fileName>
<TimeRange>
    <!--opt, object, time range-->
    <beginTime>
        <!--opt, time, start time-->00:00:00+08:00
    </beginTime>
    <endTime>
        <!--opt, time, end time-->00:00:00+08:00
    </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
    <!--opt, enum, whether to enable sub stream, subType:string, desc:whether to enable sub stream-->true
</streamEncryptEnable>
<streamPassword>
    <!--opt, string, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}-->test
</streamPassword>
</StreamInput>
</SubWindowParam>
</SubWindow>
</SubwindowList>
<wndType>
    <!--opt, enum, window type, subType:string, desc>window type-->signalSource

```



```

</wndType>
<zoomEnabled>
  <!--opt, bool-->true
</zoomEnabled>
<audioEnabled>
  <!--opt, bool, whether to enable audio-->true
</audioEnabled>
<wndLockKeep>
  <!--opt, bool-->true
</wndLockKeep>
<Graphic>
  <!--opt, object-->
  <windowRegisterID>
    <!--opt, int-->0
  </windowRegisterID>
</Graphic>
<Subtitle>
  <!--opt, object-->
  <moveDirection>
    <!--opt, enum, subType:string-->left
  </moveDirection>
  <moveSpeed>
    <!--opt, enum, moving speed, subType:string, desc:moving speed-->fast
  </moveSpeed>
</Subtitle>
</WallWindow>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <ID>
    <!--ro, opt, int, range:[1,2147483647]-->1
  </ID>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	descri
4	Invalid Operation	failToDisplayPcScreenOnWallForSplicingSourceIsDisplayedOnWall	0x4000A1F7	--
4	Invalid Operation	theNumberOfIntelligentDecodingChannelsHasReachedItsLimit	0x4000A29A	--
4	Invalid Operation	theSplicingSourceHasBeenDisplayedOnSmartWall	0x4000A2A3	--
4	Invalid Operation	numberOfLayersExceededLimit	0x4000A30E	--
4	Invalid Operation	inputChannelHasBeenLinkedToOtherOutputChannels	0x4000A3C5	--
4	Invalid Operation	timesOfDisplayingOnSmartWallHasReachedTheLimitOfInputBoard	0x4000A3C6	--
4	Invalid Operation	theConferenceModeDoesNotSupportWindowRoaming	0x4000A714	--
4	Invalid Operation	windowSplitScreenIsNotSupportedInLowLatencyMode	0x4000A7C4	--
4	Invalid Operation	lowLatencyModeOnlySupportsTheMainSourceLayerForWallDisplay	0x4000A7C5	--
4	Invalid Operation	splitScreenNotSupportedWithSyncLockOn	0x4000A81F	--
4	Invalid Operation	singleLayerOnlySupportedWithSyncLockOn	0x4000A820	--
4	Invalid Operation	onlySourceLayerAllowedWithSyncLockOn	0x4000A821	--

9.7.11.5 Get parameters configuration of a specific window

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WallWindow xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, window of the video wall, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, string, No.-->test
  </id>
  <wndOperateMode>
    <!--ro, opt, enum, "uniformCoordinate,resolutionCoordinate", subType:string, desc:"uniformCoordinate", "resolutionCoordinate"-->uniformCoordinate
  </wndOperateMode>
</WallWindow>
```

```

<rect>
  <!--ro, opt, object-->
  <Coordinate>
    <!--ro, opt, object, coordinates-->
    <x>
      <!--ro, opt, int, X-coordinate, range:[0,1920]-->1
    </x>
    <y>
      <!--ro, opt, int, Y-coordinate, range:[0,1920]-->1
    </y>
  </Coordinate>
  <width min="0" max="1920">
    <!--ro, opt, int, width, range:[0,1920], attr:min{req, int},max{req, int}-->1
  </width>
  <height min="0" max="1920">
    <!--ro, opt, int, height, range:[0,1920], attr:min{req, int},max{req, int}-->1
  </height>
</Rect>
<Coordinate>
  <!--ro, opt, object, original point coordinate, desc:the origin is the upper-left corner of the screen-->
  <x>
    <!--ro, opt, int, X-coordinate, range:[0,1920]-->1
  </x>
  <y>
    <!--ro, opt, int, Y-coordinate, range:[0,1920]-->1
  </y>
</Coordinate>
<ResolutionRect>
  <!--ro, opt, object, resolution coordinate, desc:resolution coordinate-->
  <Coordinate>
    <!--ro, opt, object, coordinates-->
    <x>
      <!--ro, opt, int, X-coordinate, desc:X-coordinate-->1
    </x>
    <y>
      <!--ro, opt, int, Y-coordinate, desc:Y-coordinate-->1
    </y>
  </Coordinate>
  <width min="0" max="1920">
    <!--ro, opt, int, width, attr:min{req, int},max{req, int}, desc:width-->1
  </width>
  <height min="0" max="1920">
    <!--ro, opt, int, height, attr:min{req, int},max{req, int}, desc:height-->1
  </height>
</ResolutionRect>
<layerIdx>
  <!--ro, opt, int-->1
</layerIdx>
<displayWinNo>
  <!--ro, opt, bool, true (enable), false (disable)-->true
</displayWinNo>
<windowMode>
  <!--ro, opt, enum, "1, 4, 9, 16", subType:int, desc:1 (1 days), 15 (15 days), 9 (9 days)-->1
</windowMode>
<wndShowMode>
  <!--ro, opt, enum, screen displaying mode, subType:string, desc:fullScreenMode"-->subWndMode
</wndShowMode>
<amplifyingSubWndNo>
  <!--ro, opt, int-->1
</amplifyingSubWndNo>
<wndTopKeep>
  <!--ro, opt, bool, "true,false"-->true
</wndTopKeep>
<wndOpenKeep>
  <!--ro, opt, bool, "true,false"-->true
</wndOpenKeep>
<SubWindowList>
  <!--ro, opt, object, sub-window List-->
  <SubWindow>
    <!--ro, opt, object, sub-window-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <SubWindowParam>
      <!--ro, opt, object-->
      <signalMode>
        <!--ro, opt, enum, subType:string-->video input
      </signalMode>
      <videoInputChannelID>
        <!--ro, opt, string, video input channel ID, desc:video input channel ID-->16842753
      </videoInputChannelID>
      <streamingChannelID>
        <!--ro, opt, string-->test
      </streamingChannelID>
      <StreamInput>
        <!--ro, opt, object-->
        <streamInputMode>
          <!--ro, req, enum, stream type, subType:string, desc:stream type-->realtime
        </streamInputMode>
        <StreamInputRealtime>
          <!--ro, opt, object, "realStream" (real-time stream), "Playback-->
          <durationInUnit>
            <!--ro, opt, int, unit:s-->1
          </durationInUnit>
        </StreamInputRealtime>
      </StreamInput>
    </SubWindowParam>
  </SubWindow>
</SubWindowList>

```

```

<StreamRealtimeUnitList>
  <!--ro, opt, object-->
  <StreamRealtimeUnit>
    <!--ro, opt, object-->
    <streamType>
      <!--ro, req, enum, stream type, subType:string, desc:stream type-->in URL
    </streamType>
    <StreamInURL>
      <!--ro, opt, object, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
      <URL>
        <!--ro, req, string, URL address-->test
      </URL>
    </StreamInURL>
    <StreamByDdns>
      <!--ro, opt, object, dep:and,
{$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
      <DdnsServerInfo>
        <!--ro, opt, object-->
        <domain>
          <!--ro, req, enum, domain name, subType:string, desc:domain name-->ipv4
        </domain>
        <port>
          <!--ro, req, int, port-->80
        </port>
        <ddnsType>
          <!--ro, req, string, ddns type-->test
        </ddnsType>
        <username>
          <!--ro, req, string, user name-->test
        </username>
        <password>
          <!--ro, req, string, password-->test
        </password>
      </DdnsServerInfo>
      <EncodeDevInfo>
        <!--ro, opt, object-->
        <domain>
          <!--ro, req, enum, domain name, subType:string, desc:domain name-->ipv4
        </domain>
        <port>
          <!--ro, req, int, port-->1
        </port>
        <transmitProtocol>
          <!--ro, req, enum, transport protocol, subType:string, desc:transport protocol-->tcp
        </transmitProtocol>
        <protocol>
          <!--ro, req, enum, protocol, subType:string, desc:protocol-->DAHUA
        </protocol>
        <username>
          <!--ro, req, string, user name-->test
        </username>
        <password>
          <!--ro, req, string, password-->test
        </password>
        <channelMode>
          <!--ro, req, enum, channel mode, subType:string, desc:channel mode-->normal
        </channelMode>
        <channelType>
          <!--ro, req, enum, channel type, subType:string, desc:channel type-->main
        </channelType>
        <channelZero>
          <!--ro, opt, int-->1
        </channelZero>
        <channelNormal>
          <!--ro, opt, int-->1
        </channelNormal>
        <channelStreaming>
          <!--ro, opt, int-->1
        </channelStreaming>
        <channelDistributed>
          <!--ro, opt, int-->1
        </channelDistributed>
      </EncodeDevInfo>
      <MediaGatewayInfo>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <domain>
          <!--ro, req, enum, domain name, subType:string, desc:domain name-->ipv4
        </domain>
        <port>
          <!--ro, req, int, port-->1
        </port>
        <transmitProtocol>
          <!--ro, req, enum, transport protocol, subType:string, desc:transport protocol-->tcp
        </transmitProtocol>
      </MediaGatewayInfo>
    </StreamByDdns>
    <StreamByDomain>
      <!--ro, opt, object-->
      <EncodeDevInfo>

```

```

<!--ro, opt, object-->
<domain>
  <!--ro, req, enum, domain name, subType:string, desc:domain name-->ipv4
</domain>
<port>
  <!--ro, req, int, port-->1
</port>
<transmitProtocol>
  <!--ro, req, enum, transport protocol, subType:string, desc:transport protocol-->tcp
</transmitProtocol>
<protocol>
  <!--ro, req, enum, protocol, subType:string, desc:protocol-->DAHUA
</protocol>
<username>
  <!--ro, req, string, user name-->test
</username>
<password>
  <!--ro, req, string, password-->test
</password>
<channelMode>
  <!--ro, req, enum, channel mode, subType:string, desc:channel mode-->normal
</channelMode>
<channelType>
  <!--ro, req, enum, channel type, subType:string, desc:channel type-->main
</channelType>
<channelZero>
  <!--ro, opt, int-->1
</channelZero>
<channelNormal>
  <!--ro, opt, int-->1
</channelNormal>
<channelStreaming>
  <!--ro, opt, int-->1
</channelStreaming>
<channelDistributed>
  <!--ro, opt, int-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, enum, whether to enable the function, subType:string, desc:whether to enable the function-->true
  </enabled>
  <domain>
    <!--ro, req, enum, domain name, subType:string, desc:domain name-->ipv4
  </domain>
  <port>
    <!--ro, req, int, port-->1
  </port>
  <transmitProtocol>
    <!--ro, req, enum, transport protocol, subType:string, desc:transport protocol-->tcp
  </transmitProtocol>
</MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
  <!--ro, opt, object-->
  <playbackMode>
    <!--ro, opt, enum, playback mode, subType:string, desc:playback mode-->file name
  </playbackMode>
</EncodeDevInfo>
  <!--ro, opt, object-->
  <domain>
    <!--ro, req, enum, domain name, subType:string, desc:domain name-->ipv4
  </domain>
  <port>
    <!--ro, req, int, port-->1
  </port>
  <transmitProtocol>
    <!--ro, req, enum, transport protocol, subType:string, desc:transport protocol-->tcp
  </transmitProtocol>
  <protocol>
    <!--ro, req, enum, protocol, subType:string, desc:protocol-->DAHUA
  </protocol>
  <username>
    <!--ro, req, string, user name-->test
  </username>
  <password>
    <!--ro, req, string, password-->test
  </password>
  <channelMode>
    <!--ro, req, enum, channel mode, subType:string, desc:channel mode-->normal
  </channelMode>
  <channelType>
    <!--ro, req, enum, channel type, subType:string, desc:channel type-->main
  </channelType>
  <channelZero>
    <!--ro, opt, int-->1
  </channelZero>
  <channelNormal>
    <!--ro, opt, int-->1
  </channelNormal>
</channelNormal>

```

```

<channelStreaming>
  <!--ro, opt, int-->1
</channelStreaming>
<channelDistributed>
  <!--ro, opt, int-->1
</channelDistributed>
</EncodeDevInfo>
<fileName>
  <!--ro, opt, string, file name-->test
</fileName>
<TimeRange>
  <!--ro, opt, object, time range-->
  <beginTime>
    <!--ro, opt, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--ro, opt, time, end time-->00:00:00+08:00
  </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--ro, opt, bool, whether to enable stream encryption-->true
</streamEncryptEnable>
<streamPassword>
  <!--ro, opt, string, stream password, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}, desc:stream password-->test
</streamPassword>
</StreamInput>
<pictureFormat>
  <!--ro, opt, enum, subType:string-->BMP
</pictureFormat>
<signalSourceName>
  <!--ro, opt, string-->test
</signalSourceName>
</SubWindowParam>
</SubWindow>
</SubWindowList>
<wndType>
  <!--ro, opt, enum, window type, subType:string, desc:window type-->signalSource
</wndType>
<zoomEnabled>
  <!--ro, opt, bool, whether to enable the function-->true
</zoomEnabled>
<audioEnabled>
  <!--ro, opt, bool, whether to enable audio-->true
</audioEnabled>
<wndLockKeep>
  <!--ro, opt, bool-->true
</wndLockKeep>
<Graphic>
  <!--ro, opt, object-->
  <windowRegisterID>
    <!--ro, opt, int-->0
  </windowRegisterID>
</Graphic>
<Subtitle>
  <!--ro, opt, object-->
  <moveDirection>
    <!--ro, opt, enum, subType:string-->left
  </moveDirection>
  <moveSpeed>
    <!--ro, opt, enum, moving speed, subType:string, desc:moving speed-->fast
  </moveSpeed>
</Subtitle>
</WallWindow>

```

9.7.11.6 Set parameters of a specific window

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<WallWindow xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version(req, string, protocolVersion)-->
  <id>

```

```

<!--opt, string, No.-->test
</id>
<wndOperateMode>
  <!--opt, enum, "uniformCoordinate,resolutionCoordinate", subType:string, desc:"uniformCoordinate", "resolutionCoordinate"-->uniformCoordinate
</wndOperateMode>
<Rect>
  <!--opt, object-->
  <Coordinate>
    <!--opt, object, coordinates-->
    <x>
      <!--opt, int, X-coordinate, range:[0,1920]-->1
    </x>
    <y>
      <!--opt, int, Y-coordinate, range:[0,1920]-->1
    </y>
  </Coordinate>
  <width>
    <!--opt, int, width, range:[0,1920]-->1
  </width>
  <height>
    <!--opt, int, length, range:[0,1920]-->1
  </height>
</Rect>
<Coordinate>
  <!--opt, object, desc:the origin is the upper-left corner of the screen-->
  <x>
    <!--opt, int, X-coordinate, range:[0,1920]-->1
  </x>
  <y>
    <!--opt, int, Y-coordinate, range:[0,1920]-->1
  </y>
</Coordinate>
<ResolutionRect>
  <!--opt, object-->
  <Coordinate>
    <!--opt, object, coordinates-->
    <x>
      <!--opt, int, X-coordinate-->1
    </x>
    <y>
      <!--opt, int, Y-coordinate-->1
    </y>
  </Coordinate>
  <width>
    <!--opt, int-->1
  </width>
  <height>
    <!--opt, int-->1
  </height>
</ResolutionRect>
<layerIdx>
  <!--opt, int-->1
</layerIdx>
<displayWinNo>
  <!--opt, enum, subType:object, desc:true (enable), false (disable)-->>true
</displayWinNo>
<windowMode>
  <!--opt, enum, "1,4,9,16", subType:int, desc:"1,4,9,16"-->1
</windowMode>
<wndShowMode>
  <!--opt, enum, fullScreenMode", subType:string, desc:fullScreenMode"-->subWndMode
</wndShowMode>
<amplifyingSubWndNo>
  <!--opt, int-->1
</amplifyingSubWndNo>
<wndTopKeep>
  <!--opt, enum, "true,false", subType:bool, desc:true (enable), false (disable)-->true
</wndTopKeep>
<wndOpenKeep>
  <!--opt, enum, "true,false", subType:bool, desc:true (enable), false (disable)-->true
</wndOpenKeep>
<SubWindowList>
  <!--opt, object-->
  <SubWindow>
    <!--opt, object-->
    <id>
      <!--opt, int-->1
    </id>
    <SubWindowParam>
      <!--opt, object-->
      <signalMode>
        <!--opt, enum, subType:string-->video input
      </signalMode>
      <videoInputChannelID>
        <!--opt, string-->1
      </videoInputChannelID>
      <streamingChannelID>
        <!--opt, string-->test
      </streamingChannelID>
      <StreamInput>
        <!--opt, object-->
        <streamInputMode>
          <!--req, enum, stream type, subType:string-->realtime
        </streamInputMode>

```

```

<!--opt, object-->
<StreamInputRealtime>
  <!--opt, object-->
  <durationInUnit>
    <!--opt, int, unit:s-->1
  </durationInUnit>
  <StreamRealtimeUnitList>
    <!--opt, object-->
    <StreamRealtimeUnit>
      <!--opt, object-->
      <streamType>
        <!--req, enum, subType:string-->in URL
      </streamType>
      <StreamInURL>
        <!--opt, object, dep:and,
        {$.SubWindow.SubWindowParam.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
        <URL>
          <!--req, string-->test
        </URL>
      </StreamInURL>
    </StreamByDdns>
    <!--opt, object, dep:and,
    {$.SubWindow.SubWindowParam.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
    <DdnsServerInfo>
      <!--opt, object-->
      <domain>
        <!--req, enum, subType:string-->ipv4
      </domain>
      <port>
        <!--req, int-->80
      </port>
      <ddnsType>
        <!--req, string-->test
      </ddnsType>
      <username>
        <!--req, string-->test
      </username>
      <password>
        <!--req, string-->test
      </password>
    </DdnsServerInfo>
    <EncodeDevInfo>
      <!--opt, object-->
      <domain>
        <!--req, enum, subType:string-->ipv4
      </domain>
      <port>
        <!--req, int-->1
      </port>
      <transmitProtocol>
        <!--req, enum, subType:string-->tcp
      </transmitProtocol>
      <protocol>
        <!--req, enum, subType:string-->
      </protocol>
      <username>
        <!--req, string-->test
      </username>
      <password>
        <!--req, string-->test
      </password>
      <channelMode>
        <!--req, enum, subType:string-->normal
      </channelMode>
      <channelType>
        <!--req, enum, subType:string-->main
      </channelType>
      <channelZero>
        <!--opt, int-->1
      </channelZero>
      <channelNormal>
        <!--opt, int-->1
      </channelNormal>
      <channelStreaming>
        <!--opt, int-->1
      </channelStreaming>
      <channelDistributed>
        <!--opt, int-->1
      </channelDistributed>
    </EncodeDevInfo>
    <MediaGatewayInfo>
      <!--opt, object-->
      <enabled>
        <!--req, enum, whether to enable the function, subType:string, desc:true (enable), false (disable)-->true
      </enabled>
      <domain>
        <!--req, enum, domain name, subType:string-->ipv4
      </domain>
      <port>
        <!--req, int, port-->1
      </port>
      <transmitProtocol>
        <!--req, enum, transport protocol, subType:string-->tcp
      </transmitProtocol>

```



```

</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--opt, object-->
  <EncodeDevInfo>
    <!--opt, object-->
    <domain>
      <!--req, enum, domain name, subType:string-->ipv4
    </domain>
    <port>
      <!--req, int, port-->1
    </port>
    <transmitProtocol>
      <!--req, enum, transport protocol, subType:string-->tcp
    </transmitProtocol>
    <protocol>
      <!--req, enum, protocol, subType:string-->
    </protocol>
    <username>
      <!--req, string, user name, range:[1,32]-->test
    </username>
    <password>
      <!--req, string, password-->test
    </password>
    <channelMode>
      <!--req, enum, channel mode, subType:string-->normal
    </channelMode>
    <channelType>
      <!--req, enum, channel type, subType:string-->main
    </channelType>
    <channelZero>
      <!--opt, int-->1
    </channelZero>
    <channelNormal>
      <!--opt, int-->1
    </channelNormal>
    <channelStreaming>
      <!--opt, int-->1
    </channelStreaming>
    <channelDistributed>
      <!--opt, int-->1
    </channelDistributed>
  </EncodeDevInfo>
</MediaGatewayInfo>
<MediaGatewayInfo>
  <!--opt, object-->
  <enabled>
    <!--req, enum, whether to enable the function, subType:string, desc:true (enable), false (disable)-->true
  </enabled>
  <domain>
    <!--req, enum, subType:string-->ipv4
  </domain>
  <port>
    <!--req, int, port-->1
  </port>
  <transmitProtocol>
    <!--req, enum, transport protocol, subType:string-->tcp
  </transmitProtocol>
</MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeUnit>
</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
  <!--opt, object-->
  <playbackMode>
    <!--opt, enum, playback mode, subType:string-->file name
  </playbackMode>
</StreamInputPlayback>
<EncodeDevInfo>
  <!--opt, object-->
  <domain>
    <!--req, enum, domain name, subType:string-->ipv4
  </domain>
  <port>
    <!--req, int, port-->1
  </port>
  <transmitProtocol>
    <!--req, enum, transport protocol, subType:string-->tcp
  </transmitProtocol>
  <protocol>
    <!--req, enum, protocol, subType:string-->
  </protocol>
  <username>
    <!--req, string, user name-->test
  </username>
  <password>
    <!--req, string, password-->test
  </password>
  <channelMode>
    <!--req, enum, channel mode, subType:string-->normal
  </channelMode>
  <channelType>
    <!--req, enum, channel type, subType:string-->main
  </channelType>
  <channelZero>
    <!--opt, int-->1
  </channelZero>
  <channelNormal>
    <!--opt, int-->1
  </channelNormal>
  <channelStreaming>
    <!--opt, int-->1
  </channelStreaming>
  <channelDistributed>
    <!--opt, int-->1
  </channelDistributed>
</EncodeDevInfo>

```

```

<channelZero>
  <!--opt, int-->1
</channelZero>
<channelNormal>
  <!--opt, int-->1
</channelNormal>
<channelStreaming>
  <!--opt, int-->1
</channelStreaming>
<channelDistributed>
  <!--opt, int-->1
</channelDistributed>
</EncodeDevInfo>
<fileName>
  <!--opt, string, file name-->test
</fileName>
<TimeRange>
  <!--opt, object, time range-->
  <beginTime>
    <!--opt, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--opt, time, end time-->00:00:00+08:00
  </endTime>
</TimeRange>
</StreamInputPlayback>
<streamEncryptEnable>
  <!--opt, enum, subType:string, desc:true (enable), false (disable)-->true
</streamEncryptEnable>
<streamPassword>
  <!--opt, string, dep:and,{$.SubWindow.SubWindowParam.StreamInput.streamEncryptEnable,eq,true}-->test
</streamPassword>
</StreamInput>
</SubWindowParam>
</SubWindow>
</SubWindowList>
<wndType>
  <!--opt, enum, subType:string-->signalSource
</wndType>
<zoomEnabled>
  <!--opt, bool-->true
</zoomEnabled>
<audioEnabled>
  <!--opt, bool, whether to enable audio-->true
</audioEnabled>
<wndLockKeep>
  <!--opt, bool-->true
</wndLockKeep>
<Graphic>
  <!--opt, object-->
  <windowRegisterID>
    <!--opt, int-->0
  </windowRegisterID>
</Graphic>
<Subtitle>
  <!--opt, object-->
  <moveDirection>
    <!--opt, enum, subType:string-->left
  </moveDirection>
  <moveSpeed>
    <!--opt, enum, moving speed, subType:string-->fast
  </moveSpeed>
</Subtitle>
</WallWindow>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode
4	Invalid Operation	inLongDistanceTransmissionModeTheLocalSignalSourcesOfSubBoardInSlot6CannotBeDisp
4	Invalid Operation	theConferenceModeDoesNotSupportWindowRoaming
4	Invalid Operation	oneTouchSubScreenSyncDoesNotSupportSplicedSourceSynchronization
4	Invalid Operation	conferenceModeProhibitsEmptyOutputsOnMainDisplayWall
4	Invalid Operation	conferenceModeExceedsMaxSubScreenSplitsLimit
4	Invalid Operation	networkSourcesUnsupportedInConferenceMode
4	Invalid Operation	splitScreenNotSupportedWithSyncLockOn
4	Invalid Operation	singleLayerOnlySupportedWithSyncLockOn
4	Invalid Operation	onlySourceLayerAllowedWithSyncLockOn

9.7.11.7 Delete a specific window

Request URL

DELETE /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

9.7.11.8 Bottom the window

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/bottom

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

9.7.11.9 Get single configuration capabilities of sub-windows

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SubWindow xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, sub-window, attr:version{req, string, protocolVersion}-->
  <id min="1" max="10">
    <!--ro, opt, int, sequence number, attr:min{req, int},max{req, int}-->1
  </id>
  <SubWindowParam>
    <!--ro, opt, object, sub-window parameter-->
    <signalMode opt="video input, stream id, stream setting, alarm linkage">
      <!--ro, opt, enum, signal mode, subType:string, attr:opt{req, string}, desc:"videoInput" (video input), "streamId" (stream ID), "streamSetting" (stream settings), "alarmLinkage" (alarm linkage), "sharedSignal" (shared signal source)-->video input
    </signalMode>
    <videoInputChannelID min="1" max="10">
      <!--ro, opt, string, video input channel ID, attr:min{req, int},max{req, int}-->1
    </videoInputChannelID>
    <streamingChannelID min="1" max="10">
      <!--ro, opt, string, stream channel ID, attr:min{req, int},max{req, int}-->test
    </streamingChannelID>
    <StreamInput>
      <!--ro, opt, object, stream input-->
      <streamInputMode opt="realtime, playback">
        <!--ro, req, enum, stream type, subType:string, attr:opt{req, string}, desc:"realtime" (real-time stream), "playback" (playback stream)-->realtime
      </streamInputMode>
      <StreamInputRealtime>
        <!--ro, opt, object, real-time stream information-->
        <durationInUnit min="1" max="10">
          <!--ro, opt, int, decoding display interval of real-time stream, unit:s, attr:min{req, int},max{req, int}-->1
        </durationInUnit>
        <StreamRealtimeUnitList>
          <!--ro, opt, array, real-time stream list, subType:object-->
          <StreamRealtimeUnit>
            <!--ro, opt, object, single real-time stream unit-->
            <streamType opt="in URL, by ddns, by domain">
              <!--ro, req, enum, stream type, subType:string, attr:opt{req, string}, desc:"in URL", "by ddns", "by domain"-->in URL
            </streamType>
            <StreamInURL>
              <!--ro, opt, object, URL stream, dep:and,
              {$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,in URL}-->
              <URL min="1" max="10">
                <!--ro, req, string, URL address, attr:min{req, int},max{req, int}-->test
              </URL>
            </StreamInURL>
            <StreamByDdns>
              <!--ro, opt, object, DDNS stream, dep:and,
              {$.SubWindow.SubWindowParam.StreamInput.StreamInputRealtime.StreamRealtimeUnitList.StreamRealtimeUnit.streamType,eq,by ddns}-->
              <DdnsServerInfo>
                <!--ro, opt, object, DDNS server information-->
                <domain opt="ipv4,ipv6,domain">
                  <!--ro, req, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
                </domain>
                <port min="1" max="10">
                  <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->80
                </port>
                <ddnsType opt="hiDdns,noIp,...">
                  <!--ro, req, string, DDNS type, attr:opt{req, string}-->test
                </ddnsType>
                <username min="1" max="10">
                  <!--ro, req, string, user name., attr:min{req, int},max{req, int}-->test
                </username>
                <password min="1" max="10">
                  <!--ro, req, string, password, attr:min{req, int},max{req, int}-->test
                </password>
              </DdnsServerInfo>
            <EncodeDevInfo>
              <!--ro, opt, object, information of device in playback-->
              <domain opt="ipv4,ipv6,domain">
                <!--ro, req, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
              </domain>
              <port min="1" max="10">
                <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
              </port>
              <transmitProtocol opt="tcp,udp,mcast">
                <!--ro, req, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "multicast" (multicast protocol)-->tcp
              </transmitProtocol>
            </EncodeDevInfo>
          </StreamRealtimeUnit>
        </StreamRealtimeUnitList>
      </StreamInputRealtime>
    </StreamInput>
  </SubWindowParam>
</SubWindow>
```

```

</transmitProtocol>
<username min="1" max="10">
  <!--ro, req, string, user name., attr:min{req, int},max{req, int}-->test
</username>
<password min="1" max="10">
  <!--ro, req, string, password, attr:min{req, int},max{req, int}-->test
</password>
<channelMode opt="normal,zero,streaming,distributed">
  <!--ro, req, enum, channel mode, subType:string, attr:opt{req, string}, desc:"normal", "zero", "streaming", "distributed"-->normal
</channelMode>
<channelZero min="1" max="10">
  <!--ro, opt, int, channel-zero, attr:min{req, int},max{req, int}-->1
</channelZero>
<channelNormal min="1" max="10">
  <!--ro, opt, int, normal channel, attr:min{req, int},max{req, int}-->1
</channelNormal>
<channelStreaming min="1" max="10">
  <!--ro, opt, int, stream channel, attr:min{req, int},max{req, int}-->1
</channelStreaming>
<channelDistributed min="1" max="10">
  <!--ro, opt, int, distributed channel, attr:min{req, int},max{req, int}-->1
</channelDistributed>
</EncodeDevInfo>
<MediaGatewayInfo>
  <!--ro, opt, object, media gateway information-->
  <enabled opt="true,false">
    <!--ro, req, enum, whether to enable, subType:string, attr:opt{req, string}, desc:true (enable), false (disable)-->true
  </enabled>
  <domain opt="ipv4,ipv6,domain">
    <!--ro, req, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
  </domain>
  <port min="1" max="10">
    <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
  </port>
  <transmitProtocol opt="tcp,udp,mcast">
    <!--ro, req, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
  </transmitProtocol>
</MediaGatewayInfo>
</StreamByDdns>
<StreamByDomain>
  <!--ro, opt, object, domain stream-->
  <EncodeDevInfo>
    <!--ro, opt, object, information of device in playback-->
    <domain opt="ipv4,ipv6,domain">
      <!--ro, req, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port min="1" max="10">
      <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <transmitProtocol opt="tcp,udp,mcast">
      <!--ro, req, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
    </transmitProtocol>
    <username min="1" max="10">
      <!--ro, req, string, user name., attr:min{req, int},max{req, int}-->test
    </username>
    <password min="1" max="10">
      <!--ro, req, string, password, attr:min{req, int},max{req, int}-->test
    </password>
    <channelMode opt="normal,zero,streaming,distributed">
      <!--ro, req, enum, channel mode, subType:string, attr:opt{req, string}, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelZero min="1" max="10">
      <!--ro, opt, int, channel-zero, attr:min{req, int},max{req, int}-->1
    </channelZero>
    <channelNormal min="1" max="10">
      <!--ro, opt, int, normal channel, attr:min{req, int},max{req, int}-->1
    </channelNormal>
    <channelStreaming min="1" max="10">
      <!--ro, opt, int, stream channel, attr:min{req, int},max{req, int}-->1
    </channelStreaming>
    <channelDistributed min="1" max="10">
      <!--ro, opt, int, distributed channel, attr:min{req, int},max{req, int}-->1
    </channelDistributed>
  </EncodeDevInfo>
  <MediaGatewayInfo>
    <!--ro, opt, object, media gateway information-->
    <enabled opt="true,false">
      <!--ro, req, enum, whether to enable, subType:string, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
    <domain opt="ipv4,ipv6,domain">
      <!--ro, req, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port min="1" max="10">
      <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <transmitProtocol opt="tcp,udp,mcast">
      <!--ro, req, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast"
(multicast protocol)-->tcp
    </transmitProtocol>
  </MediaGatewayInfo>
</StreamByDomain>
</StreamRealtimeInit>

```

```

</StreamRealtimeUnitList>
</StreamInputRealtime>
<StreamInputPlayback>
  <!--ro, opt, object, playback stream information-->
  <playbackMode>
    <!--ro, opt, enum, playback mode, subType:string, desc:"file name" (playback by file name), "time range" (playback by time range)-->file name
  </playbackMode>
  <EncodeDevInfo>
    <!--ro, opt, object, information of device in playback-->
    <domain opt="ipv4,ipv6,domain">
      <!--ro, req, enum, domain name, subType:string, attr:opt{req, string}, desc:"ipv4", "ipv6", "domain"-->ipv4
    </domain>
    <port min="1" max="10">
      <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <transmitProtocol opt="tcp,udp,mcast">
      <!--ro, req, enum, transport protocol, subType:string, attr:opt{req, string}, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "multicast"
      (multicast protocol)-->tcp
    </transmitProtocol>
    <username min="1" max="10">
      <!--ro, req, string, user name., attr:min{req, int},max{req, int}-->test
    </username>
    <password min="1" max="10">
      <!--ro, req, string, password, attr:min{req, int},max{req, int}-->test
    </password>
    <channelMode opt="normal,zero,streaming,distributed">
      <!--ro, req, enum, channel mode, subType:string, attr:opt{req, string}, desc:"normal", "zero", "streaming", "distributed"-->normal
    </channelMode>
    <channelZero min="1" max="10">
      <!--ro, opt, int, channel-zero, attr:min{req, int},max{req, int}-->1
    </channelZero>
    <channelNormal min="1" max="10">
      <!--ro, opt, int, normal channel, attr:min{req, int},max{req, int}-->1
    </channelNormal>
    <channelStreaming min="1" max="10">
      <!--ro, opt, int, stream channel, attr:min{req, int},max{req, int}-->1
    </channelStreaming>
    <channelDistributed min="1" max="10">
      <!--ro, opt, int, distributed channel, attr:min{req, int},max{req, int}-->1
    </channelDistributed>
  </EncodeDevInfo>
  <fileName min="1" max="10">
    <!--ro, opt, string, file name, attr:min{req, int},max{req, int}-->test
  </fileName>
  <TimeRange>
    <!--ro, opt, object, time range-->
    <beginTime>
      <!--ro, opt, time, start time-->00:00:00+08:00
    </beginTime>
    <endTime>
      <!--ro, opt, time, end time-->00:00:00+08:00
    </endTime>
  </TimeRange>
</StreamInputPlayback>
<streamEncryptEnable opt="true,false">
  <!--ro, opt, enum, whether to enable stream encryption, subType:string, attr:opt{req, string}, desc:true (enable), false (disable)-->true
</streamEncryptEnable>
<streamPassword min="1" max="10">
  <!--ro, opt, string, stream password, attr:min{req, int},max{req, int}, desc:sensitive information encryption. It is valid when the value of
  streamEncryptEnable is true-->test
</streamPassword>
</StreamInput>
<AlarmLinkageInfoList size="10">
  <!--ro, opt, array, alarm linkage information list, subType:object, attr:size{req, int}-->
  <AlarmLinkageInfo>
    <!--ro, req, object, alarm linkage information-->
    <alarmLinkageID min="1" max="10">
      <!--ro, req, int, alarm linkage No., attr:min{req, int},max{req, int}-->1
    </alarmLinkageID>
  </AlarmLinkageInfo>
</AlarmLinkageInfoList>
<subWindowMode opt="normal,polling">
  <!--ro, opt, string, sub-window mode, attr:opt{req, string}-->test
</subWindowMode>
<hostname opt="ipv4Address,ipv6Address">
  <!--ro, opt, enum, address type, subType:string, attr:opt{req, string}, desc:"ipv4Address", "ipv6Address"-->ipv4Address
</hostname>
<ipv4Address min="1" max="16">
  <!--ro, opt, string, IPv4 address, attr:min{req, int},max{req, int}, desc:it is valid when value of signalMode is sharedSingal-->192.168.1.1
</ipv4Address>
<ipv6Address min="1" max="128">
  <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}, desc:it is valid when value of signalMode is sharedSingal-->fe80::4ba5:790e
</ipv6Address>
<port min="0" max="65535">
  <!--ro, opt, int, port No., attr:min{req, int},max{req, int}, desc:it is valid when value of signalMode is sharedSingal-->1
</port>
<trackNum>
  <!--ro, opt, object, track count, desc:the ultra HD source is split into multiple tracks, requiring the acquisition and splicing of multiple signals.
  It is valid when value of signalMode is sharedSingal-->
  <x min="0" max="16">
    <!--ro, req, int, track row, attr:min{req, int},max{req, int}-->1
  </x>
  <y min="0" max="16">

```

```

        <!--ro, req, int, track column, attr:min{req, int},max{req, int}-->1
    </y>
</trackNum>
</SubwindowParam>
</Subwindow>

```

9.7.11.10 Get parameters of decoding delay

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/decodeDelay

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<DecodeDelayParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <param>
    <!--ro, opt, enum, "default,mostRealTime,moreRealTime,balance,moreFluency,mostFluency,auto", subType:string,
    desc:"default,mostRealTime,moreRealTime,balance,moreFluency,mostFluency,auto"-->default
  </param>
</DecodeDelayParam>

```

9.7.11.11 Get decoding delay capability

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/decodeDelay/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<DecodeDelayParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <param>
    <!--ro, opt, enum, "default,mostRealTime,moreRealTime,balance,moreFluency,mostFluency,auto", subType:string,
    desc:"default,mostRealTime,moreRealTime,balance,moreFluency,mostFluency,auto"-->default
  </param>
</DecodeDelayParam>

```

9.7.11.12 Get the configuration capability of full-frame-rate fluent video mode

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/sub/<VWSWID>/fullFrame/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--
VWSWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FullFrameParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration capability of full-frame-rate fluent video mode, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether the device supports hot spare exception, attr:opt{req, string}, desc:whether the device supports hot spare exception-->true
  </enabled>
</FullFrameParam>
```

9.7.11.13 Top the window

Request URL

PUT /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/<VWMWID>/top

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--
VWMWID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	multipleVideowallClientConflict	0x4000A4F8	--

9.7.11.14 Get the window configuration capability of the video wall

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

hie.u.tu@tacorp.vn
CNG TY C PHN U T CNG NGH THIN N

```

<?xml version="1.0" encoding="UTF-8"?>

<WallWindowCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, window configuration capability of the video wall, attr:version{req, string, protocolVersion}-->
  <windowMode opt="1,4,6,8,9,16,25,36,49,64">
    <!--ro, opt, enum, supported split mode, subType:int, attr:opt{req, string}, desc:1 (1 window), 4 (4 windows), 6 (6 windows), 8 (8 windows), 9 (9 windows), 16 (16 windows), 25 (25 windows), 36 (36 windows), 49 (49 windows), 64 (64 windows)-->1
  </windowMode>
  <CycleCap>
    <!--ro, opt, object, auto-switch capability-->
    <maxCycleNum>
      <!--ro, opt, int, maximum number of auto-switches-->0
    </maxCycleNum>
    <singleCycleMaxSignalNum>
      <!--ro, opt, int, maximum number of signal sources per auto-switch-->0
    </singleCycleMaxSignalNum>
    <duration min="0" max="10">
      <!--ro, opt, int, time duration, attr:min{req, int},max{req, int}-->0
    </duration>
    <maxMonitorNum>
      <!--ro, opt, int, maximum number of cameras per auto-switch-->64
    </maxMonitorNum>
    <isSupportCycleCtrl>
      <!--ro, opt, int, whether the device supports auto-switch control-->0
    </isSupportCycleCtrl>
  </CycleCap>
  <isSupportWinTopBottom>
    <!--ro, opt, bool, whether the device supports window always-on-top while windowing-->true
  </isSupportWinTopBottom>
  <isSupportPlayBack>
    <!--ro, opt, bool, streaming mode of sub window-->true
  </isSupportPlayBack>
  <isSupportPicCapture>
    <!--ro, opt, bool, whether the sub window supports capture-->true
  </isSupportPicCapture>
  <isSupportDecodeDelay>
    <!--ro, opt, bool, whether the device supports decoding delay-->true
  </isSupportDecodeDelay>
  <wndWidthAlignUnit>
    <!--ro, opt, int, width alignment unit of physical resolution window (Netra output board)-->1
  </wndWidthAlignUnit>
  <wndHeightAlignUnit>
    <!--ro, opt, int, height alignment unit of physical resolution window (Netra output board)-->1
  </wndHeightAlignUnit>
  <resWndWidth min="1" max="10">
    <!--ro, opt, int, physical resolution window width, attr:min{req, int},max{req, int}-->1
  </resWndWidth>
  <resWndHeight min="1" max="10">
    <!--ro, opt, int, physical resolution window height, attr:min{req, int},max{req, int}-->1
  </resWndHeight>
  <isSupportSubWndAmplify>
    <!--ro, opt, bool, whether the device supports sub-window zooming-->true
  </isSupportSubWndAmplify>
  <isSupportFullFrame>
    <!--ro, opt, bool, whether the device supports sub-window fluent video-->true
  </isSupportFullFrame>
  <isSptMultiScreenGetSubStream>
    <!--ro, opt, bool, whether it supports getting sub-stream in multi-screen mode-->true
  </isSptMultiScreenGetSubStream>
  <isSptCatchStreamAlarmHint>
    <!--ro, opt, bool, whether it supports configuring streaming alarm notification-->true
  </isSptCatchStreamAlarmHint>
  <hisiWndWidthAlignUnit>
    <!--ro, opt, int, width alignment unit of physical resolution window (output board)-->1
  </hisiWndWidthAlignUnit>
  <hisiWndHeightAlignUnit>
    <!--ro, opt, int, height alignment unit of physical resolution window (output board)-->1
  </hisiWndHeightAlignUnit>
  <isSupportClearStreamCfg>
    <!--ro, opt, bool, whether the device supports clearing streaming configurations, desc:it is enabled by default application scenario: when the client switches window layout from 9-window to 4-window, and then back to 9-window, the streaming parameters of 5 newly opened windows will be cleared and the streaming will not restore-->true
  </isSupportClearStreamCfg>
  <subScreenWindowMode opt="1,4">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </subScreenWindowMode>
</WallWindowCap>

```

9.7.11.15 Get the parameters configuration capability of sub-stream in multi-screen mode

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/subSteam/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
{
  "MutiScreenSubStreamCap": {
    /*ro, opt, object, the configuration of sub-stream parameters*/
    "enabled": {
      /*ro, opt, object, whether to enable the function or not*/
      "@opt": [true, false]
      /*ro, opt, array, whether to enable the function or not, subType:bool*/
    },
    "winConutLimit": {
      /*ro, req, object, threshold number of divided windows*/
      "@opt": ["1", "4", "6", "8", "9", "16", "32", "64"]
      /*ro, req, array, threshold number of divided windows, subType:string, desc:threshold number of divided windows*/
    },
    "streamType": {
      /*ro, opt, object*/
      "@opt": ["main", "sub"]
      /*ro, opt, array, subType:string*/
    }
  }
}
```

9.7.11.16 Get the configuration parameters of the stream type for streaming when the number of windows exceeds the limit

Request URL

GET /ISAPI/DisplayDev/VideoWall/<videoWallID>/windows/subSteam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
videoWallID	string	--

Request Message

None

Response Message

```
{
  "MutiScreenSubStream": {
    /*ro, opt, object*/
    "enabled": true,
    /*ro, opt, bool, whether to enable the function*/
    "winConutLimit": "test",
    /*ro, req, string, threshold number of divided windows, desc:When the number of divided windows exceeds the threshold, streams of low performance will be used*/
    "streamType": "main"
    /*ro, opt, string*/
  }
}
```

9.7.11.17 Get the pre-editing capability of video wall

Request URL

GET /ISAPI/DisplayDev/VideoWall/preEdit/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "wallID": {
    /*ro, req, object, video wall number*/
    "@min": 0,
    /*ro, opt, int, minimum value*/
    "@max": 0
    /*ro, req, int, maximum value*/
  },
  "operateType": {
    /*ro, opt, object, operation type*/
    "@opt": ["start", "save", "forward", "back", "up", "undo", "quit"]
    /*ro, req, array, options, subType:string*/
  }
}

```

9.8 IoT Central Control

9.8.1 IoT Gateway

9.8.1.1 Get general IoT gateway capability

Request URL

GET /ISAPI/IoTGateway/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "IoTGatewayCap": {
    /*ro, req, object, general IoT gateway capability*/
    "isSptChannelRelatedInfo": true,
    /*ro, opt, bool*/
    "isSptRS485Cfg": true,
    /*ro, opt, bool*/
    "isSptGetExternalDeviceInfo": true,
    /*ro, opt, bool*/
    "isSptOsdManualPageTurn": true,
    /*ro, opt, bool*/
    "isSptAirControl": true,
    /*ro, opt, bool*/
    "isSptgeneralOutputControl": true,
    /*ro, opt, bool*/
    "isSptAlarmLinkage": true,
    /*ro, opt, bool*/
    "isSptAlarmVariableInfo": true,
    /*ro, opt, bool*/
    "isSptAlarmPoint": true,
    /*ro, opt, bool*/
    "isSptDataUploadMode": true,
    /*ro, opt, bool*/
    "isSptModbusFile": true,
    /*ro, opt, bool*/
    "isSptRS485File": true,
    /*ro, opt, bool*/
    "isSptSensorInfo": true,
    /*ro, opt, bool*/
    "isSptSensorData": true,
    /*ro, opt, bool*/
    "isSptGatewayDeviceInfo": true,
    /*ro, opt, bool*/
    "isSptVoiceBroadcast": true,
    /*ro, opt, bool*/
    "isSptSafetyHelmetCfg": true,
    /*ro, opt, bool*/
    "isSptSafetyHelmetAdvancedCfg": true,
    /*ro, opt, bool*/
    "isSptNetworkSensorCfg": true,
    /*ro, opt, bool*/
    "isSptVariableCfg": true,
    /*ro, opt, bool*/
    "isSptEnergyManagement": true,
    /*ro, opt, bool*/
    "isSptPlanCfg": true,
    /*ro, opt, bool*/
    "isSptAlarmSOSCfg": true,
    /*ro, opt, bool*/
    "isSptLightControl": true
  }
}

```

```

        "isSupportRealTimeTracking": true,
        /*ro, opt, bool*/
        "isSptCloudPlatformCfg": true,
        /*ro, opt, bool*/
        "isSptRealTimeTracking": true,
        /*ro, opt, bool*/
        "isSupportDataAlarmInterval": true,
        /*ro, opt, bool*/
        "isSupportSubscribeVariable": true,
        /*ro, opt, bool*/
        "isSupportVariableLinkage": true,
        /*ro, opt, bool*/
        "isSupport433SensorParam": true,
        /*ro, opt, bool*/
        "isSupport433SensorRFParmTemplate": true,
        /*ro, opt, bool*/
        "isSupport433WirelessTransfer": true,
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        "isSupportIoTGatewayConnect": true,
        /*ro, opt, bool*/
        "isSupportDataCollectSite": true,
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        "isSupportDataProcessSite": true,
        /*ro, opt, bool*/
        "isSupportAutoMapDataProcessSite": true,
        /*ro, opt, bool*/
        "isSupportMovingRing": true,
        /*ro, opt, bool*/
        "isSupportVariableAlarmScriptRule": true,
        /*ro, opt, bool*/
        "isSupportProductionDadaBordInfo": true,
        /*ro, opt, bool*/
        "isSupportVariableUploadPlatform": true,
        /*ro, opt, bool*/
        "isSupportVariableDataSearch": true,
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        "isSupportVariableAlarmDataSearch": true,
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        "isSupportProcessVariableCurrentData": true,
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        "isSupportVariableAlarmNumStatistics": true,
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        "isSupportVariableAlarmDistributionStatistics": true,
        /*ro, opt, bool*/
        "isSupportSwitchLinkage": true,
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        "isSupportWiFiCfg": true,
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        "isSupportUploadPlugInMgr": true,
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        "isSupportCollectionPlugInMgr": true,
        /*ro, opt, bool*/
        "isSupportDatacollectScript": true,
        /*ro, opt, bool*/
        "isSupportDatacollectSiteSynchronize": true,
        /*ro, opt, bool*/
        "AIOTType": "B81",
        /*ro, opt, enum, subType:string*/
        "IOTActionCap": {
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            "isSupportPDUMgr": true,
            /*ro, opt, bool*/
            "isSupportProjectionScreenMgr": true,
            /*ro, opt, bool*/
            "isSupportRelayMgr": true,
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            "isSupportProjectionSerialMgr": true,
            /*ro, opt, bool*/
            "isSupportRecordingMgr": true,
            /*ro, opt, bool*/
            "isSupportInfraredTransponderMgr": true,
            /*ro, opt, bool*/
            "isSupportDimmableDriverMgr": true,
            /*ro, opt, bool*/
            "isSupportTempAndHumSensorMgr": true,
            /*ro, opt, bool*/
            "isSupportSmartPlugMgr": true,
            /*ro, opt, bool*/
            "isSupportCurtainMgr": true,
            /*ro, opt, bool*/
            "isSupportAllInOneSensorMgr": true,
            /*ro, opt, bool*/
            "isSupportSwitchSocketMgr": true,
            /*ro, opt, bool*/
            "isSupportSwitchInMgr": true,
            /*ro, opt, bool*/
        },
        "PlaceIOTActionCap": {
        /*ro, opt, object*/
            "isSupportPlaceAction": true,

```

```

        /*ro, opt, bool*/
        "isSupportPlaceDeviceMgr": true,
        /*ro, opt, bool*/
        "isSupportPlaceScene": true,
        /*ro, opt, bool*/
        "isSupportPlaceEnergyManagement": true
        /*ro, opt, bool*/
    },
    "AppBackstageCap": {
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        "isSupportLightBasicMgr": true,
        /*ro, opt, bool*/
        "isSupportAirConditionerBasicMgr": true,
        /*ro, opt, bool*/
        "isSupportProjectionScreenBasicMgr": true,
        /*ro, opt, bool*/
        "isSupportCurtainBasicMgr": true,
        /*ro, opt, bool*/
        "isSupportDisplayBasicMgr": true,
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        "isSupportPlaceSceneBasicMgr": true,
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        "isSupportAuthorityGlobalFunction": true,
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        "isSupportAuthorityControl": true,
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        "isSupportAudioBasicMgr": true,
        /*ro, opt, bool*/
        "isSupportPlaceEnvironmentMgr": true,
        /*ro, opt, bool*/
        "isSupportCustomSwitchBasicMgr": true
        /*ro, opt, bool*/
    },
    "IOTControlCap": {
        /*ro, opt, object*/
        "isSupportLightGlobalControl": true,
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        "isSupportProjectionScreenControl": true,
        /*ro, opt, bool*/
        "isSupportAirConditionerControl": true,
        /*ro, opt, bool*/
        "isSupportAudioControl": true,
        /*ro, opt, bool*/
        "isSupportDisplayControl": true,
        /*ro, opt, bool*/
        "isSupportCustomSwitchControl": true
        /*ro, opt, bool*/
    },
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    /*ro, opt, bool*/
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    /*ro, opt, bool*/
    "isSupportSceneLink": true,
    /*ro, opt, bool*/
    "isSupportVideoMatrix": true,
    /*ro, opt, bool*/
    "isSupportModifyFDCInfo": true,
    /*ro, opt, bool*/
    "isSupportFDCPriceCfg": true,
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    "isSupportFDCEnergyConsumption": true,
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    "isSupportSearchFDCEnergyConsumption": true,
    /*ro, opt, bool*/
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    "isSupportSearchAlarmRules": true,
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    "isSupportModifyAlarmRules": true,
    /*ro, opt, bool*/
    "isSupportDeleteAlarmRules": true,
    /*ro, opt, bool*/
    "isSupportSensorDataReportInterval": true,
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    "isSupportSensorDataCollectionInterval": true,
    /*ro, opt, bool*/
    "isSupportChildDevCollectionInterval": true,
    /*ro, opt, bool*/
    "isSupportGateCollectionInterval": true,
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    "isSupportGateReportInterval": true,
    /*ro, opt, bool*/
    "isSupportDeleteSensorList": true,
    /*ro, opt, bool*/

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/*ro, opt, bool*/
"isSupportGetSensorBasicControl": true,
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"isSupportSetSensorBasicControl": true,
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"isSupportGetUnclearAlarmList": true,
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"isSupportEnvironmentalInformationCfg": true,
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"isSupportEquipmentAccessCfg": true,
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"isSupportSearchSingleFDCCconsumption": true,
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"isSupportGetLoadingActionPlugInInfo": true,
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"isSupportLoadingActionPlugIn": true,
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"isSupportImportActionPlugInConfigFile": true,
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"isSupportDeleteActionPlugIn": true,
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"isSupportDeleteTrial": true,
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"isSupportAddTrial": true,
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"isSupportGetAppBackstageTrial": true,
/*ro, opt, bool*/
"isSupportDeleteFDCInfo": true,
/*ro, opt, bool*/
"isSupportGetSensorRealTimeData": true,
/*ro, opt, bool*/
"isSupportSearchLiveChannelAction": true,
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"isSupportDeleteLiveChannelAction": true,
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"isSupportModifyLiveChannelAction": true,
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"isSupportGetLiveChannelInfoList": true,
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"isSupportSearchLiveChannelInfo": true,
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"isSupportModifyCorrectionCfg": true,
/*ro, opt, bool*/
"isSupportGetCorrectionCfg": true,
```



```
/*ro, opt, bool*/
"isSupportGetAddedPCList": true,
/*ro, opt, bool*/
"isSupportGetAppBackstagePCList": true,
/*ro, opt, bool, POST /ISAPI/IoTGateway/PCControl/GetAppBackstagePCList?format=json, desc:POST /ISAPI/IoTGateway/PCControl/GetAppBackstagePCList?
format=json*/
"isSupportAlarmRules": true,
/*ro, opt, bool, /ISAPI/System/AlarmRules/AlarmRules?format=json, desc:/ISAPI/System/AlarmRules/AlarmRules?format=json*/
"childmanageCap": {
/*ro, opt, object, gateway sub device management*/
  "isSupportAddChild": true,
  /*ro, opt, bool, whether it supports adding sub-device, desc:whether it supports adding sub-device*/
  "isSupportModifyChild": true,
  /*ro, opt, bool*/
  "isSupportRemoveChild": true,
  /*ro, opt, bool*/
  "isSupportScanControl": true,
  /*ro, opt, bool*/
  "isSupportSeachDevice": true,
  /*ro, opt, bool*/
  "isSupportSearchChild": true,
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  "isSupportFullchild": true,
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  "isSupportchangechild": true
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"isSupportSearchSensorRuleCap": true,
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"isSupportSearchChildScheduleCfg": true,
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"isSupportSetChildScheduleCfg": true,
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"isSupportSearchSensorDetailInfo": true,
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"isSupportGetActionPlugInInfo": true,
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"isSupportSenseDataUpdate": true,
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"isSupportCalibrateInclinationAccelerometer": true,
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"isSupportGetGNSSInfo": true,
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"isSupportSearchGasSenseDetectionResult": true,
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"isSupportGetMicUnitRouterInfo": true,
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"isSupportProductionDadaVedioTrigger": true,
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```

```
"isSupportSearchHistorySensorData": true,
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"isSupportSensorTelemeterDataUploadMode": true,
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"isSupportSearchGasSenseDetectionRuleParams": true,
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"isSupportGetUWBLabelInfoList": true,
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"isSupportUWBBaseStationPilotLampParam": true,
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"isSupportAntiCollisionFilterLabelList": true,
/*ro, opt, bool*/
"isSupportSearchGasSenseDetectionStatus": true,
/*ro, opt, bool*/
"isSupportGetAsynUWBLabelInfo": true,
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"isSupportUWBLabelPositioningParams": true,
/*ro, opt, bool*/
"isSupportUWBBaseStationBasicInfo": true,
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"isSupportPrecisePositioningDataExport": true,
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"isSupportTOFDetectParams": true,
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"isSupportDeviceLocationAsyncParams": true,
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"isSupportModifyProductionDataTriggerParams": true,
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"isSupportGetProductionDataTriggerParams": true,
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"isSupportModifyProductionDataScheduleParams": true,
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"isSupportTranscodeModeParams": true,
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"isSupportSetAlarmRules": true,
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"isSupportAlarmRulesTriggerParams": true,
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"isSupportChildmanage": true,
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"isSupportDataUploadBasicParam": true,
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"isSupportTemperatureRealTimeData": true,
/*ro, opt, bool*/
"isSupportWeighingSetUpParams": true,
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"isSupportWeighingModeParams": true,
/*ro, opt, bool*/
"isSupportSearchVehicleWeighingResult": true,
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"isSupportGetCorrectionStatus": true,
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"isSupportExportAuthDevInfoTemplate": true,
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"isSupportModifyCustomVariable": true,
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"isSupportSearchCustomVariableCurrentData": true,
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"isSupportAddCustomVariable": true,
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"isSupportSearchCustomVariable": true,
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"isSupportGetSensorDevTypeMappings": true,
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```

```

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"isSupportModifyCustomVariableTriggerParams": true,
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"isSupportModifyCustomVariableGroup": true,
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"isSupportAddCustomVariableGroup": true,
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"isSupportDeleteCustomVariableGroupList": true,
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"isSupportAddOperateRegion": true,
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"isSupportModifyNetSerialPort": true,
/*ro, opt, bool*/
"isSupportGetNetSerialPortList": true,
/*ro, opt, bool*/
"isSupportSerialNetModeParams": true,
/*ro, opt, bool*/
"isSupportGetSensorCollectDataFile": true,
/*ro, opt, bool*/
"isSupportSearchOpticalFiberAlarmRecord": true,
/*ro, opt, bool, whether it supports searching for alarm records of vibration fiber-optic sensing servers, desc:whether it supports searching for
alarm records of vibration fiber-optic sensing servers*/
"isSupportExecutePlaceScene": true,
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"isSupportDeleteAirConditionerAction": true,
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"isSupportSearchAirConditionerAction": true,
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"isSupportAddAirConditionerAction": true,
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"isSupportModifyAirConditionerAction": true,
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"isSupportModifySensorParams": true,
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"isSupportModifyLightControlInfo": true,
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"isSupportAppSceneCategoryParams": true,
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"isSupportDeleteNetSerialPortList": true,
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"isSupportSearchAirConditionerControlInfo": true,
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"isSupportSearchLightControlInfoList": true,
/*ro, opt, bool*/
"isSupportTemperatureMaxThreshold": true,
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"isSupportGetAudioControlInfoList": true,
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```

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"isSupportAddDevCtrlOperation": true,
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"isSupportDeleteDevCtrlOperation": true,
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"isSupportSearchSingleDevCtrlOperation": true,
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"isSupportGetThermometryInfo": true,
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"isSupportExportHistoryTemperatureData": true,
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"isSupportGetHistoryTemperatureData": true,
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"isSupportThermometryOutputParams": true,
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"isSupportThermometryAdvancedParams": true,
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format=json*/
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10 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on <https://tpp.hikvision.com> and visit our Training Center: <https://tpp.hikvision.com/tpp/Training>. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.

hie.u.tu@tacorp.vn
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